

Processor Programming Reference (PPR) for AMD Family 19h Model 11h, Revision B2 Processors Volume 5 of 9

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7 Security

[ASP](#) firmware may program the RSMU in an IP to conditionally block read or write requests that originate from SMN initiators that do not meet specific Trust, Initiator ID, or Unit ID requirements. The SMN Trust Level for Originating IPs is defined in Table 150 [Trust Level Descriptions]. All SMN traffic originating from IP not included in the table is considered untrusted.

If a security policy for a register exists, it is expressed through a ternary expression that describes the registers accessibility as a function of the initiator trust level. An exception to this method of security policy expression exists for all DBGU registers. All DBGU registers are only accessible by implicitly trusted initiators.

If a part has been unlocked through the secure unlock process, the security policy is not applied, and the accessibility of a register is defined by its hardware accessibility alone.

Also refer to 11.8 [Data Fabric Security].

7.1 Trust Level Definitions

Table 150: Trust Level Descriptions

SMN Trust Level (InitiatorTrust)	Description	Originating IPs	InitiatorID/UnitID/ SecFlag
0	Implicitly Trusted	Secure MP0 (PSP)	MP0.* WAFLC.MP0.*
1	Reserved		
2	Highly Trusted	MP1 x86 microcode RSMU DF	MP1.* MP5 .* CCXSEC0.* CCXSEC1.* WAFLC.MP1.* RSMU.* DF.* MPIQ
3	Trusted	Trusted IP	MPDMA
4	Reserved		
5	Somewhat Trusted	x86 SMM Code	
6	Reserved		
7	Untrusted	All other IPs including HDT TST_AXI_MASTER	

When HDT is making a request on behalf of an MP or x86 core and not through TST_AXI_MASTER, the trust level of the request matches the trust level that the MP or core has at the time of the request.

Table 151: IOD Initiator ID and Unit ID Map

Initiator ID	Initiator Name	Unit ID	Unit Name
255	MP0	55	CPU Sec
		52	CCP1
		5	SysHub
		4	SMN

1	MP1	3	CCP0
		2	CPU
		1	JTAG
		6	DMAC
		5	SysHub
		4	SMN
		2	CPU
46	DF	1	JTAG
		2	PRQ0
		3	PRQ1
		4	PwrMgt0
		5	PwrMgt1
		6	Pint0
		7	Pint1
66	WAFLC	0	Initiators (except ASP /MP1) in BSP socket
		1	Initiators (except ASP/MP1) in AP socket
		2-31	Reserved
		64	ASP is BSP socket
		65	ASP in AP socket
		66-95	Reserved
		96	MP1 in BSP socket
		97	MP1 in AP socket
		98-127	Reserved

Table 152: CCD Initiator ID Map

CCD Number	Initiator Name	Initiator ID
0	MP5	783
	CCXSEC0	768
1	MP5	799
	CCXSEC0	784
2	MP5	815
	CCXSEC0	800
3	MP5	831
	CCXSEC0	816
4	MP5	847
	CCXSEC0	832
5	MP5	863
	CCXSEC0	848
6	MP5	879
	CCXSEC0	864
7	MP5	895
	CCXSEC0	880
8	MP5	911
	CCXSEC0	896
9	MP5	927
	CCXSEC0	912
10	MP5	943
	CCXSEC0	928

11	MP5	959
	CCXSEC0	944

8 Advanced Platform Management Link (APML)

8.1 Overview

The Advanced Platform Management Link I3C ([APML-I3C](#)) is an I3C based 2-wire processor target interface. APML-I3C is also referred as side band interface (SBI). At the application layer, APML has an expanded set of mailbox commands compared to the previous generation. At the data link layer, the protocol has been upgraded from [SMBus](#) 2.0 to I3C. As a fall back option the I3C target can operate in I2C mode.

APML-I3C is used to communicate with the Remote Management Interface (see SBI Remote Management Interface (SB-RMI) and SBI Temperature Sensor Interface (SB-TSI)). For I3C specification, see 1.2 [[Reference Documents](#)].

8.1.1 Definitions

Table 153: APML Definitions

Term	Description
ARA	Alert response address.
ARP	Address Resolution Protocol
CCC	Common Command Code for I3C
Controller or I3C Controller	The device that initiates and terminates all communication and drives the clock, SCL.
EC	Embedded Controller.
ENTDAA	Enter Dynamic Address Assignment CCC for I3C
I2C	Inter-Integrated Circuit
I3C	Improved Inter-Integrated Circuit
KBC	Keyboard Controller.
MDC	Mega Data Center
OOB	Out of Band interface, typically referring to APML as opposed to In Band interface which refers to HSMP
ODTS	On die temperature sensing. UMC could be configured to manage DIMM thermals via command throttling and appropriate refresh rate based on temperature range reported by inband MR4 reading from DRAM devices.
PEC	Packet Error Checking or Packet Error Code.
POR	Power on reset.
RTS	Remote temperature sensor, typical examples are ADM1032, LM99, MAX6657, EMC1002.
SBI	Sideband interface.
SB-RMI	Remote Management interface.
SDR	Single Data Rate
SDU	Scan Dump Utility
SETDASA	Set Dynamic Address from Static Address CCC for I3C
Target or I3C target	The target cannot initiate I3C communication and cannot drive the clock but can drive the data signal SDA and the alert signal ALERT_L.
TSI	Temperature sensor interface.
TSOD	Temperature sensor mounted on DIMM PCB.

8.2 SBI Bus Characteristics

The SBI largely follows MIPI I3C Basic Specification 1.0; see [docI3C](#).

8.2.1 I3C Protocol Support

The SBI follows I3C protocol mainly for:

- The processor does not implement I3C controller functionality, but acts as target only.
- Max of 12.5Mhz SDR mode is supported.
- The SBI mainly uses the I3C private Write and Read transfers to support Single or Block Write and Read.
- Target can only drive SDA, but not SCL to controller.
- Dynamic address assigned based on Static address can be supported for I3C target.
- PEC is not supported in I2C or I3C modes by target.

8.3 SBI Processor Information

8.3.1 SBI Processor Pins

Up to five processor pins are used for SBI support: two for data transfer, two for address determination and one for an interrupt output. Of the two address pins, one bit is socket_id used to determine the static address for each socket. These pins do not have changeable pinstrap. The Serial Interface Clock (SIC) and Serial Interface Data (SID) pins function as the I3C clock and data pins respectively. The I3C alert output pin (ALERT_L) is used to signal interrupts to the I3C controller.

8.3.2 Processor States

SBI responds to I3C traffic except when PWROK is de-asserted (and for a brief period after it is de-asserted). Access to internal processor state using SB-RMI is not supported under the following conditions:

- During cold and warm resets.
- During the APIC spin loop.

In order to allow [MP1](#) firmware to perform initialization of certain SB-RMI and SB-TSI registers, BMC should hold off conducting any transaction to either of these targets for at least 2 sec after cold or warm reset of the CPU.

8.4 SBI Remote Management Interface (SB-RMI)

SB-RMI interface uses the SB-RMI protocol to provide functions such as access to [CPUID](#) registers, processor [MCA](#) registers and [APML](#) mailbox commands which are described in details later. The SB-RMI protocol described below matches the version number encoded in the SBRMI::Revision register.

8.4.1 SB-RMI Transmission Protocol

SBI uses a standard I3C protocol to provide Private Write and Private Read Transfers. Figure below shows the I3C spec defined Private Write and Private Read transaction protocol. Each byte in the protocol is sent with the most significant bit first (bit[7]). The ninth data bit of each SDR Data Word written by the I3C controller (also referred to as the T-Bit) is a Parity bit, calculated using odd parity. The ninth (T) Data bit of each SDR Data Word returned for Private Read indicates if it is the End-of-Data. The controller or the target may drive the ninth bit to 0 (SDA Low) to indicate the end of Message.

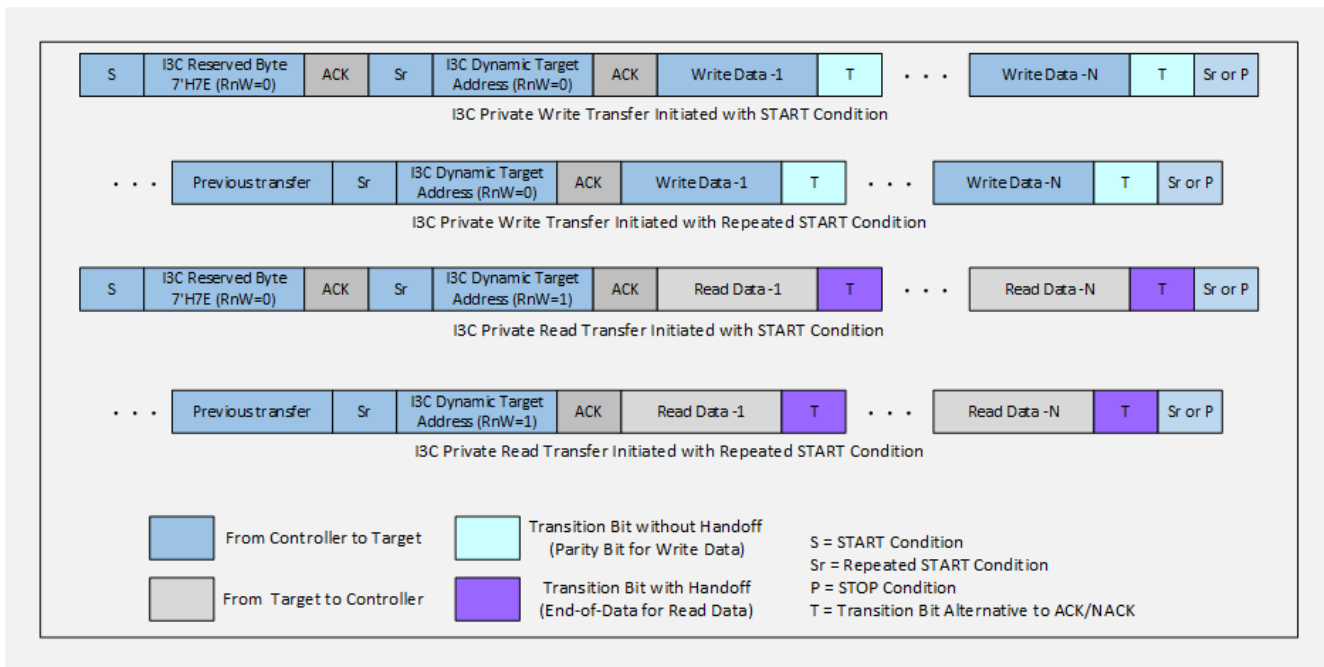


Figure 33: I3C Protocol for Private Write and Read

SB-RMI Write and Read transactions are conducted using I3C defined Private Write and Read protocol or standard I2C Write and Read. The "Command Code" provides the register offset to which the write or read is targeted. Command codes 71h, 72h and 73h are special because they are used for accessing [CPUID](#) and [MCA](#) registers. All other values represent SBRMI registers. In SB-RMI revision 2.0 the command code is one byte long.

Figure below shows the transaction protocol for Single Write and Read. Single Write consists of the "Command Code" followed by one byte of "Write Data". Single Read consists of two parts, a Private Write for providing the "Command Code", followed by a Private Read transaction to read one byte of data with Repeated START or START condition.

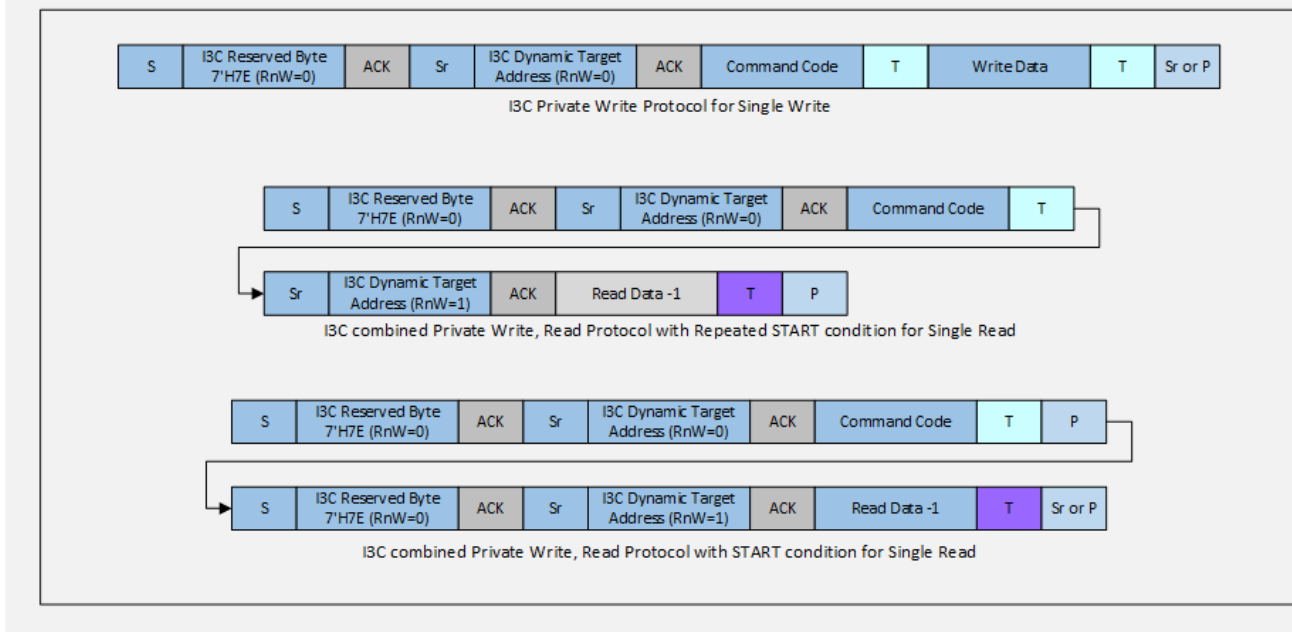


Figure 34: SBI Transmission Protocol for Single Write and Read

Figure below shows the transaction protocol for Block Write and Read. For Block Write, the "Command Code" is followed by multiple bytes of "Write Data". The interpretation of data bytes is based on the value of "Command Code". Block Read consists of two parts, a Private Write for providing the "Command Code", followed by multiple bytes of read data stream with Repeated START or START condition.

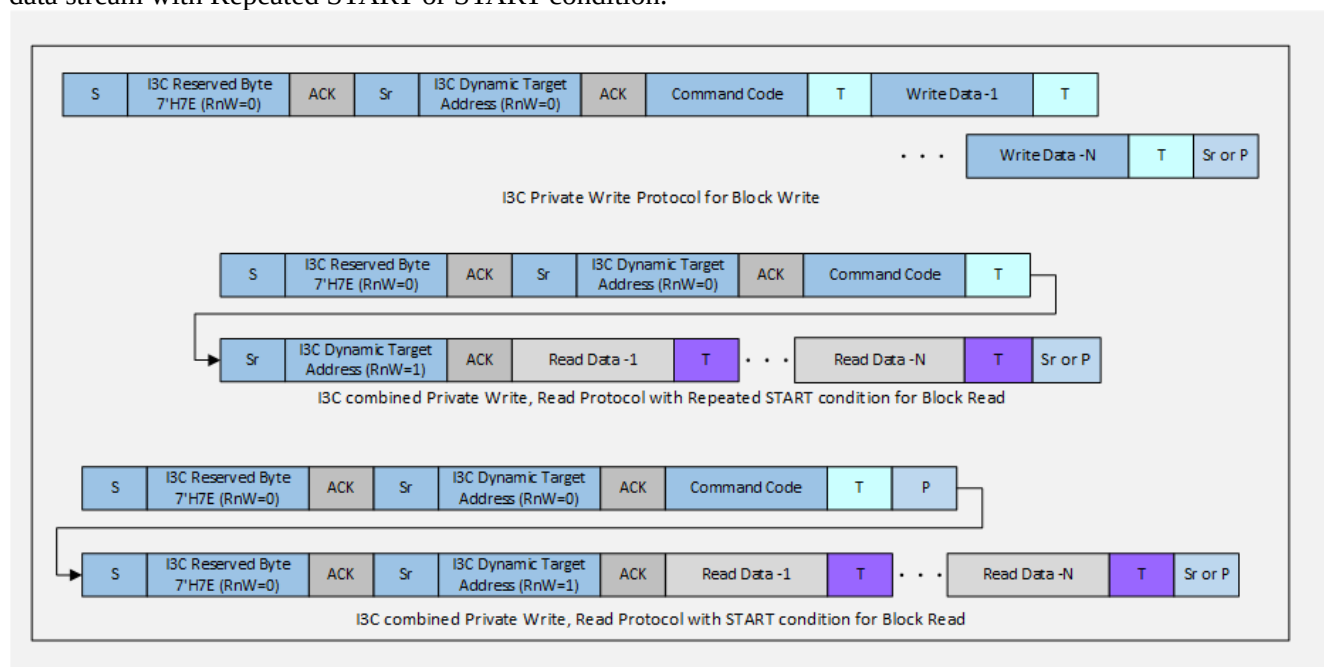


Figure 35: SBI Transmission Protocol for Block Write and Read

8.4.2 SB-RMI Functions

SB-RMI provides an interface for an external I3C controller that can be used to perform tasks such as monitoring the processor [MCA](#) registers, reading processor [CPUID](#) registers and power management or [RAS](#) actions using mailbox interface. SB-RMI supports signaling ALERT_L when a MCE is received by any thread or when software sets SBRMI::Status[SwAlertSts]. Another situation to assert ALERT_L is when hardware sets SBRMI::Status[HwAlertSts] upon completion of SBRMIx71, SBRMIx72 and SBRMIx73 commands which is only valid for I3C interface. Each package has an independent I3C target device. See 8.5.1 [SBI Static Address].

Each package is required to contain the same number of logical threads. The I3C target port attached to each package may access only the logical threads within the package. Core::X86::Cpuid::SizeId identifies the number of logical threads available in a package.

8.4.2.1 SB-RMI Processor State Access

The SB-RMI Functions table describes the functions for accessing processor state. See the [Processor](#) Programming Reference of the processor family for additional information about the processor registers. [MSR](#) not listed in below table is not accessible, will get "Unsupported Command" status.

Table 154: SB-RMI Functions

Function	Description	Thread Specific
CPUID	Access to CPUID registers. General purpose registers are not altered unlike a	Y

	processor CPUID instruction. Use Read CPUID Command Protocol described in 8.4.2.1.2 [SB-RMI Read CPUID Command Protocol]. Access is Read-only.	
MCA Registers	Register read command using the register address to access Core::X86::Msr:: MCG_CAP determines the number of MCA banks. See 3.1.2.2.1 [Legacy MCA Registers] and 3.1.2.2.3 [MCAX Registers] for the MCA registers within this range. Use Read Processor Register Command Protocol described in 8.4.2.1.1 [SB-RMI Read Processor Register Command Protocol]. Access is Read-only.	Y
DRAM Throttle	Register read or write command access through legacy MSR C001_0079 SMN address to the DRAM Controller ThrottleCtrl register UMC::ThrottleCtrl has been discontinued. DRAM throttling is available via "Write DRAM Throttle" command described in 8.4.2.2 [SB-RMI Mailbox Service] section. DRAM throttling can also occur due to Thermal Management Event based on UMC periodic polling of DDR5 MR4. See 12.2.21 [DDR5 Thermal Management and Bandwidth Capping].	N
Mailbox Service	Soft mailbox service request to firmware for power management purposes. Past implementations allowed for mailbox operations to X86 software. No usage models for communication with x86 software exists and x86 software messaging is not supported. Access is Read-write. See 8.4.2.2 [SB-RMI Mailbox Service] for details.	N
Boot Status	Boot Status is placed in outbound message register SBRMI::MP0OutBndMsg. Access is Read-only.	N

8.4.2.1.1 SB-RMI Read Processor Register Command Protocol

SB-RMI read processor register command is performed by using the SBI Block Write to load register address first. After the command is performed completely by the processor, hardware will set SBRMI::Status[HwAlertSts] to trigger ALERT_L pin asserted. After that, a read data/status can be issued from the controller side to check if the command is successfully completed or not as well as reading data by using a Block Read. The previous command must be complete before a new command can be issued.

Table 155: SB-RMI Read Processor Register (Write Phase)

Byte	Byte Name	Value	Notes
1	Dynamic Address	XXXXXXXX0b	Target Address (Write).
2	Command	73h	Command code.
3	WrDataLen	07h	7 Bytes.
4	WrData1	0Xh	Number of bytes to read from register (NumRdB). Valid values are 1 through 8.
5	WrData2	86h	Read Register command.
6	WrData3	XXXX_XXXXb	Bit [0] is reserved. Bits [7:1] represent thread[6:0]. Thread [7] is indicated by SBRMI::Thread128CS
7	WrData4	XXh	Register Address [7:0] from the SB-RMI Functions table.
8	WrData5	XXh	Register Address [15:8] from the SB-RMI Functions table.
9	WrData6	XXh	Register Address [23:16] from the SB-RMI Functions table.
10	WrData7	XXh	Register Address [31:24] from the SB-RMI Functions table.

After the above command is completed, hardware will set SBRMI::Status[HwAlertSts] to trigger ALERT_L pin asserted. Before ALERT_L pin is asserted, controller side can do other accesses for write or read. After assertion, a Block Read for data/status can be issued to read the data/status for the previous command. Please note, a new Private Write to reload the "Command" first must be issued first before the Block Read command.

Table 156: SB-RMI Read Processor Register (Reload Read Address Phase)

Byte	Byte Name	Value	Notes
1	Dynamic Address	XXXXXXXX0b	Target Address (Write).
2	Command	73h	Command code.

Table 157: SB-RMI Read Processor Register (Read Data/Status Phase)

Byte	Byte Name	Value	Notes
1	Dynamic Address	XXXXXXXX1b	Target Address (Read).
2	RdDataLen	0Xh	Number of bytes returned = NumRdB+1.
3	Status	XXh	Status Code.
4	RdData1	XXh	Register Data [7:0].
5	RdData2	XXh	Register Data [15:8]. Optional.
6	RdData3	XXh	Register Data [23:16]. Optional.
7	RdData4	XXh	Register Data [31:24]. Optional.
8	RdData5	XXh	Register Data [39:32]. Optional.
9	RdData6	XXh	Register Data [47:40]. Optional.
10	RdData7	XXh	Register Data [55:48]. Optional.
11	RdData8	XXh	Register Data [63:56]. Optional.

8.4.2.1.2 SB-RMI Read CPUID Command Protocol

SB-RMI read [CPUID](#) command is performed by using the SBI Block Write based on Private Write Process Call to load CPUID request info first. After the command is completed by the processor, hardware will set SBRMI::Status[HwAlertSts] to trigger ALERT_L pin asserted. After that, a read data/status can be issued from the controller side to check if the command is successfully completed or not as well as reading data by using a Block Read based on Private Read Process Call command. The previous command must be complete before a new command can be issued.

Table 158: SB-RMI Read CPUID (Write Phase)

Byte	Byte Name	Value	Notes
1	Dynamic Address	XXXXXXXX0b	Target Address (Write).
2	Command	73h	Command code.
3	WrDataLen	08h	8 Bytes.
4	WrData1	08h	Number of CPUID bytes to read.
5	WrData2	91h	Read CPUID command.
6	WrData3	XXXX_XXXXb	Bit [0] is reserved. Bits [7:1] represent thread[6:0]. Thread [7] is indicated by SBRMI::Thread128CS
7	WrData4	XXh	CPUID Function [7:0].
8	WrData5	XXh	CPUID Function [15:8].
9	WrData6	XXh	CPUID Function [23:16].
10	WrData7	XXh	CPUID Function [31:24].
11	WrData8	ECX[3:0]_000Xb	ECX[3:0] is the initial ECX value for extended CPUID

			operations. Must be 0h for non-extended operations. X: 0b=Return ebx:eax; 1b=Return edx:ecx.
--	--	--	---

After the above command is completed, hardware will set SBRMI::Status[HwAlertSts] to trigger ALERT_L pin asserted. Before ALERT_L pin is asserted, controller side can do other accesses for write or read. After assertion, a Block Read for data/status can be issued to read the data/status for the previous command. Please note, a new Private Write to reload the "Command" first must be issued first before the Block Read command.

Table 159: SB-RMI Read CPUID (Reload Read Address Phase)

Byte	Byte Name	Value	Notes
1	Dynamic Address	XXXXXXXX0b	Target Address (Write).
2	Command	73h	Read CPUID/Read Register Command Format.

Table 160: SB-RMI Read CPUID (Read Status/Data Phase)

Byte	Byte Name	Value	Notes
1	Dynamic Address	XXXXXXXX1b	Target Address (Read).
2	RdDataLen	09h	Number of CPUID bytes+1
3	Status	XXh	Status Code.
4	RdData1	XXh	eax or ecx [7:0].
5	RdData2	XXh	eax or ecx [15:8].
6	RdData3	XXh	eax or ecx [23:16].
7	RdData4	XXh	eax or ecx [31:24].
8	RdData5	XXh	ebx or edx [7:0].
9	RdData6	XXh	ebx or edx [15:8].
10	RdData7	XXh	ebx or edx [23:16].
11	RdData8	XXh	ebx or edx [31:24].

8.4.2.1.3 SB-RMI Write Processor Register Command

In prior processors, Write Register/Load Address command was only used for DRAM throttle register in UMC using legacy [MSR](#) address C001_0079. This feature is now discontinued. So SB-RMI Write [Processor](#) register command is no longer necessary.

8.4.2.1.4 SB-RMI Protocol Status Codes

The legal values for the Status byte of the SB-RMI processor state accesses are shown in the following table. For each of the status code which should be set after a command is processed, hardware will set SBRMI::Status[HwAlertSts] to trigger ALERT_L pin asserted first. After that, a read data/status can be issued to check the status code.

Table 161: SB-RMI Status Codes

Status Code	Name	Description
00h	Success	Command.
11h	Command Timeout	Command did not complete before a configured timeout event occurred, which indicates MP hasn't cleared the pending interrupt before timeout event occurs. This status code will never occur if (SBRMIx01[TimeoutDis]==1). MP has not sent the request to CPU/NB.

22h	Warm reset	A warm reset occurred during the transaction.
40h	Unknown Command Format	The value in Command Format field is not recognized.
41h	Invalid Read Length	The value in RdDataLen is less than 1 or greater than 32.
44h	Invalid thread	Invalid thread selected, the data length 0x09 and dummy data will be returned on invalid thread.
45h	Unsupported Command	Command not supported by the processor.
81h	Command Aborted	The processor core targeted by the command could not start the command and was aborted by the processor.

8.4.2.2 SB-RMI Mailbox Service

SB-RMI supports soft mailbox service request to [MP1](#) (power management firmware) through SBRMI inbound/outbound message registers. The message type is defined in the following table.

Table 162: SB-RMI Soft Mailbox Message

Command	Message	Description	Command Data In	Command Data Out
01h	Read Package Power Consumption	Read the average package power consumption. This is estimated per part power capped to cTDP. This data may be useful to assess relative variations in power under different operating conditions and workloads. This data should not be used for system power budgeting.	None	PkgPwr (mWatts), 32-bit integer.
02h	Write Package Power Limit	Set maximum power consumption for SOC package. Note: there is a limit on the minimum power that the processor can operate at; no further socket power reduction occurs if the socket power limit is set below that minimum.	PkgPwrLimit (mWatts), 32-bit integer.	None
03h	Read Package Power Limit	Read the maximum power consumption for SOC package	None	PkgPwrLimit (mWatts), 32-bit integer
04h	Read Max Package Power Limit	Read the maximum package power limit.	None	ReadMaxPackagePowerLimit (mWatts), 32-bit integer.
05h	Read TDP	Read current Thermal Design Power Limit	None	ReadTDP (mWatts), 32-bit integer.
06h	Read Max cTDP	Read Max configured Thermal Design Power	None	ReadMaxcTDP (mWatts), 32-bit integer.
07h	Read Min cTDP	Read Min configured	None	ReadMincTDP

		Thermal Design Power		(mWatts), 32-bit integer.
08h	Read BIOS Boost Fmax	Read BIOS Boost Fmax	[31:16]=Logical Core ID. [15:0]=Reserved.	ReadBIOSBoostFmax (MHz), 32-bit integer.
09h	Read APML Boost Limit	Read APML Boost Limit	[31:16]=Logical Core ID. [15:0]=Reserved.	ReadAPMLBoostLimit (MHz), 32-bit integer.
0Ah	Write APML Boost Limit	Write APML Boost Limit Operates on logical cores and not threads.	[31:16]=Logical Core ID. [15:0]=Frequency in MHz.	None
0Bh	Write APML Boost Limit All Cores	Write APML Boost Limit operates on all logical cores and no logical core requires specifying.	[15:0]=Frequency in MHz.	None
0Ch	Read Dram Throttle	Read Dram Throttle will always report the highest percentage value imposed by PROCHOT induced throttle or Write Dram Throttle or ODTs thermal event. This command does not factor in throttling due to TSOD (if enabled and triggered). The value returned is the average across all channels.	None	ReadDramThrottle (% , 0-100)
0Dh	Write Dram Throttle	Write Dram Throttle command provides a software path to throttle the command rate on all channels immediately. Once this command is issued, the software based throttle is disabled by writing a throttle value of 0. MP1 FW caps the throttle percentage to 80 when the input value is > 80.	WriteDramThrottle (% , 0-80).	None
0Eh	Read Prochot Status	Read PROCHOT Status.	None	ReadProchotStatus: 0=Not PROCHOT. 1=PROCHOT.
0Fh	Read Prochot Residency	Read PROCHOT Residency (since the boot time or last read of Prochot Residency). Return value is expressed as 16 fractional bits. Value of FFFFh represents 100% and 0000h represents 0% of time the socket spends in PROCHOT.	None	ReadProchotResidency (Percentage of time), 16 fractional bits.
10h	Reserved	N/A	N/A	N/A
11h	Read NBIO Error	Read NBIO Error Logging	[31:24]=NBIO	NBIOErrorLoggingRe

	Logging Register	Register.	quadrant. [23:0]=4-byte aligned Register Offset. The lowest 2 bits will be forced to 0 by MP1 firmware	gister value.
12h	Reserved	N/A	N/A	N/A
13h	Read IOD Bist Result	Read IOD BIST status.	None	ReadIODBistResult: 0=BIST pass. 1=BIST fail.
14h	Read CCD Bist Result	Read CCD BIST status. Results are read for each CCD present in the system. The number of logical CCD instances may be determined by use of CPUID reported CoreId[7:5].	Logical CCD instance number, 32-bit integer.	ReadCCDBistResult: 0=BIST pass. 1=BIST fail.
15h	Read CCX Bist Result	Read the BIST results of Cpu Core Complex. Results represent a bit vector of the BIST status for each physical core in the complex(n:0) and L3. The number of CCX in the socket and number of cores per CCX can be determined using CPUID functions.	Logical CCX instance number (0..K-1) where K is the number of CCXs, 32-bit integer.	ReadCCXBist Result: [23:16]=Core[n:0]. [1]=L3 X3D if X3D variant else 1 [0]=L3. Each bit is BIST result where 0=BIST fail 1=BIST pass Cores that are disabled or not present will report a default value of 0.
16h	Read CCLK Frequency Limit	Provides the socket's CPU core clock (CCLK) frequency limit from the most restrictive infrastructure limit at the time of the request.	None	Frequency (MHz).
17h	Read Socket C0 Residency	Provides the average C0 residency across all cores in the socket. 100% specifies that all enabled cores in the socket are running in C0.	None	Socket C0 residency [%].
18h	Get Max DDR Bandwidth and Utilization	Provides per socket: 1. Theoretical maximum DDR Bandwidth in GB/s. 2. Current utilized DDR Bandwidth (Read+Write) in GB/s. 3. Current utilized DDR Bandwidth as a percentage of theoretical maximum.	None	[31:20]=Max bandwidth in GBps. [19:8]=Utilized bandwidth (R+W) in GBps. [7:0]=Utilized bandwidth in %.
1Fh	Read PPIN fuse	Return the 64b PPIN value.	[31:1]=Reserved	[31:0]=PPIN[31:0] if

		MP1 firmware may return an abort code if the data cannot be accessed. Refer to 2.1.7 [Processor Inventory Management] . BMC firmware must enforce appropriate policy controls for allowing PPIN access through APLM.	[0]=Hi or Lo 32b requested	DataIn[0]==0 else PPIN[63:32]
20h	Read Postcode	Read most recent postcode at specified offset. MP1 caches the 8 most recent postcodes. When input parameter is 0 MP1 will refresh the cache before returning the latest postcode. Index 0 is the most recent postcode; higher index refers to correspondingly older postcode.	[31:3]=Reserved [2:0]=Postcode offset	[31:0]=postcode for given offset
21h	Reserved			
22h	Read Public Serial Number of Die	Read per die public serial number. The chiplets (or die) are categorized as either CCD or IOD. The chiplet number is 0-based index up to the maximum number of chiplets (CCD chiplets if category is CCD, or IOD chiplets if category is IOD). The serial number is a 64b value, so the Dword offset is specified in the input parameter.	[31:17]=Reserved [16]=Dword offset 0=LoDword 1=HiDword [15:8]=Chiplet number [7:0]=Category 0=CCD 1=IOD	[31:0]=Serial Number Dword
31h ¹	WRITE_FAST_PT_LIMIT	Set APU power limit for system power supply peak control.	[31:0]=fPPT in mW.	None
34h ¹	WRITE_THERM_CTL_LIMIT	Set the thermal throttling limit.	[31:0]=Therm limit in degree Celsius.	None
35h ¹	WRITE_VRM_VDD_CURRENT_LIMIT	Set VDDCR_VDD TDC .	[31:0]=VDD TDC in mA.	None
36h ¹	WRITE_VRM_VDD_MAXIMUM_CURRENT_LIMIT	Set VDDCR_VDD EDC.	[31:0]=VDD EDC in mA.	None
40h	BMC_REPORT_DIMM_POWER_CONSUMPTION	Per-DIMM Power Consumption reported periodically by BMC (at rate of 10ms or less) when BMC owns the SPD side-band bus.	[31:17]=DIMM Power (mWatt). [16:8]=Update Rate (ms). [7:0]=DIMM_ADDRE	None See notes under Table 168 [APML DIMM ADDRESS Encodings]

			SS.	for DataIn and DataOut field details
41h	BMC_REPORT_DIMM_THERMAL_SENSOR	Per-DIMM Thermal Sensor (2 Sensors per DIMM) reported periodically by BMC (at rate of 10ms or less) when BMC owns the SPD side-band bus.	[31:21]=Temperature (Degrees C). [16:8]=Update Rate (ms). [7:0]=DIMM_ADDRESS. SS.	None See notes under Table 168 [APML DIMM ADDRESS Encodings] for DataIn and DataOut field details
42h	BMC_RAS_PCIE_CONFIG_ACCESS	Read 32-bit Data from Extended PCI Config Space. Note: This command allows access to PCIe® Config Space, except for Seg0:Bus0:Dev18 and Seg0:Bus0:Dev19.	PCI address: [31:28]=Segment [27:16]=4-Byte aligned Offset (0..4092). [15:8]=Bus. [7:3]=Device. [2:0]=Function.	[31:0]=Data from Offset in PCIe Config Space.
43h	BMC_RAS_MCA_VALIDITY_CHECK	Return number of MCA Banks with Valid (non-zero) Status.	None	[31:16]=Unsigned Integer for number of bytes per MCA bank. [15:0]=Unsigned Integer for number of MCA Banks with Valid Status.
44h	BMC_RAS_MCA_MSR_DUMP	Read 32-bit of Data from MCA Bank reported by BMC_RAS_MCA_VALIDITY_CHECK.	[31:16]=Unsigned Integer for 0-based Index of MCA Bank. [15:0]=4-Byte aligned Offset within MCA Bank (0..124).	[31:0]=Data from Offset in MCA Bank.
45h	BMC_RAS_FCH_RESET_REASON	Get FCH Reason Code from previous RESET. This command is typically used to get source of previous reset from BSP (socket 0). BMC can optionally choose to read the reason code from AP (socket 1).	[31:0]=Unsigned Integer for ID of FCH Register: 0=FCH::PM:: S5_RESET_STATUS . 1=FCH::PM:: BREAK_EVENT . Other values are Reserved.	[31:0]=Data from corresponding FCH Register.
46h	GET_DIMM_TEMP_RANGE_AND_REFRESH_RATE	Get Per-DIMM Temperature Range and Refresh Rate for a given channel based on inband MR4 polling. The refresh rate is determined by the warmest DRAM device across all DIMMs in the channel (highest MR4.OP[2:0] value) in accordance to the threshold specified in JEDEC DDR5 SDRAM specification (JESD79). Due to hardware	[7:0]=DIMM_ADDRESS. See notes under Table 168 [APML DIMM ADDRESS Encodings] for DataIn and DataOut field details	[31:4]=Reserved [3]=Refresh Rate 0=1x 1=2x. [2:0]=Temperature Range (an approximation) 001b, if refresh rate is 1x. 101b if refresh rate is 2x.

		limitations, MP1 has visibility to only the refresh rate (1x or 2x) but not the actual MR4.OP[2:0] value. Hence the refresh rate (DataOut[3]) is accurate but the temperature range (DataOut[2:0]) is an approximation.		
47h	GET_DIMM_POWER_CONSUMPTION	Get Per-DIMM Power Consumption. This is applicable when BMC does not own the SPD side-band bus.	[7:0]=DIMM_ADDRESS See notes under Table 168 [APML DIMM ADDRESS Encodings] for DataIn and DataOut field details	[31:17] DIMM Power (mWatt), [16:8] Update Rate (ms), [7:0] DIMM_ADDRESS
48h	GET_DIMM_THERMAL_SENSOR	Get Per-DIMM Thermal Sensor (2 Sensors per DIMM). This is applicable when BMC does not own the SPD side-band bus.	[7:0]=DIMM_ADDRESS. See notes under Table 168 [APML DIMM ADDRESS Encodings] for DataIn and DataOut field details	[31:21] Temperature (Degrees C), [16:8] Update Rate (ms), [7:0] DIMM_ADDRESS
49h	PWR_CURRENT_ACTIVE_FREQUENCY_LIMIT_SOCKET	Get Current Active Frequency Limit per Socket.	None	[31:16]=Frequency (MHz). [15:0]=Source of Limit. As shown in Table 163 [APML Frequency Limit Source].
4Ah	PWR_CURRENT_ACTIVE_FREQUENCY_LIMIT_CORE	Get Current CCLK limit set per Core.	Logical Core ID, 32-bit integer	[31:16]=Reserved. [15:0]=BaseFreq (MHz).
4Bh	PWR_SVI_TELEMETRY_ALL_RAILS	Get SVI-based Telemetry for all rails.	None	[31:0]=Power (mW).
4Ch	GET_SOCKET_FREQ_RANGE	Get Fmax and Fmin per Socket.	None	[31:16]=Fmax (MHz). [15:0]=Fmin (MHz).
4Dh	GET_CURRENT_IO_BANDWIDTH	Get Current Bandwidth on IO Link.	[15:8]=Link ID. As shown in Table 166 [APML Link ID Encoding]. [7:3]=Reserved. [2:0]=Bandwidth Type. As shown in Table 164 [APML IO Bandwidth Encoding].	[31:0]=IO BandWidth (Mbps).
4Eh	GET_CURRENT_XGMI_BAND	Get Current Bandwidth on xGMI Link.	[15:8]=Link ID. As shown in Table 166	[31:0]=xGMI bandwidth (Mbps).

	WIDTH		[APML Link ID Encoding]. [7:3]=Reserved. [2:0]=Bandwidth Type. As shown in Table 165 [APML XGMI Bandwidth Encoding].	
4Fh	GMI3_LINK_WIDTH_RANGE	Set Max and Min width of GMI3 Link as follows: 0=Quarter Width. 1=Half Width. 2=Full Width.	[31:16]=Reserved [15:8]=Min link width. [7:0]=Max link width. Note: Max value must be >= Min value. Invalid configurations result in a Failed response.	None.
50h	XGMI_LINK_WIDTH_RANGE	Set Max and Min width of xGMI Link as follows: 0=x4. 1=x8. 2=x16.	[31]=Reserved [15:8]=Min link width. [7:0]=Max link width. Note1: Max value must be >= Min value. Invalid configurations result in a Failed response. Note2: If this function is called from both Primary and Secondary sockets, the largest Width values from either call are used.	None.
51h	APB_DISABLE	Messages APBEnable and APBDisable specify DF (Infinity Fabric™) P-State behavior. DF P-States specify the frequency of clock domains from the CPU core boundary through to and including system memory, where DF P0 has highest performance and higher numbered P-states have lower performance. By default, an algorithm adjusts DF P-States automatically in order to optimize performance. However, this default may be changed to a fixed DF P-	[31]:8]=Reserved [7:0]=DF P-state Valid values: 0, 1, 2	None

		State through a CBS option at boot-time. APBDisable may also be used to disable this algorithm and force a fixed DF P-State. Note: While the socket is in PC6 or if PROCHOT_L is asserted, the lowest DF P-State (highest value) is enforced regardless of the APBEnable/APBDisable state. See Note 2 at the end of this table about arbitration of the active DF Pstate range.		
52h	APB_ENABLE	This function enables the DF P-State Performance Boost algorithm. See the APBDisable function for more information. See Note 2 at the end of this table about arbitration of the active DF Pstate range.	None	None
53h	GET_CURRENT_DFPSTATE_FREQUENCY	Provides Infinity Fabric Clock (FCLK), DRAM Memory Clock (MEMCLK), and UMC Clock Divider (UCLK) for the current socket DF P-state.	None	[31:16]=MEMCLK (MHz). [15]=UCLK. 0=Divide by 1. 1=Divide by 2. [14:12]=Reserved. [11:0]=FCLK (MHz).
54h	SET_LCLK_DPM_LEVEL_RANGE	Sets the Max & Min LCLK DPM Level on a given NBIO per socket. The DPM Level is an encoding to represent the PCIe Link Frequency (LCLK) under a root complex (NBIO), as shown in Table 167 [APML LCLK Frequency Per DPM Level]. Note: These Levels can also be set from HSMP; the last value received from either APML or HSMP is enforced.	[23:16]=NBIO ID (0 to 3). [15:8]=Max DPM Level (0 to 3), [7:0]=Min DPM Level (0 to 3). Note: Max value must be $\geq$ Min value.	None
55h	BMC_GET_RAPL_UNITS	Get RAPL (Running Average Power Limit) Units. Scaling factor for energy (in Joules) is $1/(2^{ESU})$, and for time (in seconds) is $1/(2^{TU})$. ESU: energy status units	None	[31:20]=Reserved. [19:16]=TU Value. [15:13]=Reserved. [12:8]=ESU Value. [7:0]=Reserved.

		TU: time units		
56h	BMC_GET_RAPL_CORE_LO_COUNTER	Get RAPL (Running Average Power Limit) Lo-Word of Energy Counter per Core. Energy = Counter * ESU Multiplier (Joules). See 8.4.2.2.3 [Reading RAPL Energy Counters].	Logical Core ID, 32-bit integer.	[31:0]=Core Lo_Count Register Value.
57h	BMC_GET_RAPL_CORE_HI_COUNTER	Get RAPL (Running Average Power Limit) Hi-Word of Energy Counter per Core. Energy = Counter * ESU Multiplier (Joules). See 8.4.2.2.3 [Reading RAPL Energy Counters].	Logical Core ID, 32-bit integer.	[31:0]=Core Hi_Count Register Value.
58h	BMC_GET_RAPL_PKG_COUNTER	Get RAPL (Running Average Power Limit) Lo/Hi-Word of Energy Counter per Package. Energy = Counter * ESU Multiplier (Joules). See 8.4.2.2.3 [Reading RAPL Energy Counters].	0=Read Lo_Count Register. 1=Read Hi_Count Register.	[31:0]=Pkg Lo/Hi_Count Register Value.
59h	BMC_GET_CPU_BASE_FREQUENCY	Get Base Frequency per CPU socket.	None	[31:16]=Reserved. [15:0]=BaseFreq (MHz).
5Ah	BMC_CONTROL_PCIE_LINK_RATE	Set the Link Rate on PCIe devices. Note that if CXL™ device operating at Gen5 rate is present on any link then PCIe rate is locked at Gen5 rate and request to change to Gen4 rate will be ignored.	[31:8]=Reserved [7:0]=Link Rate encoding: 0=Auto-Detect bandwidth utilization, and set Link Rate accordingly. 1=Limit at Gen4 Rate. 2=Limit at Gen5 Rate.	Previous mode: 0=Auto-Detect. 1=Limit at Gen4. 2=Limit at Gen5.
5Bh	BMC_RAS_DBG_LOG_VALIDITY_CHECK	Return the number of Dbg Log instances of type Dbg Log Block ID that have an error log to report. Every instance of a given Block ID will have the same length of error log. The length of error log can vary between Block IDs. See Table 169 [BMC RAS DBG Log Block ID Encoding] for block instance and offset encoding	[31:8]=Reserved. [7:0]=Dbg Log Block ID.	[31:25]=Reserved. [24:16]=Unsigned integer, length of error log in bytes per instance (0, 4, 8, 12, ... 256). [15:9]=Reserved. [8:0]=Unsigned integer, number of Dbg Log instances that have an error log to report (0..256). 0 means no instance has anything to report.
5Ch	BMC_RAS_DBG_LOG_DUMP	Read 32b of Data from Dbg Log instance if BMC_RAS_DBG_LOG_VA	[31:24]=Reserved. [23:16]=Unsigned integer, 0-based index	[31:0]=4 byte data from requested offset of Dbg Log Block

		LIDITY_CHECK reported non-zero instances. See Table 169 [BMC RAS DBG Log Block ID Encoding] for block instance and offset encoding	of Dbg Log instance (0..255). [15:8]=Dbg Log Block ID [7:0]=4-byte aligned offset in error log (0, 4, 8, 12, ...).	instance
5Dh	PWR_EFFICIENCY_MODE_SELECTION	Select current power efficiency profile policy. See Note 2 at the end of this table about arbitration of the active DF Pstate range.	[31:3]=Reserved. [2:0]=Mode Selection: 000b=High Performance Mode. 001b=Power Efficiency Mode. 010b=IO Performance Mode. 011b=Balanced Memory Performance Mode. Other=Reserved. See 8.4.2.2.1 [APML Performance Modes].	None
5Eh	DF_PSTATE_RANGE	Set the Max and Min DF P-State. Given the DF_PSTATE encodings, Max value must be <= Min value. See Note 2 at the end of this table about arbitration of the active DF Pstate range.	[31:16]=Reserved. [15:8]=Min DF P-State. [7:0]=Max DF P-State. P_State values are: 0 – DFP0 1 – DFP1 2 – DFP2	None.
5Fh	GET_LCLK_DPM_LEVEL_RANGE	Gets the Max & Min LCLK DPM Level on a given NBIO per socket. The DPM Level is an encoding to represent the PCIe Link Frequency (LCLK) under a root complex (NBIO), as shown in Table 167 [APML LCLK Frequency Per DPM Level]. Note: These Levels can also be set from HSMP; this message returns the current Levels which may have been set from either APML or HSMP.	[23:16]=NBIO ID (0 to 3).	[23:16]=NBIO ID (0 to 3) [15:8]=Max DPM Level (0 to 3), [7:0]=Min DPM Level (0 to 3).
60h	GET_UCODE_REVISION	Get microcode revision number. This is revision number reported from IA32_BIOS_SIGN_ID MSR using CUID instruction. The command may return an	N/A	[31:0]=Microcode revision value from IA32_BIOS_SIGN_ID MSR

		Abort error code if the MSR is not accessible (eg. if core is not out of reset)		
61h-67h	Reserved	N/A	N/A	N/A
68h	BMC_PCIE_CONFIG_WRITE	Write 32-bit DataIn to Extended PCI Config Space of target specified in previous BMC_RAS_PCI_CONFIG_ACCESS command. MP1 firmware will restrict the writes to SFI (System Firmware Intermediary) extended capability registers and return SB-RMI error code "Invalid input parameter" if offset does not target SFI extended capability. BMC is expected to know the offset of the SFI extended capability in the configuration space of the PCIe target.	[31:0]=Data value to be written. Note that byte enables are not supported. To modify a subset of bytes, BMC must read, modify and write at 4 byte granularity.	Return code. 0=Success any other value=Reserved
69h	Reserved	N/A	N/A	N/A
6Ah	BMC_RAS_DELAY_RESET_ON_SYNCFLOOD_OVERRIDE	Overrides the delay value set via BIOS for the current boot instance. Delay value reverts to BIOS config setting after reboot. Limited to 5 override requests per boot instance. Command will be rejected if sent more than once per minute.	[31:10]=Reserved. [9]=StopDelayCounter . If set, stops the active delay countdown, which extends the DelayResetCpuOnSyncFlood indefinitely; system will not reset. The CPU processes this bit only during the SyncFlood condition. This action is final; counter cannot be resumed. Use mailbox command BMC_RAS_RESET_ON_SYNCFLOOD or toggle KBRESET_L pin if system reset is desired at any point after stopping ResetCpuOnSyncFlood counter. [8]=DisableDelayCounter. If set,	0=NACK. MP1 firmware will not honor the override request 1=ACK. MP1 firmware will honor the override request.

			ResetCpuOnSyncFlood response will not be delayed in the next SyncFlood, regardless of the delay value. Overriding this value takes effect at runtime; before a SyncFlood condition. [7:0]=Override delay value. Range=5-255 mins. Overriding this value only takes effect at runtime; before a SyncFlood condition.	
6Bh	BMC_RAS_RESET_ON_SYNCFLOOD	Requests warm reset after syncflood. Reset only works during syncflood condition	N/A	0=NACK. MP1 firmware will not proceed with reset 1=ACK. MP1 firmware will proceed with reset

Notes:

1. Family 19h Model 18h processors support only commands in the command code range 30h-3Fh, all other commands are unsupported. Rest of the Family 19h Models 10h-1Fh processors (i.e. non-18h) support all commands except those in range 30h-3Fh.
2. APML commands APB_DISABLE, APB_ENABLE, PWR_EFFICIENCY_MODE_SELECTION, SET_DF_PSTATE_RANGE, as well as HSMP commands APBDisable, APBEnable, PwrEfficiencyModeSelection, SetDfPstateRange all influence the active DF Pstate range. The last one of these HSMP or APML commands sent will override any previous setting of the active DF Pstate range, including overriding any active DF Pstate range set by BIOS options.

Table 163: APML Frequency Limit Source

Source Type Encoding (DataOut[15:0])	Description
DataOut[0]=1	cHTC-Active
DataOut[1]=1	PROCHOT
DataOut[2]=1	Thermal Designed Current (TDC) Limit
DataOut[3]=1	Package Power Tracking (PPT) Limit
DataOut[4]=1	OPN Max
DataOut[5]=1	Reliability Limit (Fused Max or Reliability Monitor Fmax@Vmax)
DataOut[6]=1	APML Agent
DataOut[7]=1	HSMP Agent
DataOut[15:8]	Reserved

Note: There may be more than one active limiter.

Table 164: APML IO Bandwidth Encoding

Encoding	Bandwidth Type
001b	Aggregate bandwidth
Other	Reserved

Table 165: APML XGMI Bandwidth Encoding

Encoding	Bandwidth Type
001b	Aggregate bandwidth (bi-directional bandwidth per link)
010b	Read bandwidth (uni-directional bandwidth per link)
100b	Write bandwidth (uni-directional bandwidth per link)
Other	Reserved

Note: max theoretical bandwidth per link = xGMI speed * link width * MULT, where MULT = 1 (uni-dir) or 2 (bi-dir).
For a x16 link operating at 32Gbps, max theoretical aggregate bandwidth = 1024 Gbps (32*16*2)

Table 166: APML Link ID Encoding

Encoding	Link Type
00000001b	P0
00000010b	P1
00000100b	P2
00001000b	P3
00010000b	G0
00100000b	G1
01000000b	G2
10000000b	G3

Table 167: APML LCLK Frequency Per DPM Level

DPM Level	LCLK Frequency
0	Lowest Freq.
1..3	Highest Freq.

Table 168: APML DIMM ADDRESS Encodings

DIMM_ADDRESS[7:0] (Mode = 0)	DIMM_ADDRESS[7:0] (Mode = 1)
Bit[7]=0 (for Mode 0) Bit[6]=Temperature Sensor#: TS0/1, else Reserved. Bit[5:4]=I2/I3C Port# Bit[3]=Reserved. Bit[2:0]=DIMM SPD Host ID. Refer to JEDEC SPD5118, SPD5108 Hub and Serial Presence Detect Device Standard (JESD300-5).	Bit[7]=1 (for Mode 1) Bit[6]=Temperature Sensor#: TS0/1, else Reserved. Bit[5]=Reserved. Bit[4]=DIMM#: DIMM0/1. Bit[3:0]=UMC/DDRPHY Instance ID Refer to 12.1.2.1 [UMC and DDR Phy Logical Mapping].

Notes:

- 1: DIMM Power is a 15-bit unsigned value representing power consumed in mW (0-32767)
- 2: Refresh rate is the time between successive sampling from given DIMM (0-511ms). 0 means sampling time < 1ms, and 511 means sampling time >= 511ms.
- 3: DIMM_ADDRESS is encoded per above Table 168 [APML DIMM ADDRESS Encodings].
- 4: Temperature (Deg C) is an 11-bit signed value, with a scaling factor of 0.25.
Thus 3FFh=255.75 (1023*0.25), 400h= -256 (-1024*0.25), 1h=0.25 and 7FFh= -0.25 (-1*0.25).

Table 169: BMC RAS DBG Log Block ID Encoding

Debug Log ID	Log length (bytes)	Offset	Value
0=DF OrigWDT	16	0	CCM ID
		4	DF::OrigWdtAddrLogStat

		8	DF::OrigWdtAddrLogLo
		12	DF::OrigWdtAddrLogHi
33=PcieDump	196	0	Root Port Instance ID: [7:0] = NBIO instance [15:8] = PCIe Controller Instance [23:16] = Root Port Instance [31:24] = Reserved (00h)
		4	PCIERCCFG:: PCIE_ADV_ERR_RPT_ENH_CAP_LIST
		8	PCIERCCFG:: PCIE_UNCORR_ERR_STATUS
		12	PCIERCCFG:: PCIE_UNCORR_ERR_MASK
		16	PCIERCCFG:: PCIE_UNCORR_ERR_SEVERITY
		20	PCIERCCFG:: PCIE_CORR_ERR_STATUS
		24	PCIERCCFG:: PCIE_CORR_ERR_MASK
		28	PCIERCCFG:: PCIE_ADV_ERR_CAP_CNTL
		32	PCIERCCFG:: PCIE_HDR_LOG0
		36	PCIERCCFG:: PCIE_HDR_LOG1
		40	PCIERCCFG:: PCIE_HDR_LOG2
		44	PCIERCCFG:: PCIE_HDR_LOG3
		48	PCIERCCFG:: PCIE_ROOT_ERR_CMD
		52	PCIERCCFG:: PCIE_ROOT_ERR_STATUS
		56	PCIERCCFG:: PCIE_ERR_SRC_ID
		60	PCIERCCFG:: PCIE_TLP_PREFIX_LOG0
		64	PCIERCCFG:: PCIE_TLP_PREFIX_LOG1
		68	PCIERCCFG:: PCIE_TLP_PREFIX_LOG2
		72	PCIERCCFG:: PCIE_TLP_PREFIX_LOG3
		76	PCIERCCFG:: DEVICE_CNTL + PCIERCCFG:: DEVICE_STATUS
		80	PCIERCCFG:: LINK_CNTL + PCIERCCFG:: LINK_STATUS
		84	PCIERCCFG:: SLOT_CNTL + PCIERCCFG:: SLOT_STATUS
		88	PCIERCCFG:: LINK_CNTL2 + PCIERCCFG:: LINK_STATUS2
		92	PCIERCCFG:: PCIE_LANE_ERROR_STATUS
		96	PCIERCCFG:: PCIE_RP_PIO_STATUS
		100	PCIERCCFG:: LINK_STATUS_16GT
		104	PCIERCCFG:: LINK_STATUS_32GT
		108	PCIEPORT:: PCIE_ERR_CNTL
		112, 116	Reserved
		120	PCIEPORT:: PCIE_LC_STATE0
		124	PCIEPORT:: PCIE_LC_STATE1
		128	PCIEPORT:: PCIE_LC_STATE2
		132	PCIEPORT:: PCIE_LC_STATE3
		136	PCIEPORT:: PCIE_LC_STATE4
		140	PCIEPORT:: PCIE_LC_STATE5
		144	PCIECORE:: PCIE_LC_STATE6
		146	PCIECORE:: PCIE_LC_STATE7
		152	PCIECORE:: PCIE_LC_STATE8

		156	PCIECORE:: PCIE_LC_STATE9
		160	PCIECORE:: PCIE_LC_STATE10
		164	PCIECORE:: PCIE_LC_STATE11
		168, 172, ... 192	Reserved

1. Refer to docRAS for details of each Dbg Log ID dump

8.4.2.2.1 APLM Performance Modes

Performance modes can be set through [APML](#) or HSMP; the value that was last set is enforced.

- High Performance Mode: this mode will choose the default [DF](#) P-State algorithm and will try to maximize application performance using the application characteristics. This mode favors core performance and the system boots into this mode. In this mode all DF P-States are available and the default DF P-State and DLWM algorithms are active.
- Power Efficiency Mode: this mode configures the system for power efficiency at the expense of some performance. This mode limits the boost frequency available to the cores and restricts the DF P-States available in the system. This mode also monitors the system load to dynamically adjust performance for maximum power efficiency.
- IO Performance Mode: this mode sets up the Infinity Fabric to maximize the IO sub-system performance. This mode can result in lower core performance in some cases, to maximize IO throughput.
- Balanced Memory Performance Mode: this mode biases the memory subsystem and Infinity Fabric performance towards efficiency, by lowering the frequency of the fabric and the width of the [xGMI](#) links under light traffic conditions. Core behavior is unaffected. There may be a performance impact under lightly loaded conditions for memory-bound applications compared to the default high performance mode. With higher memory and fabric load, the system becomes similar in performance to the default high performance mode.

8.4.2.2.2 SB-RMI Mailbox Sequence

The sequence is as follows:

1. The initiator (BMC) indicates that command is to be serviced by firmware by writing 80h to SBRMI::InBndMsg_inst7 (SBRMIx3F). This register must be set to 80h after reset.
2. The initiator (BMC) writes the command to SBRMI::InBndMsg_inst0 (SBRMIx38).
3. For Write operations or Read operations, which require additional addressing information as shown in Table 162 [SB-RMI Soft Mailbox Message] above, the initiator (BMC) writes Command Data In[31:0] to SBRMI::InBndMsg_inst[4:1] {SBRMIx3C(MSB);SBRMIx39(LSB)}.
4. The initiator (BMC) writes 01h to SBRMI::SoftwareInterrupt to notify firmware to perform the requested Read or Write command.
5. Firmware reads the message and performs the defined action.
6. Firmware writes the original command to outbound message register SBRMI::OutBndMsg_inst0 (SBRMIx30).
7. Firmware writes SBRMI::Status[SwAlertSts] = 1 to generate an ALERT (if enabled) to initiator (BMC) to indicate completion of the requested command. Firmware must (if applicable) put the message data into the message registers SBRMI::OutBndMsg_inst[4:1] {SBRMIx34(MSB);SBRMIx31(LSB)}. If the ALERT is not connected to BMC or external Southbridge, SBRMI::Status[SwAlertSts] can be polled to monitor the completion of the requested command.
8. Firmware clears the interrupt on SBRMI::SoftwareInterrupt.
9. For a Read operation, the initiator (BMC) reads the firmware response Command Data Out[31:0] from SBRMI::OutBndMsg_inst[4:1] {SBRMIx34(MSB);SBRMIx31(LSB)}.
10. BMC must write 1'b1 to SBRMI::Status[SwAlertSts] to clear the ALERT to initiator (BMC). It is recommended to clear the ALERT upon completion of the current mailbox command.

Table 170: SB-RMI Soft Mailbox Error Code

Error Type	Description	Code
No error	Mailbox message command executed successfully without an error.	00h
Command Aborted	Mailbox message command was aborted due to internal error. DataOut must be ignored.	01h
Unknown Command	Unknown mailbox message.	02h
Invalid Core	Invalid core is specified in mailbox message parameters.	03h

The mailbox error code is written by Firmware in SBRMI::OutBndMsg_inst7 (SBRMIx37).

8.4.2.2.3 Reading RAPL Energy Counters

The following procedure is recommended to detect and correct a potential overflow when reading 64-bit RAPL counters with separate 32-bit Lo-Word and 32-bit Hi-Word reads via SB-RMI Mailbox messages:

1. Read the Hi-Word.
2. Read the Lo-Word.
3. Read the Hi-Word again and compare it to the previous Hi-Word value.
4. If the Hi-Word values are equal, keep the previous Lo-Word value.
5. Else, read the Lo-Word again and keep the last Hi-Word value.

8.4.2.3 SB-RMI Boot code status

Boot code will dump the dynamic boot status into SBRMI::MP0OutBndMsg. BMC can then just read this status through SBI interface to determine progress through the boot flow.

8.4.2.4 SB-RMI Register Access

The SB-RMI registers can be read or written from the I3C interface by using the I3C Private Write or Private Read protocols with the SB-RMI register address indicated in the command code, which follows the I3C dynamic target address. Both Block Read/Write and Single Read/Write can also be supported.

Block Read/Write protocol access for SB-RMI registers is controlled by SBRMI::Control[BlkRWEn]. The SB-RMI interface supports Block Writes of up to 32 bytes, and Block Reads of up to 32 bytes as specified by SBRMI::ReadSize[RdSize]. Bytes are returned in ascending register order starting with the first SB-RMI register in the command byte. Some rules for write and read are listed as follows.

1. SB-RMI register offsets within x00h-x0Fh, can only support for Single Write if it is writable.
2. SB-RMI register offsets within x10h-x5Fh or x80h-xDFh, if it is writable it can support for Single Write and Block Write based on SBRMI::Control[BlkRWEn].
3. SB-RMI register offsets within x00h-x5Fh or x80h-xDFh, if it is readable it can support for Single Read and Block Read.
 - If SBRMI::Control[BlkRWEn] is zero or SBRMI::ReadSize[RdSize] is zero or 1, only supports Single read by return 1 byte of read data.
 - If SBRMI::Control[BlkRWEn] is set and SBRMI::ReadSize[RdSize] is greater than 1, supports Block read by return SBRMI::ReadSize[RdSize] bytes of read data with extra 1 byte of "byte count".
4. SB-RMI register offsets within SBRMIx[7F:70], Single Read/Write access to these offsets is not supported:
 - Offsets SBRMIx71, SBRMIx72 and SBRMIx73 are used for processor state access; refer to 8.4.2.1 [SB-RMI Processor State Access] for the access sequence.
 - Offsets SBRMIx70, SBRMIx[7F:74] are reserved for future expansion; access to these offsets is undefined.

8.4.2.4.1 SB-RMI Register Block Access

The following example shows a write from SBRMIx18 to SBRMIx1F using SB-RMI block write with SBRMIx01[BlkRWEn] set to 1.

Table 171: SB-RMI Register Block Write Protocol

Byte	Byte Name	Value	Notes
1	Dynamic Address	XXXXXXXX0b	Target Address (Write).
2	Command	18h	Indicates starting register SBRMIx18.
3	Byte Count	08h	Number of bytes returned.
4	Data Byte 1	00h	Write a value to SBRMIx18h.
5	Data Byte 2	00h	Write a value to SBRMIx19h.
6	Data Byte 3	00h	Write a value to SBRMIx1Ah.
7	Data Byte 4	00h	Write a value to SBRMIx1Bh.
8	Data Byte 5	00h	Write a value to SBRMIx1Ch.
9	Data Byte 6	00h	Write a value to SBRMIx1Dh.
10	Data Byte 7	00h	Write a value to SBRMIx1Eh.
11	Data Byte 8	00h	Write a value to SBRMIx1Fh.

The following example shows a read from SBRMIx10 to SBRMIx17 using SBI block read with SBRMIx01[BlkRWEn] set to 1 and SBRMIx03[RdSize] set to 8.

Table 172: SB-RMI Register Block Read Protocol

Byte	Byte Name	Value	Notes
1	Dynamic Address	XXXXXXXX0b	Target Address (Write).
2	Command	10h	Indicates starting register SBRMIx10.
3	Dynamic Address	XXXXXXXX1b	Target Address (Read).
4	Byte Count	08h	Number of bytes returned.
5	Data Byte 1	00h	Value of SBRMIx10h.
6	Data Byte 2	00h	Value of SBRMIx11h.
7	Data Byte 3	00h	Value of SBRMIx12h.
8	Data Byte 4	00h	Value of SBRMIx13h.
9	Data Byte 5	00h	Value of SBRMIx14h.
10	Data Byte 6	00h	Value of SBRMIx15h.
11	Data Byte 7	00h	Value of SBRMIx16h.
12	Data Byte 8	00h	Value of SBRMIx17h.

8.4.2.4.2 SB-RMI Register Byte Access

The following example shows a write to SBRMIx03 using the SBI single write protocol.

Table 173: SB-RMI Register Write Byte Protocol

Byte	Byte Name	Value	Notes
1	Dynamic Address	XXXXXXXX0b	Target Address (Write).
2	Command	03h	Indicates SB-RMI register 03.
3	Data Byte	04h	Write a value of 04h.

The following example shows a read from SBRMIx03 using the SBI single read with SBRMIx01[BlkRWEn] set to 0.

Table 174: SB-RMI Register Read Byte Protocol

Byte	Byte Name	Value	Notes
1	Dynamic Address	XXXXXXXX0b	Target Address (Write).
2	Command	03h	Indicates register SBRMI0x03.
3	Dynamic Address	XXXXXXXX1b	Target Address (Read).
4	Data Byte	04h	Value of SBRMIx03h.

8.4.2.5 SB-RMI Alert

The processor alerts the SBI when a Machine Check Exception occurs within the system. The Machine Check Exception status is reflected in registers SBRMIx[1F:10][MceStat], or SBRMIx[5F:50][MceStat]. To enable this functionality, SBRMIx[2F:20][MceAlertMsk], or SBRMIx[CF:C0][MceAlertMsk] must be clear.

The processor alerts the SBI on system fatal error event. This status is reflected in SBRMIx02[SwAlertSts]. To enable this functionality, SBRMIx01[SwAlertMask] must be clear. For this case, SBRMI::RasStatus[fatal_err] is set. controller can issue a Read command to dump this register after Alert detection, clears it by writing 1'b1 to the corresponding bit. For the SBRMI::RasStatus register definition, please see SB-RMI Registers section for the details.

The processor alerts the SBI on SBRMIx71, SBRMIx72 and SBRMIx73 command completion. This status is reflected in SBRMIx02[HwAlertSts]. To enable this functionality, SBRMIx01[HwAlertMask] must be clear.

8.4.3 SBI Error Detection and Recovery

This section describes the various error detection and recovery methods that can be used on the SBI bus. The important item in providing a high reliability SBI connection is the ability to detect when an error occurs and to gracefully recover from that error. When the SBI connections are noisy, messages can become garbled which, in turn, may cause undefined behavior on the SBI bus. The most common noise sources are cross-talk and clock skew. Cross-talk results when the SBI connections are routed too close to other signal carrying lines. Clock skew is usually a result of higher than expected capacitance, between the SBI signals (clock and / or data) and ground, which causes the controller and target devices to disagree on when data should be stable and when it is allowed to be changing.

8.4.3.1 Error Detection

SBI provides several methods of error detection: Transition from address ACK/NACK to SDR controller write data, ninth bit of SDR controller written data as parity checking. For SBRMIx71, SBRMIx72 and SBRMIx73 command, timeout is also protected. The ACK/NAK mechanism and parity checking is always active in SBI, but timeout is optional.

8.4.3.1.1 ACK/NAK Mechanism

When performing an SDR Write, the hand off from the target's Address Header ACK to the controller's first Data bit is different in I3C. The end of any Address Header is an ACK or NACK by the one or more addressed targets, using Open Drain on SDA. If 7'h7E is received, then it is the ACK of all I3C targets on the bus. If a single target address is received, then it is ACK or NACK of the addressed target. When the Address Header results in a NACK, the controller may choose to continue the transaction by generating a Repeated START or relinquish the bus by generating a STOP.

8.4.3.1.2 Ninth Bit of SDR Controller Written Data as Parity

The ninth data bit of each SDR Data Word written by the I3C controller (also referred to as the T-Bit) is a Parity bit, calculated using odd parity. Parity can help in detecting noise-caused errors on the line.

8.4.3.1.3 Bus Timeouts

The I3C protocol itself has no timeout mechanism. For commands such as SBRMIx71, SBRMIx72 and SBRMIx73, hardware will wait for MP to complete the operation by clearing pending interrupt, and then assert ALERT_L to controller to trigger following status/ data read. If MP has no response within a configured timeout period, then "Status Code" as "Command Timeout" is asserted as well as ALERT_L pin, to prevent a device waiting indefinitely on a message that may not be coming.

8.4.3.2 Error Recovery

The simplest form of error recovery is a retry. When the bus controller detects an unexpected NACK, it should abort the current transfer and retry the message sequence. In some cases, however, a message can be so garbled that a simple retry is insufficient. This can occur, if there are multiple devices on the bus and a garbled address byte has caused the wrong target device to be selected. That target device may even continue to transmit during the retry. In those cases, it will be necessary to force a reset of all devices on the SBI bus, before retrying the message transfer.

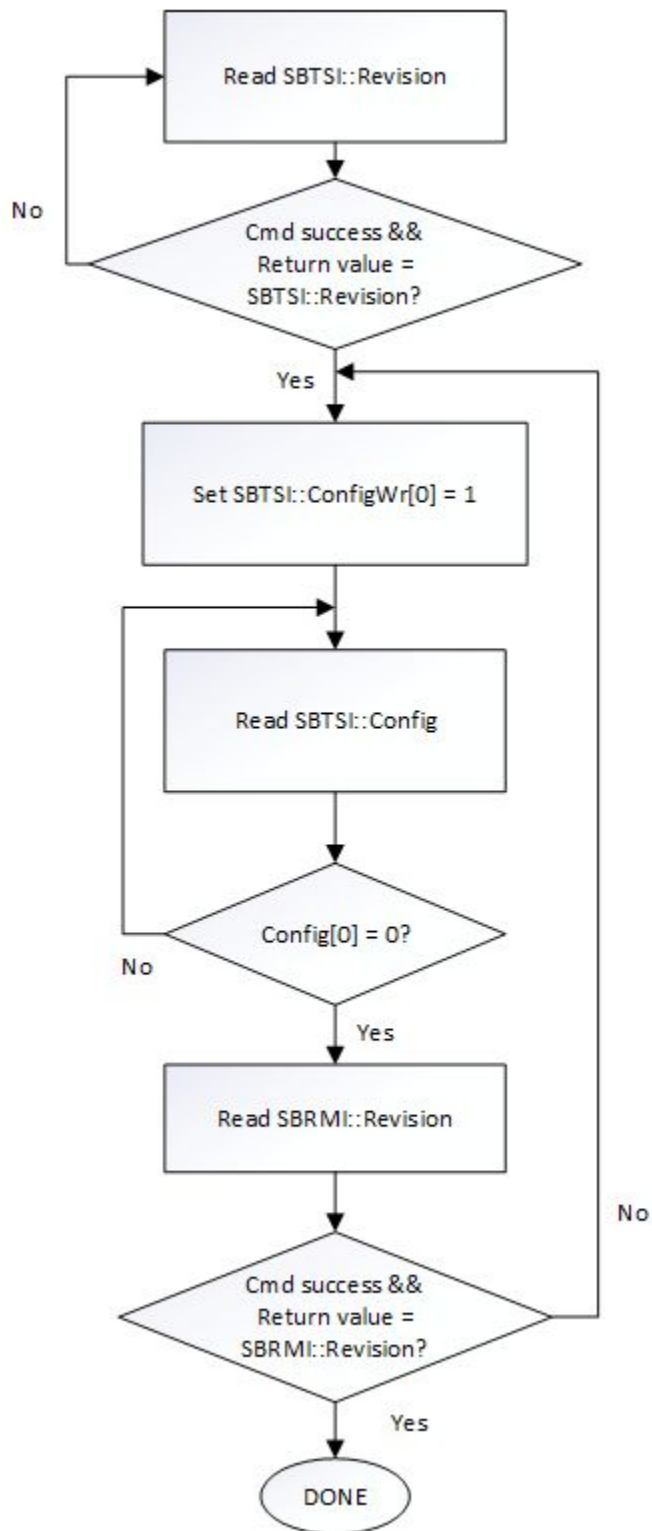
Unexpected reset of BMC in the middle of SBI transactions can potentially leave the target device controllers in a hung state or RX, TX FIFOs in non-empty state that could skew the data bytes associated with SBI requests. Therefore it is recommended that both SB-RMI and SB-TSI reset recovery flows should be executed by BMC in its boot flow to clear off stale state in the SBI targets.

8.4.3.2.1 SB-RMI Soft Reset Recovery

This section describes the method to recover the SB-RMI peripheral using a soft reset and avoid an expensive CPU socket reset. The soft reset recovery flow should be triggered by the SBI controller such as a BMC when the SB-RMI peripheral gets into a state such as a hang or perpetual NACK for all attempted transactions or read data that is out of sync with read requests (e.g. sequence of reads for registers <R1, R2, R3> results in data sequence of <Indeterminate, R1value, R2value>).

To trigger a soft reset recovery flow of the SB-RMI, assuming that the SB-TSI peripheral is functional, the following sequence should be followed:

1. SBI controller writes 1 to SBTSI::ConfigWr[RMISoftReset].
2. [MP1](#) firmware polls on that bit and when set, executes the SB-RMI peripheral recovery flow through a soft reset of the I3C hardware and clears it upon completion.
3. BMC polls for SBTSI::Config[RMISoftReset] to clear. Once satisfied, BMC can start sending I2C/I3C commands to SB-RMI again.

*Figure 36: SB-RMI Soft Reset Recovery*

8.5 SBI Physical Interface

8.5.1 SBI Static Address

The I3C controller should assign dynamic addresses to any I3C target devices with or without a known static address based on I3C protocol. If it is assigned based on static address, then SETDASA CCC is sent from controller to target, otherwise ENTDAAC CCC is sent from controller to target. Currently static address is supported.

The I3C dynamic or static address are really 7 bits: bits[7:1] as the left-justified address, and bit[0] as the Read/Write flag, where 0 indicates a Write and 1 indicates a Read.

The table below shows the SBI Address Encoding.

Table 175: SBI Address Encoding

Socket ID / Socket number	SB-RMI Address	Comment
0000b	3Ch	socket 0 SB-RMI address
0001b	38h	socket 1 SB-RMI address

8.5.2 SBI Bus Timing

SBI supports 12.5MHz SDR I3C operation. Refer to the timing parameters in the I3C specification.

8.6 SB-RMI Registers

Reads to unimplemented registers return 00h. Writes to unimplemented registers are discarded.

SBRMIx00 [Revision] (SBRMI::Revision)	
Read-only. Reset: 20h.	
Bits	Description
7:0	Revision: SB-RMI revision. Read-only. Reset: 20h. This field specifies the APML -I3C specification revision that the product is compliant to. 0x20 = 2.0x Revision.

SBRMIx01 [Control] (SBRMI::Control)

Read-write. Reset: 01h.

Bits	Description
7	HwAlertMask: hardware alert mask. Read-write. Reset: 0. 0=ALERT_L signaling is enabled when SBRMIx02[HwAlertSts] is set. 1=ALERT_L signaling is disabled when SBRMIx02[HwAlertSts] is set.
6:5	Reserved.
4	SwAlertMask: software alert mask. Read-write. Reset: 0. 0=ALERT_L signaling is enabled when SBRMIx02[SwAlertSts] is set. 1=ALERT_L signaling is disabled when SBRMIx02[SwAlertSts] is set.
3	BlkRWEn: block read/write enable. Read-write. Reset: 0. 0=I3C accesses can only use the byte read/write protocol. 1=I3C accesses can only use the block read/write protocol. Description: Controls block read/write access to register ranges SBRMIx[5F:10][MceStat] and SBRMIx[DE:80] which encompassed a register set of SBRMIx[87:80][MP0OutBndMsg], SBRMIx[98:91][threadEnStat], SBRMIx[DF:D8][threadEnStat]. NOTE: For register ranges SBRMIx[0F:00], it only support Single Write independent of the state of the BlkRWEn control bit, but its read still depends on BlkRWEn control bit.
2	TimeoutDis: SB-RMI timeout disable. Read-write. Reset: 0. 1=I3C defined timeouts are disabled. It is only used for SB-RMI to process SBRMIx71, SBRMIx72 and SBRMIx73 commands to wait for Firmware to complete the commands by clearing pending interrupt.
1	TSISoftReset: trigger TSI soft reset flow. Read-write. Reset: 0. 1=SBI controller triggers MP1 firmware to soft reset SB-TSI peripheral. 0=soft reset of SB-TSI I3C hardware completed and BMC may proceed communication with the SB-TSI peripheral. See 9.2.7 [SB-TSI Error Detection and Recovery] for details.
0	AlertMask: SB-RMI alert mask. Read-write. Reset: 1. Read-write. 1=ALERT_L signaling disabled. 0=ALERT_L is asserted if any unmasked event is present in the SBRMI::AlertStatus registers SBRMIx[1F:10][MceStat], or SBRMIx[5F:50], or if SBRMIx02[SwAlertSts] = 1 and SBRMIx01[SwAlertMask] = 0, or if SBRMIx02[HwAlertSts] = 1 and SBRMIx01[HwAlertMask] = 0.

SBRMIx02 [Status] (SBRMI::Status)

Reset: 00h.

Bits	Description
7	HwAlertSts: SB-RMI hardware alert status. Read-write, Volatile. Reset: 0. Write-one-to-clear from the I3C interface; read-write from the processor. Set by hardware as a result of a SBRMIx71, SBRMIx72 and SBRMIx73 command completion and ready to assert ALERT_L for this completion.
6	MP0AlertSts: Alert from MP0 for MP0 outbound message. Read-write, Volatile. Reset: 0. Write-one-to-clear from the I3C interface; set by hardware when MP0 firmware writes to an internal register after setting up Mp0OutBndMsg registers. It indicates availability of MP0 data to BMC. This bit was formerly named PSPAlertStatus.
5:2	Reserved.
1	SwAlertSts: SB-RMI software alert status. Read-write, Volatile. Reset: 0. Write-one-to-clear from the I3C interface; read-write from the processor. Set by firmware to indicate the completion of a mailbox operation or due to an asynchronous event, such as those reported by the SBRMI::RasStatus register.
0	AlertSts: SB-RMI alert status. Read-only, Volatile. Reset: 0. This bit is set to 1 by hardware when an alert event is present in at least one bit of SBRMIx[1F:10][MceStat], or SBRMIx[5F:50][MceStat]. It is cleared by hardware when all those registers are clear.

SBRMIx03 [Read Size] (SBRMI::ReadSize)

Read-write. Reset: 01h.

This register specifies the number of bytes to return when using the block read protocol.

Bits	Description
7:6	Reserved.
5:0	RdSize: read size. Read-write. Reset: 01h. Specifies the number of bytes to return when using the block read protocol.
ValidValues:	
Value	Description
00h	Reserved.
20h-01h	<Value> bytes.
3Fh-21h	Reserved.

SBRMIx[04...98] [Thread Enable Status] (SBRMI::ThreadEnableStatus)

Read-only.

_inst[23:0]; SBRMIx[9[8:1],4[A:3],0[D:8],05,04]

Bits	Description		
7:0	threadEnStat: thread Enable Status. Read-only.		
	Description: 1=thread is enabled.		
	Offset[7:0]	inst	Description
	04h	0	threads [7:0]
	05h	1	threads [15:8]
	08h	2	threads [23:16]
	09h	3	threads [31:24]
	0Ah	4	threads [39:32]
	0Bh	5	threads [47:40]
	0Ch	6	threads [55:48]
	0Dh	7	threads [63:56]
	43h	8	threads [71:64]
	44h	9	threads [79:72]
	45h	10	threads [87:80]
	46h	11	threads [95:88]
	47h	12	threads [103:96]
	48h	13	threads [111:104]
	49h	14	threads [119:112]
	4Ah	15	threads [127:120]
	91h	16	threads [135:128]
	92h	17	threads [143:136]
	93h	18	threads [151:144]
	94h	19	threads [159:152]
	95h	20	threads [167:160]
	96h	21	threads [175:168]
97h	22	threads [183:176]	
98h	23	threads [191:184]	

SBRMIx[10...5F] [Alert Status] (SBRMI::AlertStatus)

Read,Write-1-to-clear,Volatile.

_inst[31:0]; SBRMIx[5[F:0],1[F:0]]

Bits	Description		
7:0	MceStat: MCE status. Read,Write-1-to-clear,Volatile.		
	Description: Bit vector for threads. 1=MCE occurred for thread. Set by hardware. Some register instances hold only 4 status bits, in which case the upper 4 bits are reserved.		
	Offset[7:0]	inst	Description
	10h	0	threads [176,160,144,128,48,32,16,0]
	11h	1	threads [177,161,145,129,49,33,17,1]
	12h	2	threads [178,162,146,130,50,34,18,2]
	13h	3	threads [179,163,147,131,51,35,19,3]
	14h	4	threads [180,164,148,132,52,36,20,4]
	15h	5	threads [181,165,149,133,53,37,21,5]
	16h	6	threads [182,166,150,134,54,38,22,6]
	17h	7	threads [183,167,151,135,55,39,23,7]
	18h	8	threads [184,168,152,136,56,40,24,8]
	19h	9	threads [185,169,153,137,57,41,25,9]
	1Ah	10	threads [186,170,154,138,58,42,26,10]
	1Bh	11	threads [187,171,155,139,59,43,27,11]
	1Ch	12	threads [188,172,156,140,60,44,28,12]
	1Dh	13	threads [189,173,157,141,61,45,29,13]
	1Eh	14	threads [190,174,158,142,62,46,30,14]
	1Fh	15	threads [191,175,159,143,63,47,31,15]
	50h	16	{0,0,0,0}, threads [112,96,80,64]
	51h	17	{0,0,0,0}, threads [113,97,81,65]
	52h	18	{0,0,0,0}, threads [114,98,82,66]
	53h	19	{0,0,0,0}, threads [115,99,83,67]
	54h	20	{0,0,0,0}, threads [116,100,84,68]
	55h	21	{0,0,0,0}, threads [117,101,85,69]
	56h	22	{0,0,0,0}, threads [118,102,86,70]
	57h	23	{0,0,0,0}, threads [119,103,87,71]
	58h	24	{0,0,0,0}, threads [120,104,88,72]
	59h	25	{0,0,0,0}, threads [121,105,89,73]
	5Ah	26	{0,0,0,0}, threads [122,106,90,74]
	5Bh	27	{0,0,0,0}, threads [123,107,91,75]
	5Ch	28	{0,0,0,0}, threads [124,108,92,76]
	5Dh	29	{0,0,0,0}, threads [125,109,93,77]
	5Eh	30	{0,0,0,0}, threads [126,110,94,78]
	5Fh	31	{0,0,0,0}, threads [127,111,95,79]

SBRMIx[20...CF] [Alert Mask] (SBRMI::AlertMask)

Read-write.

_inst[31:0]; SBRMIx[C[F:0],2[F:0]]

Bits	Description	
7:0	MceAlertMsk: MCE alert mask. Read-write.	
	Description: Bit vector for threads. 1=Alert signaling disabled for corresponding SBRMI::AlertStatus[MceStat] for thread. Some register instances hold only 4 mask bits, in which case the upper 4 bits are reserved.	
	Offset[7:0]	inst Description
	20h	0 threads [176,160,144,128,48,32,16,0]
	21h	1 threads [177,161,145,129,49,33,17,1]
	22h	2 threads [178,162,146,130,50,34,18,2]
	23h	3 threads [179,163,147,131,51,35,19,3]
	24h	4 threads [180,164,148,132,52,36,20,4]
	25h	5 threads [181,165,149,133,53,37,21,5]
	26h	6 threads [182,166,150,134,54,38,22,6]
	27h	7 threads [183,167,151,135,55,39,23,7]
	28h	8 threads [184,168,152,136,56,40,24,8]
	29h	9 threads [185,169,153,137,57,41,25,9]
	2Ah	10 threads [186,170,154,138,58,42,26,10]
	2Bh	11 threads [187,171,155,139,59,43,27,11]
	2Ch	12 threads [188,172,156,140,60,44,28,12]
	2Dh	13 threads [189,173,157,141,61,45,29,13]
	2Eh	14 threads [190,174,158,142,62,46,30,14]
	2Fh	15 threads [191,175,159,143,63,47,31,15]
	C0h	16 {0,0,0,0}, threads [112,96,80,64]
	C1h	17 {0,0,0,0}, threads [113,97,81,65]
	C2h	18 {0,0,0,0}, threads [114,98,82,66]
	C3h	19 {0,0,0,0}, threads [115,99,83,67]
	C4h	20 {0,0,0,0}, threads [116,100,84,68]
	C5h	21 {0,0,0,0}, threads [117,101,85,69]
	C6h	22 {0,0,0,0}, threads [118,102,86,70]
	C7h	23 {0,0,0,0}, threads [119,103,87,71]
	C8h	24 {0,0,0,0}, threads [120,104,88,72]
	C9h	25 {0,0,0,0}, threads [121,105,89,73]
	CAh	26 {0,0,0,0}, threads [122,106,90,74]
	CBh	27 {0,0,0,0}, threads [123,107,91,75]
	CCh	28 {0,0,0,0}, threads [124,108,92,76]
	CDh	29 {0,0,0,0}, threads [125,109,93,77]
	CEh	30 {0,0,0,0}, threads [126,110,94,78]
	CFh	31 {0,0,0,0}, threads [127,111,95,79]

SBRMIx3[0...7] [Out-Bound Message] (SBRMI::OutBndMsg)

Read-write. Reset: 00h.

_inst[7:0]; SBRMIx3[7:0]

Bits	Description
7:0	OutBndMsg: outbound message data. Read-write. Reset: 00h.
	Description: Read-write from the processor; read-only from the I3C interface.
	Usage convention is:
	- SBRMI::OutBndMsg_inst0 is command copied by firmware from SBRMI::InBndMsg_inst0. - SBRMI::OutBndMsg_inst[4:1] are 32 bit data. - SBRMI::OutBndMsg_inst[5] reserved. - SBRMI::OutBndMsg_inst[6] is count of RMI soft reset flows executed by MP1 firmware since CPU reset - SBRMI::OutBndMsg_inst[7] contains Mailbox Error Code, per Table 170 [SB-RMI Soft Mailbox Error Code].

Offset[7:0]	inst	Description
30h	0	Outbound message 0
31h	1	Outbound message 1
32h	2	Outbound message 2
33h	3	Outbound message 3
34h	4	Outbound message 4
35h	5	Outbound message 5
36h	6	Outbound message 6
37h	7	Outbound message 7

SBRMIx3[8...F] [In-Bound Message] (SBRMI::InBndMsg)

Read-write. Reset: 00h.

_inst[7:0]; SBRMIx3[F:8]

Bits	Description																											
7:0	InBndMsg: inbound message data. Read-write. Reset: 00h.																											
	Description: Read-write from the I3C interface; read-only from the processor. These registers are used for communicating 32 bit messages from BMC to firmware.																											
	Usage convention is:																											
	- SBRMI::InBndMsg_inst0 is command - SBRMI::InBndMsg_inst[4:1] are 32 bit data - SBRMI::InBndMsg_inst[6:5] are reserved. - SBRMI::InBndMsg_inst7: [7] Must be 1'b1 to send message to firmware																											
	<table><tr><th>Offset[7:0]</th><th>inst</th><th>Description</th></tr><tr><td>38h</td><td>0</td><td>Inbound message 0</td></tr><tr><td>39h</td><td>1</td><td>Inbound message 1</td></tr><tr><td>3Ah</td><td>2</td><td>Inbound message 2</td></tr><tr><td>3Bh</td><td>3</td><td>Inbound message 3</td></tr><tr><td>3Ch</td><td>4</td><td>Inbound message 4</td></tr><tr><td>3Dh</td><td>5</td><td>Inbound message 5</td></tr><tr><td>3Eh</td><td>6</td><td>Inbound message 6</td></tr><tr><td>3Fh</td><td>7</td><td>Inbound message 7</td></tr></table>	Offset[7:0]	inst	Description	38h	0	Inbound message 0	39h	1	Inbound message 1	3Ah	2	Inbound message 2	3Bh	3	Inbound message 3	3Ch	4	Inbound message 4	3Dh	5	Inbound message 5	3Eh	6	Inbound message 6	3Fh	7	Inbound message 7
	Offset[7:0]	inst	Description																									
	38h	0	Inbound message 0																									
	39h	1	Inbound message 1																									
	3Ah	2	Inbound message 2																									
	3Bh	3	Inbound message 3																									
3Ch	4	Inbound message 4																										
3Dh	5	Inbound message 5																										
3Eh	6	Inbound message 6																										
3Fh	7	Inbound message 7																										

SBRMIx40 [Software Interrupt] (SBRMI::SoftwareInterrupt)

Read,Write-1-only. Reset: 00h.

This register is used by the I3C controller to generate an interrupt to the processor to indicate that a message is available.

Bits	Description
7:1	Reserved.
0	SwInt: firmware interrupt. Read,Write-1-only. Reset: 0. Read, write-1-only from the I3C interface; read, write-1-to-clear from firmware. 1=Indicates a firmware mailbox service request.

SBRMIx4B [Thread128 CS register] (SBRMI::Thread128CS)

Read-write. Reset: 00h.

This register is used to extend the threads from 128 to 256 for SBRMIx71, SBRMIx72 and SBRMIx73 commands. Setting as 1'b1 to select high 128 threads while setting as 1'b0 to select low 128 threads. It is set by I3C interface.

Bits	Description
7:1	Reserved.
0	thread128cs: high or low thread 128 selection register. Read-write. Reset: 0. Read-Write from the I3C interface; specifies the high or low thread 128 indicated in SBRMIx71, SBRMIx72 and SBRMIx73 commands. Setting as 1'b0 to indicate the low 128 threads; setting as 1'b1 to indicate the high 128 threads.

SBRMIx4C [RAS Status register] (SBRMI::RasStatus)

Read-write. Reset: 00h.

This register is used to indicate the [APML](#) ALERT source. It is set by processor, and write-1-to-clear by I3C interface.

Bits	Description
7:3	Reserved.
2	reset_ctrl_err: reset control error. Read-write. Reset: 0. Read-Write from the I3C interface; specifies the APML ALERT source is due to a reset controller error; warm-reset requests may not complete. It is set by software, and write-1-to-clear by I3C interface.
1	fch_err: FCH error. Read-write. Reset: 0. Read-Write from the I3C interface; specifies the APML ALERT source is due to a FCH error; warm- or cold-reset requests may not complete. It is set by software, and write-1-to-clear by I3C interface.
0	fatal_err: fatal error. Read-write. Reset: 0. Read-Write from the I3C interface; specifies the APML ALERT source is due to a fatal error. It is set by software, and write-1-to-clear by I3C interface.

SBRMIx4E [Thread Number Low] (SBRMI::ThreadNumberLow)

Read-write. Reset: 00h.

This register indicates the low part of maximum number of threads present, starting from 1.

Bits	Description
7:0	threadNumlow: thread number low. Read-write. Reset: 00h. Read-only from the I3C interface; specifies the low part of the maximum number of threads present. Firmware loads the initial value based on the maximum number of threads available after any fused off or soft-down-coring is complete. 'd64 means max of 64 threads, 'd128 means max of 128 threads.

SBRMIx4F [Thread Number High] (SBRMI::ThreadNumberHigh)

Read-write. Reset: 00h.

This register indicates the high part of maximum number of threads present. It is only valid when 256 threads needs to be supported.

Bits	Description
7:0	threadNumhigh: thread number high. Read-write. Reset: 00h. Read-only from the I3C interface; specifies the high part of the maximum number of threads present. This register is only needed when thread number is equal or larger than 256. Firmware loads the initial value based on the maximum number of threads available after any fused off or soft-down-coring is complete.

SBRMIx8[0...7] [ASP Out-Bound Message] (SBRMI::MP0OutBndMsg)

Read-write. Reset: 00h.

_inst[7:0]; SBRMIx8[7:0]

Bits	Description																											
7:0	MP0OutBndMsg: outbound message data. Read-write. Reset: 00h. Description: Read-write from the processor; read-only from the I3C interface. These registers are used for sending messages from ASP firmware running on the ASP to the I3C controller. ASP boot status is dynamically written to this register during the boot process. <table><tr><th>Offset[7:0]</th><th>inst</th><th>Description</th></tr><tr><td>80h</td><td>0</td><td>ASP Outbound message 0</td></tr><tr><td>81h</td><td>1</td><td>ASP Outbound message 1</td></tr><tr><td>82h</td><td>2</td><td>ASP Outbound message 2</td></tr><tr><td>83h</td><td>3</td><td>ASP Outbound message 3</td></tr><tr><td>84h</td><td>4</td><td>ASP Outbound message 4</td></tr><tr><td>85h</td><td>5</td><td>ASP Outbound message 5</td></tr><tr><td>86h</td><td>6</td><td>ASP Outbound message 6</td></tr><tr><td>87h</td><td>7</td><td>ASP Outbound message 7</td></tr></table>	Offset[7:0]	inst	Description	80h	0	ASP Outbound message 0	81h	1	ASP Outbound message 1	82h	2	ASP Outbound message 2	83h	3	ASP Outbound message 3	84h	4	ASP Outbound message 4	85h	5	ASP Outbound message 5	86h	6	ASP Outbound message 6	87h	7	ASP Outbound message 7
Offset[7:0]	inst	Description																										
80h	0	ASP Outbound message 0																										
81h	1	ASP Outbound message 1																										
82h	2	ASP Outbound message 2																										
83h	3	ASP Outbound message 3																										
84h	4	ASP Outbound message 4																										
85h	5	ASP Outbound message 5																										
86h	6	ASP Outbound message 6																										
87h	7	ASP Outbound message 7																										

SBRMIx90 [ASP Alert Mask] (SBRMI::MP0AlertMask)

Read-write. Reset: 00h.

Bits	Description
7:2	Reserved.
1	Mp0AlertMask. Read-write. Reset: 0. 0=ALERT_L signaling is enabled when MP0AlertSts is set. 1=ALERT_L signaling is disabled when MP0AlertSts is set.
0	Reserved.

9 SB Temperature Sensor Interface (SB-TSI)

9.1 Overview

The SBI temperature sensor interface (SB-TSI) integrates the functionality of a typical 8-pin remote temperature sensor (RTS) shown in Figure 37 [RTS Thermal Management Example], into the CPU socket. The goal is to resemble a typical RTS so that KBC or BMC firmware requires minimal changes for future AMD products as shown in Figure 38 [SB-TSI Thermal Management Example]. SB-TSI supports the I3C protocols that typical RTS supports.

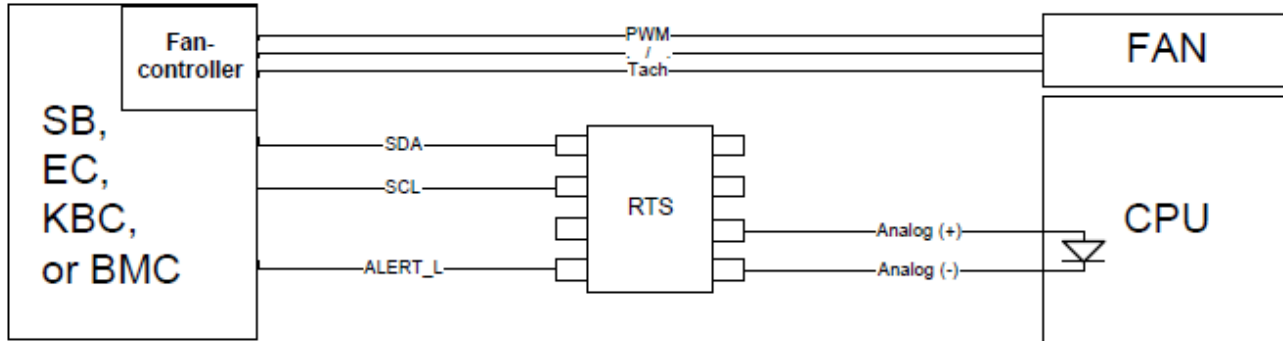


Figure 37: RTS Thermal Management Example

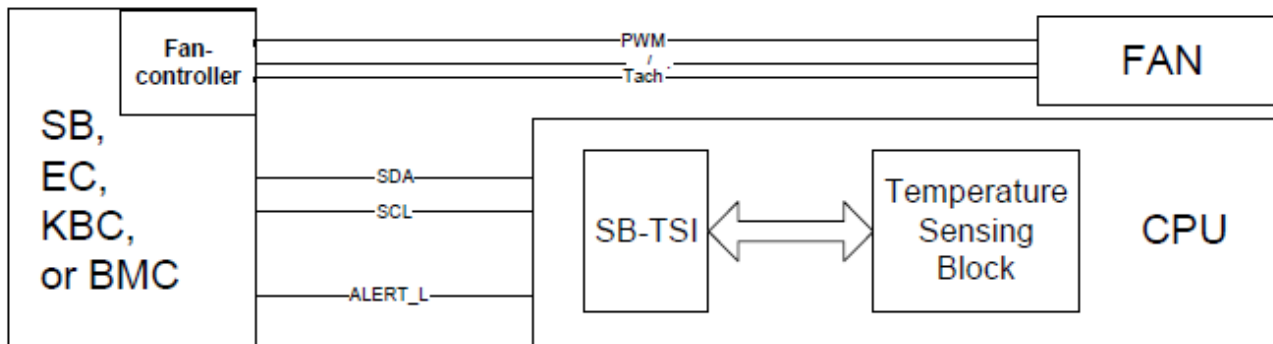


Figure 38: SB-TSI Thermal Management Example

Refer to the following external sources for additional information.

- MIPI I3C specification Version 1.0, see [docI3C](#).

9.1.1 Definitions

Table 176: SB-TSI Definitions

Term	Description
BMC	Base management controller.
TCC	Temperature calculation circuit.
Tctl	Processor temperature control value.
TSM	Temperature sensor macro.
SB-TSI	Sideband Internal Temperature Sensor Interface. See APML .

9.1.2 Tctl

Tctl is a processor temperature control value used for processor thermal management. In mission mode it reports the maximum temperature amongst multiple on-die temperature sensors located across all the constituent chiplets in a socket. Tctl is accessible externally on the platform through SBTSI interface. Tctl does not represent the real measured temperature but instead an offset adjusted value such that a value of 100 represents the max sustainable operating die temperature depending on the processor family and model. This enables the abstraction of the real temperature to a standard scale where fan speed profile can be specified against standardized Tctl values. Refer to [docTDG](#) for further details.

9.2 SB-TSI Protocol

The SB-TSI largely follows I3C protocol with the following characteristics:

- The processor does not implement I3C controller functionality. It only functions as a target.
- Max of 12.5Mhz SDR mode is supported.
- The SBI mainly uses the I3C private Write and Read transfers to support Single Write and Read.
- As a target device, it can only drive SDA, but not SCL to controller.
- Dynamic address assigned based on Static address can be supported for I3C target.
- Packet Error Correction (PEC) is not supported in I2C or I3C modes by target.

In order to allow [MP1](#) firmware to perform initialization of certain SB-RMI and SB-TSI registers, BMC should hold off conducting any transaction to either of these targets for at least 2 secs after cold or warm reset of the CPU.

9.2.1 I3C Private Write and Read Protocol

SB-TSI uses a standard I3C protocol to provide Private Write and Private Read Transfers. Figure below shows the I3C spec defined Private Write and Private Read transmission protocol. Each byte in the protocol is sent with the most significant bit first (bit[7]). The ninth data bit of each SDR Data Word written by the I3C controller (also referred to as the T-Bit) is a Parity bit, calculated using odd parity. The ninth (T) Data bit of each SDR Data Word returned by the I3C target indicates if it is the End-of-Data, it returns the ninth bit as 0 (SDA Low) to end the Message.

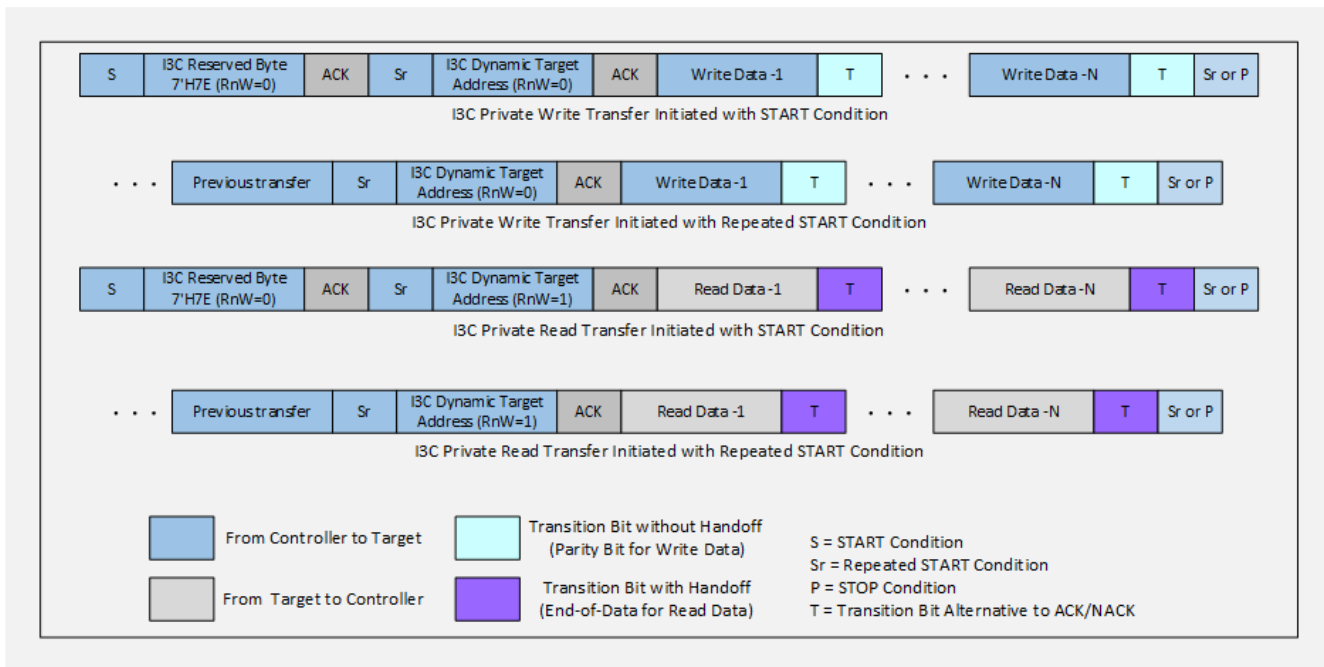


Figure 39: I3C Transmission Protocol for Private Write and Read

9.2.2 SB-TSI Read/Write Byte Protocol

An I3C controller can Read or Write SB-TSI internal registers by issuing a Private Read or a Write command with register address, written data filled in the transmission. SB-TSI only supports Single Read or Write. Figure below shows the transmission protocol for Single Write and Read based on I3C defined Private Write and Read protocol. For Single Write, one byte of "Command Code" is provided for register offset and one byte of "Write Data" is provided for written data. For Single Read, one byte of "Command Code" is provided for register offset by using a Private Write first, and followed by a Private Read to read one byte of data using Repeated START or START condition.

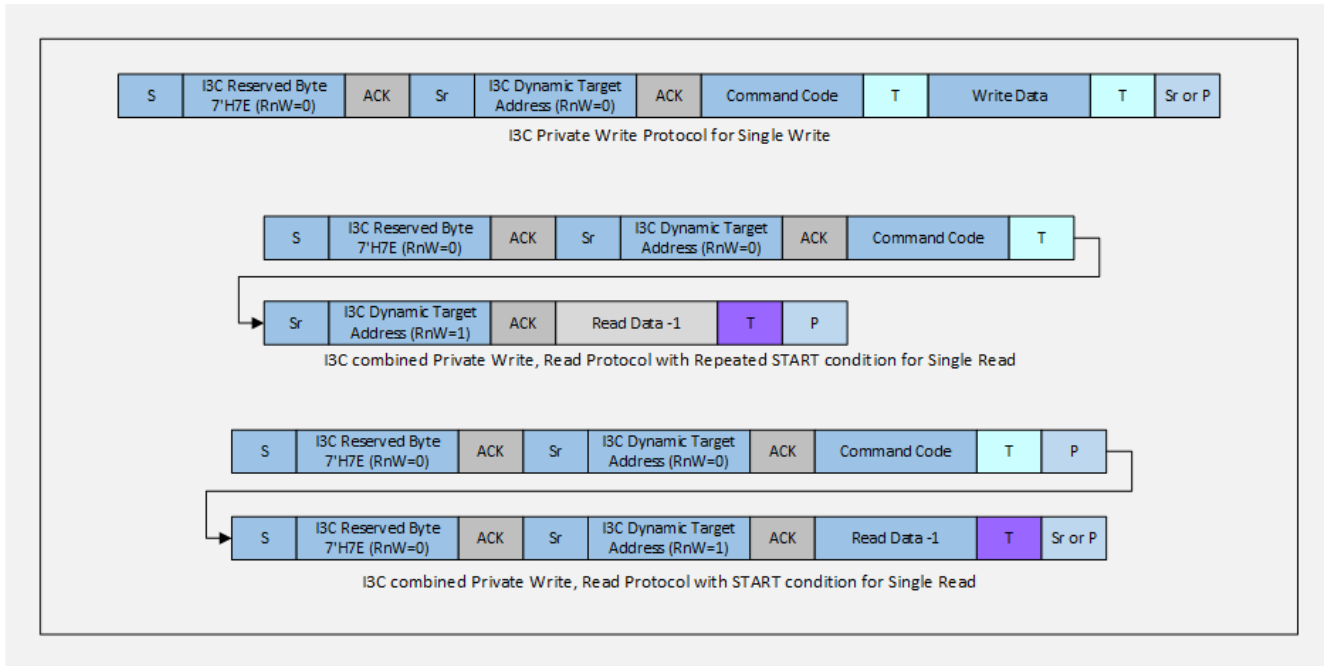


Figure 40: SB-TSI Transmission Protocol for Single Write and Read

9.2.3 Alert Behavior

The ALERT_L pin is asserted if (SBTSI::Status[TempHighAlert] || SBTSI::Status[TempLowAlert]) && ~SBTSI::Config[AlertMask] as shown in Figure 3. The following registers also affect temperature alert behavior.

- SBTSI::UpdateRate[UpRate]: Specifies rate at which temperature thresholds are checked.
- {SBTSI::HiTempInt[HiTempInt], SBTSI::HiTempDec[HiTempDec]}: Sets high temperature threshold.
- {SBTSI::LoTempInt[LoTempInt], SBTSI::LoTempDec[LoTempDec]}: Sets low temperature threshold.
- SBTSI::AlertThreshold[AlertThr]: Specifies number of consecutive temperature samples to assert an alert.

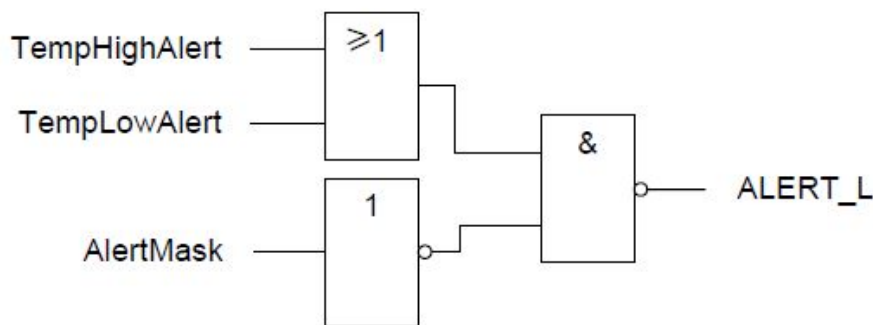


Figure 41: Alert Assertion Diagram

9.2.4 Atomic Read Mechanism

Atomic reading procedures are required to ensure the two required Reads (integer and decimal) parts of CPU temperature always originate from one temperature value. SB-TSI offers functions to maintain atomicity between the temperature integer and decimal bytes.

SBTSI::Config[ReadOrder] specifies the order for reading the following registers:

- Integer and decimal part of SBTSI::CpuTempInt and SBTSI::CpuTempDec

If SBTSI::Config[ReadOrder] is 0, then a read of the integer part triggers a latch of the decimal part until the next read of the integer part. This latching syncs the decimal part with the integer part. The integer part is updated continuously.

If SBTSI::Config[ReadOrder] is 1, the read order to ensure atomicity is reversed, i.e., the read of the decimal part triggers a latch of the integer part until the next read of the decimal part. This latching syncs the integer part with the decimal part. The decimal part is updated continuously.

If it is not possible to specify a read order as described above, the Run/Stop bit (SBTSI::Config[RunStop]) may be used to provide atomicity of reading the CPU temperature. If this bit is 0, the CPU temperature registers are updated continuously. If this bit is 1, the CPU temperature registers are no longer updated and any subsequent read requests will deliver the last latched value.

- Set SBTSI::Config[RunStop].
- Read the integer (SBTSI::CpuTempInt) or the decimal (SBTSI::CpuTempDec) part of the CPU temperature.
- Read the remaining part of the CPU temperature.
- Clear SBTSI::Config[RunStop].

9.2.5 SB-TSI Temperature and Threshold Encodings

SB-TSI CPU temperature readings and limit registers encode the temperature in increments of 0.125 from 0 to 255.875. The high byte represents the integer portion of the temperature from 0 to 255. One increment in the high byte is equivalent to a step of one. The upper three bits of the low byte represent the decimal portion of the temperature. One increment of these bits is equivalent to a step of 0.125.

Table 177: SB-TSI CPU Temperature and Threshold Encoding Examples

Temperature	Temperature High Byte SBTSI::CpuTempInt[CpuTempInt] SBTSI::HiTempInt[HiTempInt] SBTSI::LoTempInt[LoTempInt]	Temperature Low Byte SBTSI::CpuTempDec[CpuTempDec] SBTSI::HiTempDec[HiTempDec] SBTSI::LoTempDec[LoTempDec]
0.000 °C	0000_0000b	0000_0000b
1.000 °C	0000_0001b	0000_0000b
25.125 °C	0001_1001b	0010_0000b
50.875 °C	0011_0010b	1110_0000b
90.000 °C	0101_1010b	0000_0000b

9.2.6 SB-TSI Temperature Offset Encoding

By default, SBTSI::CpuTempInt and SBTSI::CpuTempDec provide Tctl from the processor. The temperature offset registers allow the system to adjust the SB-TSI temperature from Tctl.

The SB-TSI temperature offset registers use a different encoding in order to provide negative temperature values. SBTSI::CpuTempOffInt[CpuTempOffInt] and SBTSI::CpuTempOffDec[CpuTempOffDec] form an 11-bit, 2's complement value representing the temperature offset. The high byte encodes the integer portion of the temperature and the upper three bits of the low byte represent the fractional portion of the temperature offset. One increment of these bits is equivalent to a step of 0.125 °C. After reset the offset is always set to 0 °C. Software needs to adjust the offset to the appropriate level.

Table 178: SB-TSI Temperature Offset Encoding Examples

Temperature	Temperature High Byte SBTSI::CpuTempOffInt[CpuTempOffInt]	Temperature Low Byte SBTSI::CpuTempOffDec[CpuTempOffDec]
-10.375 °C	1111_0101b	1010_0000b
-0.250 °C	1111_1111b	1100_0000b
0.000 °C	0000_0000b	0000_0000b
0.875 °C	0000_0000b	1110_0000b
10.000 °C	0000_1010b	0000_0000b

9.2.7 SB-TSI Error Detection and Recovery

SB-TSI is one of the two peripheral SBI devices on the SBI bus. The section 8.4.3 [SBI Error Detection and Recovery] describes the overall error detection and recovery methods available for SBI bus. This section describes the method to recover the SB-TSI peripheral using a soft reset and avoid an expensive CPU socket reset. The soft reset recovery flow should be triggered by the SBI controller such as a BMC when the SB-TSI peripheral gets into a state such as a hang or perpetual NACK for all attempted transactions or read data that is out of sync with read requests (e.g. sequence of reads for registers <R1, R2, R3> results in data sequence of <Indeterminate, R1value, R2value>).

To trigger a soft reset recovery flow of the SB-TSI, assuming that the SB-RMI peripheral is functional, the following sequence should be followed:

1. SBI controller writes 1 to SBRMI::Control[TSISoftReset].
2. [MP1](#) firmware polls on that bit and when set, executes the SB-TSI peripheral recovery flow through a soft reset of the I3C hardware and clears it upon completion.
3. BMC polls for TSISoftReset to clear. Once satisfied, BMC can start sending I2C/I3C commands to TSI again.

Unexpected reset of BMC in the middle of SBI transactions can potentially leave the target device in a hung state or RX, TX FIFOs in non-empty state that could skew the data bytes associated with SBI requests. Therefore it is recommended that both SB-RMI and SB-TSI reset recovery flows should be executed by BMC in its boot flow to clear off stale state in the SBI targets.

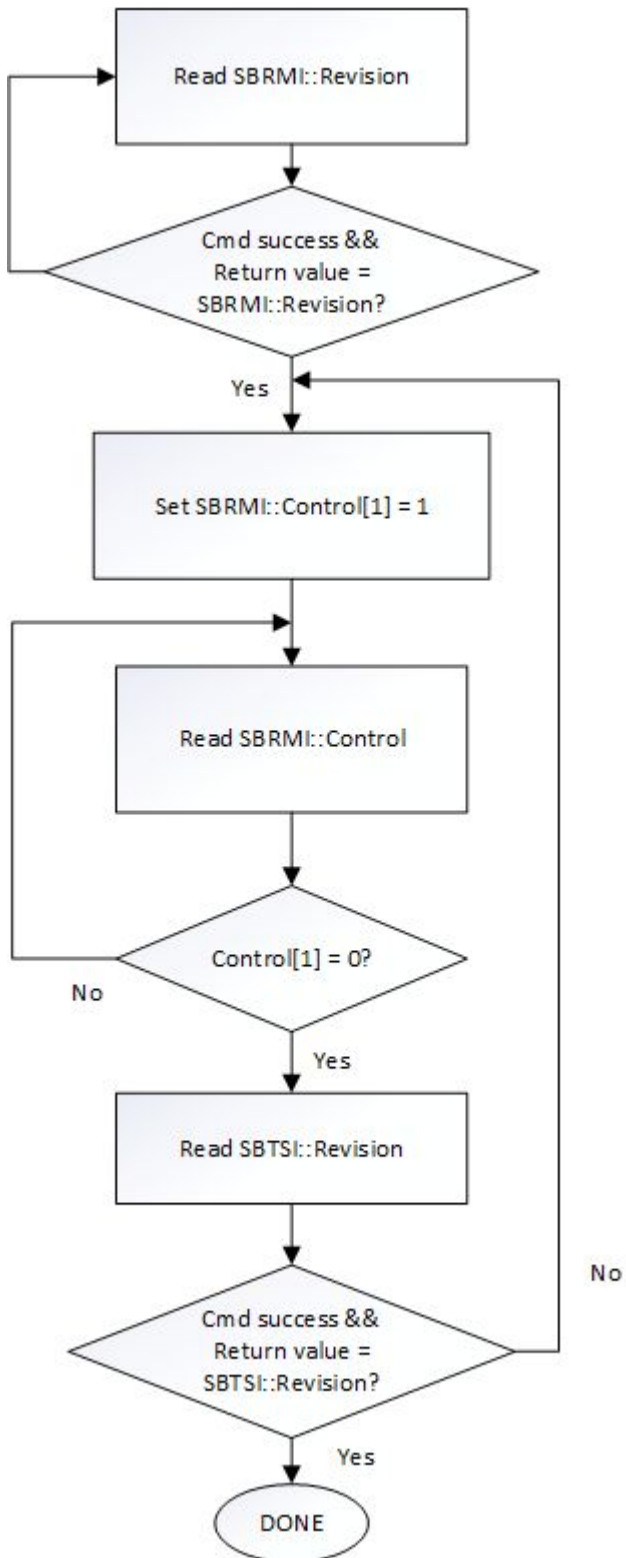


Figure 42: SB-TSI Soft Reset Recovery

9.3 SB-TSI Physical Interface

This chapter describes the physical interface of the SB-TSI.

9.3.1 SB-TSI I3C Static Address

The I3C controller should assign dynamic addresses to any I3C target devices with or without a known static address based on I3C protocol. If it is assigned based on static address, then SETDASA CCC is sent from controller to target, otherwise ENTDAAC CCC is sent from controller to target. Currently static address is supported.

The I3C dynamic or static address are really 7 bits, and left with the 1 bit of Read/Write indication as a WRITE (0) making bit[0].

Table 179: SB-TSI Static Address Encodings

Socket ID	SB-TSI Address
0b	4Ch for 7 bit.
1b	48h for 7 bit.

9.3.2 SB-TSI Bus Timing

SB-TSI supports 12.5MHz SDR mode according to the I3C-bus Specification and User Manual.

9.3.3 SB-TSI Bus Electrical Parameters

SB-TSI conforms to most of the I3C fast-mode electrical parameters. See the Electrical Data Sheet for the processor family for electrical parameters.

9.3.4 Pass-FET Option

The KBC may not have the capability to directly interface to SB-TSI. Pass FETs may be used to create two [SMBus](#) segments, see Figure 4.

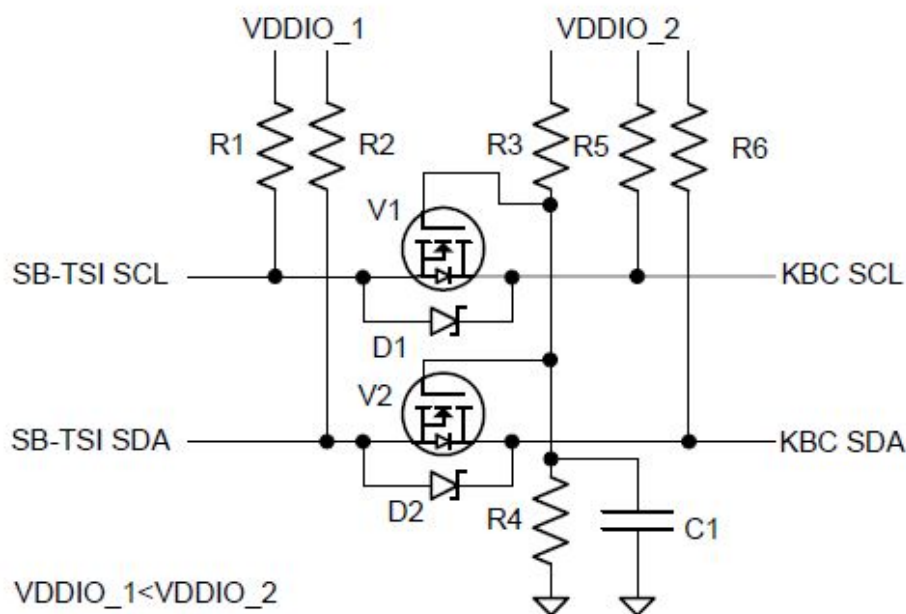


Figure 43: Pass FET Implementation

Notes:

- SCL and SDA pull-up resistors (R5 and R6, respectively) are the normal pull-up resistors for an SMBus segment

and are not part of the translation circuit. They are shown for completeness.

- The gates of the FETs are tied to a voltage approximately V_{gs} above the lower rail voltage. A resistive divider is shown, but a convenient power rail would do nicely.
- Care must be taken to install the FETs so that any body diode does not conduct.
- The key requirement is that the high side drive low enough to register as a low on the low side. (High side $V_{ol} < V_{il}$ on low side)

9.4 SB-TSI Registers

Reads to unimplemented registers return 00h. Writes to unimplemented registers are discarded.

SBTSIx01 [CPU Integer Temperature] (SBTSI::CpuTempInt)	
Read-only.	
The CPU temperature is calculated by adding the CPU temperature offset (SBTSI::CpuTempOffInt, SBTSI::CpuTempOffDec) to the processor control temperature (Tctl). SBTSI::CpuTempInt and SBTSI::CpuTempDec combine to return the CPU temperature. For the temperature encoding, see 9.2.5 [SB-TSI Temperature and Threshold Encodings]	
Bits	Description
7:0	CpuTempInt: integer CPU temperature value. Read-only. Reset: Cold,XXh. This field returns the integer portion of the CPU temperature.

SBTSIx02 [SB-TSI Status] (SBTSI::Status)	
Read-only, Volatile.	
the alert is cleared when the temperature does not meet the threshold conditions for temperature and number of samples. See 9.2.3 [Alert Behavior].	
Bits	Description
7:5	Reserved.
4	TempHighAlert: temperature high alert. Read-only, Volatile. Reset: Cold,X. 1=Indicates that the CPU temperature is greater than or equal to the high temperature threshold (SBTSI::HiTempInt, SBTSI::HiTempDec) for SBTSI::AlertThreshold[AlertThr] consecutive samples. 0=Indicates that the CPU temperature is less than the high temperature threshold (SBTSI::HiTempInt, SBTSI::HiTempDec) for SBTSI::AlertThreshold[AlertThr] samples.
3	TempLowAlert: temperature low alert. Read-only, Volatile. Reset: Cold,X. 1=Indicates that the CPU temperature is less than or equal to the low temperature threshold (SBTSI::LoTempInt, SBTSI::LoTempDec) for SBTSI::AlertThreshold[AlertThr] consecutive samples. 0=Indicates the CPU temperature is greater than the low temperature threshold (SBTSI::LoTempInt, SBTSI::LoTempDec) for SBTSI::AlertThreshold[AlertThr] samples.
2:0	Reserved.

SBTSIx03 [SB-TSI Configuration] (SBTSI::Config)

Reset: Cold,00h.

The bits in this register are Read-only and can be written by Writing to the corresponding bits in SBTSI::ConfigWr. See 9.2.3 [Alert Behavior] and 9.2.4 [Atomic Read Mechanism].

Bits	Description
7	AlertMask: alert mask. Read-only, Volatile. Reset: Cold,0. 0=ALERT_L pin enabled. 1=ALERT_L pin disabled and does not assert. Read-only; set-by-hardware. Either SBTSI::Status[TempHighAlert] == 1 or SBTSI::Status[TempLowAlert] == 1.
6	RunStop: run stop. Read-only. Reset: Cold,0. 0=Updates to SBTSI::CpuTempInt and SBTSI::CpuTempDec and the alert comparisons are enabled; alert history counters (specified by SBTSI::AlertThreshold[AlertThr]) and the corresponding timer (specified by SBTSI::UpdateRate[UpRate]) continue to update. 1=Updates to SBTSI::CpuTempInt and SBTSI::CpuTempDec and the alert comparisons are disabled; alert history counters (specified by SBTSI::AlertThreshold[AlertThr]) and the corresponding timer (specified by SBTSI::UpdateRate[UpRate]) are stopped. See 9.2.4 [Atomic Read Mechanism] for further details.
5	ReadOrder: atomic read order. Read-only. Reset: Cold,0. 0=Reading SBTSI::CpuTempInt causes the state of SBTSI::CpuTempDec to be latched. 1=Reading SBTSI::CpuTempDec causes the state of SBTSI::CpuTempInt to be latched. See 9.2.4 [Atomic Read Mechanism] for further details.
4	Reserved.
3	SBRMIAddrMode: Addressing mode for SBRMI register space. Read-only. Reset: Cold,0. 0=1 byte required for SBRMI register address.
2:1	Reserved.
0	RMISoftReset: trigger RMI soft reset flow. Read-only. Reset: Cold,0. 0=soft reset of SB-RMI I3C hardware completed and BMC may proceed communication with the SB-RMI peripheral. 1=SBI controller triggers MP1 firmware to soft reset the SB-RMI peripheral. See 8.4.3.2.1 [SB-RMI Soft Reset Recovery] for further details.

SBTSIx04 [Update Rate] (SBTSI::UpdateRate)

Read-write. Reset: Cold,08h.

Bits	Description																										
7:0	UpRate: update rate. Read-write. Reset: Cold,08h. This field specifies the rate at which CPU temperature is compared against the temperature thresholds to determine if an alert event has occurred. Write access causes a reset of the alert history counters (specified by SBTSI::AlertThreshold[AlertThr]) and the corresponding timer (specified by SBTSI::UpdateRate[UpRate]). Valid Values: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>0.0625 Hz</td></tr> <tr> <td>01h</td><td>0.125 Hz</td></tr> <tr> <td>02h</td><td>0.25 Hz</td></tr> <tr> <td>03h</td><td>0.5 Hz</td></tr> <tr> <td>04h</td><td>1 Hz</td></tr> <tr> <td>05h</td><td>2 Hz</td></tr> <tr> <td>06h</td><td>4 Hz</td></tr> <tr> <td>07h</td><td>8 Hz</td></tr> <tr> <td>08h</td><td>16 Hz</td></tr> <tr> <td>09h</td><td>32 Hz</td></tr> <tr> <td>0Ah</td><td>64 Hz</td></tr> <tr> <td>FFh-0Bh</td><td>Reserved.</td></tr> </table>	Value	Description	00h	0.0625 Hz	01h	0.125 Hz	02h	0.25 Hz	03h	0.5 Hz	04h	1 Hz	05h	2 Hz	06h	4 Hz	07h	8 Hz	08h	16 Hz	09h	32 Hz	0Ah	64 Hz	FFh-0Bh	Reserved.
Value	Description																										
00h	0.0625 Hz																										
01h	0.125 Hz																										
02h	0.25 Hz																										
03h	0.5 Hz																										
04h	1 Hz																										
05h	2 Hz																										
06h	4 Hz																										
07h	8 Hz																										
08h	16 Hz																										
09h	32 Hz																										
0Ah	64 Hz																										
FFh-0Bh	Reserved.																										

SBTSIx07 [High Temperature Integer Threshold] (SBTSI::HiTempInt)

Read-write. Reset: Cold,46h.

The high temperature threshold specifies the CPU temperature that causes ALERT_L to assert if the CPU temperature is greater than or equal to the threshold. SBTSI::HiTempInt and SBTSI::HiTempDec combine to specify the high temperature threshold. See 9.2.5 [SB-TSI Temperature and Threshold Encodings]. Reset value equals 70 °C. Write access causes a reset of the alert history counters (specified by SBTSI::AlertThreshold[AlertThr]) and the corresponding timer (specified by SBTSI::UpdateRate[UpRate]). See 9.2.3 [Alert Behavior].

Bits	Description
7:0	HiTempInt: high temperature integer threshold. Read-write. Reset: Cold,46h. This field specifies the integer portion of the high temperature threshold.

SBTSIx08 [Low Temperature Integer Threshold] (SBTSI::LoTempInt)

Read-write. Reset: Cold,00h.

The low temperature threshold specifies the CPU temperature that causes ALERT_L to assert if the CPU temperature is less than or equal to the threshold. SBTSI::LoTempInt and SBTSI::LoTempDec combine to specify the low temperature threshold. See 9.2.5 [SB-TSI Temperature and Threshold Encodings]. Write access causes a reset of the alert history counters (specified by SBTSI::AlertThreshold[AlertThr]) and the corresponding timer (specified by SBTSI::UpdateRate[UpRate]). See 9.2.3 [Alert Behavior].

Bits	Description
7:0	LoTempInt: low temperature integer threshold. Read-write. Reset: Cold,00h. This field specifies the integer portion of the low temperature threshold.

SBTSIx09 [SB-TSI Configuration Write] (SBTSI::ConfigWr)

Read-write. Reset: Cold,00h.

This register provides write access to SBTSI::Config.

Bits	Description
7	AlertMask: alert mask. Read-write. Reset: Cold,0. See SBTSI::Config[AlertMask].
6	RunStop: run stop. Read-write. Reset: Cold,0. See SBTSI::Config[RunStop].
5	ReadOrder: atomic read order. Read-write. Reset: Cold,0. See SBTSI::Config[ReadOrder].
4:1	Reserved.
0	RMISoftReset: trigger RMI soft reset flow. Read-write. Reset: Cold,0. See SBTSI::Config[RMISoftReset].

SBTSIx10 [CPU Decimal Temperature] (SBTSI::CpuTempDec)

Read-only.

See SBTSI::CpuTempInt.

Bits	Description
7:5	CpuTempDec: decimal CPU temperature value. Read-only. Reset: Cold,XXXb. Read-only. This field returns the decimal portion of the CPU temperature.
4:0	Reserved.

SBTSIx11 [CPU Temperature Offset High Byte] (SBTSI::CpuTempOffInt)

Read-write. Reset: Cold,00h.

SBTSI::CpuTempOffInt and SBTSI::CpuTempOffDec combine to specify the CPU temperature offset. See 9.2.6 [SB-TSI Temperature Offset Encoding] for encoding details.

Bits	Description
7:0	CpuTempOffInt: CPU temperature integer offset. Read-write. Reset: Cold,00h. This field specifies the integer portion of the CPU temperature offset added to Tctl to calculate the CPU temperature. Write access causes a reset of the alert history counters (specified by SBTSI::AlertThreshold[AlertThr]) and the corresponding timer (specified by SBTSI::UpdateRate[UpRate]).

SBTSIx12 [CPU Temperature Decimal Offset] (SBTSI::CpuTempOffDec)

Read-write. Reset: Cold,00h.

See SBTSI::CpuTempOffInt.

Bits	Description
7:5	CpuTempOffDec: CPU temperature decimal offset. Read-write. Reset: Cold,0h. This field specifies the decimal/fractional portion of the CPU temperature offset added to Tctl to calculate the CPU temperature. Write access causes a reset of the alert history counters (specified by SBTSI::AlertThreshold[AlertThr]) and the corresponding timer (specified by SBTSI::UpdateRate[UpRate]).
4:0	Reserved.

SBTSIx13 [High Temperature Decimal Threshold] (SBTSI::HiTempDec)

Read-write. Reset: Cold,00h.

See SBTSI::HiTempInt.

Bits	Description
7:5	HiTempDec: high temperature decimal threshold. Read-write. Reset: Cold,0h. This field specifies the decimal portion of the high temperature threshold.
4:0	Reserved.

SBTSIx14 [Low Temperature Decimal Threshold] (SBTSI::LoTempDec)

Read-write. Reset: Cold,00h.

See SBTSI::LoTempInt.

Bits	Description
7:5	LoTempDec: low temperature decimal threshold. Read-write. Reset: Cold,0h. This field specifies the decimal portion of the low temperature threshold.
4:0	Reserved.

SBTSIx22 [Reserved Register] (SBTSI::Reserved)

Read-write. Reset: 00h.

Bits	Description
7:1	Reserved.
0	Reserve: Reserved register. Read-write. Reset: 0. This is reserved register bit.

SBTSIx32 [Alert Threshold Register] (SBTSI::AlertThreshold)

Read-write. Reset: Cold,00h.

See 9.2.3 [Alert Behavior].

Bits	Description
7:3	Reserved.
2:0	AlertThr: alert threshold. Read-write. Reset: Cold,0h. Specifies the number of consecutive CPU temperature samples for which a temperature alert condition needs to remain valid before the corresponding alert bit is set. it specifies the number of consecutive CPU temperature samples for which a temperature alert condition need to remain not valid before the corresponding alert bit gets cleared. Write access resets the alert history counters (specified by SBTSI::AlertThreshold[AlertThr]) and the corresponding timer (specified by SBTSI::UpdateRate[UpRate]). Details in SBTSI::Status.
ValidValues:	
Value	Description
0h	1 Sample
6h-1h	<Value+1> Samples
7h	8 Samples

SBTSIxFE [Manufacture ID] (SBTSI::ManId)

Read-only. Reset: Cold,00h.

Bits	Description
7:1	Reserved.
0	ManId: Manufacture ID. Read-only. Reset: Cold,0. Returns the AMD manufacture ID.

SBTSIxFF [Revision] (SBTSI::Revision)

Read-only. Reset: Cold,04h.

Bits	Description
7:0	Revision: SB-TSI revision. Read-only. Reset: Cold,04h. Specifies the SBI temperature sensor interface revision.

10 Host System Management Port (HSMP)

Table 180: Definitions

Term	Description
HSMP	Host System Management Port
SMN	System Management Network

10.1 Overview

The Host System Management Port (HSMP) is an interface to provide OS-level software with access to system management functions via a set of mailbox registers.

10.2 SMN Mailbox Registers

The set of HSMP mailbox registers can be accessed in SMN space via a pair of SMN_INDEX and SMN_DATA registers:

- SMN_INDEX = Table 191 [BXXD00F0x0C4 (IOHC::NB_SMN_INDEX_3)].
- SMN_DATA = Table 192 [BXXD00F0x0C8 (IOHC::NB_SMN_DATA_3)].

To access a specific HSMP mailbox register, software writes the SMN Address of the HSMP mailbox register into the SMN_INDEX register, and then reads from or writes to the SMN_DATA register.

Table 181: SMN Address for HSMP Mailbox Registers

HSMP Mailbox Register	SMN Address (SMN_INDEX)
Message ID	3B10534h
Message Response	3B10980h
Message Argument_0	3B109E0h
Message Argument_1	3B109E4h
Message Argument_2	3B109E8h
Message Argument_3	3B109ECh
Message Argument_4	3B109F0h
Message Argument_5	3B109F4h
Message Argument_6	3B109F8h
Message Argument_7	3B109FCh

10.2.1 Message ID

A 32-bit read-write register to specify the value for the requested system management function. Writes to this register initiate the command message sequence.

10.2.2 Message Arguments

A set of eight 32-bit read-write registers to specify the input data and to capture the output data for each command message. Software writes input values to these register prior to writing to the Message ID register.

10.2.3 Message Response

An 8-bit read-write register that provides the response when the command message has completed. Software writes zero to this register to clear the Response value prior to writing to the Message ID register.

10.3 Mailbox Protocol

To request a given HSMP function, software executes the following sequence:

- Write zero (0) to the Message Response register.
- Write function arguments into the Message Argument registers.
- Write the function ID into the Message ID register.
- Wait (poll) until the Message Response register returns a non-zero value.

10.3.1 Response Codes

Upon completion of the command sequence, if the Message Response contains a RESULT_OK (01h) value, the command completed successfully and results may be read from the Message Argument registers. Otherwise an error occurred and results from the Message Argument registers are invalid.

Table 182: HSMP Response Codes

Response Value	Description
01h	RESULT_OK. Command completed successfully.
FFh	Command failed – Invalid input Arguments.
FEh	Command failed – Invalid or unsupported message ID.
FDh	Command rejected – Prerequisite to execute the command not satisfied
FCh	Command rejected – MP1 is busy
02h..FBh	Command failed – Other reasons.

10.4 HSMP Functions

Table 183 [HSMP Functions] below describes the supported HSMP functions. Unless otherwise noted, Input and Output Arguments are specified via the Message Argument_0 (Arg0) register. When needed, other message registers such as Argument_1 (Arg1) etc. may be used.

Table 183: HSMP Functions

Function ID	Function Name	Description	Input Arguments	Output Arguments
01h	TestMessage	Increments the input argument value by 1. This is used to check if the HSMP interface is functioning correctly.	[31:0]=Any value	[Arg0] + 1
02h	GetSmuVersion or GetMP1FwVersion	Provides the MP1 firmware version.	None	MP1 firmware version
03h	GetInterfaceVersion	Provides the HSMP interface version. Each version supports varying Function IDs as specified in the Table 184 [HSMP Supported Functions Per Interface Version].	None	Interface Version
04h	ReadSocketPower	Provides the average package	None	Socket power

		power consumption. This data may be useful to assess relative variations in power under different operating conditions and workloads. This data should not be used for system power budgeting.		(mWatts)
05h	WriteSocketPowerLimit	Sets the socket power limit. The value written is clipped to the maximum cTDP range for the processor. Note: there is a limit on the minimum power that the processor can operate at; no further socket power reduction occurs if the socket power limit is set below that minimum. Note: There are independent Power Limit registers through HSMP and APML ; whichever is the most constraining between the HSMP and APML limit at any given time is enforced.	[31:0]=Socket Power Limit (mWatts)	None
06h	ReadSocketPowerLimit	Provides the current socket power limit.	None	Socket power limit (mWatts)
07h	ReadMaxSocketPowerLimit	Provides the maximum socket power limit. This specifies the clip value for the WriteSocketPowerLimit function.	None	Max Socket power limit (mWatts)
08h	WriteBoostLimit	Sets a Maximum Frequency Limit on a CPU core defined by the specified ApicId. 1. For processors with SMT enabled, writes to different ApicIds that map to the same physical core overwrite the previous write to that core. 2. Values written are constrained to the supported frequency range of the processor. 3. Writes that contain an invalid ApicId result in a Failed response. Note: There are independent Boost Limit registers through HSMP and APML; whichever is the most constraining between the HSMP and APML limit at any given time is enforced. See 10.4.1 [HSMP Boost Limit].	[31:16]=ApicId. [15:0]=Max Frequency (MHz).	None

09h	WriteBoostLimitAllCores	Similar to WriteBoostLimit, however the Frequency Limit applies to all cores in the socket. Note: There are independent Boost Limit registers through HSMP and APML; whichever is the most constraining between the HSMP and APML limit at any given time is enforced. See 10.4.1 [HSMP Boost Limit].	[15:0]=Max Frequency (MHz).	None
0Ah	ReadBoostLimit	Provides the frequency limit currently enforced through WriteBoostLimit and WriteBoostLimitAllCores. If no boost limits have been specified, Fmax is returned. Writes that contain an invalid ApicId result in a Failed response. Note: There are independent Boost Limit registers through HSMP and APML; this message provides only the boost limit associated with HSMP.	[15:0]=ApicId.	[15:0]=Frequency (MHz).
0Bh	ReadProchotStatus	Provides the current PROCHOT status: 0=PROCHOT not asserted. 1=PROCHOT asserted.	None	PROCHOT Status
0Ch	SetXgmiLinkWidthRange	Sets Max and Min width of xGMI Link as follows: 0=x4 1=x8 2=x16	[31:16]=Reserved. [15:8]=MinLinkWidth. [7:0]=MaxLinkWidth. Note1: Max value must be >= Min value. Invalid configurations result in Failed response. Note2: If this function is called from both Primary and Secondary sockets, the largest Width values from either call are used.	None.
0Dh	APBDisable	Messages APBEnable and APBDisable specify DF (Infinity Fabric™) P-State behavior. DF P-States specify the frequency of clock domains from the CPU	[31:8]=Reserved [7:0]=DF P-state (0 to 2).	None.

		core boundary through to and including system memory, where DF P0 has highest performance and higher numbered P-states have lower performance. By default, an algorithm adjusts DF P-States automatically in order to optimize performance. However, this default may be changed to a fixed DF P-State through a CBS option at boot-time. APBDisable may also be used to disable this algorithm and force a fixed DF P-State. Note: While the socket is in PC6 or if PROCHOT_L is asserted, the lowest DF P-State (highest numerically) is enforced regardless of the APBEnable/APBDisable state.		
0Eh	APBEnable	This function enables the DF P-State Performance Boost algorithm. See the APBDisable function for more information.	None	None
0Fh	ReadCurrentFclkMemclk	Provides Infinity Fabric Clock (FCLK) and DRAM Memory Clock (MEMCLK) for the current socket DF P-state.	None	Arg0=FCLK (MHz), Arg1=MEMCLK (MHz)
10h	ReadCclkFrequencyLimit	Provides the CPU core clock (CCLK) frequency limit for the socket, from the most restrictive infrastructure limit at the time of the message.	None	Frequency (MHz)
11h	ReadSocketC0Residency	Provides the average C0 residency across all cores in the socket. 100% specifies that all enabled cores in the socket are running in C0.	None	Socket C0 Residency (%)
12h	SetLclkDpmLevelRange	Sets the Max & Min LCLK DPM Level on a given NBIO per socket. The DPM Level is an encoding to represent the PCIe® Link Frequency (LCLK) under a root complex (NBIO), as shown in Table 185 [HSMP LCLK Frequency Per DPM Level]. Note: These Levels can also be set from APML; the last value received from either HSMP or APML is enforced.	[23:16]=NBIO ID (0 to 3). [15:8]=Max DPM Level (0 to 3). [7:0]=Min DPM Level (0 to 3). Note: Max value must be >= Min value.	None

13h	GetLclkDpmLevelRange	Gets the Max & Min LCLK DPM Level on a given NBIO per socket. The DPM Level is an encoding to represent the PCIe Link Frequency (LCLK) under a root complex (NBIO), as shown in Table 185 [HSMP LCLK Frequency Per DPM Level]. Note: These Levels can also be set from APML; this message returns the current Levels which may have been set from either HSMP or APML.	[23:16]=NBIO ID (0 to 3).	[15:8]=Max DPM Level (0 to 3). [7:0]=Min DPM Level (0 to 3).
14h	GetMaxDDRBandwidthAndUtilization	Provides per socket: 1. Theoretical maximum DDR Bandwidth in GB/s. 2. Current utilized DDR Bandwidth (Read+Write) in GB/s. 3. Current utilized DDR Bandwidth as a percentage of theoretical maximum.	None	[31:20]=Max bandwidth in Gbps. [19:8]=Utilized (R+W) bandwidth in Gbps. [7:0]=Utilized bandwidth in %.
15h	Reserved	N/A	N/A	N/A
16h	GetDIMMTempRangeAndRefreshRate	Get Per-DIMM Temperature Range and Refresh Rate for a given channel based on inband MR4 polling. The refresh rate is determined by the warmest DRAM device across all DIMMs in the channel (highest MR4.OP[2:0] value) in accordance to the threshold specified in JEDEC DDR5 SDRAM specification (JESD79). Due to hardware limitations, MP1 has visibility to only the refresh rate (1x or 2x) but not the actual MR4.OP[2:0] value. Hence the refresh rate (DataOut[3]) is accurate but the temperature range (DataOut[2:0]) is an approximation.	[7:0]=DIMM_ADDRESS. See notes under Table 190 [HSMP DIMM ADDRESS Encodings] for DataIn and DataOut field details.	[3]=Refresh rate: 0=Refresh rate is 1x. 1=Refresh rate is 2x. [2:0]=Temperature Range=001b, if refresh rate is 1x; 101b if refresh rate is 2x.
17h	GetDIMMPowerConsumption	Get Per-DIMM Power Consumption.	[7:0]=DIMM_ADDRESS. See notes under Table 190 [HSMP DIMM ADDRESS Encodings] for DataIn and DataOut field details.	[31:17]=DIMM Power ¹ (mWatt). [16:8]=Update Rate (ms). [7:0]=DIMM_ADDRESS.

18h	GetDIMMThermalSensor	Get Per-DIMM Thermal Sensor (2 Sensors per DIMM).	[7:0]=DIMM_ADDRESS. See notes under Table 190 [HSMP DIMM ADDRESS Encodings] for DataIn and DataOut field details.	[31:21]=Temperature (Degrees C). [16:8]=Update Rate (ms). [7:0]=DIMM_ADDRESS.
19h	PwrCurrentActiveFreqLimitSocket	Get Current Active Frequency Limit per Socket.	None	[31:16]=Frequency (MHz). [15:0]=Source of Limit: As shown in Table 186 [HSMP Frequency Limit Source].
1Ah	PwrCurrentActiveFreqLimitCore	Get Current CCLK limit set per Core.	[15:0]=ApicId	[31:0]=Frequency (MHz).
1Bh	PwrSviTelemetryAllRails	Get SVI-based Telemetry for all rails.	None	[31:0]=Power (mW).
1Ch	GetSocketFreqRange	Get Fmax and Fmin per Socket.	None	[31:16]=Fmax (MHz). [15:0]=Fmin (MHz).
1Dh	GetCurrentIoBandwidth	Get Current Bandwidth on IO Link.	[17:8]=Link ID: As shown in Table 189 [HSMP Link ID Encoding]. [7:3]=Reserved. [2:0]=Bandwidth Type. As shown in Table 187 [HSMP IO Bandwidth Encoding].	[31:0]=IO BandWidth (Mbps).
1Eh	GetCurrentXgmiBandwidth	Get Current Bandwidth on xGMI Link.	[15:8]=Link ID. As shown in Table 189 [HSMP Link ID Encoding]. [7:3]=Reserved [2:0]=Bandwidth Type. As shown in Table 188 [HSMP XGMI Bandwidth Encoding].	[31:0]=xGMI BandWidth (Mbps).
1Fh	GMI3LinkWidthRange	Set or Get Max and Min GMI3 Link width as follows: 0=Quarter Width 1=Half Width	[31:16]=Reserved. [15:8]=MinLinkWidth [7:0]=MaxLinkWidth	None.

		2=Full Width	h Note: Max value must be $\geq$ Min value. Invalid configurations result in a Failed response.	
20h	ControlPcieLinkRate	Set the Link Rate on PCIe devices. Note that if CXL™ device operating at Gen5 rate is present on any link then PCIe rate is locked at Gen5 rate and request to change to Gen4 rate will be ignored.	[31:8]=Reserved. [7:0]=Link Rate encoding : 0=Auto-Detect bandwidth utilization, and set Link Rate accordingly. 1=Limit at Gen4 Rate. 2=Limit at Gen5 Rate	Previous Mode: 0=Auto-Detect. 1=Limit at Gen4. 2=Limit at Gen5.
21h	PwrEfficiencyModeSelection	Select power efficiency profile policy.	[31:3]=Reserved. [2:0]=Mode Selection: 000b=High Performance Mode. 001b=Power Efficiency Mode. 010b=IO Performance Mode. 011b=Balanced Memory Performance Mode. Other=Reserved. See 10.4.3 [HSMP Performance Modes] .	None
22h	DfPstateRange	Set the Max and Min DF P-State. P_State values are: 0 – DFP0 (high performance) 1 – DFP1 2 – DFP2 (low performance) Max value must be $\leq$ Min value. DF P-States can be set through APML or HSMP; the value that was last set is enforced.	[31:16]=Reserved. [15:8]=Min DF P-State. [7:0]=Max DF P-State.	None.

Table 184: HSMP Supported Functions Per Interface Version

Interface Version	Supported Function IDs
0001h	01h through 11h

0002h	01h through 12h
0003h	01h through 14h
0004h	01h through 15h
0005h	01h through 2Fh

Table 185: HSMP LCLK Frequency Per DPM Level

DPM Level	LCLK Frequency
0	Lowest Freq.
1..3	Highest Freq.

Table 186: HSMP Frequency Limit Source

Source Type Encoding (Arg0[15:0])	Description
Arg0[0]	cHTC-Active
Arg0[1]	PROCHOT
Arg0[2]	Thermal Designed Current (TDC) Limit
Arg0[3]	Package Power Tracking (PPT) Limit
Arg0[4]	OPN Max
Arg0[5]	Reliability Limit (Fused Max or Reliability Monitor Fmax@Vmax)
Arg0[6]	APML Agent
Arg0[7]	HSMP Agent
Arg0[15..8]	Reserved

Note: There may be more than one active limiter.

Table 187: HSMP IO Bandwidth Encoding

Encoding	Bandwidth Type
001b	Aggregate bandwidth
Other	Reserved

Table 188: HSMP XGMI Bandwidth Encoding

Encoding	Bandwidth Type
001b	Aggregate bandwidth
010b	Read bandwidth
100b	Write bandwidth
Other	Reserved

Table 189: HSMP Link ID Encoding

Encoding	Link Type
00000001b	P0
00000010b	P1
00000100b	P2
00001000b	P3
00010000b	G0
00100000b	G1
01000000b	G2
10000000b	G3

Table 190: HSMP DIMM ADDRESS Encodings

DIMM_ADDRESS[7:0] (Mode = 0)	DIMM_ADDRESS[7:0] (Mode = 1)
Bit[7]=0 (for Mode 0) Bit[6]=Temperature Sensor#: TS0/1, else Reserved. Bit[5:4]=I2/I3C Port# Bit[3]=Reserved. Bit[2:0]=DIMM SPD Host ID. Refer to JEDEC SPD5118, SPD5108 Hub and Serial Presence Detect Device Standard (JESD300-5).	Bit[7]=1 (for Mode 1) Bit[6]=Temperature Sensor#: TS0/1, else Reserved. Bit[5]=Reserved. Bit[4]=DIMM#: DIMM0/1. Bit[3:0]=UMC/DDRPHY Instance ID Refer to 12.1.2.1 [UMC and DDR Phy Logical Mapping].

Notes:

- 1: DIMM Power is a 15-bit unsigned value representing power consumed in mW (0-32767)
- 2: Refresh rate is time since last update (0-511ms). 0 means last update was < 1ms, and 511 means update was >= 511ms.
- 3: DIMM_ADDRESS is encoded per above Table 190 [HSMP DIMM ADDRESS Encodings].
- 4: Temperature (Deg C) is an 11-bit signed value, with a scaling factor of 0.25.
Thus 3FFh=255.75 (1023*0.25), 400h= -256 (-1024*0.25), 1h=0.25 and 7FFh= -0.25 (-1*0.25).

10.4.1 HSMP Boost Limit

The WriteBoostLimit function (ID 08h), provides the ability to limit frequency on a per core basis. However, processors are constrained to frequency resolution per the following requirements:

- Cores are grouped into Core Complexes (CCXes), each of which contains 1 or more cores, depending on the OPN.
- Each [CCX](#) includes a clock generator that supports a resolution of 25 MHz.
- If all cores of a CCX are not programmed for the same boost limit frequency, then the core with the lowest requested frequency will get assigned a frequency that is up to 15-20% lower than the cores with the highest requested frequency.
- If the specified boost limit frequency of a core is not supported, then the processor selects the next lower supported frequency.

Example 1: If all cores in a CCX are programmed for a boost limit of 3.024 GHz, then all cores of the CCX are limited to 3.000 GHz (next lower supported frequency, with a 25MHz resolution).

Example 2: If three out of four cores of a CCX are programmed for a boost limit of 3.000 GHz, and the last core for 2.700 GHz, then the three high-frequency cores remain set to 3.000 GHz, while the low-frequency core is limited to 2.550 GHz! (~15% lower than highest requested frequency).

10.4.2 ApicId Mapping

The WriteBoostLimit (ID 08h) and ReadBoostLimit (ID 0Ah) functions are directed at logical cores, as specified by their ApicId. Hence, this section describes how to determine which cores are grouped into the same Core Complex ([CCX](#)), and therefore abide by the boost limit frequency resolution rules in section 10.4.1 [HSMP Boost Limit] above.

All cores grouped into a common CCX share a Last-Level (L3) cache, hence

Core::X86::CpuId::CachePropEax3[NumSharingCache] can be used to first determine all logical processors (threads) in the cores that share an L3 cache, as follows:

Calculate L3ShareId from the ApicId of each thread as follows:

$$L3ShareId = \text{ApicId} \gg \log_2 (\text{NumSharingCache} + 1)$$

If (NumSharingCache + 1) is not a power of two, round it up to the next power of two. Then, threads with the same L3ShareId share the L3 cache.

Once all threads grouped into a given CCX have been determined, a representative ApicId per core can be derived by filtering out redundant ApicIds when there are multiple threads enabled per core ([SMT](#)). The number of threads per core can be determined from Core::X86::Cpuid::[CoreId\[ThreadsPerCore\]](#), as follows:

Calculate CoreSharedId from the ApicId of each thread as follows:

$$\text{CoreSharedId} = \text{ApicId} \gg \log_2 (\text{ThreadsPerCore} + 1)$$

If (ThreadsPerCore + 1) is not a power of two, round it up to the next power of two. Then, threads with the same CoreSharedId belong to the same core.

10.4.3 HSMP Performance Modes

Performance modes can be set through HSMP or [APML](#); the value that was last set is enforced.

- High Performance Mode: this mode will choose the default [DF](#) P-State algorithm and will try to maximize application performance using the application characteristics. This mode favors core performance and the system boots into this mode. In this mode all DF P-States are available and the default DF P-State and DLWM algorithms are active.
- Power Efficiency Mode: this mode configures the system for power efficiency at the expense of some performance. This mode limits the boost frequency available to the cores and restricts the DF P-States available in the system. This mode also monitors the system load to dynamically adjust performance for maximum power efficiency.
- IO Performance Mode: this mode sets up the Infinity Fabric to maximize the IO sub-system performance. This mode can result in lower core performance in some cases, to maximize IO throughput.
- Balanced Memory Performance Mode: this mode biases the memory subsystem and Infinity Fabric performance towards efficiency, by lowering the frequency of the fabric and the width of the [xGMI](#) links under light traffic conditions. Core behavior is unaffected. There may be a performance impact under lightly loaded conditions for memory-bound applications compared to the default high performance mode. With higher memory and fabric load, the system becomes similar in performance to the default high performance mode.

10.5 Registers

10.5.1 IOHC Registers

Table 191: BXXD00F0x0C4 (IOHC::NB_SMN_INDEX_3)

Read-write. Reset: 0000_0000h.	
_iohub0_nbio0_aliasHOST; BXXD00F0x0C4; BXX=_iohub0_nbio0 instance of NB_BUS_NUM from Table 193 [IOHCMISC[0...3]x00000044 (IOHC::NB_BUS_NUM_CNTL)]	
_iohub0_nbio1_aliasHOST; BXXD00F0x0C4; BXX=_iohub0_nbio1 instance of NB_BUS_NUM from Table 193 [IOHCMISC[0...3]x00000044 (IOHC::NB_BUS_NUM_CNTL)]	
_iohub1_nbio0_aliasHOST; BXXD00F0x0C4; BXX=_iohub1_nbio0 instance of NB_BUS_NUM from Table 193 [IOHCMISC[0...3]x00000044 (IOHC::NB_BUS_NUM_CNTL)]	
_iohub1_nbio1_aliasHOST; BXXD00F0x0C4; BXX=_iohub1_nbio1 instance of NB_BUS_NUM from Table 193 [IOHCMISC[0...3]x00000044 (IOHC::NB_BUS_NUM_CNTL)]	
_iohub[1:0]_nbio0_aliasSMN; NBCFG[1:0]x000000C4; NBCFG[1:0]=13[C:B]0_0000h	
_iohub[1:0]_nbio1_aliasSMN; NBCFG[3:2]x000000C4; NBCFG[3:2]=13[E:D]0_0000h	
Bits	Description

31:0	NB_SMN_INDEX_3. Read-write. Reset: 0000_0000h. Index value SMN Index/Data pair access. Register access using these reg pairs will have security level 7.
------	--

Table 192: BXXD00F0x0C8 (IOHC::NB_SMN_DATA_3)

Read-write. Reset: 0000_0000h.	
_iohub0_nbio0_aliasHOST; BXXD00F0x0C8; BXX=_iohub0_nbio0 instance of NB_BUS_NUM from Table 193 [IOHCMISC[0...3]x00000044 (IOHC::NB_BUS_NUM_CNTL)]	
_iohub0_nbio1_aliasHOST; BXXD00F0x0C8; BXX=_iohub0_nbio1 instance of NB_BUS_NUM from Table 193 [IOHCMISC[0...3]x00000044 (IOHC::NB_BUS_NUM_CNTL)]	
_iohub1_nbio0_aliasHOST; BXXD00F0x0C8; BXX=_iohub1_nbio0 instance of NB_BUS_NUM from Table 193 [IOHCMISC[0...3]x00000044 (IOHC::NB_BUS_NUM_CNTL)]	
_iohub1_nbio1_aliasHOST; BXXD00F0x0C8; BXX=_iohub1_nbio1 instance of NB_BUS_NUM from Table 193 [IOHCMISC[0...3]x00000044 (IOHC::NB_BUS_NUM_CNTL)]	
_iohub[1:0]_nbio0_aliasSMN; NBCFG[1:0]x000000C8; NBCFG[1:0]=13[C:B]0_0000h	
_iohub[1:0]_nbio1_aliasSMN; NBCFG[3:2]x000000C8; NBCFG[3:2]=13[E:D]0_0000h	
Bits	Description
31:0	NB_SMN_DATA_3. Read-write. Reset: 0000_0000h. Index value SMN Index/Data pair access. Register access using these reg pairs will have security level 7.

Table 193: IOHCMISC[0...3]x00000044 (IOHC::NB_BUS_NUM_CNTL)

Read-write. Reset: 0000_0000h.	
GNB Bus Number Control.	
_iohub[1:0]_nbio0_aliasSMN; IOHCMISC[1:0]x00000044; IOHCMISC[1:0]=13[C:B]1_0000h	
_iohub[1:0]_nbio1_aliasSMN; IOHCMISC[3:2]x00000044; IOHCMISC[3:2]=13[E:D]1_0000h	
Bits	Description
31:24	Reserved
23:16	NB_SEGMENT. Read-write. Reset: 00h. Specifies the number of the NBIO segment in a multi-segmented system.
15:9	Reserved
8	NB_BUS_LAT_Mode. Read-write. Reset: 0. Description: NBIO bus number is specified by NB_BUS_NUM. 0 = Local bus number of NBIO is capture from any type 0 configuration request. 1= Use the NB_BUS_NUM to decode for configuration cycles targeting the NBIO bus.
7:0	NB_BUS_NUM. Read-write. Reset: 00h. Specifies the number of the NBIO local bus when NB_BUS_LAT_mode is set.

11 Data Fabric (DF)

The Scalable Data Fabric (SDF) is a coherent fabric connecting CPU, I/O, and DRAM.

11.1 Definitions

Table 194: Definitions

Term	Description
DataPoisonEnabled	Data Poisoning is enabled.
DiePresentMask	Die Present Mask.
InitiatorTrust	Initiator Trust Level
Is1NodeSystem	1 node is connected in the system.
Is2NodeSystem	2 nodes are connected in the system.
Is2Node2LinkSystem	2 nodes are connected in the system with two links.
NumNodesInSystem	Number of nodes in the system.
IsCcm0Enabled	CCM0 is enabled.
IsCcm1Enabled	CCM1 is enabled.
IsCcm2Enabled	CCM2
IsCcm3Enabled	CCM3
IsCcm4Enabled	CCM4
IsCcm5Enabled	CCM5
IsCcm6Enabled	CCM6
IsCcm7Enabled	CCM7
IsEccEnabled	ECC is enabled.
IsMultiDie	Multi-die.
IsMultiSocket	Multi-socket.
IsMultiNode	Multi-node.
IsNode0	Node0 value
IsNode1	Node1 value
IsProbeFilterEnabled	Probe Filter is enabled.
IsSlitMode	SLT mode.
IsSmtEnabled	SMT is enabled.
MyDieId	Identifier of this die.
MySocketId	Identifier of this socket.
NumChannels	Number of DRAM channels.
NumDiesInSystem	Number of dies in the system.
NumEnabledThreadsInSystem	Number of enabled CPU threads in the system.
SecurePart	Fuse state that determines if security settings are applied. All production parts are "secure".
SmmBios	A trust level (and thus register access security) while the BIOS is running in System Management Mode (SMM).
WdtControl	Watchdog Timer Control.
XgmiDlwmEnabled	XGMI DLWM is enabled.

Table 195: List of Acronyms used in Data Fabric (DF)

Term	Definition
AALU	Atomic ALU
ACE	AXI Coherency Extensions.
APCB	AGESA™ Platform Control Block.
Cac	Dynamic switching Capacitance. Actively weighted amount of capacitance switching per cycle. Dynamic power is: $\text{dynPwr} = \text{Cac} * V^2 * f$.
CCM	CPU Core Coherent Module.
CDMA	Cacheable DMA.
CS	Coherent Slave.
CSD	Coherent Slave Data (CSD) buffer.
CSQ	Coherent Slave Queue.
DCE	Display Controller Engine.
EA	Effective Address, or Efficiency Arbiter.
FCAC	Fabric Configuration Access Control.
FCI	Fabric Configuration Interface.
FHI	Fabric coherent HyperTransport™ Interface.
FIT	Failures In Time. Defines a rate of errors or failures in one billion device hours.
FTI	Fabric Transport Interface.
HWA	Hardware Assert.
IOM	IO Master.
IOS	IO Slave.
IPI	InterProcessor Interrupt. Interrupt targeted to a specific Thread of a CPU or can be broadcast to all CPU Threads for synchronization.
LFSR	Linear Feedback Shift Register. A shift register whose input is a linear function of its previous state. The initial value is called the seed, and because the operation of the register is deterministic, the stream of values is completely determined by its current (or previous) state.
LINT	Local Interrupt.
MCA	Machine Check Architecture.
MCM	Multi-Chip Module.
MGCG	Medium Grain Clock Gating.
NCM or nCM	Non-Coherent Master.
NMI	Non-Maskable Interrupt.
PG	Power Gating.
PIE	Power management, Interrupts, Etc.
QoS	Quality of Service.
RAS	Reliability, Availability, and Serviceability.
RAW	Read-After-Write data hazard. Occurs when the pipeline creates the potential for an instruction to read an operand before a prior instruction write to it. It is a hazard to the intended Read-After-Write operand access.
RSPD	Response Data Buffer.
SCM	Single-Chip Module.
SDP	Scalable Data Port.
SPF	System Probe Filter.
TCDX	Transport Command and Data Crossbar switch. Handles the buffering and routing of command and data packets from the source to a destination port.
TMR	Trusted Memory Region.
TPR	Task Priority.

TSC	Time Stamp Counter
ULV	Ultra-Low Voltage.
UMC	Unified Memory Controller or DRAM Controller.
VC	Virtual Channel.
VCE	Video Compression Engine.
VDCI	Voltage Domain Crossing Interface.
VCO	Voltage Controlled Oscillator.
WAR	Write After-Read data hazard. Occurs when the pipeline creates the potential for an instruction to write to a register before a prior instruction reads it. It is a hazard to the intended Write-After-Read operand access.
WB	Write Back.
WAW	Write-After-Write data hazard. Occurs when the pipeline creates the potential for an instruction to write to a register before a prior instruction writes to it. It is a hazard to the intended Write-After-Write operand access.

11.2 Data Fabric Overview

The Data Fabric is assembled using a collection of modular building block components. Components consist of masters, slaves and transport as well as a management unit. Masters are provisioned where requests enter the Data Fabric from other IP blocks. Slaves are egress ports from the Data Fabric, where transactions are sent to other IP for service. The Management Unit, known as PIE, takes care of interrupts, fabric power management and other support functions. Transport components provide the connectivity among all of the other components.

There are two types of components, transport-layer and protocol-agents.

11.2.1 Data Fabric Transport Layer Components

Data Fabric Transport-layer components connect only to each other and Data Fabric protocol agents. Transport components provide the connectivity among all of the other components.

11.2.1.1 Transport Command and Data Crossbar (TCDX)

The data fabric transport routes command and data packets from a source to a destination. It is built out of NxN TCDX switches. An NxN switch allows many different topologies, such as a ring, mesh or a tree. The fabric transport is the central piece of the data fabric which provides the scalability and the ease of assembly of the chips. TCDX connects all [DF](#) components in the system to each other, including other TCDXs. It routes command and data packets from any source to any destination.

11.2.1.2 Coherent AMD socKet Extender (CAKE)

CAKE extends the [DF](#) transport layer off-chip by connecting to a CAKE in another die. Each CAKE provides an egress and ingress path. CAKE egress drives a [PCS](#) interface which drives a GMI (same socket) or [xGMI](#) (different socket) physical link off-chip. CAKE ingress receives from PCS and forwards to the transport-layer.

11.2.2 Data Fabric Protocol Agents

[DF](#) protocol agents connect external clients to the DF transport layer. Protocol agents package client data and forward the data to the client-designated destination. Each protocol agent is tailored for a specific type of client.

11.2.2.1 Core Coherent Master (CCM)

CPU Coherent Master (aka CCM) provides the interface for any client which caches data. It is the interface block which connects to the CPU complex (aka [CCX](#)). It receives requests from the CCX over SDP and forwards it to the transport over FTI. Probes received from CS are sent to CCX and probe responses received are returned back to CS.

11.2.2.2 Non-Coherent Master (NCM)

NCM interfaces to clients which do not cache data. NCM provides interfaces for DCE.

The NCM differs from the CCM or GCM by supporting masters which do not contain caches which need to be kept coherent with memory and with other caches.

Non-coherent Masters have two main queues: request and response. None of the clients which use this block require PCI type ordering, where requests have to not only be ordered globally but may have to also be visible in order. Hence there is no need to track the requests in the queue for its lifetime. It is primarily a transient queue where the depth has to only cover the roundtrip latency from SDP client to NCM. As soon the request is forwarded to the destination the request can be de-allocated. Similarly, there is no need to return a SrcDone back to the Coherent Slave. The response may simply be forwarded to the destination over SDP and the response does not even have to wait for an Ack from the client.

NCM has a probe interface on the FTI side to receive bus lock requests. It also has debug, register read/write, and other miscellaneous interfaces. Request queue supports multiple virtual channels and is able to pick requests in different virtual channels out of order. It supports full QoS with priority forwarding and priority based pickers.

11.2.2.3 IO Master and IO Slave

IO Master (IOM)

The IOM component has an SDP port and a FTI connection to the fabric transport layer, just like IOS, which interfaces to IOHUB.

- SDP port, Master: Accesses from I/O device into the system.

IO Slave (IOS)

The IOS component has an SDP port and a FTI connection to the fabric transport layer, just like IOM, which interfaces to IOHUB.

- SDP port, Slave: Accesses from system to I/O device.

The IOM is similar to the NCM, but it possesses the ability to enforce "[PCIe®](#)-style" ordering. The IOS is a NonCoherent Slave. For the IOS, refer to the section below where the NCS (NonCoherent Slave) is described.

11.2.2.4 Coherent Slave (CS)

The Coherent Slave (CS) is designed to receive requests from the clients through the transport layer switches and is responsible for maintaining ordering and coherency for those requests. The Coherent Slave design implements the cHT+ coherency protocol. It interfaces to UMC through an SDP port and uses Fabric Transport Interface (FTI) to communicate with the transport layer. The CS interfaces with the system probe filter (SPF) to limit or eliminate probes.

CS has several functions:

- CS acts as the global ordering point and works with CCM to ensure global visibility.
- CS issues probes for coherent transactions.
- The system probe filter, when present, attaches to a single CS.

The Coherent Slave is the Data Fabric's interface to the memory controller (UMC). System memory may be spread across

multiple coherent slaves on a single die or multiple dies. Each CS services transactions to the memory addresses located 'behind' that CS. Each CS operates independently from other CSs.

A Coherent Slave is the system's memory ordering point for all of the memory addresses behind that CS. The order in which requests reach the CS defines the order in which operations to a given address are serviced.

For a coherent transaction, the Coherent Slave makes the decision on which of the system caches to probe, based on a property of the transaction, SDP_ReqDomain.

Table 196: Domain and Probe Eligibility

NCMInGpuDomain	SDP_ReqDomain	CPU Cache Probe Eligible	GPU Cache Probe Eligible
0	0	Yes	No
0	1	Yes	Yes
1	0	No	Yes
1	1	Yes	Yes

11.2.2.4.1 Coherent Slave System Probe Filter

The Coherent Slave optionally includes a system probe filter (SPF). The probe filter helps limit the number of coherency probes that must be sent. In addition, the probe filter can limit the number of targets a coherency probe gets sent to, including the possibility that the multicast probe is converted to a directed probe to a single complex. The availability of the probe filter is product specific, and is reported in DF::DfCapability[Spf].

The probe filter is a directory for all cache lines; specifically, it has an entry for each and every cache line that may be cached in an L1, L2, or L3 cache and includes the expected state of the line. Each CS tracks only the cache lines for its associated memory. The directory has its own storage and does not affect available cache.

When the probe filter reaches capacity, the CS invalidates cache lines from CPU cores to make room for new allocations. The probe filter includes ECC on the indices. Correctable ECC errors can be corrected on the fly. Uncorrectable ECC errors in the directory are not fatal, but the directory entry is no longer reliable for tracking coherency. The CS "corrects" the ECC error by reverting to a multicast probe to all core complexes. Once an uncorrectable error has been detected in a probe filter index, all accesses that use this index cause a multicast probe until a system reset occurs. These errors are reported to the coherent slave's [MCA](#) bank as corrected errors since the system functionality is not affected.

The probe filter also detects, through coherency protocol checks, some failures in the coherency state transitions and logs them to the coherent slave's MCA bank as uncorrected errors.

11.2.2.5 Power Management, Interrupts, Etc. (PIE)

The two main functions of the PIE block are power management and interrupts. In addition this block supports other miscellaneous functions which falls under the "etcetera" category. Some of the functions hosted in this module include:

- Fabric power management
 - Controls power zones within [DF](#)
 - Coordinates clock gating
- Sequences DF clock frequency changes
- Coordinates bus lock
- Handles Interrupts: detects, distributes notifications, coordinates DF with other IP
- Register access and register security
- [RAS](#) functions
- Interrupt controller interface

- Interrupt controllers (APIC)
- Performance monitors
- [MCA](#)

11.3 Accessing Data Fabric Configuration Registers

Each node in the [DF](#) contains an 8KB range of register addresses (2048 32-bit registers) logically organized into groups of 256 registers referred to as "function spaces", and numbered "function0" through "functionN". Within a function space, registers are assigned at offsets from the base address. Not all offsets are used in every function space.

Registers can be accessed by CPUs through PCI space (see 11.3.1 [CPU Sourced DF Configuration Register Access]).

11.3.1 CPU Sourced DF Configuration Register Access

CPUs access [DF](#) configuration registers through PCI space. DF has registers in PCI Base Configuration Space and Extended Configuration Space, accessed through Bus 0, devices 24 to 31, where device 24 corresponds to DF on node 0 (sometimes called node 0), device 25 to node 1, and so on.

11.3.2 DF Configuration Register Broadcast and Instance Mode Access

When [DF](#) contains multiple copies of a register in the various components, they reside at the same function and offset in the DF address space. Although the value of this register is often the same for all instances of that component, in some cases the system configuration may require different values in different component instances. Therefore, DF provides a mechanism to access all instances (broadcast) or single instances (single-instance) of a register. The usage of which access method (broadcast versus single-instance) depends on the targeted register. Some registers must use single-instance because they have unique values. Other registers may use broadcast when software knows that all components are required to have a uniform value. Finally, some registers exist only in a single component and can use broadcast or single-instance since there is no difference between the two accesses for this type of register. This allows many registers (power management registers, indirect access registers, etc.) to be accessed in broadcast mode.

The following registers are known to require single-instance accesses. Broadcast accesses to these registers cause unpredictable and difficult to diagnose behavior.

- DRAM address mapping registers
- Privileged address registers
- TCDX Routing tables
- Credit registers (e.g. to optimize credit values for a downbinned topology)
- [MCA](#) registers
- DF Performance Monitors

All single-instance accesses must be indirect. That is, the software must select an indirect-access register pair and use the index register to select which instance of the register is being used.

Broadcast accesses may be direct or indirect.

See 11.3.3 [DF Configuration Register Direct and Indirect Access] for more information on direct and indirect access.

A broadcast mode read result is the OR of all the instances with matching function space/offset. Thus, a broadcast read provides unreliable results unless to a register that has only one instance or has the same value in all instances. This method is not supported for the MCA registers – a broadcast read of MCA registers does not return the OR reduction of all instances.

A broadcast on one node does not affect any other node.

11.3.3 DF Configuration Register Direct and Indirect Access

[DF](#) registers can be accessed directly or indirectly. Direct access uses the register's function and offset in the access. Indirect access uses an index/data pair register in the DF to access the desired register.

11.3.3.1 Indirect Access Registers

Indirect access to [DF](#) configuration registers requires setup of a specific registers:

DF::FabricIndirectConfigAccessAddress register (FICAA), and the use of associated

DF::FabricIndirectConfigAccessDataLo, DF::FabricIndirectConfigAccessDataHi registers (FICADL and FICADH; FICAD for both).

There are six sets of indirect access registers, with three registers in each set: one for address and two for data (to support 64-bit accesses). DF provides six sets of FICAA/FICAD (FICAA[5:0], and paired FICADL[5:0], FICADH[5:0]) and associates them by index. For example, a write through FICADL[2] is routed to the register specified in FICAA[2]. Using one FICAA/FICAD to access a different instance of FICAA/FICAD has undefined results.

The sets are assigned to specific sources:

set 0: [ASP](#)

set 1: [MP1](#)

set 2: Microcode

set 3: BIOS, operating-system, and drivers

set 4: LW-[SMI](#)

set 5: Reserved

Accesses using single-instance and 64-bit registers requires use of indirect access.

11.3.3.2 DF Indirect Access Procedure

Indirect access procedure is:

1. Select the appropriate FICAA/FICAD according to the source.
2. Write FICAA, specifying the following fields. Note that the fully concatenated value can be found within the register description of the targeted register.
 1. CfgRegInstID: The instance ID..
 2. IndCfgAccFuncNum: [DF](#) Function number.
 3. IndCfgAccRegNum : DF Offset number.
 4. CfgRegInstAccEn:
 1. Set to 1: Single-instance access.
 2. Set to 0: broadcast.
 5. SixtyFourBitRegEn: set to 1 when accessing a 64-bit register.
3. For reads:
 1. Read FICADL. If reading 64-bit registers, the request must be made with a 64-bit register access to DF::FabricIndirectConfigAccessDataLo.
4. For writes:
 1. Write FICADL. If writing 64-bit registers, the request must be made with a 64-bit access to DF::FabricIndirectConfigAccessDataLo.

11.3.4 DF Configuration Register 64 bit Atomic Access

11.3.4.1 CPU 64-bit Access

CPU's are allowed to issue 64-bit requests directly to [DF](#) configuration register space. The address must be 8-byte aligned.

When PIE receives a 64-bit access request from CCM, it generates the beats necessary to fulfill the atomicity requirements for the access.

11.3.5 DF Configuration Register Access Examples

For Instance and Fabric IDs, see:

- DF::FabricBlockInstanceInformation0
- DF::FabricBlockInstanceInformation1
- DF::FabricBlockInstanceInformation2
- DF::FabricBlockInstanceInformation3_CSNCSPiEALLM
- DF::FabricBlockInstanceInformation3_IOS
- DF::FabricBlockInstanceInformation3_TCDXPFXCAKE

Notes:

For clarity, the examples place FCAC at D18F4x040.

It is assumed the reader understands how to get to a PCI device and function (i.e. through CF8/CFC, or through the [MMIO](#) port). This method can be used from any node.

These examples use these instance IDs, for illustration only:

- CS0: 0
- CS1: 1
- CCM0: 2
- CCM1: 3
- IOMS0: 4
- PIE: 5
- TCDX0: 6
- TCDX1: 7
- TCDX2: 8
- TCDX3: 9
- TCDX4: 10
- TCDX5: 11
- TCDX6: 12

11.3.5.1 Example: Broadcast Write Through PCI

Broadcast write to F3x104 (CCM configuration register) on node 5 through PCI configuration (direct).

1. The device number for node 5 is device $(24+5) = 29 = 0x1D$
2. The FCAC is already be locked to broadcast-only mode, as this is done by the [ASP](#).
3. Write bus 0, device 0x1D, function 3, offset 0x104 to the expected value. This broadcasts only to the components on node 5. It does not affect other nodes.

11.3.5.2 Example: Instance Read Via Indirect Access

Instance read of node 5's CS1 FabricBlockInstanceInformation0 register (for instance, F0x44) through an indirect PCI access register. Most of the FICAA fields can be derived from the register description.

Note: To use an indirect access register, put FCAC in broadcast mode. The FCAC affects the write to the FICAA, so if FCAC was pointing to an instance other than PIE, the write to FICAA is a NOP.

1. Node 5 device number (24+5) = 29 = 0x1D (see 11.3.1 [CPU Sourced DF Configuration Register Access])
2. Select a fabric indirect register set. This example uses register set 2 (F4x058 FICAA; F4x088 FICADL), so all the accesses are to device 0x1D, function 4, offset 0x058/0x088.
3. Write FICAA2 register fields as follows:
 1. CfgRegInstID = 1. CS1 instance ID is 1 in this example.
 2. IndCfgAccFuncNum = 0. This example targets function 0.
 3. IndCfgAccRegNum = 0x11 (0x44 >> 2). This example targets offset 0x44.
 4. CfgRegInstAccEn = 1. This example is for single-instance.
4. Read FICADL2 and it returns the value in node 5 CS1's FabricBlockInstanceInformation0 register.

11.3.6 DF Configuration Registers Common to All Function Spaces

Since [DF](#) registers reside in PCI configuration space, each implemented function space has the following registers:

- Offset 0x00 – Device/Vendor ID Registers
 - DF::DeviceVendorId0; DF::DeviceVendorId1; DF::DeviceVendorId2; DF::DeviceVendorId3; DF::DeviceVendorId4; DF::DeviceVendorId5; DF::DeviceVendorId6
 - The Device ID is product specific and also varies by DF function space.
- Offset 0x04 – Status/Command Registers
 - DF::StatusCommand0; DF::StatusCommand1; DF::StatusCommand2; DF::StatusCommand3; DF::StatusCommand4; DF::StatusCommand5; DF::StatusCommand6
- Offset 0x08 – Class Code/Revision ID Registers
 - DF::ClassCodeRevId0; DF::ClassCodeRevId1; DF::ClassCodeRevId2; DF::ClassCodeRevId3; DF::ClassCodeRevId4; DF::ClassCodeRevId5; DF::ClassCodeRevId6
- Offset 0x0C – Header Type Registers
 - DF::HeaderType0; DF::HeaderType1; DF::HeaderType2; DF::HeaderType3; DF::HeaderType4; DF::HeaderType5; DF::HeaderType6
- Offset 0x34 – Capabilities Pointer Registers
 - DF::CapabilitiesPtr0; DF::CapabilitiesPtr1; DF::CapabilitiesPtr2; DF::CapabilitiesPtr3; DF::CapabilitiesPtr4; DF::CapabilitiesPtr5; DF::CapabilitiesPtr6
- Offset 0x100 – Extended Capabilities Registers
 - DF::EnhCapabilityHdr0; DF::EnhCapabilityHdr1; DF::EnhCapabilityHdr2; DF::EnhCapabilityHdr3; DF::EnhCapabilityHdr4; DF::EnhCapabilityHdr5; DF::EnhCapabilityHdr6

11.3.7 DF Response to Access of Unimplemented Registers

[DF](#) drops all writes to unimplemented registers. Reads to unimplemented registers return all zeroes. DF does not log any errors for accesses to unimplemented registers or functions.

11.4 Data Fabric Instance and Fabric IDs

Every data fabric component within a node is identified by a unique instance identifier ("InstanceID") used to access component registers. For example, to program a register in TCDX0 only, specify the instance ID for TCDX0 in the register access (see 11.3.5 [DF Configuration Register Access Examples]).

For address mapping and packet routing, each Data Fabric protocol component is assigned a unique Data Fabric Identifier ("FabricID"). The FabricID consists of a NodeID section and a ComponentID section. Refer to SystemFabricIdMask to understand how to determine the NodeID from the ComponentID. The value for a component's FabricID is assigned at reset. Transport-layer components (TCDXs and CAKE) do not have a FabricID. Address maps are populated with the targeted component's FabricID.

FabricBlockInstanceInformation3[BlockFabricID] and FabricBlockInstanceInformation3[BlockInstanceID] self-report each component's identifiers.

11.4.1 System InstanceID and FabricID Assignment

The table below displays the [DF](#) Component identifiers. The reset values are the ComponentIDs (FabricIDs). In some cases the values are modified during the initialization process. The DF block component FabricID should be read to obtain the actual ComponentID (and actual FabricID).

Table 197: InstanceID and ComponentID assignment

Component	InstanceID	ComponentID
Coherent Slave 0 (UMC0)	00h	00h
Coherent Slave 1 (UMC1)	01h	01h
Coherent Slave 2 (UMC2)	02h	02h
Coherent Slave 3 (UMC3)	03h	03h
Coherent Slave 4 (UMC4)	04h	04h
Coherent Slave 5 (UMC5)	05h	05h
Coherent Slave 6 (UMC6)	06h	06h
Coherent Slave 7 (UMC7)	07h	07h
Coherent Slave 8 (UMC8)	08h	08h
Coherent Slave 9 (UMC9)	09h	09h
Coherent Slave 10 (UMC10)	0Ah	0Ah
Coherent Slave 11 (UMC11)	0Bh	0Bh
Coherent Slave 12 (CMP0)	0Ch	0Ch
Coherent Slave 13 (CMP1)	0Dh	0Dh
Coherent Slave 14 (CMP2)	0Eh	0Eh
Coherent Slave 15 (CMP3)	0Fh	0Fh
Coherent Core Master 0 (CCM0)	10h	60h
Coherent Core Master 1 (CCM1)	11h	61h
Coherent Core Master 2 (CCM2)	12h	62h
Coherent Core Master 3 (CCM3)	13h	63h
Coherent Core Master 4 (CCM4)	14h	64h
Coherent Core Master 5 (CCM5)	15h	65h
Coherent Core Master 6 (CCM6)	16h	66h
Coherent Core Master 7 (CCM7)	17h	67h
Accelerator Coherent Master 0 (ACM0)	18h	6Ch
Accelerator Coherent Master 1 (ACM1)	19h	6Dh
Accelerator Coherent Master 2 (ACM2)	1Ah	6Eh
Accelerator Coherent Master 3 (ACM3)	1Bh	6Fh
Non-coherent Master 0 (NCM0_IOMMU0)	1Ch	70h

Non-coherent Master 1 (NCM1_IOMMU1)	1Dh	71h
Non-coherent Master 2 (NCM2_IOMMU2)	1Eh	72h
Non-coherent Master 3 (NCM3_IOMMU3)	1Fh	73h
I/O Master 0 (IOM0_IOHUBM0)	20h	78h
I/O Master 1 (IOM1_IOHUBM1)	21h	79h
I/O Master 2 (IOM2_IOHUBM2)	22h	7Ah
I/O Master 3 (IOM3_IOHUBM3)	23h	7Bh
IOHUBS0 (IOHUBS0)	24h	20h
IOHUBS1 (IOHUBS1)	25h	21h
IOHUBS2 (IOHUBS2)	26h	22h
IOHUBS3 (IOHUBS3)	27h	23h
ICNG0 (ICNG0)	28h	7Ch
ICNG1 (ICNG1)	29h	7Dh
ICNG2 (ICNG2)	2Ah	7Eh
ICNG3 (ICNG3)	2Bh	7Fh
PIE (PIE0)	2Ch	77h
CAKE0	2Dh	N/A
CAKE1	2Eh	N/A
CAKE2	2Fh	N/A
CAKE3	30h	N/A
CAKE4	31h	N/A
CAKE5	32h	N/A
CAKE6	33h	N/A
CAKE7	34h	N/A
CNLI0 (CNLI0)	35h	N/A
CNLI1 (CNLI1)	36h	N/A
CNLI2 (CNLI2)	37h	N/A
CNLI3 (CNLI3)	38h	N/A
PFX0 (PFX0)	39h	N/A
PFX1 (PFX1)	3Ah	N/A
PFX2 (PFX2)	3Bh	N/A
PFX3 (PFX3)	3Ch	N/A
PFX4 (PFX4)	3Dh	N/A
PFX5 (PFX5)	3Eh	N/A
PFX6 (PFX6)	3Fh	N/A
PFX7 (PFX7)	40h	N/A
SPF0 (SPF0)	41h	N/A
SPF1 (SPF1)	42h	N/A
SPF2 (SPF2)	43h	N/A
SPF3 (SPF3)	44h	N/A
SPF4 (SPF4)	45h	N/A
SPF5 (SPF5)	46h	N/A
SPF6 (SPF6)	47h	N/A
SPF7 (SPF7)	48h	N/A
SPF8 (SPF8)	49h	N/A
SPF9 (SPF9)	4Ah	N/A
SPF10 (SPF10)	4Bh	N/A
SPF11 (SPF11)	4Ch	N/A

SPF12 (SPF12)	4Dh	N/A
SPF13 (SPF13)	4Eh	N/A
SPF14 (SPF14)	4Fh	N/A
SPF15 (SPF15)	50h	N/A
TCDX0	51h	N/A
TCDX1	52h	N/A
TCDX2	53h	N/A
TCDX3	54h	N/A
TCDX4	55h	N/A
TCDX5	56h	N/A
TCDX6	57h	N/A
TCDX7	58h	N/A
TCDX8	59h	N/A
TCDX9	5Ah	N/A
TCDX10	5Bh	N/A
TCDX11	5Ch	N/A
TCDX12	5Dh	N/A
TCDX13	5Eh	N/A
TCDX14	5Fh	N/A
TCDX15	60h	N/A

11.5 Data Fabric Initialization

Data Fabric Initialization is done by software during boot.

11.5.1 Address Map Initialization

11.5.1.1 DRAM Address Map Initialization

See 11.7 [DF DRAM Address Mapping Rules for SoC].

11.5.1.2 CXL™ Memory Address Map Initialization

[CXL™](#) initialization is performed by firmware during boot. CXL memory devices (Type 2 and Type 3) are treated as part of DRAM Address Map. See 11.7 [DF DRAM Address Mapping Rules for SoC].

11.5.2 Decoding Last I/O Addresses

The Last I/O Address Decode Table provides decoding guidance for registers DF::OrigWdtAddrLogLo and DF::OrigWdtAddrLogHi. These registers log a "last I/O address" after a fatal error, and is typically read by diagnosis software.

Table 198: Last I/O Address Decode Table

Address Range	I/O Address Type	Decoding Guidance
---------------	------------------	-------------------

0 to FFFFFFFC_FFFFFFFF	DRAM or MMIO	The address must be looked up in the address map to determine if (a) it is a DRAM address or (b) a MMIO address, and then to decode what target it is. In the "OrigWDTAddrLog" feature, the registers provide the destination FabricID.
FFFFE_10000000 to FFFFE_1FFFFFFF Note: Upper address bit range: "FFFFFFFE_xxxxxxx"; is not accessible, and therefore, not made visible.	PCIe	Decoded as PCIe configuration accesses where bits 27-20=bus, bits 19-15=device, bits 14-12=function, bits 11-0=register offset. When PCIe segments are enabled, DstFabricID indicates the IOHUB.
FFFFFFFD_FC000000 to FFFFFFFD_FC0FFFFF	PCIe	Decoded as PCIe I/O cycles (x86 IN/OUT instructions) where bits 15:0 are the I/O address offset.
ALL OTHER ADDRESSES	System Functions (Internal)	Can be used internally for system functions. Observations of this would decode to an error within the processor.

11.5.3 Register Values Written Based on Configuration

Table 199: System Configuration Dependent Register Settings

Register	Fields	Comments
DF::NodeToLinkSet7 DF::NodeToLinkSet6 DF::NodeToLinkSet5 DF::NodeToLinkSet4 DF::NodeToLinkSet3 DF::NodeToLinkSet2 DF::NodeToLinkSet1 DF::NodeToLinkSet0	All fields	The values are based on the links that are enabled (multinode only).
DF::LinkTgtMode_CCM DF::LinkTgtMode_CAKE DF::LinkTgtMode_CS DF::LinkTgtMode_ICNG DF::LinkTgtMode_IOM DF::LinkTgtMode_NCM DF::LinkTgtMode_NCSIOS DF::LinkTgtMode_PIE	All fields	The values are based on the number of CCDs, memory population, and link configuration.
DF::LinkTgtCfg_CCM DF::LinkTgtCfg_CS DF::LinkTgtCfg_GCMNCM DF::LinkTgtCfg_ICNG DF::LinkTgtCfg_IOM DF::LinkTgtCfg_NCSIOS DF::LinkTgtCfg_PIE	All fields	The values are based on the number of CCDs, memory population, and link configuration.
DF::RspNeighborNodeIdCmpTblRegister	All fields	The values are based on how many nodes are enabled (multinode only).

DF::ReqNeighborNodeIdCmpTblRegister	All fields	The values are based on how many nodes are enabled (multinode only).
DF::LocalRoutingTable	All fields	The values are based on the configuration of enabled devices.
DF::RemoteRoutingTable	All fields	The values are based on the link configuration (multinode only).
DF::LocalMcastRoutingTable	All fields	The values are based on the configuration of enabled devices.
DF::RemoteMcastRoutingTable	All fields	The values are based on the link configuration (multinode only).
DF::McastTypeMask9 DF::McastTypeMask8 DF::McastTypeMask7 DF::McastTypeMask6 DF::McastTypeMask5 DF::McastTypeMask4 DF::McastTypeMask3 DF::McastTypeMask2 DF::McastTypeMask1 DF::McastTypeMask0	All fields	The values are based on the configuration of enabled devices, the number of nodes, and the link configuration.
DF::FtiResponseClumpVCMask0	All fields	The values are based on the link configuration (multinode only).
DF::CakeVCRemap	All fields	The values are based on the link configuration (multinode only).
DF::CakeVCRemap1	All fields	The values are based on the link configuration (multinode only).
DF::FuseDisable	All fields	ComponentDisable may be set for unused components.
DF::SystemCfg	MyNodeId	Set to 2 for socket 1.
	OtherSocketPresent	Set to 1 on multi-socket systems.
DF::SystemComponentCnt	All fields except GCMCount	The values are based on the IOHUB configuration and the multinode configuration.
DF::PIEFabricCompCnt	All fields	The values are based on the configuration of enabled devices and number of nodes.

DF::SpecialRegister1_MASTERS DF::SpecialRegister1_PIE	LclCSFabricID	The NodeID of LclCSFabricID is adjusted in multinode. The code always uses logical CS0 on the local socket and the target. If no memory is attached to physical CS0, the first used CS is made logical CS0. If no memory is attached to any CS on the socket, physical CS0 is kept enabled just for this feature. LclNCSFabricID is unused in servers.
DF::SpecialSysFunctionFabricID1	LclPIEFabricID	The NodeID of LclPIEFabricID is adjusted in multinode.
DF::SpecialSysFunctionFabricID2	FchIOSFabricID	The value is adjusted based on the SOC.
DF::SpecialSysFunctionFabricID3	LclICNGFabricID	The NodeID of LclICNGFabricID is adjusted in multinode. IOM copy: Adjusted for the IONUMA domain mapping; MSIs are sent to the ICNG according to the intended IONUMA setting: - IOM0: ICNG2 - IOM1: ICNG0 - IOM2: ICNG1 - IOM3: ICNG3 Must also account for downbin conditions and not send to an ICNG that has been removed because there are no attached CCDs. NodeID added on multi-socket systems. CCM Copy: NodeID added on multi-socket systems. Each CCM and IOM can have a unique value. Access by per-instance only.
DF::LclNodeTypeMap	NodeCount	The value is based on the multinode configuration (multinode only).
DF::RmtNodeTypeMap	BaseNodeID	The field is set to 8. This avoids aliasing in the node type map

DF::McaIntCtrl	All fields	The NodeID of McaIntPIEFabricID is adjusted in multinode.
DF::FabricBlockInstanceInformation3_CSN CSPIEALLM DF::FabricBlockInstanceInformation3_IOS DF::FabricBlockInstanceInformation3_TC DXPFXCAKE	BlockFabricID	Certain memory configurations require the BlockFabricID of Coherent Slaves to be adjusted. In these scenarios, this register is reprogrammed to provide a logical FabricID of a coherent slave. This field is also altered on any block that is not part of the fabric but has been removed from the configuration.
DF::InterruptControl	ApicIdNumSkt	The value is based on the configuration. Ignored if the system does not enable X2APIC mode.
	ApicIdSktIdLsb	The value is based on the configuration. Ignored if the system does not enable X2APIC mode. In X2APIC mode, program this to the location of the SocketID bit in the Initial APIC ID algorithm of the core.
	SMTEnable	The value is based on whether the cores enable SMT mode.
	ApcSwizzleType	Set to 1b.
DF::ApicClusterIcngMap3 DF::ApicClusterIcngMap2 DF::ApicClusterIcngMap1 DF::ApicClusterIcngMap0	All fields	The values are based on the configuration. In X2APIC mode, program this reverse map of the Initial APICID logical clustering. Ignored if the system does not enable X2APIC mode or if DF::InterruptConfig1[IntBcstOptDis]=1.
DF::InterruptConfig1	IntBcstOptDis	This bit is normally set to zero for X2APIC mode. Setting this bit disables the optimization to target interrupts based on an X2APIC cluster definition.
DF::ClusterCfg	NumCpuNodes	The value is based on the number of sockets and the platform configuration.

	MyCpuNodeID	The value is based on the number of sockets and the platform configuration.
	ExpPrbRspCls[7:0]	The value is based on the core configuration. ExpPrbRspCls[3:0] is programmed for socket0 and ExpPrbRspCls[7:4] is programmed for socket1. Programming setup assumes DF::DfGlobalCtrl l[McastMaskPerNodeCLV] = 1. The value in the field includes the count of probe responses from each quadrant, for dual-headed CCMs, wide-mode CCMs and CCDs with multiple CCX/CCD. The programming includes the case where the quadrant has been removed from the probe network and cases when one or more of CCM0, CCM1, CCM2 and CCM3 are left as part of the probe network to spoof probe responses. In this last case, a CCM that has no enabled and part of the probe network. Each CCM may have a unique value. Access by per-instance only.
	LclClusterEn	The value is set to 0xF.
DF::CCDEnable	All fields	The value is based on the core configuration.
DF::CcdUnitIdMask	CcxIdShft and CcxIdMask	The value is based on the core configuration . Other fields in this register do not change from the reset value.
DF::CoherentSlaveConfigB2	DisCdmaAdjLastSteerTag	Partial initialization of this register. Remaining bits in the register set by RIB. The value is based on the core configuration.
DF::DfGlobalCtrl	McastMaskPerNodeCLV	Unconditionally set to 1.
DF::AcmEnable	All fields	The value is based on the presence and type of CXL devices.

DF::CcmSdpEnable	All fields	The value is based on the core configuration.
DF::CcmSltModeCfg	FarCsBaseCompId and FarCsLimitCompId	Only programmed with SltModeEn=1, adjusted based on the actual memory population.
DF::CnliCxlRxCrdControlHigh3 DF::CnliCxlRxCrdControlHigh2 DF::CnliCxlRxCrdControlHigh1 DF::CnliCxlRxCrdControlHigh0	All Fields	CNLI request and data token register.

11.6 Data Fabric Address Maps

Data Fabric maps various types of address ranges:

- DRAM address map registers (11.6.2 [DRAM Address Maps]) control interleave configuration and associate address ranges with a coherent slave.
- PCI I/O address maps associate an IO slave with an address range that is accessible through legacy IN/OUT instructions.
- PCI configuration address maps associate an IO slave with a PCI configuration space.

11.6.1 Reserved Address Range

11.6.1.1 Special Address Range

For system purposes, the processor and data fabric define a 12GB "special address range" at the very top of the maximum physical address space. This address range is not accessible to I/O or CPU clients. For instance, if the processor supports a 48-bit physical address, the memory from (2*48-12GB) to (2*48-1) is not accessible. CPU page tables that reference this address result in a page-fault (#PF) exception and I/O accesses are aborted. The BIOS must map this region as inaccessible to the operating system.

11.6.1.2 IOMMU Reserved Region

If [IOMMU](#) is enabled, the processor must reserve the following addresses:

- 0x00FD_0000_0000->0x00FD_F8FF_FFFF (3.9GB)
- 0x00FD_F910_0000->0x00FD_F91F_FFFF (1MB)
- 0x00FD_FB00_0000->0x00FD_FFFF_FFFF (80MB)

[MMIO](#) addresses must not be assigned in these regions.

If the first DRAM address map (the one starting at address 0) includes these regions, the BIOS must map these regions as inaccessible to the operating system. In addition, the firmware must use these DRAM addresses for any privileged memory regions that would be taken out of this same address map.

If any other DRAM address map includes these regions, the firmware must boost this address map so that it starts at 1TB. The entire region between the previous DRAM address map, and 1TB, is then marked as inaccessible to the operating system.

11.6.2 DRAM Address Maps

The processor routes DRAM accesses using DRAM address mapping registers, which must be accessed per-instance as each specifies a different address range. Software programs DRAM Base/Limit pairs for each component as summarized below:

Table 200: DF DRAM Address Map Ranges

Masters	Coherent Slave (CS)	PIE
16	4	8

List of applicable registers:

- DF::DramBaseAddress
- DF::DramLimitAddress
- DF::DramHoleControl
- DF::DramOffset
- DF::DramAddressCtl[HashIntlvCtl1G]
- DF::DramAddressCtl[HashIntlvCtl2M]
- DF::DramAddressCtl[HashIntlvCtl64K]

11.6.2.1 DRAM Address Normalization and Interleave

DRAM Address Interleaving configuration is specified by:

- DF::DramAddressIntlv[IntLvAddrSel]
- DF::DramAddressIntlv[IntLvNumDies]
- DF::DramAddressIntlv[IntLvNumSockets]
- DF::DramAddressIntlv[IntLvNumChan]

11.7 DF DRAM Address Mapping Rules for SoC

See 11.6.2.1 [DRAM Address Normalization and Interleave]

11.7.1 PCI I/O Address Maps

The processor routes PCI I/O accesses using 12 sets of PCI I/O address mapping registers. In addition, the VGA-compatible PCI I/O address space can be mapped using DF::VGAEn.

A PCI I/O access that does not hit on a valid PCI IO/VGA address mapping register is sent to the compatibility (subtractive) address space.

The Data Fabric only allows CPU clients to access PCI I/O.

List of applicable registers:

- DF::X86IOBaseAddress
- DF::X86IOLimitAddress

11.7.2 PCI Configuration Address Maps

The processor routes PCI configuration accesses using 12 sets of PCI configuration address mapping registers. The Data Fabric requires that every PCI bus is mapped in the registers and does not support compatibility (subtractive) address space.

List of applicable registers:

- DF::CfgAddressCntl
- DF::CfgBus0Type1DstFabricIDReg0, DF::CfgBus0Type1DstFabricIDReg1

The first 256 bytes of every PCI configuration space function are called "Base Configuration Space" ([BCS](#)) and bytes 256 through 4095 are called "Extended Configuration Space" ([ECS](#)).

CPU software accesses PCI configuration space (including Data Fabric registers) in two ways:

- Through IO-space:
 - IO instructions to addresses CF8h and CFCh.
 - Only PCI-defined segment 0 is accessible.
- Through [MMIO](#) space:
 - Configuration space is a region of (system) memory space.
 - The base address and size of this range is specified in the MMIO Configuration Base/Limit Address Register DF::X86IOBaseAddress.
 - The size is determined by the number of configuration-space bus numbers supported by the system and by the number of segments; each segment contains a separate set of 256 buses.
- Up to 128 segments are supported.
 - Accesses to this range are converted to configuration space accesses as follows:
 - Address[31:0] = {segment [3:0], bus[7:0], device[4:0], function[2:0], offset[11:0]}.
- The BIOS may use either configuration space access mechanism during boot. Before booting the OS, BIOS is expected to disable IO access to ECS and enable MMIO access.

The Data Fabric only allows CPU clients to access PCI configuration space.

11.8 Data Fabric Security

The Data Fabric supports memory encryption.

11.8.1 Data Fabric Register Security

The [ASP](#) controls all data fabric register security features, based on the security policy set by the ASP. This section describes the base security policy but other security policies may be used by the ASP.

During BIOS boot, the DRAM address map registers are read-only. After BIOS boot, all data fabric registers are read-only.

11.8.2 Data Fabric Memory Security

The data fabric hardware can be programmed by the [ASP](#) to protect memory contents.

Data fabric security features are designed to protect:

- Memory used for system purposes, such as private regions for ASP, [MP1](#) or for blocks to save and restore state over [C-states](#). These are referred to as "privileged memory regions".
- The ASEG regions.
- The TSEG region, as long as it is 1MB aligned.
- VMGuard memory.

The privileged memory regions are protected from inappropriate access by any unauthorized requestor. Any attempt to

access these regions from unauthorized requestors is fenced by the CS. The authorized requestors for each region are as follows:

- ASP region: Only the originator matching the ASP FabricID/UnitID is allowed.
- MP1 region: Only the originator matching the MP1 FabricID/UnitID is allowed.
- CC6 region: Only CPUs are allowed.
- [DF](#) region: Only the Data Fabric PIE block is allowed.

In addition, the originator matching the ASP FabricID/UnitID is allowed access to all regions.

The additional data fabric ASEG and TSEG region protection, when enabled by BIOS completing a call to the ASP after initialization, creates an additional fence to protect these regions from access from any source other than a CPU operating in [SMM](#). In order to utilize the TSEG protection, it is required that platform BIOS sets the following bits:

Core::X86::Msrb::[SMMAddr](#)[[TSegBase](#)[51:17]] bits[19:17] and Core::X86::Msrb::[SMMMask](#)[[TSegMask](#)[51:17]] bits[19:17] to 000b. The TSEG must be a multiple of 1MB and aligned to a 1MB boundary.

11.9 DF Performance Monitors

11.9.1 Fabric Performance Monitor Counter (PMC) Events

11.9.1.1 Background

The Infinity Fabric™ (also known as "Fabric", or as "[DF](#)") is a scalable and modular coherent interconnect for the EPYC™ Server SOC (System On Chip).

The following abstract block diagram of an [IOD](#) shows the some of the key interfaces of the Infinity Fabric. In this conceptual rendering, there are a certain number of CPU Moderator blocks (CCMs), IO Moderator blocks (IOMs), Coherent Stations (CSs), and Infinity Link Controller blocks (LINKs). While the Infinity Fabric stretches across IODs over the [xGMI](#) Infinity Links, the Fabric interface blocks are associated with a given IOD.

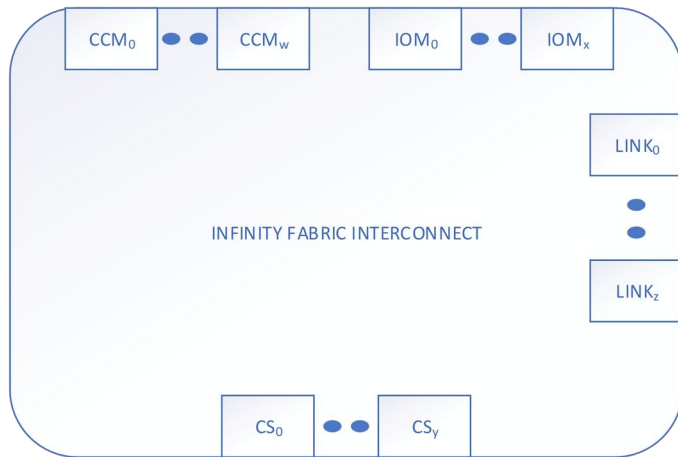


Figure 44: Infinity Fabric™ Interconnect

Table 201: Interface Descriptions

Interface	Descriptions
-----------	--------------

CS	Coherent Station (Interface to one UMC instance)
CCM	CPU Moderator (interface to one or two CCD instances)
IOM	IO Moderator (interface to one NBIO instance)
LINK	Interface to inter-socket Infinity Fabric xGMI link

Hardware counter events are implemented to provide visibility into the flow of transaction traffic. There are up to 16 counters available that can be programmed to count events concurrently. The number of available Fabric PMCs can be determined by accessing CPUID_Fn80000022_EBX [Extended Performance Monitoring and Debug EBX]. The Fabric [PMC](#) counters can be programmed to select the desired instance and events via [MSR Core::X86::Msrr::DF_PERF_CTL](#). The count values captured in the Fabric PMCs can subsequently be accessed via [MSR Core::X86::Msrr::DF_PERF_CTR](#). The counts are also accessible by software via the RDPMC instruction.

11.9.1.2 Interfaces and Instance IDs

Each CCM can interface to either one or two Core/Cache Complex Dies (CCDs). Each IOM interfaces to one Northbridge IO (NBIO) instance. Each CS interfaces with one Unified Memory Controller (UMC) instance, each of which supports a DDR5 memory channel. The LINK blocks are applicable only to Dual Socket (2P) systems, and interface with the respective [xGMI](#) links.

Each [PMC](#) event is selected by means of a 14-bit Event Selection field (EventSel[13:0]).

- Bits [13:6] are used to specify an Instance ID, which corresponds to a unique identifier of one of the interface blocks (CS, CCM, IOM, LINK).
- Bits [5:0] are used to specify the Event Encoding (the specific counter event within the block)

The available Instance IDs for the interface blocks on Family 19h Models 10h-1Fh are shown in the table below. On a given processor and platform configuration, not all interfaces may be connected or active.

Table 202: Instance IDs for the Interface Blocks

Interface Block	Instance ID (hexadecimal)
CS0 through CS11	0x0 through 0xB
CCM0 through CCM7	0x10 through 0x17
IOM0 through IOM3	0x20 through 0x23
LINK0 through LINK7	0x2D through 0x34

11.9.1.3 Event Definitions

This section describes the available Fabric PMCs.

11.9.1.3.1 DATA_BW (Data Bandwidth)

The DATA_BW (data bandwidth) event is intended to help software measure data movement at the key interfaces of the Fabric. The event is designed to count the number of beats of data at the interface being measured. In some cases, there is a degree of selectivity on what beats to measure (for example, beats associated with read transactions versus those associated with write transactions).

The Event Encoding for the DATA_BW [PMC](#) event is 0x1F. A subset of the CCM blocks have two interfaces active. Depending on the processor and platform configuration, these can be used either to connect to two [CCD](#) instances (one interface per CCD), or to connect to a single CCD instance (both interfaces to the same CCD). In such cases, the Event Encoding for the DATA_BW PMC event for the first interface on that CCM is 0x1E, and that for the second interface on that CCM is 0x1F.

Table 203: DATA_BW PMC Event Unitmask for CCM, IOM and CS Interfaces

UnitMask Bit Field	UnitMask Field Name	Description
11:10	SrcDstDieProx	Select proximity of the source or destination of the data beat selected by TxnType. Must be programmed to 0x3 when measuring CS read bandwidth. 0x0: Reserved 0x1: Count only beats with SrcDst on the same processor die 0x2: Count only beats with SrcDst on a remote processor die 0x3: Count all beats regardless of SrcDst die proximity
9:1	Reserved	Required to be programmed to 0x1FF
0	TxnType	Select type of data beat 0: Data beat in response to a Read request 1: Data beat accompanying a Write request

Table 204: DATA_BW PMC Event Unitmask for LINK Controller Interface

UnitMask Bit Field	UnitMask Field Name	Description
11:1	Reserved	Required to be programmed to 0x79F
0	Dirn	Select Direction of Data Transfer 0: Outbound Data beat dispatched from this controller to the link 1: Inbound Data beat accepted by this controller from the link

11.9.1.3.2 Unit of Measurement

The count represents the number of data beats that were monitored at the selected interface over the measurement time window. The number of bytes transferred are computed by the product of the number of beats and the size of the data beat. Dividing the number of bytes transferred by the measurement time window will allow the user to compute the data bandwidth at the selected interface. The following table specifies the size (unit) of the data beats at each interface in bytes for Family 19h Models 10h-1Fh.

Table 205: DATA_BW PMC Data Beat Size per Interface and TxnType

Interface	TxnType	Size of data beat counted by DATA_BW
CS	READ	64B
CS	WRITE	64B
CCM	READ	32B
CCM	WRITE	64B
IOM	READ	64B
IOM	WRITE	64B
LINK	READ	64B
LINK	WRITE	64B

11.9.1.3.3 Usage Notes for SrcDst field

SrcDst captures the notion of either the source of a beat of data, or the destination. The interface being monitored determines whether the field represents the source or the destination.

The general flow for a transaction involves the movement of requests and data between originators (source) and completers (destination).

The Coherent Station (CS) interface is a natural sink point or destination for write memory transactions originating from the [CCD](#) or the NBIO. At the same time, it is the natural originator or source for read data being returned to the CCD or NBIO from the memory controller. Therefore, when monitoring at the CS interface, SrcDst refers to the Source of write data beats (when TxnType == 1), and to the destination of read response data beats (when TxnType == 0).

This perspective is inverted when monitoring the CCM or IOM interfaces. The CCM or IOM are natural originators (or sources) for write transactions (to either memory or [MMIO](#)) originating from the CPU or NBIO respectively. They are also natural sink points or destinations for read data being returned to the CPU or NBIO respectively. Therefore, when monitoring at the CCM or IOM interfaces, SrcDst refers to the eventual destination of write data beats (when TxnType == 1), and to the source of read response data beats (when TxnType == 0).

Thus, SrcDstDieProx allows the user to select the source of write data beats, or the destination of read response data beats, when being monitored at a natural destination like the CS. Alternatively, it allows the user to select the destination of write data beats, or the source of read response data beats, when being monitored at a natural originator like the CCM or IOM. The SrcDstDieProx field allows selecting based on whether the source or destination is on the local processor node or a remote processor node. This field is only applicable for systems with more than one socket.

11.9.1.4 Algorithm for Data Bandwidth measurement

- Write to the appropriate instance of Core::X86::Msrr::[DF_PERF_CTL](#) with desired value of EventSelect[13:0] and UnitMask[11:0]
- Write zero to the appropriate instance of Core::X86::Msrr::[DF_PERF_CTR](#) (to initialize the counter)
- Have software wait a reasonable/appropriate amount of time.
- Read the Core::X86::Msrr::[DF_PERF_CTR](#) count value with the RDPMC instruction.
- Convert the count to the number of bytes using the Beat Size table above.
- Divide the number of bytes by the amount of time that occurred between the write of zero to Core::X86::Msrr::[DF_PERF_CTR](#) and the read of Core::X86::Msrr::[DF_PERF_CTR](#), to obtain a throughput value in bytes/second.

11.9.1.5 Examples for EventSelect and UnitMask

11.9.1.5.1 Examples 1:

At the Coherent Station 0, count the number of write data beats received from all sources on the Local processor node.

EventSelect[13:0] = {InstanceId [13:6], Event[5:0]} = {8'h0, 6'h1F} = 0x1F

UnitMask[11:0] = {SrcDstDieProx[11:10], Reserved[9:1], TxnType[0]} = {2'h1, 9'h1FF, 1'h1} = 0x7FF

11.9.1.5.2 Examples 2:

At CPU Moderator 3 Interface 0, count the number of read response data beats received from all destinations on all processor nodes

EventSelect[13:0] = {InstanceId[13:6], Event[5:0]} = {8'h13, 6'h1E} = 0x4DE

UnitMask[11:0] = {SrcDstDieProx[11:10], Reserved[9:1], TxnType[0]} = {2'h3, 9'h1FF, 1'h0} = 0xFFE

11.9.1.5.3 Examples 3:

At LINK Controller 2, count the number of outbound data beats (from the controller towards the link)

EventSelect[13:0] = {InstanceId[13:6], Event[5:0]} = {8'h2F, 6'h1F} = 0xBDF

UnitMask[11:0] = {Reserved[11:1], Dirn[0]} = {11'h79F, 1'h0} = 0xF3E

11.9.2 Data Fabric Performance Monitor Events**11.9.2.1 Filter Implementation**

Filters are implemented for the following perfmons

Table 206: DF Performance Monitor Filter Implementation by Component

CS	CAKE/CCM/IOM/NCM	
DF::PMC::CS::CS_DATA_BW	DF::PMC::CAKE::CAKE_DATA_BW DF::PMC::CCM::CCM_DATA_BW0 DF::PMC::CCM::CCM_DATA_BW1 DF::PMC::IOM::IOM_DATA_BW	

11.9.2.2 Data Fabric CAKE Performance Monitor Events

DFPMCx00000[B5...D1]F [CAKE DATA_BW Data Bandwidth] (DF::PMC::CAKE::CAKE_DATA_BW)		
Read-write. Reset: 0000h.		
_inst[CAKE[7:0]]; DFPMCx0000_0[D1,CD,C9,C5,C1,BD,B9,B5]F		
Bits	Description	
15:12	Reserved.	
11:10	ReqSrcDstMode . Read-write. Reset: 0h. SrcDst mode selection for Req beats	
	ValidValues:	
	Value	Description
	0h	Disabled. No Req beats counted
	1h	Count Req beats where receiver (FabRcv) matches SrcDst field
	2h	Count Req beats where sender (FabXmt) matches SrcDst field
	3h	Count all Req beats, regardless of SrcDst field
9:8	RspSrcDstMode . Read-write. Reset: 0h. SrcDst mode selection for Rsp beats. (FabXmt is invalid for beats from CAKE to FTI)	
	ValidValues:	
	Value	Description
	0h	Disabled. No Rsp beats counted
	1h	Count Rsp beats where receiver (FabRcv) matches SrcDst field
	2h	Count Rsp beats where sender (FabXmt) matches SrcDst field
	3h	Count all Rsp beats, regardless of SrcDst field
7:6	Reserved.	
5:1	SrcDst . Read-write. Reset: 00h. Select transactions based on the source/dest of the data - UMC, CXL , IO, GPU, or CPU. Multi-bit selects are also possible. See Mode fields for more options	
	ValidValues:	
	Value	Description
	00h	Disabled. No transactions selected
	01h	Count only beats to/from UMC
	02h	Count only beats to/from CPU
	03h	Reserved.
	04h	Count only beats to/from GPU
	07h-05h	Reserved.
	08h	Count only beats to/from IO
	0Fh-09h	Reserved.
	10h	Count only beats to/from CXL
	1Eh-11h	Reserved.
	1Fh	Count any beat
0	Dirn . Read-write. Reset: 0. 0=Count Data Beats from FTI to CAKE (outbound to link). 1=Count Data Beats from CAKE to FTI (inbound from link). Select Direction of Data Movement	

11.9.2.3 Data Fabric CCM Performance Monitor Events

DFPMCx00000[41...6D]E [CCM DATA_BW0 SDP Bandwidth] (DF::PMC::CCM::CCM_DATA_BW0)																					
Read-write. Reset: 0000h.																					
_inst[ACM[3:0],CCM[7:0]]; DFPMCx0000_0[6D,69,65,61,5D,59,55,51,4D,49,45,41]E																					
Bits	Description																				
15:12	Reserved.																				
11:10	SrcDstDieProx. Read-write. Reset: 0h. Select transactions based on the die proximity of the source/destination																				
	ValidValues:																				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Disabled. No transactions selected</td></tr> <tr> <td>1h</td><td>Count only transactions with source/destination on Local cluster</td></tr> <tr> <td>2h</td><td>Count only transactions with source/destination on Remote cluster</td></tr> <tr> <td>3h</td><td>Count any transaction, regardless of source/destination cluster proximity</td></tr> </table>	Value	Description	0h	Disabled. No transactions selected	1h	Count only transactions with source/destination on Local cluster	2h	Count only transactions with source/destination on Remote cluster	3h	Count any transaction, regardless of source/destination cluster proximity										
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1h	Count only transactions with source/destination on Local cluster																				
2h	Count only transactions with source/destination on Remote cluster																				
3h	Count any transaction, regardless of source/destination cluster proximity																				
9:6	Reserved.																				
5:2	SrcDst. Read-write. Reset: 0h. Select transactions based on the source/dest of the data - UMC, CXL , IO, CPU (cache). Multi-bit selects are also possible.																				
	ValidValues:																				
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4h	Count only data beats to/from IO																				
7h-5h	Reserved.																				
8h	Count only data beats to/from CXL																				
Eh-9h	Reserved.																				
Fh	Count any transaction, regardless of source																				
1	ModeSel. Read-write. Reset: 0. 0=Source mode: Attributes in [11:2] are used to match the actual source of data (originator of the consumed response). 1=Home mode: Attributes in [11:2] are used to match the home of data (destination of the read). Select between Source mode and Home mode for Read Responses. For Writes (TxnType == 1), only Home mode is applicable																				
0	TxnType. Read-write. Reset: 0. 0=Count Read Response Data Beats. 1=Count Write Data Beats. Select Transaction Type.																				

DFPMCx00000[41...5D]F [CCM DATA_BW1 SDP Bandwidth] (DF::PMC::CCM::CCM_DATA_BW1)

Read-write. Reset: 0000h.

_inst[CCM[7:0]]; DFPMCx0000_0[5D,59,55,51,4D,49,45,41]F

Bits	Description																				
15:12	Reserved.																				
11:10	SrcDstDieProx. Read-write. Reset: 0h. Select transactions based on the die proximity of the source/destination																				
	ValidValues:																				
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3h	Reserved.																				
4h	Count only data beats to/from IO																				
7h-5h	Reserved.																				
8h	Count only data beats to/from CXL																				
Eh-9h	Reserved.																				
Fh	Count any transaction, regardless of source																				
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0	TxnType. Read-write. Reset: 0. 0=Count Read Response Data Beats. 1=Count Write Data Beats. Select Transaction Type.																				

11.9.2.4 Data Fabric CS Performance Monitor Events

DFPMCx00000[01...3D]F [CS DATA_BW Data Bandwidth] (DF::PMC::CS::CS_DATA_BW)		
Read-write. Reset: 0000h.		
This PMC functionality is fully operational for CS_UMC instances. It is only operational for Reads in CS_CMP instances. This PMC should not be used for measuring Write BW for CS_CMP.		
<code>_inst[CMP[3:0],CS[11:0]];</code> DFPMCx0000_0[3D,39,35,31,2D,29,25,21,1D,19,15,11,0D,09,05,01]F		
Bits	Description	
15:12	Reserved.	
11:10	SrcDstDieProx. Read-write. Reset: 0h. Select transactions based on the die proximity of the source	
	ValidValues:	
	Value	Description
	0h	Disabled. No transactions selected
	1h	Count only transactions with source on Local die
	2h	Count only transactions with source on Remote die
	3h	Count any transaction, regardless of source die proximity
9:8	Reserved.	
7:6	SrcDstDieType. Read-write. Reset: 0h. Select transactions based on die type of the source	
	ValidValues:	
	Value	Description
	0h	Disabled. No transactions selected
	1h	Count only transactions with source on the same die type
	2h	Count only transactions with source on a different die type
	3h	Count any transaction, regardless of source die type
5:2	SrcDst. Read-write. Reset: 0h. Select transactions based on the type of the requestor - CXL , IO, GPU, or CPU. Multi-bit selects are also possible	
	ValidValues:	
	Value	Description
	0h	Disabled. No transactions selected
	1h	Count only transactions from CPU source
	2h	Count only transactions from GPU source
	3h	Reserved.
	4h	Count only transactions from IO source
	7h-5h	Reserved.
	8h	Count only transactions from CXL source
	Eh-9h	Reserved.
	Fh	Count any transaction, regardless of source
1	Reserved.	
0	TxnType. Read-write. Reset: 0. 0=Count Read Response Data Beats. 1=Count Write Data Beats. Select Transaction Type	

11.9.2.5 Data Fabric IOM Performance Monitor Events

DFPMCx000008[1...D]F [IOM DATA_BW DATA BANDWIDTH] (DF::PMC::IOM::IOM_DATA_BW)		
Read-write. Reset: 0000h.		
_inst[IOM[3:0]]; DFPMCx0000_08[D,9,5,1]F		
Bits	Description	
15:12	Reserved.	
11:10	SrcDstDieProx. Read-write. Reset: 0h. Select transactions based on the die proximity of the source/destination	
	ValidValues:	
	Value	Description
	0h	Disabled. No transactions selected
	1h	Count only transactions with source/destination on Local die
	2h	Count only transactions with source/destination on Remote die
	3h	Count any transaction, regardless of source/destination die proximity
9:6	Reserved.	
5:2	SrcDst. Read-write. Reset: 0h. Select transactions based on the source/dest of the data - UMC, CXL , IO, CPU (cache). Multi-bit selects are also possible.	
	ValidValues:	
	Value	Description
	0h	Disabled. No transactions selected
	1h	Count only data beats from CPU (cache). For Reads only.
	2h	Count only data beats to/from UMC
	3h	Reserved.
	4h	Count only data beats to/from IO
	7h-5h	Reserved.
	8h	Count only data beats to/from CXL
	Eh-9h	Reserved.
	Fh	Count any transaction, regardless of source
1	Reserved.	
0	TxnType. Read-write. Reset: 0. 0=Count Read Response Data Beats. 1=Count Write Data Beats. Select Transaction Type	

11.10 Registers

11.10.1 Function 0 Registers

D18F0x000 [Device/Vendor ID Function 0] (DF::DeviceVendorId0)	
Read-only. Reset: Fixed,14AD_1022h.	
PCI Device/Vendor ID 0 register	
_aliasHOST; D18F0x000	
_aliasSMN; D18F0x00000000; D18F0=4900_0000h	
Bits	Description
31:16	DeviceID. Read-only. Reset: Fixed,14ADh. Device ID
15:0	VendorID. Read-only. Reset: Fixed,1022h. Vendor ID

D18F0x004 [Status Command Function 0] (DF::StatusCommand0)

Read-only. Reset: Fixed,0000_0000h.

PCI Status/Command register

_aliasHOST; D18F0x004_aliasSMN; D18F0x00000004; D18F0=4900_0000h

Bits	Description
31:16	Status . Read-only. Reset: Fixed,0000h. Status
15:0	Command . Read-only. Reset: Fixed,0000h. Command

D18F0x008 [Class Code Revision ID Function 0] (DF::ClassCodeRevId0)

Read-only. Reset: Fixed,0600_0000h.

PCI Class Code/Revision ID register

_aliasHOST; D18F0x008_aliasSMN; D18F0x00000008; D18F0=4900_0000h

Bits	Description
31:8	ClassCode . Read-only. Reset: Fixed,06_0000h. Class Code. Provides the hostbridge class code as defined in the PCI specification.
7:0	RevisionId . Read-only. Reset: Fixed,00h. RevID: Revision ID.

D18F0x00C [Header Type Function 0] (DF::HeaderType0)

Read-only. Reset: Fixed,0080_0000h.

PCI Header Type register

_aliasHOST; D18F0x00C_aliasSMN; D18F0x0000000C; D18F0=4900_0000h

Bits	Description
31:0	HeaderType . Read-only. Reset: Fixed,0080_0000h. Header Type: These bits are fixed at their default values. The header type field indicates that there are multiple functions present in this device.

D18F0x034 [Capabilities Pointer Function 0] (DF::CapabilitiesPtr0)

Read-only. Reset: Fixed,0000_0000h.

PCI Capabilities Pointer register

_aliasHOST; D18F0x034_aliasSMN; D18F0x00000034; D18F0=4900_0000h

Bits	Description
31:8	Reserved.
7:0	CapPtr . Read-only. Reset: Fixed,00h. Capabilities Pointer.

D18F0x040 [Fabric Block Instance Count] (DF::FabricBlockInstanceCount)

Fabric Block Instance Count Register provides information on the total number of [DF](#) components; the components included in the count are listed in FabricBlockInstanceInformation0[InstanceType].

_instBCST_aliasHOST; D18F0x040

_instPIE0_aliasHOST; D18F0x040_x2C; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=0, IndCfgAccRegNum=0x010]

_instBCST_aliasSMN; D18F0x00000040; D18F0=4900_0000h

_instPIE0_aliasSMN; D18F0x00000040_x2C; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=0, IndCfgAccRegNum=0x010]

Bits	Description
31:28	Reserved.
27:24	MajorRevision. Reset: 4h. Specifies the major revision of the Data Fabric hardware, MajorRevision.MinorRevision. AccessType: SecurePart ? Read-only : Read-write.
23:16	MinorRevision. Reset: 00h. Specifies the minor revision of the Data Fabric hardware, MajorRevision.MinorRevision. AccessType: SecurePart ? Read-only : Read-write.
15:8	Reserved.
7:0	BlkInstCount. Read-only. Provides the total number of instances of all the blocks in the Data Fabric. Reset: _instBCST_alias[SMN,HOST]: Fixed,61h Reset: _instPIE0_alias[SMN,HOST]: 61h

D18F0x044 [Fabric Block Instance Information 0] (DF::FabricBlockInstanceInformation0)

Fabric Block Instance Information 0 provides information on the block capabilities. See also DF::FabricBlockInstanceCount.

_instBCST_aliasHOST; D18F0x044															
_inst[TCDX[15:0],SPF[15:0],PFX[7:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F0x044_x[60:00]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:00], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x011]															
_instBCST_aliasSMN; D18F0x00000044; D18F0=4900_0000h															
_inst[TCDX[15:0],SPF[15:0],PFX[7:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F0x00000044_x[60:00]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:00], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x011]															
Bits	Description														
31:27	Reserved.														
26:24	InstanceSubType. Read-only. Component sub-types.														
	Reset: _inst[ACM[3:0]]_alias[SMN,HOST]: 1h														
	Reset: _inst[CAKE[7:0]]_alias[SMN,HOST]: 2h														
	Reset: _inst[CCM[7:0]]_alias[SMN,HOST]: 0h														
	Reset: _inst[CMP[3:0]]_alias[SMN,HOST]: 2h														
	Reset: _inst[CNLI[3:0]]_alias[SMN,HOST]: 0h														
	Reset: _inst[ICNG[3:0],CS[11:00]]_alias[SMN,HOST]: 1h														
	Reset: _inst[IOM[3:0]]_alias[SMN,HOST]: 0h														
	Reset: _inst[IOS[3:0]]_alias[SMN,HOST]: 2h														
	Reset: _inst[NCM[3:0]]_alias[SMN,HOST]: 4h														
	Reset: _inst[PFX[7:0]]_alias[SMN,HOST]: 1h														
	Reset: _inst[TCDX[15:00],SPF[15:00],PIE0]_alias[SMN,HOST]: 0h														
	ValidValues:														
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>CNLI: CNLI IOM: IOHUBM; PIE: PIE; SPF: SFP; TCDX: TCDX</td></tr> <tr> <td>1h</td><td>CCM: ACM; CS: UMC; PIE: ICNG; TCDX: PFX</td></tr> <tr> <td>2h</td><td>CAKE: xGMI; CS: CMP IOS: IOHUBS</td></tr> <tr> <td>3h</td><td>No sub-type</td></tr> <tr> <td>4h</td><td>NCM: IOMMU</td></tr> <tr> <td>7h-5h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	CNLI: CNLI IOM: IOHUBM; PIE: PIE; SPF: SFP; TCDX: TCDX	1h	CCM: ACM; CS: UMC; PIE: ICNG; TCDX: PFX	2h	CAKE: xGMI ; CS: CMP IOS: IOHUBS	3h	No sub-type	4h	NCM: IOMMU	7h-5h	Reserved.
Value	Description														
0h	CNLI: CNLI IOM: IOHUBM; PIE: PIE; SPF: SFP; TCDX: TCDX														
1h	CCM: ACM; CS: UMC; PIE: ICNG; TCDX: PFX														
2h	CAKE: xGMI ; CS: CMP IOS: IOHUBS														
3h	No sub-type														
4h	NCM: IOMMU														
7h-5h	Reserved.														
23	McaBankPresent. Read-only. Set by hardware if the Instance has an MCA bank.														
	Reset: _inst[ACM[3:0]]_alias[SMN,HOST]: 0														
	Reset: _instBCST_alias[SMN,HOST]: Fixed,0														
	Reset: _inst[CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: 0														
	Reset: _inst[CMP[3:0]]_alias[SMN,HOST]: 1														
	Reset: _inst[CNLI[3:0]]_alias[SMN,HOST]: 0														
	Reset: _inst[CS[11:00]]_alias[SMN,HOST]: 1														
	Reset: _inst[PFX[7:0],NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0]]_alias[SMN,HOST]: 0														
	Reset: _inst[SPF[15:00],PIE0]_alias[SMN,HOST]: 1														
	Reset: _inst[TCDX[15:00]]_alias[SMN,HOST]: 0														
22	Reserved.														
21:20	FtiPortDatBusCnt. Read-only. Number of data buses attached to an FTI port.														
	Reset: _inst[ACM[3:0]]_alias[SMN,HOST]: 1h														
	Reset: _instBCST_alias[SMN,HOST]: Fixed,1h														
	Reset: _inst[NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: 1h														
	Reset: _inst[PFX[7:0]]_alias[SMN,HOST]: 0h														
	Reset: _inst[TCDX[15:00],SPF[15:00],PIE0]_alias[SMN,HOST]: 1h														
19	Reserved.														
18:16	FtiPortCnt. Read-only. Number of Fabric transport interfaces connected to instance.														
	Reset: _inst[ACM[3:0]]_alias[SMN,HOST]: 1h														
	Reset: _instBCST_alias[SMN,HOST]: Fixed,1h														
	Reset: _inst[NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: 1h														
	Reset: _inst[PFX[7:0]]_alias[SMN,HOST]: 6h														
	Reset: _inst[SPF[15:00],PIE0]_alias[SMN,HOST]: 1h														
	Reset: _inst[TCDX[2:0]]_alias[SMN,HOST]: 6h														
	Reset: _inst[TCDX[4:3]]_alias[SMN,HOST]: 5h														
	Reset: _inst[TCDX[15:05]]_alias[SMN,HOST]: 6h														

15	Reserved.										
14	SdpRdRespBuses. Read-only. 0=1 response bus. 1=2 response buses. Number of SDP Read response buses. Reset: _inst[ACM[3:0]]_alias[SMN,HOST]: 0 Reset: _instBCST_alias[SMN,HOST]: Fixed,0 Reset: _inst[TCDX[15:00],SPF[15:00],PIE0,PFX[7:0],NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: 0										
13:12	SdpPortCnt. Read-only. Number of SDP ports connected to the instance. Reset: _inst[ACM[3:0]]_alias[SMN,HOST]: 1h Reset: _instBCST_alias[SMN,HOST]: Fixed,1h Reset: _inst[CAKE[7:0]]_alias[SMN,HOST]: 0h Reset: _inst[CMP[3:0],CCM[7:0]]_alias[SMN,HOST]: 1h Reset: _inst[CNLI[3:0]]_alias[SMN,HOST]: 2h Reset: _inst[CS[11:00]]_alias[SMN,HOST]: 1h Reset: _inst[ICNG[3:0]]_alias[SMN,HOST]: 0h Reset: _inst[NCM[3:0],IOS[3:0],IOM[3:0]]_alias[SMN,HOST]: 1h Reset: _inst[PIE0,PFX[7:0]]_alias[SMN,HOST]: 0h Reset: _inst[SPF[15:00]]_alias[SMN,HOST]: 1h Reset: _inst[TCDX[15:00]]_alias[SMN,HOST]: 0h										
11:10	Reserved.										
9:8	FtiDataBusWidth. Read-only. FTI data bus width. Reset: _inst[ACM[3:0]]_alias[SMN,HOST]: 2h Reset: _instBCST_alias[SMN,HOST]: Fixed,2h Reset: _inst[TCDX[15:00],SPF[15:00],PIE0,PFX[7:0],NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: 2h ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>64 bits</td></tr> <tr> <td>1h</td><td>128 bits</td></tr> <tr> <td>2h</td><td>256 bits</td></tr> <tr> <td>3h</td><td>512 bits</td></tr> </table>	Value	Description	0h	64 bits	1h	128 bits	2h	256 bits	3h	512 bits
Value	Description										
0h	64 bits										
1h	128 bits										
2h	256 bits										
3h	512 bits										
7	Reserved.										
6	Enabled. Read-only. Indicates component is enabled for responding to register accesses. Reset: _inst[ACM[3:0]]_alias[SMN,HOST]: 1 Reset: _instBCST_alias[SMN,HOST]: Fixed,1 Reset: _inst[TCDX[15:00],SPF[15:00],PIE0,PFX[7:0],NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: 1										
5:4	SdpDataBusWidth. SDP data bus width. AccessType: _inst[IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: Read-only AccessType: _inst[NCM[3:0]]_alias[SMN,HOST]: Read-only, Volatile AccessType: _inst[TCDX[15:00],SPF[15:00],PIE0,PFX[7:0]]_alias[SMN,HOST]: Read-only Reset: _inst[ACM[3:0]]_alias[SMN,HOST]: 2h Reset: _instBCST_alias[SMN,HOST]: Fixed,2h Reset: _inst[IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: 2h Reset: _inst[NCM[3:0]]_alias[SMN,HOST]: 3h Reset: _inst[TCDX[15:00],SPF[15:00],PIE0,PFX[7:0]]_alias[SMN,HOST]: 2h ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>64 bits</td></tr> <tr> <td>1h</td><td>128 bits</td></tr> <tr> <td>2h</td><td>256 bit</td></tr> <tr> <td>3h</td><td>512 bits</td></tr> </table>	Value	Description	0h	64 bits	1h	128 bits	2h	256 bit	3h	512 bits
Value	Description										
0h	64 bits										
1h	128 bits										
2h	256 bit										
3h	512 bits										
3:0	InstanceType. Read-only. Instance type register. Reset: _inst[ACM[3:0]]_alias[SMN,HOST]: 0h Reset: _instBCST_alias[SMN,HOST]: Fixed,0h Reset: _inst[CAKE[7:0]]_alias[SMN,HOST]: Ah										

Reset: _inst[CCM[7:0]]_alias[SMN,HOST]: 0h
 Reset: _inst[CMPI[3:0]]_alias[SMN,HOST]: 4h
 Reset: _inst[CNLI[3:0]]_alias[SMN,HOST]: Dh
 Reset: _inst[CS[11:00]]_alias[SMN,HOST]: 4h
 Reset: _inst[ICNG[3:0]]_alias[SMN,HOST]: 7h
 Reset: _inst[IOM[3:0]]_alias[SMN,HOST]: 3h
 Reset: _inst[IOS[3:0]]_alias[SMN,HOST]: 5h
 Reset: _inst[NCM[3:0]]_alias[SMN,HOST]: 2h
 Reset: _inst[PFX[7:0]]_alias[SMN,HOST]: 6h
 Reset: _instPIE0_alias[SMN,HOST]: 7h
 Reset: _inst[SPF[15:00]]_alias[SMN,HOST]: 8h
 Reset: _inst[TCDX[15:00]]_alias[SMN,HOST]: 6h

ValidValues:

Value	Description
0h	CCM
1h	GCM
2h	NCM
3h	IOM
4h	CS
5h	NCS
6h	TCDX
7h	PIE
8h	SPF
9h	LLC
Ah	CAKE
Ch-Bh	Reserved.
Dh	CNLI
Fh-Eh	Reserved.

D18F0x048 [Fabric Block Instance Information 1] (DF::FabricBlockInstanceInformation1)

Read-only.

Fabric Block Instance Information 1 provides topology information by providing the block's neighbors instance IDs. See also DF::FabricBlockInstanceCount.

_instBCST_aliasHOST; D18F0x048

_inst[TCDX[15:0],SPF[15:0],PFX[7:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F0x048_x[60:00]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:00], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x012]

_instBCST_aliasSMN; D18F0x00000048; D18F0=4900_0000h

_inst[TCDX[15:0],SPF[15:0],PFX[7:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F0x00000048_x[60:00]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:00], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x012]

Bits	Description
31:24	FtiPort3NeighborInstId. Read-only. Valid if FabricBlockInstanceInformation0[FtiPortCnt] > 3
	Reset: _inst[NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 00h
	Reset: _instPFX0_alias[SMN,HOST]: 42h
	Reset: _instPFX1_alias[SMN,HOST]: 44h
	Reset: _instPFX2_alias[SMN,HOST]: 46h
	Reset: _instPFX3_alias[SMN,HOST]: 48h
	Reset: _instPFX4_alias[SMN,HOST]: 4Ah
	Reset: _instPFX5_alias[SMN,HOST]: 4Ch
	Reset: _instPFX6_alias[SMN,HOST]: 4Eh
	Reset: _instPFX7_alias[SMN,HOST]: 50h
	Reset: _inst[SPF[15:00],PIE0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_alias[SMN,HOST]: 01h
	Reset: _instTCDX1_alias[SMN,HOST]: 2Ah
	Reset: _instTCDX2_alias[SMN,HOST]: 04h
	Reset: _instTCDX3_alias[SMN,HOST]: 21h
	Reset: _instTCDX4_alias[SMN,HOST]: 23h
	Reset: _instTCDX5_alias[SMN,HOST]: 57h
	Reset: _instTCDX6_alias[SMN,HOST]: 56h
	Reset: _instTCDX7_alias[SMN,HOST]: 24h
	Reset: _instTCDX8_alias[SMN,HOST]: 26h
	Reset: _instTCDX9_alias[SMN,HOST]: 58h
	Reset: _instTCDX10_alias[SMN,HOST]: 34h
	Reset: _instTCDX11_alias[SMN,HOST]: 59h
	Reset: _instTCDX12_alias[SMN,HOST]: 07h
	Reset: _instTCDX13_alias[SMN,HOST]: 29h
	Reset: _instTCDX14_alias[SMN,HOST]: 0Ah
	Reset: _instTCDX15_alias[SMN,HOST]: 2Fh
23:16	FtiPort2NeighborInstId. Read-only. Valid if FabricBlockInstanceInformation0[FtiPortCnt] > 2
	Reset: _inst[NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 00h
	Reset: _instPFX0_alias[SMN,HOST]: 41h
	Reset: _instPFX1_alias[SMN,HOST]: 43h
	Reset: _instPFX2_alias[SMN,HOST]: 45h
	Reset: _instPFX3_alias[SMN,HOST]: 47h
	Reset: _instPFX4_alias[SMN,HOST]: 49h
	Reset: _instPFX5_alias[SMN,HOST]: 4Bh
	Reset: _instPFX6_alias[SMN,HOST]: 4Dh
	Reset: _instPFX7_alias[SMN,HOST]: 4Fh
	Reset: _inst[TCDX0,SPF[15:00],PIE0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_alias[SMN,HOST]: 28h
	Reset: _instTCDX2_alias[SMN,HOST]: 03h
	Reset: _instTCDX3_alias[SMN,HOST]: 25h
	Reset: _instTCDX4_alias[SMN,HOST]: 27h
	Reset: _instTCDX5_alias[SMN,HOST]: 57h
	Reset: _instTCDX6_alias[SMN,HOST]: 56h
	Reset: _instTCDX7_alias[SMN,HOST]: 20h
	Reset: _instTCDX8_alias[SMN,HOST]: 22h
	Reset: _instTCDX9_alias[SMN,HOST]: 0Dh
	Reset: _instTCDX10_alias[SMN,HOST]: 33h
	Reset: _instTCDX11_alias[SMN,HOST]: 0Fh
	Reset: _instTCDX12_alias[SMN,HOST]: 06h
	Reset: _instTCDX13_alias[SMN,HOST]: 2Bh

	Reset: _instTCDX14_alias[SMN,HOST]: 09h
	Reset: _instTCDX15_alias[SMN,HOST]: 30h
15:8	FtiPort1NeighborInstId. Read-only. Valid if FabricBlockInstanceInformation0[FtiPortCnt] > 1
	Reset: _inst[NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 00h
	Reset: _instPFX0_alias[SMN,HOST]: 01h
	Reset: _instPFX1_alias[SMN,HOST]: 0Ch
	Reset: _instPFX2_alias[SMN,HOST]: 04h
	Reset: _instPFX3_alias[SMN,HOST]: 0Dh
	Reset: _instPFX4_alias[SMN,HOST]: 07h
	Reset: _instPFX5_alias[SMN,HOST]: 0Eh
	Reset: _instPFX6_alias[SMN,HOST]: 0Ah
	Reset: _instPFX7_alias[SMN,HOST]: 0Fh
	Reset: _inst[SPF[15:00],PIE0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_alias[SMN,HOST]: 14h
	Reset: _instTCDX1_alias[SMN,HOST]: 56h
	Reset: _instTCDX2_alias[SMN,HOST]: 16h
	Reset: _inst[TCDX[4:3]]_alias[SMN,HOST]: 56h
	Reset: _inst[TCDX[6:5]]_alias[SMN,HOST]: 52h
	Reset: _instTCDX7_alias[SMN,HOST]: 1Ch
	Reset: _instTCDX8_alias[SMN,HOST]: 1Eh
	Reset: _instTCDX9_alias[SMN,HOST]: 0Ch
	Reset: _instTCDX10_alias[SMN,HOST]: 32h
	Reset: _instTCDX11_alias[SMN,HOST]: 0Eh
	Reset: _instTCDX12_alias[SMN,HOST]: 17h
	Reset: _instTCDX13_alias[SMN,HOST]: 56h
	Reset: _instTCDX14_alias[SMN,HOST]: 15h
	Reset: _instTCDX15_alias[SMN,HOST]: 2Dh
7:0	FtiPort0NeighborInstId. Read-only. Valid if FabricBlockInstanceInformation0[FtiPortCnt] > 0
	Reset: _inst[ACM[1:0]]_alias[SMN,HOST]: 5Ah
	Reset: _inst[ACM[3:2]]_alias[SMN,HOST]: 5Ch
	Reset: _inst[CAKE[3:0]]_alias[SMN,HOST]: 60h
	Reset: _inst[CAKE[7:4]]_alias[SMN,HOST]: 5Bh
	Reset: _instCCM0_alias[SMN,HOST]: 51h
	Reset: _instCCM1_alias[SMN,HOST]: 5Fh
	Reset: _instCCM2_alias[SMN,HOST]: 53h
	Reset: _instCCM3_alias[SMN,HOST]: 5Dh
	Reset: _instCCM4_alias[SMN,HOST]: 51h
	Reset: _instCCM5_alias[SMN,HOST]: 5Fh
	Reset: _instCCM6_alias[SMN,HOST]: 53h
	Reset: _instCCM7_alias[SMN,HOST]: 5Dh
	Reset: _inst[CMP[1:0]]_alias[SMN,HOST]: 5Ah
	Reset: _inst[CMP[3:2]]_alias[SMN,HOST]: 5Ch
	Reset: _instCNLI0_alias[SMN,HOST]: 35h
	Reset: _instCNLI1_alias[SMN,HOST]: 36h
	Reset: _instCNLI2_alias[SMN,HOST]: 37h
	Reset: _instCNLI3_alias[SMN,HOST]: 38h
	Reset: _inst[CS[2:0]]_alias[SMN,HOST]: 51h
	Reset: _inst[CS[5:3]]_alias[SMN,HOST]: 53h
	Reset: _inst[CS[8:6]]_alias[SMN,HOST]: 5Dh
	Reset: _inst[CS[11:09]]_alias[SMN,HOST]: 5Fh
	Reset: _instICNG0_alias[SMN,HOST]: 52h
	Reset: _instICNG1_alias[SMN,HOST]: 5Eh
	Reset: _instICNG2_alias[SMN,HOST]: 52h
	Reset: _instICNG3_alias[SMN,HOST]: 5Eh
	Reset: _instIOM0_alias[SMN,HOST]: 58h
	Reset: _instIOM1_alias[SMN,HOST]: 54h
	Reset: _instIOM2_alias[SMN,HOST]: 59h
	Reset: _instIOM3_alias[SMN,HOST]: 55h
	Reset: _instIOS0_alias[SMN,HOST]: 58h
	Reset: _instIOS1_alias[SMN,HOST]: 54h
	Reset: _instIOS2_alias[SMN,HOST]: 59h
	Reset: _instIOS3_alias[SMN,HOST]: 55h
	Reset: _instNCM0_alias[SMN,HOST]: 58h
	Reset: _instNCM1_alias[SMN,HOST]: 54h

Reset: _instNCM2_alias[SMN,HOST]: 59h
Reset: _instNCM3_alias[SMN,HOST]: 55h
Reset: _instPFX0_alias[SMN,HOST]: 02h
Reset: _instPFX1_alias[SMN,HOST]: 00h
Reset: _instPFX2_alias[SMN,HOST]: 05h
Reset: _instPFX3_alias[SMN,HOST]: 03h
Reset: _instPFX4_alias[SMN,HOST]: 08h
Reset: _instPFX5_alias[SMN,HOST]: 06h
Reset: _instPFX6_alias[SMN,HOST]: 0Bh
Reset: _instPFX7_alias[SMN,HOST]: 09h
Reset: _instPIE0_alias[SMN,HOST]: 5Ah
Reset: _inst[SPF[15:00]]_alias[SMN,HOST]: 00h
Reset: _instTCDX0_alias[SMN,HOST]: 10h
Reset: _instTCDX1_alias[SMN,HOST]: 51h
Reset: _instTCDX2_alias[SMN,HOST]: 12h
Reset: _inst[TCDX[4:3]]_alias[SMN,HOST]: 60h
Reset: _instTCDX5_alias[SMN,HOST]: 54h
Reset: _instTCDX6_alias[SMN,HOST]: 58h
Reset: _inst[TCDX[8:7]]_alias[SMN,HOST]: 57h
Reset: _instTCDX9_alias[SMN,HOST]: 2Ch
Reset: _instTCDX10_alias[SMN,HOST]: 31h
Reset: _instTCDX11_alias[SMN,HOST]: 00h
Reset: _instTCDX12_alias[SMN,HOST]: 13h
Reset: _instTCDX13_alias[SMN,HOST]: 5Dh
Reset: _instTCDX14_alias[SMN,HOST]: 11h
Reset: _instTCDX15_alias[SMN,HOST]: 2Eh

D18F0x04C [Fabric Block Instance Information 2] (DF::FabricBlockInstanceInformation2)

Read-only.

Fabric Block Instance Information 2 provides topology information by providing the block's neighbors instance IDs. See also DF::FabricBlockInstanceCount.

_instBCST_aliasHOST; D18F0x04C

_inst[TCDX[15:0],SPF[15:0],PFX[7:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F0x04C_x[60:00]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:00], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x013]

_instBCST_aliasSMN; D18F0x0000004C; D18F0=4900_0000h

_inst[TCDX[15:0],SPF[15:0],PFX[7:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F0x0000004C_x[60:00]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:00], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x013]

Bits Description

31:16 Reserved.

15:8 **FtiPort5NeighborInstId.** Read-only. Valid if FabricBlockInstanceInformation0[FtiPortCnt] > 5

Reset: _inst[NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 00h

Reset: _instPFX0_alias[SMN,HOST]: 3Ah

Reset: _instPFX1_alias[SMN,HOST]: 39h

Reset: _instPFX2_alias[SMN,HOST]: 3Ch

Reset: _instPFX3_alias[SMN,HOST]: 3Bh

Reset: _instPFX4_alias[SMN,HOST]: 3Eh

Reset: _instPFX5_alias[SMN,HOST]: 3Dh

Reset: _instPFX6_alias[SMN,HOST]: 40h

Reset: _instPFX7_alias[SMN,HOST]: 3Fh

Reset: _inst[SPF[15:00],PIE0]_alias[SMN,HOST]: 00h

Reset: _instTCDX0_alias[SMN,HOST]: 52h

Reset: _instTCDX1_alias[SMN,HOST]: 53h

Reset: _instTCDX2_alias[SMN,HOST]: 52h

Reset: _inst[TCDX[4:3]]_alias[SMN,HOST]: 00h

Reset: _instTCDX5_alias[SMN,HOST]: 55h

Reset: _instTCDX6_alias[SMN,HOST]: 59h

Reset: _inst[TCDX[8:7]]_alias[SMN,HOST]: 5Bh

Reset: _instTCDX9_alias[SMN,HOST]: 19h

Reset: _instTCDX10_alias[SMN,HOST]: 59h

Reset: _instTCDX11_alias[SMN,HOST]: 1Bh

Reset: _instTCDX12_alias[SMN,HOST]: 5Eh

Reset: _instTCDX13_alias[SMN,HOST]: 5Fh

Reset: _instTCDX14_alias[SMN,HOST]: 5Eh

Reset: _instTCDX15_alias[SMN,HOST]: 55h

7:0 **FtiPort4NeighborInstId.** Read-only. Valid if FabricBlockInstanceInformation0[FtiPortCnt] > 4

Reset: _inst[NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 00h

Reset: _instPFX0_alias[SMN,HOST]: 3Ah

Reset: _instPFX1_alias[SMN,HOST]: 39h

Reset: _instPFX2_alias[SMN,HOST]: 3Ch

Reset: _instPFX3_alias[SMN,HOST]: 3Bh

Reset: _instPFX4_alias[SMN,HOST]: 3Eh

Reset: _instPFX5_alias[SMN,HOST]: 3Dh

Reset: _instPFX6_alias[SMN,HOST]: 40h

Reset: _instPFX7_alias[SMN,HOST]: 3Fh

Reset: _inst[SPF[15:00],PIE0]_alias[SMN,HOST]: 00h

Reset: _instTCDX0_alias[SMN,HOST]: 02h

Reset: _instTCDX1_alias[SMN,HOST]: 57h

Reset: _instTCDX2_alias[SMN,HOST]: 05h

Reset: _instTCDX3_alias[SMN,HOST]: 1Dh

Reset: _instTCDX4_alias[SMN,HOST]: 1Fh

Reset: _inst[TCDX[6:5]]_alias[SMN,HOST]: 5Eh

Reset: _instTCDX7_alias[SMN,HOST]: 5Ah

Reset: _instTCDX8_alias[SMN,HOST]: 5Ch

Reset: _instTCDX9_alias[SMN,HOST]: 18h

Reset: _instTCDX10_alias[SMN,HOST]: 58h

Reset: _instTCDX11_alias[SMN,HOST]: 1Ah

Reset: _instTCDX12_alias[SMN,HOST]: 08h

Reset: _instTCDX13_alias[SMN,HOST]: 57h

	Reset: _instTCDX14_alias[SMN,HOST]: 0Bh
	Reset: _instTCDX15_alias[SMN,HOST]: 54h

D18F0x050 [Fabric Block Instance Information 3] (DF::FabricBlockInstanceInformation3_CSNCSPIEALLM)

This register provides the InstanceID and, when applicable, the FabricID of the block in which it is instanced. Not all the block instances require FabricID. See also DF::FabricBlockInstanceCount.

_instBCST_aliasHOST; D18F0x050

_inst[SPF[15:0],CNLI[3:0],PIE0,ICNG[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F0x050_x[[50:41],3[8:5],2[C:8],[23:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],3[8:5],2[C:8],[23:00]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x014]

_instBCST_aliasSMN; D18F0x00000050; D18F0=4900_0000h

_inst[SPF[15:0],CNLI[3:0],PIE0,ICNG[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F0x00000050_x[[50:41],3[8:5],2[C:8],[23:00]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],3[8:5],2[C:8],[23:00]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x014]

Bits	Description
31:20	Reserved.
19:8	BlockFabricID. Init: 11.5.3 [Register Values Written Based on Configuration].
	AccessType: _inst[CMP[3:0],CCM[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write
	AccessType: _inst[CNLI[3:0]]_alias[SMN,HOST]: Read-only
	AccessType: _inst[SPF[15:00],PIE0,NCM[3:0],IOM[3:0],ICNG[3:0],CS[11:00]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write
	Reset: _instACM0_alias[SMN,HOST]: 06Ch
	Reset: _instACM1_alias[SMN,HOST]: 06Dh
	Reset: _instACM2_alias[SMN,HOST]: 06Eh
	Reset: _instACM3_alias[SMN,HOST]: 06Fh
	Reset: _instCCM0_alias[SMN,HOST]: 060h
	Reset: _instCCM1_alias[SMN,HOST]: 061h
	Reset: _instCCM2_alias[SMN,HOST]: 062h
	Reset: _instCCM3_alias[SMN,HOST]: 063h
	Reset: _instCCM4_alias[SMN,HOST]: 064h
	Reset: _instCCM5_alias[SMN,HOST]: 065h
	Reset: _instCCM6_alias[SMN,HOST]: 066h
	Reset: _instCCM7_alias[SMN,HOST]: 067h
	Reset: _instCMP0_alias[SMN,HOST]: 00Ch
	Reset: _instCMP1_alias[SMN,HOST]: 00Dh
	Reset: _instCMP2_alias[SMN,HOST]: 00Eh
	Reset: _instCMP3_alias[SMN,HOST]: 00Fh
	Reset: _inst[CS0,CNLI[3:0]]_alias[SMN,HOST]: 000h
	Reset: _instCS1_alias[SMN,HOST]: 001h
	Reset: _instCS2_alias[SMN,HOST]: 002h
	Reset: _instCS3_alias[SMN,HOST]: 003h
	Reset: _instCS4_alias[SMN,HOST]: 004h
	Reset: _instCS5_alias[SMN,HOST]: 005h
	Reset: _instCS6_alias[SMN,HOST]: 006h
	Reset: _instCS7_alias[SMN,HOST]: 007h
	Reset: _instCS8_alias[SMN,HOST]: 008h
	Reset: _instCS9_alias[SMN,HOST]: 009h
	Reset: _instCS10_alias[SMN,HOST]: 00Ah
	Reset: _instCS11_alias[SMN,HOST]: 00Bh
	Reset: _instCNG0_alias[SMN,HOST]: 07Ch
	Reset: _instCNG1_alias[SMN,HOST]: 07Dh
	Reset: _instCNG2_alias[SMN,HOST]: 07Eh
	Reset: _instCNG3_alias[SMN,HOST]: 07Fh
	Reset: _instIOM0_alias[SMN,HOST]: 078h
	Reset: _instIOM1_alias[SMN,HOST]: 079h
	Reset: _instIOM2_alias[SMN,HOST]: 07Ah
	Reset: _instIOM3_alias[SMN,HOST]: 07Bh
	Reset: _instNCM0_alias[SMN,HOST]: 070h
	Reset: _instNCM1_alias[SMN,HOST]: 071h
	Reset: _instNCM2_alias[SMN,HOST]: 072h
	Reset: _instNCM3_alias[SMN,HOST]: 073h
	Reset: _instPIE0_alias[SMN,HOST]: 077h
	Reset: _inst[SPF[15:00]]_alias[SMN,HOST]: 000h
	Description: CCM0 component instance Fabric ID. This field may be updated by ASP through SMN after boot up if there are no pins allocated on the package to identify the socket and die. Allocation of bits in this field for socket, die, and component is specified in registers DF::SystemFabricIdMask0 and DF::SystemFabricIdMask1. Component instances of CNLI: TCDX and CAKE do not have Fabric IDs.

7:0	BlockInstanceID. Read-only. This component's Instance ID
	Reset: _instACM0_alias[SMN,HOST]: 18h
	Reset: _instACM1_alias[SMN,HOST]: 19h
	Reset: _instACM2_alias[SMN,HOST]: 1Ah
	Reset: _instACM3_alias[SMN,HOST]: 1Bh
	Reset: _instCCM0_alias[SMN,HOST]: 10h
	Reset: _instCCM1_alias[SMN,HOST]: 11h
	Reset: _instCCM2_alias[SMN,HOST]: 12h
	Reset: _instCCM3_alias[SMN,HOST]: 13h
	Reset: _instCCM4_alias[SMN,HOST]: 14h
	Reset: _instCCM5_alias[SMN,HOST]: 15h
	Reset: _instCCM6_alias[SMN,HOST]: 16h
	Reset: _instCCM7_alias[SMN,HOST]: 17h
	Reset: _instCMP0_alias[SMN,HOST]: 0Ch
	Reset: _instCMP1_alias[SMN,HOST]: 0Dh
	Reset: _instCMP2_alias[SMN,HOST]: 0Eh
	Reset: _instCMP3_alias[SMN,HOST]: 0Fh
	Reset: _instCNLI0_alias[SMN,HOST]: 35h
	Reset: _instCNLI1_alias[SMN,HOST]: 36h
	Reset: _instCNLI2_alias[SMN,HOST]: 37h
	Reset: _instCNLI3_alias[SMN,HOST]: 38h
	Reset: _instCS0_alias[SMN,HOST]: 00h
	Reset: _instCS1_alias[SMN,HOST]: 01h
	Reset: _instCS2_alias[SMN,HOST]: 02h
	Reset: _instCS3_alias[SMN,HOST]: 03h
	Reset: _instCS4_alias[SMN,HOST]: 04h
	Reset: _instCS5_alias[SMN,HOST]: 05h
	Reset: _instCS6_alias[SMN,HOST]: 06h
	Reset: _instCS7_alias[SMN,HOST]: 07h
	Reset: _instCS8_alias[SMN,HOST]: 08h
	Reset: _instCS9_alias[SMN,HOST]: 09h
	Reset: _instCS10_alias[SMN,HOST]: 0Ah
	Reset: _instCS11_alias[SMN,HOST]: 0Bh
	Reset: _instICNG0_alias[SMN,HOST]: 28h
	Reset: _instICNG1_alias[SMN,HOST]: 29h
	Reset: _instICNG2_alias[SMN,HOST]: 2Ah
	Reset: _instICNG3_alias[SMN,HOST]: 2Bh
	Reset: _instIOM0_alias[SMN,HOST]: 20h
	Reset: _instIOM1_alias[SMN,HOST]: 21h
	Reset: _instIOM2_alias[SMN,HOST]: 22h
	Reset: _instIOM3_alias[SMN,HOST]: 23h
	Reset: _instNCM0_alias[SMN,HOST]: 1Ch
	Reset: _instNCM1_alias[SMN,HOST]: 1Dh
	Reset: _instNCM2_alias[SMN,HOST]: 1Eh
	Reset: _instNCM3_alias[SMN,HOST]: 1Fh
	Reset: _instPIE0_alias[SMN,HOST]: 2Ch
	Reset: _instSPF0_alias[SMN,HOST]: 41h
	Reset: _instSPF1_alias[SMN,HOST]: 42h
	Reset: _instSPF2_alias[SMN,HOST]: 43h
	Reset: _instSPF3_alias[SMN,HOST]: 44h
	Reset: _instSPF4_alias[SMN,HOST]: 45h
	Reset: _instSPF5_alias[SMN,HOST]: 46h
	Reset: _instSPF6_alias[SMN,HOST]: 47h
	Reset: _instSPF7_alias[SMN,HOST]: 48h
	Reset: _instSPF8_alias[SMN,HOST]: 49h
	Reset: _instSPF9_alias[SMN,HOST]: 4Ah
	Reset: _instSPF10_alias[SMN,HOST]: 4Bh
	Reset: _instSPF11_alias[SMN,HOST]: 4Ch
	Reset: _instSPF12_alias[SMN,HOST]: 4Dh
	Reset: _instSPF13_alias[SMN,HOST]: 4Eh
	Reset: _instSPF14_alias[SMN,HOST]: 4Fh
	Reset: _instSPF15_alias[SMN,HOST]: 50h

D18F0x050 [Fabric Block Instance Information 3] (DF::FabricBlockInstanceInformation3_IOS)

This register provides the InstanceID and, when applicable, the FabricID of the block in which it is instanced. Not all the block instances require FabricID. See also DF::FabricBlockInstanceCount.

_instBCST_aliasHOST; D18F0x050

_inst[IOS[3:0]]_aliasHOST; D18F0x050_x2[7:4]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[7:4], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x014]

_instBCST_aliasSMN; D18F0x00000050; D18F0=4900_0000h

_inst[IOS[3:0]]_aliasSMN; D18F0x00000050_x2[7:4]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[7:4], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x014]

Bits	Description
31:20	Reserved.
19:8	BlockFabricID. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Reset: _instIOS0_alias[SMN,HOST]: 020h Reset: _instIOS1_alias[SMN,HOST]: 021h Reset: _instIOS2_alias[SMN,HOST]: 022h Reset: _instIOS3_alias[SMN,HOST]: 023h Description: CCM0 component instance Fabric ID. This field may be updated by ASP through SMN after boot up if there are no pins allocated on the package to identify the socket and die. Allocation of bits in this field for socket, die, and component is specified in registers DF::SystemFabricIdMask0 and DF::SystemFabricIdMask1. Component instances of CNLI: TCDX and CAKE do not have Fabric IDs.
7:0	BlockInstanceID. Read-only. This component's Instance ID Reset: _instIOS0_alias[SMN,HOST]: 24h Reset: _instIOS1_alias[SMN,HOST]: 25h Reset: _instIOS2_alias[SMN,HOST]: 26h Reset: _instIOS3_alias[SMN,HOST]: 27h

D18F0x050 [Fabric Block Instance Information 3] (DF::FabricBlockInstanceInformation3_TCDXPFXCAKE)

Read-only.

This register provides the InstanceID and, when applicable, the FabricID of the block in which it is instanced. Not all the block instances require FabricID. See also DF::FabricBlockInstanceCount.

_instBCST_aliasHOST; D18F0x050

_inst[TCDX[15:0],PFX[7:0],CAKE[7:0]]_aliasHOST; D18F0x050_x[[60:51],[40:39],[34:2D]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:51],[40:39],[34:2D]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x014]

_instBCST_aliasSMN; D18F0x00000050; D18F0=4900_0000h

_inst[TCDX[15:0],PFX[7:0],CAKE[7:0]]_aliasSMN; D18F0x00000050_x[[60:51],[40:39],[34:2D]]; D18F0=4900_0000h;
FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:51],[40:39],[34:2D]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x014]

Bits	Description
31:20	Reserved.
19:8	BlockFabricID. Read-only. Init: 11.5.3 [Register Values Written Based on Configuration]. Reset: _instBCST_alias[SMN,HOST]: Fixed,000h Reset: _inst[TCDX[15:00],PFX[7:0],CAKE[7:0]]_alias[SMN,HOST]: 000h Description: CCM0 component instance Fabric ID. This field may be updated by ASP through SMN after boot up if there are no pins allocated on the package to identify the socket and die. Allocation of bits in this field for socket, die, and component is specified in registers DF::SystemFabricIdMask0 and DF::SystemFabricIdMask1. Component instances of CNLI: TCDX and CAKE do not have Fabric IDs.
7:0	BlockInstanceID. Read-only. This component's Instance ID Reset: _instCAKE0_alias[SMN,HOST]: 2Dh Reset: _instCAKE1_alias[SMN,HOST]: 2Eh Reset: _instCAKE2_alias[SMN,HOST]: 2Fh Reset: _instCAKE3_alias[SMN,HOST]: 30h Reset: _instCAKE4_alias[SMN,HOST]: 31h Reset: _instCAKE5_alias[SMN,HOST]: 32h Reset: _instCAKE6_alias[SMN,HOST]: 33h Reset: _instCAKE7_alias[SMN,HOST]: 34h Reset: _instPFX0_alias[SMN,HOST]: 39h Reset: _instPFX1_alias[SMN,HOST]: 3Ah Reset: _instPFX2_alias[SMN,HOST]: 3Bh Reset: _instPFX3_alias[SMN,HOST]: 3Ch Reset: _instPFX4_alias[SMN,HOST]: 3Dh Reset: _instPFX5_alias[SMN,HOST]: 3Eh Reset: _instPFX6_alias[SMN,HOST]: 3Fh Reset: _instPFX7_alias[SMN,HOST]: 40h Reset: _instTCDX0_alias[SMN,HOST]: 51h Reset: _instTCDX1_alias[SMN,HOST]: 52h Reset: _instTCDX2_alias[SMN,HOST]: 53h Reset: _instTCDX3_alias[SMN,HOST]: 54h Reset: _instTCDX4_alias[SMN,HOST]: 55h Reset: _instTCDX5_alias[SMN,HOST]: 56h Reset: _instTCDX6_alias[SMN,HOST]: 57h Reset: _instTCDX7_alias[SMN,HOST]: 58h Reset: _instTCDX8_alias[SMN,HOST]: 59h Reset: _instTCDX9_alias[SMN,HOST]: 5Ah Reset: _instTCDX10_alias[SMN,HOST]: 5Bh Reset: _instTCDX11_alias[SMN,HOST]: 5Ch Reset: _instTCDX12_alias[SMN,HOST]: 5Dh Reset: _instTCDX13_alias[SMN,HOST]: 5Eh Reset: _instTCDX14_alias[SMN,HOST]: 5Fh Reset: _instTCDX15_alias[SMN,HOST]: 60h

D18F0x090 [DF Capability] (DF::DfCapability)

The Fabric Capabilities Register0 and the Capabilities Pointer Register have different purposes.

_instBCST_aliasHOST; D18F0x090

_instPIE0_aliasHOST; D18F0x090_x2C; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=0, IndCfgAccRegNum=0x024]

_instBCST_aliasSMN; D18F0x00000090; D18F0=4900_0000h

_instPIE0_aliasSMN; D18F0x00000090_x2C; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=0, IndCfgAccRegNum=0x024]

Bits	Description
31:2	Reserved.
1	Spf. Read-only, Volatile. 0=Probe filter support is not present on this processor. 1=This processor supports probe filtering. Probe filter capability. Reset: _instPIE0_alias[SMN,HOST]: 1
0	DataPoison. Read-only. 0=The processor does not support SUCCOR. 1=The processor supports software containment of uncorrectable errors through context synchronization data poisoning and deferred error interrupts. Software uncorrectable error containment and recovery capability (SUCCOR). Reset: _instBCST_alias[SMN,HOST]: Fixed,1 Reset: _instPIE0_alias[SMN,HOST]: 1

D18F0x100 [Enhanced Capability Header Function 0] (DF::EnhCapabilityHdr0)

Reset: 0000_0000h.

PCI Enhanced Capability Header register

_aliasHOST; D18F0x100

_aliasSMN; D18F0x00000100; D18F0=4900_0000h

Bits	Description
31:0	Reserved.

D18F0x384 [Secure Loader Configuration] (DF::SecureLoaderConfig)

Reset: 0000_0000h.

Secure Loader (SLDEV) configuration. During initialization of the secure kernel, the secure loader region (supporting SKINIT) must be protected from unauthorized access. This secure loader region is enabled by microcode setting up this register with SLRgnEnable = 1. Only accesses from CCMs are allowed when the region is enabled. When initialization completes, SLRgnEnable must be cleared to 0 to make the memory available for normal access; the secure load region does not exist when SLRgnEnable is zero.

Presence of this register is product specific. For products which do not instantiate this register, this offset is reserved.

_instBCST_aliasHOST; D18F0x384

_inst[CMP[3:0],CS[11:0]]_aliasHOST; D18F0x384_x0[F:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x0E1]

_instBCST_aliasSMN; D18F0x00000384; D18F0=4900_0000h

_inst[CMP[3:0],CS[11:0]]_aliasSMN; D18F0x00000384_x0[F:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x0E1]

Bits	Description
31:16	SLAddr. Reset: 0000h. Secure Loader Region Address [31:16]. Address [55:32] are defined to be 0. AccessType: SecurePart ? Read-only : Read-write.
15:1	Reserved.
0	SLRgnEnable. Reset: 0. 0=Secure Loader Region does not exist. 1=Secure Loader Region enabled. Secure Loader Region Enable AccessType: SecurePart ? Read-only : Read-write.

D18F0xC00 [Core Coherent Master Configuration Access Control] (DF::CoreMasterAccessCtrl)

Read-write. Reset: 0000_0000h.

Control bits for configuration accesses.

_instBCST_aliasHOST; D18F0xC00_inst[ACM[3:0],CCM[7:0]]_aliasHOST; D18F0xC00_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x300]_instBCST_aliasSMN; D18F0x00000C00; D18F0=4900_0000h_inst[ACM[3:0],CCM[7:0]]_aliasSMN; D18F0x00000C00_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x300]**Bits Description**

31:2	Reserved.
1	DisPciCfgReg. Read-write. Reset: 0. Disable CFC/CF8 Accesses to IO space. When set to 1, CFC/CF8 accesses are treated as PCI IO space accesses and not PCI configuration accesses.
0	EnableCf8ExtCfg. Read-write. Reset: 0. Enable PCI extended configuration register access for CFC/CF8 accesses.

D18F0xC04 [Config Address Control] (DF::CfgAddressCntl)

Read-write. Reset: 0000_0000h.

Configuration Address Control Register.

_instBCST_aliasHOST; D18F0xC04_inst[IOS[3:0]]_aliasHOST; D18F0xC04_x2[7:4]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[7:4], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x301]_instBCST_aliasSMN; D18F0x00000C04; D18F0=4900_0000h_inst[IOS[3:0]]_aliasSMN; D18F0x00000C04_x2[7:4]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[7:4], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x301]**Bits Description**

31:8	Reserved.
7:0	SecBusNum. Read-write. Reset: 00h. Specifies the configuration-space bus number of the IO Link.

D18F0xC08 [VGA Enable] (DF::VGAEn)

Read-write. Reset: 0000_0000h.

VGA Enable register for VGA compatibility space. [MMIO](#) Range A_0000 through B_FFFF. X86IO space from 3B0-3BB and 3C0-3DF.[_instBCST_aliasHOST](#); D18F0xC08[_inst\[ACM\[3:0\],CCM\[7:0\]\]_aliasHOST](#); D18F0xC08_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x302][_instBCST_aliasSMN](#); D18F0x00000C08; D18F0=4900_0000h[_inst\[ACM\[3:0\],CCM\[7:0\]\]_aliasSMN](#); D18F0x00000C08_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x302]

Bits	Description
31:16	Reserved.
15:4	VgaEn_DstFabricID . Read-write. Reset: 000h. For transactions which hit in this range, this field specifies the destination FabricID which may be local or on a remote die.
3	Reserved.
2	VgaEn_CpuDis . Read-write. Reset: 0. 1=DF::VGAEn[VgaEn_VE]-selected MMIO range is disregarded for CPU accesses (i.e., CPU accesses to this range are treated as if DF::VGAEn[VgaEn_VE]=0). VGA MMIO access by CPU control.
1	VgaEn_NP . Read-write. Reset: 0. 0=CPU write requests go through the downstream posted write channel. 1=CPU write requests to the MMIO range selected by DF::VGAEn[VgaEn_VE] go through the non-posted channel. VGA Enable register Non-Posted configuration bit. Steers CPU Write requests to posted or non-posted write channel.
0	VgaEn_VE . Read-write. Reset: 0. 0=Transactions targeting the VGA compatible address space are not affected by the state of this register. 1=Transactions targeting the VGA-compatible address space are routed and controlled as specified by this register. Description: VGA Enable. The VGA-compatible address space is: (1) the MMIO range A_0000h through B_FFFFh (2) IO space accesses in which address bits[9:0] range from 3B0h to 3BBh or 3C0h to 3DFh. Address bits[15:10] are not decoded; this only applies to the first 64 KB of IO space (i.e., address bits[24:16] must be zero.)

D18F0xC20 [Configuration Bus-0 Type-1 Destination Fabric ID Register-0]**(DF::CfgBus0Type1DstFabricIDReg0)**

Read-write. Reset: 0023_0077h.

Specifies the Destination Fabric ID (PIE/[FCH](#) IOMS) for Extended [PCIe](#) configuration accesses of Type-1 targeting Bus-0. Register-0 specifies the Destination Fabric IDs for Device numbers 24 and 25.[_instBCST_aliasHOST](#); D18F0xC20[_inst\[ACM\[3:0\],CCM\[7:0\]\]_aliasHOST](#); D18F0xC20_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x308][_instBCST_aliasSMN](#); D18F0x00000C20; D18F0=4900_0000h[_inst\[ACM\[3:0\],CCM\[7:0\]\]_aliasSMN](#); D18F0x00000C20_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x308]

Bits	Description
31:28	Reserved.
27:16	CfgBus0Type1_Dev25DstFabricID . Read-write. Reset: 023h. Specifies the destination for Device Number 25, which can be PIE or IOMS Fabric ID for the Node
15:12	Reserved.
11:0	CfgBus0Type1_Dev24DstFabricID . Read-write. Reset: 077h. Specifies the destination for Device Number 24, which can be PIE or IOMS Fabric ID for the Node

D18F0xC24 [Configuration Bus0 Type1 Destination Fabric ID Register-1] (DF::CfgBus0Type1DstFabricIDReg1)

Read-write. Reset: 0023_0023h.

Specifies the Destination Fabric ID (PIE/[FCH](#) IOMS) for Extended [PCIe](#) configuration accesses of Type-1 targeting Bus-0. Register-1 specifies the Destination Fabric IDs for Device numbers 26 and 27.`_instBCST_aliasHOST; D18F0xC24``_inst[ACM[3:0],CCM[7:0]]_aliasHOST; D18F0xC24_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x309]``_instBCST_aliasSMN; D18F0x00000C24; D18F0=4900_0000h``_inst[ACM[3:0],CCM[7:0]]_aliasSMN; D18F0x00000C24_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x309]`

Bits	Description
31:28	Reserved.
27:16	CfgBus0Type1_Dev27DstFabricID . Read-write. Reset: 023h. Specifies the destination for Device Number 27, which can be PIE or IOMS Fabric ID for the Node
15:12	Reserved.
11:0	CfgBus0Type1_Dev26DstFabricID . Read-write. Reset: 023h. Specifies the destination for Device Number 26, which can be PIE or IOMS Fabric ID for the Node

D18F0xC28 [Configuration Bus-0 Type-1 Destination Fabric ID Register-2] (DF::CfgBus0Type1DstFabricIDReg2)

Read-write. Reset: 0023_0023h.

Specifies the Destination Fabric ID (PIE/[FCH](#) IOMS) for Extended [PCIe](#) configuration accesses of Type-1 targeting Bus-0. Register-0 specifies the Destination Fabric IDs for Device numbers 28 and 29.`_instBCST_aliasHOST; D18F0xC28``_inst[ACM[3:0],CCM[7:0]]_aliasHOST; D18F0xC28_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x30A]``_instBCST_aliasSMN; D18F0x00000C28; D18F0=4900_0000h``_inst[ACM[3:0],CCM[7:0]]_aliasSMN; D18F0x00000C28_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x30A]`

Bits	Description
31:28	Reserved.
27:16	CfgBus0Type1_Dev29DstFabricID . Read-write. Reset: 023h. Specifies the destination for Device Number 29, which can be PIE or IOMS Fabric ID for the Node
15:12	Reserved.
11:0	CfgBus0Type1_Dev28DstFabricID . Read-write. Reset: 023h. Specifies the destination for Device Number 28, which can be PIE or IOMS Fabric ID for the Node

D18F0xC2C [Configuration Bus0 Type1 Destination Fabric ID Register-3] (DF::CfgBus0Type1DstFabricIDReg3)

Read-write. Reset: 0023_0023h.

Specifies the Destination Fabric ID (PIE/[FCH](#) IOMS) for Extended [PCIe](#) configuration accesses of Type-1 targeting Bus-0. Register-1 specifies the Destination Fabric IDs for Device numbers 30 and 31.`_instBCST_aliasHOST; D18F0xC2C``_inst[ACM[3:0],CCM[7:0]]_aliasHOST; D18F0xC2C_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x30B]``_instBCST_aliasSMN; D18F0x00000C2C; D18F0=4900_0000h``_inst[ACM[3:0],CCM[7:0]]_aliasSMN; D18F0x00000C2C_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x30B]`

Bits	Description
31:28	Reserved.
27:16	CfgBus0Type1_Dev31DstFabricID . Read-write. Reset: 023h. Specifies the destination for Device Number 31, which can be PIE or IOMS Fabric ID for the Node
15:12	Reserved.
11:0	CfgBus0Type1_Dev30DstFabricID . Read-write. Reset: 023h. Specifies the destination for Device Number 30, which can be PIE or IOMS Fabric ID for the Node

D18F0xC80...D18F0xCB8 [Configuration Base Address Maps] (DF::CfgBaseAddress)

Read-write. Reset: 0000_0000h.

Configuration Address Map registers specify mapping from configuration addresses to the corresponding FabricID of the IOS where they need to be routed. Configuration addresses are within the defined range if $DF::CfgBaseAddress[BusNumBase] \leq BusNo \leq DF::CfgLimitAddress[BusNumLimit]$ and match $SegmentNo == DF::CfgBaseAddress[SegmentNum]$. Configuration regions must not overlap. The appropriate RE or WE bit(s) must be set.

_instBCST_n[7:0]_aliasHOST; D18F0xC[[B:8][8,0]]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n0_aliasHOST; D18F0xC80_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x320]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n1_aliasHOST; D18F0xC88_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x322]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n2_aliasHOST; D18F0xC90_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x324]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n3_aliasHOST; D18F0xC98_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x326]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n4_aliasHOST; D18F0xCA0_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x328]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n5_aliasHOST; D18F0xCA8_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x32A]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n6_aliasHOST; D18F0xCB0_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x32C]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n7_aliasHOST; D18F0xCB8_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x32E]

_instBCST_n[7:0]_aliasSMN; D18F0x0000_0C[B8,B0,A8,A0,98,90,88,80]; D18F0=4900_0000h

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n0_aliasSMN; D18F0x0000C80_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x320]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n1_aliasSMN; D18F0x0000C88_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x322]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n2_aliasSMN; D18F0x0000C90_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x324]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n3_aliasSMN; D18F0x0000C98_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x326]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n4_aliasSMN; D18F0x0000CA0_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x328]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n5_aliasSMN; D18F0x0000CA8_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x32A]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n6_aliasSMN; D18F0x0000CB0_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x32C]

_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n7_aliasSMN; D18F0x0000CB8_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x32E]

Bits	Description
31:24	Reserved.
23:16	BusNumBase. Read-write. Reset: 00h. Bus Number Base Bits[7:0] for this configuration-space address map. See DF::CfgAddressCntl[SecBusNum].
15:8	SegmentNum. Read-write. Reset: 00h. Segment Number for this configuration-space address map.
7:2	Reserved.
1	WE. Read-write. Reset: 0. 1=Writes to this configuration-space address range are enabled. Write Enable.
0	RE. Read-write. Reset: 0. 1=Reads to this configuration-space address range are enabled. Read Enable.

D18F0xC84...D18F0xCBC [Configuration Address Map Control] (DF::CfgLimitAddress)

Read-write. Reset: 0000_0020h.

Configuration Address Map registers specify mapping from configuration addresses to the corresponding FabricID of the IOS where they need to be routed. Configuration addresses are within the defined range if $DF::CfgBaseAddress[BusNumBase] \leq BusNo \leq DF::CfgLimitAddress[BusNumLimit]$ and match $SegmentNo == DF::CfgBaseAddress[SegmentNum]$. Configuration regions must not overlap. The appropriate RE or WE bit(s) must be set.

_instBCST_n[7:0]_aliasHOST; D18F0xC[C:4]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n0_aliasHOST; D18F0xC84_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x321]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n1_aliasHOST; D18F0xC8C_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x323]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n2_aliasHOST; D18F0xC94_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x325]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n3_aliasHOST; D18F0xC9C_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x327]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n4_aliasHOST; D18F0xCA4_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x329]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n5_aliasHOST; D18F0xCAC_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x32B]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n6_aliasHOST; D18F0xCB4_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x32D]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n7_aliasHOST; D18F0xCBC_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x32F]_instBCST_n[7:0]_aliasSMN; D18F0x0000_0C[BC,B4,AC,A4,9C,94,8C,84]; D18F0=4900_0000h_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n0_aliasSMN; D18F0x0000C84_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x321]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n1_aliasSMN; D18F0x0000C8C_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x323]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n2_aliasSMN; D18F0x0000C94_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x325]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n3_aliasSMN; D18F0x0000C9C_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x327]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n4_aliasSMN; D18F0x0000CA4_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x329]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n5_aliasSMN; D18F0x0000CAC_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x32B]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n6_aliasSMN; D18F0x0000CB4_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x32D]_inst[IOM[3:0],ACM[3:0],CCM[7:0]]_n7_aliasSMN; D18F0x0000CBC_x[2[3:0],1[B:0]]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x32F]

Bits	Description
31:24	Reserved.
23:16	BusNumLimit. Read-write. Reset: 00h. Bus Number Limit Bits[7:0] for this IO Space.
15:12	Reserved.
11:0	DstFabricID. Read-write. Reset: 020h. For transactions which hit in this range, this field specifies the destination FabricID which may be local or on a remote die.

D18F0xD00...D18F0xD38 [IO Space Base Address] (DF::X86IOBaseAddress)

Read-write. Reset: 0000_0000h.

IO Space Base Address Registers specify the mapping from IO addresses to the corresponding FabricID of the IOHUB where they need to be routed. See DF::X86IOLimitAddress for Limit Address Register. These accesses are transactions resulting from x86-defined IN and OUT instructions. IO address ranges are specified by up to 8 sets of base/limit registers: DF::X86IOBaseAddress and DF::X86IOLimitAddress. Transaction addresses that are within the specified base/limit range are routed to the destination port specified by DF::X86IOLimitAddress[DstFabricID]. These requests use a fixed address for address bits [47:25] described in the Coherent HyperTransport Plus specification under System address map. Only lower address bits [24:12] are matched through the range registers.

IO mapping rules:

IO space transaction addresses are within the defined range if {DF::X86IOBaseAddress[IOBase], 000h} <= address <= {DF::X86IOLimitAddress[IOLimit], FFFh} and as specified by DF::X86IOBaseAddress[IE];

IO regions must not overlap.

The appropriate DF::X86IOBaseAddress[RE] or DF::X86IOBaseAddress[WE] bit(s) must be configured.

_instBCST_n[7:0]_aliasHOST; D18F0xD[3:0][8,0]

_inst[ACM[3:0],CCM[7:0]]_n0_aliasHOST; D18F0xD00_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x340]

_inst[ACM[3:0],CCM[7:0]]_n1_aliasHOST; D18F0xD08_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x342]

_inst[ACM[3:0],CCM[7:0]]_n2_aliasHOST; D18F0xD10_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x344]

_inst[ACM[3:0],CCM[7:0]]_n3_aliasHOST; D18F0xD18_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x346]

_inst[ACM[3:0],CCM[7:0]]_n4_aliasHOST; D18F0xD20_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x348]

_inst[ACM[3:0],CCM[7:0]]_n5_aliasHOST; D18F0xD28_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x34A]

_inst[ACM[3:0],CCM[7:0]]_n6_aliasHOST; D18F0xD30_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x34C]

_inst[ACM[3:0],CCM[7:0]]_n7_aliasHOST; D18F0xD38_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x34E]

_instBCST_n[7:0]_aliasSMN; D18F0x0000_0D[38,30,28,20,18,10,08,00]; D18F0=4900_0000h

_inst[ACM[3:0],CCM[7:0]]_n0_aliasSMN; D18F0x00000D00_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x340]

_inst[ACM[3:0],CCM[7:0]]_n1_aliasSMN; D18F0x00000D08_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x342]

_inst[ACM[3:0],CCM[7:0]]_n2_aliasSMN; D18F0x00000D10_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x344]

_inst[ACM[3:0],CCM[7:0]]_n3_aliasSMN; D18F0x00000D18_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x346]

_inst[ACM[3:0],CCM[7:0]]_n4_aliasSMN; D18F0x00000D20_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x348]

_inst[ACM[3:0],CCM[7:0]]_n5_aliasSMN; D18F0x00000D28_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x34A]

_inst[ACM[3:0],CCM[7:0]]_n6_aliasSMN; D18F0x00000D30_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x34C]

_inst[ACM[3:0],CCM[7:0]]_n7_aliasSMN; D18F0x00000D38_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x34E]

Bits	Description
31:29	Reserved.
28:16	IOBase. Read-write. Reset: 0000h. IO Base address[24:12] for this IO Space range.
15:6	Reserved.
5	IE. Read-write. Reset: 0. 0=The PCI IO space address window is not affected by this bit. 1=The PCI IO space address window is limited to the first 256KB of each 1KB block specified; this only applies to the first 64KB of IO space. PCI IO space address window ISA Enable.
4:2	Reserved.
1	WE. Read-write. Reset: 0. 1=Writes to this IO-space address range are enabled. Write Enable.
0	RE. Read-write. Reset: 0. 1=Reads to this IO-space address range are enabled. Read Enable.

D18F0xD04...D18F0xD3C [IO Space Limit Address] (DF::X86IOLimitAddress)

Read-write. Reset: 0000_0020h.

IO Space Limit Address Register. See DF::X86IOBaseAddress.

_instBCST_n[7:0]_aliasHOST; D18F0xD[[3:0][C,4]]

_inst[ACM[3:0],CCM[7:0]]_n0_aliasHOST; D18F0xD04_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x341]

_inst[ACM[3:0],CCM[7:0]]_n1_aliasHOST; D18F0xD0C_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x343]

_inst[ACM[3:0],CCM[7:0]]_n2_aliasHOST; D18F0xD14_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x345]

_inst[ACM[3:0],CCM[7:0]]_n3_aliasHOST; D18F0xD1C_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x347]

_inst[ACM[3:0],CCM[7:0]]_n4_aliasHOST; D18F0xD24_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x349]

_inst[ACM[3:0],CCM[7:0]]_n5_aliasHOST; D18F0xD2C_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x34B]

_inst[ACM[3:0],CCM[7:0]]_n6_aliasHOST; D18F0xD34_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x34D]

_inst[ACM[3:0],CCM[7:0]]_n7_aliasHOST; D18F0xD3C_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x34F]

_instBCST_n[7:0]_aliasSMN; D18F0x0000_0D[3C,34,2C,24,1C,14,0C,04]; D18F0=4900_0000h

_inst[ACM[3:0],CCM[7:0]]_n0_aliasSMN; D18F0x0000D04_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x341]

_inst[ACM[3:0],CCM[7:0]]_n1_aliasSMN; D18F0x0000D0C_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x343]

_inst[ACM[3:0],CCM[7:0]]_n2_aliasSMN; D18F0x0000D14_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x345]

_inst[ACM[3:0],CCM[7:0]]_n3_aliasSMN; D18F0x0000D1C_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x347]

_inst[ACM[3:0],CCM[7:0]]_n4_aliasSMN; D18F0x0000D24_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x349]

_inst[ACM[3:0],CCM[7:0]]_n5_aliasSMN; D18F0x0000D2C_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x34B]

_inst[ACM[3:0],CCM[7:0]]_n6_aliasSMN; D18F0x0000D34_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x34D]

_inst[ACM[3:0],CCM[7:0]]_n7_aliasSMN; D18F0x0000D3C_x1[B:0]; D18F0=4900_0000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=0, IndCfgAccRegNum=0x34F]

Bits	Description
31:29	Reserved.
28:16	IOLimit . Read-write. Reset: 0000h. IO Limit address[24:12] for this register's IO Space.
15:12	Reserved.
11:0	DstFabricID . Read-write. Reset: 020h. For transactions which hit in this IO Space range, this field specifies the destination FabricID which may be local to the die or on a remote die.

11.10.2 Function 1 Registers**D18F1x000 [Device/Vendor ID Function 1] (DF::DeviceVendorId1)**

Read-only. Reset: Fixed,14AE_1022h.

PCI Device/Vendor ID 1 register

_aliasHOST; D18F1x000

_aliasSMN; D18F1x00000000; D18F1=4900_1000h

Bits	Description
31:16	DeviceID . Read-only. Reset: Fixed,14AEh. Device ID
15:0	VendorID . Read-only. Reset: Fixed,1022h. Vendor ID

D18F1x004 [Status Command Function 1] (DF::StatusCommand1)

Read-only. Reset: Fixed,0000_0000h.

PCI Status/Command register

_aliasHOST; D18F1x004

_aliasSMN; D18F1x00000004; D18F1=4900_1000h

Bits	Description
31:16	Status. Read-only. Reset: Fixed,0000h. Status
15:0	Command. Read-only. Reset: Fixed,0000h. Command

D18F1x008 [Class Code Revision ID Function 1] (DF::ClassCodeRevId1)

Read-only. Reset: Fixed,0600_0000h.

PCI Class Code/Revision ID register

_aliasHOST; D18F1x008

_aliasSMN; D18F1x00000008; D18F1=4900_1000h

Bits	Description
31:8	ClassCode. Read-only. Reset: Fixed,06_0000h. Class Code. Provides the hostbridge class code as defined in the PCI specification.
7:0	RevisionId. Read-only. Reset: Fixed,00h. RevID: Revision ID.

D18F1x00C [Header Type Function 1] (DF::HeaderType1)

Read-only. Reset: Fixed,0080_0000h.

PCI Header Type register

_aliasHOST; D18F1x00C

_aliasSMN; D18F1x0000000C; D18F1=4900_1000h

Bits	Description
31:0	HeaderType. Read-only. Reset: Fixed,0080_0000h. Header Type: These bits are fixed at their default values. The header type field indicates that there are multiple functions present in this device.

D18F1x034 [Capabilities Pointer Function 1] (DF::CapabilitiesPtr1)

Read-only. Reset: Fixed,0000_0000h.

PCI Capabilities Pointer register

_aliasHOST; D18F1x034

_aliasSMN; D18F1x00000034; D18F1=4900_1000h

Bits	Description
31:8	Reserved.
7:0	CapPtr. Read-only. Reset: Fixed,00h. Capabilities Pointer.

D18F1x100 [Enhanced Capability Header Function 1] (DF::EnhCapabilityHdr1)

Reset: 0000_0000h.

PCI Enhanced Capability Header register

_aliasHOST; D18F1x100

_aliasSMN; D18F1x00000100; D18F1=4900_1000h

Bits	Description
31:0	Reserved.

D18F1x104 [CCD Enable Register] (DF::CCDEnable)

[CCD](#) Enable Register. This register has enable bit for each CCD connected to this CCM, and an enable bit for each [CCX](#) to support Dense CCDs

_instBCST_aliasHOST; D18F1x104

_inst[ACM[3:0],CCM[7:0]]_aliasHOST; D18F1x104_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=1, IndCfgAccRegNum=0x041]

_instBCST_aliasSMN; D18F1x00000104; D18F1=4900_1000h

_inst[ACM[3:0],CCM[7:0]]_aliasSMN; D18F1x00000104_x1[B:0]; D18F1=4900_1000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=1, IndCfgAccRegNum=0x041]

Bits	Description
31:18	Reserved.
17:16	CCXEn. Init: 11.5.3 [Register Values Written Based on Configuration]. For a Dense CCD , indicates which probe unit(s) are enabled. Used only if DfGlobalCtrl[DenseMode]==1 AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: Reserved AccessType: _inst[CCM[7:0],BCST]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write Reset: _inst[CCM[7:0],BCST]_alias[SMN,HOST]: 1h
15:2	Reserved.
1:0	CCDEn. Reset: 3h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies enabled CCDs. Bit position == 1: associated CCD enabled; Bit position == 0: associated CCD disabled.

D18F1x108 [UMC Channel Enable] (DF::UmcChannelEnable)

Reset: 0000_0000h.

Specifies whether UMCCH is enabled. This register is used by PIE to initiate SDP port clocking handshake. Various UMC configurations are known during DRAM training after [PSP](#) accesses SPD. This should combine memory fusing and DRAM presence detect, and match configurations programmed into UMC.

_instBCST_aliasHOST; D18F1x108

_instPIE0_aliasHOST; D18F1x108_x2C; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=1, IndCfgAccRegNum=0x042]

_instBCST_aliasSMN; D18F1x00000108; D18F1=4900_1000h

_instPIE0_aliasSMN; D18F1x00000108_x2C; D18F1=4900_1000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=1, IndCfgAccRegNum=0x042]

Bits	Description
31:12	Reserved.
11:0	UmcChnlEnable. Reset: 000h. AccessType: SecurePart ? Read-only : Read-write. Description: Each bit position associates with a UMCCH; e.g., bit 0 associates with UMCCH0. Each bit position set to 1 enables associated UMCCH. Each bit position set to 0 disables the associated UMCCH.

D18F1x114 [CCM SDP Enable] (DF::CcmSdpEnable)

Reset: 0000_0FFFh.

This register contains enable bits for each CCM SDP port in the system.

_instBCST_aliasHOST; D18F1x114_inst[CMP[3:0],CS[11:0]]_aliasHOST; D18F1x114_x0[F:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=1, IndCfgAccRegNum=0x045]_instBCST_aliasSMN; D18F1x00000114; D18F1=4900_1000h_inst[CMP[3:0],CS[11:0]]_aliasSMN; D18F1x00000114_x0[F:0]; D18F1=4900_1000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=1, IndCfgAccRegNum=0x045]

Bits	Description
31:16	Node1CcmSdpEn. Reset: 0000h. Init: 11.5.3 [Register Values Written Based on Configuration]. Enables for each CCM SDP port on Node offset 1 AccessType: SecurePart ? Read-only : Read-write.
15:0	Node0CcmSdpEn. Reset: 0FFFh. Init: 11.5.3 [Register Values Written Based on Configuration]. Enables for each CCM SDP port on Node offset 0 AccessType: SecurePart ? Read-only : Read-write.

D18F1x11C [ACM Enable] (DF::AcmEnable)

Reset: 0000_000Fh.

This register contains enable bits for each ACM in the system.

_instBCST_aliasHOST; D18F1x11C_inst[CMP[3:0],CS[11:0]]_aliasHOST; D18F1x11C_x0[F:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=1, IndCfgAccRegNum=0x047]_instBCST_aliasSMN; D18F1x0000011C; D18F1=4900_1000h_inst[CMP[3:0],CS[11:0]]_aliasSMN; D18F1x0000011C_x0[F:0]; D18F1=4900_1000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=1, IndCfgAccRegNum=0x047]

Bits	Description
31:20	Reserved.
19:16	Node1AcmEn. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Enables for each ACM on Node offset 1 AccessType: SecurePart ? Read-only : Read-write.
15:4	Reserved.
3:0	Node0AcmEn. Reset: Fh. Init: 11.5.3 [Register Values Written Based on Configuration]. Enables for each ACM on Node offset 0 AccessType: SecurePart ? Read-only : Read-write.

D18F1x3D0 [FTI Response Clumped VC Mask Register] (DF::FtiResponseClumpVCMask0)

Contains masks to specify which of the FTI Response VCs map to FTI Clumped Response VCs.

`_instBCST_aliasHOST; D18F1x3D0``_inst[TCDX[15:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F1x3D0_x[[60:51],[34:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:51],[34:00]], IndCfgAccFuncNum=1, IndCfgAccRegNum=0x0F4]``_instBCST_aliasSMN; D18F1x000003D0; D18F1=4900_1000h``_inst[TCDX[15:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F1x000003D0_x[[60:51],[34:00]]; D18F1=4900_1000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:51],[34:00]], IndCfgAccFuncNum=1, IndCfgAccRegNum=0x0F4]`**Bits Description**

31:25 Reserved.

24:16 **FtiRspVC1Mask**. Init: 11.5.3 [Register Values Written Based on Configuration]. Each bit position set to 1 maps the FTI Response VC at that bit's index to FTI Response VC1.

AccessType: SecurePart ? Read-only : Read-write.

Reset: `_inst[IOM[3:0],ICNG[3:0],CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 0ECh`Reset: `_inst[IOS[3:0]]_alias[SMN,HOST]: 0ECh`Reset: `_inst[PIE0,NCM[3:0]]_alias[SMN,HOST]: 0ECh`Reset: `_inst[TCDX[15:00]]_alias[SMN,HOST]: 0ECh`**ValidValues:**

Bit	Description
[0]	Maps FTI VC0 to Clumped VC1 for responses
[1]	Maps FTI VC1 to Clumped VC1 for responses
[2]	Maps FTI VC2 to Clumped VC1 for responses
[3]	Maps FTI VC3 to Clumped VC1 for responses
[4]	Maps FTI VC4 to Clumped VC1 for responses
[5]	Maps FTI VC5 to Clumped VC1 for responses
[6]	Maps FTI VC6 to Clumped VC1 for responses
[7]	Maps FTI VC7 to Clumped VC1 for responses
[8]	Maps FTI VC8 to Clumped VC1 for responses

15:9 Reserved.

8:0 **FtiRspVC0Mask**. Init: 11.5.3 [Register Values Written Based on Configuration]. Each bit position set to 1 maps the FTI Response VC at that bit's index to FTI Response VC0.

AccessType: SecurePart ? Read-only : Read-write.

Reset: `_inst[IOM[3:0],ICNG[3:0],CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 113h`Reset: `_inst[IOS[3:0]]_alias[SMN,HOST]: 113h`Reset: `_inst[PIE0,NCM[3:0]]_alias[SMN,HOST]: 113h`Reset: `_inst[TCDX[15:00]]_alias[SMN,HOST]: 113h`**ValidValues:**

Bit	Description
[0]	Maps FTI VC0 to Clumped VC0 for responses
[1]	Maps FTI VC1 to Clumped VC0 for responses
[2]	Maps FTI VC2 to Clumped VC0 for responses
[3]	Maps FTI VC3 to Clumped VC0 for responses
[4]	Maps FTI VC4 to Clumped VC0 for responses
[5]	Maps FTI VC5 to Clumped VC0 for responses
[6]	Maps FTI VC6 to Clumped VC0 for responses
[7]	Maps FTI VC7 to Clumped VC0 for responses
[8]	Maps FTI VC8 to Clumped VC0 for responses

11.10.3 Function 2 Registers

D18F2x000 [Device/Vendor ID Function 2] (DF::DeviceVendorId2)	
Read-only. Reset: Fixed,14AF_1022h.	
PCI Device/Vendor ID 2 register	
_aliasHOST; D18F2x000	
_aliasSMN; D18F2x00000000; D18F2=4900_2000h	
Bits	Description
31:16	DeviceID. Read-only. Reset: Fixed,14AFh. Device ID
15:0	VendorID. Read-only. Reset: Fixed,1022h. Vendor ID
D18F2x004 [Status Command Function 2] (DF::StatusCommand2)	
Read-only. Reset: Fixed,0000_0000h.	
PCI Status/Command register	
_aliasHOST; D18F2x004	
_aliasSMN; D18F2x00000004; D18F2=4900_2000h	
Bits	Description
31:16	Status. Read-only. Reset: Fixed,0000h. Status
15:0	Command. Read-only. Reset: Fixed,0000h. Command
D18F2x008 [Class Code Revision ID Function 2] (DF::ClassCodeRevId2)	
Read-only. Reset: Fixed,0600_0000h.	
PCI Class Code/Revision ID register	
_aliasHOST; D18F2x008	
_aliasSMN; D18F2x00000008; D18F2=4900_2000h	
Bits	Description
31:8	ClassCode. Read-only. Reset: Fixed,06_0000h. Class Code. Provides the hostbridge class code as defined in the PCI specification.
7:0	RevisionId. Read-only. Reset: Fixed,00h. RevID: Revision ID.
D18F2x00C [Header Type Function 2] (DF::HeaderType2)	
Read-only. Reset: Fixed,0080_0000h.	
PCI Header Type register	
_aliasHOST; D18F2x00C	
_aliasSMN; D18F2x0000000C; D18F2=4900_2000h	
Bits	Description
31:0	HeaderType. Read-only. Reset: Fixed,0080_0000h. Header Type: These bits are fixed at their default values. The header type field indicates that there are multiple functions present in this device.
D18F2x034 [Capabilities Pointer Function 2] (DF::CapabilitiesPtr2)	
Read-only. Reset: Fixed,0000_0000h.	
PCI Capabilities Pointer register	
_aliasHOST; D18F2x034	
_aliasSMN; D18F2x00000034; D18F2=4900_2000h	
Bits	Description
31:8	Reserved.
7:0	CapPtr. Read-only. Reset: Fixed,00h. Capabilities Pointer.

D18F2x068 [Coherent Slave Configuration B2] (DF::CoherentSlaveConfigB2)

This register contains configuration for the Coherent Slave that needs to remain in the Always ON power domain

_instBCST_aliasHOST; D18F2x068

_inst[COMP[3:0],CS[11:0]]_aliasHOST; D18F2x068_x0[F:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=2, IndCfgAccRegNum=0x01A]

_instBCST_aliasSMN; D18F2x00000068; D18F2=4900_2000h

_inst[COMP[3:0],CS[11:0]]_aliasSMN; D18F2x00000068_x0[F:0]; D18F2=4900_2000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=2, IndCfgAccRegNum=0x01A]

Bits	Description
31	WaitSuppMultReXmit. Reset: 1. AccessType: SecurePart ? Read-only : Read-write. Description: 1: Rest of DF doesn't support Multiple Retransmits on FTI. Temporary bit until CCM/IOM catch up. 0: Allow CS_TB to test multiple Retransmits on FTI
30	DisBeqAdpDutyCycle. Reset: 0. AccessType: SecurePart ? Read-only : Read-write. Description: 1: Disable Adaptive Duty Cycle for arbitrating between CSQ and BEQ for response picks. Uses a fixed 75:25 split CSQ:BEQ. 0: Enables Adaptive duty cycle
29	PfqProcChgToXNRSrcDn. Reset: 0. 1: Process ChgToXNR on SrcDn for SPF state correction. 0: Process ChgToXNR on Last Probe Response for SPF state correction. AccessType: SecurePart ? Read-only : Read-write. Init: ((DF::AcmEnable[Node0AcmEn] DF::AcmEnable[Node1AcmEn]) != 0) ? 1.
28	DisCdmaAdjLastSteerTag. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. 1: Disable the action of subtracting 1 from the Steering Tag for the highest valued Tag. Expected to be used when the highest numbered Dense CCD has CCX0 downbinned AccessType: SecurePart ? Read-only : Read-write.
27	DisRinseRspBeqXfer. Reset: 1. Init: 1. 1: Disallow a transaction that gets a Rinse Response to move to the BEQ AccessType: SecurePart ? Read-only : Read-write.
26	DisDngrdRspBeqXfer. Reset: 1. Init: 1. 1: Disallow a transaction that gets a Dngrd Response to move to the BEQ. Only applies when we create a CSQ downgrade based on the SPF Response Status AccessType: SecurePart ? Read-only : Read-write.
25	EnUnkwnRspBeqXfer. Reset: 0. 1: Allow Unknown Status SPF Responses to transfer writes to the BEQ 0: Do not allow Unknown Status SPF Response to transfer writes to BEQ while the count of such responses is non-zero AccessType: SecurePart ? Read-only : Read-write.
24:11	Reserved.
10	DisRmtTgtReqGO. Reset: 0. 1: Disable TgtReqGO to Remote Masters AccessType: SecurePart ? Read-only : Read-write.
9	DisACMTgtReqGO. Reset: 0. 1: Disable TgtReqGO to ACM AccessType: SecurePart ? Read-only : Read-write.
8	DisCCMTgtReqGO. Reset: 0. 1: Disable TgtReqGO to CCM AccessType: SecurePart ? Read-only : Read-write.
7	DropRinseOnLpfHit. Reset: 0. 1: Drop a rinse recommendation on an LPF Hit AccessType: SecurePart ? Read-only : Read-write.
6	DisDirPrbRinse. Reset: 0. 0: Allows Demote Super Probes to be Directed. 1: Forces Demote Super Probes to be Multicast AccessType: SecurePart ? Read-only : Read-write.
5	HoldDngrdForRinseRet. Reset: 1. 1: Holds the CSQ Downgrade Allocations until all Rinses that have finished in PFQ also finish in CSQ AccessType: SecurePart ? Read-only : Read-write.
4	DisProbeEligBankTok. Reset: 1. 0: Expands Bank Token requirements, whenever applicable to any Probe Eligible command instead of Downgrade Eligible Commands

	AccessType: SecurePart ? Read-only : Read-write.
3	DisUmcCancelPPFHitX . Reset: 1. Component CS0 instances: 1=Disable UMC Cancels on Probes issued for PPF Hits on X lines. CMP instances: Reserved and must be 1. AccessType: SecurePart ? Read-only : Read-write.
2	DisUmcCancelLPFHitX . Component instance CS0: 1=Disable UMC Cancels on Probes issued for LPF Hits on X lines. CMP instances: Reserved, and must be 1. AccessType: SecurePart ? Read-only : Read-write. Reset: <code>_instBCST_alias[SMN,HOST]: 0</code> Reset: <code>_inst[CMP[3:0]]_alias[SMN,HOST]: 1</code> Reset: <code>_inst[CS[11:00]]_alias[SMN,HOST]: 0</code> Init: <code>_inst[CS[11:00],BCST]_alias[SMN,HOST]: 1</code>
1	DisMultFtRexmit . Reset: 1. 1: Disable Multiple FTI Read Responses with RETRANSMIT Data Status. 0: Allow up to two FTI Read Responses with RETRANSMIT Data Status AccessType: SecurePart ? Read-only : Read-write.
0	DisTgtReqGO . Reset: 0. 1: Disable TgtReqGO. 0: Allow TgtReqGO to be issued AccessType: SecurePart ? Read-only : Read-write.

D18F2x100 [Enhanced Capability Header Function 2] (DF::EnhCapabilityHdr2)

Reset: 0000_0000h.

PCI Enhanced Capability Header register

`_aliasHOST; D18F2x100``_aliasSMN; D18F2x00000100; D18F2=4900_2000h`

Bits	Description
31:0	Reserved.

D18F2x8F0 [DRAM Error Injection Address Low] (DF::DramErrInjAddrLo)

Reset: 0000_0000h.

DRAM Error Injection Address Low. Software is expected to set up the Error injection vector (DF::DramErrInjAddrLo and DF::DramErrInjAddrHi) registers before triggering the command for error injection. Software may use WB memory type and set up data in a local cache before injecting errors. Hardware treats these addresses as coherent space and issues probe to copy back and invalidate cached copies. Software should poll DF::DramErrInjAddrLo[ErrInjDone] before it initiates error injection to another address. This register is locked and disabled via DF::DramErrInjAddrLo[LockAndDisErrInj].

`_instBCST_aliasHOST; D18F2x8F0``_instPIE0_aliasHOST; D18F2x8F0_x2C; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=2, IndCfgAccRegNum=0x23C]``_instBCST_aliasSMN; D18F2x000008F0; D18F2=4900_2000h``_instPIE0_aliasSMN; D18F2x000008F0_x2C; D18F2=4900_2000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=2, IndCfgAccRegNum=0x23C]`

Bits	Description
31:6	ErrInjAddrLo . Reset: 000_0000h. Error injection address low. Address bits[31:6] of the cache line where the error is injected. See DF::DramErrInjAddrHi[ErrInjAddrHi]. AccessType: SecurePart ? Read-only : Read-write.
5:3	Reserved.
2	LockAndDisErrInj . Reset: 0. Lock error injection widget. Once this field is set to 1, none of the fields in this register can be written. AccessType: SecurePart ? Read-only : Read-write.
1	ErrInjDone . Reset: 0. Set by hardware once the write is sent to CS. Software should poll this bit to ensure the error has been injected in the data. Software should clear this bit on a new request. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
0	ErrInjEn . Reset: 0. A new command to inject error is triggered by writing a 1. Software should clear DF::DramErrInjAddrLo[ErrInjDone] and DF::DramErrInjAddrLo[ErrInjEn] if there are no more errors to inject. AccessType: SecurePart ? Read-only : Read-write.

D18F2x8F4 [DRAM Error Injection Address High] (DF::DramErrInjAddrHi)

Reset: 0000_0000h.

DRAM Error Injection Address High.

_instBCST_aliasHOST; D18F2x8F4_instPIE0_aliasHOST; D18F2x8F4_x2C; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=2, IndCfgAccRegNum=0x23D]_instBCST_aliasSMN; D18F2x000008F4; D18F2=4900_2000h_instPIE0_aliasSMN; D18F2x000008F4_x2C; D18F2=4900_2000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=2, IndCfgAccRegNum=0x23D]**Bits Description**

31:24 Reserved.

23:0 **ErrInjAddrHi**. Reset: 00_0000h. Error injection address high. Address bits[55:32] of the cache line where the error is injected.

AccessType: SecurePart ? Read-only : Read-write.

D18F2x8FC [MCA Interrupt Control] (DF::McaIntCtrl)

Reset: 0000_0077h.

This register contains [MCA](#) Interrupt Controls._instBCST_aliasHOST; D18F2x8FC_inst[**CMP**[3:0],**CS**[11:0]]_aliasHOST; D18F2x8FC_x0[F:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=2, IndCfgAccRegNum=0x23F]_instBCST_aliasSMN; D18F2x000008FC; D18F2=4900_2000h_inst[**CMP**[3:0],**CS**[11:0]]_aliasSMN; D18F2x000008FC_x0[F:0]; D18F2=4900_2000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=2, IndCfgAccRegNum=0x23F]**Bits Description**

31:12 Reserved.

11:0 **McaIntPIEFabricID**. Reset: 077h. Init: 11.5.3 [Register Values Written Based on Configuration]. FabricID of the PIE which receives the [MCA](#) Interrupt Packet from CS. Reset to local PIE FabricID.

AccessType: SecurePart ? Read-only : Read-write.

11.10.4 Function 3 Registers**D18F3x000 [Device/Vendor ID Function 3] (DF::DeviceVendorId3)**

Read-only. Reset: Fixed,14B0_1022h.

PCI Device/Vendor ID 3 register

_aliasHOST; D18F3x000_aliasSMN; D18F3x00000000; D18F3=4900_3000h**Bits Description**31:16 **DeviceID**. Read-only. Reset: Fixed,14B0h. Device ID15:0 **VendorID**. Read-only. Reset: Fixed,1022h. Vendor ID**D18F3x004 [Status Command Function 3] (DF::StatusCommand3)**

Read-only. Reset: Fixed,0000_0000h.

PCI Status/Command register

_aliasHOST; D18F3x004_aliasSMN; D18F3x00000004; D18F3=4900_3000h**Bits Description**31:16 **Status**. Read-only. Reset: Fixed,0000h. Status15:0 **Command**. Read-only. Reset: Fixed,0000h. Command

D18F3x008 [Class Code Revision ID Function 3] (DF::ClassCodeRevId3)

Read-only. Reset: Fixed,0600_0000h.

PCI Class Code/Revision ID register

_aliasHOST; D18F3x008

_aliasSMN; D18F3x00000008; D18F3=4900_3000h

Bits	Description
31:8	ClassCode. Read-only. Reset: Fixed,06_0000h. Class Code. Provides the hostbridge class code as defined in the PCI specification.
7:0	RevisionId. Read-only. Reset: Fixed,00h. RevID: Revision ID.

D18F3x00C [Header Type Function 3] (DF::HeaderType3)

Read-only. Reset: Fixed,0080_0000h.

PCI Header Type register

_aliasHOST; D18F3x00C

_aliasSMN; D18F3x0000000C; D18F3=4900_3000h

Bits	Description
31:0	HeaderType. Read-only. Reset: Fixed,0080_0000h. Header Type: These bits are fixed at their default values. The header type field indicates that there are multiple functions present in this device.

D18F3x034 [Capabilities Pointer Function 3] (DF::CapabilitiesPtr3)

Read-only. Reset: Fixed,0000_0000h.

PCI Capabilities Pointer register

_aliasHOST; D18F3x034

_aliasSMN; D18F3x00000034; D18F3=4900_3000h

Bits	Description
31:8	Reserved.
7:0	CapPtr. Read-only. Reset: Fixed,00h. Capabilities Pointer.

D18F3x040 [DF Global Configuration] (DF::DfGlobalCfg)**Data Fabric Configuration Bits****_instBCST_aliasHOST; D18F3x040****_inst[TCDX[15:0],SPF[15:0],PFX[7:0],CNLI[3:0],CAKE[7:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F3x040_x[[60:2C],[27:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:2C],[27:00]], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x010]****_instBCST_aliasSMN; D18F3x00000040; D18F3=4900_3000h****_inst[TCDX[15:0],SPF[15:0],PFX[7:0],CNLI[3:0],CAKE[7:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F3x00000040_x[[60:2C],[27:00]]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:2C],[27:00]], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x010]****Bits Description****31:10 Reserved.****9 FrcFtiMscClkEnOn. Forces FMI ClkEn output signals on.**AccessType: **_inst[BCST,ACM[3:0]]_alias[SMN,HOST]: Reserved**AccessType: **_inst[CAKE[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[CCM[7:0]]_alias[SMN,HOST]: Reserved**AccessType: **_inst[PFX[7:0],NCM[3:0],IOS[3:0],IOM[3:0],CS[11:00],CNLI[3:0],CMP[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_instPIE0_alias[SMN,HOST]: Reserved**AccessType: **_inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**Reset: **_inst[TCDX[15:00],SPF[15:00],PFX[7:0],NCM[3:0],IOS[3:0],IOM[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CAKE[7:0]]_alias[SMN,HOST]: 0****8 DisEnhBusLock. Init: 1. When set, disables the Enhanced Bus Lock protocol.**AccessType: **_inst[BCST,ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[CAKE[7:0]]_alias[SMN,HOST]: Reserved**AccessType: **_inst[CCM[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[CS[11:00],CNLI[3:0],CMP[3:0]]_alias[SMN,HOST]: Reserved**AccessType: **_inst[IOM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[PFX[7:0],NCM[3:0],IOS[3:0]]_alias[SMN,HOST]: Reserved**AccessType: **_instPIE0_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: Reserved**Reset: **_inst[PIE0,IOM[3:0],CCM[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 0****7 DisTransErrRsp. Masters map all responses with TRANSERR to OKAY before SDP.**AccessType: **_inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[CAKE[7:0],BCST]_alias[SMN,HOST]: Reserved**AccessType: **_inst[CCM[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[CS[11:00],CNLI[3:0],CMP[3:0]]_alias[SMN,HOST]: Reserved**AccessType: **_inst[IOM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[IOS[3:0]]_alias[SMN,HOST]: Reserved**AccessType: **_inst[NCM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[TCDX[15:00],SPF[15:00],PIE0,PFX[7:0]]_alias[SMN,HOST]: Reserved**Reset: **_inst[NCM[3:0],IOM[3:0],CCM[7:0],ACM[3:0]]_alias[SMN,HOST]: 0****6 DisProtViolRsp. Masters map all responses with PROTBIOL to OKAY before SDP.**AccessType: **_inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[CAKE[7:0],BCST]_alias[SMN,HOST]: Reserved**AccessType: **_inst[CCM[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[CS[11:00],CNLI[3:0],CMP[3:0]]_alias[SMN,HOST]: Reserved**AccessType: **_inst[IOM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[IOS[3:0]]_alias[SMN,HOST]: Reserved**AccessType: **_inst[NCM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[TCDX[15:00],SPF[15:00],PIE0,PFX[7:0]]_alias[SMN,HOST]: Reserved**Reset: **_inst[NCM[3:0],IOM[3:0],CCM[7:0],ACM[3:0]]_alias[SMN,HOST]: 0**Init: **_inst[CCM[7:0],ACM[3:0]]_alias[SMN,HOST]: (MCA::LS::MCA_CTL_MASK_LS_ccd0_lthree0_core0[SystemReadDataErrorLoad] ||****MCA::LS::MCA_CTL_MASK_LS_ccd0_lthree0_core0[SystemReadDataErrorSch] ||****MCA::LS::MCA_CTL_MASK_LS_ccd0_lthree0_core0[SystemReadDataErrorWcb]) ? 1 : 0****5 DisSlvErrRsp. Masters map all responses with SLVERR to OKAY before SDP.**AccessType: **_inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[CAKE[7:0],BCST]_alias[SMN,HOST]: Reserved**AccessType: **_inst[CCM[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[CS[11:00],CNLI[3:0],CMP[3:0]]_alias[SMN,HOST]: Reserved**AccessType: **_inst[IOM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[IOS[3:0]]_alias[SMN,HOST]: Reserved**AccessType: **_inst[NCM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write**AccessType: **_inst[TCDX[15:00],SPF[15:00],PIE0,PFX[7:0]]_alias[SMN,HOST]: Reserved**Reset: **_inst[NCM[3:0],IOM[3:0],CCM[7:0],ACM[3:0]]_alias[SMN,HOST]: 0**

4	DisMstAbtRsp. Masters map all responses with DECERR to OKAY before SDP. AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _inst[CAKE[7:0],BCST]_alias[SMN,HOST]: Reserved AccessType: _inst[CCM[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _inst[CS[11:00],CNLI[3:0],CMP[3:0]]_alias[SMN,HOST]: Reserved AccessType: _inst[IOM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _inst[IOS[3:0]]_alias[SMN,HOST]: Reserved AccessType: _inst[NCM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _inst[TCDX[15:00],SPF[15:00],PIE0,PFX[7:0]]_alias[SMN,HOST]: Reserved Reset: _inst[NCM[3:0],IOM[3:0],CCM[7:0],ACM[3:0]]_alias[SMN,HOST]: 0 Init: _inst[NCM[3:0],IOM[3:0],CCM[7:0],ACM[3:0]]_alias[SMN,HOST]: 1
3	FrcPfxClkEnOn. Forces PFX ClkEn output signals on. AccessType: _inst[CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: Reserved AccessType: _inst[CMP[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _inst[CNLI[3:0]]_alias[SMN,HOST]: Reserved AccessType: _inst[CS[11:00]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _inst[NCM[3:0],IOS[3:0],IOM[3:0]]_alias[SMN,HOST]: Reserved AccessType: _inst[PFX[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _instPIE0_alias[SMN,HOST]: Reserved AccessType: _inst[SPF[15:00]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _inst[TCDX[15:00]]_alias[SMN,HOST]: Reserved Reset: _inst[SPF[15:00],PFX[7:0],CS[11:00],CMP[3:0]]_alias[SMN,HOST]: 0
2	FrcSdpClkEnOn. Reset: 0. Forces SDP ClkEn output signals on. AccessType: SecurePart ? Read-only : Read-write.
1	FrcFtiClkEnOn. Reset: 0. Forces FTI ClkEn output signals on. This field may only be reprogrammed after routing tables are programmed. AccessType: SecurePart ? Read-only : Read-write.
0	DatCacBitTransSampleDis. Reset: 0. Disables per-word data bus transition sampling at all Data Fabric Cac manager data sampling points. This downgrades data sampling to count just valid word transfers. AccessType: SecurePart ? Read-only : Read-write.

D18F3x044 [DF Global Control] (DF::DfGlobalCtrl)**Data Fabric Mode Bits**

_instBCST_aliasHOST; D18F3x044

_inst[TCDX[15:0],SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F3x044_x[[60:41],[38:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[38:00]], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x011]

_instBCST_aliasSMN; D18F3x00000044; D18F3=4900_3000h

_inst[TCDX[15:0],SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F3x00000044_x[[60:41],[38:00]]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[38:00]], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x011]

Bits Description

31 DenseMode. Init: 11.5.3 [Register Values Written Based on Configuration]. 1: [IOD](#) is interfaced to a Dense [CCD](#)
0: IOD is interfaced to a Classic CCD

AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _instBCST_alias[SMN,HOST]: Reserved

AccessType: _inst[CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[CNLI[3:0]]_alias[SMN,HOST]: Reserved

AccessType: _inst[ICNG[3:0],CS[11:00]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[NCM[3:0],IOS[3:0],IOM[3:0]]_alias[SMN,HOST]: Reserved

AccessType: _instPIE0_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: Reserved

Reset: _inst[PIE0,ICNG[3:0],CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 0

30:28 GNCMWDTCntSel. Data Fabric GCM and NCM Watchdog Timer Count select.

AccessType: _inst[CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[CMP[3:0]]_alias[SMN,HOST]: Reserved

AccessType: _inst[CNLI[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[CS[11:00]]_alias[SMN,HOST]: Reserved

AccessType: _inst[ICNG[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[IOM[3:0]]_alias[SMN,HOST]: Reserved

AccessType: _inst[SPF[15:00],PIE0,NCM[3:0],IOS[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[TCDX[15:00]]_alias[SMN,HOST]: Reserved

Reset: _inst[SPF[15:00],PIE0,NCM[3:0],IOS[3:0],ICNG[3:0],CNLI[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 6h

Init: _inst[SPF[15:00],PIE0,NCM[3:0],IOS[3:0],ICNG[3:0],CNLI[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: WdtControl

ValidValues:

Value	Description
0h	31
1h	127
2h	255
3h	511
4h	1023
5h	2047
6h	4095
7h	16383

27:21 Reserved.

20 DisWdtHwaSyndDbgBus. Reset: 0. 0=[DF](#) component drives HW Assert or watch dog debug information on debug bus when HW Assert or WDT error occurred. PIE [MCA](#) logs this information in MCA Bank. 1=Debug bus does not carry HW Assert or watch dog error information. Selects Debug bus information source on HW Assert/[WDT](#) error.

AccessType: SecurePart ? Read-only : Read-write.

19 Reserved.

18:16 CCMWDTCntSel. Data Fabric CCM Watchdog Timer Count select.

AccessType:

_inst[SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[TCDX[15:00]]_alias[SMN,HOST]: Reserved

Reset:

_inst[SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 6h

	Init: _inst[SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: WdtControl																		
	ValidValues:																		
	<table><tr><th>Value</th><th>Description</th></tr><tr><td>0h</td><td>31</td></tr><tr><td>1h</td><td>127</td></tr><tr><td>2h</td><td>255</td></tr><tr><td>3h</td><td>511</td></tr><tr><td>4h</td><td>1023</td></tr><tr><td>5h</td><td>2047</td></tr><tr><td>6h</td><td>4095</td></tr><tr><td>7h</td><td>16383</td></tr></table>	Value	Description	0h	31	1h	127	2h	255	3h	511	4h	1023	5h	2047	6h	4095	7h	16383
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15	Reserved.																		
14:12	IOMWDTCntSel. Data Fabric IOM Watchdog Timer Count select.																		
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10:8	PIEWDTCntSel. Data Fabric PIE SMN Watchdog Timer Count select.																		
	AccessType: _inst[SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write																		
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	Init: _inst[SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: WdtControl																		

	ValidValues:	
	Value	Description
	0h	31
	1h	127
	2h	255
	3h	511
	4h	1023
	5h	2047
	6h	4095
	7h	16383
7:6	Reserved.	
5:4	WDTBaseSel. Data Fabric Watchdog Timer timebase select. The rates provided in the table below are based on a 100 MHz REFCLK , and will scale accordingly based on REFCLK changes.	
	AccessType: _inst[SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write	
	AccessType: _inst[TCDX[15:00]]_alias[SMN,HOST]: Reserved	
	Reset: _inst[SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 0h	
	Init: _inst[SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: WdtControl	
	ValidValues:	
	Value	Description
	0h	1.31ms
	1h	1.28us
	2h	10ns
	3h	Timer Disabled
3	DisImmSyncFloodOnFatalErr. Reset: 1. Init: 0. 0=When a DF component detects a fatal error, it sends a sync packet on its outbound SDP link, sends a notification to PIE, and freezes its local queues. 1=When a DF component detects a fatal error, it only sends a notification to PIE and relies on PIE to initiate sync flood. Determines whether a DF component immediately posts a sync packet to its SDP port when it detects a fatal error or only notifies PIE.	
	AccessType: SecurePart ? Read-only : Read-write.	
2:1	Reserved.	
0	McastMaskPerNodeCLV. Init: 11.5.3 [Register Values Written Based on Configuration]. 0=FTI Mcast Mask has cluster valid and socket valid bits. Each socket which as the socket valid bits set, will have the same clusters probed as per the cluster valid bits. 1=FTI Mcast Mask has per socket cluster valid bits. This is currently only supported for up to 2 node. Controls whether Mcast Mask is interpreted as per-node cluster valid bits or the legacy view.	
	AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: Reserved	
	AccessType: _inst[CAKE[7:0],BCST]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write	
	AccessType: _inst[CCM[7:0]]_alias[SMN,HOST]: Reserved	
	AccessType: _inst[CMP[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write	
	AccessType: _inst[CNLI[3:0]]_alias[SMN,HOST]: Reserved	
	AccessType: _inst[CS[11:00]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write	
	AccessType: _inst[PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0]]_alias[SMN,HOST]: Reserved	
	AccessType: _inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write	
	Reset: _inst[TCDX[15:00],SPF[15:00],CS[11:00],CMP[3:0],CAKE[7:0],BCST]_alias[SMN,HOST]: 0	

D18F3x100 [Enhanced Capability Header Function 3] (DF::EnhCapabilityHdr3)

Reset: 0000_0000h.

PCI Enhanced Capability Header register

_aliasHOST; D18F3x100_aliasSMN; D18F3x00000100; D18F3=4900_3000h**Bits** **Description**

31:0 Reserved.

D18F3x104 [Interrupt Control] (DF::InterruptControl)

Reset: 0008_0700h.

Interrupt Control Register.

_instBCST_aliasHOST; D18F3x104

_inst[ICNG[3:0]]_aliasHOST; D18F3x104_x2[B:8]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x041]

_instBCST_aliasSMN; D18F3x00000104; D18F3=4900_3000h

_inst[ICNG[3:0]]_aliasSMN; D18F3x00000104_x2[B:8]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x041]

Bits Description

31:22 Reserved.

21:20 **ApicIdNumSkt.** Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specify total number of sockets in system. ICNG will broadcast interrupt probe with destination socket id more than this value

AccessType: SecurePart ? Read-only : Read-write.

ValidValues:

Value	Description
0h	Single socket system
1h	Two socket system
3h-2h	Reserved.

19:16 **ApicIdSktIdLsb.** Reset: 8h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specify Socket ID bit position in APIC ID.

AccessType: SecurePart ? Read-only : Read-write.

ValidValues:

Value	Description
3h-0h	Reserved.
4h	APIC_ID[5:4] represents socket id
5h	APIC_ID[6:5] represents socket id
6h	APIC_ID[7:6] represents socket id
7h	APIC_ID[8:7] represents socket id
8h	APIC_ID[9:8] represents socket id
Fh-9h	Reserved.

15:12 Reserved.

11 **SMTEnable.** Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Reflects the [SMT](#) configuration status for the system. This register field must be set before the CPU exits reset.

AccessType: SecurePart ? Read-only : Read-write.

10 **ApicExtSpur.** Reset: 1. 0=The lower 4 bits of the spurious vector are read-only 1111b. 1=The lower 4 bits of the spurious vector are writable. APIC extended spurious vector enable. This enables the extended APIC spurious vector functionality; it affects APICF0[Vector].

AccessType: SecurePart ? Read-only : Read-write.

9 **ApicExtId.** Reset: 1. 0=APIC ID is 4 bits. 1=APIC ID is 8 bits. Enable extended APIC ID size.

AccessType: SecurePart ? Read-only : Read-write.

8 **ApicExtBrdCst.** Reset: 1. 0=APIC broadcast is 0Fh. 1=APIC broadcast is FFh. If ApicExtBrdCst=1 then software must assert ApicExtId. Enable the extended APIC broadcast functionality.

AccessType: SecurePart ? Read-only : Read-write.

7:4 **ArbIntPreferLclDie.** Reset: 0h. Specifies in 1/8 granularity the percentage of die level arbitrations which chooses the local die if possible.

AccessType: SecurePart ? Read-only : Read-write.

	ValidValues:	
	Value	Description
	0h	Never prefer local die
	1h	Prefer local die in 1 out of 8 arbitrations.
	2h	Prefer local die in 2 out of 8 arbitrations.
	3h	Prefer local die in 3 out of 8 arbitrations.
	4h	Prefer local die in 4 out of 8 arbitrations.
	5h	Prefer local die in 5 out of 8 arbitrations.
	6h	Prefer local die in 6 out of 8 arbitrations.
	7h	Prefer local die in 7 out of 8 arbitrations.
	8h	Prefer local die in all arbitrations.
	Fh-9h	Reserved.
3	LintEn. Reset: 0. 0=ExtInt/NMI interrupts delivered unchanged. 1=Enable the conversion of broadcast ExtInt and NMI interrupt requests to LINT0 and LINT1 local interrupts, respectively, before delivering to the local APIC. This conversion only takes place if the local APIC is hardware enabled. LINT0 and LINT1 are controlled by APIC3[60:50]. Enable conversion of ExtInt/NMI to LINT0/1. AccessType: SecurePart ? Read-only : Read-write.	
2:0	ApcSwizzleType. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Selects CPU thread number to APIC number mapping (Swizzling). AccessType: SecurePart ? Read-only : Read-write.	
	ValidValues:	
	Value	Description
	0h	No Swizzle (N[6:0] to N[6:0] Mapping)
	1h	N[6:0] will map to {N[2:1],N[6:3],N[0]}
	2h	N[6:0] will map to {N[3:2],N[6:4],N[1:0]}
	3h	N[6:0] will map to {N[4:3],N[6:5],N[2:0]}
	4h	N[6:0] will map to {N[5:4],N[6],N[3:0]}
	5h	Illegal
	6h	Illegal
	7h	Illegal

D18F3x114 [Interrupt Configuration 1] (DF::InterruptConfig1)

Reset: 0200_04C0h.

Interrupt Configuration Register 1.

_instBCST_aliasHOST; D18F3x114

_inst[ICNG[3:0]]_aliasHOST; D18F3x114_x2[B:8]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x045]

_instBCST_aliasSMN; D18F3x00000114; D18F3=4900_3000h

_inst[ICNG[3:0]]_aliasSMN; D18F3x00000114_x2[B:8]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x045]

Bits	Description
31:28	Reserved.
27:23	APCPassICISatLmt. Reset: 04h. Maximum number of APIC register picked (across all APCTOPS's) when any ICI request is pending in PRQ. Counter reach this threshold, APIC register access will wait for all APCTOP interface to be idle, so that older ICI request (if pending) can be picked. Counter gets cleared when any ICI request gets picked or APIC register access gets picked and counter is already saturated. AccessType: SecurePart ? Read-only : Read-write.
22	DisSmuApicAccess. Reset: 0. 0=SMU/PSP can access APIC register space via SDP. 1=SMU/PSP access to APIC register space is disabled. Disables SMU/ PSP access to APIC register space through SDP/FTI. AccessType: SecurePart ? Read-only : Read-write.
21	Reserved.
20	DisMultAPCParallelPick. Reset: 0. Disables APIC register pick for different APCTOP in parallel. AccessType: SecurePart ? Read-only : Read-write.
19	UseApcDoneAsAck. Reset: 0. Treat Done from APCTOP for register access as Ack. So PRQ will not be dispatching new request to APCTOP when SelfIPI write is complete in APIC but waiting for interrupt edge delivery to CPU. AccessType: SecurePart ? Read-only : Read-write.
18	Reserved.
17	IntBcstOptDis. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Disables interrupt broadcast optimization for X2APIC AccessType: SecurePart ? Read-only : Read-write.
16	EnAVICDlvryInterlock. Reset: 0. 0=Edge Delivery interlock for AVIC Doorbell is disabled. 1=Edge Delivery interlock for AVIC Doorbell is enabled. Enable Delivery interlock for AVIC Doorbell. AccessType: SecurePart ? Read-only : Read-write.
15:12	Reserved.
11	DisClrVectorEdge. Reset: 0. 0=Interrupt logic sends Clear Vector Probe when it gets higher priority interrupt and older is pending with core. 1=Interrupt logic does not send Clear Vector Probe when it gets higher priority interrupt and older is pending with core. Disable Clear Vector probe generation. AccessType: SecurePart ? Read-only : Read-write.
10	DisLegacyIntAck. Reset: 1. 0=Vector Read change interrupt state from pending to in-service. 1=Vector Read does not have any side effect. Disable Legacy IntAck Flow. AccessType: SecurePart ? Read-only : Read-write.
9:8	Reserved.
7	AdvanceRRArbOnPostArbFixInt. Reset: 1. 0=Arbiter is advanced during arbitration phase. 1=Arbiter is advanced on post arbitration Fixed interrupt. Specifies if round robin arbiter for LPA interrupt is advanced during arbitration phase or on post arbitration Fixed interrupt AccessType: SecurePart ? Read-only : Read-write.
6	PrefInterlockedThrd. Reset: 1. 0=All Interrupt edge are send using round robin pick for all threads and fixed priority for different edge types. 1=Interrupt edge participating in interrupt delivery interlock is preferred. Specifies whether interrupt edge participating in interrupt delivery interlock should be preferred over other interrupt edges AccessType: SecurePart ? Read-only : Read-write.

5	DisPriorFocusProcessor . Reset: 0. 0=A processor that is previously ran same ISR is preferred. 1=Previously ran LPA vector does not affect arbitration. Specifies whether or not a processor that previously ran the same ISR is preferred AccessType: SecurePart ? Read-only : Read-write.										
4	ForceLpaArbWhenAllInCc6 . Reset: 0. 0=Allows local arbitration. 1=Force system wide arbitration. Specifies LP interrupt arbitration behavior when all cores in PIE/ICNG cluster are in CC6. AccessType: SecurePart ? Read-only : Read-write.										
3:2	ArbIntrLclClusterMode . Reset: 0h. Local cluster arbitration mode. AccessType: SecurePart ? Read-only : Read-write. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Local cluster mode is used as long as we find a non-zero APR within this cluster.</td></tr> <tr> <td>1h</td><td>local cluster mode is used all the time for every single LPA.</td></tr> <tr> <td>2h</td><td>Reserved.</td></tr> <tr> <td>3h</td><td>Local cluster mode is disabled. This is the master chicken bit for the entire field.</td></tr> </table>	Value	Description	0h	Local cluster mode is used as long as we find a non-zero APR within this cluster.	1h	local cluster mode is used all the time for every single LPA.	2h	Reserved.	3h	Local cluster mode is disabled. This is the master chicken bit for the entire field.
Value	Description										
0h	Local cluster mode is used as long as we find a non-zero APR within this cluster.										
1h	local cluster mode is used all the time for every single LPA.										
2h	Reserved.										
3h	Local cluster mode is disabled. This is the master chicken bit for the entire field.										
1	DisMCEDeliveryOnFataError . Reset: 0. 0=MCE delivery to the CPU is Enabled. 1=MCE delivery to the CPU is disabled if a DF fatal error has been detected. Disable MCE delivery to the CPU if a DF fatal error has been detected. AccessType: SecurePart ? Read-only : Read-write.										
0	DisMCEDelivery . Reset: 0. 0=MCE delivery to the CPU is Enabled. 1=MCE delivery to the CPU is Disabled. Disable MCE delivery to the CPU AccessType: SecurePart ? Read-only : Read-write.										

D18F3x150 [APIC Cluster to ICNG Map] (DF::ApicClusterIcngMap0)

APIC Cluster to ICNG Map. Logical APIC ID [9:4] bits will be used to lookup this table

_instBCST_aliasHOST; D18F3x150_inst[ICNG[3:0]]_aliasHOST; D18F3x150_x2[B:8]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x054]_instBCST_aliasSMN; D18F3x00000150; D18F3=4900_3000h_inst[ICNG[3:0]]_aliasSMN; D18F3x00000150_x2[B:8]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x054]

Bits	Description
31:30	ApicCluster15IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 15. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
29:28	ApicCluster14IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 14. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
27:26	ApicCluster13IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 13. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
25:24	ApicCluster12IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 12. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
23:22	ApicCluster11IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 11. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
21:20	ApicCluster10IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 10. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
19:18	ApicCluster9IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 9. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h

17:16	ApicCluster8IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 8.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
15:14	ApicCluster7IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 7.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
13:12	ApicCluster6IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 6.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
11:10	ApicCluster5IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 5.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
9:8	ApicCluster4IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 4.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
7:6	ApicCluster3IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 3.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
5:4	ApicCluster2IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 2.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
3:2	ApicCluster1IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 1.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
	Reset: _instICNG3_alias[SMN,HOST]: 3h

1:0	ApicCluster0IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 0.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
	Reset: _instICNG3_alias[SMN,HOST]: 3h

D18F3x154 [APIC Cluster to ICNG Map] (DF::ApicClusterIcngMap1)

APIC Cluster to ICNG Map. Logical APIC ID [9:4] bits will be used to lookup this table

_instBCST_aliasHOST; D18F3x154_inst[ICNG[3:0]]_aliasHOST; D18F3x154_x2[B:8]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x055]_instBCST_aliasSMN; D18F3x00000154; D18F3=4900_3000h_inst[ICNG[3:0]]_aliasSMN; D18F3x00000154_x2[B:8]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x055]

Bits	Description
31:30	ApicCluster31IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 31. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
29:28	ApicCluster30IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 30. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
27:26	ApicCluster29IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 29. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
25:24	ApicCluster28IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 28. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
23:22	ApicCluster27IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 27. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
21:20	ApicCluster26IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 26. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
19:18	ApicCluster25IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 25. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h

17:16	ApicCluster24IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 24.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
15:14	ApicCluster23IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 23.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
13:12	ApicCluster22IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 22.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
11:10	ApicCluster21IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 21.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
9:8	ApicCluster20IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 20.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
7:6	ApicCluster19IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 19.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
5:4	ApicCluster18IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 18.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
3:2	ApicCluster17IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 17.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
	Reset: _instICNG3_alias[SMN,HOST]: 3h

1:0	ApicCluster16IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 16.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
	Reset: _instICNG3_alias[SMN,HOST]: 3h

D18F3x158 [APIC Cluster to ICNG Map] (DF::ApicClusterIcngMap2)

APIC Cluster to ICNG Map. Logical APIC ID [9:4] bits will be used to lookup this table

_instBCST_aliasHOST; D18F3x158_inst[ICNG[3:0]]_aliasHOST; D18F3x158_x2[B:8]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x056]_instBCST_aliasSMN; D18F3x00000158; D18F3=4900_3000h_inst[ICNG[3:0]]_aliasSMN; D18F3x00000158_x2[B:8]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x056]

Bits	Description
31:30	ApicCluster47IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 47. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
29:28	ApicCluster46IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 46. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
27:26	ApicCluster45IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 45. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
25:24	ApicCluster44IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 44. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
23:22	ApicCluster43IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 43. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
21:20	ApicCluster42IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 42. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
19:18	ApicCluster41IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 41. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h

17:16	ApicCluster40IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 40.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
15:14	ApicCluster39IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 39.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
13:12	ApicCluster38IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 38.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
11:10	ApicCluster37IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 37.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
9:8	ApicCluster36IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 36.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
7:6	ApicCluster35IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 35.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
5:4	ApicCluster34IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 34.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
3:2	ApicCluster33IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 33.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
	Reset: _instICNG3_alias[SMN,HOST]: 3h

1:0	ApicCluster32IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 32.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
	Reset: _instICNG3_alias[SMN,HOST]: 3h

D18F3x15C [APIC Cluster to ICNG Map] (DF::ApicClusterIcngMap3)

APIC Cluster to ICNG Map. Logical APIC ID [9:4] bits will be used to lookup this table

_instBCST_aliasHOST; D18F3x15C_inst[ICNG[3:0]]_aliasHOST; D18F3x15C_x2[B:8]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x057]_instBCST_aliasSMN; D18F3x0000015C; D18F3=4900_3000h_inst[ICNG[3:0]]_aliasSMN; D18F3x0000015C_x2[B:8]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x057]

Bits	Description
31:30	ApicCluster63IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 63. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
29:28	ApicCluster62IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 62. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
27:26	ApicCluster61IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 61. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
25:24	ApicCluster60IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 60. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
23:22	ApicCluster59IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 59. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
21:20	ApicCluster58IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 58. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h
19:18	ApicCluster57IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 57. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instICNG0_alias[SMN,HOST]</u> : 0h Reset: <u>_instICNG1_alias[SMN,HOST]</u> : 1h Reset: <u>_instICNG2_alias[SMN,HOST]</u> : 2h Reset: <u>_instICNG3_alias[SMN,HOST]</u> : 3h

17:16	ApicCluster56IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 56.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
15:14	ApicCluster55IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 55.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
13:12	ApicCluster54IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 54.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
11:10	ApicCluster53IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 53.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
9:8	ApicCluster52IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 52.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
7:6	ApicCluster51IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 51.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
5:4	ApicCluster50IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 50.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
3:2	ApicCluster49IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 49.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
	Reset: _instICNG3_alias[SMN,HOST]: 3h

1:0	ApicCluster48IcngId. Init: 11.5.3 [Register Values Written Based on Configuration]. Instance number of ICNG block servicing logical APIC cluster 48.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instICNG0_alias[SMN,HOST]: 0h
	Reset: _instICNG1_alias[SMN,HOST]: 1h
	Reset: _instICNG2_alias[SMN,HOST]: 2h
	Reset: _instICNG3_alias[SMN,HOST]: 3h

D18F3x180 [PIE Configuration] (DF::PIEConfig)

Reset: 0000_0000h.

PIE Configuration Register 0.

_instBCST_aliasHOST; D18F3x180

_instPIE0_aliasHOST; D18F3x180_x2C; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=3, IndCfgAccRegNum=0x060]

_instBCST_aliasSMN; D18F3x00000180; D18F3=4900_3000h

_instPIE0_aliasSMN; D18F3x00000180_x2C; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=3, IndCfgAccRegNum=0x060]

Bits	Description
31:20	Reserved.
19	DisSyncFloodProp. Reset: 0. 0=When PIE detects a sync flood condition, it propagates sync flood to all DF components. 1=Disable propagating sync flood to all DF components. Configure sync flood propagation.
	AccessType: SecurePart ? Read-only : Read-write.
18:4	Reserved.
3	EnMcaWakeOnSmnAcc. Reset: 0. 0=PIE returns Slave error during DF Cstate. 1=MCA Access triggers DF Cstate exit, register access is served once DF Cstate exit complete. Specify PIE behavior on MCA register access from SMN during DF Cstate
	AccessType: SecurePart ? Read-only : Read-write.
2:0	Reserved.

D18F3x1A0 [PIE Control] (DF::PIEControl)

Reset: 0000_0000h.

PIE Control Register.

_instBCST_aliasHOST; D18F3x1A0

_instPIE0_aliasHOST; D18F3x1A0_x2C; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=3, IndCfgAccRegNum=0x068]

_instBCST_aliasSMN; D18F3x000001A0; D18F3=4900_3000h

_instPIE0_aliasSMN; D18F3x000001A0_x2C; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=3, IndCfgAccRegNum=0x068]

Bits	Description
31	DvmCompBcstIOSEn. Reset: 0. Init: 1. Enables IOS in the DVM Completion Bcst McMask.
	AccessType: SecurePart ? Read-only : Read-write.
30	DvmSyncBcstCCMEEn. Reset: 0. Init: 1. Enables CCM in the DVM Sync Bcst McMask and in the expected completion count using the SysMgt Join/Leave count.
	AccessType: SecurePart ? Read-only : Read-write.
29	DvmSyncBcstIOSEn. Reset: 0. Init: 1. Enables IOS in the DVM Sync Bcst McMask and in the expected completion count using SystemComponentCnt[IOMCount].
	AccessType: SecurePart ? Read-only : Read-write.
28	DVMOpBcstIOSEn. Reset: 0. Init: 1. Enables IOS in the DVM Op Broadcast McMask.
	AccessType: SecurePart ? Read-only : Read-write.
27	Reserved.
26:24	MCADomainSelect. Reset: 0h. Selects the MCA Domain for the PIE MCA Bank. Interrupts will be delivered to the MDOMRegisterMDOM
	AccessType: SecurePart ? Read-only : Read-write.
23:0	Reserved.

D18F3x620 [CAKE Virtual Channel Remapping Register] (DF::CakeVCRemap)

Reset: 0000_0000h.

Specifies Remote to Local and Remote to Remote VC remapping for both requests and responses.

_instBCST_aliasHOST; D18F3x620_inst[CAKE[7:0]]_aliasHOST; D18F3x620_x[34:2D]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x188]_instBCST_aliasSMN; D18F3x00000620; D18F3=4900_3000h_inst[CAKE[7:0]]_aliasSMN; D18F3x00000620_x[34:2D]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x188]

Bits	Description
31	ReqVCRemapEn. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. 1=Request Virtual Channel remapping is enabled. 0=Request Virtual Channel remapping is disabled. AccessType: SecurePart ? Read-only : Read-write.
30	RspVCRemapEn. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. 1=Response Virtual Channel remapping is enabled. 0=Response Virtual Channel remapping is disabled. AccessType: SecurePart ? Read-only : Read-write.
29:22	Reserved.
21	VC8RemapEn. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. 1=Include VC8 in Virtual Channel remapping. 0=Exclude VC8 from Virtual Channel remapping. AccessType: SecurePart ? Read-only : Read-write.
20	VC1RemapEn. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. 1=Include VC1 in Virtual Channel remapping. 0=Exclude VC1 from Virtual Channel remapping. AccessType: SecurePart ? Read-only : Read-write.
19	VC0RemapEn. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. 1=Include VC0 in Virtual Channel remapping. 0=Exclude VC0 from Virtual Channel remapping. AccessType: SecurePart ? Read-only : Read-write.
18:15	RspR2RVcRemap. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Remote to Remote remap VC for responses. When RspVCRemapEn is set Cake will change the response VC to RspR2RVcRemap if the destination is not the neighbor node. AccessType: SecurePart ? Read-only : Read-write.
14	Reserved.
13:10	RspR2LVcRemap. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Remote to Local remap VC for responses. When RspVCRemapEn is set Cake will change the response VC to RspR2LVcRemap if the destination is the neighbor node. AccessType: SecurePart ? Read-only : Read-write.
9	Reserved.
8:5	ReqR2RVcRemap. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Remote to Remote remap VC for requests. When ReqVCRemapEn is set Cake will change the request VC to ReqR2RVcRemap if the destination is not the neighbor node. AccessType: SecurePart ? Read-only : Read-write.
4	Reserved.
3:0	ReqR2LVcRemap. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Remote to Local remap VC for requests. When ReqVCRemapEn is set Cake will change the request VC to ReqR2LVcRemap if the destination is the neighbor node. AccessType: SecurePart ? Read-only : Read-write.

D18F3x62C [CAKE Virtual Channel Remapping Register 1] (DF::CakeVCRemap1)

Reset: 0000_0000h.

Specifies Additional Remote to Local and Remote to Remote VC remapping for both requests and responses.

_instBCST_aliasHOST; D18F3x62C_inst[CAKE[7:0]]_aliasHOST; D18F3x62C_x[34:2D]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x18B]_instBCST_aliasSMN; D18F3x0000062C; D18F3=4900_3000h_inst[CAKE[7:0]]_aliasSMN; D18F3x0000062C_x[34:2D]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x18B]

Bits	Description
31	SrcRspVCRemapEn. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. 1=Source response remapping is enabled. 0=Source response remapping is disabled. Source response remapping has precedence over RspVCRemapEn/VC8RemapEn/VC1RemapEn/VC0RemapEn behavior. AccessType: SecurePart ? Read-only : Read-write.
30	PrbRspVCRemapEn. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. 1=Probe response remapping is enabled. 0=Probe response remapping is disabled. Probe response remapping has precedence over RspVCRemapEn/VC8RemapEn/VC1RemapEn/VC0RemapEn behavior. AccessType: SecurePart ? Read-only : Read-write.
29:9	Reserved.
8:5	SrcRspVcRemap. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Remap VC for source responses. When SrcRspVCRemapEn is set Cake will change all source response VCs to PrbSrcRspVcRemap. AccessType: SecurePart ? Read-only : Read-write.
4	Reserved.
3:0	PrbRspVcRemap. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Remap VC for probe responses. When PrbRspVCRemapEn is set Cake will change all probe response VCs to PrbSrcRspVcRemap. AccessType: SecurePart ? Read-only : Read-write.

D18F3x8F0 [Hardware Assert Status Low] (DF::HardwareAssertStatusLow)

Reset: Cold,0000_0000h.

The lower 32 hardware assertion status bits. Each data fabric component type maps its own assertions to this register.

_instBCST_aliasHOST; D18F3x8F0_inst[TCDX[15:0],SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F3x8F0_x[[60:41],[38:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[38:00]], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x23C]_instBCST_aliasSMN; D18F3x000008F0; D18F3=4900_3000h_inst[TCDX[15:0],SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F3x000008F0_x[[60:41],[38:00]]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[38:00]], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x23C]

Bits	Description
31	HWAssert31. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 31. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
30	HWAssert30. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 30. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
29	HWAssert29. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 29. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
28	HWAssert28. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 28. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
27	HWAssert27. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 27. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
26	HWAssert26. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 26. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
25	HWAssert25. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 25. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
24	HWAssert24. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 24. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
23	HWAssert23. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 23. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
22	HWAssert22. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 22. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
21	HWAssert21. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 21. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
20	HWAssert20. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 20. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
19	HWAssert19. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 19. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
18	HWAssert18. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 18. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
17	HWAssert17. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 17. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
16	HWAssert16. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 16. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
15	HWAssert15. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 15. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
14	HWAssert14. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 14. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
13	HWAssert13. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 13. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
12	HWAssert12. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 12.

	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
11	HWAssert11. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 11.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
10	HWAssert10. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 10.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
9	HWAssert9. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 9.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
8	HWAssert8. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 8.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
7	HWAssert7. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 7.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
6	HWAssert6. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 6.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
5	HWAssert5. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 5.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
4	HWAssert4. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 4.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
3	HWAssert3. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 3.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
2	HWAssert2. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 2.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
1	HWAssert1. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 1.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
0	HWAssert0. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 0.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.

D18F3x8F4 [Hardware Assert Status High] (DF::HardwareAssertStatusHigh)

Reset: Cold,0000_0000h.

The upper 32 hardware assertion status bits. Each data fabric component type maps its own assertions to this register.

_instBCST_aliasHOST; D18F3x8F4

_inst[TCDX[15:0],SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F3x8F4_x[[60:41],[38:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[38:00]], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x23D]

_instBCST_aliasSMN; D18F3x000008F4; D18F3=4900_3000h

_inst[TCDX[15:0],SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F3x000008F4_x[[60:41],[38:00]]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[38:00]], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x23D]

Bits	Description
31	HWAssert31. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 63. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
30	HWAssert30. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 62. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
29	HWAssert29. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 61. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
28	HWAssert28. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 60. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
27	HWAssert27. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 59. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
26	HWAssert26. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 58. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
25	HWAssert25. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 57. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
24	HWAssert24. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 56. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
23	HWAssert23. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 55. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
22	HWAssert22. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 54. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
21	HWAssert21. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 53. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
20	HWAssert20. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 52. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
19	HWAssert19. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 51. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
18	HWAssert18. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 50. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
17	HWAssert17. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 49. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
16	HWAssert16. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 48. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
15	HWAssert15. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 47. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
14	HWAssert14. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 46. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
13	HWAssert13. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 45. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
12	HWAssert12. Reset: Cold,0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 44.

	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
11	HWAssert11. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 43.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
10	HWAssert10. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 42.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
9	HWAssert9. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 41.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
8	HWAssert8. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 40.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
7	HWAssert7. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 39.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
6	HWAssert6. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 38.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
5	HWAssert5. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 37.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
4	HWAssert4. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 36.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
3	HWAssert3. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 35.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
2	HWAssert2. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 34.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
1	HWAssert1. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 33.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
0	HWAssert0. Reset: Cold, 0. 0=Assert NOT detected. 1=Assert Detected. Hardware Assert 32.
	AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.

D18F3x8F8 [Hardware Assert Mask Low] (DF::HardwareAssertMaskLow)

Reset: Cold,FFFF_FFFFh.

The lower 32 hardware assert mask bits. When a bit is set to 1, the corresponding hardware assert in the Hardware Assert Status Low is not reported. Each data fabric component has a different set of assertions mapped to the bits in this register. See HardwareAssertStatusLow.

_instBCST_aliasHOST; D18F3x8F8

_inst[TCDX[15:0],SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F3x8F8_x[[60:41],[38:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[38:00]], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x23E]

_instBCST_aliasSMN; D18F3x000008F8; D18F3=4900_3000h

_inst[TCDX[15:0],SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F3x000008F8_x[[60:41],[38:00]]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[38:00]], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x23E]

Bits	Description
31	HWAssertMsk31. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 31 AccessType: SecurePart ? Read-only : Read-write.
30	HWAssertMsk30. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 30 AccessType: SecurePart ? Read-only : Read-write.
29	HWAssertMsk29. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 29 AccessType: SecurePart ? Read-only : Read-write.
28	HWAssertMsk28. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 28 AccessType: SecurePart ? Read-only : Read-write.
27	HWAssertMsk27. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 27 AccessType: SecurePart ? Read-only : Read-write.
26	HWAssertMsk26. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 26 AccessType: SecurePart ? Read-only : Read-write.
25	HWAssertMsk25. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 25 AccessType: SecurePart ? Read-only : Read-write.
24	HWAssertMsk24. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 24 AccessType: SecurePart ? Read-only : Read-write.
23	HWAssertMsk23. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 23 AccessType: SecurePart ? Read-only : Read-write. Init: <u>_inst[CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 0</u> Init: <u>_inst[CNLI[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0</u> Init: <u>_inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00]]_alias[SMN,HOST]: 0</u>
22	HWAssertMsk22. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 22 AccessType: SecurePart ? Read-only : Read-write. Init: <u>_inst[CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 0</u> Init: <u>_inst[CNLI[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0</u> Init: <u>_inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00]]_alias[SMN,HOST]: 0</u>
21	HWAssertMsk21. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 21 AccessType: SecurePart ? Read-only : Read-write. Init: <u>_inst[BCST,ACM[3:0]]_alias[SMN,HOST]: 0</u>

	Init: _inst[CAKE[7:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0
	Init: _inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0]]_alias[SMN,HOST]: 0
20	HWAssertMsk20. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 20
	AccessType: SecurePart ? Read-only : Read-write.
19	HWAssertMsk19. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 19
	AccessType: SecurePart ? Read-only : Read-write.
18	HWAssertMsk18. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 18
	AccessType: SecurePart ? Read-only : Read-write.
	Init: _inst[CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 0
	Init: _inst[CNLI[3:0]]_alias[SMN,HOST]: 1
	Init: _inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00]]_alias[SMN,HOST]: 0
17	HWAssertMsk17. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 17
	AccessType: SecurePart ? Read-only : Read-write.
16	HWAssertMsk16. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 16
	AccessType: SecurePart ? Read-only : Read-write.
15	HWAssertMsk15. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 15
	AccessType: SecurePart ? Read-only : Read-write.
14	HWAssertMsk14. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 14
	AccessType: SecurePart ? Read-only : Read-write.
13	HWAssertMsk13. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 13
	AccessType: SecurePart ? Read-only : Read-write.
12	HWAssertMsk12. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 12
	AccessType: SecurePart ? Read-only : Read-write.
11	HWAssertMsk11. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 11
	AccessType: SecurePart ? Read-only : Read-write.
10	HWAssertMsk10. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 10
	AccessType: SecurePart ? Read-only : Read-write.
9	HWAssertMsk9. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 9
	AccessType: SecurePart ? Read-only : Read-write.
8	HWAssertMsk8. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 8
	AccessType: SecurePart ? Read-only : Read-write.
7	HWAssertMsk7. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 7
	AccessType: SecurePart ? Read-only : Read-write.
6	HWAssertMsk6. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 6
	AccessType: SecurePart ? Read-only : Read-write.

5	HWAssertMsk5. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 5 AccessType: SecurePart ? Read-only : Read-write. Init: _inst[CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 0 Init: _inst[CNLI[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00]]_alias[SMN,HOST]: 0
4	HWAssertMsk4. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 4 AccessType: SecurePart ? Read-only : Read-write. Init: _inst[CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 0 Init: _inst[CS[11:00],CNLI[3:0],CMP[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0]]_alias[SMN,HOST]: 0
3	HWAssertMsk3. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 3 AccessType: SecurePart ? Read-only : Read-write. Init: _inst[BCST,ACM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[CAKE[7:0]]_alias[SMN,HOST]: 0 Init: _inst[CMP[3:0],CCM[7:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[CNLI[3:0]]_alias[SMN,HOST]: 0 Init: _inst[CS[11:00]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[ICNG[3:0]]_alias[SMN,HOST]: 0 Init: _inst[IOM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[IOS[3:0]]_alias[SMN,HOST]: 0 Init: _inst[NCM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[TCDX[15:00],SPF[15:00],PIE0]_alias[SMN,HOST]: 0
2	HWAssertMsk2. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 2 AccessType: SecurePart ? Read-only : Read-write. Init: _inst[BCST,ACM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[CAKE[7:0]]_alias[SMN,HOST]: 0 Init: _inst[CMP[3:0],CCM[7:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[CNLI[3:0]]_alias[SMN,HOST]: 0 Init: _inst[CS[11:00]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[ICNG[3:0]]_alias[SMN,HOST]: 0 Init: _inst[IOM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[IOS[3:0]]_alias[SMN,HOST]: 0 Init: _inst[NCM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[TCDX[15:00],SPF[15:00],PIE0]_alias[SMN,HOST]: 0
1	HWAssertMsk1. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 1 AccessType: SecurePart ? Read-only : Read-write. Init: _inst[BCST,ACM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[CAKE[7:0]]_alias[SMN,HOST]: 0 Init: _inst[CMP[3:0],CCM[7:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[CNLI[3:0]]_alias[SMN,HOST]: 0 Init: _inst[CS[11:00]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[ICNG[3:0]]_alias[SMN,HOST]: 0 Init: _inst[IOM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[IOS[3:0]]_alias[SMN,HOST]: 0 Init: _inst[NCM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[TCDX[15:00],SPF[15:00],PIE0]_alias[SMN,HOST]: 0
0	HWAssertMsk0. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 0 AccessType: SecurePart ? Read-only : Read-write. Init: _inst[BCST,ACM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[CAKE[7:0]]_alias[SMN,HOST]: 0 Init: _inst[CMP[3:0],CCM[7:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0 Init: _inst[CNLI[3:0]]_alias[SMN,HOST]: 0 Init: _inst[CS[11:00]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0

	Init: _inst[ICNG[3:0]]_alias[SMN,HOST]: 0
	Init: _inst[IOM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0
	Init: _inst[IOS[3:0]]_alias[SMN,HOST]: 0
	Init: _inst[NCM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0
	Init: _inst[TCDX[15:00],SPF[15:00],PIE0]_alias[SMN,HOST]: 0

D18F3x8FC [Hardware Assert Mask High] (DF::HardwareAssertMaskHigh)

Reset: Cold,FFFF_FFFFh.

The upper 32 hardware assert mask bits. When a bit is set to 1, the corresponding hardware assert in the Hardware Assert Status High is not reported. Each data fabric component has a different set of assertions mapped to the bits in this register. See HardwareAssertStatusHigh.

_instBCST_aliasHOST; D18F3x8FC
_inst[TCDX[15:0],SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST;
D18F3x8FC_x[[60:41],[38:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[38:00]], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x23F]
_instBCST_aliasSMN; D18F3x000008FC; D18F3=4900_3000h
_inst[TCDX[15:0],SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN;
D18F3x000008FC_x[[60:41],[38:00]]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[38:00]], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x23F]

Bits	Description
31	HWAssertMsk31. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 63. AccessType: SecurePart ? Read-only : Read-write. Init: <u>_inst[BCST,ACM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0</u> Init: <u>_inst[CAKE[7:0]]_alias[SMN,HOST]: 0</u> Init: <u>_inst[CCM[7:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0</u> Init: <u>_inst[IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0]]_alias[SMN,HOST]: 0</u> Init: <u>_inst[IOS[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0</u> Init: <u>_inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0]]_alias[SMN,HOST]: 0</u>
30	HWAssertMsk30. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 62. AccessType: SecurePart ? Read-only : Read-write. Init: <u>_inst[BCST,ACM[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0</u> Init: <u>_inst[CAKE[7:0]]_alias[SMN,HOST]: 0</u> Init: <u>_inst[CCM[7:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0</u> Init: <u>_inst[IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0]]_alias[SMN,HOST]: 0</u> Init: <u>_inst[IOS[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0</u> Init: <u>_inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0]]_alias[SMN,HOST]: 0</u>
29	HWAssertMsk29. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 61. AccessType: SecurePart ? Read-only : Read-write. Init: <u>_inst[BCST,ACM[3:0]]_alias[SMN,HOST]: 1</u> Init: <u>_inst[CAKE[7:0]]_alias[SMN,HOST]: 0</u> Init: <u>_inst[CCM[7:0]]_alias[SMN,HOST]: 1</u> Init: <u>_inst[IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0]]_alias[SMN,HOST]: 0</u> Init: <u>_inst[IOS[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0</u> Init: <u>_inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0]]_alias[SMN,HOST]: 0</u>
28	HWAssertMsk28. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 60. AccessType: SecurePart ? Read-only : Read-write. Init: <u>_inst[IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 0</u> Init: <u>_inst[IOS[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0</u> Init: <u>_inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0]]_alias[SMN,HOST]: 0</u>
27	HWAssertMsk27. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 59. AccessType: SecurePart ? Read-only : Read-write. Init: <u>_inst[IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 0</u> Init: <u>_inst[IOS[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0</u> Init: <u>_inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0]]_alias[SMN,HOST]: 0</u>
26	HWAssertMsk26. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 58. AccessType: SecurePart ? Read-only : Read-write.
25	HWAssertMsk25. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 57.

	AccessType: SecurePart ? Read-only : Read-write.
24	HWAssertMsk24. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 56.
	AccessType: SecurePart ? Read-only : Read-write.
23	HWAssertMsk23. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 55.
	AccessType: SecurePart ? Read-only : Read-write.
22	HWAssertMsk22. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 54.
	AccessType: SecurePart ? Read-only : Read-write.
21	HWAssertMsk21. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 53.
	AccessType: SecurePart ? Read-only : Read-write.
20	HWAssertMsk20. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 52.
	AccessType: SecurePart ? Read-only : Read-write.
	Init: _inst[BCST,ACM[3:0]]_alias[SMN,HOST]: 1
	Init: _inst[CAKE[7:0]]_alias[SMN,HOST]: 0
	Init: _inst[CCM[7:0]]_alias[SMN,HOST]: 1
	Init: _inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0]]_alias[SMN,HOST]: 0
19	HWAssertMsk19. Reset: Cold,1. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 51.
	AccessType: SecurePart ? Read-only : Read-write.
	Init: _inst[IOM[3:0],ICNG[3:0],CS[11:00],CNLI[3:0],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 0
	Init: _inst[IOS[3:0]]_alias[SMN,HOST]: DataPoisonEnabled ? 1 : 0
	Init: _inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0]]_alias[SMN,HOST]: 0
18	HWAssertMsk18. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 50.
	AccessType: SecurePart ? Read-only : Read-write.
17	HWAssertMsk17. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 49.
	AccessType: SecurePart ? Read-only : Read-write.
16	HWAssertMsk16. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 48.
	AccessType: SecurePart ? Read-only : Read-write.
15	HWAssertMsk15. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 47.
	AccessType: SecurePart ? Read-only : Read-write.
14	HWAssertMsk14. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 46.
	AccessType: SecurePart ? Read-only : Read-write.
13	HWAssertMsk13. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 45.
	AccessType: SecurePart ? Read-only : Read-write.
12	HWAssertMsk12. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 44.
	AccessType: SecurePart ? Read-only : Read-write.
11	HWAssertMsk11. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 43.
	AccessType: SecurePart ? Read-only : Read-write.

10	HWAssertMsk10. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 42. AccessType: SecurePart ? Read-only : Read-write.
9	HWAssertMsk9. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 41. AccessType: SecurePart ? Read-only : Read-write.
8	HWAssertMsk8. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 40. AccessType: SecurePart ? Read-only : Read-write.
7	HWAssertMsk7. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 39. AccessType: SecurePart ? Read-only : Read-write.
6	HWAssertMsk6. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 38. AccessType: SecurePart ? Read-only : Read-write.
5	HWAssertMsk5. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 37. AccessType: SecurePart ? Read-only : Read-write.
4	HWAssertMsk4. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 36. AccessType: SecurePart ? Read-only : Read-write.
3	HWAssertMsk3. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 35. AccessType: SecurePart ? Read-only : Read-write.
2	HWAssertMsk2. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 34. AccessType: SecurePart ? Read-only : Read-write.
1	HWAssertMsk1. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 33. AccessType: SecurePart ? Read-only : Read-write.
0	HWAssertMsk0. Reset: Cold,1. Init: 0. 0=Assert enabled and reported. 1=Assert NOT reported. Hardware Assert Mask 32. AccessType: SecurePart ? Read-only : Read-write.

D18F3x908 [CCM SLT Mode Configuration] (DF::CcmSltModeCfg)

CCM SLT Mode configuration Bits.

_instBCST_aliasHOST; D18F3x908_inst[CCM[7:0]]_aliasHOST; D18F3x908_x1[7:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[7:0], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x242]_instBCST_aliasSMN; D18F3x00000908; D18F3=4900_3000h_inst[CCM[7:0]]_aliasSMN; D18F3x00000908_x1[7:0]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[7:0], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x242]

Bits	Description
31:21	Reserved.
20:16	ReqNodeId . Reset: 02h. Node ID to be set in CCM for requests being routed to CAKE in SLT mode AccessType: SecurePart ? Read-only : Read-write.
15:14	Reserved.
13:7	FarCsLimitCompId . Init: 11.5.3 [Register Values Written Based on Configuration]. Far CS Componnet ID Limit AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instCCM0_alias</u> [SMN,HOST]: 0Bh Reset: <u>_instCCM1_alias</u> [SMN,HOST]: 05h Reset: <u>_instCCM2_alias</u> [SMN,HOST]: 0Bh Reset: <u>_instCCM3_alias</u> [SMN,HOST]: 05h Reset: <u>_instCCM4_alias</u> [SMN,HOST]: 0Bh Reset: <u>_instCCM5_alias</u> [SMN,HOST]: 05h Reset: <u>_instCCM6_alias</u> [SMN,HOST]: 0Bh Reset: <u>_instCCM7_alias</u> [SMN,HOST]: 05h
6:0	FarCsBaseCompId . Init: 11.5.3 [Register Values Written Based on Configuration]. Far CS Componnet ID Base AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_instCCM0_alias</u> [SMN,HOST]: 06h Reset: <u>_instCCM1_alias</u> [SMN,HOST]: 00h Reset: <u>_instCCM2_alias</u> [SMN,HOST]: 06h Reset: <u>_instCCM3_alias</u> [SMN,HOST]: 00h Reset: <u>_instCCM4_alias</u> [SMN,HOST]: 06h Reset: <u>_instCCM5_alias</u> [SMN,HOST]: 00h Reset: <u>_instCCM6_alias</u> [SMN,HOST]: 06h Reset: <u>_instCCM7_alias</u> [SMN,HOST]: 00h

D18F3x910 [Originator Watchdog Timer Status Register] (DF::OrigWdtAddrLogStat)

Reset: 0000_0000h.

Qualifies the address stored in OrigWdtAddrLogLo and OrigWdtAddrLogHi, and contains other debug information.

`_instBCST_aliasHOST; D18F3x910``_inst[ACM[3:0],CCM[7:0]]_aliasHOST; D18F3x910_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x244]``_instBCST_aliasSMN; D18F3x00000910; D18F3=4900_3000h``_inst[ACM[3:0],CCM[7:0]]_aliasSMN; D18F3x00000910_x1[B:0]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x244]`

Bits	Description
31:23	Reserved.
22:20	RSPQState . Reset: 0h. RSPQ State of the transaction which triggered the Originator WDT . AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
19:14	UnitID . Reset: 00h. Originator Unit ID of the transaction which triggered the Originator WDT . AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
13:2	DstFabricID . Reset: 000h. Destination Fabric ID of the transaction which triggered the Originator WDT . AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
1	Certainty . Reset: 0. Indicates that the address and debug information stored in the Originator WDT is likely associated with a system hang. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.
0	Valid . Reset: 0. Indicates that the Originator WDT expired and stored an address in DF::OrigWdtAddrLogLo and DF::OrigWdtAddrLogHi, and other fields in this register are valid. Any Write to this field will reset DF::OrigWdtAddrLogStat, DF::OrigWdtAddrLogLo and DF::OrigWdtAddrLogHi registers. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.

D18F3x914 [Originator Watchdog Timer Address Register] (DF::OrigWdtAddrLogLo)

Reset: 0000_0000h.

Contains the address of the transaction which triggered the Originator [WDT](#).`_instBCST_aliasHOST; D18F3x914``_inst[ACM[3:0],CCM[7:0]]_aliasHOST; D18F3x914_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x245]``_instBCST_aliasSMN; D18F3x00000914; D18F3=4900_3000h``_inst[ACM[3:0],CCM[7:0]]_aliasSMN; D18F3x00000914_x1[B:0]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x245]`

Bits	Description
31:0	WdtAddressLo . Reset: 0000_0000h. Originator WDT Transaction Address bits [33:2]. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.

D18F3x918 [Originator Watchdog Timer Address Register] (DF::OrigWdtAddrLogHi)

Reset: 0000_0000h.

Contains the address of the transaction which triggered the Originator [WDT](#).`_instBCST_aliasHOST; D18F3x918``_inst[ACM[3:0],CCM[7:0]]_aliasHOST; D18F3x918_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x246]``_instBCST_aliasSMN; D18F3x00000918; D18F3=4900_3000h``_inst[ACM[3:0],CCM[7:0]]_aliasSMN; D18F3x00000918_x1[B:0]; D18F3=4900_3000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=3, IndCfgAccRegNum=0x246]`

Bits	Description
31:0	WdtAddressHi . Reset: 0000_0000h. Originator WDT Transaction Address bits [MSB:34]. AccessType: SecurePart ? Read-only, Volatile : Read-write, Volatile.

11.10.5 Function 4 Registers**D18F4x000 [Device/Vendor ID Function 4] (DF::DeviceVendorId4)**

Read-only. Reset: Fixed,14B1_1022h.	
PCI Device/Vendor ID 4 register	
_aliasHOST; D18F4x000	
_aliasSMN; D18F4x00000000; D18F4=4900_4000h	
Bits	Description
31:16	DeviceID. Read-only. Reset: Fixed,14B1h. Device ID
15:0	VendorID. Read-only. Reset: Fixed,1022h. Vendor ID

D18F4x004 [Status Command Function 4] (DF::StatusCommand4)

Read-only. Reset: Fixed,0000_0000h.	
PCI Status/Command register	
_aliasHOST; D18F4x004	
_aliasSMN; D18F4x00000004; D18F4=4900_4000h	
Bits	Description
31:16	Status. Read-only. Reset: Fixed,0000h. Status
15:0	Command. Read-only. Reset: Fixed,0000h. Command

D18F4x008 [Class Code Revision ID Function 4] (DF::ClassCodeRevId4)

Read-only. Reset: Fixed,0600_0000h.	
PCI Class Code/Revision ID register	
_aliasHOST; D18F4x008	
_aliasSMN; D18F4x00000008; D18F4=4900_4000h	
Bits	Description
31:8	ClassCode. Read-only. Reset: Fixed,06_0000h. Class Code. Provides the hostbridge class code as defined in the PCI specification.
7:0	RevisionId. Read-only. Reset: Fixed,00h. RevID: Revision ID.

D18F4x00C [Header Type Function 4] (DF::HeaderType4)

Read-only. Reset: Fixed,0080_0000h.	
PCI Header Type register	
_aliasHOST; D18F4x00C	
_aliasSMN; D18F4x0000000C; D18F4=4900_4000h	
Bits	Description
31:0	HeaderType. Read-only. Reset: Fixed,0080_0000h. Header Type: These bits are fixed at their default values. The header type field indicates that there are multiple functions present in this device.

D18F4x034 [Capabilities Pointer Function 4] (DF::CapabilitiesPtr4)

Read-only. Reset: Fixed,0000_0000h.	
PCI Capabilities Pointer register	
_aliasHOST; D18F4x034	
_aliasSMN; D18F4x00000034; D18F4=4900_4000h	
Bits	Description
31:8	Reserved.
7:0	CapPtr. Read-only. Reset: Fixed,00h. Capabilities Pointer.

D18F4x040 [Fabric Configuration Access Control (FCAC)] (DF::FabricConfigAccessControl)

Reset: 0000_0000h.

Configuration register visible for documentation purposes only.

_n[5:0] _aliasHOST; D18F4x040_n[5:0] _aliasSMN; D18F4x00000040; D18F4=4900_4000h

Bits	Description
31:2	Reserved.
1	CfgRegInstAccRegLock . Reset: 0. Init: 1. This field is fixed to 1. AccessType: SecurePart ? Read-only : Read-write.
0	Reserved.

D18F4x0[80...94] [Fabric Indirect Configuration Access Address (FICAA)]**(DF::FabricIndirectConfigAccessAddress)**

Reset: 0000_0000h.

For more information see 11.3.3 [DF Configuration Register Direct and Indirect Access].

_n[5:0] _aliasHOST; D18F4x0[94,90,8[C,8,4,0]]_n[5:0] _aliasSMN; D18F4x0000_00[94,90,8C,88,84,80]; D18F4=4900_4000h

Bits	Description
31:24	Reserved.
23:16	CfgRegInstID . Reset: 00h. Identifies the instanceID of the register to access. AccessType: SecurePart ? Read-only : Read-write.
15	Reserved.
14	SixtyFourBitRegEn . Reset: 0. Specifies 64-bit register access using both the upper and lower 32 bit data registers. AccessType: SecurePart ? Read-only : Read-write.
13:11	IndCfgAccFuncNum . Reset: 0h. Function number of the local DF configuration register to be accessed through DF::FabricIndirectConfigAccessDataLo and DF::FabricIndirectConfigAccessDataHi. AccessType: SecurePart ? Read-only : Read-write.
10:1	IndCfgAccRegNum . Reset: 000h. AccessType: SecurePart ? Read-only : Read-write. Description: Register number of the local DF Configuration register to be accessed through DF::FabricIndirectConfigAccessDataLo and DF::FabricIndirectConfigAccessDataHi. The register offset address is {IndCfgAccRegNum[11:2], 00b}.
0	CfgRegInstAccEn . Reset: 0. 0=Target all instances. 1=Target the instance specified by CfgRegInstID. Configuration Register Instance Access Enable. AccessType: SecurePart ? Read-only : Read-write.

D18F4x0[A0...C8] [Fabric Indirect Configuration Access Data Lo (FICAD)]**(DF::FabricIndirectConfigAccessDataLo)**

Read-write, Volatile. Reset: 0000_0000h.

Accesses through this register affect the lower 32 bits of the register specified in DF::FabricIndirectConfigAccessAddress. Fabric Indirect Configuration Access Data Registers at offsets xC and x4 are to the upper 32 bits of a 64 bit access. Registers at offsets x8 and x0 are to the lower 32 bits. For more information see 11.3.3 [DF Configuration Register Direct and Indirect Access].

_n[5:0] _aliasHOST; D18F4x0[[C:A]][8,0]]_n[5:0] _aliasSMN; D18F4x0000_00[C8,C0,B8,B0,A8,A0]; D18F4=4900_4000h

Bits	Description
31:0	Data . Read-write, Volatile. Reset: 0000_0000h. Indirect Data Lo

**D18F4x0[A4...CC] [Fabric Indirect Configuration Access Data Hi (FICAD)]
(DF::FabricIndirectConfigAccessDataHi)**

Read-write, Volatile. Reset: 0000_0000h.

Accesses through this register affect the upper 32 bits of register specified in DF::FabricIndirectConfigAccessAddress. Fabric Indirect Configuration Access Data Registers at offsets xC and x4 are to the upper 32 bits for 64 bit access. Registers at offsets x8 and x0 registers are to the lower 32 bits. For more information see 11.3.3 [DF Configuration Register Direct and Indirect Access].

_n[5:0]_aliasHOST; D18F4x0[[C:A][C,4]]_n[5:0]_aliasSMN; D18F4x0000_00[CC,C4,BC,B4,AC,A4]; D18F4=4900_4000h**Bits Description**31:0 **Data.** Read-write, Volatile. Reset: 0000_0000h. Indirect Data Hi**D18F4x100 [Enhanced Capability Header Function 4] (DF::EnhCapabilityHdr4)**

Reset: 0000_0000h.

PCI Enhanced Capability Header register

_aliasHOST; D18F4x100_aliasSMN; D18F4x00000100; D18F4=4900_4000h**Bits Description**

31:0 Reserved.

D18F4x110 [Master Request Control] (DF::MasterRequestControl)

Read-write. Reset: 0000_0000h.

Throttle rates and active request limits for master components.

_instBCST_aliasHOST; D18F4x110_inst[IOM[3:0], NCM[3:0], ACM[3:0], CCM[7:0]]_aliasHOST; D18F4x110_x[23:10]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[23:10], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x044]_instBCST_aliasSMN; D18F4x00000110; D18F4=4900_4000h_inst[IOM[3:0], NCM[3:0], ACM[3:0], CCM[7:0]]_aliasSMN; D18F4x00000110_x[23:10]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[23:10], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x044]**Bits Description**

31 Reserved.

30:26 **MstIoReqThrottle.** Read-write. Reset: 00h. IO request throttle rate of the Master components.**ValidValues:**

Value	Description
00h	Throttle disabled and the master may issue an IO request every cycle.
1Fh-01h	Throttle enabled and specifies the minimum number of fabric cycles between any two requests which target IO. When changing this field during operation, always write the field to all zeroes before changing to the new values.

25:18 Reserved.

17:12 **MstIoNpActReqLmt.** Read-write. Reset: 00h. Master to IO Non-Posted Active Request Limit.**ValidValues:**

Value	Description
00h	This field does not limit the number of active IO non-posted requests.
3Fh-01h	Maximum allowed number of active or outstanding non-posted requests from this Master to IO waiting for a response.

11:0 Reserved.

D18F4x12C [Master Active Memory Request Limits] (DF::MasterActiveMemReqLimits)

Read-write. Reset: 0000_0000h.

Thresholds for active memory requests for master components.

_instBCST_aliasHOST; D18F4x12C

_inst[IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_aliasHOST; D18F4x12C_x[23:10]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[23:10], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x04B]

_instBCST_aliasSMN; D18F4x0000012C; D18F4=4900_4000h

_inst[IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_aliasSMN; D18F4x0000012C_x[23:10]; D18F4=4900_4000h;
FabricIndirectConfigAccessAddress[CfgRegInstID=0x[23:10], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x04B]

Bits	Description
31:28	Reserved.
27:20	MstActReqLmtStop . Read-write. Reset: 00h. Number of outstanding memory requests before the Master enters the Stop regime
19:18	Reserved.
17:10	MstActReqLmtHeavy . Read-write. Reset: 00h. Number of outstanding memory requests before the Master enters the Heavy regime
9:8	Reserved.
7:0	MstActReqLmtLight . Read-write. Reset: 00h. Number of outstanding memory requests before the Master enters the Light regime

D18F4x180 [System Configuration] (DF::SystemCfg)

This register holds the Die and Socket information for the system and the die on which it resides.

_instBCST_aliasHOST; D18F4x180

_inst[TCDX[15:0],SPF[15:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST;
D18F4x180_x[[60:41],[34:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[34:00]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x060]

_instBCST_aliasSMN; D18F4x00000180; D18F4=4900_4000h

_inst[TCDX[15:0],SPF[15:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN;
D18F4x00000180_x[[60:41],[34:00]]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[34:00]], IndCfgAccFuncNum=4,
IndCfgAccRegNum=0x060]

Bits	Description										
31:28	Reserved.										
27:16	MyNodeId. Init: 11.5.3 [Register Values Written Based on Configuration]. Node Id of the die this instance of the register is on. AccessType: SecurePart ? Read-only : Read-write. Reset: _inst[TCDX[15:00],SPF[15:00],PIE0,NCM[3:0],IOS[3:0],IOM[3:0],ICNG[3:0],CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 000h										
15:9	Reserved.										
8	OtherSocketPresent. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. The other socket is present and enabled in the system. This field is used by the probe/broadcast MultiCast Mask. AccessType: SecurePart ? Read-only : Read-write.										
7:5	Reserved.										
4	MyNodeMap. Reset: 0. Node Map Type for this node. 0=Local Map Type, 1=Remote Map Type. AccessType: SecurePart ? Read-only : Read-write.										
3:2	OtherDieType. Reset: 0h. Die Type [1:0] of other die(s) in this system, which is/are not the same type as this die. AccessType: SecurePart ? Read-only : Read-write. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>CPU</td></tr> <tr> <td>1h</td><td>APU</td></tr> <tr> <td>2h</td><td>dGPU</td></tr> <tr> <td>3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	CPU	1h	APU	2h	dGPU	3h	Reserved.
Value	Description										
0h	CPU										
1h	APU										
2h	dGPU										
3h	Reserved.										
1:0	MyDieType. Reset: 0h. Die Type [1:0] of the die this instance of the register is on. AccessType: SecurePart ? Read-only : Read-write. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>CPU</td></tr> <tr> <td>1h</td><td>APU</td></tr> <tr> <td>2h</td><td>dGPU</td></tr> <tr> <td>3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	CPU	1h	APU	2h	dGPU	3h	Reserved.
Value	Description										
0h	CPU										
1h	APU										
2h	dGPU										
3h	Reserved.										

D18F4x184 [System Component Count] (DF::SystemComponentCnt)

Reset: 1000_0401h.

This register holds the system-wide counts for various component types.

_instBCST_aliasHOST; D18F4x184_instPIE0_aliasHOST; D18F4x184_x2C; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=4, IndCfgAccRegNum=0x061]_instBCST_aliasSMN; D18F4x00000184; D18F4=4900_4000h_instPIE0_aliasSMN; D18F4x00000184_x2C; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=4, IndCfgAccRegNum=0x061]

Bits	Description
31:26	IOSCount. Reset: 04h. Init: 11.5.3 [Register Values Written Based on Configuration]. System count of present and enabled IOSs AccessType: SecurePart ? Read-only : Read-write.
25:16	GCMCount. Reset: 000h. System count of present and enabled GCMs AccessType: SecurePart ? Read-only : Read-write.
15:8	IOMCount. Reset: 04h. Init: 11.5.3 [Register Values Written Based on Configuration]. System count of present and enabled IOMs AccessType: SecurePart ? Read-only : Read-write.
7:0	PIECount. Reset: 01h. Init: 11.5.3 [Register Values Written Based on Configuration]. System count of present and enabled PIEs AccessType: SecurePart ? Read-only : Read-write.

D18F4x188 [CcdUnitId Mask] (DF::CcdUnitIdMask)

Reset: A0C0_01C1h.

Describes how to identify the L3, [CCX](#), cores and threads of a UnitId for the [CCD](#).[_instBCST_aliasHOST](#); D18F4x188[_inst\[CAKE\[7:0\],PIE0,ICNG\[3:0\],ACM\[3:0\],CCM\[7:0\],CMP\[3:0\],CS\[11:0\]\]_aliasHOST](#); D18F4x188_x[[34:28],[1B:00]];
FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[34:28],[1B:00]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x062][_instBCST_aliasSMN](#); D18F4x00000188; D18F4=4900_4000h[_inst\[CAKE\[7:0\],PIE0,ICNG\[3:0\],ACM\[3:0\],CCM\[7:0\],CMP\[3:0\],CS\[11:0\]\]_aliasSMN](#); D18F4x00000188_x[[34:28],[1B:00]]; D18F4=4900_4000h;
FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[34:28],[1B:00]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x062]

Bits	Description
31:29	L3IdShft. Reset: 5h. L3Id shift amount for the UnitId. Tells how many bits to shift the UnitId by to align it to bit 0. In effect, this field identifies the LSB of the L3Id. AccessType: SecurePart ? Read-only : Read-write.
28:26	CcxIdShft. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. CcxId shift amount for the UnitId. Tells how many bits to shift the UnitId by to align it to bit 0. In effect, this field identifies the LSB of the CcxId. AccessType: SecurePart ? Read-only : Read-write.
25:23	CoreIdShft. Reset: 1h. CoreId shift amount for the UnitId. Tells how many bits to shift the UnitId by to align it to bit 0. In effect, this field identifies the LSB of the CoreId. AccessType: SecurePart ? Read-only : Read-write.
22:17	L3IdMask. Reset: 20h. L3Id Mask for the UnitId. If a bit in this mask is set, then the corresponding bit in the UnitId is used to identify a L3. L3Id bits must be contiguous. AccessType: SecurePart ? Read-only : Read-write.
16:11	CcxIdMask. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. CcxId Mask for the UnitId. If a bit in this mask is set, then the corresponding bit in the UnitId is used to identify a CCX . CcxId bits must be contiguous. AccessType: SecurePart ? Read-only : Read-write.
10:5	CoreIdMask. Reset: 0Eh. CoreId Mask for the UnitId. If a bit in this mask is set, then the corresponding bit in the UnitId is used to identify a Core. CoreId bits must be contiguous. AccessType: SecurePart ? Read-only : Read-write.
4:0	ThrdIdMask. Reset: 01h. ThreadId Mask for the UnitId. If a bit in this mask is set, then the corresponding bit in the UnitId is used to identify a Thread . ThreadId bits must be contiguous. AccessType: SecurePart ? Read-only : Read-write.

D18F4x18C [PIE Fabric Component Count] (DF::PIEFabricCompCnt)

Reset: 0004_0404h.

PIE Fabric Component Count register. This register must be programmed with the correct total system component counts. PIE uses this to determine how many broadcast responses and bus lock grants to expect.

_instBCST_aliasHOST; D18F4x18C_inst[PIE0,ICNG[3:0]]_aliasHOST; D18F4x18C_x2[C:8]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[C:8], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x063]_instBCST_aliasSMN; D18F4x0000018C; D18F4=4900_4000h_inst[PIE0,ICNG[3:0]]_aliasSMN; D18F4x0000018C_x2[C:8]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[C:8], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x063]

Bits	Description
31:24	Reserved.
23:16	ACMCnt. Reset: 04h. Init: 11.5.3 [Register Values Written Based on Configuration]. System count of present and enabled ACMs AccessType: SecurePart ? Read-only : Read-write.
15:8	ICNGCount. Reset: 04h. Init: 11.5.3 [Register Values Written Based on Configuration]. System count of present and enabled ICNGs AccessType: SecurePart ? Read-only : Read-write.
7:0	SysBusLockNCMGrantCnt. Reset: 04h. Init: 11.5.3 [Register Values Written Based on Configuration]. Number of Non-Coherent Masters which send Grant in Bus Lock. AccessType: SecurePart ? Read-only : Read-write.

D18F4x190 [Special System Function Fabric ID] (DF::SpecialSysFunctionFabricID1)

Special System function Registers Holds information about the FabricIDs of Local PIE on Die, Master PIE in the system.

_instBCST_aliasHOST; D18F4x190_inst[IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_aliasHOST; D18F4x190_x[23:10]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[23:10], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x064]_instBCST_aliasSMN; D18F4x00000190; D18F4=4900_4000h_inst[IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_aliasSMN; D18F4x00000190_x[23:10]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[23:10], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x064]

Bits	Description
31:28	Reserved.
27:16	MstrPIEFabricID. Reset: 077h. FabricID of the Master PIE. This is always FabricID of PIE on Node0. AccessType: SecurePart ? Read-only : Read-write.
15:12	Reserved.
11:0	LclPIEFabricID. Init: 11.5.3 [Register Values Written Based on Configuration]. FabricID of the Local PIE on Die. Reset to local PIE FabricID. AccessType: <u>_inst[CCM[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]</u> : SecurePart ? Read-only : Read-write AccessType: <u>_inst[IOM[3:0]]_alias[SMN,HOST]</u> : Reserved AccessType: <u>_inst[NCM[3:0]]_alias[SMN,HOST]</u> : SecurePart ? Read-only : Read-write Reset: <u>_inst[NCM[3:0],CCM[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]</u> : 077h

D18F4x194 [Special System Function Fabric ID] (DF::SpecialSysFunctionFabricID2)

Reset: 0023_0000h.

Special System function Registers Holds information about the FabricIDs of Local IOMS and [FCH](#) IOMS (IOMS with SB)[_instBCST_aliasHOST](#); D18F4x194[_inst\[ACM\[3:0\],CCM\[7:0\]\]_aliasHOST](#); D18F4x194_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x065][_instBCST_aliasSMN](#); D18F4x00000194; D18F4=4900_4000h[_inst\[ACM\[3:0\],CCM\[7:0\]\]_aliasSMN](#); D18F4x00000194_x1[B:0]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x065]

Bits	Description
31:28	Reserved.
27:16	FchIOSFabricID . Reset: 023h. Init: 11.5.3 [Register Values Written Based on Configuration]. FabricID of IOS with South-bridge. AccessType: SecurePart ? Read-only : Read-write.
15:0	Reserved.

D18F4x198 [Special System Function Fabric ID] (DF::SpecialSysFunctionFabricID3)

Special System function Registers Holds information about the FabricIDs of Local PIE on Die, Master PIE in the system.

[_instBCST_aliasHOST](#); D18F4x198[_inst\[IOM\[3:0\],ACM\[3:0\],CCM\[7:0\]\]_aliasHOST](#); D18F4x198_x[2[3:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x066][_instBCST_aliasSMN](#); D18F4x00000198; D18F4=4900_4000h[_inst\[IOM\[3:0\],ACM\[3:0\],CCM\[7:0\]\]_aliasSMN](#); D18F4x00000198_x[2[3:0],1[B:0]]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2[3:0],1[B:0]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x066]

Bits	Description
31:12	Reserved.
11:0	LclICNGFabricID . Init: 11.5.3 [Register Values Written Based on Configuration]. FabricID of local ICNG. AccessType: SecurePart ? Read-only : Read-write. Reset: _inst[ACM[3:0]]_alias[SMN,HOST] : 000h Reset: _instCCM0_alias[SMN,HOST] : 07Ch Reset: _instCCM1_alias[SMN,HOST] : 07Dh Reset: _instCCM2_alias[SMN,HOST] : 07Eh Reset: _instCCM3_alias[SMN,HOST] : 07Fh Reset: _instCCM4_alias[SMN,HOST] : 07Ch Reset: _instCCM5_alias[SMN,HOST] : 07Dh Reset: _instCCM6_alias[SMN,HOST] : 07Eh Reset: _instCCM7_alias[SMN,HOST] : 07Fh Reset: _instIOM0_alias[SMN,HOST] : 07Dh Reset: _instIOM1_alias[SMN,HOST] : 07Ch Reset: _instIOM2_alias[SMN,HOST] : 07Fh Reset: _instIOM3_alias[SMN,HOST] : 07Eh

D18F4x1A0 [Special Register 1] (DF::SpecialRegister1_MASTERS)

Reset: 0000_0000h.

Special Register-1. Holds information about the FabricIDs of NCS or CS for NCM/GCM.

_instBCST_aliasHOST; D18F4x1A0_inst[IOM[3:0],NCM[3:0]]_aliasHOST; D18F4x1A0_x[23:1C]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[23:1C], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x068]_instBCST_aliasSMN; D18F4x000001A0; D18F4=4900_4000h_inst[IOM[3:0],NCM[3:0]]_aliasSMN; D18F4x000001A0_x[23:1C]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[23:1C], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x068]

Bits	Description
31:28	Reserved.
27:16	LdCSFabricID . Reset: 000h. Init: 11.5.3 [Register Values Written Based on Configuration]. FabricId of local CS where NCM/GCM/PIE forwards SDP requests that are not in the NCM/GCM/PIE's address map. AccessType: SecurePart ? Read-only : Read-write.
15:12	Reserved.
11:0	LdNCSFabricID . Reset: 000h. FabricID of NCS (e.g. ATHub). Reset to FabricID of NCS. AccessType: SecurePart ? Read-only : Read-write.

D18F4x1A0 [Special Register 1] (DF::SpecialRegister1_PIE)

Reset: 0000_0000h.

Special Register-1. Holds information about the FabricIDs of NCS or CS for NCM/GCM.

_instBCST_aliasHOST; D18F4x1A0_instPIE0_aliasHOST; D18F4x1A0_x2C; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=4, IndCfgAccRegNum=0x068]_instBCST_aliasSMN; D18F4x000001A0; D18F4=4900_4000h_instPIE0_aliasSMN; D18F4x000001A0_x2C; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=4, IndCfgAccRegNum=0x068]

Bits	Description
31:28	Reserved.
27:16	LdCSFabricID . Reset: 000h. Init: 11.5.3 [Register Values Written Based on Configuration]. FabricId of local CS where NCM/GCM/PIE forwards SDP requests that are not in the NCM/GCM/PIE's address map. AccessType: SecurePart ? Read-only : Read-write.
15:0	Reserved.

D18F4x1A8 [Cluster Configuration] (DF::ClusterCfg)

This register holds the CPU Cluster configuration information for the die on which it resides.

_instBCST_aliasHOST; D18F4x1A8

_inst[TCDX[15:0],SPF[15:0],CAKE[7:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4x1A8_x[[60:41],[34:2D],[1B:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[34:2D],[1B:00]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x06A]

_instBCST_aliasSMN; D18F4x000001A8; D18F4=4900_4000h

_inst[TCDX[15:0],SPF[15:0],CAKE[7:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x000001A8_x[[60:41],[34:2D],[1B:00]]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[60:41],[34:2D],[1B:00]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x06A]

Bits Description

31:30 NumCpuNodes. Init: 11.5.3 [Register Values Written Based on Configuration]. Number of Valid CPU Nodes in the system (0-origin)

AccessType: _inst[SPF[15:00],CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[TCDX[15:00]]_alias[SMN,HOST]: Reserved

Reset: _inst[SPF[15:00],CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],BCST,ACM[3:0]]_alias[SMN,HOST]: 0h

ValidValues:

Value	Description
0h	1 CPU Node
1h	2 CPU Nodes
2h	3 CPU Nodes
3h	4 CPU Nodes

29:28 MyCpuNodeId. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. The CPU Node Id of this node

AccessType: SecurePart ? Read-only : Read-write.

27:25 ExpPrbRspCls7. Init: 11.5.3 [Register Values Written Based on Configuration]. Expected Probe Responses from Cluster 7. This is used as Cluster 3 on Node 1 when DfGlobalCtrl[McastMaskPerNodeCLV] == 1]

AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _instBCST_alias[SMN,HOST]: Reserved

AccessType: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: Reserved

Reset: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 2h

24:22 ExpPrbRspCls6. Init: 11.5.3 [Register Values Written Based on Configuration]. Expected Probe Responses from Cluster 6. This is used as Cluster 2 on Node 1 when DfGlobalCtrl[McastMaskPerNodeCLV] == 1]

AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _instBCST_alias[SMN,HOST]: Reserved

AccessType: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: Reserved

Reset: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 2h

21:19 ExpPrbRspCls5. Init: 11.5.3 [Register Values Written Based on Configuration]. Expected Probe Responses from Cluster 5. This is used as Cluster 1 on Node 1 when DfGlobalCtrl[McastMaskPerNodeCLV] == 1]

AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _instBCST_alias[SMN,HOST]: Reserved

AccessType: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: Reserved

Reset: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 2h

18:16 ExpPrbRspCls4. Init: 11.5.3 [Register Values Written Based on Configuration]. Expected Probe Responses from Cluster 4. This is used as Cluster 0 on Node 1 when DfGlobalCtrl[McastMaskPerNodeCLV] == 1]

AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _instBCST_alias[SMN,HOST]: Reserved

AccessType: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: Reserved

Reset: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 2h

15:13 ExpPrbRspCls3. Init: 11.5.3 [Register Values Written Based on Configuration]. Expected Probe Responses from Cluster 3. This field is aliased across Nodes when DfGlobalCtrl[McastMaskPerNodeCLV] == 0]

AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _instBCST_alias[SMN,HOST]: Reserved

AccessType: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write

AccessType: _inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: Reserved

Reset: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 2h

12:10	ExpPrbRspCls2. Init: 11.5.3 [Register Values Written Based on Configuration]. Expected Probe Responses from Cluster 2. This field is aliased across Nodes when DfGlobalCtrl[McastMaskPerNodeCLV] == 0] AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _instBCST_alias[SMN,HOST]: Reserved AccessType: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: Reserved Reset: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 2h
9:7	ExpPrbRspCls1. Init: 11.5.3 [Register Values Written Based on Configuration]. Expected Probe Responses from Cluster 1. This field is aliased across Nodes when DfGlobalCtrl[McastMaskPerNodeCLV] == 0] AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _instBCST_alias[SMN,HOST]: Reserved AccessType: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: Reserved Reset: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 2h
6:4	ExpPrbRspCls0. Init: 11.5.3 [Register Values Written Based on Configuration]. Expected Probe Responses from Cluster 0. This field is aliased across Nodes when DfGlobalCtrl[McastMaskPerNodeCLV] == 0] AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _instBCST_alias[SMN,HOST]: Reserved AccessType: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: Reserved Reset: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: 2h
3:0	LclClusterEn. Init: 11.5.3 [Register Values Written Based on Configuration]. CCM Cluster Enables AccessType: _inst[ACM[3:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _instBCST_alias[SMN,HOST]: Reserved AccessType: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0]]_alias[SMN,HOST]: SecurePart ? Read-only : Read-write AccessType: _inst[TCDX[15:00],SPF[15:00]]_alias[SMN,HOST]: Reserved Reset: _inst[CS[11:00],CMP[3:0],CCM[7:0],CAKE[7:0],ACM[3:0]]_alias[SMN,HOST]: Fh

D18F4x1B0 [System Fabric ID Mask 0] (DF::SystemFabricIdMask0)

Reset: 0F80_007Fh.

Describes how to identify the socket / die / component portions of a FabricId for the entire system. See 11.4 [Data Fabric Instance and Fabric IDs].

_instBCST_aliasHOST; D18F4x1B0

_inst[SPF[15:0],CNLI[3:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4x1B0_x[[50:41],3[8:5],[2C:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],3[8:5],[2C:00]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x06C]

_instBCST_aliasSMN; D18F4x000001B0; D18F4=4900_4000h

_inst[SPF[15:0],CNLI[3:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x000001B0_x[[50:41],3[8:5],[2C:00]]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],3[8:5],[2C:00]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x06C]

Bits	Description
31:16	NodeIdMask. Reset: 0F80h. NodeId Mask for the system. If a bit in this mask is set, then the corresponding bit in the FabricId is used to identify a node. NodeId bits must be contiguous. AccessType: SecurePart ? Read-only : Read-write.
15:0	CompIdMask. Reset: 007Fh. ComponentId Mask for the system. If a bit in this mask is set, then the corresponding bit in the FabricId is used to identify a component. ComponentId bits must be contiguous. AccessType: SecurePart ? Read-only : Read-write.

D18F4x1B4 [System Fabric ID Mask 1] (DF::SystemFabricIdMask1)

Reset: 0000_0107h.

Describes how to identify the socket / die / component portions of a FabricId for the entire system. See 11.4 [Data Fabric Instance and Fabric IDs].

_instBCST_aliasHOST; D18F4x1B4_inst[SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4x1B4_x[[50:41],[38:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],[38:00]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x06D]_instBCST_aliasSMN; D18F4x000001B4; D18F4=4900_4000h_inst[SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x000001B4_x[[50:41],[38:00]]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],[38:00]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x06D]

Bits	Description
31:12	Reserved.
11:8	SocketIdShft. Reset: 1h. SocketId shift amount for the system. Tells how many bits to shift the SocketId by to align it to bit 0. In effect, this field identifies the LSB of the SocketId. AccessType: SecurePart ? Read-only : Read-write.
7:4	Reserved.
3:0	NodeIdShft. Reset: 7h. NodeId shift amount for the system. Tells how many bits to shift the NodeId by to align it to bit 0. In effect, this field identifies the LSB of the NodeId. AccessType: SecurePart ? Read-only : Read-write.

D18F4x1B8 [System Fabric ID Mask 2] (DF::SystemFabricIdMask2)

Reset: 001E_0001h.

Describes how to identify the socket / die / component portions of a FabricId for the entire system. See 11.4 [Data Fabric Instance and Fabric IDs].

_instBCST_aliasHOST; D18F4x1B8_inst[SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4x1B8_x[[50:41],[38:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],[38:00]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x06E]_instBCST_aliasSMN; D18F4x000001B8; D18F4=4900_4000h_inst[SPF[15:0],CNLI[3:0],CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x000001B8_x[[50:41],[38:00]]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],[38:00]], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x06E]

Bits	Description
31:16	SocketIdMask. Reset: 001Eh. SocketId Mask for the system. If a bit in this mask is set, then the corresponding bit in the NodeId is used to identify a socket. SocketId bits must be contiguous. AccessType: SecurePart ? Read-only : Read-write.
15:0	DieIdMask. Reset: 0001h. DieId Mask for the system. If a bit in this mask is set, then the corresponding bit in the NodeId is used to identify a die. DieId bits must be contiguous. AccessType: SecurePart ? Read-only : Read-write.

D18F4x1E0 [Local Node Type Map] (DF::LclNodeTypeMap)

Reset: 0001_0000h.

Node count and base node ID info for a node of Type Local.

_instBCST_aliasHOST; D18F4x1E0_inst[PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4x1E0_x[2C:00];
FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x078]_instBCST_aliasSMN; D18F4x000001E0; D18F4=4900_4000h_inst[PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x000001E0_x[2C:00]; D18F4=4900_4000h;
FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x078]**Bits Description**

31:28	Reserved.
27:16	NodeCount. Reset: 001h. Init: 11.5.3 [Register Values Written Based on Configuration]. Count of Node Type Local nodes in the system. AccessType: SecurePart ? Read-only : Read-write.
15:12	Reserved.
11:0	BaseNodeId. Reset: 000h. Lowest-numbered node ID of Node Type Local. AccessType: SecurePart ? Read-only : Read-write.

D18F4x1E4 [Remote Node Type Map] (DF::RmtNodeTypeMap)

Reset: 0000_0000h.

Node count and base node ID info for a node of Type Remote.

_instBCST_aliasHOST; D18F4x1E4_inst[PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4x1E4_x[2C:00];
FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x079]_instBCST_aliasSMN; D18F4x000001E4; D18F4=4900_4000h_inst[PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x000001E4_x[2C:00]; D18F4=4900_4000h;
FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x079]**Bits Description**

31:28	Reserved.
27:16	NodeCount. Reset: 000h. Count of Node Type Remote nodes in the system. AccessType: SecurePart ? Read-only : Read-write.
15:12	Reserved.
11:0	BaseNodeId. Reset: 000h. Init: 11.5.3 [Register Values Written Based on Configuration]. Lowest-numbered node ID of Node Type Remote. AccessType: SecurePart ? Read-only : Read-write.

D18F4x1E8 [Link Target Configuration for CAKE] (DF::LinkTgtCfg_CAKE)

Reset: 0000_0000h.

This register determines the link target destination for remote FTI transactions when components are in register, LinkTgtMode. All components except TCDX instantiate the link target config register. Components only populate the field(s) that are necessary.

_instBCST_aliasHOST; D18F4x1E8_inst[CAKE[7:0]]_aliasHOST; D18F4x1E8_x[34:2D]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]_instBCST_aliasSMN; D18F4x000001E8; D18F4=4900_4000h_inst[CAKE[7:0]]_aliasSMN; D18F4x000001E8_x[34:2D]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]

Bits	Description
31:27	Reserved.
26:24	BrdcstPrbLnkId. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the LinkID that components use for all outgoing broadcast probe/system management packets. AccessType: SecurePart ? Read-only : Read-write.
23	Reserved.
22:20	CakeLnkId. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the LinkID that CAKE uses for all outgoing FTI packets. For packets that are at the destination node, This field is used as-is for the LinkId field. For packets that are at an intermediate node, this field is OR'ed with the Link Set that is determined by the lookup of the destination node ID. AccessType: SecurePart ? Read-only : Read-write.
19	Reserved.
18:16	DirPrbRspLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for directed probe responses when the component instance of LinkTgtMode register, and its DirPrbRspLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
15	Reserved.
14:12	DirPrbLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for directed probes when the component instance of LinkTgtMode register, and its DirPrbLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
11	Reserved.
10:8	SrcRspLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for source responses when the component instance of LinkTgtMode register, and its SrcRspLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
7	Reserved.
6:4	TgtRspLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for target responses when the component instance of LinkTgtMode register, and its TgtRspLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
3	Reserved.
2:0	ReqLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for requests when the component instance of LinkTgtMode register, and its ReqLnkSelMode field, selects Register Mode AccessType: SecurePart ? Read-only : Read-write.

D18F4x1E8 [Link Target Configuration for CCM] (DF::LinkTgtCfg_CCM)

Reset: 0000_0000h.

This register determines the link target destination for remote FTI transactions when components are in register, LinkTgtMode. All components except TCDX instantiate the link target config register. Components only populate the field(s) that are necessary.

_instBCST_aliasHOST; D18F4x1E8_inst[ACM[3:0],CCM[7:0]]_aliasHOST; D18F4x1E8_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]_instBCST_aliasSMN; D18F4x000001E8; D18F4=4900_4000h_inst[ACM[3:0],CCM[7:0]]_aliasSMN; D18F4x000001E8_x1[B:0]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]

Bits	Description
31	EnableIoBasedLinkRouting. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. 0=All requests use the first two NodeToLinkSet routing tables and will XOR in DF::LinkTgtCfg_CCM[ReqLnkIndx] when DF::LinkTgtMode_CCM[ReqLnkSelMode] specifies FabricRcv mode. 1=IO requests use the second two NodeToLinkSet routing tables and do not enable the XOR of DF::LinkTgtCfg_CCM[ReqLnkIndx] when DF::LinkTgtMode_CCM[ReqLnkSelMode] specifies FabricRcv mode. Does not affect CS_CMP, CS_UMC, or PIE requests. Does not affect non-request. Specifies the NodeToLinkSet behavior for requests destined to IOS. This field does not affect non-requests. AccessType: SecurePart ? Read-only : Read-write.
30:19	Reserved.
18:16	DirPrbRspLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for directed probe responses when the component instance of LinkTgtMode register, and its DirPrbRspLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
15:11	Reserved.
10:8	SrcRspLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for source responses when the component instance of LinkTgtMode register, and its SrcRspLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
7:3	Reserved.
2:0	ReqLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for requests when the component instance of LinkTgtMode register, and its ReqLnkSelMode field, selects Register Mode AccessType: SecurePart ? Read-only : Read-write.

D18F4x1E8 [Link Target Configuration for CS] (DF::LinkTgtCfg_CS)

Reset: 0000_0000h.

This register determines the link target destination for remote FTI transactions when components are in register, LinkTgtMode. All components except TCDX instantiate the link target config register. Components only populate the field(s) that are necessary.

_instBCST_aliasHOST; D18F4x1E8

_inst[CMP[3:0],CS[11:0]]_aliasHOST; D18F4x1E8_x0[F:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]

_instBCST_aliasSMN; D18F4x000001E8; D18F4=4900_4000h

_inst[CMP[3:0],CS[11:0]]_aliasSMN; D18F4x000001E8_x0[F:0]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]

Bits	Description
31:27	Reserved.
26:24	BrdcstPrbLnkId. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the LinkID that components use for all outgoing broadcast probe/system management packets. AccessType: SecurePart ? Read-only : Read-write.
23:15	Reserved.
14:12	DirPrbLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for directed probes when the component instance of LinkTgtMode register, and its DirPrbLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
11:7	Reserved.
6:4	TgtRspLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for target responses when the component instance of LinkTgtMode register, and its TgtRspLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
3	Reserved.
2:0	ReqLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for requests when the component instance of LinkTgtMode register, and its ReqLnkSelMode field, selects Register Mode AccessType: SecurePart ? Read-only : Read-write.

D18F4x1E8 [Link Target Configuration for GCM/NCM] (DF::LinkTgtCfg_GCMNCM)

Reset: 0000_0000h.

This register determines the link target destination for remote FTI transactions when components are in register, LinkTgtMode. All components except TCDX instantiate the link target config register. Components only populate the field(s) that are necessary.

_instBCST_aliasHOST; D18F4x1E8

_inst[NCM[3:0]]_aliasHOST; D18F4x1E8_x1[F:C]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[F:C], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]

_instBCST_aliasSMN; D18F4x000001E8; D18F4=4900_4000h

_inst[NCM[3:0]]_aliasSMN; D18F4x000001E8_x1[F:C]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[F:C], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]

Bits	Description
31:3	Reserved.
2:0	ReqLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for requests when the component instance of LinkTgtMode register, and its ReqLnkSelMode field, selects Register Mode AccessType: SecurePart ? Read-only : Read-write.

D18F4x1E8 [Link Target Configuration for ICNG] (DF::LinkTgtCfg_ICNG)

Reset: 0000_0000h.

This register determines the link target destination for remote FTI transactions when components are in register, LinkTgtMode. All components except TCDX instantiate the link target config register. Components only populate the field(s) that are necessary.

_instBCST_aliasHOST; D18F4x1E8_inst[ICNG[3:0]]_aliasHOST; D18F4x1E8_x2[B:8]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]_instBCST_aliasSMN; D18F4x000001E8; D18F4=4900_4000h_inst[ICNG[3:0]]_aliasSMN; D18F4x000001E8_x2[B:8]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]

Bits	Description
31:27	Reserved.
26:24	BrdcstPrbLnkIdx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the LinkID that components use for all outgoing broadcast probe/system management packets. AccessType: SecurePart ? Read-only : Read-write.
23:19	Reserved.
18:16	DirPrbRspLnkIdx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for directed probe responses when the component instance of LinkTgtMode register, and its DirPrbRspLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
15	Reserved.
14:12	DirPrbLnkIdx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for directed probes when the component instance of LinkTgtMode register, and its DirPrbLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
11:7	Reserved.
6:4	TgtRspLnkIdx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for target responses when the component instance of LinkTgtMode register, and its TgtRspLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
3:0	Reserved.

D18F4x1E8 [Link Target Configuration for IOM] (DF::LinkTgtCfg_IOM)

Reset: 0000_0000h.

This register determines the link target destination for remote FTI transactions when components are in register, LinkTgtMode. All components except TCDX instantiate the link target config register. Components only populate the field(s) that are necessary.

_instBCST_aliasHOST; D18F4x1E8_inst[IOM[3:0]]_aliasHOST; D18F4x1E8_x2[3:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[3:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]_instBCST_aliasSMN; D18F4x000001E8; D18F4=4900_4000h_inst[IOM[3:0]]_aliasSMN; D18F4x000001E8_x2[3:0]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[3:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]

Bits	Description
31	EnableIoBasedLinkRouting. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. 0=All requests use the first two NodeToLinkSet routing tables and will XOR in DF::LinkTgtCfg_CCM[ReqLnkIndx] when DF::LinkTgtMode_CCM[ReqLnkSelMode] specifies FabricRcv mode. 1=IO requests use the second two NodeToLinkSet routing tables and do not enable the XOR of DF::LinkTgtCfg_CCM[ReqLnkIndx] when DF::LinkTgtMode_CCM[ReqLnkSelMode] specifies FabricRcv mode. Does not affect CS_CMP, CS_UMC, or PIE requests. Does not affect non-request. Specifies the NodeToLinkSet behavior for requests destined to IOS. This field does not affect non-requests. AccessType: SecurePart ? Read-only : Read-write.
30:11	Reserved.
10:8	SrcRspLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for source responses when the component instance of LinkTgtMode register, and its SrcRspLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
7:3	Reserved.
2:0	ReqLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for requests when the component instance of LinkTgtMode register, and its ReqLnkSelMode field, selects Register Mode AccessType: SecurePart ? Read-only : Read-write.

D18F4x1E8 [Link Target Configuration for NCS and IOS] (DF::LinkTgtCfg_NCSIOS)

Reset: 0000_0000h.

This register determines the link target destination for remote FTI transactions when components are in register, LinkTgtMode. All components except TCDX instantiate the link target config register. Components only populate the field(s) that are necessary.

_instBCST_aliasHOST; D18F4x1E8_inst[IOS[3:0]]_aliasHOST; D18F4x1E8_x2[7:4]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[7:4], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]_instBCST_aliasSMN; D18F4x000001E8; D18F4=4900_4000h_inst[IOS[3:0]]_aliasSMN; D18F4x000001E8_x2[7:4]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[7:4], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]

Bits	Description
31:7	Reserved.
6:4	TgtRspLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for target responses when the component instance of LinkTgtMode register, and its TgtRspLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
3	Reserved.
2:0	ReqLnkIndx. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for requests when the component instance of LinkTgtMode register, and its ReqLnkSelMode field, selects Register Mode AccessType: SecurePart ? Read-only : Read-write.

D18F4x1E8 [Link Target Configuration for PIE] (DF::LinkTgtCfg_PIE)

Reset: 0000_0000h.

This register determines the link target destination for remote FTI transactions when components are in register, LinkTgtMode. All components except TCDX instantiate the link target config register. Components only populate the field(s) that are necessary.

_instBCST_aliasHOST; D18F4x1E8_instPIE0_aliasHOST; D18F4x1E8_x2C; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]_instBCST_aliasSMN; D18F4x000001E8; D18F4=4900_4000h_instPIE0_aliasSMN; D18F4x000001E8_x2C; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07A]

Bits	Description
31:27	Reserved.
26:24	BrdcstPrbLnkId . Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the LinkID that components use for all outgoing broadcast probe/system management packets. AccessType: SecurePart ? Read-only : Read-write.
23:19	Reserved.
18:16	DirPrbRspLnkIndx . Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for directed probe responses when the component instance of LinkTgtMode register, and its DirPrbRspLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
15	Reserved.
14:12	DirPrbLnkIndx . Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for directed probes when the component instance of LinkTgtMode register, and its DirPrbLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
11:7	Reserved.
6:4	TgtRspLnkIndx . Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for target responses when the component instance of LinkTgtMode register, and its TgtRspLnkSelMode field, selects Register Mode. AccessType: SecurePart ? Read-only : Read-write.
3	Reserved.
2:0	ReqLnkIndx . Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the link index to use for requests when the component instance of LinkTgtMode register, and its ReqLnkSelMode field, selects Register Mode AccessType: SecurePart ? Read-only : Read-write.

D18F4x1EC [Link Target Mode for CAKE] (DF::LinkTgtMode_CAKE)

Reset: 0000_0000h.

This register is used to determine how components derive the FTI LinkId for remote destinations. All components, except CAKE and TCDX, implement the Link Target Mode Register. On systems that support multiple link sets, CAKE is required to implement LnkIndexMask.

_instBCST_aliasHOST; D18F4x1EC_inst[CAKE[7:0]]_aliasHOST; D18F4x1EC_x[34:2D]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]_instBCST_aliasSMN; D18F4x000001EC; D18F4=4900_4000h_inst[CAKE[7:0]]_aliasSMN; D18F4x000001EC_x[34:2D]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]

Bits	Description
31:29	Reserved.
28	LnkIndxShft. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the number of bits that a FabricID is right-shifted such that the bits which determine the link index are put in bits 2:0. The component instance of register LinkTgtMode, and its LnkIndxMask field, is used (after the shift) to mask out unused bits. AccessType: SecurePart ? Read-only : Read-write.
27	Reserved.
26:24	LnkIndxMask. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. The mask of bits within a LinkID which are the link index. AccessType: SecurePart ? Read-only : Read-write.
23:21	Reserved.
20	BrdcstPrbRspLnkIdMode. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link ID for broadcast probe responses. 1b: Probe LinkId Mode The link index is taken from the broadcast probe's LinkID. Unless there is only one link index set, using this mode may cause the LinkID to mismatch what is in the routing tables for the response node. The use of this mode is intended for only single link-index systems. 0b: DirPrb Mode The LinkID is selected as if this was a directed probe response (using DirPrbRspLnkSelMode). This mode is used when there is more than one link set and the probe response packet FabricRcv has to be masked.
19:18	DirPrbRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for directed probe responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
17:16	DirPrbRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for directed probe responses, and, when BrdcstPrbRspLnkSelMode is 1b, the link index for broadcast probe responses. The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId. 00b: Register Mode The link index is taken from LinkTargetConfig[DirPrbRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkIndxShft and then ANDed with LnkIndxMask). 10b : Reserved 11b : Reserved
15:14	DirPrbLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for directed probes when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
13:12	DirPrbLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write.

	Description: Specifies the mode that is used to select the link index for directed probes. The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId 00b: Register Mode The link index is taken from LinkTargetConfig[DirPrbLnkIndx]. 01b: FabricRcv Mode The link index is taken from the directed probe's FabricRcv (shifted by LnkJndxShft and then ANDed with LnkJndxMask). 10b : Reserved 11b : Reserved
11:10	SrcRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for source responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
9:8	SrcRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for source responses (SrcDone, MemFetch, MemCancel). The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId. 00b: Register Mode The link index is taken from LinkTargetConfig[SrcRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkJndxShft and then ANDed with LnkJndxMask) 10b : Reserved 11b : Reserved
7:6	TgtRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for target responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
5:4	TgtRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for target responses (TgtDone(D)). The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId. 00b: Register Mode The link index is taken from LinkTargetConfig[TgtRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkJndxShft and then ANDed with LnkJndxMask) 10b : ReqSrcLnkId Mode
3:2	ReqLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for requests when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
1:0	ReqLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for requests 00b: Register Mode The link index is taken from the register LinkTargetConfig and its field, ReqLnkIndx. 01b: FabricRcv Mode The link index is taken from the request packet's FabricRcv (shifted by LnkJndxShft and then ANDed with LnkJndxMask) 10b : Reserved 11b : Reserved

D18F4x1EC [Link Target Mode for CCM] (DF::LinkTgtMode_CCM)

Reset: 0000_0000h.

This register is used to determine how components derive the FTI LinkId for remote destinations. All components, except CAKE and TCDX, implement the Link Target Mode Register. On systems that support multiple link sets, CAKE is required to implement LnkIndexMask.

_instBCST_aliasHOST; D18F4x1EC_inst[ACM[3:0],CCM[7:0]]_aliasHOST; D18F4x1EC_x1[B:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]_instBCST_aliasSMN; D18F4x000001EC; D18F4=4900_4000h_inst[ACM[3:0],CCM[7:0]]_aliasSMN; D18F4x000001EC_x1[B:0]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[B:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]

Bits	Description
31:29	Reserved.
28	LnkIndxShft. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the number of bits that a FabricID is right-shifted such that the bits which determine the link index are put in bits 2:0. The component instance of register LinkTgtMode, and its LnkIndxMask field, is used (after the shift) to mask out unused bits. AccessType: SecurePart ? Read-only : Read-write.
27	Reserved.
26:24	LnkIndxMask. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. The mask of bits within a LinkID which are the link index. AccessType: SecurePart ? Read-only : Read-write.
23:21	Reserved.
20	BrdcstPrbRspLnkIdMode. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link ID for broadcast probe responses. 1b: Probe LinkId Mode The link index is taken from the broadcast probe's LinkID. Unless there is only one link index set, using this mode may cause the LinkID to mismatch what is in the routing tables for the response node. The use of this mode is intended for only single link-index systems. 0b: DirPrb Mode The LinkID is selected as if this was a directed probe response (using DirPrbRspLnkSelMode). This mode is used when there is more than one link set and the probe response packet FabricRcv has to be masked.
19:18	DirPrbRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for directed probe responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
17:16	DirPrbRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for directed probe responses, and, when BrdcstPrbRspLnkSelMode is 1b, the link index for broadcast probe responses. The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId. 00b: Register Mode The link index is taken from LinkTargetConfig[DirPrbRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkIndxShft and then ANDed with LnkIndxMask). 10b : Reserved 11b : Reserved
15:12	Reserved.
11:10	SrcRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for source responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
9:8	SrcRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration].

	AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for source responses (SrcDone, MemFetch, MemCancel). The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId. 00b: Register Mode The link index is taken from LinkTargetConfig[SrcRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkJndxShft and then ANDed with LnkJndxMask) 10b : Reserved 11b : Reserved
7:4	Reserved.
3:2	ReqLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for requests when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
1:0	ReqLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for requests 00b: Register Mode The link index is taken from the register LinkTargetConfig and its field, ReqLnkJndx. 01b: FabricRcv Mode The link index is taken from the request packet's FabricRcv (shifted by LnkJndxShft and then ANDed with LnkJndxMask) 10b : Reserved 11b : Reserved

D18F4x1EC [Link Target Mode for CS] (DF::LinkTgtMode_CS)

Reset: 0000_0000h.

This register is used to determine how components derive the FTI LinkId for remote destinations. All components, except CAKE and TCDX, implement the Link Target Mode Register. On systems that support multiple link sets, CAKE is required to implement LnkIndexMask.

_instBCST_aliasHOST; D18F4x1EC_inst[CMP[3:0],CS[11:0]]_aliasHOST; D18F4x1EC_x0[F:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]_instBCST_aliasSMN; D18F4x000001EC; D18F4=4900_4000h_inst[CMP[3:0],CS[11:0]]_aliasSMN; D18F4x000001EC_x0[F:0]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]

Bits	Description
31:29	Reserved.
28	LnkIndxShft. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the number of bits that a FabricID is right-shifted such that the bits which determine the link index are put in bits 2:0. The component instance of register LinkTgtMode, and its LnkIndxMask field, is used (after the shift) to mask out unused bits. AccessType: SecurePart ? Read-only : Read-write.
27	Reserved.
26:24	LnkIndxMask. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. The mask of bits within a LinkID which are the link index. AccessType: SecurePart ? Read-only : Read-write.
23:16	Reserved.
15:14	DirPrbLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for directed probes when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
13:12	DirPrbLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for directed probes. The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId 00b: Register Mode The link index is taken from LinkTargetConfig[DirPrbLnkIndx]. 01b: Reserved. 10b : Reserved 11b : Reserved
11:8	Reserved.
7:6	TgtRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for target responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
5:4	TgtRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for target responses (TgtDone(D)). The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId. 00b: Register Mode The link index is taken from LinkTargetConfig[TgtRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkIndxShft and then ANDed with LnkIndxMask) 10b : ReqSrcLnkId Mode
3:2	ReqLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for requests when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.

1:0	ReqLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration].
	AccessType: SecurePart ? Read-only : Read-write.
	Description: Specifies the mode that is used to select the link index for requests 00b: Register Mode The link index is taken from the register LinkTargetConfig and its field, ReqLnkIndx. 01b: FabricRcv Mode The link index is taken from the request packet's FabricRcv (shifted by LnkIndxShft and then ANDed with LnkIndxMask) 10b : Reserved 11b : Reserved

D18F4x1EC [Link Target Mode for ICNG] (DF::LinkTgtMode_ICNG)

Reset: 0000_0000h.

This register is used to determine how components derive the FTI LinkId for remote destinations. All components, except CAKE and TCDX, implement the Link Target Mode Register. On systems that support multiple link sets, CAKE is required to implement LnkIndexMask.

_instBCST_aliasHOST; D18F4x1EC_inst[ICNG[3:0]]_aliasHOST; D18F4x1EC_x2[B:8]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]_instBCST_aliasSMN; D18F4x000001EC; D18F4=4900_4000h_inst[ICNG[3:0]]_aliasSMN; D18F4x000001EC_x2[B:8]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[B:8], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]

Bits	Description
31:29	Reserved.
28	LnkIndxShft. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the number of bits that a FabricID is right-shifted such that the bits which determine the link index are put in bits 2:0. The component instance of register LinkTgtMode, and its LnkIndxMask field, is used (after the shift) to mask out unused bits. AccessType: SecurePart ? Read-only : Read-write.
27	Reserved.
26:24	LnkIndxMask. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. The mask of bits within a LinkID which are the link index. AccessType: SecurePart ? Read-only : Read-write.
23:21	Reserved.
20	BrdcstPrbRspLnkIdMode. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link ID for broadcast probe responses. 1b: Probe LinkId Mode The link index is taken from the broadcast probe's LinkID. Unless there is only one link index set, using this mode may cause the LinkID to mismatch what is in the routing tables for the response node. The use of this mode is intended for only single link-index systems. 0b: DirPrb Mode The LinkID is selected as if this was a directed probe response (using DirPrbRspLnkSelMode). This mode is used when there is more than one link set and the probe response packet FabricRcv has to be masked.
19:18	DirPrbRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for directed probe responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
17:16	DirPrbRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for directed probe responses, and, when BrdcstPrbRspLnkSelMode is 1b, the link index for broadcast probe responses. The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId. 00b: Register Mode The link index is taken from LinkTargetConfig[DirPrbRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkIndxShft and then ANDed with LnkIndxMask). 10b : Reserved 11b : Reserved
15:14	DirPrbLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for directed probes when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
13:12	DirPrbLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write.

	Description: Specifies the mode that is used to select the link index for directed probes. The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId 00b: Register Mode The link index is taken from LinkTargetConfig[DirPrbLnkIndx]. 01b: FabricRcv Mode The link index is taken from the directed probe's FabricRcv (shifted by LnkJndxShft and then ANDed with LnkJndxMask). 10b : Reserved 11b : Reserved
11:8	Reserved.
7:6	TgtRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for target responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
5:4	TgtRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for target responses (TgtDone(D)). The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId. 00b: Register Mode The link index is taken from LinkTargetConfig[TgtRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkJndxShft and then ANDed with LnkJndxMask) 10b : ReqSrcLnkId Mode
3:0	Reserved.

D18F4x1EC [Link Target Mode for IOM] (DF::LinkTgtMode_IOM)

Reset: 0000_0000h.

This register is used to determine how components derive the FTI LinkId for remote destinations. All components, except CAKE and TCDX, implement the Link Target Mode Register. On systems that support multiple link sets, CAKE is required to implement LnkIndexMask.

_instBCST_aliasHOST; D18F4x1EC_inst[IOM[3:0]]_aliasHOST; D18F4x1EC_x2[3:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[3:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]_instBCST_aliasSMN; D18F4x000001EC; D18F4=4900_4000h_inst[IOM[3:0]]_aliasSMN; D18F4x000001EC_x2[3:0]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[3:0], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]

Bits	Description
31:29	Reserved.
28	LnkIndxShft. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the number of bits that a FabricID is right-shifted such that the bits which determine the link index are put in bits 2:0. The component instance of register LinkTgtMode, and its LnkIndxMask field, is used (after the shift) to mask out unused bits. AccessType: SecurePart ? Read-only : Read-write.
27	Reserved.
26:24	LnkIndxMask. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. The mask of bits within a LinkID which are the link index. AccessType: SecurePart ? Read-only : Read-write.
23:12	Reserved.
11:10	SrcRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for source responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
9:8	SrcRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for source responses (SrcDone, MemFetch, MemCancel). The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId. 00b: Register Mode The link index is taken from LinkTargetConfig[SrcRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkIndxShft and then ANDed with LnkIndxMask) 10b : Reserved 11b : Reserved
7:4	Reserved.
3:2	ReqLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for requests when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
1:0	ReqLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for requests 00b: Register Mode The link index is taken from the register LinkTargetConfig and its field, ReqLnkIndx. 01b: FabricRcv Mode The link index is taken from the request packet's FabricRcv (shifted by LnkIndxShft and then ANDed with LnkIndxMask) 10b : Reserved 11b : Reserved

D18F4x1EC [Link Target Mode for NCM] (DF::LinkTgtMode_NCM)

Reset: 0000_0000h.

This register is used to determine how components derive the FTI LinkId for remote destinations. All components, except CAKE and TCDX, implement the Link Target Mode Register. On systems that support multiple link sets, CAKE is required to implement LnkIndexMask.

_instBCST_aliasHOST; D18F4x1EC_inst[NCM[3:0]]_aliasHOST; D18F4x1EC_x1[F:C]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[F:C], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]_instBCST_aliasSMN; D18F4x000001EC; D18F4=4900_4000h_inst[NCM[3:0]]_aliasSMN; D18F4x000001EC_x1[F:C]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x1[F:C], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]

Bits	Description
31:29	Reserved.
28	LnkIndxShft. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the number of bits that a FabricID is right-shifted such that the bits which determine the link index are put in bits 2:0. The component instance of register LinkTgtMode, and its LnkIndxMask field, is used (after the shift) to mask out unused bits. AccessType: SecurePart ? Read-only : Read-write.
27	Reserved.
26:24	LnkIndxMask. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. The mask of bits within a LinkID which are the link index. AccessType: SecurePart ? Read-only : Read-write.
23:4	Reserved.
3:2	ReqLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for requests when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
1:0	ReqLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for requests 00b: Register Mode The link index is taken from the register LinkTargetConfig and its field, ReqLnkIndx. 01b: FabricRcv Mode The link index is taken from the request packet's FabricRcv (shifted by LnkIndxShft and then ANDed with LnkIndxMask) 10b : Reserved 11b : Reserved

D18F4x1EC [Link Target Mode for NCS and IOS] (DF::LinkTgtMode_NCSIOS)

Reset: 0000_0000h.

This register is used to determine how components derive the FTI LinkId for remote destinations. All components, except CAKE and TCDX, implement the Link Target Mode Register. On systems that support multiple link sets, CAKE is required to implement LnkIndexMask.

_instBCST_aliasHOST; D18F4x1EC_inst[IOS[3:0]]_aliasHOST; D18F4x1EC_x2[7:4]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[7:4], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]_instBCST_aliasSMN; D18F4x000001EC; D18F4=4900_4000h_inst[IOS[3:0]]_aliasSMN; D18F4x000001EC_x2[7:4]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2[7:4], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]

Bits	Description
31:29	Reserved.
28	LnkIndxShft. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the number of bits that a FabricID is right-shifted such that the bits which determine the link index are put in bits 2:0. The component instance of register LinkTgtMode, and its LnkIndxMask field, is used (after the shift) to mask out unused bits. AccessType: SecurePart ? Read-only : Read-write.
27	Reserved.
26:24	LnkIndxMask. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. The mask of bits within a LinkID which are the link index. AccessType: SecurePart ? Read-only : Read-write.
23:8	Reserved.
7:6	TgtRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for target responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
5:4	TgtRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for target responses (TgtDone(D)). The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId. 00b: Register Mode The link index is taken from LinkTargetConfig[TgtRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkIndxShft and then ANDed with LnkIndxMask) 10b : ReqSrcLnkId Mode
3:2	ReqLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for requests when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
1:0	ReqLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for requests 00b: Register Mode The link index is taken from the register LinkTargetConfig and its field, ReqLnkIndx. 01b: FabricRcv Mode The link index is taken from the request packet's FabricRcv (shifted by LnkIndxShft and then ANDed with LnkIndxMask) 10b : Reserved 11b : Reserved

D18F4x1EC [Link Target Mode for PIE] (DF::LinkTgtMode_PIE)

Reset: 0000_0000h.

This register is used to determine how components derive the FTI LinkId for remote destinations. All components, except CAKE and TCDX, implement the Link Target Mode Register. On systems that support multiple link sets, CAKE is required to implement LnkIndexMask.

_instBCST_aliasHOST; D18F4x1EC_instPIE0_aliasHOST; D18F4x1EC_x2C; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]_instBCST_aliasSMN; D18F4x000001EC; D18F4=4900_4000h_instPIE0_aliasSMN; D18F4x000001EC_x2C; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x2C, IndCfgAccFuncNum=4, IndCfgAccRegNum=0x07B]

Bits	Description
31:29	Reserved.
28	LnkIndxShft. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the number of bits that a FabricID is right-shifted such that the bits which determine the link index are put in bits 2:0. The component instance of register LinkTgtMode, and its LnkIndxMask field, is used (after the shift) to mask out unused bits. AccessType: SecurePart ? Read-only : Read-write.
27	Reserved.
26:24	LnkIndxMask. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. The mask of bits within a LinkID which are the link index. AccessType: SecurePart ? Read-only : Read-write.
23:21	Reserved.
20	BrdcstPrbRspLnkIdMode. Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link ID for broadcast probe responses. 1b: Probe LinkId Mode The link index is taken from the broadcast probe's LinkID. Unless there is only one link index set, using this mode may cause the LinkID to mismatch what is in the routing tables for the response node. The use of this mode is intended for only single link-index systems. 0b: DirPrb Mode The LinkID is selected as if this was a directed probe response (using DirPrbRspLnkSelMode). This mode is used when there is more than one link set and the probe response packet FabricRcv has to be masked.
19:18	DirPrbRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for directed probe responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
17:16	DirPrbRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for directed probe responses, and, when BrdcstPrbRspLnkSelMode is 1b, the link index for broadcast probe responses. The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId.00b: Register Mode The link index is taken from LinkTargetConfig[DirPrbRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkIndxShft and then ANDed with LnkIndxMask). 10b : Reserved 11b : Reserved
15:14	DirPrbLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for directed probes when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
13:12	DirPrbLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write.

	Description: Specifies the mode that is used to select the link index for directed probes. The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId 00b: Register Mode The link index is taken from LinkTargetConfig[DirPrbLnkIndx]. 01b: FabricRcv Mode The link index is taken from the directed probe's FabricRcv (shifted by LnkJndxShft and then ANDed with LnkJndxMask). 10b : Reserved 11b : Reserved
11:10	SrcRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for source responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
9:8	SrcRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for source responses (SrcDone, MemFetch, MemCancel). The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId. 00b: Register Mode The link index is taken from LinkTargetConfig[SrcRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkJndxShft and then ANDed with LnkJndxMask) 10b : Reserved 11b : Reserved
7:6	TgtRspLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for target responses when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
5:4	TgtRspLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for target responses (TgtDone(D)). The link set is selected from the FabricRcv and then OR'd with the link index to form the LinkId. 00b: Register Mode The link index is taken from LinkTargetConfig[TgtRspLnkIndx]. 01b: FabricRcv Mode The link index is taken from the response packet's FabricRcv (shifted by LnkJndxShft and then ANDed with LnkJndxMask) 10b : ReqSrcLnkId Mode
3:2	ReqLnkBiasMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. Bias mode used to select link ID for requests when the number of links per linkset is not a power of 2. AccessType: SecurePart ? Read-only : Read-write.
1:0	ReqLnkSelMode. Reset: 0h. Init: 11.5.3 [Register Values Written Based on Configuration]. AccessType: SecurePart ? Read-only : Read-write. Description: Specifies the mode that is used to select the link index for requests 00b: Register Mode The link index is taken from the register LinkTargetConfig and its field, ReqLnkJndx. 01b: FabricRcv Mode The link index is taken from the request packet's FabricRcv (shifted by LnkJndxShft and then ANDed with LnkJndxMask) 10b : Reserved 11b : Reserved

D18F4xBF0 [Multicast Type Mask 6] (DF::McastTypeMask6)

Reset: 3F3F_3F3Fh.

Mcast type mask 6. Each register field corresponds to a TCDX port vector that is the allowed set of mcast outputs, both coherency and sys mgt, per input port.

_instBCST_aliasHOST; D18F4xBF0_inst[TCDX[15:0]]_aliasHOST; D18F4xBF0_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x2FC]_instBCST_aliasSMN; D18F4x00000BF0; D18F4=4900_4000h_inst[TCDX[15:0]]_aliasSMN; D18F4x00000BF0_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x2FC]

Bits	Description
31:30	Reserved.
29:24	McastMaskP3LclDst. Reset: 3Fh. Specifies the superset of all output ports for mcast probes entering P3 and there is a local destination AccessType: SecurePart ? Read-only : Read-write. Init: <u>_inst[TCDX[2,0]]_alias[SMN,HOST]</u> : 23h Init: <u>_inst[TCDX[4:3]]_alias[SMN,HOST]</u> : 00h Init: <u>_instTCDX5_alias[SMN,HOST]</u> : 22h Init: <u>_instTCDX6_alias[SMN,HOST]</u> : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 30h Init: <u>_inst[TCDX[8:7]]_alias[SMN,HOST]</u> : 00h Init: <u>_instTCDX9_alias[SMN,HOST]</u> : 31h Init: <u>_inst[TCDX[11:10]]_alias[SMN,HOST]</u> : 30h Init: <u>_inst[TCDX[14,12]]_alias[SMN,HOST]</u> : 23h Init: <u>_instTCDX15_alias[SMN,HOST]</u> : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 30h
23:22	Reserved.
21:16	McastMaskP2LclDst. Reset: 3Fh. Specifies the superset of all output ports for mcast probes entering P2 and there is a local destination AccessType: SecurePart ? Read-only : Read-write. Init: <u>_inst[TCDX[2,0]]_alias[SMN,HOST]</u> : 23h Init: <u>_inst[TCDX[4:3]]_alias[SMN,HOST]</u> : 00h Init: <u>_instTCDX5_alias[SMN,HOST]</u> : 11h Init: <u>_instTCDX6_alias[SMN,HOST]</u> : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 03h Init: <u>_inst[TCDX[8:7]]_alias[SMN,HOST]</u> : 00h Init: <u>_instTCDX9_alias[SMN,HOST]</u> : 08h Init: <u>_instTCDX10_alias[SMN,HOST]</u> : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 30h Init: <u>_instTCDX11_alias[SMN,HOST]</u> : 08h Init: <u>_inst[TCDX[14,12]]_alias[SMN,HOST]</u> : 23h Init: <u>_instTCDX15_alias[SMN,HOST]</u> : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 30h
15:14	Reserved.
13:8	McastMaskP1LclDst. Reset: 3Fh. Specifies the superset of all output ports for mcast probes entering P1 and there is a local destination AccessType: SecurePart ? Read-only : Read-write. Init: <u>_instTCDX0_alias[SMN,HOST]</u> : 00h Init: <u>_instTCDX1_alias[SMN,HOST]</u> : 05h Init: <u>_instTCDX2_alias[SMN,HOST]</u> : 00h Init: <u>_inst[TCDX[4:3]]_alias[SMN,HOST]</u> : 1Ch Init: <u>_inst[TCDX[6:5]]_alias[SMN,HOST]</u> : 31h Init: <u>_inst[TCDX[8:7]]_alias[SMN,HOST]</u> : 00h Init: <u>_instTCDX9_alias[SMN,HOST]</u> : 08h Init: <u>_instTCDX10_alias[SMN,HOST]</u> : 30h Init: <u>_instTCDX11_alias[SMN,HOST]</u> : 08h Init: <u>_instTCDX12_alias[SMN,HOST]</u> : 00h Init: <u>_instTCDX13_alias[SMN,HOST]</u> : 05h Init: <u>_instTCDX14_alias[SMN,HOST]</u> : 00h

	Init: _instTCDX15_alias[SMN,HOST]: ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 30h
7:6	Reserved.
5:0	McastMaskP0LclDst. Reset: 3Fh. Specifies the superset of all output ports for mcast probes entering P0 and there is a local destination AccessType: SecurePart ? Read-only : Read-write. Init: _instTCDX0_alias[SMN,HOST]: 00h Init: _instTCDX1_alias[SMN,HOST]: 32h Init: _instTCDX2_alias[SMN,HOST]: 00h Init: _inst[TCDX[4:3]]_alias[SMN,HOST]: ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 1Eh Init: _instTCDX5_alias[SMN,HOST]: ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 06h Init: _instTCDX6_alias[SMN,HOST]: 1Eh Init: _instTCDX7_alias[SMN,HOST]: ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 1Ch : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 1Eh Init: _instTCDX8_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 1Eh : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 0Ch Init: _instTCDX9_alias[SMN,HOST]: 39h Init: _instTCDX10_alias[SMN,HOST]: ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 30h Init: _inst[TCDX[12:11]]_alias[SMN,HOST]: 00h Init: _instTCDX13_alias[SMN,HOST]: 32h Init: _instTCDX14_alias[SMN,HOST]: 00h Init: _instTCDX15_alias[SMN,HOST]: ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 30h

D18F4xBF4 [Multicast Type Mask 7] (DF::McastTypeMask7)

Reset: 0000_3F3Fh.

Mcast type mask 7. Each register field corresponds to a TCDX port vector that is the allowed set of mcast outputs, both coherency and sys mgt, per input port.

_instBCST_aliasHOST; D18F4xBF4_inst[TCDX[15:0]]_aliasHOST; D18F4xBF4_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x2FD]_instBCST_aliasSMN; D18F4x00000BF4; D18F4=4900_4000h_inst[TCDX[15:0]]_aliasSMN; D18F4x00000BF4_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x2FD]

Bits	Description
31:14	Reserved.
13:8	McastMaskP5LclDst. Reset: 3Fh. Specifies the superset of all output ports for mcast probes entering P5 and there is a local destination AccessType: SecurePart ? Read-only : Read-write. Init: <u>_instTCDX0_alias[SMN,HOST]</u> : 03h Init: <u>_instTCDX1_alias[SMN,HOST]</u> : 13h Init: <u>_instTCDX2_alias[SMN,HOST]</u> : 03h Init: <u>_instTCDX5_alias[SMN,HOST]</u> : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 18h Init: <u>_instTCDX6_alias[SMN,HOST]</u> : 1Eh Init: <u>_instTCDX7_alias[SMN,HOST]</u> : 1Fh Init: <u>_instTCDX8_alias[SMN,HOST]</u> : 1Eh Init: <u>_inst[TCDX[11:09]]_alias[SMN,HOST]</u> : 00h Init: <u>_instTCDX12_alias[SMN,HOST]</u> : 03h Init: <u>_instTCDX13_alias[SMN,HOST]</u> : 13h Init: <u>_instTCDX14_alias[SMN,HOST]</u> : 03h Init: <u>_instTCDX15_alias[SMN,HOST]</u> : 00h
7:6	Reserved.
5:0	McastMaskP4LclDst. Reset: 3Fh. Specifies the superset of all output ports for mcast probes entering P4 and there is a local destination AccessType: SecurePart ? Read-only : Read-write. Init: <u>_instTCDX0_alias[SMN,HOST]</u> : 23h Init: <u>_instTCDX1_alias[SMN,HOST]</u> : 28h Init: <u>_instTCDX2_alias[SMN,HOST]</u> : 23h Init: <u>_inst[TCDX[4:3]]_alias[SMN,HOST]</u> : 00h Init: <u>_instTCDX5_alias[SMN,HOST]</u> : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 22h : ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 23h Init: <u>_instTCDX6_alias[SMN,HOST]</u> : 23h Init: <u>_instTCDX7_alias[SMN,HOST]</u> : 2Fh Init: <u>_instTCDX8_alias[SMN,HOST]</u> : 01h Init: <u>_instTCDX9_alias[SMN,HOST]</u> : 00h Init: <u>_instTCDX10_alias[SMN,HOST]</u> : 20h Init: <u>_instTCDX11_alias[SMN,HOST]</u> : 00h Init: <u>_instTCDX12_alias[SMN,HOST]</u> : 23h Init: <u>_instTCDX13_alias[SMN,HOST]</u> : 28h Init: <u>_instTCDX14_alias[SMN,HOST]</u> : 23h Init: <u>_instTCDX15_alias[SMN,HOST]</u> : 00h

D18F4xBF8 [Multicast Type Mask 8] (DF::McastTypeMask8)

Reset: 3F3F_3F3Fh.

Mcast type mask 8. Each register field corresponds to a TCDX port vector that is the allowed set of mcast outputs, both coherency and sys mgt, per input port.

_instBCST_aliasHOST; D18F4xBF8_inst[TCDX[15:0]]_aliasHOST; D18F4xBF8_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x2FE]_instBCST_aliasSMN; D18F4x00000BF8; D18F4=4900_4000h_inst[TCDX[15:0]]_aliasSMN; D18F4x00000BF8_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x2FE]

Bits	Description
31:30	Reserved.
29:24	McastMaskP3RmtDst . Reset: 3Fh. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the superset of all output ports for mcast probes entering P3 and there is a remote destination AccessType: SecurePart ? Read-only : Read-write.
23:22	Reserved.
21:16	McastMaskP2RmtDst . Reset: 3Fh. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the superset of all output ports for mcast probes entering P2 and there is a remote destination AccessType: SecurePart ? Read-only : Read-write.
15:14	Reserved.
13:8	McastMaskP1RmtDst . Reset: 3Fh. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the superset of all output ports for mcast probes entering P1 and there is a remote destination AccessType: SecurePart ? Read-only : Read-write.
7:6	Reserved.
5:0	McastMaskP0RmtDst . Reset: 3Fh. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the superset of all output ports for mcast probes entering P0 and there is a remote destination AccessType: SecurePart ? Read-only : Read-write.

D18F4xBFC [Multicast Type Mask 9] (DF::McastTypeMask9)

Reset: 0000_3F3Fh.

Mcast type mask 9. Each register field corresponds to a TCDX port vector that is the allowed set of mcast outputs, both coherency and sys mgt, per input port.

_instBCST_aliasHOST; D18F4xBFC_inst[TCDX[15:0]]_aliasHOST; D18F4xBFC_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x2FF]_instBCST_aliasSMN; D18F4x00000BFC; D18F4=4900_4000h_inst[TCDX[15:0]]_aliasSMN; D18F4x00000BFC_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x2FF]

Bits	Description
31:14	Reserved.
13:8	McastMaskP5RmtDst . Reset: 3Fh. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the superset of all output ports for mcast probes entering P5 and there is a remote destination AccessType: SecurePart ? Read-only : Read-write.
7:6	Reserved.
5:0	McastMaskP4RmtDst . Reset: 3Fh. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the superset of all output ports for mcast probes entering P4 and there is a remote destination AccessType: SecurePart ? Read-only : Read-write.

D18F4xC00...D18F4xCFC [Remote Multicast Routing Tables] (DF::RemoteMcastRoutingTable)

A block of 256 registers are reserved for remote Mcast routing tables. Each register corresponds to a {SocketID,DieID} that can send probe multicasts or system management broadcasts in the coherent fabric and contains the Mcast route from that {SocketID,DieID} to the masters and links. RemoteMcastRoutingTable0 corresponds to {SocketID,DieID} 0; RemoteMcastRoutingTable5 corresponds to {SocketID,DieID} 5; etc. For more about SocketID and DieID, see 11.4 [Data Fabric Instance and Fabric IDs].

_instBCST_n[63:0]_aliasHOST; D18F4xC[[F:0][C,8,4,0]]

_inst[TCDX[15:0]]_n0_aliasHOST; D18F4xC00_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x300]

_inst[TCDX[15:0]]_n1_aliasHOST; D18F4xC04_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x301]

_inst[TCDX[15:0]]_n2_aliasHOST; D18F4xC08_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x302]

_inst[TCDX[15:0]]_n3_aliasHOST; D18F4xC0C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x303]

_inst[TCDX[15:0]]_n4_aliasHOST; D18F4xC10_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x304]

_inst[TCDX[15:0]]_n5_aliasHOST; D18F4xC14_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x305]

_inst[TCDX[15:0]]_n6_aliasHOST; D18F4xC18_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x306]

_inst[TCDX[15:0]]_n7_aliasHOST; D18F4xC1C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x307]

_inst[TCDX[15:0]]_n8_aliasHOST; D18F4xC20_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x308]

_inst[TCDX[15:0]]_n9_aliasHOST; D18F4xC24_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x309]

_inst[TCDX[15:0]]_n10_aliasHOST; D18F4xC28_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x30A]

_inst[TCDX[15:0]]_n11_aliasHOST; D18F4xC2C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x30B]

_inst[TCDX[15:0]]_n12_aliasHOST; D18F4xC30_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x30C]

_inst[TCDX[15:0]]_n13_aliasHOST; D18F4xC34_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x30D]

_inst[TCDX[15:0]]_n14_aliasHOST; D18F4xC38_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x30E]

_inst[TCDX[15:0]]_n15_aliasHOST; D18F4xC3C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x30F]

_inst[TCDX[15:0]]_n16_aliasHOST; D18F4xC40_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x310]

_inst[TCDX[15:0]]_n17_aliasHOST; D18F4xC44_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x311]

_inst[TCDX[15:0]]_n18_aliasHOST; D18F4xC48_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x312]

_inst[TCDX[15:0]]_n19_aliasHOST; D18F4xC4C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x313]

_inst[TCDX[15:0]]_n20_aliasHOST; D18F4xC50_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x314]

_inst[TCDX[15:0]]_n21_aliasHOST; D18F4xC54_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x315]

_inst[TCDX[15:0]]_n22_aliasHOST; D18F4xC58_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x316]

_inst[TCDX[15:0]]_n23_aliasHOST; D18F4xC5C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x317]

_inst[TCDX[15:0]]_n24_aliasHOST; D18F4xC60_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x318]

_inst[TCDX[15:0]]_n25_aliasHOST; D18F4xC64_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x319]

_inst[TCDX[15:0]]_n26_aliasHOST; D18F4xC68_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x31A]

_inst[TCDX[15:0]]_n27_aliasHOST; D18F4xC6C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x31B]

_inst[TCDX[15:0]]_n28_aliasHOST; D18F4xC70_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x31C]

[illegible]

_inst[TCDX[15:0]]_n63_aliasSMN; D18F4x0000CFC_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x33F]
_instBCST_n[63:0]_aliasSMN; D18F4x0000_0C[FC,F8,F4,F0,EC,E8,E4,E0,DC,D8,D4,D0,CC,C8,C4,C0,BC,B8,B4,B0,AC,A8,A4,A0,9C,98,94,90,8C,88,84,80,7C,78,74,70,6C,68,64,60,5C,58,54,50,4C,48,44,40,3C,38,34,30,2C,28,24,20,1C,18,14,10,0C,08,04,00]; D18F4=4900_4000h
_inst[TCDX[15:0]]_n0_aliasSMN; D18F4x00000C00_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x300]
_inst[TCDX[15:0]]_n1_aliasSMN; D18F4x00000C04_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x301]
_inst[TCDX[15:0]]_n2_aliasSMN; D18F4x00000C08_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x302]
_inst[TCDX[15:0]]_n3_aliasSMN; D18F4x00000C0C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x303]
_inst[TCDX[15:0]]_n4_aliasSMN; D18F4x00000C10_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x304]
_inst[TCDX[15:0]]_n5_aliasSMN; D18F4x00000C14_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x305]
_inst[TCDX[15:0]]_n6_aliasSMN; D18F4x00000C18_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x306]
_inst[TCDX[15:0]]_n7_aliasSMN; D18F4x00000C1C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x307]
_inst[TCDX[15:0]]_n8_aliasSMN; D18F4x00000C20_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x308]
_inst[TCDX[15:0]]_n9_aliasSMN; D18F4x00000C24_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x309]
_inst[TCDX[15:0]]_n10_aliasSMN; D18F4x00000C28_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x30A]
_inst[TCDX[15:0]]_n11_aliasSMN; D18F4x00000C2C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x30B]
_inst[TCDX[15:0]]_n12_aliasSMN; D18F4x00000C30_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x30C]
_inst[TCDX[15:0]]_n13_aliasSMN; D18F4x00000C34_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x30D]
_inst[TCDX[15:0]]_n14_aliasSMN; D18F4x00000C38_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x30E]
_inst[TCDX[15:0]]_n15_aliasSMN; D18F4x00000C3C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x30F]
_inst[TCDX[15:0]]_n16_aliasSMN; D18F4x00000C40_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x310]
_inst[TCDX[15:0]]_n17_aliasSMN; D18F4x00000C44_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x311]
_inst[TCDX[15:0]]_n18_aliasSMN; D18F4x00000C48_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x312]
_inst[TCDX[15:0]]_n19_aliasSMN; D18F4x00000C4C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x313]
_inst[TCDX[15:0]]_n20_aliasSMN; D18F4x00000C50_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x314]
_inst[TCDX[15:0]]_n21_aliasSMN; D18F4x00000C54_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x315]
_inst[TCDX[15:0]]_n22_aliasSMN; D18F4x00000C58_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x316]
_inst[TCDX[15:0]]_n23_aliasSMN; D18F4x00000C5C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x317]
_inst[TCDX[15:0]]_n24_aliasSMN; D18F4x00000C60_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x318]
_inst[TCDX[15:0]]_n25_aliasSMN; D18F4x00000C64_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x319]
_inst[TCDX[15:0]]_n26_aliasSMN; D18F4x00000C68_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x31A]
_inst[TCDX[15:0]]_n27_aliasSMN; D18F4x00000C6C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x31B]
_inst[TCDX[15:0]]_n28_aliasSMN; D18F4x00000C70_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x31C]
_inst[TCDX[15:0]]_n29_aliasSMN; D18F4x00000C74_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x31D]
_inst[TCDX[15:0]]_n30_aliasSMN; D18F4x00000C78_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x31E]
_inst[TCDX[15:0]]_n31_aliasSMN; D18F4x00000C7C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x31F]

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29:24 **RmtMcastRoute3**. Init: 11.5.3 [Register Values Written Based on Configuration]. RmtMcastRoute3[7:0]: Specifies the Mcast route from this {SocketID,DieID} to the masters and links in this SOC.

AccessType: SecurePart ? Read-only : Read-write.

Reset: _instTCDX0_n[1:0]_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n2_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n3_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n4_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n5_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n6_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n7_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n8_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n9_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n10_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n11_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n12_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n13_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n14_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n15_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n16_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n17_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n18_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n19_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n20_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n21_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n22_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n23_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n24_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n25_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n26_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n27_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n28_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n29_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n30_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n31_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n32_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n33_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n34_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n35_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n36_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n37_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n38_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n39_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n40_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n41_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n42_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n43_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n44_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n45_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n46_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n47_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n48_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n49_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n50_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n51_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n52_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n53_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n54_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n55_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n56_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n57_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n58_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n59_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n60_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n61_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n63_alias[SMN,HOST]
_instTCDX1_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX1_n2_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n4_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n6_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n8_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n10_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n12_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n14_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n16_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n18_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n20_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n22_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n24_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n26_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n28_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n30_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n32_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n34_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n36_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n38_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n40_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n42_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n44_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n46_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n48_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n50_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n52_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n54_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n56_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n58_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n60_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n62_alias[SMN,HOST]: 2Dh

Reset: _instTCDX1_n63_alias[SMN,HOST]
_instTCDX2_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX2_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n4_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n6_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n8_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n10_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n12_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n14_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n16_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n20_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n22_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n24_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n54_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n63_alias[SMN,HOST]

<u>_instTCDX3_n[1:0]_alias[SMN,HOST]</u>
00h
Reset: <u>_instTCDX3_n2_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n3_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n4_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n5_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n6_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n7_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n8_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n9_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n10_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n11_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n12_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n13_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n14_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n15_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n16_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n17_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n18_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n19_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n20_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n21_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n22_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n23_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n24_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n25_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n26_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n27_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n28_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n29_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n30_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n31_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n32_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n33_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n34_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n35_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n36_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n37_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n38_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n39_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n40_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n41_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n42_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n43_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n44_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n45_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n46_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n47_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n48_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n49_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n50_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n51_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n52_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n53_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n54_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n55_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n56_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n57_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n58_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n59_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n60_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n61_alias[SMN,HOST]</u> : 00h
Reset: <u>_instTCDX3_n62_alias[SMN,HOST]</u> : 1Eh
Reset: <u>_instTCDX3_n63_alias[SMN,HOST]</u>
<u>_instTCDX4_n[1:0]_alias[SMN,HOST]</u>

00h
Reset: _instTCDX4_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n16_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n18_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n20_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n22_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n24_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n26_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n28_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n30_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n32_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n34_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n36_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n38_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n40_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n42_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n44_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n46_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n52_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n54_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n63_alias[SMN,HOST]
_instTCDX5_n[1:0]_alias[SMN,HOST]
00h

Reset: _instTCDX5_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n16_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n18_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n20_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n22_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n24_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n26_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n28_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n30_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n32_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n34_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n36_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n38_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n40_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n42_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n44_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n46_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n52_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n54_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n63_alias[SMN,HOST]
Reset: _instTCDX6_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX6_n2_alias[SMN,HOST]: 33h

Reset: _instTCDX6_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n4_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n6_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n8_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n10_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n12_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n14_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n16_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n18_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n20_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n22_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n24_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n26_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n28_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n30_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n32_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n34_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n36_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n38_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n40_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n42_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n44_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n46_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n48_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n50_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n52_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n54_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n56_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n58_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n60_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n62_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n63_alias[SMN,HOST]
Reset: _instTCDX7_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX7_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n3_alias[SMN,HOST]: 00h

Reset: _instTCDX7_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n16_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n18_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n20_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n22_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n24_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n26_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n28_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n30_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n32_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n34_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n36_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n38_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n40_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n42_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n44_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n46_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n52_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n54_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n63_alias[SMN,HOST]
Reset: _instTCDX8_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX8_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n4_alias[SMN,HOST]: 1Eh

Reset: _instTCDX8_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n16_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n18_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n20_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n22_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n24_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n26_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n28_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n30_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n32_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n34_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n36_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n38_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n40_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n42_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n44_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n46_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n52_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n54_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n63_alias[SMN,HOST]
Reset: _instTCDX9_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX9_n2_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n4_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n5_alias[SMN,HOST]: 00h

Reset: _instTCDX9_n6_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n8_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n10_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n12_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n14_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n16_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n18_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n20_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n22_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n24_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n26_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n28_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n30_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n32_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n34_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n36_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n38_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n40_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n42_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n44_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n46_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n48_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n50_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n52_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n53_alias[SMN,HOST]: 00h
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Reset: _instTCDX9_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n56_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n58_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n60_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n62_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n63_alias[SMN,HOST]
Reset: _instTCDX10_n[63:0]_alias[SMN,HOST]
Reset: _instTCDX11_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX11_n2_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n4_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n5_alias[SMN,HOST]: 00h

Reset: _instTCDX11_n6_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n8_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n10_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n12_alias[SMN,HOST]: 30h
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Reset: _instTCDX11_n32_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n34_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n36_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n38_alias[SMN,HOST]: 30h
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Reset: _instTCDX11_n40_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n41_alias[SMN,HOST]: 00h
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Reset: _instTCDX11_n43_alias[SMN,HOST]: 00h
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Reset: _instTCDX11_n50_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n52_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n54_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n56_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n58_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n60_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n62_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n63_alias[SMN,HOST]
Reset: _instTCDX12_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX12_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n4_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n6_alias[SMN,HOST]: 03h

Reset: _instTCDX12_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n8_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n10_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n12_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n14_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n16_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n20_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n22_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n23_alias[SMN,HOST]: 00h
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Reset: _instTCDX12_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n54_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n63_alias[SMN,HOST]
Reset: _instTCDX13_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX13_n2_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n4_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n6_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n7_alias[SMN,HOST]: 00h

Reset: _instTCDX13_n8_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n10_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n12_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n14_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n16_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n18_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n20_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n22_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n24_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n26_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n28_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n30_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n32_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n34_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n36_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n38_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n40_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n42_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n44_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n46_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n48_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n50_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n52_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n54_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n56_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n58_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n60_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n62_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n63_alias[SMN,HOST]
Reset: _instTCDX14_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX14_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n4_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n6_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n8_alias[SMN,HOST]: 03h

Reset: _instTCDX14_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n10_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n12_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n14_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n16_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n20_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n22_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n24_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n54_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n63_alias[SMN,HOST]
Reset: _instTCDX15_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX15_n2_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n4_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n6_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n8_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n9_alias[SMN,HOST]: 00h

	Reset: _instTCDX15_n10_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n11_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n12_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n13_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n14_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n15_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n16_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n17_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n18_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n19_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n20_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n21_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n22_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n23_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n24_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n25_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n26_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n27_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n28_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n29_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n30_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n31_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n32_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n33_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n34_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n35_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n36_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n37_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n38_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n39_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n40_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n41_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n42_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n43_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n44_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n45_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n46_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n47_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n48_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n49_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n50_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n51_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n52_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n53_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n54_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n55_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n56_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n57_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n58_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n59_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n60_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n61_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n62_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n63_alias[SMN,HOST]: 00h
23:22	Reserved.
21:16	RmtMcastRoute2. Init: 11.5.3 [Register Values Written Based on Configuration]. RmtMcastRoute2[7:0]: Specifies the Mcast route from this {SocketID,DieID} to the masters and links in this SOC. AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_n[1:0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n2_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n3_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n4_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n5_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n6_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n8_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n10_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n12_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n14_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n16_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n20_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n22_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n24_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n54_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n63_alias[SMN,HOST]
Reset: _instTCDX1_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX1_n2_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n4_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n6_alias[SMN,HOST]: 2Dh

Reset: _instTCDX1_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n8_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n10_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n12_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n14_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n16_alias[SMN,HOST]: 2Dh
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Reset: _instTCDX1_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n36_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n38_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n39_alias[SMN,HOST]: 00h
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Reset: _instTCDX1_n41_alias[SMN,HOST]: 00h
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Reset: _instTCDX1_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n44_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n46_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n48_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n50_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n51_alias[SMN,HOST]: 00h
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Reset: _instTCDX1_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n56_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n58_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n60_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n62_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n63_alias[SMN,HOST]
Reset: _instTCDX2_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX2_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n4_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n6_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n7_alias[SMN,HOST]: 00h

Reset: _instTCDX2_n8_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n10_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n12_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n14_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n16_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n20_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n22_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n24_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n54_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n63_alias[SMN,HOST]
Reset: _instTCDX3_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX3_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n8_alias[SMN,HOST]: 1Eh

Reset: _instTCDX3_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n16_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n18_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n20_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n22_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n23_alias[SMN,HOST]: 00h
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Reset: _instTCDX3_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n32_alias[SMN,HOST]: 1Eh
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Reset: _instTCDX3_n36_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n38_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n40_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n41_alias[SMN,HOST]: 00h
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Reset: _instTCDX3_n51_alias[SMN,HOST]: 00h
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Reset: _instTCDX3_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n63_alias[SMN,HOST]
Reset: _instTCDX4_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX4_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n5_alias[SMN,HOST]: 00h
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Reset: _instTCDX4_n9_alias[SMN,HOST]: 00h

Reset: _instTCDX4_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n16_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n18_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n20_alias[SMN,HOST]: 1Eh
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Reset: _instTCDX4_n36_alias[SMN,HOST]: 1Eh
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Reset: _instTCDX4_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n46_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n52_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n54_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n63_alias[SMN,HOST]
Reset: _instTCDX5_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX5_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n5_alias[SMN,HOST]: 00h
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Reset: _instTCDX5_n7_alias[SMN,HOST]: 00h
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Reset: _instTCDX5_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n10_alias[SMN,HOST]: 1Eh

Reset: _instTCDX5_n11_alias[SMN,HOST]: 00h
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Reset: _instTCDX5_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n16_alias[SMN,HOST]: 1Eh
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Reset: _instTCDX5_n19_alias[SMN,HOST]: 00h
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Reset: _instTCDX5_n23_alias[SMN,HOST]: 00h
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Reset: _instTCDX5_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n26_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n28_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n30_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n31_alias[SMN,HOST]: 00h
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Reset: _instTCDX5_n33_alias[SMN,HOST]: 00h
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Reset: _instTCDX5_n36_alias[SMN,HOST]: 1Eh
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Reset: _instTCDX5_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n52_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n54_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n63_alias[SMN,HOST]
Reset: _instTCDX6_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX6_n2_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n4_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n6_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n7_alias[SMN,HOST]: 00h
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Reset: _instTCDX6_n10_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n11_alias[SMN,HOST]: 00h

Reset: _instTCDX6_n12_alias[SMN,HOST]: 33h
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Reset: _instTCDX6_n36_alias[SMN,HOST]: 33h
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Reset: _instTCDX6_n60_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n62_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n63_alias[SMN,HOST]
Reset: _instTCDX7_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX7_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n11_alias[SMN,HOST]: 00h
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Reset: _instTCDX7_n13_alias[SMN,HOST]: 00h
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Reset: _instTCDX7_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n32_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n34_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n36_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n38_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n40_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n42_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n44_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n46_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n52_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n54_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n63_alias[SMN,HOST]
Reset: _instTCDX8_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX8_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n13_alias[SMN,HOST]: 00h

Reset: _instTCDX8_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n16_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n18_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n20_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n22_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n24_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n26_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n28_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n30_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n32_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n34_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n36_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n38_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n40_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n42_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n44_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n46_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n52_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n54_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n63_alias[SMN,HOST]
Reset: _instTCDX9_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX9_n2_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n4_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n6_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n8_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n10_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n12_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n14_alias[SMN,HOST]: 31h

Reset: _instTCDX9_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n16_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n18_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n20_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n22_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n24_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n26_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n28_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n30_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n32_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n34_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n36_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n38_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n40_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n42_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n44_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n46_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n48_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n50_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n52_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n54_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n56_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n58_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n60_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n62_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n63_alias[SMN,HOST]
Reset: _instTCDX10_n[63:0]_alias[SMN,HOST]
Reset: _instTCDX11_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX11_n2_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n4_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n6_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n8_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n10_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n12_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n14_alias[SMN,HOST]: 30h

Reset: _instTCDX11_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n16_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n18_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n20_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n22_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n24_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n26_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n28_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n30_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n32_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n34_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n36_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n38_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n40_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n42_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n44_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n46_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n48_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n50_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n52_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n54_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n56_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n58_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n60_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n62_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n63_alias[SMN,HOST]
Reset: _instTCDX12_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX12_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n4_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n6_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n8_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n10_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n12_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n14_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n15_alias[SMN,HOST]: 00h

Reset: _instTCDX12_n16_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n20_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n22_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n23_alias[SMN,HOST]: 00h
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Reset: _instTCDX12_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n54_alias[SMN,HOST]: 03h
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Reset: _instTCDX12_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n63_alias[SMN,HOST]
Reset: _instTCDX13_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX13_n2_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n4_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n6_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n8_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n10_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n12_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n14_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n16_alias[SMN,HOST]: 2Dh

Reset: _instTCDX13_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n18_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n19_alias[SMN,HOST]: 00h
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Reset: _instTCDX13_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n36_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n38_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n40_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n41_alias[SMN,HOST]: 00h
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Reset: _instTCDX13_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n44_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n46_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n48_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n50_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n52_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n54_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n56_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n58_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n60_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n62_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n63_alias[SMN,HOST]
Reset: _instTCDX14_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX14_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n4_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n6_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n8_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n10_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n12_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n14_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n16_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n17_alias[SMN,HOST]: 00h

Reset: _instTCDX14_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n20_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n22_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n24_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n54_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n63_alias[SMN,HOST]
Reset: _instTCDX15_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX15_n2_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n4_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n6_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n8_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n10_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n12_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n14_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n16_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n18_alias[SMN,HOST]: 30h

	Reset: _instTCDX15_n19_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n20_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n21_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n22_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n23_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n24_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n25_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n26_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n27_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n28_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n29_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n30_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n31_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n32_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n33_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n34_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n35_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n36_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n37_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n38_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n39_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n40_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n41_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n42_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n43_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n44_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n45_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n46_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n47_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n48_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n49_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n50_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n51_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n52_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n53_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n54_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n55_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n56_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n57_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n58_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n59_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n60_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n61_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n62_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n63_alias[SMN,HOST]: 00h
15:14	Reserved.
13:8	RmtMcastRoute1. Init: 11.5.3 [Register Values Written Based on Configuration]. RmtMcastRoute1[7:0]: Specifies the Mcast route from this {SocketID,DieID} to the masters and links in this SOC. AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_n[1:0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n2_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n3_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n4_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n5_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n6_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n7_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n8_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n9_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n10_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n11_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n12_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n13_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n14_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n16_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n20_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n22_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n24_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n54_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n63_alias[SMN,HOST]
Reset: _instTCDX1_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX1_n2_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n4_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n6_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n8_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n10_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n12_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n14_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n15_alias[SMN,HOST]: 00h

Reset: _instTCDX1_n16_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n18_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n20_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n22_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n24_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n26_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n28_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n30_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n32_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n34_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n36_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n38_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n40_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n42_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n44_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n46_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n48_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n50_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n52_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n54_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n56_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n58_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n60_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n62_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n63_alias[SMN,HOST]
Reset: _instTCDX2_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX2_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n4_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n6_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n8_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n10_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n12_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n14_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n16_alias[SMN,HOST]: 03h

Reset: _instTCDX2_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n20_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n22_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n24_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n53_alias[SMN,HOST]: 00h
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Reset: _instTCDX2_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n63_alias[SMN,HOST]
Reset: _instTCDX3_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX3_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n16_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n17_alias[SMN,HOST]: 00h

Reset: _instTCDX3_n18_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n20_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n22_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n24_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n26_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n28_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n30_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n32_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n34_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n36_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n38_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n40_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n42_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n44_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n46_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n52_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n54_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n63_alias[SMN,HOST]
Reset: _instTCDX4_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX4_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n16_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n18_alias[SMN,HOST]: 1Eh

Reset: _instTCDX4_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n20_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n22_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n24_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n26_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n27_alias[SMN,HOST]: 00h
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Reset: _instTCDX4_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n30_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n32_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n34_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n36_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n38_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n40_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n41_alias[SMN,HOST]: 00h
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Reset: _instTCDX4_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n46_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n52_alias[SMN,HOST]: 1Eh
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Reset: _instTCDX4_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n63_alias[SMN,HOST]
Reset: _instTCDX5_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX5_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n14_alias[SMN,HOST]: 1Eh
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Reset: _instTCDX5_n18_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n19_alias[SMN,HOST]: 00h

Reset: _instTCDX5_n20_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n22_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n23_alias[SMN,HOST]: 00h
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Reset: _instTCDX5_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n63_alias[SMN,HOST]
Reset: _instTCDX6_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX6_n2_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n4_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n6_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n8_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n10_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n12_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n13_alias[SMN,HOST]: 00h
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Reset: _instTCDX6_n18_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n20_alias[SMN,HOST]: 33h

Reset: _instTCDX6_n21_alias[SMN,HOST]: 00h
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Reset: _instTCDX6_n23_alias[SMN,HOST]: 00h
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Reset: _instTCDX6_n62_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n63_alias[SMN,HOST]
Reset: _instTCDX7_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX7_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n11_alias[SMN,HOST]: 00h
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Reset: _instTCDX7_n63_alias[SMN,HOST]
Reset: _instTCDX8_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX8_n2_alias[SMN,HOST]: 1Eh
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Reset: _instTCDX8_n57_alias[SMN,HOST]: 00h
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Reset: _instTCDX8_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n63_alias[SMN,HOST]
Reset: _instTCDX9_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX9_n2_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n4_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n6_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n8_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n10_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n12_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n14_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n16_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n18_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n20_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n22_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n23_alias[SMN,HOST]: 00h

Reset: _instTCDX9_n24_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n26_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n28_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n30_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n32_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n34_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n36_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n38_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n40_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n42_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n44_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n46_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n48_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n50_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n52_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n54_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n56_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n58_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n60_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n62_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n63_alias[SMN,HOST]
Reset: _instTCDX10_n[63:0]_alias[SMN,HOST]
Reset: _instTCDX11_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX11_n2_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n4_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n6_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n8_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n10_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n12_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n14_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n16_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n18_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n20_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n22_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n23_alias[SMN,HOST]: 00h

Reset: _instTCDX11_n24_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n26_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n28_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n30_alias[SMN,HOST]: 30h
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Reset: _instTCDX11_n32_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n34_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n36_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n38_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n40_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n42_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n44_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n46_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n47_alias[SMN,HOST]: 00h
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Reset: _instTCDX11_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n50_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n52_alias[SMN,HOST]: 30h
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Reset: _instTCDX11_n55_alias[SMN,HOST]: 00h
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Reset: _instTCDX11_n58_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n60_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n62_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n63_alias[SMN,HOST]
Reset: _instTCDX12_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX12_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n4_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n6_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n8_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n10_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n12_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n14_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n16_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n20_alias[SMN,HOST]: 03h
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Reset: _instTCDX12_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n24_alias[SMN,HOST]: 03h

Reset: _instTCDX12_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n28_alias[SMN,HOST]: 03h
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Reset: _instTCDX12_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n31_alias[SMN,HOST]: 00h
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Reset: _instTCDX12_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n41_alias[SMN,HOST]: 00h
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Reset: _instTCDX12_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n50_alias[SMN,HOST]: 03h
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Reset: _instTCDX12_n52_alias[SMN,HOST]: 03h
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Reset: _instTCDX12_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n63_alias[SMN,HOST]
Reset: _instTCDX13_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX13_n2_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n4_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n6_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n8_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n10_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n12_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n13_alias[SMN,HOST]: 00h
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Reset: _instTCDX13_n15_alias[SMN,HOST]: 00h
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Reset: _instTCDX13_n18_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n19_alias[SMN,HOST]: 00h
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Reset: _instTCDX13_n25_alias[SMN,HOST]: 00h

Reset: _instTCDX13_n26_alias[SMN,HOST]: 2Dh
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Reset: _instTCDX13_n40_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n41_alias[SMN,HOST]: 00h
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Reset: _instTCDX13_n45_alias[SMN,HOST]: 00h
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Reset: _instTCDX13_n49_alias[SMN,HOST]: 00h
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Reset: _instTCDX13_n58_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n60_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n62_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n63_alias[SMN,HOST]
Reset: _instTCDX14_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX14_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n4_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n6_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n8_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n9_alias[SMN,HOST]: 00h
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Reset: _instTCDX14_n26_alias[SMN,HOST]: 03h

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Reset: _instTCDX14_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n63_alias[SMN,HOST]
Reset: _instTCDX15_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX15_n2_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n4_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n6_alias[SMN,HOST]: 30h
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Reset: _instTCDX15_n8_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n10_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n11_alias[SMN,HOST]: 00h
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Reset: _instTCDX15_n16_alias[SMN,HOST]: 30h
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Reset: _instTCDX15_n27_alias[SMN,HOST]: 00h

	Reset: _instTCDX15_n28_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n29_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n30_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n31_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n32_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n33_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n34_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n35_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n36_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n37_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n38_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n39_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n40_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n41_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n42_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n43_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n44_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n45_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n46_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n47_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n48_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n49_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n50_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n51_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n52_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n53_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n54_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n55_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n56_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n57_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n58_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n59_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n60_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n61_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n62_alias[SMN,HOST]: 30h
	Reset: _instTCDX15_n63_alias[SMN,HOST]: 00h
7:6	Reserved.
5:0	RmtMcastRoute0 . Init: 11.5.3 [Register Values Written Based on Configuration]. RmtMcastRoute0[7:0]: Specifies the Mcast route from this {SocketID,DieID} to the masters and links in this SOC. AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_n[1:0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n2_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n3_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n4_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n5_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n6_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n7_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n8_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n9_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n10_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n11_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n12_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n13_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n14_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n15_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n16_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n17_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n18_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n19_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n20_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n21_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n22_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n23_alias[SMN,HOST]: 00h

Reset: _instTCDX0_n24_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n54_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX0_n63_alias[SMN,HOST]
Reset: _instTCDX1_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX1_n2_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n4_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n6_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n8_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n10_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n12_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n14_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n16_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n18_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n20_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n22_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n24_alias[SMN,HOST]: 2Dh

Reset: _instTCDX1_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n26_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n28_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n30_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n32_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n34_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n36_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n38_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n40_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n42_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n44_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n46_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n48_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n50_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n52_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n54_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n56_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n58_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n60_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n62_alias[SMN,HOST]: 2Dh
Reset: _instTCDX1_n63_alias[SMN,HOST]
Reset: _instTCDX2_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX2_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n4_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n6_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n8_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n10_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n12_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n14_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n16_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n20_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n22_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n24_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n25_alias[SMN,HOST]: 00h

Reset: _instTCDX2_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n54_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n63_alias[SMN,HOST]
Reset: _instTCDX3_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX3_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n16_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n18_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n20_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n22_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n24_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n26_alias[SMN,HOST]: 1Eh

Reset: _instTCDX3_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n28_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n30_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n32_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n34_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n36_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n38_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n40_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n42_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n44_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n46_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n52_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n54_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX3_n63_alias[SMN,HOST]
Reset: _instTCDX4_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX4_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n16_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n18_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n20_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n22_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n24_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n26_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n27_alias[SMN,HOST]: 00h

Reset: _instTCDX4_n28_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n30_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n32_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n34_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n36_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n38_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n40_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n42_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n44_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n46_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n52_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n54_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX4_n63_alias[SMN,HOST]
Reset: _instTCDX5_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX5_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n10_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n12_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n14_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n16_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n18_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n20_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n22_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n24_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n26_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n28_alias[SMN,HOST]: 1Eh

Reset: _instTCDX5_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n30_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n32_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n34_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n36_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n38_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n40_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n42_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n44_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n46_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n50_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n52_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n53_alias[SMN,HOST]: 00h
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Reset: _instTCDX5_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX5_n63_alias[SMN,HOST]
Reset: _instTCDX6_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX6_n2_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n4_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n6_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n8_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n10_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n12_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n14_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n16_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n18_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n20_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n22_alias[SMN,HOST]: 33h
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Reset: _instTCDX6_n28_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n29_alias[SMN,HOST]: 00h

Reset: _instTCDX6_n30_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n31_alias[SMN,HOST]: 00h
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Reset: _instTCDX6_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n62_alias[SMN,HOST]: 33h
Reset: _instTCDX6_n63_alias[SMN,HOST]
Reset: _instTCDX7_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX7_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n5_alias[SMN,HOST]: 00h
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Reset: _instTCDX7_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n8_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n10_alias[SMN,HOST]: 1Eh
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Reset: _instTCDX7_n51_alias[SMN,HOST]: 00h
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Reset: _instTCDX7_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX7_n63_alias[SMN,HOST]
Reset: _instTCDX8_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX8_n2_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n4_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n6_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n8_alias[SMN,HOST]: 1Eh
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Reset: _instTCDX8_n31_alias[SMN,HOST]: 00h

Reset: _instTCDX8_n32_alias[SMN,HOST]: 1Eh
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Reset: _instTCDX8_n48_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n49_alias[SMN,HOST]: 00h
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Reset: _instTCDX8_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n52_alias[SMN,HOST]: 1Eh
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Reset: _instTCDX8_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n56_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n58_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n60_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n62_alias[SMN,HOST]: 1Eh
Reset: _instTCDX8_n63_alias[SMN,HOST]
Reset: _instTCDX9_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX9_n2_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n4_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n6_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n8_alias[SMN,HOST]: 31h
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Reset: _instTCDX9_n60_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n62_alias[SMN,HOST]: 31h
Reset: _instTCDX9_n63_alias[SMN,HOST]
Reset: _instTCDX10_n[63:0]_alias[SMN,HOST]
Reset: _instTCDX11_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX11_n2_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n4_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n5_alias[SMN,HOST]: 00h
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Reset: _instTCDX11_n8_alias[SMN,HOST]: 30h
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Reset: _instTCDX11_n48_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n50_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n52_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n54_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n56_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n58_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n60_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n62_alias[SMN,HOST]: 30h
Reset: _instTCDX11_n63_alias[SMN,HOST]
_instTCDX12_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX12_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n4_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n6_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n8_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n10_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n12_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n14_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n16_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n20_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n22_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n24_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n33_alias[SMN,HOST]: 00h

Reset: _instTCDX12_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n54_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n63_alias[SMN,HOST]
_instTCDX13_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX13_n2_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n4_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n6_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n8_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n10_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n12_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n14_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n16_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n18_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n20_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n22_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n24_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n26_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n28_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n30_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n32_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n34_alias[SMN,HOST]: 2Dh

Reset: _instTCDX13_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n36_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n38_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n40_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n42_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n44_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n46_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n48_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n50_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n52_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n54_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n56_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n58_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n60_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n62_alias[SMN,HOST]: 2Dh
Reset: _instTCDX13_n63_alias[SMN,HOST]
Reset: _instTCDX14_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX14_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n4_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n6_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n8_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n10_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n12_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n14_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n16_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n18_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n20_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n22_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n24_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n26_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n28_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n30_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n32_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n34_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n35_alias[SMN,HOST]: 00h

Reset: _instTCDX14_n36_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n38_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n40_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n42_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n44_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n46_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n48_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n50_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n52_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n54_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n56_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n58_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n60_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n62_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n63_alias[SMN,HOST]
Reset: _instTCDX15_n[1:0]_alias[SMN,HOST]
00h
Reset: _instTCDX15_n2_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n3_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n4_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n5_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n6_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n7_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n8_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n9_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n10_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n11_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n12_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n13_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n14_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n15_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n16_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n17_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n18_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n19_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n20_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n21_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n22_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n23_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n24_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n25_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n26_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n27_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n28_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n29_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n30_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n31_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n32_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n33_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n34_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n35_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n36_alias[SMN,HOST]: 30h

Reset: _instTCDX15_n37_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n38_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n39_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n40_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n41_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n42_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n43_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n44_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n45_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n46_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n47_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n48_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n49_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n50_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n51_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n52_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n53_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n54_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n55_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n56_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n57_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n58_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n59_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n60_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n61_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n62_alias[SMN,HOST]: 30h
Reset: _instTCDX15_n63_alias[SMN,HOST]: 00h

D18F4xE00 [Node to Link Set Lookup Register 0] (DF::NodeToLinkSet0)

Reset: 0000_0000h.

This register provides a list of links per destination node as defined in the Multi-die routing section of the FTI specification.

_instBCST_aliasHOST; D18F4xE00

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4xE00_x[34:00]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x380]

_instBCST_aliasSMN; D18F4x00000E00; D18F4=4900_4000h

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x00000E00_x[34:00]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x380]

Bits	Description
31:24	Node3LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 3. AccessType: SecurePart ? Read-only : Read-write.
23:16	Node2LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 2. AccessType: SecurePart ? Read-only : Read-write.
15:8	Node1LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 1. AccessType: SecurePart ? Read-only : Read-write.
7:0	Node0LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 0. AccessType: SecurePart ? Read-only : Read-write.

D18F4xE04 [Node to Link Set Lookup Register 1] (DF::NodeToLinkSet1)

Reset: 0000_0000h.

This register provides a list of links per destination node as defined in the Multi-die routing section of the FTI specification.

_instBCST_aliasHOST; D18F4xE04

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4xE04_x[34:00]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x381]

_instBCST_aliasSMN; D18F4x00000E04; D18F4=4900_4000h

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x00000E04_x[34:00]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x381]

Bits	Description
31:24	Node7LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 7. AccessType: SecurePart ? Read-only : Read-write.
23:16	Node6LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 6. AccessType: SecurePart ? Read-only : Read-write.
15:8	Node5LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 5. AccessType: SecurePart ? Read-only : Read-write.
7:0	Node4LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 4. AccessType: SecurePart ? Read-only : Read-write.

D18F4xE08 [Node to Link Set Lookup Register 2] (DF::NodeToLinkSet2)

Reset: 0000_0000h.

This register provides a list of links per destination node as defined in the Multi-die routing section of the FTI specification.

_instBCST_aliasHOST; D18F4xE08

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4xE08_x[34:00]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x382]

_instBCST_aliasSMN; D18F4x00000E08; D18F4=4900_4000h

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x00000E08_x[34:00]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x382]

Bits	Description
31:24	Node11LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 11. AccessType: SecurePart ? Read-only : Read-write.
23:16	Node10LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 10. AccessType: SecurePart ? Read-only : Read-write.
15:8	Node9LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 9. AccessType: SecurePart ? Read-only : Read-write.
7:0	Node8LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 8. AccessType: SecurePart ? Read-only : Read-write.

D18F4xE0C [Node to Link Set Lookup Register 3] (DF::NodeToLinkSet3)

Reset: 0000_0000h.

This register provides a list of links per destination node as defined in the Multi-die routing section of the FTI specification.

_instBCST_aliasHOST; D18F4xE0C

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4xE0C_x[34:00]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x383]

_instBCST_aliasSMN; D18F4x00000E0C; D18F4=4900_4000h

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x00000E0C_x[34:00]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x383]

Bits	Description
31:24	Node15LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 15. AccessType: SecurePart ? Read-only : Read-write.
23:16	Node14LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 14. AccessType: SecurePart ? Read-only : Read-write.
15:8	Node13LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 13. AccessType: SecurePart ? Read-only : Read-write.
7:0	Node12LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 12. AccessType: SecurePart ? Read-only : Read-write.

D18F4xE10 [Node to Link Set Lookup Register 4] (DF::NodeToLinkSet4)

Reset: 0000_0000h.

This register provides a list of links per destination node as defined in the Multi-die routing section of the FTI specification.

_instBCST_aliasHOST; D18F4xE10

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4xE10_x[34:00]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x384]

_instBCST_aliasSMN; D18F4x00000E10; D18F4=4900_4000h

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x00000E10_x[34:00]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x384]

Bits	Description
31:24	Node19LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 19. AccessType: SecurePart ? Read-only : Read-write.
23:16	Node18LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 18. AccessType: SecurePart ? Read-only : Read-write.
15:8	Node17LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 17. AccessType: SecurePart ? Read-only : Read-write.
7:0	Node16LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 16. AccessType: SecurePart ? Read-only : Read-write.

D18F4xE14 [Node to Link Set Lookup Register 5] (DF::NodeToLinkSet5)

Reset: 0000_0000h.

This register provides a list of links per destination node as defined in the Multi-die routing section of the FTI specification.

_instBCST_aliasHOST; D18F4xE14

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4xE14_x[34:00]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x385]

_instBCST_aliasSMN; D18F4x00000E14; D18F4=4900_4000h

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x00000E14_x[34:00]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x385]

Bits	Description
31:24	Node23LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 23. AccessType: SecurePart ? Read-only : Read-write.
23:16	Node22LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 22. AccessType: SecurePart ? Read-only : Read-write.
15:8	Node21LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 21. AccessType: SecurePart ? Read-only : Read-write.
7:0	Node20LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 20. AccessType: SecurePart ? Read-only : Read-write.

D18F4xE18 [Node to Link Set Lookup Register 6] (DF::NodeToLinkSet6)

Reset: 0000_0000h.

This register provides a list of links per destination node as defined in the Multi-die routing section of the FTI specification.

_instBCST_aliasHOST; D18F4xE18

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4xE18_x[34:00]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x386]

_instBCST_aliasSMN; D18F4x00000E18; D18F4=4900_4000h

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x00000E18_x[34:00]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x386]

Bits	Description
31:24	Node27LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 27. AccessType: SecurePart ? Read-only : Read-write.
23:16	Node26LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 26. AccessType: SecurePart ? Read-only : Read-write.
15:8	Node25LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 25. AccessType: SecurePart ? Read-only : Read-write.
7:0	Node24LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 24. AccessType: SecurePart ? Read-only : Read-write.

D18F4xE1C [Node to Link Set Lookup Register 7] (DF::NodeToLinkSet7)

Reset: 0000_0000h.

This register provides a list of links per destination node as defined in the Multi-die routing section of the FTI specification.

_instBCST_aliasHOST; D18F4xE1C

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F4xE1C_x[34:00]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x387]

_instBCST_aliasSMN; D18F4x00000E1C; D18F4=4900_4000h

_inst[CAKE[7:0],PIE0,ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F4x00000E1C_x[34:00]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:00], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x387]

Bits	Description
31:24	Node31LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 31. AccessType: SecurePart ? Read-only : Read-write.
23:16	Node30LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 30. AccessType: SecurePart ? Read-only : Read-write.
15:8	Node29LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 29. AccessType: SecurePart ? Read-only : Read-write.
7:0	Node28LnkSet. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Link Set for destination Node 28. AccessType: SecurePart ? Read-only : Read-write.

D18F4xE40...D18F4xE4C [Remote Routing Tables] (DF::RemoteRoutingTable)

A block of 4 registers are reserved for remote routing tables. Each register contains directed routes for two NodeID's in the data fabric and contains the routing vectors for request and response channels. The probe channel uses the request directed routes. RemoteRoutingTable0 contains routes for NodeID 0,1; RemoteRoutingTable1 contains routes for NodeID 2,3; etc.

_instBCST_n[3:0]_aliasHOST; D18F4xE4[C,8,4,0]	
_inst[TCDX[15:0]]_n0_aliasHOST; D18F4xE40_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x390]	
_inst[TCDX[15:0]]_n1_aliasHOST; D18F4xE44_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x391]	
_inst[TCDX[15:0]]_n2_aliasHOST; D18F4xE48_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x392]	
_inst[TCDX[15:0]]_n3_aliasHOST; D18F4xE4C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x393]	
_instBCST_n[3:0]_aliasSMN; D18F4x0000_0E4[C,8,4,0]; D18F4=4900_4000h	
_inst[TCDX[15:0]]_n0_aliasSMN; D18F4x00000E40_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x390]	
_inst[TCDX[15:0]]_n1_aliasSMN; D18F4x00000E44_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x391]	
_inst[TCDX[15:0]]_n2_aliasSMN; D18F4x00000E48_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x392]	
_inst[TCDX[15:0]]_n3_aliasSMN; D18F4x00000E4C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x393]	
Bits	Description
31:30	Reserved.
29:24	RmtRspRoute1. Init: 11.5.3 [Register Values Written Based on Configuration]. RmtRspRoute[7:0]: Specifies the destination port for the response packets.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_n[1:0]_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n[1:0]_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n[1:0]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n[1:0]_alias[SMN,HOST]: 01h
	Reset: _instTCDX3_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX4_n[1:0]_alias[SMN,HOST]: 01h
	Reset: _instTCDX4_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX5_n0_alias[SMN,HOST]: 01h
	Reset: _instTCDX5_n1_alias[SMN,HOST]: 20h
	Reset: _instTCDX5_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX6_n0_alias[SMN,HOST]: 04h
	Reset: _instTCDX6_n1_alias[SMN,HOST]: 08h
	Reset: _instTCDX6_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX7_n[1:0]_alias[SMN,HOST]: 01h
	Reset: _instTCDX7_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX8_n[1:0]_alias[SMN,HOST]: 01h
	Reset: _instTCDX8_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX9_n[1:0]_alias[SMN,HOST]: 08h
	Reset: _instTCDX9_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX10_n0_alias[SMN,HOST]: 10h
	Reset: _instTCDX10_n1_alias[SMN,HOST]: 20h
	Reset: _instTCDX10_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX11_n[1:0]_alias[SMN,HOST]: 08h
	Reset: _instTCDX11_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX12_n[1:0]_alias[SMN,HOST]: 20h
	Reset: _instTCDX12_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX13_n[1:0]_alias[SMN,HOST]: 10h
	Reset: _instTCDX13_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX14_n[1:0]_alias[SMN,HOST]: 20h
	Reset: _instTCDX14_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n0_alias[SMN,HOST]: 01h
	Reset: _instTCDX15_n1_alias[SMN,HOST]: 04h

	Reset: _instTCDX15_n[3:2]_alias[SMN,HOST]: 00h
23:22	Reserved.
21:16	RmtReqRoute1. Init: 11.5.3 [Register Values Written Based on Configuration]. RmtReqRoute[7:0]: Specifies the destination port for the request and probe packets. AccessType: SecurePart ? Read-only : Read-write. Reset: _instTCDX0_n[3:0]_alias[SMN,HOST]: 20h Reset: _instTCDX1_n[3:0]_alias[SMN,HOST]: 10h Reset: _instTCDX2_n[3:0]_alias[SMN,HOST]: 20h Reset: _instTCDX3_n[1:0]_alias[SMN,HOST]: 01h Reset: _instTCDX3_n[3:2]_alias[SMN,HOST]: 02h Reset: _instTCDX4_n[1:0]_alias[SMN,HOST]: 01h Reset: _instTCDX4_n[3:2]_alias[SMN,HOST]: 02h Reset: _instTCDX5_n0_alias[SMN,HOST]: 01h Reset: _instTCDX5_n1_alias[SMN,HOST]: 20h Reset: _instTCDX5_n[3:2]_alias[SMN,HOST] _instTCDX6_n0_alias[SMN,HOST] 04h Reset: _instTCDX6_n1_alias[SMN,HOST]: 08h Reset: _instTCDX6_n2_alias[SMN,HOST]: 01h Reset: _instTCDX6_n3_alias[SMN,HOST]: 20h Reset: _instTCDX7_n[1:0]_alias[SMN,HOST]: 01h Reset: _instTCDX7_n[3:2]_alias[SMN,HOST]: 20h Reset: _instTCDX8_n[1:0]_alias[SMN,HOST]: 01h Reset: _instTCDX8_n[3:2]_alias[SMN,HOST]: 20h Reset: _instTCDX9_n[3:0]_alias[SMN,HOST]: 08h Reset: _instTCDX10_n[1:0]_alias[SMN,HOST]: 10h Reset: _instTCDX10_n2_alias[SMN,HOST]: 02h Reset: _instTCDX10_n3_alias[SMN,HOST] _instTCDX11_n[3:0]_alias[SMN,HOST] 08h Reset: _instTCDX12_n[3:0]_alias[SMN,HOST]: 20h Reset: _instTCDX13_n[3:0]_alias[SMN,HOST]: 10h Reset: _instTCDX14_n[3:0]_alias[SMN,HOST]: 20h Reset: _instTCDX15_n0_alias[SMN,HOST]: 01h Reset: _instTCDX15_n1_alias[SMN,HOST]: 04h Reset: _instTCDX15_n2_alias[SMN,HOST]: 10h Reset: _instTCDX15_n3_alias[SMN,HOST]: 20h
15:14	Reserved.
13:8	RmtRspRoute0. Init: 11.5.3 [Register Values Written Based on Configuration]. RmtRspRoute[7:0]: Specifies the destination port for the response packets. AccessType: SecurePart ? Read-only : Read-write. Reset: _instTCDX0_n[1:0]_alias[SMN,HOST]: 20h Reset: _instTCDX0_n[3:2]_alias[SMN,HOST]: 00h Reset: _instTCDX1_n[1:0]_alias[SMN,HOST]: 02h Reset: _instTCDX1_n[3:2]_alias[SMN,HOST]: 00h Reset: _instTCDX2_n[1:0]_alias[SMN,HOST]: 20h Reset: _instTCDX2_n[3:2]_alias[SMN,HOST]: 00h Reset: _instTCDX3_n[1:0]_alias[SMN,HOST]: 01h Reset: _instTCDX3_n[3:2]_alias[SMN,HOST]: 00h Reset: _instTCDX4_n[1:0]_alias[SMN,HOST]: 01h Reset: _instTCDX4_n[3:2]_alias[SMN,HOST]: 00h Reset: _instTCDX5_n0_alias[SMN,HOST]: 01h Reset: _instTCDX5_n1_alias[SMN,HOST]: 20h Reset: _instTCDX5_n[3:2]_alias[SMN,HOST]: 00h Reset: _instTCDX6_n0_alias[SMN,HOST]: 04h Reset: _instTCDX6_n1_alias[SMN,HOST]: 08h Reset: _instTCDX6_n[3:2]_alias[SMN,HOST]: 00h Reset: _instTCDX7_n[1:0]_alias[SMN,HOST]: 01h Reset: _instTCDX7_n[3:2]_alias[SMN,HOST]: 00h Reset: _instTCDX8_n[1:0]_alias[SMN,HOST]: 01h Reset: _instTCDX8_n[3:2]_alias[SMN,HOST]: 00h Reset: _instTCDX9_n[1:0]_alias[SMN,HOST]: 08h

	Reset: _instTCDX9_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX10_n0_alias[SMN,HOST]: 10h
	Reset: _instTCDX10_n1_alias[SMN,HOST]: 20h
	Reset: _instTCDX10_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX11_n[1:0]_alias[SMN,HOST]: 08h
	Reset: _instTCDX11_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX12_n[1:0]_alias[SMN,HOST]: 20h
	Reset: _instTCDX12_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX13_n[1:0]_alias[SMN,HOST]: 02h
	Reset: _instTCDX13_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX14_n[1:0]_alias[SMN,HOST]: 20h
	Reset: _instTCDX14_n[3:2]_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n0_alias[SMN,HOST]: 02h
	Reset: _instTCDX15_n1_alias[SMN,HOST]: 08h
	Reset: _instTCDX15_n[3:2]_alias[SMN,HOST]: 00h
7:6	Reserved.
5:0	RmtReqRoute0. Init: 11.5.3 [Register Values Written Based on Configuration]. RmtReqRoute[7:0]: Specifies the destination port for the request and probe packets. <u>AccessType: SecurePart ? Read-only : Read-write.</u>
	Reset: _instTCDX0_n[3:0]_alias[SMN,HOST]: 20h
	Reset: _instTCDX1_n[3:0]_alias[SMN,HOST]: 02h
	Reset: _instTCDX2_n[3:0]_alias[SMN,HOST]: 20h
	Reset: _instTCDX3_n[1:0]_alias[SMN,HOST]: 01h
	Reset: _instTCDX3_n[3:2]_alias[SMN,HOST]: 02h
	Reset: _instTCDX4_n[1:0]_alias[SMN,HOST]: 01h
	Reset: _instTCDX4_n[3:2]_alias[SMN,HOST]: 02h
	Reset: _instTCDX5_n0_alias[SMN,HOST]: 01h
	Reset: _instTCDX5_n1_alias[SMN,HOST]: 20h
	Reset: _instTCDX5_n[3:2]_alias[SMN,HOST]
	_instTCDX6_n0_alias[SMN,HOST]
	04h
	Reset: _instTCDX6_n1_alias[SMN,HOST]: 08h
	Reset: _instTCDX6_n2_alias[SMN,HOST]: 01h
	Reset: _instTCDX6_n3_alias[SMN,HOST]: 20h
	Reset: _instTCDX7_n[1:0]_alias[SMN,HOST]: 01h
	Reset: _instTCDX7_n[3:2]_alias[SMN,HOST]: 20h
	Reset: _instTCDX8_n[1:0]_alias[SMN,HOST]: 01h
	Reset: _instTCDX8_n[3:2]_alias[SMN,HOST]: 20h
	Reset: _instTCDX9_n[3:0]_alias[SMN,HOST]: 08h
	Reset: _instTCDX10_n[1:0]_alias[SMN,HOST]: 10h
	Reset: _instTCDX10_n2_alias[SMN,HOST]: 01h
	Reset: _instTCDX10_n3_alias[SMN,HOST]: 04h
	Reset: _instTCDX11_n[3:0]_alias[SMN,HOST]: 08h
	Reset: _instTCDX12_n[3:0]_alias[SMN,HOST]: 20h
	Reset: _instTCDX13_n[3:0]_alias[SMN,HOST]: 02h
	Reset: _instTCDX14_n[3:0]_alias[SMN,HOST]: 20h
	Reset: _instTCDX15_n0_alias[SMN,HOST]: 02h
	Reset: _instTCDX15_n1_alias[SMN,HOST]: 08h
	Reset: _instTCDX15_n2_alias[SMN,HOST]: 10h
	Reset: _instTCDX15_n3_alias[SMN,HOST]: 20h

D18F4xE60 [Multicast Type Mask 0] (DF::McastTypeMask0)

Mcast type mask 0. Each register field corresponds to a TCDX port vector leading to the master(s), or link(s) that may lead to those masters specified by the Mcast mask in the probe multicast or system management broadcast.

_instBCST_aliasHOST; D18F4xE60	
_inst[TCDX[15:0]]_aliasHOST; D18F4xE60_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x398]	
_instBCST_aliasSMN; D18F4x00000E60; D18F4=4900_4000h	
_inst[TCDX[15:0]]_aliasSMN; D18F4x00000E60_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x398]	
Bits	Description
31:30	Reserved.
29:24	McastMaskPie. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the Mcast mask for local PIE.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_alias[SMN,HOST]: 20h
	Reset: _instTCDX1_alias[SMN,HOST]: 10h
	Reset: _instTCDX2_alias[SMN,HOST]: 20h
	Reset: _inst[TCDX[4:3]]_alias[SMN,HOST]: 02h
	Reset: _instTCDX5_alias[SMN,HOST]: 04h
	Reset: _instTCDX6_alias[SMN,HOST]: 01h
	Reset: _instTCDX7_alias[SMN,HOST]: 10h
	Reset: _instTCDX8_alias[SMN,HOST]: 20h
	Reset: _instTCDX9_alias[SMN,HOST]: 01h
	Reset: _instTCDX10_alias[SMN,HOST]: 10h
	Reset: _instTCDX11_alias[SMN,HOST]: 08h
	Reset: _instTCDX12_alias[SMN,HOST]: 20h
	Reset: _instTCDX13_alias[SMN,HOST]: 10h
	Reset: _instTCDX14_alias[SMN,HOST]: 20h
	Reset: _instTCDX15_alias[SMN,HOST]: 10h
23:22	Reserved.
21:16	McastMaskIoms. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the Mcast mask for local IOMS.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_alias[SMN,HOST]: 12h
	Reset: _instTCDX2_alias[SMN,HOST]: 00h
	Reset: _inst[TCDX[4:3]]_alias[SMN,HOST]: 0Eh
	Reset: _inst[TCDX[7:5]]_alias[SMN,HOST]: 2Dh
	Reset: _instTCDX8_alias[SMN,HOST]: 0Ch
	Reset: _instTCDX9_alias[SMN,HOST]: 08h
	Reset: _instTCDX10_alias[SMN,HOST]: 30h
	Reset: _inst[TCDX[12:11]]_alias[SMN,HOST]: 00h
	Reset: _instTCDX13_alias[SMN,HOST]: 12h
	Reset: _instTCDX14_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_alias[SMN,HOST]: 30h
15:14	Reserved.
13:8	McastMaskCcm. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the Mcast mask for local CCMs.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_alias[SMN,HOST]: 23h
	Reset: _instTCDX1_alias[SMN,HOST]: 33h
	Reset: _instTCDX2_alias[SMN,HOST]: 23h
	Reset: _inst[TCDX[4:3]]_alias[SMN,HOST]: 02h
	Reset: _inst[TCDX[6:5]]_alias[SMN,HOST]: 1Eh
	Reset: _inst[TCDX[8:7]]_alias[SMN,HOST]: 01h
	Reset: _instTCDX9_alias[SMN,HOST]: 08h
	Reset: _instTCDX10_alias[SMN,HOST]: 10h
	Reset: _instTCDX11_alias[SMN,HOST]: 08h
	Reset: _instTCDX12_alias[SMN,HOST]: 23h
	Reset: _instTCDX13_alias[SMN,HOST]: 33h

	Reset: _instTCDX14_alias[SMN,HOST]: 23h
	Reset: _instTCDX15_alias[SMN,HOST]: 30h
7:6	Reserved.
5:0	McastMaskGcm. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the Mcast mask for local GCMs.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _inst[TCDX[15:00]]_alias[SMN,HOST]: 00h

D18F4xE64 [Multicast Type Mask 1] (DF::McastTypeMask1)

Reset: 0000_0000h.

Mcast type mask 1. Each register field corresponds to a TCDX port vector leading to the master(s), or link(s) that may lead to those masters specified by the Mcast mask in the probe multicast or system management broadcast.

_instBCST_aliasHOST; D18F4xE64

_inst[TCDX[15:0]]_aliasHOST; D18F4xE64_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x399]

_instBCST_aliasSMN; D18F4x00000E64; D18F4=4900_4000h

_inst[TCDX[15:0]]_aliasSMN; D18F4x00000E64_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x399]

Bits	Description
31:14	Reserved.
13:8	McastMaskRmtSysXgmi. Reset: 00h. Specifies the Mcast mask for local broadcasts targetting system management xGMI links (i.e. used by PIE).
	AccessType: SecurePart ? Read-only : Read-write.
7:6	Reserved.
5:0	McastMaskLclSysXgmi. Reset: 00h. Specifies the Mcast mask for local broadcasts targetting system management xGMI links (i.e. used by PIE).
	AccessType: SecurePart ? Read-only : Read-write.
	Init: _instTCDX1_alias[SMN,HOST] (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 10h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 02h
	Init: _inst[TCDX[5,3]]_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 01h
	Init: _instTCDX6_alias[SMN,HOST] (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 01h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 04h
	Init: _instTCDX7_alias[SMN,HOST] (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 01h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 20h
	Init: _instTCDX9_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 08h
	Init: _instTCDX10_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 02h
	Init: _instTCDX13_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 10h
	Init: _instTCDX15_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 03h

D18F4xE68 [Multicast Type Mask 2] (DF::McastTypeMask2)

Mcast type mask 2. Each register field corresponds to a TCDX port vector leading to the master(s), or link(s) that may lead to those masters specified by the Mcast mask in the probe multicast or system management broadcast.

_instBCST_aliasHOST; D18F4xE68

_inst[TCDX[15:0]]_aliasHOST; D18F4xE68_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x39A]

_instBCST_aliasSMN; D18F4x00000E68; D18F4=4900_4000h

_inst[TCDX[15:0]]_aliasSMN; D18F4x00000E68_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x39A]

Bits	Description
31:30	Reserved.
29:24	McastMaskCl3. Specifies the Mcast mask for TCDX ports towards Cluster 3. AccessType: SecurePart ? Read-only : Read-write. Reset: _instTCDX0_alias[SMN,HOST]: 20h Reset: _instTCDX1_alias[SMN,HOST]: 02h Reset: _instTCDX2_alias[SMN,HOST]: 20h Reset: _inst[TCDX[4:3]]_alias[SMN,HOST]: 02h Reset: _instTCDX5_alias[SMN,HOST]: 10h Reset: _instTCDX6_alias[SMN,HOST]: 04h Reset: _inst[TCDX[8:7]]_alias[SMN,HOST]: 01h Reset: _instTCDX9_alias[SMN,HOST]: 08h Reset: _instTCDX10_alias[SMN,HOST]: 10h Reset: _instTCDX11_alias[SMN,HOST]: 08h Reset: _instTCDX12_alias[SMN,HOST]: 03h Reset: _instTCDX13_alias[SMN,HOST]: 01h Reset: _inst[TCDX[15:14]]_alias[SMN,HOST]: 20h
23:22	Reserved.
21:16	McastMaskCl2. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the Mcast mask for TCDX ports towards Cluster 2. AccessType: SecurePart ? Read-only : Read-write. Reset: _inst[TCDX[1:0]]_alias[SMN,HOST]: 20h Reset: _instTCDX2_alias[SMN,HOST]: 03h Reset: _inst[TCDX[4:3]]_alias[SMN,HOST]: 02h Reset: _instTCDX5_alias[SMN,HOST]: 04h Reset: _instTCDX6_alias[SMN,HOST]: 02h Reset: _inst[TCDX[8:7]]_alias[SMN,HOST]: 01h Reset: _instTCDX9_alias[SMN,HOST]: 08h Reset: _instTCDX10_alias[SMN,HOST]: 10h Reset: _instTCDX11_alias[SMN,HOST]: 08h Reset: _instTCDX12_alias[SMN,HOST]: 20h Reset: _instTCDX13_alias[SMN,HOST]: 10h Reset: _instTCDX14_alias[SMN,HOST]: 20h Reset: _instTCDX15_alias[SMN,HOST]: 10h
15:14	Reserved.
13:8	McastMaskCl1. Specifies the Mcast mask for TCDX ports towards Cluster 1. AccessType: SecurePart ? Read-only : Read-write. Reset: _instTCDX0_alias[SMN,HOST]: 20h Reset: _instTCDX1_alias[SMN,HOST]: 10h Reset: _instTCDX2_alias[SMN,HOST]: 20h Reset: _inst[TCDX[4:3]]_alias[SMN,HOST]: 02h Reset: _instTCDX5_alias[SMN,HOST]: 08h Reset: _instTCDX6_alias[SMN,HOST]: 10h Reset: _inst[TCDX[8:7]]_alias[SMN,HOST]: 01h Reset: _instTCDX9_alias[SMN,HOST]: 08h Reset: _instTCDX10_alias[SMN,HOST]: 10h Reset: _instTCDX11_alias[SMN,HOST]: 08h Reset: _inst[TCDX[13:12]]_alias[SMN,HOST]: 20h Reset: _instTCDX14_alias[SMN,HOST]: 03h Reset: _instTCDX15_alias[SMN,HOST]: 20h
7:6	Reserved.

5:0	McastMaskC10 . Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the Mcast mask for TCDX ports towards Cluster 0.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_alias[SMN,HOST]: 03h
	Reset: _instTCDX1_alias[SMN,HOST]: 01h
	Reset: _instTCDX2_alias[SMN,HOST]: 20h
	Reset: _inst[TCDX[5:3]]_alias[SMN,HOST]: 02h
	Reset: _instTCDX6_alias[SMN,HOST]: 08h
	Reset: _inst[TCDX[8:7]]_alias[SMN,HOST]: 01h
	Reset: _instTCDX9_alias[SMN,HOST]: 08h
	Reset: _instTCDX10_alias[SMN,HOST]: 10h
	Reset: _instTCDX11_alias[SMN,HOST]: 08h
	Reset: _instTCDX12_alias[SMN,HOST]: 20h
	Reset: _instTCDX13_alias[SMN,HOST]: 02h
	Reset: _instTCDX14_alias[SMN,HOST]: 20h
	Reset: _instTCDX15_alias[SMN,HOST]: 10h

D18F4xE6C [Multicast Type Mask 3] (DF::McastTypeMask3)

Mcast type mask 3. Each register field corresponds to a TCDX port vector leading to the master(s), or link(s) that may lead to those masters specified by the Mcast mask in the probe multicast or system management broadcast.

_instBCST_aliasHOST; D18F4xE6C

_inst[TCDX[15:0]]_aliasHOST; D18F4xE6C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x39B]

_instBCST_aliasSMN; D18F4x00000E6C; D18F4=4900_4000h

_inst[TCDX[15:0]]_aliasSMN; D18F4x00000E6C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x39B]

Bits	Description
31:14	Reserved.
13:8	McastMaskIcng. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the Mcast mask for local ICNG. AccessType: SecurePart ? Read-only : Read-write. Reset: _instTCDX0_alias[SMN,HOST]: 20h Reset: _instTCDX1_alias[SMN,HOST]: 1Eh Reset: _instTCDX2_alias[SMN,HOST]: 20h Reset: _inst[TCDX[4:3]]_alias[SMN,HOST]: 02h Reset: _inst[TCDX[6:5]]_alias[SMN,HOST]: 1Eh Reset: _inst[TCDX[8:7]]_alias[SMN,HOST]: 01h Reset: _instTCDX9_alias[SMN,HOST]: 08h Reset: _instTCDX10_alias[SMN,HOST]: 10h Reset: _instTCDX11_alias[SMN,HOST]: 08h Reset: _instTCDX12_alias[SMN,HOST]: 20h Reset: _instTCDX13_alias[SMN,HOST]: 1Eh Reset: _instTCDX14_alias[SMN,HOST]: 20h Reset: _instTCDX15_alias[SMN,HOST]: 30h
7:6	Reserved.
5:0	McastMaskNcm. Specifies the Mcast mask for local NCMs. AccessType: SecurePart ? Read-only : Read-write. Reset: _inst[TCDX[2:0]]_alias[SMN,HOST]: 00h Reset: _inst[TCDX[4:3]]_alias[SMN,HOST]: 10h Reset: _instTCDX5_alias[SMN,HOST]: 21h Reset: _instTCDX6_alias[SMN,HOST]: 0Ch Reset: _instTCDX7_alias[SMN,HOST]: 23h Reset: _instTCDX8_alias[SMN,HOST]: 12h Reset: _instTCDX9_alias[SMN,HOST]: 38h Reset: _instTCDX10_alias[SMN,HOST]: 20h Reset: _instTCDX11_alias[SMN,HOST]: 30h Reset: _inst[TCDX[15:12]]_alias[SMN,HOST]: 00h Init: _inst[TCDX[4:3]]_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 10h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 12h Init: _instTCDX5_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 21h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 2Dh Init: _instTCDX6_alias[SMN,HOST]: ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 2Dh : (Is1NodeSystem && IsNode0) ? 0Ch Init: _instTCDX7_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 23h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 33h Init: _instTCDX10_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 20h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 30h Init: _instTCDX15_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 30h

D18F4xE70 [Multicast Type Mask 4] (DF::McastTypeMask4)

Reset: 0000_0000h.

Mcast type mask 4. Each register field corresponds to a TCDX port vector leading to the master(s), or link(s) that may lead to those masters specified by the Mcast mask in the probe multicast or system management broadcast.

`_instBCST_aliasHOST; D18F4xE70``_inst[TCDX[15:0]]_aliasSMN; D18F4xE70_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x39C]``_instBCST_aliasSMN; D18F4x00000E70; D18F4=4900_4000h``_inst[TCDX[15:0]]_aliasSMN; D18F4x00000E70_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x39C]`

Bits	Description
31:30	Reserved.
29:24	McastMaskLclToRmtCn3. Reset: 00h. Specifies the Mcast mask for TCDX ports towards Node 3. Used for local routing AccessType: SecurePart ? Read-only : Read-write. Init: <code>_instTCDX0_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 20h</code> Init: <code>_instTCDX1_alias[SMN,HOST]</code> <code>(Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 10h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 12h</code> Init: <code>_instTCDX2_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 20h</code> Init: <code>_inst[TCDX[4:3]]_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 01h</code> Init: <code>_instTCDX5_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 21h</code> Init: <code>_instTCDX6_alias[SMN,HOST]</code> <code>(Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 21h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 0Ch</code> Init: <code>_inst[TCDX[8:7]]_alias[SMN,HOST]</code> <code>(Is1NodeSystem && IsNode0) ? 00h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 01h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 20h</code> Init: <code>_instTCDX9_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 08h</code> Init: <code>_instTCDX10_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 0Ah</code> Init: <code>_instTCDX11_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 08h</code> Init: <code>_instTCDX12_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 20h</code> Init: <code>_instTCDX13_alias[SMN,HOST]</code> <code>(Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 10h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 12h</code> Init: <code>_instTCDX14_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 20h</code> Init: <code>_instTCDX15_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 0Fh</code>
23:22	Reserved.
21:16	McastMaskLclToRmtCn2. Reset: 00h. Specifies the Mcast mask for TCDX ports towards Node 2. Used for local routing AccessType: SecurePart ? Read-only : Read-write. Init: <code>_instTCDX0_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 20h</code> Init: <code>_instTCDX1_alias[SMN,HOST]</code> <code>(Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 10h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 12h</code> Init: <code>_instTCDX2_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 20h</code> Init: <code>_inst[TCDX[4:3]]_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 01h</code> Init: <code>_instTCDX5_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 21h</code> Init: <code>_instTCDX6_alias[SMN,HOST]</code>

	(Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 21h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 0Ch Init: _inst[TCDX[8:7]]_alias[SMN,HOST] (Is1NodeSystem && IsNode0) ? 00h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 01h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 20h Init: _instTCDX9_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 08h Init: _instTCDX10_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 0Ah Init: _instTCDX11_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 08h Init: _instTCDX12_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 20h Init: _instTCDX13_alias[SMN,HOST] (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 10h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 12h Init: _instTCDX14_alias[SMN,HOST]: (Is1NodeSystem && IsNode0) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 20h Init: _instTCDX15_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 00h : ((Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 0Fh
15:14	Reserved.
13:8	McastMaskLclToRmtCn1. Reset: 00h. Specifies the Mcast mask for TCDX ports towards Node 1. Used for local routing AccessType: SecurePart ? Read-only : Read-write. Init: _instTCDX0_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2NodeSystem && IsNode0)) ? 20h Init: _instTCDX1_alias[SMN,HOST] ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : (Is2Node2LinkSystem && IsNode0) ? 10h : (Is2NodeSystem && IsNode0) ? 12h Init: _instTCDX2_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2NodeSystem && IsNode0)) ? 20h Init: _inst[TCDX[4:3]]_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : (Is2NodeSystem && IsNode0) ? 01h Init: _instTCDX5_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : (Is2NodeSystem && IsNode0) ? 21h Init: _instTCDX6_alias[SMN,HOST] ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : (Is2Node2LinkSystem && IsNode0) ? 21h : (Is2NodeSystem && IsNode0) ? 0Ch Init: _inst[TCDX[8:7]]_alias[SMN,HOST] ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : (Is2NodeSystem && IsNode0) ? 01h : (Is2Node2LinkSystem && IsNode0) ? 20h Init: _instTCDX9_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2NodeSystem && IsNode0)) ? 08h Init: _instTCDX10_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 00h : (Is2Node2LinkSystem && IsNode0) ? 0Ah Init: _instTCDX11_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2NodeSystem && IsNode0)) ? 08h Init: _instTCDX12_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2NodeSystem && IsNode0)) ? 20h Init: _instTCDX13_alias[SMN,HOST] ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : (Is2Node2LinkSystem && IsNode0) ? 10h : (Is2NodeSystem && IsNode0) ? 12h Init: _instTCDX14_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : ((Is2Node2LinkSystem && IsNode0) (Is2NodeSystem && IsNode0)) ? 20h Init: _instTCDX15_alias[SMN,HOST]: ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 00h : (Is2NodeSystem && IsNode0) ? 0Fh
7:6	Reserved.
5:0	McastMaskLclToRmtCn0. Reset: 00h. Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the Mcast mask for TCDX ports towards Node 0. Used for local routing AccessType: SecurePart ? Read-only : Read-write.

D18F4xE74 [Multicast Type Mask 5] (DF::McastTypeMask5)

Reset: 0000_0000h.

Mcast type mask 5. Each register field corresponds to a TCDX port vector leading to the master(s), or link(s) that may lead to those masters specified by the Mcast mask in the probe multicast or system management broadcast.

_instBCST_aliasHOST; D18F4xE74

_inst[TCDX[15:0]]_aliasHOST; D18F4xE74_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x39D]

_instBCST_aliasSMN; D18F4x00000E74; D18F4=4900_4000h

_inst[TCDX[15:0]]_aliasSMN; D18F4x00000E74_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x39D]

Bits	Description
31:30	Reserved.
29:24	McastMaskRmtToRmtCn3. Reset: 00h. Specifies the Mcast mask for TCDX ports towards Node 3. Used for remote routing. AccessType: SecurePart ? Read-only : Read-write.
23:22	Reserved.
21:16	McastMaskRmtToRmtCn2. Reset: 00h. Specifies the Mcast mask for TCDX ports towards Node 2. Used for remote routing. AccessType: SecurePart ? Read-only : Read-write.
15:14	Reserved.
13:8	McastMaskRmtToRmtCn1. Reset: 00h. Specifies the Mcast mask for TCDX ports towards Node 1. Used for remote routing. AccessType: SecurePart ? Read-only : Read-write.
7:6	Reserved.
5:0	McastMaskRmtToRmtCn0. Reset: 00h. Specifies the Mcast mask for TCDX ports towards Node 0. Used for remote routing. AccessType: SecurePart ? Read-only : Read-write.

D18F4xE80...D18F4xEFC [Local Multicast Routing Tables] (DF::LocalMcastRoutingTable)

A block of 128 registers are reserved for local Mcast routing tables. Each register corresponds to the local FabricID that can send probe multicasts or system management broadcasts in the coherent fabric and contains the MCast route from that FabricID to the masters and links. Each register contains routes for up to 4 FabricIDs with LocalMcastRoutingTable0 containing to the lowest 4 FabricIDs, LocalMcastRoutingTable1 containing the next 4 FabricIDs and so on.

_instBCST_n[31:0]_aliasHOST; D18F4xE[[F:8][C,8,4,0]]

_inst[TCDX[15:0]]_n0_aliasHOST; D18F4xE80_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3A0]

_inst[TCDX[15:0]]_n1_aliasHOST; D18F4xE84_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3A1]

_inst[TCDX[15:0]]_n2_aliasHOST; D18F4xE88_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3A2]

_inst[TCDX[15:0]]_n3_aliasHOST; D18F4xE8C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3A3]

_inst[TCDX[15:0]]_n4_aliasHOST; D18F4xE90_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3A4]

_inst[TCDX[15:0]]_n5_aliasHOST; D18F4xE94_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3A5]

_inst[TCDX[15:0]]_n6_aliasHOST; D18F4xE98_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3A6]

_inst[TCDX[15:0]]_n7_aliasHOST; D18F4xE9C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3A7]

_inst[TCDX[15:0]]_n8_aliasHOST; D18F4xEA0_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3A8]

_inst[TCDX[15:0]]_n9_aliasHOST; D18F4xEA4_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3A9]

_inst[TCDX[15:0]]_n10_aliasHOST; D18F4xEA8_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3AA]

_inst[TCDX[15:0]]_n11_aliasHOST; D18F4xEAC_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3AB]

_inst[TCDX[15:0]]_n12_aliasHOST; D18F4xEB0_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3AC]

_inst[TCDX[15:0]]_n13_aliasHOST; D18F4xEB4_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3AD]

_inst[TCDX[15:0]]_n14_aliasHOST; D18F4xEB8_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3AE]

_inst[TCDX[15:0]]_n15_aliasHOST; D18F4xEBC_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3AF]

_inst[TCDX[15:0]]_n16_aliasHOST; D18F4xEC0_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3B0]

_inst[TCDX[15:0]]_n17_aliasHOST; D18F4xEC4_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3B1]

_inst[TCDX[15:0]]_n18_aliasHOST; D18F4xEC8_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3B2]

_inst[TCDX[15:0]]_n19_aliasHOST; D18F4xECC_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3B3]

_inst[TCDX[15:0]]_n20_aliasHOST; D18F4xED0_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3B4]

_inst[TCDX[15:0]]_n21_aliasHOST; D18F4xED4_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3B5]

_inst[TCDX[15:0]]_n22_aliasHOST; D18F4xED8_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3B6]

_inst[TCDX[15:0]]_n23_aliasHOST; D18F4xEDC_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3B7]

_inst[TCDX[15:0]]_n24_aliasHOST; D18F4xEE0_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3B8]

_inst[TCDX[15:0]]_n25_aliasHOST; D18F4xEE4_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3B9]

_inst[TCDX[15:0]]_n26_aliasHOST; D18F4xEE8_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3BA]

_inst[TCDX[15:0]]_n27_aliasHOST; D18F4xEEC_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3BB]

_inst[TCDX[15:0]]_n28_aliasHOST; D18F4xEF0_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3BC]

[illegible]

_inst[TCDX[15:0]]_n30_aliasSMN; D18F4x00000EF8_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3BE]	
_inst[TCDX[15:0]]_n31_aliasSMN; D18F4x00000EFC_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3BF]	
Bits	Description
31:30	Reserved.
29:24	LclMcastRoute3 . Init: 11.5.3 [Register Values Written Based on Configuration]. LclMcastRoute3[7:0]: Specifies the Mcast route from this FabricID to the masters and links in this SOC.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_n[3:0]_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n[28:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n29_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n31_alias[SMN,HOST]: 03h
	Reset: _instTCDX1_n0_alias[SMN,HOST]: 13h
	Reset: _instTCDX1_n[3:1]_alias[SMN,HOST]: 21h
	Reset: _instTCDX1_n[28:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n29_alias[SMN,HOST]: 2Dh
	Reset: _instTCDX1_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n31_alias[SMN,HOST]: 2Dh
	Reset: _instTCDX2_n0_alias[SMN,HOST]: 23h
	Reset: _instTCDX2_n[3:1]_alias[SMN,HOST]: 03h
	Reset: _instTCDX2_n[28:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n29_alias[SMN,HOST]: 03h
	Reset: _instTCDX2_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n31_alias[SMN,HOST]: 03h
	Reset: _instTCDX3_n[1:0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n2_alias[SMN,HOST]: 01h
	Reset: _instTCDX3_n[28:3]_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n29_alias[SMN,HOST]: 1Dh
	Reset: _instTCDX3_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n31_alias[SMN,HOST]: 05h
	Reset: _instTCDX4_n[1:0]_alias[SMN,HOST]: 01h
	Reset: _instTCDX4_n2_alias[SMN,HOST]: 00h
	Reset: _instTCDX4_n3_alias[SMN,HOST]: 01h
	Reset: _instTCDX4_n[28:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX4_n29_alias[SMN,HOST]: 1Ch
	Reset: _instTCDX4_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX4_n31_alias[SMN,HOST]: 04h
	Reset: _instTCDX5_n0_alias[SMN,HOST]: 30h
	Reset: _instTCDX5_n1_alias[SMN,HOST]: 22h
	Reset: _instTCDX5_n2_alias[SMN,HOST]: 03h
	Reset: _instTCDX5_n3_alias[SMN,HOST]: 32h
	Reset: _instTCDX5_n[28:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX5_n29_alias[SMN,HOST]: 33h
	Reset: _instTCDX5_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX5_n31_alias[SMN,HOST]: 27h
	Reset: _instTCDX6_n0_alias[SMN,HOST]: 18h
	Reset: _instTCDX6_n1_alias[SMN,HOST]: 02h
	Reset: _instTCDX6_n2_alias[SMN,HOST]: 06h
	Reset: _instTCDX6_n3_alias[SMN,HOST]: 1Eh
	Reset: _instTCDX6_n[28:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX6_n29_alias[SMN,HOST]: 1Eh
	Reset: _instTCDX6_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX6_n31_alias[SMN,HOST]: 27h
	Reset: _instTCDX7_n[28:0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX7_n29_alias[SMN,HOST]: 2Fh
	Reset: _instTCDX7_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX7_n31_alias[SMN,HOST]: 18h
	Reset: _instTCDX8_n[2:0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX8_n3_alias[SMN,HOST]: 01h
	Reset: _instTCDX8_n[28:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX8_n29_alias[SMN,HOST]: 1Eh

	Reset: _instTCDX8_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX8_n31_alias[SMN,HOST]: 18h
	Reset: _instTCDX9_n[28:0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX9_n29_alias[SMN,HOST]: 39h
	Reset: _instTCDX9_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX9_n31_alias[SMN,HOST]: 31h
	Reset: _instTCDX10_n[28:0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX10_n29_alias[SMN,HOST]: 20h
	Reset: _instTCDX10_n[31:30]_alias[SMN,HOST]
	_instTCDX11_n[2:0]_alias[SMN,HOST]
	00h
	Reset: _instTCDX11_n3_alias[SMN,HOST]: 08h
	Reset: _instTCDX11_n[28:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX11_n29_alias[SMN,HOST]: 30h
	Reset: _instTCDX11_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX11_n31_alias[SMN,HOST]: 30h
	Reset: _instTCDX12_n0_alias[SMN,HOST]: 03h
	Reset: _instTCDX12_n1_alias[SMN,HOST]: 23h
	Reset: _instTCDX12_n[3:2]_alias[SMN,HOST]: 03h
	Reset: _instTCDX12_n[28:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX12_n29_alias[SMN,HOST]: 03h
	Reset: _instTCDX12_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX12_n31_alias[SMN,HOST]: 03h
	Reset: _instTCDX13_n0_alias[SMN,HOST]: 21h
	Reset: _instTCDX13_n1_alias[SMN,HOST]: 32h
	Reset: _instTCDX13_n2_alias[SMN,HOST]: 13h
	Reset: _instTCDX13_n3_alias[SMN,HOST]: 21h
	Reset: _instTCDX13_n[28:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX13_n29_alias[SMN,HOST]: 2Dh
	Reset: _instTCDX13_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX13_n31_alias[SMN,HOST]: 3Fh
	Reset: _instTCDX14_n[1:0]_alias[SMN,HOST]: 03h
	Reset: _instTCDX14_n2_alias[SMN,HOST]: 23h
	Reset: _instTCDX14_n3_alias[SMN,HOST]: 03h
	Reset: _instTCDX14_n[28:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX14_n29_alias[SMN,HOST]: 03h
	Reset: _instTCDX14_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX14_n31_alias[SMN,HOST]: 03h
	Reset: _instTCDX15_n0_alias[SMN,HOST]: 06h
	Reset: _instTCDX15_n1_alias[SMN,HOST]: 0Ah
	Reset: _instTCDX15_n2_alias[SMN,HOST]: 03h
	Reset: _instTCDX15_n3_alias[SMN,HOST]: 06h
	Reset: _instTCDX15_n[28:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n29_alias[SMN,HOST]: 02h
	Reset: _instTCDX15_n30_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n31_alias[SMN,HOST]: 01h
23:22	Reserved.
21:16	LclMcastRoute2. Init: 11.5.3 [Register Values Written Based on Configuration]. LclMcastRoute2[7:0]: Specifies the Mcast route from this FabricID to the masters and links in this SOC.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_n0_alias[SMN,HOST]: 23h
	Reset: _instTCDX0_n[3:1]_alias[SMN,HOST]: 03h
	Reset: _instTCDX0_n[30:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n31_alias[SMN,HOST]: 03h
	Reset: _instTCDX1_n0_alias[SMN,HOST]: 32h
	Reset: _instTCDX1_n[3:1]_alias[SMN,HOST]: 21h
	Reset: _instTCDX1_n[30:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n31_alias[SMN,HOST]: 3Fh
	Reset: _instTCDX2_n[3:0]_alias[SMN,HOST]: 03h
	Reset: _instTCDX2_n[30:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n31_alias[SMN,HOST]: 03h
	Reset: _instTCDX3_n0_alias[SMN,HOST]: 01h
	Reset: _instTCDX3_n1_alias[SMN,HOST]: 00h

	Reset: _instTCDX3_n2_alias[SMN,HOST]: 01h
	Reset: _instTCDX3_n[30:3]_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n31_alias[SMN,HOST]: 05h
	Reset: _instTCDX4_n0_alias[SMN,HOST]: 00h
	Reset: _instTCDX4_n1_alias[SMN,HOST]: 01h
	Reset: _instTCDX4_n2_alias[SMN,HOST]: 00h
	Reset: _instTCDX4_n3_alias[SMN,HOST]: 01h
	Reset: _instTCDX4_n[30:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX4_n31_alias[SMN,HOST]: 04h
	Reset: _instTCDX5_n0_alias[SMN,HOST]: 11h
	Reset: _instTCDX5_n1_alias[SMN,HOST]: 22h
	Reset: _instTCDX5_n2_alias[SMN,HOST]: 03h
	Reset: _instTCDX5_n3_alias[SMN,HOST]: 32h
	Reset: _instTCDX5_n[30:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX5_n31_alias[SMN,HOST]: 31h
	Reset: _instTCDX6_n0_alias[SMN,HOST]: 10h
	Reset: _instTCDX6_n1_alias[SMN,HOST]: 02h
	Reset: _instTCDX6_n2_alias[SMN,HOST]: 06h
	Reset: _instTCDX6_n3_alias[SMN,HOST]: 1Eh
	Reset: _instTCDX6_n[30:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX6_n31_alias[SMN,HOST]: 31h
	Reset: _instTCDX7_n[30:0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX7_n31_alias[SMN,HOST]: 18h
	Reset: _instTCDX8_n[2:0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX8_n3_alias[SMN,HOST]: 01h
	Reset: _instTCDX8_n[30:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX8_n31_alias[SMN,HOST]: 18h
	Reset: _instTCDX9_n[30:0]_alias[SMN,HOST]: 00h
	Reset: _instTCDX9_n31_alias[SMN,HOST]: 31h
	Reset: _instTCDX10_n[31:0]_alias[SMN,HOST]
	_instTCDX11_n[2:0]_alias[SMN,HOST]
	00h
	Reset: _instTCDX11_n3_alias[SMN,HOST]: 08h
	Reset: _instTCDX11_n[30:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX11_n31_alias[SMN,HOST]: 30h
	Reset: _instTCDX12_n0_alias[SMN,HOST]: 03h
	Reset: _instTCDX12_n1_alias[SMN,HOST]: 23h
	Reset: _instTCDX12_n[3:2]_alias[SMN,HOST]: 03h
	Reset: _instTCDX12_n[30:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX12_n31_alias[SMN,HOST]: 03h
	Reset: _instTCDX13_n0_alias[SMN,HOST]: 21h
	Reset: _instTCDX13_n1_alias[SMN,HOST]: 32h
	Reset: _instTCDX13_n2_alias[SMN,HOST]: 13h
	Reset: _instTCDX13_n3_alias[SMN,HOST]: 21h
	Reset: _instTCDX13_n[30:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX13_n31_alias[SMN,HOST]: 2Dh
	Reset: _instTCDX14_n[1:0]_alias[SMN,HOST]: 03h
	Reset: _instTCDX14_n2_alias[SMN,HOST]: 23h
	Reset: _instTCDX14_n3_alias[SMN,HOST]: 03h
	Reset: _instTCDX14_n[30:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX14_n31_alias[SMN,HOST]: 03h
	Reset: _instTCDX15_n0_alias[SMN,HOST]: 02h
	Reset: _instTCDX15_n1_alias[SMN,HOST]: 0Ah
	Reset: _instTCDX15_n2_alias[SMN,HOST]: 03h
	Reset: _instTCDX15_n3_alias[SMN,HOST]: 0Ah
	Reset: _instTCDX15_n[30:4]_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n31_alias[SMN,HOST]: 02h
15:14	Reserved.
13:8	LclMcastRoute1 . Init: 11.5.3 [Register Values Written Based on Configuration]. LclMcastRoute1[7:0]: Specifies the Mcast route from this FabricID to the masters and links in this SOC.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_n0_alias[SMN,HOST]: 23h
	Reset: _instTCDX0_n[3:1]_alias[SMN,HOST]: 03h

Reset: _instTCDX0_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n31_alias[SMN,HOST]: 03h
Reset: _instTCDX1_n0_alias[SMN,HOST]: 32h
Reset: _instTCDX1_n1_alias[SMN,HOST]: 13h
Reset: _instTCDX1_n[3:2]_alias[SMN,HOST]: 21h
Reset: _instTCDX1_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n31_alias[SMN,HOST]: 2Dh
Reset: _instTCDX2_n0_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n1_alias[SMN,HOST]: 23h
Reset: _instTCDX2_n[3:2]_alias[SMN,HOST]: 03h
Reset: _instTCDX2_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n31_alias[SMN,HOST]: 03h
Reset: _instTCDX3_n0_alias[SMN,HOST]: 01h
Reset: _instTCDX3_n1_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n[3:2]_alias[SMN,HOST]: 01h
Reset: _instTCDX3_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n31_alias[SMN,HOST]: 05h
Reset: _instTCDX4_n0_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n1_alias[SMN,HOST]: 01h
Reset: _instTCDX4_n[30:2]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n31_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n0_alias[SMN,HOST]: 11h
Reset: _instTCDX5_n1_alias[SMN,HOST]: 30h
Reset: _instTCDX5_n2_alias[SMN,HOST]: 03h
Reset: _instTCDX5_n3_alias[SMN,HOST]: 13h
Reset: _instTCDX5_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n31_alias[SMN,HOST]: 23h
Reset: _instTCDX6_n0_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n1_alias[SMN,HOST]: 18h
Reset: _instTCDX6_n2_alias[SMN,HOST]: 06h
Reset: _instTCDX6_n3_alias[SMN,HOST]: 1Eh
Reset: _instTCDX6_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n31_alias[SMN,HOST]: 27h
Reset: _instTCDX7_n[2:0]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n3_alias[SMN,HOST]: 01h
Reset: _instTCDX7_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n31_alias[SMN,HOST]: 18h
Reset: _instTCDX8_n[30:0]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n31_alias[SMN,HOST]: 18h
Reset: _instTCDX9_n[2:0]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n3_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n31_alias[SMN,HOST]: 31h
Reset: _instTCDX10_n[31:0]_alias[SMN,HOST]
Reset: _instTCDX11_n[30:0]_alias[SMN,HOST]
00h
Reset: _instTCDX11_n31_alias[SMN,HOST]: 30h
Reset: _instTCDX12_n[3:0]_alias[SMN,HOST]: 03h
Reset: _instTCDX12_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n31_alias[SMN,HOST]: 03h
Reset: _instTCDX13_n[1:0]_alias[SMN,HOST]: 21h
Reset: _instTCDX13_n2_alias[SMN,HOST]: 13h
Reset: _instTCDX13_n3_alias[SMN,HOST]: 21h
Reset: _instTCDX13_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n31_alias[SMN,HOST]: 3Fh
Reset: _instTCDX14_n[1:0]_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n2_alias[SMN,HOST]: 23h
Reset: _instTCDX14_n3_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n31_alias[SMN,HOST]: 03h
Reset: _instTCDX15_n0_alias[SMN,HOST]: 02h
Reset: _instTCDX15_n1_alias[SMN,HOST]: 06h
Reset: _instTCDX15_n[3:2]_alias[SMN,HOST]: 03h
Reset: _instTCDX15_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n31_alias[SMN,HOST]: 01h

7:6	Reserved.
5:0	LclMcastRoute0. Init: 11.5.3 [Register Values Written Based on Configuration]. LclMcastRoute0[7:0]: Specifies the Mcast route from this FabricID to the masters and links in this SOC. <u>AccessType:</u> SecurePart ? Read-only : Read-write. Reset: <u>_instTCDX0_n0_alias</u>[SMN,HOST]: 23h Reset: <u>_instTCDX0_n[3:1]_alias</u>[SMN,HOST]: 03h Reset: <u>_instTCDX0_n[30:4]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX0_n31_alias</u>[SMN,HOST]: 03h Reset: <u>_instTCDX1_n0_alias</u>[SMN,HOST]: 32h Reset: <u>_instTCDX1_n1_alias</u>[SMN,HOST]: 13h Reset: <u>_instTCDX1_n[3:2]_alias</u>[SMN,HOST]: 21h Reset: <u>_instTCDX1_n[30:4]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX1_n31_alias</u>[SMN,HOST]: 3Fh Reset: <u>_instTCDX2_n0_alias</u>[SMN,HOST]: 03h Reset: <u>_instTCDX2_n1_alias</u>[SMN,HOST]: 23h Reset: <u>_instTCDX2_n[3:2]_alias</u>[SMN,HOST]: 03h Reset: <u>_instTCDX2_n[30:4]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX2_n31_alias</u>[SMN,HOST]: 03h Reset: <u>_instTCDX3_n0_alias</u>[SMN,HOST]: 01h Reset: <u>_instTCDX3_n[2:1]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX3_n3_alias</u>[SMN,HOST]: 01h Reset: <u>_instTCDX3_n[30:4]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX3_n31_alias</u>[SMN,HOST]: 05h Reset: <u>_instTCDX4_n0_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX4_n[2:1]_alias</u>[SMN,HOST]: 01h Reset: <u>_instTCDX4_n[30:3]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX4_n31_alias</u>[SMN,HOST]: 04h Reset: <u>_instTCDX5_n0_alias</u>[SMN,HOST]: 11h Reset: <u>_instTCDX5_n1_alias</u>[SMN,HOST]: 30h Reset: <u>_instTCDX5_n2_alias</u>[SMN,HOST]: 22h Reset: <u>_instTCDX5_n3_alias</u>[SMN,HOST]: 13h Reset: <u>_instTCDX5_n[30:4]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX5_n31_alias</u>[SMN,HOST]: 35h Reset: <u>_instTCDX6_n0_alias</u>[SMN,HOST]: 10h Reset: <u>_instTCDX6_n1_alias</u>[SMN,HOST]: 18h Reset: <u>_instTCDX6_n2_alias</u>[SMN,HOST]: 02h Reset: <u>_instTCDX6_n3_alias</u>[SMN,HOST]: 1Eh Reset: <u>_instTCDX6_n[30:4]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX6_n31_alias</u>[SMN,HOST]: 31h Reset: <u>_instTCDX7_n[2:0]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX7_n3_alias</u>[SMN,HOST]: 01h Reset: <u>_instTCDX7_n[30:4]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX7_n31_alias</u>[SMN,HOST]: 18h Reset: <u>_instTCDX8_n[30:0]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX8_n31_alias</u>[SMN,HOST]: 18h Reset: <u>_instTCDX9_n[2:0]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX9_n3_alias</u>[SMN,HOST]: 08h Reset: <u>_instTCDX9_n[30:4]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX9_n31_alias</u>[SMN,HOST]: 31h Reset: <u>_instTCDX10_n[31:0]_alias</u>[SMN,HOST] <u>_instTCDX11_n[30:0]_alias</u>[SMN,HOST] 00h Reset: <u>_instTCDX11_n31_alias</u>[SMN,HOST]: 30h Reset: <u>_instTCDX12_n[1:0]_alias</u>[SMN,HOST]: 03h Reset: <u>_instTCDX12_n2_alias</u>[SMN,HOST]: 23h Reset: <u>_instTCDX12_n3_alias</u>[SMN,HOST]: 03h Reset: <u>_instTCDX12_n[30:4]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX12_n31_alias</u>[SMN,HOST]: 03h Reset: <u>_instTCDX13_n[1:0]_alias</u>[SMN,HOST]: 21h Reset: <u>_instTCDX13_n2_alias</u>[SMN,HOST]: 32h Reset: <u>_instTCDX13_n3_alias</u>[SMN,HOST]: 21h Reset: <u>_instTCDX13_n[30:4]_alias</u>[SMN,HOST]: 00h Reset: <u>_instTCDX13_n31_alias</u>[SMN,HOST]: 2Dh

Reset: _instTCDX14_n[3:0]_alias[SMN,HOST]: 03h
Reset: _instTCDX14_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n31_alias[SMN,HOST]: 03h
Reset: _instTCDX15_n0_alias[SMN,HOST]: 02h
Reset: _instTCDX15_n1_alias[SMN,HOST]: 06h
Reset: _instTCDX15_n2_alias[SMN,HOST]: 0Ah
Reset: _instTCDX15_n3_alias[SMN,HOST]: 02h
Reset: _instTCDX15_n[30:4]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n31_alias[SMN,HOST]: 02h

D18F4xF00...D18F4xFFC [Local Routing Tables] (DF::LocalRoutingTable)

A block of 128 registers are reserved for local routing tables. Each register contains directed routes for two ComponentID's in the data fabric and contains the routing vectors for request and response channels. The probe channel uses the request directed routes. LocalRoutingTable0 contains routes for ComponentID 0,1; LocalRoutingTable1 contains routes for ComponentID 2,3; etc.

_inst[BCST_n[63:0]_aliasHOST; D18F4xF[[F:0][C,8,4,0]]
_inst[TCDX[15:0]]_n0_aliasHOST; D18F4xF00_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C0]
_inst[TCDX[15:0]]_n1_aliasHOST; D18F4xF04_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C1]
_inst[TCDX[15:0]]_n2_aliasHOST; D18F4xF08_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C2]
_inst[TCDX[15:0]]_n3_aliasHOST; D18F4xF0C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C3]
_inst[TCDX[15:0]]_n4_aliasHOST; D18F4xF10_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C4]
_inst[TCDX[15:0]]_n5_aliasHOST; D18F4xF14_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C5]
_inst[TCDX[15:0]]_n6_aliasHOST; D18F4xF18_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C6]
_inst[TCDX[15:0]]_n7_aliasHOST; D18F4xF1C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C7]
_inst[TCDX[15:0]]_n8_aliasHOST; D18F4xF20_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C8]
_inst[TCDX[15:0]]_n9_aliasHOST; D18F4xF24_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C9]
_inst[TCDX[15:0]]_n10_aliasHOST; D18F4xF28_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3CA]
_inst[TCDX[15:0]]_n11_aliasHOST; D18F4xF2C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3CB]
_inst[TCDX[15:0]]_n12_aliasHOST; D18F4xF30_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3CC]
_inst[TCDX[15:0]]_n13_aliasHOST; D18F4xF34_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3CD]
_inst[TCDX[15:0]]_n14_aliasHOST; D18F4xF38_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3CE]
_inst[TCDX[15:0]]_n15_aliasHOST; D18F4xF3C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3CF]
_inst[TCDX[15:0]]_n16_aliasHOST; D18F4xF40_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D0]
_inst[TCDX[15:0]]_n17_aliasHOST; D18F4xF44_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D1]
_inst[TCDX[15:0]]_n18_aliasHOST; D18F4xF48_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D2]
_inst[TCDX[15:0]]_n19_aliasHOST; D18F4xF4C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D3]
_inst[TCDX[15:0]]_n20_aliasHOST; D18F4xF50_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D4]
_inst[TCDX[15:0]]_n21_aliasHOST; D18F4xF54_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D5]
_inst[TCDX[15:0]]_n22_aliasHOST; D18F4xF58_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D6]
_inst[TCDX[15:0]]_n23_aliasHOST; D18F4xF5C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D7]
_inst[TCDX[15:0]]_n24_aliasHOST; D18F4xF60_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D8]
_inst[TCDX[15:0]]_n25_aliasHOST; D18F4xF64_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D9]
_inst[TCDX[15:0]]_n26_aliasHOST; D18F4xF68_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3DA]
_inst[TCDX[15:0]]_n27_aliasHOST; D18F4xF6C_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3DB]
_inst[TCDX[15:0]]_n28_aliasHOST; D18F4xF70_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3DC]
_inst[TCDX[15:0]]_n29_aliasHOST; D18F4xF74_x[60:51]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3DD]

[illegible]

_instBCST_n[63:0]_aliasSMN; D18F4x0000_0F[FC,F8,F4,F0,EC,E8,E4,E0,DC,D8,D4,D0,CC,C8,C4,C0,BC,B8,B4,B0,AC,A8,A4,A0,9C,98,94,90,8C,88,84,80,7C,78,74,70,6C,68,64,60,5C,58,54,50,4C,48,44,40,3C,38,34,30,2C,28,24,20,1C,18,14,10,0C,08,04,00]; D18F4=4900_4000h
_inst[TCDX[15:0]]_n0_aliasSMN; D18F4x00000F00_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C0]
_inst[TCDX[15:0]]_n1_aliasSMN; D18F4x00000F04_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C1]
_inst[TCDX[15:0]]_n2_aliasSMN; D18F4x00000F08_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C2]
_inst[TCDX[15:0]]_n3_aliasSMN; D18F4x00000F0C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C3]
_inst[TCDX[15:0]]_n4_aliasSMN; D18F4x00000F10_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C4]
_inst[TCDX[15:0]]_n5_aliasSMN; D18F4x00000F14_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C5]
_inst[TCDX[15:0]]_n6_aliasSMN; D18F4x00000F18_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C6]
_inst[TCDX[15:0]]_n7_aliasSMN; D18F4x00000F1C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C7]
_inst[TCDX[15:0]]_n8_aliasSMN; D18F4x00000F20_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C8]
_inst[TCDX[15:0]]_n9_aliasSMN; D18F4x00000F24_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3C9]
_inst[TCDX[15:0]]_n10_aliasSMN; D18F4x00000F28_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3CA]
_inst[TCDX[15:0]]_n11_aliasSMN; D18F4x00000F2C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3CB]
_inst[TCDX[15:0]]_n12_aliasSMN; D18F4x00000F30_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3CC]
_inst[TCDX[15:0]]_n13_aliasSMN; D18F4x00000F34_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3CD]
_inst[TCDX[15:0]]_n14_aliasSMN; D18F4x00000F38_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3CE]
_inst[TCDX[15:0]]_n15_aliasSMN; D18F4x00000F3C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3CF]
_inst[TCDX[15:0]]_n16_aliasSMN; D18F4x00000F40_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D0]
_inst[TCDX[15:0]]_n17_aliasSMN; D18F4x00000F44_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D1]
_inst[TCDX[15:0]]_n18_aliasSMN; D18F4x00000F48_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D2]
_inst[TCDX[15:0]]_n19_aliasSMN; D18F4x00000F4C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D3]
_inst[TCDX[15:0]]_n20_aliasSMN; D18F4x00000F50_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D4]
_inst[TCDX[15:0]]_n21_aliasSMN; D18F4x00000F54_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D5]
_inst[TCDX[15:0]]_n22_aliasSMN; D18F4x00000F58_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D6]
_inst[TCDX[15:0]]_n23_aliasSMN; D18F4x00000F5C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D7]
_inst[TCDX[15:0]]_n24_aliasSMN; D18F4x00000F60_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D8]
_inst[TCDX[15:0]]_n25_aliasSMN; D18F4x00000F64_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3D9]
_inst[TCDX[15:0]]_n26_aliasSMN; D18F4x00000F68_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3DA]
_inst[TCDX[15:0]]_n27_aliasSMN; D18F4x00000F6C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3DB]
_inst[TCDX[15:0]]_n28_aliasSMN; D18F4x00000F70_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3DC]
_inst[TCDX[15:0]]_n29_aliasSMN; D18F4x00000F74_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3DD]
_inst[TCDX[15:0]]_n30_aliasSMN; D18F4x00000F78_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3DE]
_inst[TCDX[15:0]]_n31_aliasSMN; D18F4x00000F7C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3DF]
_inst[TCDX[15:0]]_n32_aliasSMN; D18F4x00000F80_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3E0]

_inst[TCDX[15:0]]_n33_aliasSMN; D18F4x00000F84_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3E1]	
_inst[TCDX[15:0]]_n34_aliasSMN; D18F4x00000F88_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3E2]	
_inst[TCDX[15:0]]_n35_aliasSMN; D18F4x00000F8C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3E3]	
_inst[TCDX[15:0]]_n36_aliasSMN; D18F4x00000F90_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3E4]	
_inst[TCDX[15:0]]_n37_aliasSMN; D18F4x00000F94_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3E5]	
_inst[TCDX[15:0]]_n38_aliasSMN; D18F4x00000F98_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3E6]	
_inst[TCDX[15:0]]_n39_aliasSMN; D18F4x00000F9C_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3E7]	
_inst[TCDX[15:0]]_n40_aliasSMN; D18F4x00000FA0_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3E8]	
_inst[TCDX[15:0]]_n41_aliasSMN; D18F4x00000FA4_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3E9]	
_inst[TCDX[15:0]]_n42_aliasSMN; D18F4x00000FA8_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3EA]	
_inst[TCDX[15:0]]_n43_aliasSMN; D18F4x00000FAC_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3EB]	
_inst[TCDX[15:0]]_n44_aliasSMN; D18F4x00000FB0_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3EC]	
_inst[TCDX[15:0]]_n45_aliasSMN; D18F4x00000FB4_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3ED]	
_inst[TCDX[15:0]]_n46_aliasSMN; D18F4x00000FB8_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3EE]	
_inst[TCDX[15:0]]_n47_aliasSMN; D18F4x00000FBC_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3EF]	
_inst[TCDX[15:0]]_n48_aliasSMN; D18F4x00000FC0_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3F0]	
_inst[TCDX[15:0]]_n49_aliasSMN; D18F4x00000FC4_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3F1]	
_inst[TCDX[15:0]]_n50_aliasSMN; D18F4x00000FC8_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3F2]	
_inst[TCDX[15:0]]_n51_aliasSMN; D18F4x00000FCC_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3F3]	
_inst[TCDX[15:0]]_n52_aliasSMN; D18F4x00000FD0_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3F4]	
_inst[TCDX[15:0]]_n53_aliasSMN; D18F4x00000FD4_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3F5]	
_inst[TCDX[15:0]]_n54_aliasSMN; D18F4x00000FD8_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3F6]	
_inst[TCDX[15:0]]_n55_aliasSMN; D18F4x00000FDC_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3F7]	
_inst[TCDX[15:0]]_n56_aliasSMN; D18F4x00000FE0_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3F8]	
_inst[TCDX[15:0]]_n57_aliasSMN; D18F4x00000FE4_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3F9]	
_inst[TCDX[15:0]]_n58_aliasSMN; D18F4x00000FE8_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3FA]	
_inst[TCDX[15:0]]_n59_aliasSMN; D18F4x00000FEC_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3FB]	
_inst[TCDX[15:0]]_n60_aliasSMN; D18F4x00000FF0_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3FC]	
_inst[TCDX[15:0]]_n61_aliasSMN; D18F4x00000FF4_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3FD]	
_inst[TCDX[15:0]]_n62_aliasSMN; D18F4x00000FF8_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3FE]	
_inst[TCDX[15:0]]_n63_aliasSMN; D18F4x00000FFC_x[60:51]; D18F4=4900_4000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[60:51], IndCfgAccFuncNum=4, IndCfgAccRegNum=0x3FF]	
Bits	Description
31:30	Reserved.
29:24	LclRspRoute1 . Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the destination port for the response packets.
	AccessType: SecurePart ? Read-only : Read-write.

Reset: _instTCDX0_n0_alias[SMN,HOST]: 08h
Reset: _instTCDX0_n[7:1]_alias[SMN,HOST]: 20h
Reset: _instTCDX0_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n[17:16]_alias[SMN,HOST]: 20h
Reset: _instTCDX0_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n[51:48]_alias[SMN,HOST]: 20h
Reset: _instTCDX0_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n[57:54]_alias[SMN,HOST]: 20h
Reset: _instTCDX0_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX0_n[63:59]_alias[SMN,HOST]: 20h
Reset: _instTCDX1_n0_alias[SMN,HOST]: 01h
Reset: _instTCDX1_n[2:1]_alias[SMN,HOST]: 20h
Reset: _instTCDX1_n3_alias[SMN,HOST]: 02h
Reset: _instTCDX1_n[5:4]_alias[SMN,HOST]: 10h
Reset: _instTCDX1_n6_alias[SMN,HOST]: 02h
Reset: _instTCDX1_n7_alias[SMN,HOST]: 10h
Reset: _instTCDX1_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n[17:16]_alias[SMN,HOST]: 02h
Reset: _instTCDX1_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n48_alias[SMN,HOST]: 10h
Reset: _instTCDX1_n49_alias[SMN,HOST]: 02h
Reset: _instTCDX1_n50_alias[SMN,HOST]: 10h
Reset: _instTCDX1_n51_alias[SMN,HOST]: 02h
Reset: _instTCDX1_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n54_alias[SMN,HOST]: 02h
Reset: _instTCDX1_n55_alias[SMN,HOST]: 10h
Reset: _instTCDX1_n[57:56]_alias[SMN,HOST]: 02h
Reset: _instTCDX1_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX1_n59_alias[SMN,HOST]: 10h
Reset: _instTCDX1_n[61:60]_alias[SMN,HOST]: 02h
Reset: _instTCDX1_n62_alias[SMN,HOST]: 10h
Reset: _instTCDX1_n63_alias[SMN,HOST]: 02h
Reset: _instTCDX2_n0_alias[SMN,HOST]: 20h
Reset: _instTCDX2_n1_alias[SMN,HOST]: 04h
Reset: _instTCDX2_n2_alias[SMN,HOST]: 10h
Reset: _instTCDX2_n[7:3]_alias[SMN,HOST]: 20h
Reset: _instTCDX2_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n[17:16]_alias[SMN,HOST]: 20h
Reset: _instTCDX2_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n[51:48]_alias[SMN,HOST]: 20h
Reset: _instTCDX2_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n[57:54]_alias[SMN,HOST]: 20h
Reset: _instTCDX2_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX2_n[63:59]_alias[SMN,HOST]: 20h
Reset: _instTCDX3_n[7:0]_alias[SMN,HOST]: 02h
Reset: _instTCDX3_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n16_alias[SMN,HOST]: 04h
Reset: _instTCDX3_n17_alias[SMN,HOST]: 01h
Reset: _instTCDX3_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n[51:48]_alias[SMN,HOST]: 02h
Reset: _instTCDX3_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n[55:54]_alias[SMN,HOST]: 02h
Reset: _instTCDX3_n56_alias[SMN,HOST]: 10h
Reset: _instTCDX3_n[58:57]_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n59_alias[SMN,HOST]: 02h
Reset: _instTCDX3_n60_alias[SMN,HOST]: 08h
Reset: _instTCDX3_n61_alias[SMN,HOST]: 01h
Reset: _instTCDX3_n[63:62]_alias[SMN,HOST]
Reset: _instTCDX4_n[7:0]_alias[SMN,HOST]
02h
Reset: _instTCDX4_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n16_alias[SMN,HOST]: 01h
Reset: _instTCDX4_n17_alias[SMN,HOST]: 04h
Reset: _instTCDX4_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n[51:48]_alias[SMN,HOST]: 02h

Reset: _instTCDX4_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n[55:54]_alias[SMN,HOST]: 02h
Reset: _instTCDX4_n56_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n57_alias[SMN,HOST]: 10h
Reset: _instTCDX4_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n59_alias[SMN,HOST]: 02h
Reset: _instTCDX4_n60_alias[SMN,HOST]: 01h
Reset: _instTCDX4_n61_alias[SMN,HOST]: 08h
Reset: _instTCDX4_n[63:62]_alias[SMN,HOST]
_instTCDX5_n0_alias[SMN,HOST]
02h
Reset: _instTCDX5_n[2:1]_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n3_alias[SMN,HOST]: 10h
Reset: _instTCDX5_n[5:4]_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n[7:6]_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n16_alias[SMN,HOST]: 01h
Reset: _instTCDX5_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX5_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n48_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n49_alias[SMN,HOST]: 10h
Reset: _instTCDX5_n50_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n51_alias[SMN,HOST]: 10h
Reset: _instTCDX5_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n[55:54]_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n56_alias[SMN,HOST]: 01h
Reset: _instTCDX5_n57_alias[SMN,HOST]: 20h
Reset: _instTCDX5_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n59_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n60_alias[SMN,HOST]: 01h
Reset: _instTCDX5_n61_alias[SMN,HOST]: 20h
Reset: _instTCDX5_n62_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n63_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n0_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n[2:1]_alias[SMN,HOST]: 02h
Reset: _instTCDX6_n3_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n[5:4]_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n6_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n16_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n17_alias[SMN,HOST]: 08h
Reset: _instTCDX6_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n48_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n49_alias[SMN,HOST]: 08h
Reset: _instTCDX6_n50_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n51_alias[SMN,HOST]: 08h
Reset: _instTCDX6_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n54_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n56_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n57_alias[SMN,HOST]: 08h
Reset: _instTCDX6_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n59_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n60_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n61_alias[SMN,HOST]: 08h
Reset: _instTCDX6_n62_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n63_alias[SMN,HOST]: 08h
Reset: _instTCDX7_n[5:0]_alias[SMN,HOST]: 01h
Reset: _instTCDX7_n6_alias[SMN,HOST]: 10h
Reset: _instTCDX7_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n[17:16]_alias[SMN,HOST]: 01h
Reset: _instTCDX7_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n[51:48]_alias[SMN,HOST]: 01h

Reset: _instTCDX7_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX7_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n[57:56]_alias[SMN,HOST]: 01h
Reset: _instTCDX7_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n59_alias[SMN,HOST]: 10h
Reset: _instTCDX7_n[63:60]_alias[SMN,HOST]
Reset: _instTCDX8_n[5:0]_alias[SMN,HOST]
01h
Reset: _instTCDX8_n6_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n7_alias[SMN,HOST]: 10h
Reset: _instTCDX8_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n[17:16]_alias[SMN,HOST]: 01h
Reset: _instTCDX8_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n[51:48]_alias[SMN,HOST]: 01h
Reset: _instTCDX8_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n54_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n55_alias[SMN,HOST]: 10h
Reset: _instTCDX8_n[57:56]_alias[SMN,HOST]: 01h
Reset: _instTCDX8_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n59_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n[63:60]_alias[SMN,HOST]: 01h
Reset: _instTCDX9_n[5:0]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n6_alias[SMN,HOST]: 04h
Reset: _instTCDX9_n7_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n[17:16]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n[51:48]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n54_alias[SMN,HOST]: 20h
Reset: _instTCDX9_n[57:55]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n59_alias[SMN,HOST]: 01h
Reset: _instTCDX9_n[63:60]_alias[SMN,HOST]: 08h
Reset: _instTCDX10_n[2:0]_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n[5:3]_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n6_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n16_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n[51:48]_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n56_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n57_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n[60:59]_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n[63:61]_alias[SMN,HOST]: 20h
Reset: _instTCDX11_n[6:0]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n7_alias[SMN,HOST]: 04h
Reset: _instTCDX11_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n[17:16]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n[51:48]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n54_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX11_n[57:56]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n[63:59]_alias[SMN,HOST]: 08h
Reset: _instTCDX12_n[2:0]_alias[SMN,HOST]: 20h

Reset: _instTCDX12_n3_alias[SMN,HOST]: 08h
Reset: _instTCDX12_n[7:4]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[17:16]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n48_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n49_alias[SMN,HOST]: 01h
Reset: _instTCDX12_n50_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n51_alias[SMN,HOST]: 02h
Reset: _instTCDX12_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[57:54]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[63:59]_alias[SMN,HOST]: 20h
Reset: _instTCDX13_n0_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n[2:1]_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n3_alias[SMN,HOST]: 01h
Reset: _instTCDX13_n[5:4]_alias[SMN,HOST]: 20h
Reset: _instTCDX13_n6_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n7_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n[17:16]_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n48_alias[SMN,HOST]: 20h
Reset: _instTCDX13_n49_alias[SMN,HOST]: 01h
Reset: _instTCDX13_n50_alias[SMN,HOST]: 20h
Reset: _instTCDX13_n51_alias[SMN,HOST]: 01h
Reset: _instTCDX13_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n54_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n55_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[57:56]_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n59_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[61:60]_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n62_alias[SMN,HOST]: 08h
Reset: _instTCDX13_n63_alias[SMN,HOST]: 04h
Reset: _instTCDX14_n[3:0]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n4_alias[SMN,HOST]: 04h
Reset: _instTCDX14_n5_alias[SMN,HOST]: 10h
Reset: _instTCDX14_n[7:6]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[17:16]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n48_alias[SMN,HOST]: 01h
Reset: _instTCDX14_n49_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n50_alias[SMN,HOST]: 02h
Reset: _instTCDX14_n51_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[57:54]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[63:59]_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[2:0]_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n[5:3]_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n6_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n16_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n[51:48]_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n56_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n57_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n58_alias[SMN,HOST]: 00h

	Reset: _instTCDX15_n[60:59]_alias[SMN,HOST]: 10h
	Reset: _instTCDX15_n[63:61]_alias[SMN,HOST]: 20h
23:22	Reserved.
21:16	LclReqRoute1 . Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the destination port for the request and probe packets.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_n0_alias[SMN,HOST]: 08h
	Reset: _instTCDX0_n[7:1]_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n[15:8]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n[17:16]_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n[47:18]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n[51:48]_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n[53:52]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n[57:54]_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n58_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n[63:59]_alias[SMN,HOST]: 20h
	Reset: _instTCDX1_n0_alias[SMN,HOST]: 01h
	Reset: _instTCDX1_n[2:1]_alias[SMN,HOST]: 20h
	Reset: _instTCDX1_n3_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n[5:4]_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n6_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n7_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n[15:8]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n[17:16]_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n[47:18]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n48_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n49_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n50_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n51_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n[53:52]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n54_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n55_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n[57:56]_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n58_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n59_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n[61:60]_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n62_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n63_alias[SMN,HOST]: 02h
	Reset: _instTCDX2_n0_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n1_alias[SMN,HOST]: 04h
	Reset: _instTCDX2_n2_alias[SMN,HOST]: 10h
	Reset: _instTCDX2_n[7:3]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n[15:8]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n[17:16]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n[47:18]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n[51:48]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n[53:52]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n[57:54]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n58_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n[63:59]_alias[SMN,HOST]: 20h
	Reset: _instTCDX3_n[7:0]_alias[SMN,HOST]: 02h
	Reset: _instTCDX3_n[15:8]_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n16_alias[SMN,HOST]: 04h
	Reset: _instTCDX3_n17_alias[SMN,HOST]: 01h
	Reset: _instTCDX3_n[47:18]_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n[51:48]_alias[SMN,HOST]: 02h
	Reset: _instTCDX3_n[53:52]_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n[55:54]_alias[SMN,HOST]: 02h
	Reset: _instTCDX3_n56_alias[SMN,HOST]: 10h
	Reset: _instTCDX3_n57_alias[SMN,HOST]: 01h
	Reset: _instTCDX3_n58_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n59_alias[SMN,HOST]: 02h
	Reset: _instTCDX3_n60_alias[SMN,HOST]: 08h

Reset: _instTCDX3_n61_alias[SMN,HOST]: 01h
Reset: _instTCDX3_n[63:62]_alias[SMN,HOST]
_instTCDX4_n[7:0]_alias[SMN,HOST]
02h
Reset: _instTCDX4_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n16_alias[SMN,HOST]: 01h
Reset: _instTCDX4_n17_alias[SMN,HOST]: 04h
Reset: _instTCDX4_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n[51:48]_alias[SMN,HOST]: 02h
Reset: _instTCDX4_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n[55:54]_alias[SMN,HOST]: 02h
Reset: _instTCDX4_n56_alias[SMN,HOST]: 01h
Reset: _instTCDX4_n57_alias[SMN,HOST]: 10h
Reset: _instTCDX4_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n59_alias[SMN,HOST]: 02h
Reset: _instTCDX4_n60_alias[SMN,HOST]: 01h
Reset: _instTCDX4_n61_alias[SMN,HOST]: 08h
Reset: _instTCDX4_n[63:62]_alias[SMN,HOST]
_instTCDX5_n0_alias[SMN,HOST]
02h
Reset: _instTCDX5_n[2:1]_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n3_alias[SMN,HOST]: 10h
Reset: _instTCDX5_n[5:4]_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n[7:6]_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n16_alias[SMN,HOST]: 01h
Reset: _instTCDX5_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX5_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n48_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n49_alias[SMN,HOST]: 10h
Reset: _instTCDX5_n50_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n51_alias[SMN,HOST]: 10h
Reset: _instTCDX5_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n[55:54]_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n56_alias[SMN,HOST]: 01h
Reset: _instTCDX5_n57_alias[SMN,HOST]: 20h
Reset: _instTCDX5_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n59_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n60_alias[SMN,HOST]: 01h
Reset: _instTCDX5_n61_alias[SMN,HOST]: 20h
Reset: _instTCDX5_n62_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n63_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n0_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n[2:1]_alias[SMN,HOST]: 02h
Reset: _instTCDX6_n3_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n[5:4]_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n6_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n16_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n17_alias[SMN,HOST]: 08h
Reset: _instTCDX6_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n48_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n49_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n50_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n51_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n54_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n56_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n57_alias[SMN,HOST]: 08h
Reset: _instTCDX6_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n59_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n60_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n61_alias[SMN,HOST]: 08h

Reset: _instTCDX6_n62_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n63_alias[SMN,HOST]: 08h
Reset: _instTCDX7_n[5:0]_alias[SMN,HOST]: 01h
Reset: _instTCDX7_n6_alias[SMN,HOST]: 10h
Reset: _instTCDX7_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n[17:16]_alias[SMN,HOST]: 01h
Reset: _instTCDX7_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n[51:48]_alias[SMN,HOST]: 01h
Reset: _instTCDX7_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX7_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n[57:56]_alias[SMN,HOST]: 01h
Reset: _instTCDX7_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n59_alias[SMN,HOST]: 10h
Reset: _instTCDX7_n[63:60]_alias[SMN,HOST]
Reset: _instTCDX8_n[5:0]_alias[SMN,HOST]
01h
Reset: _instTCDX8_n6_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n7_alias[SMN,HOST]: 10h
Reset: _instTCDX8_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n[17:16]_alias[SMN,HOST]: 01h
Reset: _instTCDX8_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n[51:48]_alias[SMN,HOST]: 01h
Reset: _instTCDX8_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n54_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n55_alias[SMN,HOST]: 10h
Reset: _instTCDX8_n[57:56]_alias[SMN,HOST]: 01h
Reset: _instTCDX8_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n59_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n[63:60]_alias[SMN,HOST]: 01h
Reset: _instTCDX9_n[5:0]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n6_alias[SMN,HOST]: 04h
Reset: _instTCDX9_n7_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n[17:16]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n[51:48]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n54_alias[SMN,HOST]: 20h
Reset: _instTCDX9_n[57:55]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n59_alias[SMN,HOST]: 01h
Reset: _instTCDX9_n[63:60]_alias[SMN,HOST]: 08h
Reset: _instTCDX10_n[6:0]_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n[17:16]_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n[51:48]_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[57:56]_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n[63:59]_alias[SMN,HOST]: 10h
Reset: _instTCDX11_n[6:0]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n7_alias[SMN,HOST]: 04h
Reset: _instTCDX11_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n[17:16]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n[51:48]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n54_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n55_alias[SMN,HOST]: 20h

Reset: _instTCDX11_n[57:56]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n[63:59]_alias[SMN,HOST]: 08h
Reset: _instTCDX12_n[2:0]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n3_alias[SMN,HOST]: 08h
Reset: _instTCDX12_n[7:4]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[17:16]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n48_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n49_alias[SMN,HOST]: 01h
Reset: _instTCDX12_n50_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n51_alias[SMN,HOST]: 02h
Reset: _instTCDX12_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[57:54]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[63:59]_alias[SMN,HOST]: 20h
Reset: _instTCDX13_n0_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n[2:1]_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n3_alias[SMN,HOST]: 01h
Reset: _instTCDX13_n[5:4]_alias[SMN,HOST]: 20h
Reset: _instTCDX13_n6_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n7_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n[17:16]_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n48_alias[SMN,HOST]: 20h
Reset: _instTCDX13_n49_alias[SMN,HOST]: 01h
Reset: _instTCDX13_n50_alias[SMN,HOST]: 20h
Reset: _instTCDX13_n51_alias[SMN,HOST]: 01h
Reset: _instTCDX13_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n54_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n55_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[57:56]_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n59_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[61:60]_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n62_alias[SMN,HOST]: 08h
Reset: _instTCDX13_n63_alias[SMN,HOST]: 04h
Reset: _instTCDX14_n[3:0]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n4_alias[SMN,HOST]: 04h
Reset: _instTCDX14_n5_alias[SMN,HOST]: 10h
Reset: _instTCDX14_n[7:6]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[17:16]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n48_alias[SMN,HOST]: 01h
Reset: _instTCDX14_n49_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n50_alias[SMN,HOST]: 02h
Reset: _instTCDX14_n51_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[57:54]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n58_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[63:59]_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[2:0]_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n[5:3]_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n6_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n16_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n[51:48]_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n54_alias[SMN,HOST]: 10h

	Reset: _instTCDX15_n55_alias[SMN,HOST]: 20h
	Reset: _instTCDX15_n56_alias[SMN,HOST]: 10h
	Reset: _instTCDX15_n57_alias[SMN,HOST]: 20h
	Reset: _instTCDX15_n58_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n[60:59]_alias[SMN,HOST]: 10h
	Reset: _instTCDX15_n[63:61]_alias[SMN,HOST]: 20h
15:14	Reserved.
13:8	LclRspRoute0 . Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the destination port for the response packets.
	AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_n0_alias[SMN,HOST]: 04h
	Reset: _instTCDX0_n1_alias[SMN,HOST]: 10h
	Reset: _instTCDX0_n[7:2]_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n[15:8]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n[17:16]_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n[47:18]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n48_alias[SMN,HOST]: 01h
	Reset: _instTCDX0_n49_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n50_alias[SMN,HOST]: 02h
	Reset: _instTCDX0_n51_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n[53:52]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n[57:54]_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n[59:58]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n[63:60]_alias[SMN,HOST]: 20h
	Reset: _instTCDX1_n[1:0]_alias[SMN,HOST]: 01h
	Reset: _instTCDX1_n2_alias[SMN,HOST]: 20h
	Reset: _instTCDX1_n[4:3]_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n5_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n6_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n7_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n[15:8]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n[17:16]_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n[47:18]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n48_alias[SMN,HOST]: 01h
	Reset: _instTCDX1_n49_alias[SMN,HOST]: 20h
	Reset: _instTCDX1_n50_alias[SMN,HOST]: 01h
	Reset: _instTCDX1_n51_alias[SMN,HOST]: 20h
	Reset: _instTCDX1_n[53:52]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n54_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n[57:55]_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n[59:58]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n[61:60]_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n62_alias[SMN,HOST]: 04h
	Reset: _instTCDX1_n63_alias[SMN,HOST]: 08h
	Reset: _instTCDX2_n[1:0]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n2_alias[SMN,HOST]: 08h
	Reset: _instTCDX2_n[7:3]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n[15:8]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n[17:16]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n[47:18]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n48_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n49_alias[SMN,HOST]: 01h
	Reset: _instTCDX2_n50_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n51_alias[SMN,HOST]: 02h
	Reset: _instTCDX2_n[53:52]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n[57:54]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n[59:58]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n[63:60]_alias[SMN,HOST]: 20h
	Reset: _instTCDX3_n[7:0]_alias[SMN,HOST]: 02h
	Reset: _instTCDX3_n[15:8]_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n[17:16]_alias[SMN,HOST]: 02h
	Reset: _instTCDX3_n[47:18]_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n[51:48]_alias[SMN,HOST]: 02h

Reset: _instTCDX3_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n[56:54]_alias[SMN,HOST]: 02h
Reset: _instTCDX3_n[59:57]_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n[63:60]_alias[SMN,HOST]
_instTCDX4_n[7:0]_alias[SMN,HOST]
02h
Reset: _instTCDX4_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n[17:16]_alias[SMN,HOST]: 02h
Reset: _instTCDX4_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n[51:48]_alias[SMN,HOST]: 02h
Reset: _instTCDX4_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n[55:54]_alias[SMN,HOST]: 02h
Reset: _instTCDX4_n56_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n57_alias[SMN,HOST]: 02h
Reset: _instTCDX4_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n[63:60]_alias[SMN,HOST]
_instTCDX5_n[1:0]_alias[SMN,HOST]
02h
Reset: _instTCDX5_n2_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n[4:3]_alias[SMN,HOST]: 10h
Reset: _instTCDX5_n5_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n[7:6]_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n16_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n17_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n48_alias[SMN,HOST]: 02h
Reset: _instTCDX5_n49_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n50_alias[SMN,HOST]: 02h
Reset: _instTCDX5_n51_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n[56:54]_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n57_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n60_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n61_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n62_alias[SMN,HOST]: 02h
Reset: _instTCDX5_n63_alias[SMN,HOST]
_instTCDX6_n[1:0]_alias[SMN,HOST]
04h
Reset: _instTCDX6_n2_alias[SMN,HOST]: 02h
Reset: _instTCDX6_n[4:3]_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n5_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n6_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n16_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n48_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n49_alias[SMN,HOST]: 02h
Reset: _instTCDX6_n50_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n51_alias[SMN,HOST]: 02h
Reset: _instTCDX6_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n54_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n56_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n57_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n60_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n61_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n62_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n63_alias[SMN,HOST]: 02h
Reset: _instTCDX7_n[5:0]_alias[SMN,HOST]: 01h
Reset: _instTCDX7_n6_alias[SMN,HOST]: 10h

Reset: _instTCDX7_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n16_alias[SMN,HOST]: 08h
Reset: _instTCDX7_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n[51:48]_alias[SMN,HOST]: 01h
Reset: _instTCDX7_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX7_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n56_alias[SMN,HOST]: 02h
Reset: _instTCDX7_n57_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n60_alias[SMN,HOST]: 04h
Reset: _instTCDX7_n61_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n[63:62]_alias[SMN,HOST]
Reset: _instTCDX8_n[5:0]_alias[SMN,HOST]
01h
Reset: _instTCDX8_n6_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n7_alias[SMN,HOST]: 10h
Reset: _instTCDX8_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n16_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n17_alias[SMN,HOST]: 08h
Reset: _instTCDX8_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n[51:48]_alias[SMN,HOST]: 01h
Reset: _instTCDX8_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n54_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n55_alias[SMN,HOST]: 10h
Reset: _instTCDX8_n56_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n57_alias[SMN,HOST]: 02h
Reset: _instTCDX8_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n60_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n61_alias[SMN,HOST]: 04h
Reset: _instTCDX8_n[63:62]_alias[SMN,HOST]: 01h
Reset: _instTCDX9_n[5:0]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n6_alias[SMN,HOST]: 02h
Reset: _instTCDX9_n7_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n[17:16]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n[51:48]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX9_n[57:55]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n[63:60]_alias[SMN,HOST]: 08h
Reset: _instTCDX10_n[2:0]_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n[5:3]_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n6_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n16_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n[51:48]_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n56_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n57_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n60_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n61_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[63:62]_alias[SMN,HOST]: 10h
Reset: _instTCDX11_n[6:0]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n7_alias[SMN,HOST]: 02h

Reset: _instTCDX11_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n[17:16]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n[51:48]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n54_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n55_alias[SMN,HOST]: 10h
Reset: _instTCDX11_n[57:56]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n[63:60]_alias[SMN,HOST]: 08h
Reset: _instTCDX12_n[2:0]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n3_alias[SMN,HOST]: 04h
Reset: _instTCDX12_n4_alias[SMN,HOST]: 10h
Reset: _instTCDX12_n[7:5]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[17:16]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[51:48]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[57:54]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[63:60]_alias[SMN,HOST]: 20h
Reset: _instTCDX13_n[1:0]_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n2_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[4:3]_alias[SMN,HOST]: 01h
Reset: _instTCDX13_n5_alias[SMN,HOST]: 20h
Reset: _instTCDX13_n6_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n7_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n[17:16]_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n48_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n49_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n50_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n51_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n54_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n[57:55]_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n[61:60]_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n62_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n63_alias[SMN,HOST]: 10h
Reset: _instTCDX14_n[4:0]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n5_alias[SMN,HOST]: 08h
Reset: _instTCDX14_n[7:6]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[17:16]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[51:48]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[57:54]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[63:60]_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[2:0]_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n[5:3]_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n6_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n16_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n[51:48]_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n55_alias[SMN,HOST]: 20h

	Reset: _instTCDX15_n56_alias[SMN,HOST]: 10h
	Reset: _instTCDX15_n57_alias[SMN,HOST]: 20h
	Reset: _instTCDX15_n[59:58]_alias[SMN,HOST]: 00h
	Reset: _instTCDX15_n60_alias[SMN,HOST]: 10h
	Reset: _instTCDX15_n61_alias[SMN,HOST]: 20h
	Reset: _instTCDX15_n[63:62]_alias[SMN,HOST]: 10h
7:6	Reserved.
5:0	LclReqRoute0 . Init: 11.5.3 [Register Values Written Based on Configuration]. Specifies the destination port for the request and probe packets. AccessType: SecurePart ? Read-only : Read-write.
	Reset: _instTCDX0_n0_alias[SMN,HOST]: 04h
	Reset: _instTCDX0_n1_alias[SMN,HOST]: 10h
	Reset: _instTCDX0_n[7:2]_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n[15:8]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n[17:16]_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n[47:18]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n48_alias[SMN,HOST]: 01h
	Reset: _instTCDX0_n49_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n50_alias[SMN,HOST]: 02h
	Reset: _instTCDX0_n51_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n[53:52]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n[57:54]_alias[SMN,HOST]: 20h
	Reset: _instTCDX0_n[59:58]_alias[SMN,HOST]: 00h
	Reset: _instTCDX0_n[63:60]_alias[SMN,HOST]: 20h
	Reset: _instTCDX1_n[1:0]_alias[SMN,HOST]: 01h
	Reset: _instTCDX1_n2_alias[SMN,HOST]: 20h
	Reset: _instTCDX1_n[4:3]_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n5_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n6_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n7_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n[15:8]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n[17:16]_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n[47:18]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n48_alias[SMN,HOST]: 01h
	Reset: _instTCDX1_n49_alias[SMN,HOST]: 20h
	Reset: _instTCDX1_n50_alias[SMN,HOST]: 01h
	Reset: _instTCDX1_n51_alias[SMN,HOST]: 20h
	Reset: _instTCDX1_n[53:52]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n54_alias[SMN,HOST]: 02h
	Reset: _instTCDX1_n[57:55]_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n[59:58]_alias[SMN,HOST]: 00h
	Reset: _instTCDX1_n[61:60]_alias[SMN,HOST]: 10h
	Reset: _instTCDX1_n62_alias[SMN,HOST]: 04h
	Reset: _instTCDX1_n63_alias[SMN,HOST]: 08h
	Reset: _instTCDX2_n[1:0]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n2_alias[SMN,HOST]: 08h
	Reset: _instTCDX2_n[7:3]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n[15:8]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n[17:16]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n[47:18]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n48_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n49_alias[SMN,HOST]: 01h
	Reset: _instTCDX2_n50_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n51_alias[SMN,HOST]: 02h
	Reset: _instTCDX2_n[53:52]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n[57:54]_alias[SMN,HOST]: 20h
	Reset: _instTCDX2_n[59:58]_alias[SMN,HOST]: 00h
	Reset: _instTCDX2_n[63:60]_alias[SMN,HOST]: 20h
	Reset: _instTCDX3_n[7:0]_alias[SMN,HOST]: 02h
	Reset: _instTCDX3_n[15:8]_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n[17:16]_alias[SMN,HOST]: 02h
	Reset: _instTCDX3_n[47:18]_alias[SMN,HOST]: 00h
	Reset: _instTCDX3_n[51:48]_alias[SMN,HOST]: 02h

Reset: _instTCDX3_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n[57:54]_alias[SMN,HOST]: 02h
Reset: _instTCDX3_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX3_n[63:60]_alias[SMN,HOST]
Reset: _instTCDX4_n[7:0]_alias[SMN,HOST]
02h
Reset: _instTCDX4_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n[17:16]_alias[SMN,HOST]: 02h
Reset: _instTCDX4_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n[51:48]_alias[SMN,HOST]: 02h
Reset: _instTCDX4_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n[57:54]_alias[SMN,HOST]: 02h
Reset: _instTCDX4_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX4_n[63:60]_alias[SMN,HOST]
Reset: _instTCDX5_n[1:0]_alias[SMN,HOST]
02h
Reset: _instTCDX5_n2_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n[4:3]_alias[SMN,HOST]: 10h
Reset: _instTCDX5_n5_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n[7:6]_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n16_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n17_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n48_alias[SMN,HOST]: 02h
Reset: _instTCDX5_n49_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n50_alias[SMN,HOST]: 02h
Reset: _instTCDX5_n51_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n[56:54]_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n57_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX5_n60_alias[SMN,HOST]: 04h
Reset: _instTCDX5_n61_alias[SMN,HOST]: 08h
Reset: _instTCDX5_n62_alias[SMN,HOST]: 02h
Reset: _instTCDX5_n63_alias[SMN,HOST]
Reset: _instTCDX6_n[1:0]_alias[SMN,HOST]
04h
Reset: _instTCDX6_n2_alias[SMN,HOST]: 02h
Reset: _instTCDX6_n[4:3]_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n5_alias[SMN,HOST]: 10h
Reset: _instTCDX6_n6_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n16_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n48_alias[SMN,HOST]: 08h
Reset: _instTCDX6_n49_alias[SMN,HOST]: 02h
Reset: _instTCDX6_n50_alias[SMN,HOST]: 08h
Reset: _instTCDX6_n51_alias[SMN,HOST]: 02h
Reset: _instTCDX6_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n54_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n56_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n57_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX6_n60_alias[SMN,HOST]: 01h
Reset: _instTCDX6_n61_alias[SMN,HOST]: 20h
Reset: _instTCDX6_n62_alias[SMN,HOST]: 04h
Reset: _instTCDX6_n63_alias[SMN,HOST]: 02h
Reset: _instTCDX7_n[5:0]_alias[SMN,HOST]: 01h
Reset: _instTCDX7_n6_alias[SMN,HOST]: 10h
Reset: _instTCDX7_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n[15:8]_alias[SMN,HOST]: 00h

Reset: _instTCDX7_n16_alias[SMN,HOST]: 08h
Reset: _instTCDX7_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n[51:48]_alias[SMN,HOST]: 01h
Reset: _instTCDX7_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX7_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n56_alias[SMN,HOST]: 02h
Reset: _instTCDX7_n57_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX7_n60_alias[SMN,HOST]: 04h
Reset: _instTCDX7_n61_alias[SMN,HOST]: 20h
Reset: _instTCDX7_n[63:62]_alias[SMN,HOST]
Reset: _instTCDX8_n[5:0]_alias[SMN,HOST]
01h
Reset: _instTCDX8_n6_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n7_alias[SMN,HOST]: 10h
Reset: _instTCDX8_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n16_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n17_alias[SMN,HOST]: 08h
Reset: _instTCDX8_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n[51:48]_alias[SMN,HOST]: 01h
Reset: _instTCDX8_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n54_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n55_alias[SMN,HOST]: 10h
Reset: _instTCDX8_n56_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n57_alias[SMN,HOST]: 02h
Reset: _instTCDX8_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX8_n60_alias[SMN,HOST]: 20h
Reset: _instTCDX8_n61_alias[SMN,HOST]: 04h
Reset: _instTCDX8_n[63:62]_alias[SMN,HOST]: 01h
Reset: _instTCDX9_n[5:0]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n6_alias[SMN,HOST]: 02h
Reset: _instTCDX9_n7_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n[17:16]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n[51:48]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX9_n[57:55]_alias[SMN,HOST]: 08h
Reset: _instTCDX9_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX9_n[63:60]_alias[SMN,HOST]: 08h
Reset: _instTCDX10_n[6:0]_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n16_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n[51:48]_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n56_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n57_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX10_n60_alias[SMN,HOST]: 10h
Reset: _instTCDX10_n61_alias[SMN,HOST]: 20h
Reset: _instTCDX10_n[63:62]_alias[SMN,HOST]: 10h
Reset: _instTCDX11_n[6:0]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n7_alias[SMN,HOST]: 02h
Reset: _instTCDX11_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n[17:16]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n[51:48]_alias[SMN,HOST]: 08h

Reset: _instTCDX11_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n54_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n55_alias[SMN,HOST]: 10h
Reset: _instTCDX11_n[57:56]_alias[SMN,HOST]: 08h
Reset: _instTCDX11_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX11_n[63:60]_alias[SMN,HOST]: 08h
Reset: _instTCDX12_n[2:0]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n3_alias[SMN,HOST]: 04h
Reset: _instTCDX12_n4_alias[SMN,HOST]: 10h
Reset: _instTCDX12_n[7:5]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[17:16]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[51:48]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[57:54]_alias[SMN,HOST]: 20h
Reset: _instTCDX12_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX12_n[63:60]_alias[SMN,HOST]: 20h
Reset: _instTCDX13_n[1:0]_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n2_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[4:3]_alias[SMN,HOST]: 01h
Reset: _instTCDX13_n5_alias[SMN,HOST]: 20h
Reset: _instTCDX13_n6_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n7_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n[17:16]_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n48_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n49_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n50_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n51_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n54_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n[57:55]_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX13_n[61:60]_alias[SMN,HOST]: 10h
Reset: _instTCDX13_n62_alias[SMN,HOST]: 02h
Reset: _instTCDX13_n63_alias[SMN,HOST]: 10h
Reset: _instTCDX14_n[4:0]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n5_alias[SMN,HOST]: 08h
Reset: _instTCDX14_n[7:6]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[17:16]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[51:48]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[57:54]_alias[SMN,HOST]: 20h
Reset: _instTCDX14_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX14_n[63:60]_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[2:0]_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n[5:3]_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n6_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n7_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[15:8]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n16_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n17_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[47:18]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n[51:48]_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n[53:52]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n54_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n55_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n56_alias[SMN,HOST]: 10h
Reset: _instTCDX15_n57_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[59:58]_alias[SMN,HOST]: 00h
Reset: _instTCDX15_n60_alias[SMN,HOST]: 10h

Reset: _instTCDX15_n61_alias[SMN,HOST]: 20h
Reset: _instTCDX15_n[63:62]_alias[SMN,HOST]: 10h

11.10.6 Function 5 Registers

D18F5x000 [Device/Vendor ID Function 5] (DF::DeviceVendorId5)

Read-only. Reset: Fixed,14B2_1022h.

PCI Device/Vendor ID 5 register

_aliasHOST; D18F5x000

_aliasSMN; D18F5x00000000; D18F5=4900_5000h

Bits	Description
31:16	DeviceID. Read-only. Reset: Fixed,14B2h. Device ID
15:0	VendorID. Read-only. Reset: Fixed,1022h. Vendor ID

D18F5x004 [Status Command Function 5] (DF::StatusCommand5)

Read-only. Reset: Fixed,0000_0000h.

PCI Status/Command register

_aliasHOST; D18F5x004

_aliasSMN; D18F5x00000004; D18F5=4900_5000h

Bits	Description
31:16	Status. Read-only. Reset: Fixed,0000h. Status
15:0	Command. Read-only. Reset: Fixed,0000h. Command

D18F5x008 [Class Code Revision ID Function 5] (DF::ClassCodeRevId5)

Read-only. Reset: Fixed,0600_0000h.

PCI Class Code/Revision ID register

_aliasHOST; D18F5x008

_aliasSMN; D18F5x00000008; D18F5=4900_5000h

Bits	Description
31:8	ClassCode. Read-only. Reset: Fixed,06_0000h. Class Code. Provides the hostbridge class code as defined in the PCI specification.
7:0	RevisionId. Read-only. Reset: Fixed,00h. RevID: Revision ID.

D18F5x00C [Header Type Function 5] (DF::HeaderType5)

Read-only. Reset: Fixed,0080_0000h.

PCI Header Type register

_aliasHOST; D18F5x00C

_aliasSMN; D18F5x0000000C; D18F5=4900_5000h

Bits	Description
31:0	HeaderType. Read-only. Reset: Fixed,0080_0000h. Header Type: These bits are fixed at their default values. The header type field indicates that there are multiple functions present in this device.

D18F5x034 [Capabilities Pointer Function 5] (DF::CapabilitiesPtr5)

Read-only. Reset: Fixed,0000_0000h.

PCI Capabilities Pointer register

_aliasHOST; D18F5x034

_aliasSMN; D18F5x00000034; D18F5=4900_5000h

Bits	Description
31:8	Reserved.
7:0	CapPtr. Read-only. Reset: Fixed,00h. Capabilities Pointer.

D18F5x100 [Enhanced Capability Header Function 5] (DF::EnhCapabilityHdr5)

Reset: 0000_0000h.	
PCI Enhanced Capability Header register	
_aliasHOST; D18F5x100	
_aliasSMN; D18F5x00000100; D18F5=4900_5000h	
Bits	Description
31:0	Reserved.

11.10.7 Function 6 Registers**D18F6x000 [Device/Vendor ID Function 6] (DF::DeviceVendorId6)**

Read-only. Reset: Fixed,14B3_1022h.	
PCI Device/Vendor ID 6 register	
_aliasHOST; D18F6x000	
_aliasSMN; D18F6x00000000; D18F6=4900_6000h	
Bits	Description
31:16	DeviceID. Read-only. Reset: Fixed,14B3h. Device ID
15:0	VendorID. Read-only. Reset: Fixed,1022h. Vendor ID

D18F6x004 [Status Command Function 6] (DF::StatusCommand6)

Read-only. Reset: Fixed,0000_0000h.	
PCI Status/Command register	
_aliasHOST; D18F6x004	
_aliasSMN; D18F6x00000004; D18F6=4900_6000h	
Bits	Description
31:16	Status. Read-only. Reset: Fixed,0000h. Status
15:0	Command. Read-only. Reset: Fixed,0000h. Command

D18F6x008 [Class Code Revision ID Function 6] (DF::ClassCodeRevId6)

Read-only. Reset: Fixed,0600_0000h.	
PCI Class Code/Revision ID register	
_aliasHOST; D18F6x008	
_aliasSMN; D18F6x00000008; D18F6=4900_6000h	
Bits	Description
31:8	ClassCode. Read-only. Reset: Fixed,06_0000h. Class Code. Provides the hostbridge class code as defined in the PCI specification.
7:0	RevisionId. Read-only. Reset: Fixed,00h. RevID: Revision ID.

D18F6x00C [Header Type Function 6] (DF::HeaderType6)

Read-only. Reset: Fixed,0080_0000h.	
PCI Header Type register	
_aliasHOST; D18F6x00C	
_aliasSMN; D18F6x0000000C; D18F6=4900_6000h	
Bits	Description
31:0	HeaderType. Read-only. Reset: Fixed,0080_0000h. Header Type: These bits are fixed at their default values. The header type field indicates that there are multiple functions present in this device.

D18F6x034 [Capabilities Pointer Function 6] (DF::CapabilitiesPtr6)

Read-only. Reset: Fixed,0000_0000h.

PCI Capabilities Pointer register

_aliasHOST; D18F6x034

_aliasSMN; D18F6x00000034; D18F6=4900_6000h

Bits	Description
31:8	Reserved.
7:0	CapPtr. Read-only. Reset: Fixed,00h. Capabilities Pointer.

D18F6x0A0 [CAKE NodeId Compression Table Index] (DF::CakeNodeIdCmpTblIndex)

Reset: 0000_0000h.

Index register for NodeId Compression Table indirect access programming interface.

_instBCST_aliasHOST; D18F6x0A0

_inst[CAKE[7:0]]_aliasHOST; D18F6x0A0_x[34:2D]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x028]

_instBCST_aliasSMN; D18F6x000000A0; D18F6=4900_6000h

_inst[CAKE[7:0]]_aliasSMN; D18F6x000000A0_x[34:2D]; D18F6=4900_6000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x028]

Bits	Description
31:8	Reserved.
7:0	NodeIdCmpTblIndex. Reset: 00h. Specifies the offset of the NodeId Compression Table entry register which is accessed when the NodeId Compression Table Data Port Register is read or written. Indirect register offsets are at multiples of 0x4. AccessType: SecurePart ? Read-only : Read-write.

D18F6x0A4_nodeidcmptblindex20...D18F6x0A4_nodeidcmptblindex3C [ReqNeighborNodeId Compression Table] (DF::ReqNeighborNodeIdCmpTblRegister)

Reset: 3F3F_3F3Fh.

Register in block of 8 32-bit Request Neighbor NodeId Compression Table registers.

_instBCST_n[7:0]_aliasHOST; D18F6x0A4_nodeidcmptblindex[[3:2][C,8,4,0]]; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]	
_inst[CAKE[7:0]]_n0_aliasHOST; D18F6x0A4_nodeidcmptblindex20_x[34:2D]; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n1_aliasHOST; D18F6x0A4_nodeidcmptblindex24_x[34:2D]; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n2_aliasHOST; D18F6x0A4_nodeidcmptblindex28_x[34:2D]; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n3_aliasHOST; D18F6x0A4_nodeidcmptblindex2C_x[34:2D]; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n4_aliasHOST; D18F6x0A4_nodeidcmptblindex30_x[34:2D]; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n5_aliasHOST; D18F6x0A4_nodeidcmptblindex34_x[34:2D]; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n6_aliasHOST; D18F6x0A4_nodeidcmptblindex38_x[34:2D]; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n7_aliasHOST; D18F6x0A4_nodeidcmptblindex3C_x[34:2D]; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_instBCST_n[7:0]_aliasSMN; D18F6x000000A4_nodeidcmptblindex[3C,38,34,30,2C,28,24,20]; D18F6=4900_6000h; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]	
_inst[CAKE[7:0]]_n0_aliasSMN; D18F6x000000A4_nodeidcmptblindex20_x[34:2D]; D18F6=4900_6000h; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n1_aliasSMN; D18F6x000000A4_nodeidcmptblindex24_x[34:2D]; D18F6=4900_6000h; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n2_aliasSMN; D18F6x000000A4_nodeidcmptblindex28_x[34:2D]; D18F6=4900_6000h; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n3_aliasSMN; D18F6x000000A4_nodeidcmptblindex2C_x[34:2D]; D18F6=4900_6000h; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n4_aliasSMN; D18F6x000000A4_nodeidcmptblindex30_x[34:2D]; D18F6=4900_6000h; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n5_aliasSMN; D18F6x000000A4_nodeidcmptblindex34_x[34:2D]; D18F6=4900_6000h; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n6_aliasSMN; D18F6x000000A4_nodeidcmptblindex38_x[34:2D]; D18F6=4900_6000h; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	
_inst[CAKE[7:0]]_n7_aliasSMN; D18F6x000000A4_nodeidcmptblindex3C_x[34:2D]; D18F6=4900_6000h; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]	

Bits	Description
31:30	Reserved.
29:24	SrcNode1 . Reset: 3Fh. Init: 11.5.3 [Register Values Written Based on Configuration]. This field specifies a compressible Source NodeId (FabXmt) when paired with the Destination NodeId (FabRcv) in the DstNode1 field. AccessType: SecurePart ? Read-only : Read-write.
23:22	Reserved.
21:16	DstNode1 . Reset: 3Fh. Init: 11.5.3 [Register Values Written Based on Configuration]. This field specifies a compressible Destination NodeId (FabRcv) when paired with the Source NodeId (FabXmt) in the SrcNode1 field. AccessType: SecurePart ? Read-only : Read-write.
15:14	Reserved.
13:8	SrcNode0 . Reset: 3Fh. Init: 11.5.3 [Register Values Written Based on Configuration]. This field specifies a compressible Source NodeId (FabXmt) when paired with the Destination NodeId (FabRcv) in the DstNode0 field. AccessType: SecurePart ? Read-only : Read-write.
7:6	Reserved.

5:0	DstNode0 . Reset: 3Fh. Init: 11.5.3 [Register Values Written Based on Configuration]. This field specifies a compressible Destination NodeId (FabRcv) when paired with the Source NodeId (FabXmt) in the SrcNode0 field. AccessType: SecurePart ? Read-only : Read-write.
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D18F6x0A4_nodeidcmptblindex48...D18F6x0A4_nodeidcmptblindex4C [RspNeighborNodeId Compression Table] (DF::RspNeighborNodeIdCmpTblRegister)

Reset: 3F3F_3F3Fh.

Register in block of 4 32-bit Response Neighbor NodeId Compression Table registers.

_instBCST_n[1:0]_aliasHOST; D18F6x0A4_nodeidcmptblindex4[C,8]; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]

_inst[CAKE[7:0]]_n0_aliasHOST; D18F6x0A4_nodeidcmptblindex48_x[34:2D]; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]

_inst[CAKE[7:0]]_n1_aliasHOST; D18F6x0A4_nodeidcmptblindex4C_x[34:2D]; DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]

_instBCST_n[1:0]_aliasSMN; D18F6x000000A4_nodeidcmptblindex4[C,8]; D18F6=4900_6000h;

DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]

_inst[CAKE[7:0]]_n0_aliasSMN; D18F6x000000A4_nodeidcmptblindex48_x[34:2D]; D18F6=4900_6000h;

DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]

_inst[CAKE[7:0]]_n1_aliasSMN; D18F6x000000A4_nodeidcmptblindex4C_x[34:2D]; D18F6=4900_6000h;

DataPortWrite=DF::CakeNodeIdCmpTblIndex[NodeIdCmpTblIndex]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x029]

Bits	Description
31:30	Reserved.
29:24	DstNode3 . Reset: 3Fh. This field specifies a compressible Destination NodeId (FabRcv). AccessType: SecurePart ? Read-only : Read-write.
23:22	Reserved.
21:16	DstNode2 . Reset: 3Fh. This field specifies a compressible Destination NodeId (FabRcv). AccessType: SecurePart ? Read-only : Read-write.
15:14	Reserved.
13:8	DstNode1 . Reset: 3Fh. This field specifies a compressible Destination NodeId (FabRcv). AccessType: SecurePart ? Read-only : Read-write.
7:6	Reserved.
5:0	DstNode0 . Reset: 3Fh. This field specifies a compressible Destination NodeId (FabRcv). AccessType: SecurePart ? Read-only : Read-write. Init: _instCAKE0_n0_alias[SMN,HOST] ((Is2Node2LinkSystem && IsNode0) (Is2NodeSystem && IsNode0)) ? 00h : ((Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 02h : (Is1NodeSystem && IsNode0) ? 3Fh Init: _instCAKE1_n0_alias[SMN,HOST] (Is2NodeSystem && IsNode0) ? 00h : (Is2NodeSystem && IsNode1) ? 02h : ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 3Fh Init: _instCAKE2_n0_alias[SMN,HOST] ((Is2Node2LinkSystem && IsNode0) (Is2NodeSystem && IsNode0)) ? 00h : ((Is2Node2LinkSystem && IsNode1) (Is2NodeSystem && IsNode1)) ? 02h : (Is1NodeSystem && IsNode0) ? 3Fh Init: _instCAKE3_n0_alias[SMN,HOST] (Is2NodeSystem && IsNode0) ? 00h : (Is2NodeSystem && IsNode1) ? 02h : ((Is1NodeSystem && IsNode0) (Is2Node2LinkSystem && IsNode0) (Is2Node2LinkSystem && IsNode1)) ? 3Fh Init: _inst[CAKE[7,5]]_n0_alias[SMN,HOST] (Is2Node2LinkSystem && IsNode0) ? 00h : (Is2Node2LinkSystem && IsNode1) ? 02h : ((Is1NodeSystem && IsNode0) (Is2NodeSystem && IsNode0) (Is2NodeSystem && IsNode1)) ? 3Fh

D18F6x0D8 [CAKE Encryption Control] (DF::CakeEncrCntl)

Reset: 0000_0000h.

CAKE Encryption Control/Status Register_instBCST_aliasHOST; D18F6x0D8_inst[CAKE[7:0]]_aliasHOST; D18F6x0D8_x[34:2D]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x036]_instBCST_aliasSMN; D18F6x000000D8; D18F6=4900_6000h_inst[CAKE[7:0]]_aliasSMN; D18F6x000000D8_x[34:2D]; D18F6=4900_6000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x036]**Bits Description**

31:1	Reserved.
0	EncrEn. Reset: 0. 0=Encryption disabled. 1=Encryption enabled. Encryptions stops immediately when this bit is set to 0. Encryption Enable control. Writing a 1 to this bit, even when already set, causes CAKE to restart encryption (reset counter; clear pipeline and prime with new nonce and key) and send an encryption_start command to the companion CAKE. AccessType: SecurePart ? Read-only : Read-write.

D18F6x100 [Enhanced Capability Header Function 6] (DF::EnhCapabilityHdr6)

Reset: 0000_0000h.

PCI Enhanced Capability Header register_aliasHOST; D18F6x100_aliasSMN; D18F6x00000100; D18F6=4900_6000h**Bits Description**

31:0	Reserved.
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D18F6x4A0 [Fuse Disable Register] (DF::FuseDisable)**Fuse Disable Register**_instBCST_aliasHOST; D18F6x4A0_inst[SPF[15:0],PFX[7:0],ICNG[3:0],IOS[3:0],IOM[3:0],ACM[3:0],CCM[7:0]]_aliasHOST; D18F6x4A0_x[[50:39],2[B:0],1[B:0]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:39],2[B:0],1[B:0]], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x128]_instBCST_aliasSMN; D18F6x000004A0; D18F6=4900_6000h_inst[SPF[15:0]]_aliasSMN; D18F6x000004A0_x[50:41]; D18F6=4900_6000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[50:41], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x128]**Bits Description**

31:17	Reserved.
16	SpfDisable. Init: 11.5.3 [Register Values Written Based on Configuration]. Disable the probe filter. AccessType: <u>_inst[BCST,ACM[3:0]]_aliasHOST</u> : Reserved AccessType: <u>_instBCST_aliasSMN</u> : SecurePart ? Read-only : Read-write AccessType: <u>_inst[PFX[7:0],IOS[3:0],IOM[3:0],ICNG[3:0],CCM[7:0]]_aliasHOST</u> : Reserved AccessType: <u>_inst[SPF[15:00]]_alias[SMN,HOST]</u> : SecurePart ? Read-only : Read-write Reset: <u>_instBCST_aliasSMN</u> <u>_inst[SPF[15:00]]_alias[SMN,HOST]</u> 0
15:1	Reserved.
0	ComponentDisable. Init: 11.5.3 [Register Values Written Based on Configuration]. Disable this component. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_inst[ACM[3:0]]_aliasHOST</u> : 0 Reset: <u>_instBCST_alias[SMN,HOST]</u> : 0 Reset: <u>_inst[CCM[7:0]]_aliasHOST</u> : 0 Reset: <u>_inst[ICNG[3:0]]_aliasHOST</u> : 0 Reset: <u>_inst[IOM[3:0]]_aliasHOST</u> : 0 Reset: <u>_inst[PFX[7:0],IOS[3:0]]_aliasHOST</u> <u>_inst[SPF[15:00]]_alias[SMN,HOST]</u> 0

D18F6x4A0 [Fuse Disable Register] (DF::FuseDisable_CAKE)

Reset: 0000_0000h.

Fuse Disable Register_instBCST_aliasHOST; D18F6x4A0_inst[CAKE[7:0]]_aliasHOST; D18F6x4A0_x[34:2D]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x128]_instBCST_aliasSMN; D18F6x000004A0; D18F6=4900_6000h_inst[CAKE[7:0]]_aliasSMN; D18F6x000004A0_x[34:2D]; D18F6=4900_6000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[34:2D], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x128]

Bits	Description
31:18	Reserved.
17	EncryptDisable . Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Disable encryption AccessType: SecurePart ? Read-only : Read-write.
16:1	Reserved.
0	ComponentDisable . Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Disable this component. AccessType: SecurePart ? Read-only : Read-write.

D18F6x4A0 [Fuse Disable Register] (DF::FuseDisable_CS)

Reset: 0000_0000h.

Fuse Disable Register_instBCST_aliasHOST; D18F6x4A0_inst[CMPI[3:0],CS[11:0]]_aliasHOST; D18F6x4A0_x0[F:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x128]_instBCST_aliasSMN; D18F6x000004A0; D18F6=4900_6000h_inst[CMPI[3:0],CS[11:0]]_aliasSMN; D18F6x000004A0_x0[F:0]; D18F6=4900_6000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x128]

Bits	Description
31:17	Reserved.
16	SpfDisable . Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Disable the probe filter. AccessType: SecurePart ? Read-only : Read-write.
15:1	Reserved.
0	ComponentDisable . Reset: 0. Init: 11.5.3 [Register Values Written Based on Configuration]. Disable this component. AccessType: SecurePart ? Read-only : Read-write.

D18F6x4A0 [Fuse Disable Register] (DF::FuseDisable_1)**Fuse Disable Register**_inst[CNLI[3:0],NCM[3:0]]_aliasHOST; D18F6x4A0_x[3[8:5],1[F:C]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[3[8:5],1[F:C]], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x128]_instBCST_aliasSMN; D18F6x000004A0; D18F6=4900_6000h_inst[PFX[7:0],CNLI[3:0],ICNG[3:0],IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_aliasSMN; D18F6x000004A0_x[[40:35],[2B:10]]; D18F6=4900_6000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[40:35],[2B:10]], IndCfgAccFuncNum=6, IndCfgAccRegNum=0x128]

Bits	Description
31:1	Reserved.
0	ComponentDisable . Init: 11.5.3 [Register Values Written Based on Configuration]. Disable this component. AccessType: SecurePart ? Read-only : Read-write. Reset: <u>_inst[PFX[7:0],IOS[3:0],ICNG[3:0],BCST]_aliasSMN</u> <u>_inst[NCM[3:0],CNLI[3:0]]_alias[SMN,HOST]</u> 0

11.10.8 Function 7 Registers**D18F7x000 [Device/Vendor ID Function 7] (DF::DeviceVendorId7)**

Read-only. Reset: Fixed,14B4_1022h.

PCI Device/Vendor ID 7 register

_aliasHOST; D18F7x000_aliasSMN; D18F7x00000000; D18F7=4900_7000h

Bits	Description
31:16	DeviceID . Read-only. Reset: Fixed,14B4h. Device ID
15:0	VendorID . Read-only. Reset: Fixed,1022h. Vendor ID

D18F7x004 [Status Command Function 7] (DF::StatusCommand7)

Read-only. Reset: Fixed,0000_0000h.

PCI Status/Command register

_aliasHOST; D18F7x004_aliasSMN; D18F7x00000004; D18F7=4900_7000h

Bits	Description
31:16	Status . Read-only. Reset: Fixed,0000h. Status
15:0	Command . Read-only. Reset: Fixed,0000h. Command

D18F7x008 [Class Code Revision ID Function 7] (DF::ClassCodeRevId7)

Read-only. Reset: Fixed,0600_0000h.

PCI Class Code/Revision ID register

_aliasHOST; D18F7x008_aliasSMN; D18F7x00000008; D18F7=4900_7000h

Bits	Description
31:8	ClassCode . Read-only. Reset: Fixed,06_0000h. Class Code. Provides the hostbridge class code as defined in the PCI specification.
7:0	RevisionId . Read-only. Reset: Fixed,00h. RevID: Revision ID.

D18F7x00C [Header Type Function 7] (DF::HeaderType7)

Read-only. Reset: Fixed,0080_0000h.

PCI Header Type register

_aliasHOST; D18F7x00C_aliasSMN; D18F7x0000000C; D18F7=4900_7000h

Bits	Description
31:0	HeaderType . Read-only. Reset: Fixed,0080_0000h. Header Type: These bits are fixed at their default values. The header type field indicates that there are multiple functions present in this device.

D18F7x034 [Capabilities Pointer Function 7] (DF::CapabilitiesPtr7)

Read-only. Reset: Fixed,0000_0000h.

PCI Capabilities Pointer register

_aliasHOST; D18F7x034_aliasSMN; D18F7x00000034; D18F7=4900_7000h

Bits	Description
31:8	Reserved.
7:0	CapPtr . Read-only. Reset: Fixed,00h. Capabilities Pointer.

D18F7x100 [Enhanced Capability Header Function 7] (DF::EnhCapabilityHdr7)

Reset: 0000_0000h.

PCI Enhanced Capability Header register

_aliasHOST; D18F7x100

_aliasSMN; D18F7x00000100; D18F7=4900_7000h

Bits Description

31:0 Reserved.

D18F7x104 [DRAM Hole Control] (DF::DramHoleControl)

Reset: 0000_0000h.

This register is setup by software as a per-instance register, as the value may differ for each coherent slave instance. See DF::DramAddressCtl[LgcyMmioHoleEn].

_instBCST_aliasHOST; D18F7x104

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F7x104_x[[50:41],2C,[27:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x041]

_instBCST_aliasSMN; D18F7x00000104; D18F7=4900_7000h

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F7x00000104_x[[50:41],2C,[27:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x041]

Bits Description

31:24 **DramHoleBase**. Reset: 00h. Init: 11.5.1.1 [DRAM Address Map Initialization]. Dram Hole Base[31:24]. Specifies the base address of the IO hole below the 4GB address level for legacy 32-bit devices. [MMIO](#) hole cannot span multiple DRAM ranges; it must lie within a single DRAM address range. DRAM range which spans the IO hole needs to be hoisted (added) to compensate for the hole.

AccessType: SecurePart ? Read-only : Read-write.

23:1 Reserved.

0 **DramHoleValid**. Reset: 0. Init: 11.5.1.1 [DRAM Address Map Initialization]. 1=Dram Hole Register is valid. Dram Hole Valid.

AccessType: SecurePart ? Read-only : Read-write.

D18F7x140...D18F7x148 [DRAM Offset] (DF::DramOffset)

Reset: 0000_0000h.

In the CS, these registers are associated with DRAM Address Maps 1 through N; see DF::DramBaseAddress and DF::DramLimitAddress. There is no 'DRAM Offset Register 0' because the first DRAM address in range 0 of every CS's DRAM space is zero; hence, the 'offset' for range 0 is always zero.

_instBCST_n[3:1]_aliasHOST; D18F7x14[8,4,0]

_inst[CMP[3:0],CS[11:0]]_n1_aliasHOST; D18F7x140_x0[F:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x050]

_inst[CMP[3:0],CS[11:0]]_n2_aliasHOST; D18F7x144_x0[F:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x051]

_inst[CMP[3:0],CS[11:0]]_n3_aliasHOST; D18F7x148_x0[F:0]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x052]

_instBCST_n[3:1]_aliasSMN; D18F7x0000_014[8,4,0]; D18F7=4900_7000h

_inst[CMP[3:0],CS[11:0]]_n1_aliasSMN; D18F7x00000140_x0[F:0]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x050]

_inst[CMP[3:0],CS[11:0]]_n2_aliasSMN; D18F7x00000144_x0[F:0]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x051]

_inst[CMP[3:0],CS[11:0]]_n3_aliasSMN; D18F7x00000148_x0[F:0]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x0[F:0], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x052]

Bits Description

31:25 Reserved.

24:1 **HiAddrOffset**. Reset: 00_0000h. Init: 11.5.1.1 [DRAM Address Map Initialization]. Offset address [51:28] for CS DRAM range number corresponds to its component access instance number. Specifies the normalized (DRAM) address at the base of the associated range.

AccessType: SecurePart ? Read-only : Read-write.

0 **HiAddrOffsetEn**. Reset: 0. Init: 11.5.1.1 [DRAM Address Map Initialization]. 1=The offset specified by HiAddrOffset is added when forming the normalized address. Control addition of HiAddrOffset when forming normalized address. This field must be set to one when HiAddrOffset is non-zero.

AccessType: SecurePart ? Read-only : Read-write.

D18F7x180 [CS Target Remap 0 Register A] (DF::CsTargetRemap0A)

Reset: 7654_3210h.

Logical to Physical CS Target Remap 0 Register A

_instBCST_aliasHOST; D18F7x180

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F7x180_x[[50:41],2C,[27:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x060]

_instBCST_aliasSMN; D18F7x00000180; D18F7=4900_7000h

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F7x00000180_x[[50:41],2C,[27:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x060]

Bits	Description
31:28	RemapCS7. Reset: 7h. Physical CS mapped to logical CS7 AccessType: SecurePart ? Read-only : Read-write.
27:24	RemapCS6. Reset: 6h. Physical CS mapped to logical CS6 AccessType: SecurePart ? Read-only : Read-write.
23:20	RemapCS5. Reset: 5h. Physical CS mapped to logical CS5 AccessType: SecurePart ? Read-only : Read-write.
19:16	RemapCS4. Reset: 4h. Physical CS mapped to logical CS4 AccessType: SecurePart ? Read-only : Read-write.
15:12	RemapCS3. Reset: 3h. Physical CS mapped to logical CS3 AccessType: SecurePart ? Read-only : Read-write.
11:8	RemapCS2. Reset: 2h. Physical CS mapped to logical CS2 AccessType: SecurePart ? Read-only : Read-write.
7:4	RemapCS1. Reset: 1h. Physical CS mapped to logical CS1 AccessType: SecurePart ? Read-only : Read-write.
3:0	RemapCS0. Reset: 0h. Physical CS mapped to logical CS0 AccessType: SecurePart ? Read-only : Read-write.

D18F7x184 [CS Target Remap 0 Register B] (DF::CsTargetRemap0B)

Reset: FEDC_BA98h.

Logical to Physical CS Target Remap 0 Register B

_instBCST_aliasHOST; D18F7x184

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F7x184_x[[50:41],2C,[27:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x061]

_instBCST_aliasSMN; D18F7x00000184; D18F7=4900_7000h

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F7x00000184_x[[50:41],2C,[27:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x061]

Bits	Description
31:28	RemapCS15. Reset: Fh. Physical CS mapped to logical CS15 AccessType: SecurePart ? Read-only : Read-write.
27:24	RemapCS14. Reset: Eh. Physical CS mapped to logical CS14 AccessType: SecurePart ? Read-only : Read-write.
23:20	RemapCS13. Reset: Dh. Physical CS mapped to logical CS13 AccessType: SecurePart ? Read-only : Read-write.
19:16	RemapCS12. Reset: Ch. Physical CS mapped to logical CS12 AccessType: SecurePart ? Read-only : Read-write.
15:12	RemapCS11. Reset: Bh. Physical CS mapped to logical CS11 AccessType: SecurePart ? Read-only : Read-write.
11:8	RemapCS10. Reset: Ah. Physical CS mapped to logical CS10 AccessType: SecurePart ? Read-only : Read-write.
7:4	RemapCS9. Reset: 9h. Physical CS mapped to logical CS9 AccessType: SecurePart ? Read-only : Read-write.
3:0	RemapCS8. Reset: 8h. Physical CS mapped to logical CS8 AccessType: SecurePart ? Read-only : Read-write.

D18F7x188 [CS Target Remap 1 Register A] (DF::CsTargetRemap1A)

Reset: 7654_3210h.

Logical to Physical CS Target Remap 1 Register A

_instBCST_aliasHOST; D18F7x188

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F7x188_x[[50:41],2C,[27:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x062]

_instBCST_aliasSMN; D18F7x00000188; D18F7=4900_7000h

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F7x00000188_x[[50:41],2C,[27:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x062]

Bits	Description
31:28	RemapCS7. Reset: 7h. Physical CS mapped to logical CS7 AccessType: SecurePart ? Read-only : Read-write.
27:24	RemapCS6. Reset: 6h. Physical CS mapped to logical CS6 AccessType: SecurePart ? Read-only : Read-write.
23:20	RemapCS5. Reset: 5h. Physical CS mapped to logical CS5 AccessType: SecurePart ? Read-only : Read-write.
19:16	RemapCS4. Reset: 4h. Physical CS mapped to logical CS4 AccessType: SecurePart ? Read-only : Read-write.
15:12	RemapCS3. Reset: 3h. Physical CS mapped to logical CS3 AccessType: SecurePart ? Read-only : Read-write.
11:8	RemapCS2. Reset: 2h. Physical CS mapped to logical CS2 AccessType: SecurePart ? Read-only : Read-write.
7:4	RemapCS1. Reset: 1h. Physical CS mapped to logical CS1 AccessType: SecurePart ? Read-only : Read-write.
3:0	RemapCS0. Reset: 0h. Physical CS mapped to logical CS0 AccessType: SecurePart ? Read-only : Read-write.

D18F7x18C [CS Target Remap 1 Register B] (DF::CsTargetRemap1B)

Reset: FEDC_BA98h.

Logical to Physical CS Target Remap 1 Register B

_instBCST_aliasHOST; D18F7x18C

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F7x18C_x[[50:41],2C,[27:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x063]

_instBCST_aliasSMN; D18F7x0000018C; D18F7=4900_7000h

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F7x0000018C_x[[50:41],2C,[27:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x063]

Bits	Description
31:28	RemapCS15. Reset: Fh. Physical CS mapped to logical CS15 AccessType: SecurePart ? Read-only : Read-write.
27:24	RemapCS14. Reset: Eh. Physical CS mapped to logical CS14 AccessType: SecurePart ? Read-only : Read-write.
23:20	RemapCS13. Reset: Dh. Physical CS mapped to logical CS13 AccessType: SecurePart ? Read-only : Read-write.
19:16	RemapCS12. Reset: Ch. Physical CS mapped to logical CS12 AccessType: SecurePart ? Read-only : Read-write.
15:12	RemapCS11. Reset: Bh. Physical CS mapped to logical CS11 AccessType: SecurePart ? Read-only : Read-write.
11:8	RemapCS10. Reset: Ah. Physical CS mapped to logical CS10 AccessType: SecurePart ? Read-only : Read-write.
7:4	RemapCS9. Reset: 9h. Physical CS mapped to logical CS9 AccessType: SecurePart ? Read-only : Read-write.
3:0	RemapCS8. Reset: 8h. Physical CS mapped to logical CS8 AccessType: SecurePart ? Read-only : Read-write.

D18F7x190 [CS Target Remap 2 Register A] (DF::CsTargetRemap2A)

Reset: 7654_3210h.

Logical to Physical CS Target Remap 2 Register A

_instBCST_aliasHOST; D18F7x190

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F7x190_x[[50:41],2C,[27:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x064]

_instBCST_aliasSMN; D18F7x00000190; D18F7=4900_7000h

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F7x00000190_x[[50:41],2C,[27:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x064]

Bits	Description
31:28	RemapCS7. Reset: 7h. Physical CS mapped to logical CS7 AccessType: SecurePart ? Read-only : Read-write.
27:24	RemapCS6. Reset: 6h. Physical CS mapped to logical CS6 AccessType: SecurePart ? Read-only : Read-write.
23:20	RemapCS5. Reset: 5h. Physical CS mapped to logical CS5 AccessType: SecurePart ? Read-only : Read-write.
19:16	RemapCS4. Reset: 4h. Physical CS mapped to logical CS4 AccessType: SecurePart ? Read-only : Read-write.
15:12	RemapCS3. Reset: 3h. Physical CS mapped to logical CS3 AccessType: SecurePart ? Read-only : Read-write.
11:8	RemapCS2. Reset: 2h. Physical CS mapped to logical CS2 AccessType: SecurePart ? Read-only : Read-write.
7:4	RemapCS1. Reset: 1h. Physical CS mapped to logical CS1 AccessType: SecurePart ? Read-only : Read-write.
3:0	RemapCS0. Reset: 0h. Physical CS mapped to logical CS0 AccessType: SecurePart ? Read-only : Read-write.

D18F7x194 [CS Target Remap 2 Register B] (DF::CsTargetRemap2B)

Reset: FEDC_BA98h.

Logical to Physical CS Target Remap 2 Register B

_instBCST_aliasHOST; D18F7x194

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F7x194_x[[50:41],2C,[27:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x065]

_instBCST_aliasSMN; D18F7x00000194; D18F7=4900_7000h

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F7x00000194_x[[50:41],2C,[27:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x065]

Bits	Description
31:28	RemapCS15. Reset: Fh. Physical CS mapped to logical CS15 AccessType: SecurePart ? Read-only : Read-write.
27:24	RemapCS14. Reset: Eh. Physical CS mapped to logical CS14 AccessType: SecurePart ? Read-only : Read-write.
23:20	RemapCS13. Reset: Dh. Physical CS mapped to logical CS13 AccessType: SecurePart ? Read-only : Read-write.
19:16	RemapCS12. Reset: Ch. Physical CS mapped to logical CS12 AccessType: SecurePart ? Read-only : Read-write.
15:12	RemapCS11. Reset: Bh. Physical CS mapped to logical CS11 AccessType: SecurePart ? Read-only : Read-write.
11:8	RemapCS10. Reset: Ah. Physical CS mapped to logical CS10 AccessType: SecurePart ? Read-only : Read-write.
7:4	RemapCS9. Reset: 9h. Physical CS mapped to logical CS9 AccessType: SecurePart ? Read-only : Read-write.
3:0	RemapCS8. Reset: 8h. Physical CS mapped to logical CS8 AccessType: SecurePart ? Read-only : Read-write.

D18F7x198 [CS Target Remap 3 Register A] (DF::CsTargetRemap3A)

Reset: 7654_3210h.

Logical to Physical CS Target Remap 3 Register A

`_instBCST_aliasHOST; D18F7x198``_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F7x198_x[[50:41],2C,[27:00]];
FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x066]``_instBCST_aliasSMN; D18F7x00000198; D18F7=4900_7000h``_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F7x00000198_x[[50:41],2C,[27:00]];
D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x066]`

Bits	Description
31:28	RemapCS7. Reset: 7h. Physical CS mapped to logical CS7 AccessType: SecurePart ? Read-only : Read-write.
27:24	RemapCS6. Reset: 6h. Physical CS mapped to logical CS6 AccessType: SecurePart ? Read-only : Read-write.
23:20	RemapCS5. Reset: 5h. Physical CS mapped to logical CS5 AccessType: SecurePart ? Read-only : Read-write.
19:16	RemapCS4. Reset: 4h. Physical CS mapped to logical CS4 AccessType: SecurePart ? Read-only : Read-write.
15:12	RemapCS3. Reset: 3h. Physical CS mapped to logical CS3 AccessType: SecurePart ? Read-only : Read-write.
11:8	RemapCS2. Reset: 2h. Physical CS mapped to logical CS2 AccessType: SecurePart ? Read-only : Read-write.
7:4	RemapCS1. Reset: 1h. Physical CS mapped to logical CS1 AccessType: SecurePart ? Read-only : Read-write.
3:0	RemapCS0. Reset: 0h. Physical CS mapped to logical CS0 AccessType: SecurePart ? Read-only : Read-write.

D18F7x19C [CS Target Remap 3 Register B] (DF::CsTargetRemap3B)

Reset: FEDC_BA98h.

Logical to Physical CS Target Remap 3 Register B

_instBCST_aliasHOST; D18F7x19C

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasHOST; D18F7x19C_x[[50:41],2C,[27:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x067]

_instBCST_aliasSMN; D18F7x0000019C; D18F7=4900_7000h

_inst[SPF[15:0],PIE0,IOS[3:0],IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_aliasSMN; D18F7x0000019C_x[[50:41],2C,[27:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x067]

Bits	Description
31:28	RemapCS15. Reset: Fh. Physical CS mapped to logical CS15 AccessType: SecurePart ? Read-only : Read-write.
27:24	RemapCS14. Reset: Eh. Physical CS mapped to logical CS14 AccessType: SecurePart ? Read-only : Read-write.
23:20	RemapCS13. Reset: Dh. Physical CS mapped to logical CS13 AccessType: SecurePart ? Read-only : Read-write.
19:16	RemapCS12. Reset: Ch. Physical CS mapped to logical CS12 AccessType: SecurePart ? Read-only : Read-write.
15:12	RemapCS11. Reset: Bh. Physical CS mapped to logical CS11 AccessType: SecurePart ? Read-only : Read-write.
11:8	RemapCS10. Reset: Ah. Physical CS mapped to logical CS10 AccessType: SecurePart ? Read-only : Read-write.
7:4	RemapCS9. Reset: 9h. Physical CS mapped to logical CS9 AccessType: SecurePart ? Read-only : Read-write.
3:0	RemapCS8. Reset: 8h. Physical CS mapped to logical CS8 AccessType: SecurePart ? Read-only : Read-write.

D18F7xD80...D18F7xDB0 [CXL™ Memory Base Address] (DF::CxlMemoryBaseAddress)

Reset: 0000_0000h.

Base, limit, and interleave registers provide mapping to identify the target [CXL](#) sublink of the address.[_instBCST_n\[3:0\]_aliasHOST](#); D18F7xD[B:8]0[_inst\[CNLI\[3:0\]\]_n0_aliasHOST](#); D18F7xD80_x3[8:5]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x360][_inst\[CNLI\[3:0\]\]_n1_aliasHOST](#); D18F7xD90_x3[8:5]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x364][_inst\[CNLI\[3:0\]\]_n2_aliasHOST](#); D18F7xDA0_x3[8:5]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x368][_inst\[CNLI\[3:0\]\]_n3_aliasHOST](#); D18F7xDB0_x3[8:5]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x36C][_instBCST_n\[3:0\]_aliasSMN](#); D18F7x0000_0D[B:8]0; D18F7=4900_7000h[_inst\[CNLI\[3:0\]\]_n0_aliasSMN](#); D18F7x0000D80_x3[8:5]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x360][_inst\[CNLI\[3:0\]\]_n1_aliasSMN](#); D18F7x0000D90_x3[8:5]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x364][_inst\[CNLI\[3:0\]\]_n2_aliasSMN](#); D18F7x0000DA0_x3[8:5]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x368][_inst\[CNLI\[3:0\]\]_n3_aliasSMN](#); D18F7x0000DB0_x3[8:5]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x36C]

Bits	Description
31:26	Reserved.
25:2	CxlMemBaseAddr. Reset: 00_0000h. Init: 11.5.1.2 [CXL™ Memory Address Map Initialization]. CXL Memory Base address[51:28] AccessType: SecurePart ? Read-only : Read-write.
1	NPAEn. Reset: 0. Init: 11.5.1.2 [CXL™ Memory Address Map Initialization]. 0=Do not remove interleaving bits. 1=Remove interleaving bits. remove bits used for interleaving before sent out AccessType: SecurePart ? Read-only : Read-write.
0	AddrRngVal. Reset: 0. Init: 11.5.1.2 [CXL™ Memory Address Map Initialization]. 0=Address range is disabled. 1=Address range is valid. Address range valid (enabled/disabled) AccessType: SecurePart ? Read-only : Read-write.

D18F7xD84...D18F7xDB4 [CXL™ Memory Limit Address] (DF::CxlMemoryLimitAddress)

Reset: 0000_0000h.

Base, limit, and interleave registers provides mapping to identify the target [CXL](#) sublink of the address.[_instBCST_n\[3:0\]_aliasHOST](#); D18F7xD[B:8]4[_inst\[CNLI\[3:0\]\]_n0_aliasHOST](#); D18F7xD84_x3[8:5]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x361][_inst\[CNLI\[3:0\]\]_n1_aliasHOST](#); D18F7xD94_x3[8:5]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x365][_inst\[CNLI\[3:0\]\]_n2_aliasHOST](#); D18F7xDA4_x3[8:5]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x369][_inst\[CNLI\[3:0\]\]_n3_aliasHOST](#); D18F7xDB4_x3[8:5]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x36D][_instBCST_n\[3:0\]_aliasSMN](#); D18F7x0000_0D[B:8]4; D18F7=4900_7000h[_inst\[CNLI\[3:0\]\]_n0_aliasSMN](#); D18F7x0000D84_x3[8:5]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x361][_inst\[CNLI\[3:0\]\]_n1_aliasSMN](#); D18F7x0000D94_x3[8:5]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x365][_inst\[CNLI\[3:0\]\]_n2_aliasSMN](#); D18F7x0000DA4_x3[8:5]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x369][_inst\[CNLI\[3:0\]\]_n3_aliasSMN](#); D18F7x0000DB4_x3[8:5]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x36D]

Bits	Description
31:24	Reserved.
23:0	CxlMemLimitAddr. Reset: 00_0000h. Init: 11.5.1.2 [CXL™ Memory Address Map Initialization]. CXL Memory limit address[51:28] AccessType: SecurePart ? Read-only : Read-write.

D18F7xD88 [CXL™ Memory Address Interleave] (DF::CxlMemAddressIntlv0)

Reset: 0000_0000h.

[CXL](#) Memory address interleaving information.[_instBCST_aliasHOST](#); D18F7xD88[_inst\[CNLI\[3:0\]\]_aliasHOST](#); D18F7xD88_x3[8:5]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x362][_instBCST_aliasSMN](#); D18F7x00000D88; D18F7=4900_7000h[_inst\[CNLI\[3:0\]\]_aliasSMN](#); D18F7x00000D88_x3[8:5]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x362]

Bits	Description
31:8	Reserved.
7:4	IntLvLinkEn. Reset: 0h. Init: 11.5.1.2 [CXL™ Memory Address Map Initialization]. Enabled sub-links for this address range which specifies the CXL links across which addresses are interleaved. Non-zero if the address range is enabled. 0xF means interleaving across all 4 sub-links AccessType: SecurePart ? Read-only : Read-write.
3:1	Reserved.
0	IntLvAddrSel. Reset: 0. Init: 11.5.1.1 [DRAM Address Map Initialization]. 0=Addr[6]. 1=Addr[8]. Encoded value for this address range which specifies the starting address bit used for interleaving. The number of address bits used for interleaving depends on the number of links across which they are interleaved. Values not listed are RESERVED AccessType: SecurePart ? Read-only : Read-write.

D18F7xD98 [CXL™ Memory Address Interleave] (DF::CxlMemAddressIntlv1)

Reset: 0000_0000h.

[CXL](#) Memory address interleaving information.[_instBCST_aliasHOST](#); D18F7xD98[_inst\[CNLI\[3:0\]\]_aliasHOST](#); D18F7xD98_x3[8:5]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x366][_instBCST_aliasSMN](#); D18F7x00000D98; D18F7=4900_7000h[_inst\[CNLI\[3:0\]\]_aliasSMN](#); D18F7x00000D98_x3[8:5]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x366]

Bits	Description
31:8	Reserved.
7:4	IntLvLinkEn. Reset: 0h. Init: 11.5.1.2 [CXL™ Memory Address Map Initialization]. Enabled sub-links for this address range which specifies the CXL links across which addresses are interleaved. Non-zero if the address range is enabled. 0xF means interleaving across all 4 sub-links AccessType: SecurePart ? Read-only : Read-write.
3:1	Reserved.
0	IntLvAddrSel. Reset: 0. Init: 11.5.1.1 [DRAM Address Map Initialization]. 0=Addr[6]. 1=Addr[8]. Encoded value for this address range which specifies the starting address bit used for interleaving. The number of address bits used for interleaving depends on the number of links across which they are interleaved. Values not listed are RESERVED AccessType: SecurePart ? Read-only : Read-write.

D18F7xDA8 [CXL™ Memory Address Interleave] (DF::CxlMemAddressIntlv2)

Reset: 0000_0000h.

[CXL](#) Memory address interleaving information.[_instBCST_aliasHOST](#); D18F7xDA8[_inst\[CNLI\[3:0\]\]_aliasHOST](#); D18F7xDA8_x3[8:5]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x36A][_instBCST_aliasSMN](#); D18F7x00000DA8; D18F7=4900_7000h[_inst\[CNLI\[3:0\]\]_aliasSMN](#); D18F7x00000DA8_x3[8:5]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x36A]

Bits	Description
31:8	Reserved.
7:4	IntLvLinkEn. Reset: 0h. Init: 11.5.1.2 [CXL™ Memory Address Map Initialization]. Enabled sub-links for this address range which specifies the CXL links across which addresses are interleaved. Non-zero if the address range is enabled. 0xF means interleaving across all 4 sub-links AccessType: SecurePart ? Read-only : Read-write.
3:1	Reserved.
0	IntLvAddrSel. Reset: 0. Init: 11.5.1.1 [DRAM Address Map Initialization]. 0=Addr[6]. 1=Addr[8]. Encoded value for this address range which specifies the starting address bit used for interleaving. The number of address bits used for interleaving depends on the number of links across which they are interleaved. Values not listed are RESERVED AccessType: SecurePart ? Read-only : Read-write.

D18F7xDB8 [CXL™ Memory Address Interleave] (DF::CxlMemAddressIntlv3)

Reset: 0000_0000h.

[CXL](#) Memory address interleaving information.[_instBCST_aliasHOST](#); D18F7xDB8[_inst\[CNLI\[3:0\]\]_aliasHOST](#); D18F7xDB8_x3[8:5]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x36E][_instBCST_aliasSMN](#); D18F7x00000DB8; D18F7=4900_7000h[_inst\[CNLI\[3:0\]\]_aliasSMN](#); D18F7x00000DB8_x3[8:5]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x3[8:5], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x36E]

Bits	Description
31:8	Reserved.
7:4	IntLvLinkEn. Reset: 0h. Init: 11.5.1.2 [CXL™ Memory Address Map Initialization]. Enabled sub-links for this address range which specifies the CXL links across which addresses are interleaved. Non-zero if the address range is enabled. 0xF means interleaving across all 4 sub-links AccessType: SecurePart ? Read-only : Read-write.
3:1	Reserved.
0	IntLvAddrSel. Reset: 0. Init: 11.5.1.1 [DRAM Address Map Initialization]. 0=Addr[6]. 1=Addr[8]. Encoded value for this address range which specifies the starting address bit used for interleaving. The number of address bits used for interleaving depends on the number of links across which they are interleaved. Values not listed are RESERVED AccessType: SecurePart ? Read-only : Read-write.

D18F7xE00...D18F7xF30 [DRAM Base Address] (DF::DramBaseAddress)

Reset: 0000_0000h.

limit, ctl, and interleave registers provide mapping to identify the target coherent slave of the address. See 11.6.2 [DRAM Address Maps].

_instBCST_n[19:0]_aliasHOST; D18F7x[F3:E0]0
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n0_aliasHOST; D18F7xE00_x[[50:41],2C,[27:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x380]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n1_aliasHOST; D18F7xE10_x[[50:41],2C,[23:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x384]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n2_aliasHOST; D18F7xE20_x[[50:41],2C,[23:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x388]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n3_aliasHOST; D18F7xE30_x[[50:41],2C,[23:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x38C]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n4_aliasHOST; D18F7xE40_x[[50:41],2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x390]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n5_aliasHOST; D18F7xE50_x[[50:41],2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x394]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n6_aliasHOST; D18F7xE60_x[[50:41],2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x398]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n7_aliasHOST; D18F7xE70_x[[50:41],2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x39C]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n8_aliasHOST; D18F7xE80_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3A0]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n9_aliasHOST; D18F7xE90_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3A4]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n10_aliasHOST; D18F7xEA0_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3A8]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n11_aliasHOST; D18F7xEB0_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3AC]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n12_aliasHOST; D18F7xEC0_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B0]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n13_aliasHOST; D18F7xED0_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B4]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n14_aliasHOST; D18F7xEE0_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B8]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n15_aliasHOST; D18F7xEF0_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3BC]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n16_aliasHOST; D18F7xF00_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C0]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n17_aliasHOST; D18F7xF10_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C4]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n18_aliasHOST; D18F7xF20_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C8]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n19_aliasHOST; D18F7xF30_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3CC]
_instBCST_n[19:0]_aliasSMN; D18F7x0000_0[F3:E0]0; D18F7=4900_7000h
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n0_aliasSMN; D18F7x00000E00_x[[50:41],2C,[27:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x380]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n1_aliasSMN; D18F7x00000E10_x[[50:41],2C,[23:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x384]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n2_aliasSMN; D18F7x00000E20_x[[50:41],2C,[23:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x388]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n3_aliasSMN; D18F7x00000E30_x[[50:41],2C,[23:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x38C]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n4_aliasSMN; D18F7x00000E40_x[[50:41],2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x390]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n5_aliasSMN; D18F7x00000E50_x[[50:41],2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x394]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n6_aliasSMN; D18F7x00000E60_x[[50:41],2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x398]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n7_aliasSMN; D18F7x00000E70_x[[50:41],2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x39C]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n8_aliasSMN; D18F7x00000E80_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3A0]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n9_aliasSMN; D18F7x00000E90_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3A4]

_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n10_aliasSMN; D18F7x00000EA0_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3A8]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n11_aliasSMN; D18F7x00000EB0_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3AC]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n12_aliasSMN; D18F7x00000EC0_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B0]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n13_aliasSMN; D18F7x00000ED0_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B4]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n14_aliasSMN; D18F7x00000EE0_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B8]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n15_aliasSMN; D18F7x00000EF0_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3BC]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n16_aliasSMN; D18F7x00000F00_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C0]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n17_aliasSMN; D18F7x00000F10_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C4]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n18_aliasSMN; D18F7x00000F20_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C8]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n19_aliasSMN; D18F7x00000F30_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3CC]	
Bits	Description
31:28	Reserved.
27:0	DramBaseAddr. Reset: 000_0000h. Init: 11.5.1.1 [DRAM Address Map Initialization]. Dram Base address[55:28]
	AccessType: SecurePart ? Read-only : Read-write.

D18F7xE04...D18F7xF34 [DRAM Limit Address] (DF::DramLimitAddress)

Reset: 0000_0000h.

The 8 DRAM base, limit, ctl, and interleave registers provides mapping to identify the target coherent slave of the address. See DF::DramBaseAddress.

_instBCST_n[19:0]_aliasHOST; D18F7x[F3:E0]4
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n0_aliasHOST; D18F7xE04_x[[50:41],2C,[27:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x381]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n1_aliasHOST; D18F7xE14_x[[50:41],2C,[23:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x385]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n2_aliasHOST; D18F7xE24_x[[50:41],2C,[23:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x389]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n3_aliasHOST; D18F7xE34_x[[50:41],2C,[23:00]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x38D]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n4_aliasHOST; D18F7xE44_x[[50:41],2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x391]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n5_aliasHOST; D18F7xE54_x[[50:41],2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x395]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n6_aliasHOST; D18F7xE64_x[[50:41],2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x399]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n7_aliasHOST; D18F7xE74_x[[50:41],2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x39D]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n8_aliasHOST; D18F7xE84_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3A1]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n9_aliasHOST; D18F7xE94_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3A5]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n10_aliasHOST; D18F7xEA4_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3A9]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n11_aliasHOST; D18F7xEB4_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3AD]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n12_aliasHOST; D18F7xEC4_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B1]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n13_aliasHOST; D18F7xED4_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B5]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n14_aliasHOST; D18F7xEE4_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B9]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n15_aliasHOST; D18F7xEF4_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3BD]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n16_aliasHOST; D18F7xF04_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C1]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n17_aliasHOST; D18F7xF14_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C5]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n18_aliasHOST; D18F7xF24_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C9]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n19_aliasHOST; D18F7xF34_x[2C,[23:10]]; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3CD]
_instBCST_n[19:0]_aliasSMN; D18F7x0000_0[F3:E0]4; D18F7=4900_7000h
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n0_aliasSMN; D18F7x00000E04_x[[50:41],2C,[27:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[27:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x381]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n1_aliasSMN; D18F7x00000E14_x[[50:41],2C,[23:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x385]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n2_aliasSMN; D18F7x00000E24_x[[50:41],2C,[23:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x389]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0],CMP[3:0],CS[11:0]]_n3_aliasSMN; D18F7x00000E34_x[[50:41],2C,[23:00]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:00]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x38D]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n4_aliasSMN; D18F7x00000E44_x[[50:41],2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x391]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n5_aliasSMN; D18F7x00000E54_x[[50:41],2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x395]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n6_aliasSMN; D18F7x00000E64_x[[50:41],2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x399]
_inst[SPF[15:0],PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n7_aliasSMN; D18F7x00000E74_x[[50:41],2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[[50:41],2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x39D]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n8_aliasSMN; D18F7x00000E84_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3A1]
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n9_aliasSMN; D18F7x00000E94_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3A5]

_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n10_aliasSMN; D18F7x00000EA4_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3A9]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n11_aliasSMN; D18F7x00000EB4_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3AD]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n12_aliasSMN; D18F7x00000EC4_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B1]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n13_aliasSMN; D18F7x00000ED4_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B5]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n14_aliasSMN; D18F7x00000EE4_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B9]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n15_aliasSMN; D18F7x00000EF4_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3BD]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n16_aliasSMN; D18F7x00000F04_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C1]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n17_aliasSMN; D18F7x00000F14_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C5]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n18_aliasSMN; D18F7x00000F24_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C9]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n19_aliasSMN; D18F7x00000F34_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3CD]	
Bits	Description
31:28	Reserved.
27:0	DramLimitAddr. Reset: 000_0000h. Init: 11.5.1.1 [DRAM Address Map Initialization]. DRAM limit address[55:28]
	AccessType: SecurePart ? Read-only : Read-write.

```
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n10_aliasSMN; D18F7x00000EA8_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3AA]
```

_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n11_aliasSMN; D18F7x00000EB8_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3AE]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n12_aliasSMN; D18F7x00000EC8_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B2]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n13_aliasSMN; D18F7x00000ED8_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B6]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n14_aliasSMN; D18F7x00000EE8_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3BA]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n15_aliasSMN; D18F7x00000EF8_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3BE]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n16_aliasSMN; D18F7x00000F08_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C2]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n17_aliasSMN; D18F7x00000F18_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C6]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n18_aliasSMN; D18F7x00000F28_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3CA]	
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n19_aliasSMN; D18F7x00000F38_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3CE]	
Bits	Description
31:28	Reserved.
27:16	DstFabricID. Reset: 000h. Init: 11.5.1.1 [DRAM Address Map Initialization]. AccessType: SecurePart ? Read-only : Read-write. Description: When CS interleaving is disabled (see DF::DramAddressIntlv[IntLvNumChan]) this field provides the target CS FabricId for this address map. When CS interleaving is enabled, this field provides the CS Fabric ID where interleave starts.
15:11	Reserved.
10	HashIntlvCtl1G. Reset: 1. Init: 11.5.1.1 [DRAM Address Map Initialization]. Enables inclusion of address bits on the 1GB boundary (bits 30, 31, 32) in a hashed interleave computation. AccessType: SecurePart ? Read-only : Read-write.
9	HashIntlvCtl2M. Reset: 1. Init: 11.5.1.1 [DRAM Address Map Initialization]. Enables inclusion of address bits on the 2MB boundary (bits 21, 22, 23) in a hashed interleave computation. AccessType: SecurePart ? Read-only : Read-write.
8	HashIntlvCtl64K. Reset: 1. Init: 11.5.1.1 [DRAM Address Map Initialization]. Enables inclusion of address bits on the 64KB boundary (bits 16, 17, 18) in a hashed interleave computation. AccessType: SecurePart ? Read-only : Read-write.
7:5	RemapSel. Reset: 0h. Remap Select AccessType: SecurePart ? Read-only : Read-write.
4	RemapEn. Reset: 0. Remap Enable AccessType: SecurePart ? Read-only : Read-write.
3	Reserved.
2	SCM. Reset: 0. Init: 11.5.1.1 [DRAM Address Map Initialization]. 0=Memory type for this address map is volatile. 1=Memory type for this address map is Storage Class Memory. If set, this field indicates that the addresses described by this address map are stored in Storage Class Memory. AccessType: SecurePart ? Read-only : Read-write.
1	LgcyMmioHoleEn. Reset: 0. Init: 11.5.1.1 [DRAM Address Map Initialization]. 0=Memory hoisting is not enabled. 1=Enable memory hoisting for this address range. AccessType: SecurePart ? Read-only : Read-write. Description: This field is only permitted to be set in at most one DRAM range. BIOS sets this bit for an address range that spans the 4GB boundary and contains a hole for addresses used by legacy MMIO . Additional information needs to be programmed; see DF::DramHoleControl.
0	AddrRngVal. Reset: 0. Init: 11.5.1.1 [DRAM Address Map Initialization]. 0=Address range disabled. 1=Address range is valid. Address range valid (enabled/disabled) AccessType: SecurePart ? Read-only : Read-write.

Reset: 0000 0000h.

instBCST_n[19:0]_aliasHOST; D18F7x[F3:E0]C

[illegible]

_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n11_aliasSMN; D18F7x00000EBC_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3AF]															
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n12_aliasSMN; D18F7x00000ECC_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B3]															
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n13_aliasSMN; D18F7x00000EDC_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3B7]															
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n14_aliasSMN; D18F7x00000EEC_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3BB]															
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n15_aliasSMN; D18F7x00000EFC_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3BF]															
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n16_aliasSMN; D18F7x00000F0C_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C3]															
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n17_aliasSMN; D18F7x00000F1C_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3C7]															
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n18_aliasSMN; D18F7x00000F2C_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3CB]															
_inst[PIE0,IOM[3:0],NCM[3:0],ACM[3:0],CCM[7:0]]_n19_aliasSMN; D18F7x00000F3C_x[2C,[23:10]]; D18F7=4900_7000h; FabricIndirectConfigAccessAddress[CfgRegInstID=0x[2C,[23:10]], IndCfgAccFuncNum=7, IndCfgAccRegNum=0x3CF]															
Bits	Description														
31:19	Reserved.														
18	IntLvNumSockets. Reset: 0. Init: 11.5.1.1 [DRAM Address Map Initialization]. 0=No socket interleave. 1=2 sockets. Specifies the number of sockets across which addresses are interleaved. AccessType: SecurePart ? Read-only : Read-write.														
17:14	Reserved.														
13:12	IntLvNumDies. Reset: 0h. Init: 11.5.1.1 [DRAM Address Map Initialization]. Encoded value which specifies the number of dies across which addresses are interleaved. AccessType: SecurePart ? Read-only : Read-write. ValidValues:														
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>1 die (no interleave)</td></tr> <tr> <td>3h-1h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	1 die (no interleave)	3h-1h	Reserved.								
Value	Description														
0h	1 die (no interleave)														
3h-1h	Reserved.														
11:9	Reserved.														
8:4	IntLvNumChan. Reset: 00h. Init: 11.5.1.1 [DRAM Address Map Initialization]. AccessType: SecurePart ? Read-only : Read-write. Description: Encoded value for this address range which specifies the number of coherent slaves across which addresses are interleaved. Hash interleaves must program IntLvAddrSel to select address bit [8]. ValidValues: See: DF::DramAddressIntlv[IntLvNumChan] Valid Values.														
3	Reserved.														
2:0	IntLvAddrSel. Reset: 0h. Init: 11.5.1.1 [DRAM Address Map Initialization]. Encoded value for this address range which specifies the starting address bit used for interleaving. The number of address bits used for interleaving depends on the number of channels across which they are interleaved. Values not listed are RESERVED AccessType: SecurePart ? Read-only : Read-write. ValidValues:														
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Addr[8]</td></tr> <tr> <td>1h</td><td>Addr[9]</td></tr> <tr> <td>2h</td><td>Addr[10]</td></tr> <tr> <td>3h</td><td>Addr[11]</td></tr> <tr> <td>4h</td><td>Addr[12]</td></tr> <tr> <td>7h-5h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Addr[8]	1h	Addr[9]	2h	Addr[10]	3h	Addr[11]	4h	Addr[12]	7h-5h	Reserved.
Value	Description														
0h	Addr[8]														
1h	Addr[9]														
2h	Addr[10]														
3h	Addr[11]														
4h	Addr[12]														
7h-5h	Reserved.														

DF::DramAddressIntlv[IntLvNumChan] Valid Values

Value	Description
00h	1 channel (no interleave)
01h	2 channels
02h	Reserved.
03h	4 channels
04h	Reserved.
05h	8 channels
06h	Reserved.
07h	16 channels
08h	32 channels
0Fh-09h	Reserved.
10h	Hashed 2 channel interleaving
11h	Hashed 4 channel interleaving
12h	Hashed 8 channel interleaving
13h	Hashed 3 channel interleaving
14h	Hashed 6 channel interleaving
15h	Hashed 12 channel interleaving
16h	Hashed 5 channel interleaving
17h	Hashed 10 channel interleaving
1Fh-18h	Reserved.

11.10.9 CXL™ Registers

CXL[0...3]x00000000 [CXL™ Capability Header Register] (DF::CxlCapabilityHeader0)

Read-only, Volatile. Reset: 0311_0001h.

Specify the header for [CXL](#) Capability Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000000; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:24	ArraySize. Read-only, Volatile. Reset: 03h. defines the number of elements present in the CXL_Capability array, not including the CXL_Capability_Header element.
23:20	CxlCacheMemVersion. Read-only, Volatile. Reset: 1h. This defines the version of the CXL Cache Mem Protocol supported.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0001h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x00000004 [CXL™ Capability Header Register] (DF::CxlRASCapabilityHeader0)

Read-only, Volatile. Reset: 0401_0002h.

Specify the header for [CXL RAS](#) Capability Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000004; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:20	CxlRASCapabilityPointer. Read-only, Volatile. Reset: 040h. This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0002h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x00000008 [CXL™ Security Capability Header Register] (DF::CxlSecurityCapabilityHeader0)

Read-only, Volatile. Reset: 0101_0003h.

Specify the header for [CXL](#) Security Capability Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000008; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:20	CxlSecurityCapabilityPointer. Read-only, Volatile. Reset: 010h. This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0003h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x0000000C [CXL™ Link Capability Header Register] (DF::CxlLinkCapabilityHeader0)

Read-only, Volatile. Reset: 0141_0004h.

Specify the header for [CXL](#) Link Capability Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000000C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:20	CxlLinkCapabilityPointer. Read-only, Volatile. Reset: 014h. This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0004h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x00000014 [CXL™ Link Layer Capability Low Register] (DF::CxlLinkLayerCapabilityLow0)

Read-only, Volatile. Reset: 0000_0011h.

Specify [CXL](#) Link Layer Low Capability Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000014; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:29	NumPhyReInitRcvdLow. Read-only, Volatile. Reset: 0h. Num_Phy_ReInit value bit 2:0 reflected in the last Retry.Req message received. Used for debug.
28:24	NumRetryRcvd. Read-only, Volatile. Reset: 00h. Num_Retry value reflected in the last Retry.Req message received. Used for debug.
23:16	LLRWrapValueRcvd. Read-only, Volatile. Reset: 00h. LLR Wrap value received from INIT.Param flit. Used for debug.
15:8	LLRWrapValueSupported. Read-only, Volatile. Reset: 00h. LLR Wrap value supported by this entity. Used for debug.
7:4	CxlLinkVersionRcvd. Read-only, Volatile. Reset: 1h. Version of AL received from INIT.Param flit. Used for debug.
3:0	CxlLinkVersionSupported. Read-only, Volatile. Reset: 1h. Version of AL the port is compliant with. For CXL 1.0, this should be 'b0001.

CXL[0...3]x00000018 [CXL™ Link Layer Capability High Register] (DF::CxlLinkLayerCapabilityHigh0)

Read-only, Volatile. Reset: 0400_0000h.

Specify [CXL](#) Link Layer Capability High Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000018; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:27	Reserved.
26	No_LL_Reset_Support. Read-only, Volatile. Reset: 1. LL_Reset which is a debug only feature is not supported
25:18	NumFreeBufRcvd. Read-only, Volatile. Reset: 00h. Number of Free Buffer value received in the last Retry.Ack message received.
17:10	EchoEseqRcvd. Read-only, Volatile. Reset: 00h. Echo_Eseq value reflected in the last Retry.Ack message received.
9:2	WrPtrRcvd. Read-only, Volatile. Reset: 00h. Wr_Ptr value reflected in the last Retry.Ack message received.
1:0	NumPhyReInitRcvdHigh. Read-only, Volatile. Reset: 0h. Num_Phy_ReInit value bit 4:3 reflected in the last Retry.Reg message received. Used for debug.

CXL[0...3]x0000001C [CXL™ Link Layer Control and Status Register] (DF::CxlLinkLayerControlStatus0)

Reset: 0000_0000h.

Specify [CXL](#) Link Layer Control and Status Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000001C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:13	Reserved.
12:5	LLRetryBufConsumed. Read-only, Volatile. Reset: 00h. Snapshot of link layer retry buffer consumed
4:3	InitState. Read-only, Volatile. Reset: 0h. This field reflects the current initialization status of the Link Layer. 00: NOT_RDY_FOR_INIT (stalled or unstalled); 01: LLCTRL-INIT.Param sent and waiting to receive it; 10: Parameter exchanged successfully and Credit return is stalled; 11: Link Initialization Done
2	LLCrdStall. Read-write, Volatile. Reset: 0. If set, link layer stalls credit initialization until this bit is cleared
1	LLInitStall. Read-write, Volatile. Reset: 0. If set, link layer stalls the transmission of the LLCTRL-INIT.Param flit until this bit is cleared.
0	LLReset. Read-write, Volatile. Reset: 0. Re-initialize without resetting values in sticky registers. Write '1 to reset link - this is a destructive reset all link layer state. When link layer reset completes, HW will clear the bit to 0.

CXL[0...3]x00000020 [Cnli CXL™ .Cache/.Mem Rx Credit Control] (DF::CnliCxlRxCrdControlLow0)

Read-write. Reset: 0701_8070h.

Specify the CNLI [CXL](#) .Cache/.Mem Rx Request/Data/Response token config

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000020; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:30	CxlMemReqRspCrdCntLow. Read-write. Reset: 0h. Initially allocated credits bit 1:0 for .Mem Request/Response from Endpoint to Cnli. At least one Req token has to be allocated.
29:20	CxlCacheDatTokCnt. Read-write. Reset: 070h. Initially allocated token for .Cache Data from Endpoint to Cnli. At least one Req Dat token has to be allocated.
19:10	CxlCacheRspTokCnt. Read-write. Reset: 060h. Initially allocated token for .Cache Response from Endpoint to Cnli. At least one Rsp token has to be allocated.
9:0	CxlCacheReqTokCnt. Read-write. Reset: 070h. Initially allocated token for .Cache Request from Endpoint to Cnli. At least one Req token has to be allocated.

CXL[0...3]x00000024 [Cnli CXL™ .Cache/.Mem Rx Credit Control] (DF::CnliCxlRxCrdControlHigh0)

Read-write. Reset: 0000_1806h.

Specify the Cnli [CXL](#) .Cache/.Mem Rx Request/Data/Response token config`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00000024; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:18	Reserved.
17:8	CxlMemDatCrdCnt. Read-write. Reset: 018h. Init: 11.5.3 [Register Values Written Based on Configuration]. Initially allocated credits for .Mem Request Data from Endpoint to Cnli. At least one Req Dat token has to be allocated.
7:0	CxlMemReqRspCrdCntHigh. Read-write. Reset: 06h. Init: 11.5.3 [Register Values Written Based on Configuration]. Initially allocated credits bit 9:2 for .Mem Request/Response from Endpoint to Cnli. At least one Req token has to be allocated.

CXL[0...3]x00000028 [Cnli CXL™ .Cache/.Mem Rx Credit Return Status] (DF::CnliCxlRxCrdStatusLow0)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Rx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00000028; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:30	CxlMemReqRspCrdCntLow. Read-only, Volatile. Reset: 0h. Snapshot of accumulated .mem request or response credits bit 1:0 to be returned
29:20	CxlCacheDatCrdCnt. Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache data credits to be returned
19:10	CxlCacheRspCrdCnt. Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache response credits to be returned
9:0	CxlCacheReqCrdCnt. Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache request credits to be returned

CXL[0...3]x0000002C [Cnli CXL™ .Cache/.Mem Rx Credit Return Status] (DF::CnliCxlRxCrdStatusHigh0)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Rx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x0000002C; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:18	Reserved.
17:8	CxlMemDatCrdCnt. Read-only, Volatile. Reset: 000h. Snapshot of accumulated .mem data credits to be returned
7:0	CxlMemReqRspCrdCntHigh. Read-only, Volatile. Reset: 00h. Snapshot of accumulated .mem request or response credits bit 9:2 to be returned

CXL[0...3]x00000030 [Cnli CXL™ .Cache/.Mem Tx Credit Return Status] (DF::CnliCxlTxCrdStatusLow0)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Tx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00000030; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:30	CxlMemReqRspCrdCntLow. Read-only, Volatile. Reset: 0h. Snapshot of accumulated .mem request or response credits bit 1:0
29:20	CxlCacheDatCrdCnt. Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache data credits
19:10	CxlCacheRspCrdCnt. Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache response credits
9:0	CxlCacheReqCrdCnt. Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache request credits

CXL[0...3]x00000034 [Cnli CXL™ .Cache/.Mem Tx Credit Return Status] (DF::CnliCxlTxCrdStatusHigh0)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Tx Credit Return Status

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000034; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:18	Reserved.
17:8	CxlMemDatCrdCnt. Read-only, Volatile. Reset: 000h. Snapshot of accumulated .mem data credits
7:0	CxlMemReqRspCrdCntHigh. Read-only, Volatile. Reset: 00h. Snapshot of accumulated .mem request or response credits bit 9:2

CXL[0...3]x00000038 [Cnli CXL™ Ack Timer Control register] (DF::CnliCxlAckTimerControl0)

Read-write. Reset: 0000_2010h.

Specify CNLI [CXL](#) Ack Timer control

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000038; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:18	Reserved.
17:8	AckFlushRetimer. Read-write. Reset: 020h. Init: 380h. Specifies how many link layer clock cycles the entity should wait in case of idle, before flushing accumulated Acks using a LLCRD
7:0	AckForceThreshold. Read-write. Reset: 10h. Init: 20h. Specifies how many Flit Acks the Link Layer should accumulate before injecting a LLCRD

CXL[0...3]x0000003C [Cnli CXL™ Defeature Register] (DF::CnliCxlDefeature0)

Read-write, Volatile. Reset: 0000_0000h.

This is CNLI [CXL](#) Link Layer Defeature Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000003C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:1	Reserved.
0	MDHDisable. Read-write, Volatile. Reset: 0. Write '1 to disable MDH

CXL[0...3]x00000040 [Cnli Uncorrectable Error Status Register] (DF::CnliCxlUncorrErrStat0)

Read-write, Volatile. Reset: 0000_0000h.

This is CNLI Uncorrectable Error Status Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000040; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:12	Reserved.
11	ReceiverOverflow. Read-write, Volatile. Reset: 0. Overflow in CNLI buffers detected. Which buffer is logged in first 3 bits of Header Log register. Software Write 1 to Clear.
10	PoisonRcvd. Read-write, Volatile. Reset: 0. Received poison indication from device. Software Write 1 to Clear.
9	RsvdEncViolation. Read-write, Volatile. Reset: 0. Unrecognized encoding in received flit. Software Write 1 to Clear.
8	ReinitThreshold. Read-write, Volatile. Reset: 0. Physical layer Reinit threshold reached. Software Write 1 to Clear.
7	MemDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL .mem. Defined in CXL spec but not applicable for DF .
6	MemBEParity. Read-write, Volatile. Reset: 0. Internal Byte Enable Parity Error detected on CXL .mem. Defined in CXL spec but not applicable for DF .
5	MemAddrParity. Read-write, Volatile. Reset: 0. Internal Address Parity Error detected on CXL .mem. Defined in CXL spec but not applicable for DF .
4	MemDataParity. Read-write, Volatile. Reset: 0. Internal Data Parity Error detected on CXL .mem. Software Write 1 to Clear.
3	CacheDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL .cache. Defined in CXL spec but not applicable for DF .
2	CacheBEParity. Read-write, Volatile. Reset: 0. Internal Byte Enable Parity Error detected on CXL .cache. Defined in CXL spec but not applicable for DF .
1	CacheAddrParity. Read-write, Volatile. Reset: 0. Internal Address Parity Error detected on CXL .cache. Defined in CXL spec but not applicable for DF .
0	CacheDataParity. Read-write, Volatile. Reset: 0. Internal Data Parity Error detected on CXL .cache. Software Write 1 to Clear.

CXL[0...3]x00000044 [Cnli Uncorrectable Error Mask Register] (DF::CnliCxlUncorrErrMask0)

Read-write, Volatile. Reset: 0000_0000h.

Contains Mask bits for corresponding errors in CNLI Uncorrectable Error Register. When an error is masked, status register is updated, but Uncorrectable error (interrupt) is not sent

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000044; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:12	Reserved.
11	ReceiverOverflowMask. Read-write, Volatile. Reset: 0. Mask detection of overflow in CNLI buffers
10	PoisonRcvdMask. Read-write, Volatile. Reset: 0. Mask detection of poison from device
9	RsvdEncViolationMask. Read-write, Volatile. Reset: 0. Mask detection of unrecognized encoding in received flit
8	ReinitThresholdMask. Read-write, Volatile. Reset: 0. Mask detection of Physical layer Reinit threshold
7	MemDataECCMask. Read-write, Volatile. Reset: 0. Mask detection of Data ECC Error on CXL .mem
6	MemBEParityMask. Read-write, Volatile. Reset: 0. Mask detection of Byte Enable Parity Error on CXL .mem
5	MemAddrParityMask. Read-write, Volatile. Reset: 0. Mask detection of Address Parity Error on CXL .mem
4	MemDataParityMask. Read-write, Volatile. Reset: 0. Mask detection of Data Parity Error on CXL .mem
3	CacheDataECCMask. Read-write, Volatile. Reset: 0. Mask detection of Data ECC Error on CXL .cache
2	CacheBEParityMask. Read-write, Volatile. Reset: 0. Mask detection of Byte Enable Parity Error on CXL .cache
1	CacheAddrParityMask. Read-write, Volatile. Reset: 0. Mask detection of Address Parity Error on CXL .cache
0	CacheDataParityMask. Read-write, Volatile. Reset: 0. Mask detection of Data Parity Error on CXL .cache

CXL[0...3]x00000048 [Cnli Uncorrectable Error Severity Register] (DF::CnliCxlUncorrErrSeverity0)

Read-write, Volatile. Reset: 0000_0FFFh.

Contains severity bits for corresponding errors in CNLI Uncorrectable Error Register. When Severity=0 for an uncorrectable error, correctable error bit will be set instead of uncorrectable bit (interrupt)

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000048; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:12	Reserved.
11	ReceiverOverflowSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of overflow in CNLI buffers
10	PoisonRcvdSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of poison from device
9	RsvdEncViolationSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of unrecognized encoding in received flit
8	ReinitThresholdSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Physical layer Reinit threshold
7	MemDataECCSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data ECC Error on CXL.mem
6	MemBEParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Byte Enable Parity Error on CXL.mem
5	MemAddrParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Address Parity Error on CXL.mem
4	MemDataParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data Parity Error on CXL.mem
3	CacheDataECCSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data ECC Error on CXL.cache
2	CacheBEParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Byte Enable Parity Error on CXL.cache
1	CacheAddrParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Address Parity Error on CXL.cache
0	CacheDataParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data Parity Error on CXL.cache

CXL[0...3]x0000004C [Cnli Correctable Error Status Register] (DF::CnliCxlCorrErrStat0)

Read-write, Volatile. Reset: 0000_0000h.

This is the CNLI Correctable Error Status Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000004C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:7	Reserved.
6	PhyLayerError. Read-write, Volatile. Reset: 0. Received error on Physical Layer. Phy Reinit received
5	MemPoisonRcvd. Read-write, Volatile. Reset: 0. Received poison indication from device on CXL.mem
4	CachePoisonRcvd. Read-write, Volatile. Reset: 0. Received poison indication from device on CXL.cache
3	RetryThreshold. Read-write, Volatile. Reset: 0. Retry threshold reached
2	CrcThreshold. Read-write, Volatile. Reset: 0. CRC error threshold reached
1	MemDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL.mem
0	CacheDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL.cache

CXL[0...3]x00000050 [Cnli Correctable Error Mask Register] (DF::CnliCxlCorrErrMask0)

Read-write, Volatile. Reset: 0000_0000h.

Mask bits for CNLI Correctable Errors. When an error is masked, status register is updated, but correctable error is not sent

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000050; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:7	Reserved.
6	PhyLayerErrorMask. Read-write, Volatile. Reset: 0. Mask for error on Physical Layer. Phy Reinit received
5	MemPoisonRcvdMask. Read-write, Volatile. Reset: 0. Mask for poison indication from device on CXL.mem
4	CachePoisonRcvdMask. Read-write, Volatile. Reset: 0. Mask for poison indication from device on CXL.cache
3	RetryThresholdMask. Read-write, Volatile. Reset: 0. Mask for Retry threshold reached
2	CrcThresholdMask. Read-write, Volatile. Reset: 0. Mask for CRC error threshold reached
1	MemDataECCMask. Read-write, Volatile. Reset: 0. Mask for Internal Data ECC Error detected on CXL.mem
0	CacheDataECCMask. Read-write, Volatile. Reset: 0. Mask for Internal Data ECC Error detected on CXL.cache

CXL[0...3]x00000054 [Cnli Error Capabilities and Control Register] (DF::CnliCxlErrCapAndCtrl0)

Reset: 0000_2000h.

Cnli Error Capabilities and Control Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000054; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:14	Reserved.
13	PoisonEn. Read-write, Volatile. Reset: 1. Indicates if poison is enabled. 1: treat/log poison as correctable error; 0: treat/log poison as uncorrectable error
12:10	Reserved.
9	MultiHdrRecCap. Read-only, Volatile. Reset: 0. Multi header recording capability. Indicates if recording more than one error header is supported. This is always zero for CXL 1.1
8:4	Reserved.
3:0	FirstErrPtr. Read-only, Volatile. Reset: 0h. Bit position of the first error bit for the error reported in Uncorrectable error status register

CXL[0...3]x00000058 [Cnli Header Log Register] (DF::CnliCxlHdrLog00)

Read-only, Volatile. Reset: 0000_0000h.

0th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000058; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:0	HdrLog. Read-only, Volatile. Reset: 0000_0000h. 0th 32 bit segment of Header Log Register

CXL[0...3]x0000005C [Cnli Header Log Register] (DF::CnliCxlHdrLog10)

Read-only, Volatile. Reset: 0000_0000h.

First 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000005C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:0	HdrLog. Read-only, Volatile. Reset: 0000_0000h. First 32 bit segment of Header Log Register

CXL[0...3]x00000060 [Cnli Header Log Register] (DF::CnliCxlHdrLog20)

Read-only, Volatile. Reset: 0000_0000h.

Second 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000060; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:0	HdrLog. Read-only, Volatile. Reset: 0000_0000h. Second 32 bit segment of Header Log Register

CXL[0...3]x00000064 [Cnli Header Log Register] (DF::CnliCxlHdrLog30)

Read-only, Volatile. Reset: 0000_0000h.

Third 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00000064; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Third 32 bit segment of Header Log Register**CXL[0...3]x00000068 [Cnli Header Log Register] (DF::CnliCxlHdrLog40)**

Read-only, Volatile. Reset: 0000_0000h.

Fourth 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00000068; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Fourth 32 bit segment of Header Log Register**CXL[0...3]x0000006C [Cnli Header Log Register] (DF::CnliCxlHdrLog50)**

Read-only, Volatile. Reset: 0000_0000h.

Fifth 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x0000006C; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Fifth 32 bit segment of Header Log Register**CXL[0...3]x00000070 [Cnli Header Log Register] (DF::CnliCxlHdrLog60)**

Read-only, Volatile. Reset: 0000_0000h.

Sixth 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00000070; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Sixth 32 bit segment of Header Log Register**CXL[0...3]x00000074 [Cnli Header Log Register] (DF::CnliCxlHdrLog70)**

Read-only, Volatile. Reset: 0000_0000h.

7th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00000074; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 7th 32 bit segment of Header Log Register**CXL[0...3]x00000078 [Cnli Header Log Register] (DF::CnliCxlHdrLog80)**

Read-only, Volatile. Reset: 0000_0000h.

8th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00000078; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 8th 32 bit segment of Header Log Register**CXL[0...3]x0000007C [Cnli Header Log Register] (DF::CnliCxlHdrLog90)**

Read-only, Volatile. Reset: 0000_0000h.

9th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x0000007C; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 9th 32 bit segment of Header Log Register

CXL[0...3]x00000080 [Cnli Header Log Register] (DF::CnliCxlHdrLog100)

Read-only, Volatile. Reset: 0000_0000h.

10th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000080; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 10th 32 bit segment of Header Log Register**CXL[0...3]x00000084 [Cnli Header Log Register] (DF::CnliCxlHdrLog110)**

Read-only, Volatile. Reset: 0000_0000h.

11th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000084; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 11th 32 bit segment of Header Log Register**CXL[0...3]x00000088 [Cnli Header Log Register] (DF::CnliCxlHdrLog120)**

Read-only, Volatile. Reset: 0000_0000h.

12th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000088; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 12th 32 bit segment of Header Log Register**CXL[0...3]x0000008C [Cnli Header Log Register] (DF::CnliCxlHdrLog130)**

Read-only, Volatile. Reset: 0000_0000h.

13th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000008C; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 13th 32 bit segment of Header Log Register**CXL[0...3]x00000090 [Cnli Header Log Register] (DF::CnliCxlHdrLog140)**

Read-only, Volatile. Reset: 0000_0000h.

14th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000090; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 14th 32 bit segment of Header Log Register**CXL[0...3]x00000094 [Cnli Header Log Register] (DF::CnliCxlHdrLog150)**

Read-only, Volatile. Reset: 0000_0000h.

15th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00000094; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 15th 32 bit segment of Header Log Register

CXL[0...3]x00002000 [CXL™ Capability Header Register] (DF::CxlCapabilityHeader1)

Read-only, Volatile. Reset: 0311_0001h.

Specify the header for [CXL](#) Capability Register`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002000; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:24	ArraySize. Read-only, Volatile. Reset: 03h. defines the number of elements present in the CXL_Capability array, not including the CXL_Capability_Header element.
23:20	CxlCacheMemVersion. Read-only, Volatile. Reset: 1h. This defines the version of the CXL Cache Mem Protocol supported.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0001h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x00002004 [CXL™ Capability Header Register] (DF::CxlRASCapabilityHeader1)

Read-only, Volatile. Reset: 0401_0002h.

Specify the header for [CXL RAS](#) Capability Register`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002004; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:20	CxlRASCapabilityPointer. Read-only, Volatile. Reset: 040h. This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0002h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x00002008 [CXL™ Security Capability Header Register] (DF::CxlSecurityCapabilityHeader1)

Read-only, Volatile. Reset: 0101_0003h.

Specify the header for [CXL](#) Security Capability Register`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002008; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:20	CxlSecurityCapabilityPointer. Read-only, Volatile. Reset: 010h. This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0003h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x0000200C [CXL™ Link Capability Header Register] (DF::CxlLinkCapabilityHeader1)

Read-only, Volatile. Reset: 0141_0004h.

Specify the header for [CXL](#) Link Capability Register`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000200C; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:20	CxlLinkCapabilityPointer. Read-only, Volatile. Reset: 014h. This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0004h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x00002014 [CXL™ Link Layer Capability Low Register] (DF::CxlLinkLayerCapabilityLow1)

Read-only, Volatile. Reset: 0000_0011h.

Specify [CXL](#) Link Layer Low Capability Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002014; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:29	NumPhyReInitRcvdLow. Read-only, Volatile. Reset: 0h. Num_Phy_ReInit value bit 2:0 reflected in the last Retry.Req message received. Used for debug.
28:24	NumRetryRcvd. Read-only, Volatile. Reset: 00h. Num_Retry value reflected in the last Retry.Req message received. Used for debug.
23:16	LLRWrapValueRcvd. Read-only, Volatile. Reset: 00h. LLR Wrap value received from INIT.Param flit. Used for debug.
15:8	LLRWrapValueSupported. Read-only, Volatile. Reset: 00h. LLR Wrap value supported by this entity. Used for debug.
7:4	CxlLinkVersionRcvd. Read-only, Volatile. Reset: 1h. Version of AL received from INIT.Param flit. Used for debug.
3:0	CxlLinkVersionSupported. Read-only, Volatile. Reset: 1h. Version of AL the port is compliant with. For CXL 1.0, this should be 'b0001.

CXL[0...3]x00002018 [CXL™ Link Layer Capability High Register] (DF::CxlLinkLayerCapabilityHigh1)

Read-only, Volatile. Reset: 0400_0000h.

Specify [CXL](#) Link Layer Capability High Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002018; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:27	Reserved.
26	No_LL_Reset_Support. Read-only, Volatile. Reset: 1. LL_Reset which is a debug only feature is not supported
25:18	NumFreeBufRcvd. Read-only, Volatile. Reset: 00h. Number of Free Buffer value received in the last Retry.Ack message received.
17:10	EchoEseqRcvd. Read-only, Volatile. Reset: 00h. Echo_Eseq value reflected in the last Retry.Ack message received.
9:2	WrPtrRcvd. Read-only, Volatile. Reset: 00h. Wr_Ptr value reflected in the last Retry.Ack message received.
1:0	NumPhyReInitRcvdHigh. Read-only, Volatile. Reset: 0h. Num_Phy_ReInit value bit 4:3 reflected in the last Retry.Req message received. Used for debug.

CXL[0...3]x0000201C [CXL™ Link Layer Control and Status Register] (DF::CxlLinkLayerControlStatus1)

Reset: 0000_0000h.

Specify [CXL](#) Link Layer COntrol and Status Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000201C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:13	Reserved.
12:5	LLRetryBufConsumed. Read-only, Volatile. Reset: 00h. Snapshot of link layer retry buffer consumed
4:3	InitState. Read-only, Volatile. Reset: 0h. This field reflects the current initialization status of the Link Layer. 00: NOT_RDY_FOR_INIT (stalled or unstalled); 01: LLCTRL-INIT.Param sent and waiting to receive it; 10: Parameter exchanged successfully and Credit return is stalled; 11: Link Initialization Done
2	LLCrddStall. Read-write, Volatile. Reset: 0. If set, link layer stalls credit initialization until this bit is cleared
1	LLInitStall. Read-write, Volatile. Reset: 0. If set, link layer stalls the transmission of the LLCTRL-INIT.Param flit until this bit is cleared.
0	LLReset. Read-write, Volatile. Reset: 0. Re-initialize without resetting values in sticky registers. Write '1 to reset link - this is a destructive reset all link layer state. When link layer reset completes, HW will clear the bit to 0.

CXL[0...3]x00002020 [Cnli CXL™ .Cache/.Mem Rx Credit Control] (DF::CnliCxlRxCrdControlLow1)

Read-write. Reset: 0701_8070h.

Specify the CNLI [CXL](#) .Cache/.Mem Rx Request/Data/Response token config`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00002020; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:30	CxlMemReqRspCrdCntLow . Read-write. Reset: 0h. Initially allocated credits bit 1:0 for .Mem Request/Response from Endpoint to Cnli. At least one Req token has to be allocated.
29:20	CxlCacheDatTokCnt . Read-write. Reset: 070h. Initially allocated token for .Cache Data from Endpoint to Cnli. At least one Req Dat token has to be allocated.
19:10	CxlCacheRspTokCnt . Read-write. Reset: 060h. Initially allocated token for .Cache Response from Endpoint to Cnli. At least one Rsp token has to be allocated.
9:0	CxlCacheReqTokCnt . Read-write. Reset: 070h. Initially allocated token for .Cache Request from Endpoint to Cnli. At least one Req token has to be allocated.

CXL[0...3]x00002024 [Cnli CXL™ .Cache/.Mem Rx Credit Control] (DF::CnliCxlRxCrdControlHigh1)

Read-write. Reset: 0000_1806h.

Specify the Cnli [CXL](#) .Cache/.Mem Rx Request/Data/Response token config`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00002024; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:18	Reserved.
17:8	CxlMemDatCrdCnt . Read-write. Reset: 018h. Init: 11.5.3 [Register Values Written Based on Configuration]. Initially allocated credits for .Mem Request Data from Endpoint to Cnli. At least one Req Dat token has to be allocated.
7:0	CxlMemReqRspCrdCntHigh . Read-write. Reset: 06h. Init: 11.5.3 [Register Values Written Based on Configuration]. Initially allocated credits bit 9:2 for .Mem Request/Response from Endpoint to Cnli. At least one Req token has to be allocated.

CXL[0...3]x00002028 [Cnli CXL™ .Cache/.Mem Rx Credit Return Status] (DF::CnliCxlRxCrdStatusLow1)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Rx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00002028; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:30	CxlMemReqRspCrdCntLow . Read-only, Volatile. Reset: 0h. Snapshot of accumulated .mem request or response credits bit 1:0 to be returned
29:20	CxlCacheDatCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache data credits to be returned
19:10	CxlCacheRspCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache response credits to be returned
9:0	CxlCacheReqCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache request credits to be returned

CXL[0...3]x0000202C [Cnli CXL™ .Cache/.Mem Rx Credit Return Status] (DF::CnliCxlRxCrdStatusHigh1)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Rx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x0000202C; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:18	Reserved.
17:8	CxlMemDatCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated .mem data credits to be returned
7:0	CxlMemReqRspCrdCntHigh . Read-only, Volatile. Reset: 00h. Snapshot of accumulated .mem request or response credits bit 9:2 to be returned

CXL[0...3]x00002030 [Cnli CXL™ .Cache/.Mem Tx Credit Return Status] (DF::CnliCxlTxCrdStatusLow1)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Tx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00002030; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:30	CxlMemReqRspCrdCntLow . Read-only, Volatile. Reset: 0h. Snapshot of accumulated .mem request or response credits bit 1:0
29:20	CxlCacheDatCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache data credits
19:10	CxlCacheRspCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache response credits
9:0	CxlCacheReqCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache request credits

CXL[0...3]x00002034 [Cnli CXL™ .Cache/.Mem Tx Credit Return Status] (DF::CnliCxlTxCrdStatusHigh1)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Tx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00002034; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:18	Reserved.
17:8	CxlMemDatCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated .mem data credits
7:0	CxlMemReqRspCrdCntHigh . Read-only, Volatile. Reset: 00h. Snapshot of accumulated .mem request or response credits bit 9:2

CXL[0...3]x00002038 [Cnli CXL™ Ack Timer Control register] (DF::CnliCxlAckTimerControl1)

Read-write. Reset: 0000_2010h.

Specify CNLI [CXL](#) Ack Timer control`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00002038; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:18	Reserved.
17:8	AckFlushRetimer . Read-write. Reset: 020h. Init: 380h. Specifies how many link layer clock cycles the entity should wait in case of idle, before flushing accumulated Acks using a LLCRD
7:0	AckForceThreshold . Read-write. Reset: 10h. Init: 20h. Specifies how many Flit Acks the Link Layer should accumulate before injecting a LLCRD

CXL[0...3]x0000203C [Cnli CXL™ Defeature Register] (DF::CnliCxlDefeature1)

Read-write, Volatile. Reset: 0000_0000h.

This is CNLI [CXL](#) Link Layer Defeature Register`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x0000203C; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:1	Reserved.
0	MDHDisable . Read-write, Volatile. Reset: 0. Write '1 to disable MDH

CXL[0...3]x00002040 [Cnli Uncorrectable Error Status Register] (DF::CnliCxlUncorrErrStat1)

Read-write, Volatile. Reset: 0000_0000h.

This is CNLI Uncorrectable Error Status Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002040; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:12	Reserved.
11	ReceiverOverflow. Read-write, Volatile. Reset: 0. Overflow in CNLI buffers detected. Which buffer is logged in first 3 bits of Header Log register. Software Write 1 to Clear.
10	PoisonRcvd. Read-write, Volatile. Reset: 0. Received poison indication from device. Software Write 1 to Clear.
9	RsvdEncViolation. Read-write, Volatile. Reset: 0. Unrecognized encoding in received flit. Software Write 1 to Clear.
8	ReinitThreshold. Read-write, Volatile. Reset: 0. Physical layer Reinit threshold reached. Software Write 1 to Clear.
7	MemDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL .mem. Defined in CXL spec but not applicable for DF .
6	MemBEParity. Read-write, Volatile. Reset: 0. Internal Byte Enable Parity Error detected on CXL .mem. Defined in CXL spec but not applicable for DF .
5	MemAddrParity. Read-write, Volatile. Reset: 0. Internal Address Parity Error detected on CXL .mem. Defined in CXL spec but not applicable for DF .
4	MemDataParity. Read-write, Volatile. Reset: 0. Internal Data Parity Error detected on CXL .mem. Software Write 1 to Clear.
3	CacheDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL .cache. Defined in CXL spec but not applicable for DF .
2	CacheBEParity. Read-write, Volatile. Reset: 0. Internal Byte Enable Parity Error detected on CXL .cache. Defined in CXL spec but not applicable for DF .
1	CacheAddrParity. Read-write, Volatile. Reset: 0. Internal Address Parity Error detected on CXL .cache. Defined in CXL spec but not applicable for DF .
0	CacheDataParity. Read-write, Volatile. Reset: 0. Internal Data Parity Error detected on CXL .cache. Software Write 1 to Clear.

CXL[0...3]x00002044 [Cnli Uncorrectable Error Mask Register] (DF::CnliCxlUncorrErrMask1)

Read-write, Volatile. Reset: 0000_0000h.

Contains Mask bits for corresponding errors in CNLI Uncorrectable Error Register. When an error is masked, status register is updated, but Uncorrectable error (interrupt) is not sent

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002044; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:12	Reserved.
11	ReceiverOverflowMask. Read-write, Volatile. Reset: 0. Mask detection of overflow in CNLI buffers
10	PoisonRcvdMask. Read-write, Volatile. Reset: 0. Mask detection of poison from device
9	RsvdEncViolationMask. Read-write, Volatile. Reset: 0. Mask detection of unrecognized encoding in received flit
8	ReinitThresholdMask. Read-write, Volatile. Reset: 0. Mask detection of Physical layer Reinit threshold
7	MemDataECCMask. Read-write, Volatile. Reset: 0. Mask detection of Data ECC Error on CXL .mem
6	MemBEParityMask. Read-write, Volatile. Reset: 0. Mask detection of Byte Enable Parity Error on CXL .mem
5	MemAddrParityMask. Read-write, Volatile. Reset: 0. Mask detection of Address Parity Error on CXL .mem
4	MemDataParityMask. Read-write, Volatile. Reset: 0. Mask detection of Data Parity Error on CXL .mem
3	CacheDataECCMask. Read-write, Volatile. Reset: 0. Mask detection of Data ECC Error on CXL .cache
2	CacheBEParityMask. Read-write, Volatile. Reset: 0. Mask detection of Byte Enable Parity Error on CXL .cache
1	CacheAddrParityMask. Read-write, Volatile. Reset: 0. Mask detection of Address Parity Error on CXL .cache
0	CacheDataParityMask. Read-write, Volatile. Reset: 0. Mask detection of Data Parity Error on CXL .cache

CXL[0...3]x00002048 [Cnli Uncorrectable Error Severity Register] (DF::CnliCxlUncorrErrSeverity1)

Read-write, Volatile. Reset: 0000_0FFFh.

Contains severity bits for corresponding errors in CNLI Uncorrectable Error Register. When Severity=0 for an uncorrectable error, correctable error bit will be set instead of uncorrectable bit (interrupt)

`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002048; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:12	Reserved.
11	ReceiverOverflowSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of overflow in CNLI buffers
10	PoisonRcvdSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of poison from device
9	RsvdEncViolationSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of unrecognized encoding in received flit
8	ReinitThresholdSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Physical layer Reinit threshold
7	MemDataECCSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data ECC Error on CXL.mem
6	MemBEParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Byte Enable Parity Error on CXL.mem
5	MemAddrParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Address Parity Error on CXL.mem
4	MemDataParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data Parity Error on CXL.mem
3	CacheDataECCSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data ECC Error on CXL.cache
2	CacheBEParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Byte Enable Parity Error on CXL.cache
1	CacheAddrParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Address Parity Error on CXL.cache
0	CacheDataParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data Parity Error on CXL.cache

CXL[0...3]x0000204C [Cnli Correctable Error Status Register] (DF::CnliCxlCorrErrStat1)

Read-write, Volatile. Reset: 0000_0000h.

This is the CNLI Correctable Error Status Register

`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000204C; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:7	Reserved.
6	PhyLayerError. Read-write, Volatile. Reset: 0. Received error on Physical Layer. Phy Reinit received
5	MemPoisonRcvd. Read-write, Volatile. Reset: 0. Received poison indication from device on CXL.mem
4	CachePoisonRcvd. Read-write, Volatile. Reset: 0. Received poison indication from device on CXL.cache
3	RetryThreshold. Read-write, Volatile. Reset: 0. Retry threshold reached
2	CrcThreshold. Read-write, Volatile. Reset: 0. CRC error threshold reached
1	MemDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL.mem
0	CacheDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL.cache

CXL[0...3]x00002050 [Cnli Correctable Error Mask Register] (DF::CnliCxlCorrErrMask1)

Read-write, Volatile. Reset: 0000_0000h.

Mask bits for CNLI Correctable Errors. When an error is masked, status register is updated, but correctable error is not sent

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002050; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:7	Reserved.
6	PhyLayerErrorMask. Read-write, Volatile. Reset: 0. Mask for error on Physical Layer. Phy Reinit received
5	MemPoisonRcvdMask. Read-write, Volatile. Reset: 0. Mask for poison indication from device on CXL.mem
4	CachePoisonRcvdMask. Read-write, Volatile. Reset: 0. Mask for poison indication from device on CXL.cache
3	RetryThresholdMask. Read-write, Volatile. Reset: 0. Mask for Retry threshold reached
2	CrcThresholdMask. Read-write, Volatile. Reset: 0. Mask for CRC error threshold reached
1	MemDataECCMask. Read-write, Volatile. Reset: 0. Mask for Internal Data ECC Error detected on CXL.mem
0	CacheDataECCMask. Read-write, Volatile. Reset: 0. Mask for Internal Data ECC Error detected on CXL.cache

CXL[0...3]x00002054 [Cnli Error Capabilities and Control Register] (DF::CnliCxlErrCapAndCtrl1)

Reset: 0000_2000h.

Cnli Error Capabilities and Control Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002054; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:14	Reserved.
13	PoisonEn. Read-write, Volatile. Reset: 1. Indicates if poison is enabled. 1: treat/log poison as correctable error; 0: treat/log poison as uncorrectable error
12:10	Reserved.
9	MultiHdrRecCap. Read-only, Volatile. Reset: 0. Multi header recording capability. Indicates if recording more than one error header is supported. This is always zero for CXL 1.1
8:4	Reserved.
3:0	FirstErrPtr. Read-only, Volatile. Reset: 0h. Bit position of the first error bit for the error reported in Uncorrectable error status register

CXL[0...3]x00002058 [Cnli Header Log Register] (DF::CnliCxlHdrLog01)

Read-only, Volatile. Reset: 0000_0000h.

0th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002058; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:0	HdrLog. Read-only, Volatile. Reset: 0000_0000h. 0th 32 bit segment of Header Log Register

CXL[0...3]x0000205C [Cnli Header Log Register] (DF::CnliCxlHdrLog11)

Read-only, Volatile. Reset: 0000_0000h.

First 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000205C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:0	HdrLog. Read-only, Volatile. Reset: 0000_0000h. First 32 bit segment of Header Log Register

CXL[0...3]x00002060 [Cnli Header Log Register] (DF::CnliCxlHdrLog21)

Read-only, Volatile. Reset: 0000_0000h.

Second 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002060; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:0	HdrLog. Read-only, Volatile. Reset: 0000_0000h. Second 32 bit segment of Header Log Register

CXL[0...3]x00002064 [Cnli Header Log Register] (DF::CnliCxlHdrLog31)

Read-only, Volatile. Reset: 0000_0000h.

Third 32 bit segment of Flit Information from the flit causing Uncorrectable Error

`_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00002064; CXL[3:0]=1F1[3:0]_0000h`**Bits Description**31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Third 32 bit segment of Header Log Register**CXL[0...3]x00002068 [Cnli Header Log Register] (DF::CnliCxlHdrLog41)**

Read-only, Volatile. Reset: 0000_0000h.

Fourth 32 bit segment of Flit Information from the flit causing Uncorrectable Error

`_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00002068; CXL[3:0]=1F1[3:0]_0000h`**Bits Description**31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Fourth 32 bit segment of Header Log Register**CXL[0...3]x0000206C [Cnli Header Log Register] (DF::CnliCxlHdrLog51)**

Read-only, Volatile. Reset: 0000_0000h.

Fifth 32 bit segment of Flit Information from the flit causing Uncorrectable Error

`_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x0000206C; CXL[3:0]=1F1[3:0]_0000h`**Bits Description**31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Fifth 32 bit segment of Header Log Register**CXL[0...3]x00002070 [Cnli Header Log Register] (DF::CnliCxlHdrLog61)**

Read-only, Volatile. Reset: 0000_0000h.

Sixth 32 bit segment of Flit Information from the flit causing Uncorrectable Error

`_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00002070; CXL[3:0]=1F1[3:0]_0000h`**Bits Description**31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Sixth 32 bit segment of Header Log Register**CXL[0...3]x00002074 [Cnli Header Log Register] (DF::CnliCxlHdrLog71)**

Read-only, Volatile. Reset: 0000_0000h.

7th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

`_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00002074; CXL[3:0]=1F1[3:0]_0000h`**Bits Description**31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 7th 32 bit segment of Header Log Register**CXL[0...3]x00002078 [Cnli Header Log Register] (DF::CnliCxlHdrLog81)**

Read-only, Volatile. Reset: 0000_0000h.

8th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

`_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00002078; CXL[3:0]=1F1[3:0]_0000h`**Bits Description**31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 8th 32 bit segment of Header Log Register**CXL[0...3]x0000207C [Cnli Header Log Register] (DF::CnliCxlHdrLog91)**

Read-only, Volatile. Reset: 0000_0000h.

9th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

`_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x0000207C; CXL[3:0]=1F1[3:0]_0000h`**Bits Description**31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 9th 32 bit segment of Header Log Register

CXL[0...3]x00002080 [Cnli Header Log Register] (DF::CnliCxlHdrLog101)

Read-only, Volatile. Reset: 0000_0000h.

10th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002080; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 10th 32 bit segment of Header Log Register**CXL[0...3]x00002084 [Cnli Header Log Register] (DF::CnliCxlHdrLog111)**

Read-only, Volatile. Reset: 0000_0000h.

11th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002084; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 11th 32 bit segment of Header Log Register**CXL[0...3]x00002088 [Cnli Header Log Register] (DF::CnliCxlHdrLog121)**

Read-only, Volatile. Reset: 0000_0000h.

12th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002088; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 12th 32 bit segment of Header Log Register**CXL[0...3]x0000208C [Cnli Header Log Register] (DF::CnliCxlHdrLog131)**

Read-only, Volatile. Reset: 0000_0000h.

13th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000208C; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 13th 32 bit segment of Header Log Register**CXL[0...3]x00002090 [Cnli Header Log Register] (DF::CnliCxlHdrLog141)**

Read-only, Volatile. Reset: 0000_0000h.

14th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002090; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 14th 32 bit segment of Header Log Register**CXL[0...3]x00002094 [Cnli Header Log Register] (DF::CnliCxlHdrLog151)**

Read-only, Volatile. Reset: 0000_0000h.

15th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00002094; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 15th 32 bit segment of Header Log Register

CXL[0...3]x00004000 [CXL™ Capability Header Register] (DF::CxlCapabilityHeader2)

Read-only, Volatile. Reset: 0311_0001h.

Specify the header for [CXL](#) Capability Register`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004000; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:24	ArraySize. Read-only, Volatile. Reset: 03h. defines the number of elements present in the CXL_Capability array, not including the CXL_Capability_Header element.
23:20	CxlCacheMemVersion. Read-only, Volatile. Reset: 1h. This defines the version of the CXL Cache Mem Protocol supported.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0001h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x00004004 [CXL™ Capability Header Register] (DF::CxlRASCapabilityHeader2)

Read-only, Volatile. Reset: 0401_0002h.

Specify the header for [CXL RAS](#) Capability Register`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004004; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:20	CxlRASCapabilityPointer. Read-only, Volatile. Reset: 040h. This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0002h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x00004008 [CXL™ Security Capability Header Register] (DF::CxlSecurityCapabilityHeader2)

Read-only, Volatile. Reset: 0101_0003h.

Specify the header for [CXL](#) Security Capability Register`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004008; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:20	CxlSecurityCapabilityPointer. Read-only, Volatile. Reset: 010h. This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0003h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x0000400C [CXL™ Link Capability Header Register] (DF::CxlLinkCapabilityHeader2)

Read-only, Volatile. Reset: 0141_0004h.

Specify the header for [CXL](#) Link Capability Register`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000400C; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:20	CxlLinkCapabilityPointer. Read-only, Volatile. Reset: 014h. This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0004h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x00004014 [CXL™ Link Layer Capability Low Register] (DF::CxlLinkLayerCapabilityLow2)

Read-only, Volatile. Reset: 0000_0011h.

Specify [CXL](#) Link Layer Low Capability Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004014; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:29	NumPhyReInitRcvdLow. Read-only, Volatile. Reset: 0h. Num_Phy_ReInit value bit 2:0 reflected in the last Retry.Req message received. Used for debug.
28:24	NumRetryRcvd. Read-only, Volatile. Reset: 00h. Num_Retry value reflected in the last Retry.Req message received. Used for debug.
23:16	LLRWrapValueRcvd. Read-only, Volatile. Reset: 00h. LLR Wrap value received from INIT.Param flit. Used for debug.
15:8	LLRWrapValueSupported. Read-only, Volatile. Reset: 00h. LLR Wrap value supported by this entity. Used for debug.
7:4	CxlLinkVersionRcvd. Read-only, Volatile. Reset: 1h. Version of AL received from INIT.Param flit. Used for debug.
3:0	CxlLinkVersionSupported. Read-only, Volatile. Reset: 1h. Version of AL the port is compliant with. For CXL 1.0, this should be 'b0001.

CXL[0...3]x00004018 [CXL™ Link Layer Capability High Register] (DF::CxlLinkLayerCapabilityHigh2)

Read-only, Volatile. Reset: 0400_0000h.

Specify [CXL](#) Link Layer Capability High Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004018; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:27	Reserved.
26	No_LL_Reset_Support. Read-only, Volatile. Reset: 1. LL_Reset which is a debug only feature is not supported
25:18	NumFreeBufRcvd. Read-only, Volatile. Reset: 00h. Number of Free Buffer value received in the last Retry.Ack message received.
17:10	EchoEseqRcvd. Read-only, Volatile. Reset: 00h. Echo_Eseq value reflected in the last Retry.Ack message received.
9:2	WrPtrRcvd. Read-only, Volatile. Reset: 00h. Wr_Ptr value reflected in the last Retry.Ack message received.
1:0	NumPhyReInitRcvdHigh. Read-only, Volatile. Reset: 0h. Num_Phy_ReInit value bit 4:3 reflected in the last Retry.Req message received. Used for debug.

CXL[0...3]x0000401C [CXL™ Link Layer Control and Status Register] (DF::CxlLinkLayerControlStatus2)

Reset: 0000_0000h.

Specify [CXL](#) Link Layer COntrol and Status Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000401C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:13	Reserved.
12:5	LLRetryBufConsumed. Read-only, Volatile. Reset: 00h. Snapshot of link layer retry buffer consumed
4:3	InitState. Read-only, Volatile. Reset: 0h. This field reflects the current initialization status of the Link Layer. 00: NOT_RDY_FOR_INIT (stalled or unstalled); 01: LLCTRL-INIT.Param sent and waiting to receive it; 10: Parameter exchanged successfully and Credit return is stalled; 11: Link Initialization Done
2	LLCrdStall. Read-write, Volatile. Reset: 0. If set, link layer stalls credit initialization until this bit is cleared
1	LLInitStall. Read-write, Volatile. Reset: 0. If set, link layer stalls the transmission of the LLCTRL-INIT.Param flit until this bit is cleared.
0	LLReset. Read-write, Volatile. Reset: 0. Re-initialize without resetting values in sticky registers. Write '1 to reset link - this is a destructive reset all link layer state. When link layer reset completes, HW will clear the bit to 0.

CXL[0...3]x00004020 [Cnli CXL™ .Cache/.Mem Rx Credit Control] (DF::CnliCxlRxCrdControlLow2)

Read-write. Reset: 0701_8070h.

Specify the CNLI [CXL](#) .Cache/.Mem Rx Request/Data/Response token config`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00004020; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:30	CxlMemReqRspCrdCntLow . Read-write. Reset: 0h. Initially allocated credits bit 1:0 for .Mem Request/Response from Endpoint to Cnli. At least one Req token has to be allocated.
29:20	CxlCacheDatTokCnt . Read-write. Reset: 070h. Initially allocated token for .Cache Data from Endpoint to Cnli. At least one Req Dat token has to be allocated.
19:10	CxlCacheRspTokCnt . Read-write. Reset: 060h. Initially allocated token for .Cache Response from Endpoint to Cnli. At least one Rsp token has to be allocated.
9:0	CxlCacheReqTokCnt . Read-write. Reset: 070h. Initially allocated token for .Cache Request from Endpoint to Cnli. At least one Req token has to be allocated.

CXL[0...3]x00004024 [Cnli CXL™ .Cache/.Mem Rx Credit Control] (DF::CnliCxlRxCrdControlHigh2)

Read-write. Reset: 0000_1806h.

Specify the Cnli [CXL](#) .Cache/.Mem Rx Request/Data/Response token config`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00004024; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:18	Reserved.
17:8	CxlMemDatCrdCnt . Read-write. Reset: 018h. Init: 11.5.3 [Register Values Written Based on Configuration]. Initially allocated credits for .Mem Request Data from Endpoint to Cnli. At least one Req Dat token has to be allocated.
7:0	CxlMemReqRspCrdCntHigh . Read-write. Reset: 06h. Init: 11.5.3 [Register Values Written Based on Configuration]. Initially allocated credits bit 9:2 for .Mem Request/Response from Endpoint to Cnli. At least one Req token has to be allocated.

CXL[0...3]x00004028 [Cnli CXL™ .Cache/.Mem Rx Credit Return Status] (DF::CnliCxlRxCrdStatusLow2)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Rx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00004028; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:30	CxlMemReqRspCrdCntLow . Read-only, Volatile. Reset: 0h. Snapshot of accumulated .mem request or response credits bit 1:0 to be returned
29:20	CxlCacheDatCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache data credits to be returned
19:10	CxlCacheRspCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache response credits to be returned
9:0	CxlCacheReqCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache request credits to be returned

CXL[0...3]x0000402C [Cnli CXL™ .Cache/.Mem Rx Credit Return Status] (DF::CnliCxlRxCrdStatusHigh2)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Rx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x0000402C; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:18	Reserved.
17:8	CxlMemDatCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated .mem data credits to be returned
7:0	CxlMemReqRspCrdCntHigh . Read-only, Volatile. Reset: 00h. Snapshot of accumulated .mem request or response credits bit 9:2 to be returned

CXL[0...3]x00004030 [Cnli CXL™ .Cache/.Mem Tx Credit Return Status] (DF::CnliCxlTxCrdStatusLow2)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Tx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00004030; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:30	CxlMemReqRspCrdCntLow . Read-only, Volatile. Reset: 0h. Snapshot of accumulated .mem request or response credits bit 1:0
29:20	CxlCacheDatCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache data credits
19:10	CxlCacheRspCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache response credits
9:0	CxlCacheReqCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache request credits

CXL[0...3]x00004034 [Cnli CXL™ .Cache/.Mem Tx Credit Return Status] (DF::CnliCxlTxCrdStatusHigh2)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Tx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00004034; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:18	Reserved.
17:8	CxlMemDatCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated .mem data credits
7:0	CxlMemReqRspCrdCntHigh . Read-only, Volatile. Reset: 00h. Snapshot of accumulated .mem request or response credits bit 9:2

CXL[0...3]x00004038 [Cnli CXL™ Ack Timer Control register] (DF::CnliCxlAckTimerControl2)

Read-write. Reset: 0000_2010h.

Specify CNLI [CXL](#) Ack Timer control`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00004038; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:18	Reserved.
17:8	AckFlushRetimer . Read-write. Reset: 020h. Init: 380h. Specifies how many link layer clock cycles the entity should wait in case of idle, before flushing accumulated Acks using a LLCRD
7:0	AckForceThreshold . Read-write. Reset: 10h. Init: 20h. Specifies how many Flit Acks the Link Layer should accumulate before injecting a LLCRD

CXL[0...3]x0000403C [Cnli CXL™ Defeature Register] (DF::CnliCxlDefeature2)

Read-write, Volatile. Reset: 0000_0000h.

This is CNLI [CXL](#) Link Layer Defeature Register`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x0000403C; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:1	Reserved.
0	MDHDisable . Read-write, Volatile. Reset: 0. Write '1 to disable MDH

CXL[0...3]x00004040 [Cnli Uncorrectable Error Status Register] (DF::CnliCxlUncorrErrStat2)

Read-write, Volatile. Reset: 0000_0000h.

This is CNLI Uncorrectable Error Status Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004040; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:12	Reserved.
11	ReceiverOverflow. Read-write, Volatile. Reset: 0. Overflow in CNLI buffers detected. Which buffer is logged in first 3 bits of Header Log register. Software Write 1 to Clear.
10	PoisonRcvd. Read-write, Volatile. Reset: 0. Received poison indication from device. Software Write 1 to Clear.
9	RsvdEncViolation. Read-write, Volatile. Reset: 0. Unrecognized encoding in received flit. Software Write 1 to Clear.
8	ReinitThreshold. Read-write, Volatile. Reset: 0. Physical layer Reinit threshold reached. Software Write 1 to Clear.
7	MemDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL .mem. Defined in CXL spec but not applicable for DF .
6	MemBEParity. Read-write, Volatile. Reset: 0. Internal Byte Enable Parity Error detected on CXL .mem. Defined in CXL spec but not applicable for DF .
5	MemAddrParity. Read-write, Volatile. Reset: 0. Internal Address Parity Error detected on CXL .mem. Defined in CXL spec but not applicable for DF .
4	MemDataParity. Read-write, Volatile. Reset: 0. Internal Data Parity Error detected on CXL .mem. Software Write 1 to Clear.
3	CacheDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL .cache. Defined in CXL spec but not applicable for DF .
2	CacheBEParity. Read-write, Volatile. Reset: 0. Internal Byte Enable Parity Error detected on CXL .cache. Defined in CXL spec but not applicable for DF .
1	CacheAddrParity. Read-write, Volatile. Reset: 0. Internal Address Parity Error detected on CXL .cache. Defined in CXL spec but not applicable for DF .
0	CacheDataParity. Read-write, Volatile. Reset: 0. Internal Data Parity Error detected on CXL .cache. Software Write 1 to Clear.

CXL[0...3]x00004044 [Cnli Uncorrectable Error Mask Register] (DF::CnliCxlUncorrErrMask2)

Read-write, Volatile. Reset: 0000_0000h.

Contains Mask bits for corresponding errors in CNLI Uncorrectable Error Register. When an error is masked, status register is updated, but Uncorrectable error (interrupt) is not sent

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004044; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:12	Reserved.
11	ReceiverOverflowMask. Read-write, Volatile. Reset: 0. Mask detection of overflow in CNLI buffers
10	PoisonRcvdMask. Read-write, Volatile. Reset: 0. Mask detection of poison from device
9	RsvdEncViolationMask. Read-write, Volatile. Reset: 0. Mask detection of unrecognized encoding in received flit
8	ReinitThresholdMask. Read-write, Volatile. Reset: 0. Mask detection of Physical layer Reinit threshold
7	MemDataECCMask. Read-write, Volatile. Reset: 0. Mask detection of Data ECC Error on CXL .mem
6	MemBEParityMask. Read-write, Volatile. Reset: 0. Mask detection of Byte Enable Parity Error on CXL .mem
5	MemAddrParityMask. Read-write, Volatile. Reset: 0. Mask detection of Address Parity Error on CXL .mem
4	MemDataParityMask. Read-write, Volatile. Reset: 0. Mask detection of Data Parity Error on CXL .mem
3	CacheDataECCMask. Read-write, Volatile. Reset: 0. Mask detection of Data ECC Error on CXL .cache
2	CacheBEParityMask. Read-write, Volatile. Reset: 0. Mask detection of Byte Enable Parity Error on CXL .cache
1	CacheAddrParityMask. Read-write, Volatile. Reset: 0. Mask detection of Address Parity Error on CXL .cache
0	CacheDataParityMask. Read-write, Volatile. Reset: 0. Mask detection of Data Parity Error on CXL .cache

CXL[0...3]x00004048 [Cnli Uncorrectable Error Severity Register] (DF::CnliCxlUncorrErrSeverity2)

Read-write, Volatile. Reset: 0000_0FFFh.

Contains severity bits for corresponding errors in CNLI Uncorrectable Error Register. When Severity=0 for an uncorrectable error, correctable error bit will be set instead of uncorrectable bit (interrupt)

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004048; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:12	Reserved.
11	ReceiverOverflowSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of overflow in CNLI buffers
10	PoisonRcvdSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of poison from device
9	RsvdEncViolationSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of unrecognized encoding in received flit
8	ReinitThresholdSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Physical layer Reinit threshold
7	MemDataECCSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data ECC Error on CXL.mem
6	MemBEParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Byte Enable Parity Error on CXL.mem
5	MemAddrParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Address Parity Error on CXL.mem
4	MemDataParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data Parity Error on CXL.mem
3	CacheDataECCSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data ECC Error on CXL.cache
2	CacheBEParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Byte Enable Parity Error on CXL.cache
1	CacheAddrParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Address Parity Error on CXL.cache
0	CacheDataParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data Parity Error on CXL.cache

CXL[0...3]x0000404C [Cnli Correctable Error Status Register] (DF::CnliCxlCorrErrStat2)

Read-write, Volatile. Reset: 0000_0000h.

This is the CNLI Correctable Error Status Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000404C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:7	Reserved.
6	PhyLayerError. Read-write, Volatile. Reset: 0. Received error on Physical Layer. Phy Reinit received
5	MemPoisonRcvd. Read-write, Volatile. Reset: 0. Received poison indication from device on CXL.mem
4	CachePoisonRcvd. Read-write, Volatile. Reset: 0. Received poison indication from device on CXL.cache
3	RetryThreshold. Read-write, Volatile. Reset: 0. Retry threshold reached
2	CrcThreshold. Read-write, Volatile. Reset: 0. CRC error threshold reached
1	MemDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL.mem
0	CacheDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL.cache

CXL[0...3]x00004050 [Cnli Correctable Error Mask Register] (DF::CnliCxlCorrErrMask2)

Read-write, Volatile. Reset: 0000_0000h.

Mask bits for CNLI Correctable Errors. When an error is masked, status register is updated, but correctable error is not sent

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004050; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:7	Reserved.
6	PhyLayerErrorMask. Read-write, Volatile. Reset: 0. Mask for error on Physical Layer. Phy Reinit received
5	MemPoisonRcvdMask. Read-write, Volatile. Reset: 0. Mask for poison indication from device on CXL.mem
4	CachePoisonRcvdMask. Read-write, Volatile. Reset: 0. Mask for poison indication from device on CXL.cache
3	RetryThresholdMask. Read-write, Volatile. Reset: 0. Mask for Retry threshold reached
2	CrcThresholdMask. Read-write, Volatile. Reset: 0. Mask for CRC error threshold reached
1	MemDataECCMask. Read-write, Volatile. Reset: 0. Mask for Internal Data ECC Error detected on CXL.mem
0	CacheDataECCMask. Read-write, Volatile. Reset: 0. Mask for Internal Data ECC Error detected on CXL.cache

CXL[0...3]x00004054 [Cnli Error Capabilities and Control Register] (DF::CnliCxlErrCapAndCtrl2)

Reset: 0000_2000h.

Cnli Error Capabilities and Control Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004054; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:14	Reserved.
13	PoisonEn. Read-write, Volatile. Reset: 1. Indicates if poison is enabled. 1: treat/log poison as correctable error; 0: treat/log poison as uncorrectable error
12:10	Reserved.
9	MultiHdrRecCap. Read-only, Volatile. Reset: 0. Multi header recording capability. Indicates if recording more than one error header is supported. This is always zero for CXL 1.1
8:4	Reserved.
3:0	FirstErrPtr. Read-only, Volatile. Reset: 0h. Bit position of the first error bit for the error reported in Uncorrectable error status register

CXL[0...3]x00004058 [Cnli Header Log Register] (DF::CnliCxlHdrLog02)

Read-only, Volatile. Reset: 0000_0000h.

0th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004058; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:0	HdrLog. Read-only, Volatile. Reset: 0000_0000h. 0th 32 bit segment of Header Log Register

CXL[0...3]x0000405C [Cnli Header Log Register] (DF::CnliCxlHdrLog12)

Read-only, Volatile. Reset: 0000_0000h.

First 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000405C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:0	HdrLog. Read-only, Volatile. Reset: 0000_0000h. First 32 bit segment of Header Log Register

CXL[0...3]x00004060 [Cnli Header Log Register] (DF::CnliCxlHdrLog22)

Read-only, Volatile. Reset: 0000_0000h.

Second 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004060; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:0	HdrLog. Read-only, Volatile. Reset: 0000_0000h. Second 32 bit segment of Header Log Register

CXL[0...3]x00004064 [Cnli Header Log Register] (DF::CnliCxlHdrLog32)

Read-only, Volatile. Reset: 0000_0000h.

Third 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00004064; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Third 32 bit segment of Header Log Register**CXL[0...3]x00004068 [Cnli Header Log Register] (DF::CnliCxlHdrLog42)**

Read-only, Volatile. Reset: 0000_0000h.

Fourth 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00004068; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Fourth 32 bit segment of Header Log Register**CXL[0...3]x0000406C [Cnli Header Log Register] (DF::CnliCxlHdrLog52)**

Read-only, Volatile. Reset: 0000_0000h.

Fifth 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x0000406C; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Fifth 32 bit segment of Header Log Register**CXL[0...3]x00004070 [Cnli Header Log Register] (DF::CnliCxlHdrLog62)**

Read-only, Volatile. Reset: 0000_0000h.

Sixth 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00004070; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Sixth 32 bit segment of Header Log Register**CXL[0...3]x00004074 [Cnli Header Log Register] (DF::CnliCxlHdrLog72)**

Read-only, Volatile. Reset: 0000_0000h.

7th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00004074; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 7th 32 bit segment of Header Log Register**CXL[0...3]x00004078 [Cnli Header Log Register] (DF::CnliCxlHdrLog82)**

Read-only, Volatile. Reset: 0000_0000h.

8th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x00004078; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 8th 32 bit segment of Header Log Register**CXL[0...3]x0000407C [Cnli Header Log Register] (DF::CnliCxlHdrLog92)**

Read-only, Volatile. Reset: 0000_0000h.

9th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CnliI[3:0]]_aliasSMN; CXL[3:0]x0000407C; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 9th 32 bit segment of Header Log Register

CXL[0...3]x00004080 [Cnli Header Log Register] (DF::CnliCxlHdrLog102)

Read-only, Volatile. Reset: 0000_0000h.

10th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004080; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 10th 32 bit segment of Header Log Register**CXL[0...3]x00004084 [Cnli Header Log Register] (DF::CnliCxlHdrLog112)**

Read-only, Volatile. Reset: 0000_0000h.

11th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004084; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 11th 32 bit segment of Header Log Register**CXL[0...3]x00004088 [Cnli Header Log Register] (DF::CnliCxlHdrLog122)**

Read-only, Volatile. Reset: 0000_0000h.

12th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004088; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 12th 32 bit segment of Header Log Register**CXL[0...3]x0000408C [Cnli Header Log Register] (DF::CnliCxlHdrLog132)**

Read-only, Volatile. Reset: 0000_0000h.

13th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000408C; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 13th 32 bit segment of Header Log Register**CXL[0...3]x00004090 [Cnli Header Log Register] (DF::CnliCxlHdrLog142)**

Read-only, Volatile. Reset: 0000_0000h.

14th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004090; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 14th 32 bit segment of Header Log Register**CXL[0...3]x00004094 [Cnli Header Log Register] (DF::CnliCxlHdrLog152)**

Read-only, Volatile. Reset: 0000_0000h.

15th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00004094; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 15th 32 bit segment of Header Log Register

CXL[0...3]x00006000 [CXL™ Capability Header Register] (DF::CxlCapabilityHeader3)

Read-only, Volatile. Reset: 0311_0001h.

Specify the header for [CXL](#) Capability Register`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006000; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:24	ArraySize. Read-only, Volatile. Reset: 03h. defines the number of elements present in the CXL_Capability array, not including the CXL_Capability_Header element.
23:20	CxlCacheMemVersion. Read-only, Volatile. Reset: 1h. This defines the version of the CXL Cache Mem Protocol supported.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0001h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x00006004 [CXL™ Capability Header Register] (DF::CxlRASCapabilityHeader3)

Read-only, Volatile. Reset: 0401_0002h.

Specify the header for [CXL RAS](#) Capability Register`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006004; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:20	CxlRASCapabilityPointer. Read-only, Volatile. Reset: 040h. This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0002h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x00006008 [CXL™ Security Capability Header Register] (DF::CxlSecurityCapabilityHeader3)

Read-only, Volatile. Reset: 0101_0003h.

Specify the header for [CXL](#) Security Capability Register`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006008; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:20	CxlSecurityCapabilityPointer. Read-only, Volatile. Reset: 010h. This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0003h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x0000600C [CXL™ Link Capability Header Register] (DF::CxlLinkCapabilityHeader3)

Read-only, Volatile. Reset: 0141_0004h.

Specify the header for [CXL](#) Link Capability Register`_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000600C; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:20	CxlLinkCapabilityPointer. Read-only, Volatile. Reset: 014h. This defines the offset of the CXL_Capability relative to beginning of CXL_Capability_Header register.
19:16	CxlCapabilityVersion. Read-only, Volatile. Reset: 1h. This defines the version number of the CXL_Capability structure present.
15:0	CxlCapabilityID. Read-only, Volatile. Reset: 0004h. This defines the nature and format of the CXL_Capability register.

CXL[0...3]x00006014 [CXL™ Link Layer Capability Low Register] (DF::CxlLinkLayerCapabilityLow3)

Read-only, Volatile. Reset: 0000_0011h.

Specify [CXL](#) Link Layer Low Capability Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006014; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:29	NumPhyReInitRcvdLow. Read-only, Volatile. Reset: 0h. Num_Phy_ReInit value bit 2:0 reflected in the last Retry.Req message received. Used for debug.
28:24	NumRetryRcvd. Read-only, Volatile. Reset: 00h. Num_Retry value reflected in the last Retry.Req message received. Used for debug.
23:16	LLRWrapValueRcvd. Read-only, Volatile. Reset: 00h. LLR Wrap value received from INIT.Param flit. Used for debug.
15:8	LLRWrapValueSupported. Read-only, Volatile. Reset: 00h. LLR Wrap value supported by this entity. Used for debug.
7:4	CxlLinkVersionRcvd. Read-only, Volatile. Reset: 1h. Version of AL received from INIT.Param flit. Used for debug.
3:0	CxlLinkVersionSupported. Read-only, Volatile. Reset: 1h. Version of AL the port is compliant with. For CXL 1.0, this should be 'b0001.

CXL[0...3]x00006018 [CXL™ Link Layer Capability High Register] (DF::CxlLinkLayerCapabilityHigh3)

Read-only, Volatile. Reset: 0400_0000h.

Specify [CXL](#) Link Layer Capability High Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006018; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:27	Reserved.
26	No_LL_Reset_Support. Read-only, Volatile. Reset: 1. LL_Reset which is a debug only feature is not supported
25:18	NumFreeBufRcvd. Read-only, Volatile. Reset: 00h. Number of Free Buffer value received in the last Retry.Ack message received.
17:10	EchoEseqRcvd. Read-only, Volatile. Reset: 00h. Echo_Eseq value reflected in the last Retry.Ack message received.
9:2	WrPtrRcvd. Read-only, Volatile. Reset: 00h. Wr_Ptr value reflected in the last Retry.Ack message received.
1:0	NumPhyReInitRcvdHigh. Read-only, Volatile. Reset: 0h. Num_Phy_ReInit value bit 4:3 reflected in the last Retry.Req message received. Used for debug.

CXL[0...3]x0000601C [CXL™ Link Layer Control and Status Register] (DF::CxlLinkLayerControlStatus3)

Reset: 0000_0000h.

Specify [CXL](#) Link Layer Control and Status Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000601C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:13	Reserved.
12:5	LLRetryBufConsumed. Read-only, Volatile. Reset: 00h. Snapshot of link layer retry buffer consumed
4:3	InitState. Read-only, Volatile. Reset: 0h. This field reflects the current initialization status of the Link Layer. 00: NOT_RDY_FOR_INIT (stalled or unstalled); 01: LLCTRL-INIT.Param sent and waiting to receive it; 10: Parameter exchanged successfully and Credit return is stalled; 11: Link Initialization Done
2	LLCrdStall. Read-write, Volatile. Reset: 0. If set, link layer stalls credit initialization until this bit is cleared
1	LLInitStall. Read-write, Volatile. Reset: 0. If set, link layer stalls the transmission of the LLCTRL-INIT.Param flit until this bit is cleared.
0	LLReset. Read-write, Volatile. Reset: 0. Re-initialize without resetting values in sticky registers. Write '1 to reset link - this is a destructive reset all link layer state. When link layer reset completes, HW will clear the bit to 0.

CXL[0...3]x00006020 [Cnli CXL™ .Cache/.Mem Rx Credit Control] (DF::CnliCxlRxCrdControlLow3)

Read-write. Reset: 0701_8070h.

Specify the CNLI [CXL](#) .Cache/.Mem Rx Request/Data/Response token config

_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00006020; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:30	CxlMemReqRspCrdCntLow . Read-write. Reset: 0h. Initially allocated credits bit 1:0 for .Mem Request/Response from Endpoint to Cnli. At least one Req token has to be allocated.
29:20	CxlCacheDatTokCnt . Read-write. Reset: 070h. Initially allocated token for .Cache Data from Endpoint to Cnli. At least one Req Dat token has to be allocated.
19:10	CxlCacheRspTokCnt . Read-write. Reset: 060h. Initially allocated token for .Cache Response from Endpoint to Cnli. At least one Rsp token has to be allocated.
9:0	CxlCacheReqTokCnt . Read-write. Reset: 070h. Initially allocated token for .Cache Request from Endpoint to Cnli. At least one Req token has to be allocated.

CXL[0...3]x00006024 [Cnli CXL™ .Cache/.Mem Rx Credit Control] (DF::CnliCxlRxCrdControlHigh3)

Read-write. Reset: 0000_1806h.

Specify the Cnli [CXL](#) .Cache/.Mem Rx Request/Data/Response token config

_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00006024; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:18	Reserved.
17:8	CxlMemDatCrdCnt . Read-write. Reset: 018h. Init: 11.5.3 [Register Values Written Based on Configuration]. Initially allocated credits for .Mem Request Data from Endpoint to Cnli. At least one Req Dat token has to be allocated.
7:0	CxlMemReqRspCrdCntHigh . Read-write. Reset: 06h. Init: 11.5.3 [Register Values Written Based on Configuration]. Initially allocated credits bit 9:2 for .Mem Request/Response from Endpoint to Cnli. At least one Req token has to be allocated.

CXL[0...3]x00006028 [Cnli CXL™ .Cache/.Mem Rx Credit Return Status] (DF::CnliCxlRxCrdStatusLow3)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Rx Credit Return Status

_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00006028; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:30	CxlMemReqRspCrdCntLow . Read-only, Volatile. Reset: 0h. Snapshot of accumulated .mem request or response credits bit 1:0 to be returned
29:20	CxlCacheDatCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache data credits to be returned
19:10	CxlCacheRspCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache response credits to be returned
9:0	CxlCacheReqCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache request credits to be returned

CXL[0...3]x0000602C [Cnli CXL™ .Cache/.Mem Rx Credit Return Status] (DF::CnliCxlRxCrdStatusHigh3)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Rx Credit Return Status

_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x0000602C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:18	Reserved.
17:8	CxlMemDatCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated .mem data credits to be returned
7:0	CxlMemReqRspCrdCntHigh . Read-only, Volatile. Reset: 00h. Snapshot of accumulated .mem request or response credits bit 9:2 to be returned

CXL[0...3]x00006030 [Cnli CXL™ .Cache/.Mem Tx Credit Return Status] (DF::CnliCxlTxCrdStatusLow3)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Tx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00006030; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:30	CxlMemReqRspCrdCntLow . Read-only, Volatile. Reset: 0h. Snapshot of accumulated .mem request or response credits bit 1:0
29:20	CxlCacheDatCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache data credits
19:10	CxlCacheRspCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache response credits
9:0	CxlCacheReqCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated cache request credits

CXL[0...3]x00006034 [Cnli CXL™ .Cache/.Mem Tx Credit Return Status] (DF::CnliCxlTxCrdStatusHigh3)

Read-only, Volatile. Reset: 0000_0000h.

This is a read-only register containing bits for CNLI [CXL](#) .Cache/.Mem Tx Credit Return Status`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00006034; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:18	Reserved.
17:8	CxlMemDatCrdCnt . Read-only, Volatile. Reset: 000h. Snapshot of accumulated .mem data credits
7:0	CxlMemReqRspCrdCntHigh . Read-only, Volatile. Reset: 00h. Snapshot of accumulated .mem request or response credits bit 9:2

CXL[0...3]x00006038 [Cnli CXL™ Ack Timer Control register] (DF::CnliCxlAckTimerControl3)

Read-write. Reset: 0000_2010h.

Specify CNLI [CXL](#) Ack Timer control`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00006038; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:18	Reserved.
17:8	AckFlushRetimer . Read-write. Reset: 020h. Init: 380h. Specifies how many link layer clock cycles the entity should wait in case of idle, before flushing accumulated Acks using a LLCRD
7:0	AckForceThreshold . Read-write. Reset: 10h. Init: 20h. Specifies how many Flit Acks the Link Layer should accumulate before injecting a LLCRD

CXL[0...3]x0000603C [Cnli CXL™ Defeature Register] (DF::CnliCxlDefeature3)

Read-write, Volatile. Reset: 0000_0000h.

This is CNLI [CXL](#) Link Layer Defeature Register`_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x0000603C; CXL[3:0]=1F1[3:0]_0000h`

Bits	Description
31:1	Reserved.
0	MDHDisable . Read-write, Volatile. Reset: 0. Write '1 to disable MDH

CXL[0...3]x00006040 [Cnli Uncorrectable Error Status Register] (DF::CnliCxlUncorrErrStat3)

Read-write, Volatile. Reset: 0000_0000h.

This is CNLI Uncorrectable Error Status Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006040; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:12	Reserved.
11	ReceiverOverflow. Read-write, Volatile. Reset: 0. Overflow in CNLI buffers detected. Which buffer is logged in first 3 bits of Header Log register. Software Write 1 to Clear.
10	PoisonRcvd. Read-write, Volatile. Reset: 0. Received poison indication from device. Software Write 1 to Clear.
9	RsvdEncViolation. Read-write, Volatile. Reset: 0. Unrecognized encoding in received flit. Software Write 1 to Clear.
8	ReinitThreshold. Read-write, Volatile. Reset: 0. Physical layer Reinit threshold reached. Software Write 1 to Clear.
7	MemDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL .mem. Defined in CXL spec but not applicable for DF .
6	MemBEParity. Read-write, Volatile. Reset: 0. Internal Byte Enable Parity Error detected on CXL .mem. Defined in CXL spec but not applicable for DF .
5	MemAddrParity. Read-write, Volatile. Reset: 0. Internal Address Parity Error detected on CXL .mem. Defined in CXL spec but not applicable for DF .
4	MemDataParity. Read-write, Volatile. Reset: 0. Internal Data Parity Error detected on CXL .mem. Software Write 1 to Clear.
3	CacheDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL .cache. Defined in CXL spec but not applicable for DF .
2	CacheBEParity. Read-write, Volatile. Reset: 0. Internal Byte Enable Parity Error detected on CXL .cache. Defined in CXL spec but not applicable for DF .
1	CacheAddrParity. Read-write, Volatile. Reset: 0. Internal Address Parity Error detected on CXL .cache. Defined in CXL spec but not applicable for DF .
0	CacheDataParity. Read-write, Volatile. Reset: 0. Internal Data Parity Error detected on CXL .cache. Software Write 1 to Clear.

CXL[0...3]x00006044 [Cnli Uncorrectable Error Mask Register] (DF::CnliCxlUncorrErrMask3)

Read-write, Volatile. Reset: 0000_0000h.

Contains Mask bits for corresponding errors in CNLI Uncorrectable Error Register. When an error is masked, status register is updated, but Uncorrectable error (interrupt) is not sent

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006044; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:12	Reserved.
11	ReceiverOverflowMask. Read-write, Volatile. Reset: 0. Mask detection of overflow in CNLI buffers
10	PoisonRcvdMask. Read-write, Volatile. Reset: 0. Mask detection of poison from device
9	RsvdEncViolationMask. Read-write, Volatile. Reset: 0. Mask detection of unrecognized encoding in received flit
8	ReinitThresholdMask. Read-write, Volatile. Reset: 0. Mask detection of Physical layer Reinit threshold
7	MemDataECCMask. Read-write, Volatile. Reset: 0. Mask detection of Data ECC Error on CXL .mem
6	MemBEParityMask. Read-write, Volatile. Reset: 0. Mask detection of Byte Enable Parity Error on CXL .mem
5	MemAddrParityMask. Read-write, Volatile. Reset: 0. Mask detection of Address Parity Error on CXL .mem
4	MemDataParityMask. Read-write, Volatile. Reset: 0. Mask detection of Data Parity Error on CXL .mem
3	CacheDataECCMask. Read-write, Volatile. Reset: 0. Mask detection of Data ECC Error on CXL .cache
2	CacheBEParityMask. Read-write, Volatile. Reset: 0. Mask detection of Byte Enable Parity Error on CXL .cache
1	CacheAddrParityMask. Read-write, Volatile. Reset: 0. Mask detection of Address Parity Error on CXL .cache
0	CacheDataParityMask. Read-write, Volatile. Reset: 0. Mask detection of Data Parity Error on CXL .cache

CXL[0...3]x00006048 [Cnli Uncorrectable Error Severity Register] (DF::CnliCxlUncorrErrSeverity3)

Read-write, Volatile. Reset: 0000_0FFFh.

Contains severity bits for corresponding errors in CNLI Uncorrectable Error Register. When Severity=0 for an uncorrectable error, correctable error bit will be set instead of uncorrectable bit (interrupt)

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006048; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:12	Reserved.
11	ReceiverOverflowSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of overflow in CNLI buffers
10	PoisonRcvdSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of poison from device
9	RsvdEncViolationSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of unrecognized encoding in received flit
8	ReinitThresholdSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Physical layer Reinit threshold
7	MemDataECCSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data ECC Error on CXL.mem
6	MemBEParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Byte Enable Parity Error on CXL.mem
5	MemAddrParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Address Parity Error on CXL.mem
4	MemDataParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data Parity Error on CXL.mem
3	CacheDataECCSeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data ECC Error on CXL.cache
2	CacheBEParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Byte Enable Parity Error on CXL.cache
1	CacheAddrParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Address Parity Error on CXL.cache
0	CacheDataParitySeverity. Read-write, Volatile. Reset: 1. 0: Reduce severity of detection of Data Parity Error on CXL.cache

CXL[0...3]x0000604C [Cnli Correctable Error Status Register] (DF::CnliCxlCorrErrStat3)

Read-write, Volatile. Reset: 0000_0000h.

This is the CNLI Correctable Error Status Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000604C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:7	Reserved.
6	PhyLayerError. Read-write, Volatile. Reset: 0. Received error on Physical Layer. Phy Reinit received
5	MemPoisonRcvd. Read-write, Volatile. Reset: 0. Received poison indication from device on CXL.mem
4	CachePoisonRcvd. Read-write, Volatile. Reset: 0. Received poison indication from device on CXL.cache
3	RetryThreshold. Read-write, Volatile. Reset: 0. Retry threshold reached
2	CrcThreshold. Read-write, Volatile. Reset: 0. CRC error threshold reached
1	MemDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL.mem
0	CacheDataECC. Read-write, Volatile. Reset: 0. Internal Data ECC Error detected on CXL.cache

CXL[0...3]x00006050 [Cnli Correctable Error Mask Register] (DF::CnliCxlCorrErrMask3)

Read-write, Volatile. Reset: 0000_0000h.

Mask bits for CNLI Correctable Errors. When an error is masked, status register is updated, but correctable error is not sent

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006050; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:7	Reserved.
6	PhyLayerErrorMask. Read-write, Volatile. Reset: 0. Mask for error on Physical Layer. Phy Reinit received
5	MemPoisonRcvdMask. Read-write, Volatile. Reset: 0. Mask for poison indication from device on CXL.mem
4	CachePoisonRcvdMask. Read-write, Volatile. Reset: 0. Mask for poison indication from device on CXL.cache
3	RetryThresholdMask. Read-write, Volatile. Reset: 0. Mask for Retry threshold reached
2	CrcThresholdMask. Read-write, Volatile. Reset: 0. Mask for CRC error threshold reached
1	MemDataECCMask. Read-write, Volatile. Reset: 0. Mask for Internal Data ECC Error detected on CXL.mem
0	CacheDataECCMask. Read-write, Volatile. Reset: 0. Mask for Internal Data ECC Error detected on CXL.cache

CXL[0...3]x00006054 [Cnli Error Capabilities and Control Register] (DF::CnliCxlErrCapAndCtrl3)

Reset: 0000_2000h.

Cnli Error Capabilities and Control Register

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006054; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:14	Reserved.
13	PoisonEn. Read-write, Volatile. Reset: 1. Indicates if poison is enabled. 1: treat/log poison as correctable error; 0: treat/log poison as uncorrectable error
12:10	Reserved.
9	MultiHdrRecCap. Read-only, Volatile. Reset: 0. Multi header recording capability. Indicates if recording more than one error header is supported. This is always zero for CXL 1.1
8:4	Reserved.
3:0	FirstErrPtr. Read-only, Volatile. Reset: 0h. Bit position of the first error bit for the error reported in Uncorrectable error status register

CXL[0...3]x00006058 [Cnli Header Log Register] (DF::CnliCxlHdrLog03)

Read-only, Volatile. Reset: 0000_0000h.

0th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006058; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:0	HdrLog. Read-only, Volatile. Reset: 0000_0000h. 0th 32 bit segment of Header Log Register

CXL[0...3]x0000605C [Cnli Header Log Register] (DF::CnliCxlHdrLog13)

Read-only, Volatile. Reset: 0000_0000h.

First 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000605C; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:0	HdrLog. Read-only, Volatile. Reset: 0000_0000h. First 32 bit segment of Header Log Register

CXL[0...3]x00006060 [Cnli Header Log Register] (DF::CnliCxlHdrLog23)

Read-only, Volatile. Reset: 0000_0000h.

Second 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006060; CXL[3:0]=1F1[3:0]_0000h

Bits	Description
31:0	HdrLog. Read-only, Volatile. Reset: 0000_0000h. Second 32 bit segment of Header Log Register

CXL[0...3]x00006064 [Cnli Header Log Register] (DF::CnliCxlHdrLog33)

Read-only, Volatile. Reset: 0000_0000h.

Third 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00006064; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Third 32 bit segment of Header Log Register**CXL[0...3]x00006068 [Cnli Header Log Register] (DF::CnliCxlHdrLog43)**

Read-only, Volatile. Reset: 0000_0000h.

Fourth 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00006068; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Fourth 32 bit segment of Header Log Register**CXL[0...3]x0000606C [Cnli Header Log Register] (DF::CnliCxlHdrLog53)**

Read-only, Volatile. Reset: 0000_0000h.

Fifth 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x0000606C; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Fifth 32 bit segment of Header Log Register**CXL[0...3]x00006070 [Cnli Header Log Register] (DF::CnliCxlHdrLog63)**

Read-only, Volatile. Reset: 0000_0000h.

Sixth 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00006070; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. Sixth 32 bit segment of Header Log Register**CXL[0...3]x00006074 [Cnli Header Log Register] (DF::CnliCxlHdrLog73)**

Read-only, Volatile. Reset: 0000_0000h.

7th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00006074; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 7th 32 bit segment of Header Log Register**CXL[0...3]x00006078 [Cnli Header Log Register] (DF::CnliCxlHdrLog83)**

Read-only, Volatile. Reset: 0000_0000h.

8th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x00006078; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 8th 32 bit segment of Header Log Register**CXL[0...3]x0000607C [Cnli Header Log Register] (DF::CnliCxlHdrLog93)**

Read-only, Volatile. Reset: 0000_0000h.

9th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[Cnli[3:0]]_aliasSMN; CXL[3:0]x0000607C; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 9th 32 bit segment of Header Log Register

CXL[0...3]x00006080 [Cnli Header Log Register] (DF::CnliCxlHdrLog103)

Read-only, Volatile. Reset: 0000_0000h.

10th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006080; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 10th 32 bit segment of Header Log Register**CXL[0...3]x00006084 [Cnli Header Log Register] (DF::CnliCxlHdrLog113)**

Read-only, Volatile. Reset: 0000_0000h.

11th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006084; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 11th 32 bit segment of Header Log Register**CXL[0...3]x00006088 [Cnli Header Log Register] (DF::CnliCxlHdrLog123)**

Read-only, Volatile. Reset: 0000_0000h.

12th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006088; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 12th 32 bit segment of Header Log Register**CXL[0...3]x0000608C [Cnli Header Log Register] (DF::CnliCxlHdrLog133)**

Read-only, Volatile. Reset: 0000_0000h.

13th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x0000608C; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 13th 32 bit segment of Header Log Register**CXL[0...3]x00006090 [Cnli Header Log Register] (DF::CnliCxlHdrLog143)**

Read-only, Volatile. Reset: 0000_0000h.

14th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006090; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 14th 32 bit segment of Header Log Register**CXL[0...3]x00006094 [Cnli Header Log Register] (DF::CnliCxlHdrLog153)**

Read-only, Volatile. Reset: 0000_0000h.

15th 32 bit segment of Flit Information from the flit causing Uncorrectable Error

_inst[CNLI[3:0]]_aliasSMN; CXL[3:0]x00006094; CXL[3:0]=1F1[3:0]_0000h

Bits Description31:0 **HdrLog.** Read-only, Volatile. Reset: 0000_0000h. 15th 32 bit segment of Header Log Register

12 Unified Memory Controller (UMC)

12.1 DDR5 Overview

Each processor [IOD](#) includes twelve Unified Memory Controllers (UMC). Each UMC controls two DDR5 32-bit channels routed to the same DDR5 DIMM.

The UMC interfaces to the processor Data Fabric ([DF](#)) Coherent Slave and the System Management Network (SMN).

12.1.1 Frequency and Population Support

The tables in this section list the maximum DRAM speeds supported based on configuration. The motherboard should comply with the relevant AMD socket motherboard design guidelines (MBDG) to achieve the rated speeds. In cases where MBDG design options exist, lower-quality options may compromise the maximum achievable speed; motherboard designers should assess the tradeoffs.

The following DDR5 configuration restrictions apply to the DIMM types and populations supported by the UMC:

- Mixing of base module types within a channel is not supported. (DDR5 SPD Byte 3)
- Mixing of 3DS and non-3DS devices within a channel is not supported. (DDR5 SPD Byte 4 and SPD Byte 8)
- Mixing of x4 and x8 devices within a channel is not supported. (DDR5 SPD Byte 6 and SPD Byte 10)
- Mixing different ECC DIMM types within a channel is not supported. (DDR5 SPD Byte 235)
- Mixing ECC and non-ECC DIMM types within a channel is not supported (DDR5 SPD Byte 235)

Contact your AMD representative for more information regarding:

- Memory Population Guidelines <https://docs.amd.com/>
 - SP5 Memory Population Guidelines for AMD Family 19h Models 10h-1Fh - #56960
- AMD Validated Memory

12.1.1.1 Frequency Support

The following is the maximum DDR5 frequency supported by the processor.

Notes on JEDEC DDR5 DIMM label format for this section:

- 1R: 1 package rank of SDP DRAMs
- 2R: 2 package ranks of SDP DRAMs
- 2S2R: 2 package ranks of 2 high 3DS DRAMs
- 2S4R: 2 package ranks of 4 high 3DS DRAMs

Table 207: RDIMM Max Freq Support SP5 Packages

DIMM Slots per Channel	DIMMs Populated	DIMM		Frequency (MT/s)
		1R	2R	1.1V
1	1	1	-	4800
		-	1	4800
2	1	1	-	4800
		-	1	4800
	2	2	-	4800
		1	1	4000
		-	2	4000

Table 208: 3DS RDIMM Max Freq Support SP5 Packages

DIMM Slots per Channel	DIMMs Populated	DIMM	Frequency (MT/s)
		2S2R 2S4R	1.1V
1	1	1	4800
2	1	1	4800
	2	2	4000

LRDIMM is not supported due to lack of ecosystem enablement for full validation.

12.1.1.2 Population Support

The tables below list DIMM populations supported by the processor. DIMMs must be populated from farthest slot to closest slot to the processor on a per channel basis.

Table 209: DDR5 Population Support - Daisy Chain

Slots	DIMMs Populated	Closest DIMM	Farthest DIMM
1	1	1	N/A
2	1	-	1
	2	1	1

12.1.1.3 Voltage Support

The processor supports DDR5 DRAMs with the following nominal voltages:

- VDD 1.1V – DDR5 SPD Byte 16
- VDDQ 1.1V – DDR5 SPD Byte 17
- VPP 1.8V – DDR5 SPD Byte 18

12.1.2 UMC and DDR Phy Mapping

12.1.2.1 UMC and DDR Phy Logical Mapping

The table below shows the relationship of logical UMC instance to the processor package DDR channel.

Table 210: Instance to Package Channel Mapping for SP5, 12 DDR5 Channels

Instance	Subchannels	12acx/14acx
UMC0	CA, CB	14acx
UMC1	EA, EB	12acx
UMC2	FA, FB	12acx
UMC3	AA, AB	14acx
UMC4	BA, BB	12acx
UMC5	DA, DB	12acx
UMC6	IA, IB	14acx
UMC7	KA, KB	12acx
UMC8	LA, LB	12acx
UMC9	GA, GB	14acx

UMC10	HA, HB	12acx
UMC11	JA, JB	12acx

Note: Instance channel/subchannel names map to pin signal names. Example: Channels AA,AB mapped to UMC3 correspond to pin signal names MAA_, MAB_. Channels BA,BB mapped to UMC4 correspond to pin signal names MBA_, MBB_. And so on for other UMC instances/subchannels.

Table 211: Instance to Package Channel Mapping for TR5, 8 DDR5 Channels

Instance	Subchannels	12acx/14acx
UMC0	CA, CB	14acx
UMC1	DA, DB	12acx
UMC3	AA, AB	14acx
UMC4	BA, BB	12acx
UMC6	GA, GB	14acx
UMC7	HA, HB	12acx
UMC9	EA, EB	14acx
UMC10	FA, FB	12acx

Note: Instance channel/subchannel names map to pin signal names. Example: Subchannels AA,AB mapped to UMC3 correspond to pin signal names MAA_xxx, MAB_xxx. Subchannels BA,BB mapped to UMC4 correspond to pin signal names MBA_xxx, MBB_xxx. And so on for other UMC instances/subchannels.

12.1.2.2 DDR Phy Bump to Pin Mapping

The relationship of logical DDR5 Phy instance and bump to pin mapping varies by package.

12.1.2.2.1 SP5 DDR Phy Mapping

Table 212: Bump to Pin Mapping SP5 Package

Bump	12acx Pin	14acx Pin	Chiplet Type	Instance	Lane
A[0]	MBA0_CS_L[0]	MAA0_CS_L[0]	0	0	0
A[1]	MBA0_CS_L[1]	MAA0_CS_L[1]	0	0	1
A[2]	MBA1_CS_L[0]	MAA1_CS_L[0]	0	0	2
A[3]	MBA1_CS_L[1]	MAA1_CS_L[1]	0	0	3
A[4]	MBA_CA[0]	MAA_CA[0]	0	1	0
A[5]	MB_ALERT_L	MA_ALERT_L	0	1	1
A[6]	MBA_CA[2]	MAA_CA[2]	0	1	2
A[7]	NC	NC	0	1	3
A[8]	MBA_CA[1]	MAA_CA[1]	0	2	0
A[9]	NC	NC	0	2	1
A[10]	MBA_CA[3]	MAA_CA[3]	0	2	2
A[11]	NC	NC	0	2	3
A[12]	MBA_CA[4]	MAA_CA[4]	0	3	0
A[13]	NC	NC	0	3	1
A[14]	MBA_CA[6]	MAA_CA[6]	0	3	2
A[15]	NC	NC	0	3	3
A[16]	MBA_CA[5]	MAA_CA[5]	0	4	0
A[17]	NC	NC	0	4	1

A[18]	MBA_PAR	MAA_PAR	0	4	2
A[19]	NC	NC	0	4	3
A[20]	MB0_CLK_H	MA0_CLK_H	0	5	0
A[21]	MB0_CLK_L	MA0_CLK_L	0	5	1
A[22]	NC	NC	0	5	2
A[23]	NC	NC	0	5	3
A[24]	MB1_CLK_H	MA1_CLK_H	0	6	0
A[25]	MB1_CLK_L	MA1_CLK_L	0	6	1
A[26]	NC	NC	0	6	2
A[27]	NC	NC	0	6	3
A[28]	MBB_CA[0]	NC	0	7	0
A[29]	NC	NC	0	7	1
A[30]	MBB_CA[2]	NC	0	7	2
A[31]	NC	NC	0	7	3
A[32]	MBB_CA[1]	NC	0	8	0
A[33]	NC	NC	0	8	1
A[34]	MBB_CA[3]	NC	0	8	2
A[35]	NC	NC	0	8	3
A[36]	MBB_CA[4]	MAB_CA[0]	0	9	0
A[37]	MBA0_RSP_L	NC	0	9	1
A[38]	MBB_CA[6]	MAB_CA[2]	0	9	2
A[39]	MBA1_RSP_L	NC	0	9	3
A[40]	MBB_CA[5]	MAB_CA[1]	0	10	0
A[41]	MBB0_RSP_L	NC	0	10	1
A[42]	MBB_PAR	MAB_CA[3]	0	10	2
A[43]	MBB1_RSP_L	NC	0	10	3
A[44]	MBB0_CS_L[0]	MAB_CA[4]	0	11	0
A[45]	MBB0_CS_L[1]	MAA0_RSP_L	0	11	1
A[46]	MBB1_CS_L[0]	MAB_CA[6]	0	11	2
A[47]	MBB1_CS_L[1]	MAA1_RSP_L	0	11	3
A[48]	n/a	MAB_CA[5]	0	12	0
A[49]	n/a	MAB0_RSP_L	0	12	1
A[50]	n/a	MAB_PAR	0	12	2
A[51]	n/a	MAB1_RSP_L	0	12	3
A[52]	n/a	MAB0_CS_L[0]	0	13	0
A[53]	n/a	MAB0_CS_L[1]	0	13	1
A[54]	n/a	MAB1_CS_L[0]	0	13	2
A[55]	n/a	MAB1_CS_L[1]	0	13	3
D[7:0]	MBA_DATA[7:0]	MAA_DATA[7:0]	1	0	[7:0]
D[8]	MBA_DQS_H[5]	MAA_DQS_H[5]	1	0	8
D[9]	MBA_DQS_H[0]	MAA_DQS_H[0]	1	0	9
D[10]	MBA_DQS_L[0]	MAA_DQS_L[0]	1	0	10
D[11]	MBA_DQS_L[5]	MAA_DQS_L[5]	1	0	11
D[19:12]	MBA_DATA[15:8]	MAA_DATA[15:8]	1	1	[7:0]
D[20]	MBA_DQS_H[6]	MAA_DQS_H[6]	1	1	8
D[21]	MBA_DQS_H[1]	MAA_DQS_H[1]	1	1	9
D[22]	MBA_DQS_L[1]	MAA_DQS_L[1]	1	1	10
D[23]	MBA_DQS_L[6]	MAA_DQS_L[6]	1	1	11

D[31:24]	MBA_DATA[23:16]	MAA_DATA[23:16]	1	2	[7:0]
D[32]	MBA_DQS_H[7]	MAA_DQS_H[7]	1	2	8
D[33]	MBA_DQS_H[2]	MAA_DQS_H[2]	1	2	9
D[34]	MBA_DQS_L[2]	MAA_DQS_L[2]	1	2	10
D[35]	MBA_DQS_L[7]	MAA_DQS_L[7]	1	2	11
D[43:36]	MBA_DATA[31:24]	MAA_DATA[31:24]	1	3	[7:0]
D[44]	MBA_DQS_H[8]	MAA_DQS_H[8]	1	3	8
D[45]	MBA_DQS_H[3]	MAA_DQS_H[3]	1	3	9
D[46]	MBA_DQS_L[3]	MAA_DQS_L[3]	1	3	10
D[47]	MBA_DQS_L[8]	MAA_DQS_L[8]	1	3	11
D[55:48]	MBB_DATA[7:0]	MAB_DATA[7:0]	1	4	[7:0]
D[56]	MBB_DQS_H[5]	MAB_DQS_H[5]	1	4	8
D[57]	MBB_DQS_H[0]	MAB_DQS_H[0]	1	4	9
D[58]	MBB_DQS_L[0]	MAB_DQS_L[0]	1	4	10
D[59]	MBB_DQS_L[5]	MAB_DQS_L[5]	1	4	11
D[67:60]	MBB_DATA[15:8]	MAB_DATA[15:8]	1	5	[7:0]
D[68]	MBB_DQS_H[6]	MAB_DQS_H[6]	1	5	8
D[69]	MBB_DQS_H[1]	MAB_DQS_H[1]	1	5	9
D[70]	MBB_DQS_L[1]	MAB_DQS_L[1]	1	5	10
D[71]	MBB_DQS_L[6]	MAB_DQS_L[6]	1	5	11
D[79:72]	MBB_DATA[23:16]	MAB_DATA[23:16]	1	6	[7:0]
D[80]	MBB_DQS_H[7]	MAB_DQS_H[7]	1	6	8
D[81]	MBB_DQS_H[2]	MAB_DQS_H[2]	1	6	9
D[82]	MBB_DQS_L[2]	MAB_DQS_L[2]	1	6	10
D[83]	MBB_DQS_L[7]	MAB_DQS_L[7]	1	6	11
D[91:84]	MBB_DATA[31:24]	MAB_DATA[31:24]	1	7	[7:0]
D[92]	MBB_DQS_H[8]	MAB_DQS_H[8]	1	7	8
D[93]	MBB_DQS_H[3]	MAB_DQS_H[3]	1	7	9
D[94]	MBB_DQS_L[3]	MAB_DQS_L[3]	1	7	10
D[95]	MBB_DQS_L[8]	MAB_DQS_L[8]	1	7	11
D[103:96]	MBA_CHECK[7:0]	MAA_CHECK[7:0]	1	8	[7:0]
D[104]	MBA_DQS_H[9]	MAA_DQS_H[9]	1	8	8
D[105]	MBA_DQS_H[4]	MAA_DQS_H[4]	1	8	9
D[106]	MBA_DQS_L[4]	MAA_DQS_L[4]	1	8	10
D[107]	MBA_DQS_L[9]	MAA_DQS_L[9]	1	8	11
D[115:108]	MBB_CHECK[7:0]	MAB_CHECK[7:0]	1	9	[7:0]
D[116]	MBB_DQS_H[9]	MAB_DQS_H[9]	1	9	8
D[117]	MBB_DQS_H[4]	MAB_DQS_H[4]	1	9	9
D[118]	MBB_DQS_L[4]	MAB_DQS_L[4]	1	9	10
D[119]	MBB_DQS_L[9]	MAB_DQS_L[9]	1	9	11

Note: MAB_ mapping follows MAA_ mapping. MBB_ mapping follows MBA_ mapping. See 12.1.2.1 [UMC and DDR Phy Logical Mapping].

12.2 DDR5 UMC Initialization

DRAM initialization involves several steps in order to configure and tune the DRAM controllers and DRAM for optimal performance. Resume requires several steps to configure the UMCs to properly resume from an unpowered state such as the ACPI S3 state. The following sequence describes the steps needed after a reset for initialization or resume.

If UMC::UmcCap[MemChanDis] == 1, firmware performs programming steps to disable the UMC channel. See 12.2.15 [DDR5 Memory Channel Disable].

1. Configure the DDR supply voltage regulator.
2. If BIOS is booting from an unpowered state (ACPI S4, S5 or G3), then BIOS performs the following:
 1. Size available memory and determine MEMCLK frequency. Determine FCLK, DFCLK, and UCLK frequencies. See 12.2.1 [DDR5 UMC SPD-Based Configuration].
 1. Configure initial SOC voltages and frequencies..
 2. Program all configuration-dependent UMC registers.
 1. Program UMC::UmcConfig[DdrType].
 2. Program 12.2.1 [DDR5 UMC SPD-Based Configuration] and 12.2.2 [DDR5 UMC Control Mapping].
 3. Program "Init" values in 12.12 [UMC Registers].
 4. Program 12.2.11 [DDR5 UMC RAS]: ECC, Parity, WrCrc and RdCrc.
 5. Program 12.2.12 [DDR5 Additional UMC Configuration Dependencies].
 6. Program VDCI3 configuration registers: UMC::ClkDivCtrl, UMC::MasterVdciCtrl, and UMC::SlaveVdciCtrl .
 3. Initialize SDP Interface
 1. Set UMC::SdpCtrl[SdpInit] = 1.
 - The system should not allow DRAM traffic at this time.
 2. IF (UMC::DramScrubCtrl[DramScrubEn] == 1) THEN (CmdBufferCount = 7Fh) ELSE (CmdBufferCount = 80h).
 4. Perform DDR interface training:
 1. Program UMC::DFPM[SRSMEEn,DFPMEEn] = {0,1}.
 2. Program all configuration-dependent DDR phy registers.
 1. Program UMC::BeqCtrl0[DramRst] = 0.
 2. Program UMC::Phy::ATxOdtDrvStren[AODTStrenP].
 3. Wait 5ms.
 4. Program UMC::Phy::MAAlertControl[MALERTRxEn] = 1.
 5. Program 12.2.13 [DDR5 DDR Phy Initialization].
 3. For each Memory [P-state](#), initiate a Memory P-state change and execute PMU-based DDR training. DDR training leaves the DRAMs in self-refresh.
 1. Program UMC::Phy::PGCR[MAAlertAsync] = 1
 2. Program UMC::Phy::PGCR[DfiCtrlRxFifoRst] = 1
 3. Program UMC::Phy::AsyncAnibMode[AsyncAnibMode] = 2
 4. Program UMC registers dependent on DDR phy interface training.
 5. Program UMC registers based on enabled DDR5 [RAS](#) Recovery features.
3. If BIOS is resuming the platform from unpowered state such as the ACPI S3 state, then it restores the values programmed in step 2.
4. Initialize the UMC to DDR phy interface with a Memory P-state change to the desired startup MEMCLK frequency.
 1. Program SRSM instructions if necessary. See 12.9 [DDR5 SRSM MOP Array Encodings].
 2. Program UMC::DFPM[S3ResumeMode] = 1.
 3. Program UMC::DFPM[SRSMEEn,DFPMEEn] = {1,1}.
 4. Initiate the Memory P-state change and wait for completion.
 5. Program UMC::DFPM[S3ResumeMode] = 0.
 6. Program SRSM instructions if necessary.
5. Execute the following tasks if applicable:
 1. DDR5 DRAM MBIST PPR. See 12.2.16 [DDR5 DRAM MBIST PPR].
 2. DDR5 DRAM Post Package Repair. See 12.2.17 [DDR5 UMC Post Package Repair].
 3. DDR5 Interface Testing. See 12.2.18 [DDR5 UMC Interface Testing].
 4. UMC Data Encryption and Data Scrambling. See 12.4 [UMC Data Encryption and Data Scrambling].
 5. UMC Accelerated Memory Clear. See 12.5 [UMC Memory Clear].

1. Settings such as those found in UMC::DataCtrl must be finalized prior to Memory Clear.
2. Encryption key programming is finalized after Memory Clear.
6. DDR5 Periodic Phase Training (PPT). See 12.2.19 [DDR5 Periodic Phase Training (PPT)].
7. DDR5 DRAM On-die ECC Error Check and Scrub ([ECS](#)). See 12.2.20 [DDR5 DRAM On-die ECC Error Check and Scrub (ECS)].
8. DDR5 Thermal Management and Bandwidth Capping. See 12.2.21 [DDR5 Thermal Management and Bandwidth Capping].
9. DDR5 RAS and Recovery features must be finalized before DramReady.
6. Set UMC::UmcConfig[DramReady] = 1.

The DRAM subsystem is ready for use.

12.2.1 DDR5 UMC SPD-Based Configuration

The Serial Presence Detect (SPD) ROM is a non-volatile memory device on the DIMM encoded by the DIMM manufacturer. The description of the SPD is usually provided on a data sheet for the DIMM itself along with data describing the memory devices used. The data describes configuration and speed characteristics of the DIMM and the DRAM components mounted on the DIMM. The associated data sheet also contains the DIMM byte values that are encoded in the SPD on the DIMM.

BIOS reads the values encoded in the SPD ROM through a system-specific interface. BIOS acquires DIMM configuration information (such as the amount of memory on each DIMM) from the SPD ROM on each DIMM and uses this information to program the DRAM controller registers.

For solder-down DRAM, in the absence of an SPD ROM, platform BIOS provides an SPD data structure of the information necessary for DRAM configuration.

The SPD ROM provides values for several DRAM timing parameters that are required by the UMC. In general, BIOS should use the optimal value specified by the SPD ROM.

For DDR5, these parameters are:

UMC::DramTiming1[Tcl]: SDRAM CAS Latencies Supported, SDRAM Minimum CAS Latency Time

UMC::DramTiming1[Trcdwr,Trcdrd]: SDRAM Minimum [RAS](#) to CAS Delay Time

UMC::DramTiming2[TrpAb]: SDRAM Minimum Row Precharge Delay Time

UMC::DramTiming1[Tras]: SDRAM Minimum Active to Precharge Delay Time

UMC::DramTiming2[TrcAb]: SDRAM Minimum Active to Active/Refresh Delay Time

UMC::DramTiming6[Twr]: SDRAM Minimum Write Recovery Time

UMC::TrfcTiming[Trfc]: SDRAM Minimum Refresh Recovery Delay Time (tRFC1)

UMC::TrfcTiming[Trfc2]: SDRAM Minimum Refresh Recovery Delay Time (tRFC2)

SDRAM Minimum Refresh Recovery Delay Time (tRFCsb)

UMC::RfmCtrl: SDRAM Refresh Management, SDRAM Adaptive Refresh Management

12.2.2 DDR5 UMC Control Mapping

Program the DRAM controller mappings based on package, motherboard, and DIMM population:

- UMC::DimmCfg
 - Program the field X4Dram per DIMM SPD.
- ECC
 - If (UMC::UmcCap[EccDis] == 0) then program the following per DIMM SPD:
 - UMC::UmcConfig[DimmEccEn]
 - See 12.2.11 [DDR5 UMC RAS]
- UMC::CSMapCKE

- For each CS to 'virtual CKE' mapping program the following if the chip select is populated:
 - CSMapCKE0 = 1h
 - CSMapCKE1 = 2h
 - CSMapCKE2 = 4h
 - CSMapCKE3 = 8h
- UMC::CSMapZQ
 - For each CS to ZQ group mapping program the following by DIMM population if the chip select is populated:
 - DIMM0-only populated:
 - CSMapZQ0 = 1h
 - CSMapZQ1 = 2h
 - DIMM1-only populated:
 - CSMapZQ0 = 4h
 - CSMapZQ1 = 8h
 - DIMM0 and DIMM1 populated:
 - CSMapZQ0 = 1h
 - CSMapZQ1 = 2h
 - CSMapZQ2 = 4h
 - CSMapZQ3 = 8h
 - CKs
 - UMC::BeqCtrl2[DfiDramClkDis]:
 - RDIMM: Program per ~CS[3:0] populated.
- DFPM
 - RDIMM:
 - UMC::DFPM2[Cmd1CsEnMap] per populated CSes.
 - Program UMC::DFPM3[Cmd1DramClkDis, Cmd2DramClkDis] = {Fh, Fh}.

12.2.3 DDR5 DRAM Addressing

This section contains information for programming UMC DRAM address mappings.

12.2.3.1 DDR5 Chip Select Decoding

For each chip select there is a UMC::BaseAddr, UMC::BaseAddrSec, UMC::AddrMask and UMC::AddrMaskSec register. For each chip select pair of a DIMM module, an even chip select must be populated if the odd chip select is populated.

Each base address register specifies the starting normalized address of the block of memory associated with the chip select. Each mask register specifies the additional address bits that are consumed by the block of memory associated with the chip selects.

Adjusting the BaseAddr and AddrMask registers allows memory to be interleaved across multiple chip selects. See 12.2.3.3 [DDR5 Chip Select Interleaving].

BIOS is required to assign the largest DIMM chip select range to the lowest normalized address of the DRAM controller. As addresses increase, the chip select size is required to remain constant or decrease. This is necessary to keep chip select decoding on aligned address boundaries.

The registers UMC::BaseAddrExt, UMC::BaseAddrSecExt, UMC::AddrMaskExt, and UMC::AddrMaskSecExt are used for configurations with extended addressing to specify normalized bits [47:40].

For each UMC normalized address, a Chipselect[i] is asserted if: CSEnable[i] & (NormAddr[47:9] & ~AddrMask[i]

[47:9]).

12.2.3.2 DDR5 Row, Bank, Column Addressing

For base DDR5 operation the UMC is configured to map normalized addresses as Chip Select (CS), Row, Bank Group (BG), Bank Address (BA), and Column addressing.

The following figure describes logical addressing for standard DDR5 device types for two 32b channel interfaces per UMC:

CS Size (32b / 2 x 32b)	Device Size, width	BG			BA		Address																CH				
		2	1	0	1	0		17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	0	
2GB / 4GB	8Gb, x16	x	15	14	x	13	Row	x	x	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	6	
4GB / 8GB	8Gb, x8	16	15	14	x	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2	6	
							Row	x	x	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6	
8GB / 16GB	8Gb, x4	17	16	15	x	14	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2	6	
							Row	x	x	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6	
4GB / 8GB	16Gb, x16	x	16	15	14	13	Column	x	x	x	x	x	x	x	x	13	12	11	10	9	8	7	5	4	3	2	6
							Row	x	x	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6	
8GB / 16GB	16Gb, x8	17	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2	6	
							Row	x	x	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6	
16GB / 32GB	16Gb, x4	18	17	16	15	14	Column	x	x	x	x	x	x	x	x	13	12	11	10	9	8	7	5	4	3	2	6
							Row	x	x	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	6	
8GB / 16GB	32Gb, x16	x	16	15	14	13	Row	x	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6	
6GB ¹ / 12GB ¹	24Gb, x16 ¹	x	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2	6	
16GB / 32GB	32Gb, x8	17	16	15	14	13	Row	x	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6	
12GB ¹ / 24GB ¹	24Gb, x8 ¹	17	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2	6	
32GB / 64GB	32Gb, x4	18	17	16	15	14	Row	x	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	6	
24GB ¹ / 48GB ¹	24Gb, x4 ¹	18	17	16	15	14	Column	x	x	x	x	x	x	x	13	12	11	10	9	8	7	5	4	3	2	6	
							Row	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6	
16GB / 32GB	64Gb, x16	x	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2	6	
							Row	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6	
32GB / 64GB	64Gb, x8	17	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2	6	
							Row	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	6	
64GB / 128GB	64Gb, x4	18	17	16	15	14	Column	x	x	x	x	x	x	x	13	12	11	10	9	8	7	5	4	3	2	6	
Note 1: 24Gb-based devices: Row Address A[x:y] = 00b,01b,10b are valid, A[x:y]=11b is invalid; A[x] = row address MSB (most significant bit); A[y] = row address 2nd MSB (second most significant bit)																											

Figure 45: DDR5 DRAM Row Bank Column Address Map For Two Channels Per UMC

12.2.3.3 DDR5 Chip Select Interleaving

Chip select memory interleaving mode has the following general requirements:

- The chip selects are the same size and type.

12.2.3.3.1 DDR5 Matched DIMM Interleaving

Chip select memory interleaving mode supports 2-way and 4-way interleaving across DIMMs with matched ranks.

Examples:

- DIMM0: 8GB 2Rx8 and DIMM1: 8GB 2Rx8
 - DIMM0 and DIMM1 form one 4-way interleaved address region in the UMC normalized address space
- DIMM0: Not Populated and DIMM1: 8GB 2Rx8
 - DIMM1 form one 2-way interleaved address region in the UMC normalized address space

12.2.3.3.2 DDR5 Mixed DIMM Interleaving

Chip select memory interleaving mode supports 2-way within DIMMs with matched ranks when a DIMM of a different rank type is populated.

Examples:

- DIMM0: 16GB 1Rx8 and DIMM1: 8GB 2Rx8
 - DIMM0 forms one non-interleaved address region in the UMC normalized address space
 - DIMM1 forms a second 2-way interleaved address region in the UMC normalized address space
- DIMM0: 16GB 2Rx8 and DIMM1: 8GB 2Rx8
 - DIMM0 forms a 2-way interleaved address region in the UMC normalized address space
 - DIMM1 forms a second 2-way interleaved address region in the UMC normalized address space

12.2.3.3.3 DDR5 Three Rank Interleaving

Chip select memory interleaving mode supports a special case of mixed DIMMs to interleave 3 matching power-of-two sized ranks.

Example:

- DIMM0: 8GB 1Rx8 and DIMM1: 16GB 2Rx8; all ranks based on 8Gb, x8 DRAMs
 - DIMM0 and DIMM1 will be divided using primary and secondary decoders to form three 2-way interleaved address regions in the UMC normalized address space

Note:

- The UMC::AddrHashCS feature is not applicable to the three rank interleaving mode.

12.2.3.3.4 DDR5 Programming Chip Select Interleaving

This section describes programming for 2-way and 4-way interleaving across DIMMs with matched ranks.

A BIOS algorithm for programming chip select interleaving is as follows:

1. Program all BaseAddr and AddrMask registers using contiguous normalized address mapping.
2. Swap 1 bit for 2-way interleaving, 2 bits for 4-way interleaving:
 1. For each chip select, swap the corresponding upper BaseAddr[39:9] bits with the lower BaseAddr[39:9] bits.
 2. For each chip select pair, swap the corresponding upper AddrMask[39:9] bits with the lower AddrMask[39:9] bits.

Row/CS Interleaving Mode:

- Swap the least significant bits of CS address for the least significant bits of Row address.
- Increment UMC::AddrSel[Row] by the number of bits swapped.

The following figure describes logical addressing for standard DDR5 device types for two 32b channel interfaces per UMC:

CS Size (32b / 2 x 32b)	Device Size, width	BG			BA		Address																CS	CH			
		2	1	0	1	0		17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	0	0
2GB / 4GB	8Gb, x16	x	15	14	x	13	Row	x	x	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	6
4GB / 8GB	8Gb, x8						Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2	17	6
		16	15	14	x	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2		
8GB / 16GB	8Gb, x4						Row	x	x	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6
		17	16	15	x	14	Column	x	x	x	x	x	x	x	13	12	11	10	9	8	7	5	4	3	2		
4GB / 8GB	16Gb, x16						Row	x	x	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6
		x	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2		
8GB / 16GB	16Gb, x8						Row	x	x	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6
		17	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2		
16GB / 32GB	16Gb, x4						Row	x	x	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	6
		18	17	16	15	14	Column	x	x	x	x	x	x	x	13	12	11	10	9	8	7	5	4	3	2		
8GB / 16GB	32Gb, x16						Row	x	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6
6GB ¹ / 12GB ¹	24Gb, x16 ¹	x	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2		
16GB / 32GB	32Gb, x8						Row	x	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6
12GB ¹ / 24GB ¹	24Gb, x8 ¹	17	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2		
32GB / 64GB	32Gb, x4						Row	x	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	6
24GB ¹ / 48GB ¹	24Gb, x4 ¹	18	17	16	15	14	Column	x	x	x	x	x	x	x	13	12	11	10	9	8	7	5	4	3	2		
16GB / 32GB	64Gb, x16						Row	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6
		x	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2		
32GB / 64GB	64Gb, x8						Row	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6
		17	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2		
64GB / 128GB	64Gb, x4						Row	37	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	6
		18	17	16	15	14	Column	x	x	x	x	x	x	x	13	12	11	10	9	8	7	5	4	3	2		
Note 1: 24Gb-based devices: Row Address A[x:y] = 00b,01b,10b are valid, A[x:y]=11b is invalid; A[x] = row address MSB (most significant bit); A[y] = row address 2nd MSB (second most significant bit)																											
Chipselect Interleave		Changes from a Row/BG/BA/Col to Row/CS/BG/BA/Col Mapping																									

Figure 46: DDR5 DRAM Chip select Interleaving Address Map for Two Channels per UMC

12.2.3.4 DDR5 Bank Swap

This section describes address transformations to adjust the normalized addresses assigned to the DDR5 Bank Group (BG) and Bank Address (BA) in the DRAM address map.

CPU Address Map:

- Assign the BG0 and BG1 bits to the normalized bits starting normalized address bits 8 and 9.
- Assign the BA bits to normalized addresses immediately above the BG bit(s).
- Assign the remaining BG, if present, at the next incremental normalized bit.
- Assign the remaining Column bits to normalized addresses immediately above the BG and BA bits.

The following figure describes logical addressing for standard DDR5 device types for two 32b channel interfaces per UMC:

CS Size (32b / 2 x 32b)	Device Size, width	BG			BA		Address																CH																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
		2	1	0	1	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						

Figure 47: DDR5 DRAM CPU Bank Swap Address Map for Two Channels per UMC

12.2.3.5 DDR5 Addressing Asymmetric DIMMs

This SOC supports modules with asymmetric rank assembly. In two-DIMM slots per channel configurations, both DIMMs must be matched in size and type. Modules with asymmetric rank assembly of 3DS devices are not supported. (DDR5 SPD Byte 234)

12.2.3.6 DDR5 Addressing for Rank Multiplication

This SOC supports modules with 3DS devices. In two-DIMM slots per channel configurations, both DIMMs must be matched in size and type. (DDR5 SPD Byte 4)

BIOS can use the following guidelines to program the algorithm for 3DS DRAM addressing and UMC RM (rank multiply):

- The size of BaseAddr/AddrMask plus BaseAddrSec/AddrMaskSec for a given DIMM is programmed to reflect the total DIMM memory addressing size.
- Program UMC::AddrCfg[NumRM] per the number of multiplied ranks associated with the chip select. Program UMC::RmSel[Rm0,Rm1,Rm2] to the normalized address bits to decode the ranks associated with the chip select.
- Assign the RM bits to the normalized address bits above the BG/BA bits and below the Row bits. In chip select interleaving, assign the RM bits to the normalized address bits above the BG/BA bits and below the Row/CS bits.

Increment the UMC::AddrSel[Row] and CS position by the number of RM bits.

The following figure shows logical addressing for standard DDR5 device types for two 32-bit channel interfaces per UMC:

CS Size (32b / 2 x 32b)	Device Size, width	BG			BA		Address																CS	RM	CH			
		2	1	0	1	0		17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	0	0	0
2GB / 4GB	8Gb, x16	x	15	14	x	13	Row	x	x	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	6
4GB / 8GB	8Gb, x8	16	15	14	x	13	Row	x	x	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6
							Column	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2				
8GB / 16GB	8Gb, x4	17	16	15	x	14	Row	x	x	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6
							Column	x	x	x	x	x	x	x	13	12	11	10	9	8	7	5	4	3	2			
4GB / 8GB	16Gb, x16	x	16	15	14	13	Row	x	x	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6
							Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2			
8GB / 16GB	16Gb, x8	17	16	15	14	13	Row	x	x	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6
							Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2			
16GB / 32GB	16Gb, x4	18	17	16	15	14	Row	x	x	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	6
							Column	x	x	x	x	x	x	x	13	12	11	10	9	8	7	5	4	3	2			
8GB / 16GB	32Gb, x16	x	16	15	14	13	Row	x	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6
6GB ¹ / 12GB ¹	24Gb, x16 ¹	x	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2			
16GB / 32GB	32Gb, x8						Row	x	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6
12GB ¹ / 24GB ¹	24Gb, x8 ¹	17	16	15	14	13	Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2			
32GB / 64GB	32Gb, x4						Row	x	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	20	19	6
24GB ¹ / 48GB ¹	24Gb, x4 ¹	18	17	16	15	14	Column	x	x	x	x	x	x	x	13	12	11	10	9	8	7	5	4	3	2			
16GB / 32GB	64Gb, x16	x	16	15	14	13	Row	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6
							Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2			
32GB / 64GB	64Gb, x8	17	16	15	14	13	Row	37	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6
							Column	x	x	x	x	x	x	x	x	12	11	10	9	8	7	5	4	3	2			
64GB / 128GB	64Gb, x4	18	17	16	15	14	Row	38	37	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	6
							Column	x	x	x	x	x	x	x	x	13	12	11	10	9	8	7	5	4	3	2		
Note 1: 24Gb-based devices: Row Address A[x:y] = 00b,01b,10b are valid; A[x:y]=11b is invalid; A[x] = row address MSB (most significant bit); A[y] = row address 2nd MSB (second most significant bit)																												
Chipselect Interleave + RM		Changes from a Row/BG/BA/Col to Row/CS/RM/BG/BA/Col Mapping																										

Figure 48: DDR5 DRAM Rank Multiply Address Map for Two Channels per UMC

12.2.3.7 DDR5 Addressing for Subchannel

On this SOC, both 32-bit channels of a DDR5 module are controlled by one UMC. When UMC::BeqCtrl0[VcmEn]==1, the UMC configures the channel addressing selection.

A BIOS algorithm for programming the CH (Channel Bit) is as follows:

- The size of BaseAddr/AddrMask plus BaseAddrSec/AddrMaskSec for a given DIMM is programmed to reflect the total DIMM memory addressing size for the two UMC subchannels.
- Program UMC::RmSel[ChanBit] as the normalized address bit that decodes the A or B module DRAMs.
- Assign the channel bit to normalized address bit 6.

12.2.3.8 DDR5 Addressing for Non Power-of-Two DRAMs

Modules with non power-of-two DRAM device densities are supported by this SOC.

A BIOS algorithm for programming is as follows:

1. Split the chip select size to be represented as a function of $2^N + 2^{(N-1)}$. Assign the larger portion to the

BaseAddr/AddrMask and assign the smaller portion to BaseAddrSec/AddrMaskSec.

2. Order all chip select ranges such as to assign the largest DIMM chip select range to the lowest normalized address of the DRAM controller and as addresses increase, the chip select size remains constant or decreases.
3. Compute InvertRowMsbs and InvertRowMsbsSec for the respective even and odd chip selects of the DIMM by determining the range of assigned normalized addresses. Program such that the most significant bits of the row address do not equal 11b.
4. If performing chip select interleaving, BIOS should use up to two interleaved regions: first across BaseAddr/AddrMask and second across BaseAddrSec/AddrMaskSec.

12.2.3.9 DDR5 Address Hash Bank

A BIOS algorithm for programming the UMC::AddrHashBank feature is as follows:

CPU Address Map:

- Row Address: Program a stagger pattern of the row address bits beginning with the BA, then BG bits
- Column Address: No column bits.

The following figure shows bank address hashing for standard DDR5 device types for two 32b channel interfaces per UMC:

CS Size (32b / 2 x 32b)	Device Size, width	BG			BA		Address																CH				
		2	1	0	1	0		17	16	15	14	13	12	11	10	9	8	7	6	5	4	3		2	1	0	
2GB / 4GB	8Gb, x16	x	15	14	x	13	Row	x	x	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	6	
4GB / 8GB	8Gb, x8	16	15	14	x	13	Row	x	x	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6	
		Column	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	6	
8GB / 16GB	8Gb, x4	17	16	15	x	14	Row	x	x	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6	
		Column	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	6	
4GB / 8GB	16Gb, x16	x	16	15	14	13	Row	x	x	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6	
		Column	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	6	
8GB / 16GB	16Gb, x8	17	16	15	14	13	Row	x	x	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6	
		Column	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	6	
16GB / 32GB	16Gb, x4	18	17	16	15	14	Row	x	x	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	6	
		Column	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	6	
8GB / 16GB	32Gb, x16	x	16	15	14	13	Row	x	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6	
6GB ¹ / 12GB ¹	24Gb, x16 ¹	x	16	15	14	13	Column	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	6	
16GB / 32GB	32Gb, x8	17	16	15	14	13	Row	x	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6	
12GB ¹ / 24GB ¹	24Gb, x8 ¹	17	16	15	14	13	Column	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	6	
32GB / 64GB	32Gb, x4	18	17	16	15	14	Row	x	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	6	
24GB ¹ / 48GB ¹	24Gb, x4 ¹	18	17	16	15	14	Column	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	6	
16GB / 32GB	64Gb, x16	x	16	15	14	13	Row	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	6	
		Column	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	6	
32GB / 64GB	64Gb, x8	17	16	15	14	13	Row	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	6	
		Column	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	6	
64GB / 128GB	64Gb, x4	18	17	16	15	14	Row	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	6	
		Column	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	6	
Note 1: 24Gb-based devices: Row Address A[x:y] = 00b,01b,10b are valid, A[x:y]=11b is invalid; A[x] = row address MSB (most significant bit); A[y] = row address 2nd MSB (second most significant bit)																											
AddrHashBank n[4:0]:		4	3	2	1	0	{Green, Blue, Violet, Red, Orange}																				

Figure 49: DDR5 CPU Bank Hash DRAM Address Map for Two Channels per UMC

Note:

- For mixed-rank populations, BIOS programs this algorithm based on the minimum common number of row bit and bank bits enabled.
- For non power-of-two DRAMs, BIOS excludes the two most-significant row address bits from this algorithm.

12.2.3.10 DDR5 Address Hash CS

A BIOS algorithm for programming the UMC::AddrHashCS feature is as follows:

- Chip select interleaving with 2-way or 4-way interleaving across DIMMs with matched ranks is a prerequisite to enable address hashing with chip select.

UMC::AddrHashExtCS is used for configurations with extended addressing to specify normalized bits [47:40].

BIOS programs the function to enable the hashing mode as follows:

1. Configure XorEn per the following table.
2. Configure NormAddrXor such that the bits corresponding to the Row bits are set to ones. For two-way interleaving, set both even and odd bits. For four-way interleaving, set the even bits in the "_n0" register and the odd bits in the "_n1" register.

Note on Row bits:

- For CPU Address Map: use the first eight Row bits.

Table 213: Address Hash CS XorEn Settings

UMC::BaseAddr[CSEnable]				UMC::AddrHashCS[XorEn]	
CS0	CS1	CS2	CS3	_n0	_n1
1	1	0	0	1	0
0	0	1	1	1	0
1	0	1	0	0	1
1	1	1	1	1	1

Note:

- For non power-of-two DRAMs, BIOS excludes the two most-significant row address bits from this algorithm.
- Address hashing for chip select is not supported for the 3-rank interleaving case.

12.2.3.11 DDR5 Address Hash RM

A BIOS algorithm for programming the UMC::AddrHashRM feature is as follows:

- Rank multiply addressing is a prerequisite to enable address hashing for the rank multiply bit.

UMC::AddrHashRMExt is used for configurations with extended addressing to specify normalized bits [47:40].

BIOS programs the function to enable the hashing mode as follows:

- Disable AddrHashRM.

12.2.3.12 DDR5 Address Hash Subchannel

A BIOS algorithm for programming the UMC::AddrHashPC and UMC::AddrHashPC2 feature is as follows:

- The CH bit selection with UMC::BeqCtrl0[VcmEn]=1 is a prerequisite to enable address hashing on the channel bit.

BIOS programs the function to enable the hashing mode as follows:

- Row Address: Program all Row bits.
- Column Address: No Column bits.
- Bank Address: No BG or BA bits.

Note:

- For mixed-rank populations, BIOS programs this algorithm based on the minimum common number of row bit and bank bits enabled.
- For non power-of-two DRAMs, BIOS excludes the two most-significant row address bits from this algorithm.

12.2.4 DDR5 DRAM Data Burst Mapping

DRAM requests are mapped to data bursts on the DDR bus in the following order:

When (UMC::EccCtrl[EccBitInterleaving] == 0), a 64B request is mapped one double-word ([DW](#)) per each sequential data beat (16 data beats): DW0, DW1, ... DW15. (See example below.)

When (UMC::EccCtrl[EccBitInterleaving] == 1), a 64B request is mapped similarly except that bits are interleaved on the DRAM bus as follows:

- For x4 and x16 Symbols, bytes in the cache line are mapped:
 - Even bits map to DW0, DW2, DW4, DW6, DW8, ... DW14 on the DDR bus
 - Odd bits map to DW1, DW3, DW5, DW7, DW9, ... DW15 on the DDR bus

The following table shows an example of Full ECC EccBitInterleaving for x4 and x16 Symbols:

Table 214: Example Data Mapping: Full ECC, DDR Bus Nibble[0] Lane[3:0]

DW	EccBitInterleaving= 0	X4 EccBitInterleaving= 1	X16 EccBitInterleaving= 1
0	Data[3:0]	Data[6,4,2,0]	Data[12,8,4,0]
1	Data[35:32]	Data[7,5,3,1]	Data[13,9,5,1]
2	Data[67:64]	Data[70,68,66,64]	Data[14,10,6,2]
3	Data[99:96]	Data[71,69,67,65]	Data[15,11,7,3]
4	Data[131:128]	Data[134,132,130,128]	Data[140,136,132,128]
5	Data[163:160]	Data[135,133,131,129]	Data[141,137,133,129]
6	Data[195:192]	Data[198,196,194,192]	Data[142,138,134,130]
7	Data[227:224]	Data[199,197,195,193]	Data[143,139,135,131]
8	Data[259:256]	Data[262,260,258,256]	Data[268,264,260,256]
9	Data[291:288]	Data[263,261,259,257]	Data[269,265,261,257]
10	Data[323:320]	Data[326,324,322,320]	Data[270,266,262,258]
11	Data[355:352]	Data[327,325,323,321]	Data[271,267,263,259]
12	Data[387:384]	Data[390,388,386,384]	Data[396,392,388,384]
13	Data[419:416]	Data[391,389,387,385]	Data[397,393,389,385]
14	Data[451:448]	Data[454,452,450,448]	Data[398,394,390,386]

15	Data[483:480]	Data[455,453,451,449]	Data[399,395,391,387]
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The following table shows an example of Pin-reduced ECC, EccBitInterleaving for x4 and x16 Symbols:

Table 215: Example Data Mapping: Pin-reduced ECC, DDR Bus Nibble[0] Lane[3:0]

DW	EccBitInterleaving= 0	X4 EccBitInterleaving= 1	X16 EccBitInterleaving= 1
0	Data[3:0]	Data[12,8,4,0]	Data[24,16,8,0]
1	Data[35:32]	Data[13,9,5,1]	Data[25,17,9,1]
2	Data[67:64]	Data[14,10,6,2]	Data[26,18,10,2]
3	Data[99:96]	Data[15,11,7,3]	Data[27,19,11,3]
4	Data[131:128]	Data[140,136,132,128]	Data[28,20,12,4]
5	Data[163:160]	Data[141,137,133,129]	Data[29,21,13,5]
6	Data[195:192]	Data[142,138,134,130]	Data[30,22,14,6]
7	Data[227:224]	Data[143,139,135,131]	Data[31,23,15,7]
8	Data[259:256]	Data[268,264,260,256]	Data[280,272,264,256]
9	Data[291:288]	Data[269,265,261,257]	Data[281,273,265,257]
10	Data[323:320]	Data[270,266,262,258]	Data[282,274,266,258]
11	Data[355:352]	Data[271,267,263,259]	Data[283,275,267,259]
12	Data[387:384]	Data[396,392,388,384]	Data[284,276,268,260]
13	Data[419:416]	Data[397,393,389,385]	Data[285,277,269,261]
14	Data[451:448]	Data[398,394,390,386]	Data[286,278,270,262]
15	Data[483:480]	Data[399,395,391,387]	Data[287,279,271,263]

12.2.5 DDR5 DRAM DQ Mapping

DQ interface mapping is determined during DDR PHY interface training for this SOC.

The DDR PHY is configured to map DDR PHY DQ lane to memory DQ as 1-to-1. UMC DQ mapping registers UMC::DQMapDW0 ... UMC::DQMapDW4, UMC::DimmCfg[DqMapSwapDis] are unused.

12.2.5.1 DDR5 DRAM DQ Swizzle Detect

If PhyCfg is set, DQ swizzle discovery is performed for all populated ranks. DQ swizzle is required information for

training firmware. When DQ swizzle discovery is enabled, the training firmware finds the board and DIMM swizzle, and sets accordingly the DQ*LnSel* CSRs at the end of training.

Note that the DQ*LnSel* CSR values used during training can differ from the final values programmed at the end of training. During training, firmware relies on the DQ swizzle information to correctly program the DQ-related MR on the DRAM side and the nibble-related MR on the data buffer's side in case of LRDIMM.

12.2.6 DDR5 Command and Address Bus Configuration

This section describes the settings required for programming the timing and drive settings on the command and address pins. These tables document recommended settings for motherboards which meet the relevant AMD motherboard design guidelines.

Notes on DDR5 DIMM format for this section: SR/DR/LR correspond to electrical loads.

Table 216: DDR5 RDIMM and 3DS RDIMM CA Bus Configuration

NumDimmSlots	DdrRate	VDDIO	DIMM0	DIMM1	2N	AddrCmndDrvStren	ClkDrvStren
1	2000	V1.1	SR+DR	NP	0		
1	3200	V1.1	SR+DR	NP	0		

1	3600	V1.1	SR+DR	NP	0		
1	4000	V1.1	SR+DR	NP	0		
1	4400	V1.1	SR+DR	NP	0		
1	4800	V1.1	SR+DR	NP	0		
2	2000	V1.1	NP	SR+DR	0		

2	3200	V1.1	SR+DR	NP	0		
2	3600	V1.1	SR+DR	SR+DR	0		
2	4000	V1.1	NP	SR+DR	0		
2	4400	V1.1	SR+DR	NP	0		
2	4800	V1.1	SR	SR	0		

12.2.7 DDR5 Data Bus Configuration

This section describes the settings required for programming the drive settings and slew rates on the data bus pins. The following tables document recommended settings for motherboards which meet the relevant AMD motherboard design guidelines.

The ODT mode of DDR5 has four different termination values: RTT_WR, RTT_NOM_RD, RTT_NOM_WR and RTT(Park). Values are expressed as an integer (ohms).

Notes on DDR5 DIMM format for this section: SR/DR/LR correspond to electrical loads.

Table 217: DDR5 RDIMM Data Bus Configuration

NumDimmSlots	DdrRate	VDDIO	DIMM0	DIMM1	RTT_WR	RTT_NOM_RD	RTT_NOM_WR	RTT_PARK	DataDrvStren	DqsDrvStren	ProcOdt	PMUPhyVref	VrefDQ
1	3200+ 3600+ 4000+ 4400+ 4800+	1.1V	SR	NP	48								
1	3200+ 3600+ 4000+ 4400+ 4800+	1.1V	DR	NP	120	60	60	60					
2	3200+ 3600+ 4000+ 4400+ 4800+	1.1	NP	SR	120	60	60	60					

2	3200+ 3600+ 4000+ 4400+ 4800+	1.1	NP	DR	120	60	60	60					
2	3200+ 3600+ 4000+ 4400+ 4800+	1.1	SR	NP	48								
2	3200+ 3600+ 4000+ 4400+ 4800+	1.1	DR	NP	120	60	60	60					
2	3200+ 3600+ 4000+ 4400+ 4800+	1.1	SR	SR	48	34	34	34					
2	3200+ 3600+ 4000+ 4400+ 4800+	1.1	DR	DR	80	120	80	60					

12.2.8 DDR5 Device Data Sheet Parameters

The JEDEC specification provides values for several DRAM data sheet timing parameters that are required for UMC operation. BIOS should use the optimal value specified by JEDEC.

- UMC::DramTiming5[Tcwl] = CL – 2
- UMC::DramTiming3[TrrdL]: tRRD_L
- UMC::DramTiming3[TrrdS]: tRRD_S
- UMC::DramTiming4[Tfaw]: tFAW
- UMC::DramTiming5[TwtrL]: tWTR_L
- UMC::DramTiming5[TwtrS]: tWTR_S
- UMC::DramTiming11[TzqOperCal]: tZQCAL
- UMC::SrsTiming4[TzqOperCal]: tZQCAL
- UMC::SrsTiming7[Tvref]: VrefDQ_time
- UMC::DramTiming21[Tcpded]: tCPDED
- UMC::DramTiming11[Tzqcs]: tZQLAT
- UMC::DramTiming17[Tcsh]: tCSH
- UMC::SrsTiming0[Tcksrx]: tCKSRX
 - Additional UMC staggering: Tcksrx += (Trfc/N * (UMC_Number % 3)) where N in {1 . . . 9}.
- UMC::DramTiming14[Txs]: tXS
- UMC::SrsTiming1[Txs]: tXS
- UMC::DramTiming17[Tpd]: tPD
- UMC::DramTiming21[Txp]: tXP
- UMC::DramTiming16[Tmrr]: tMRR
- UMC::DramTiming16[Tmrw]: tMRW
- UMC::RefTiming[Trefsbrd]: tREFSBRD

12.2.9 DDR5 UMC DRAM Command Spacing

Optimal values for the following registers are platform and configuration specific. After DRAM training BIOS uses data returned in the PMU SRAM Message Block to optimally configure the registers.

The following terms are defined to simplify calculations:

- Latency Difference (LD) = UMC::DramTiming1[Tcl] – UMC::DramTiming5[Tcwl].
- Use BL16 for DDR5 calculations in this section.
- LVC = BL/2-1 = 16/2-1 = 7. UMC logic adjusts internally if CRC is enabled.
- CDD: DDR training PMU SRAM Message Block critical delay difference for this transaction type.
- Tphy_rdcsgap = [MAX](#)[0, 4 + BL/2 + CDD].
- Tphy_wrcsgap = BL/2 + [MAX](#)[0, 3 + CDD + 2].
- ODTLonWrOffset = [MIN](#) of DDR5 MR37.ODTLon_WR_Offset and DDR5 MR38.ODTLon_WR_NT_Offset enabled modes.
- ODTLoffWrOffset = [MAX](#) of DDR5 MR37.ODTLoff_WR_Offset and DDR5 MR38.ODTLoff_WR_NT_Offset enabled modes.
- ODTLonRdOffset = [MIN](#) of DDR5 MR39.ODTLon_RD_NT_Offset enabled modes.
- ODTLoffRdOffset = [MAX](#) of DDR5 MR39.ODTLoff_RD_NT_Offset enabled modes.
- For ODTLonWrOffset, ODTLoffWrOffset, ODTLonRdOffset, and ODTLoffRdOffset calculations: assume 0 MEMCLKs if the ODT mode is not in use.
- ADC = [CEILING](#)(tADC.max).
- AnyRankIs3DS = Any (UMC::DimmCfg[Dram3DS] > 0).

12.2.9.1 Read to Read

UMC::DramTiming8

- TrdrdBan = 0.
- TrdrdSc = 1.
- TrdrdScL = JEDEC tCCD_L – LVC.
- TrdrdSd = RdDrift + Tphy_rdcsgap + (UMC::DramTiming41[RdPre] MEMCLKs - 1) +

(UMC::DramTiming41[RdPost] MEMCLKs – 0.5) – LVC.

- CDD calculated using [MAX](#)(0, CS[1:0] to CS[1:0], CS[3:2] to CS[3:2]).
- $TrdrdDd = RdDrift + Tphy_rdcsgap + (UMC::DramTiming41[RdPre] MEMCLKs - 1) + (UMC::DramTiming41[RdPost] MEMCLKs - 0.5) - LVC$.
 - CDD calculated using [MAX](#)(0, CS[3:0] to CS[3:0]).
- For TrdrdSd and TrdrdDd, use the MAX of the equations above and the following TrdrdOdt:
 - $TrdrdOdt = MAX(1, BL/2 + ODTLoffRdOffset - ODTLonRdOffset + ADC - LVC)$.
- $TrdrdScDlr = IF (AnyDram3DS) THEN (JEDEC tCCD_dlr - LVC) ELSE 0 ENDIF$.

Note:

- If a single CS or single CS on each DIMM is populated, BIOS should program TrdrdSd equal to MAX(TrdrdSc, TrdrdScL).
- If a single DIMM is populated, BIOS should program TrdrdDd equal to TrdrdSd.

12.2.9.2 Write to Write

UMC::DramTiming9

- $TwrrBan = 0$.
- $TwrrSc = 1$.
- $UMC::DramTiming9[TwrrScL] = ((UMC::BeqCtrl2[DisDatMsk] \&\& UMC::UmcConfig[BurstCtrl] \&\& ! Any(UMC::DimmCfg[X4Dram] == 1)) ? JEDEC tCCD_L_WR2 : JEDEC tCCD_L_WR) - LVC$.
- $UMC::DramTiming56[TwrrScL2] = JEDEC tCCD_L_WR2 - LVC$.
- $TwrrSd = Tphy_wrcsgap + (UMC::DramTiming41[WrPre] MEMCLKs - 2) + (UMC::DramTiming41[WrPost] MEMCLKs - 0.5) - LVC$.
 - CDD calculated using [MAX](#)(0, CS[1:0] to CS[1:0], CS[3:2] to CS[3:2]).
- $TwrrDd = Tphy_wrcsgap + (UMC::DramTiming41[WrPre] MEMCLKs - 2) + (UMC::DramTiming41[WrPost] MEMCLKs - 0.5) - LVC$.
 - CDD calculated using [MAX](#)(0, CS[3:0] to CS[3:0]).
- For TwrrSd and TwrrDd, use the MAX of the equations above and the following TwrrOdt:
 - $TwrrOdt = MAX(1, BL/2 + ODTLoffWrOffset - ODTLonWrOffset + ADC - LVC)$
- $TwrrScDlr = IF (AnyDram3DS) THEN (JEDEC tCCD_WR_dlr - LVC) ELSE 0 ENDIF$.

Note:

- If a single CS or single CS on each DIMM is populated, BIOS should program TwrrSd equal to TwrrSc.
- If a single DIMM is populated, BIOS should program TwrrDd equal to TwrrSd.

12.2.9.3 Write to Read

UMC::DramTiming10

- $Twrrd = Tphy_wrcsgap + (UMC::DramTiming41[WrPost] MEMCLKs - 0.5) + (UMC::DramTiming41[RdPre] MEMCLKs) - LD - LVC$.
 - CDD calculated using [MAX](#)(0, CS[3:0] to CS[3:0]).
- For Twrrd, use the MAX of the equation above and the following TwrrdOdt:
 - $TwrrdOdt = MAX(1, BL/2 - LD + ODTLoffWrOffset - ODTLonRdOffset + ADC - LVC)$.
- $TwrrdScDlr = IF (AnyDram3DS) THEN (JEDEC tCCD_WTR_dlr - LVC) ELSE 0 ENDIF$.

12.2.9.4 Read to Write

UMC::DramTiming10

- $Trdwr = Tphy_rdcsgap + LD + (UMC::DramTiming41[RdPost] MEMCLKs - 0.5) + (UMC::DramTiming41[WrPre] MEMCLKs) + ((UMC::DramTiming41[WrPre] MEMCLKs) > 2 ? 2 : 1) - LVC$.
 - CDD calculated using [MAX](#)(0, CS[3:0] to CS[3:0]).

- For Trdwr, use the MAX of the equation above and the following TrdwrOdt:
 - $\text{TrdwrOdt} = \text{MAX}(1, \text{BL}/2 + \text{LD} + \text{ODTLoRdOffset} - \text{ODTLonWrOffset} + \text{ADC} - \text{LVC})$.

12.2.9.5 Additional Programming

Program UMC::AeCtrl5[RdBrstGap] and UMC::AeCtrl5[WrBrstGap] based on DIMM population and values for memory P-State 0 as follows:

IF (1R)

RdBrstGap = tCCD_L – tCCD_S + 3

WrBrstGap = 9

ELSE IF (>1R)

RdBrstGap = MAX(TrdrdSd, TrdrdDd) + 1

WrBrstGap = MAX(TwrwrSd, TwrwrDd) + 1

ENDIF

12.2.10 DDR5 UMC DRAM Refresh Modes

The UMC supports the following DDR5 refresh modes:

- Normal Mode (REFab)
 - UMC::SpazCtrl[AllBankRefEn] = 1
 - UMC::SpazCtrl[PerBankRefEn] = 0
 - UMC::SpazCtrl[AutoRefFineGranMode] = "0h Fixed 1X"
 - DRAM MR4[Refresh tRFC Mode] = "0B: Normal Refresh Mode (tRFC1)"
- Fine Granularity Refresh (FGR) Mode (REFab)
 - UMC::SpazCtrl[AllBankRefEn] = 1
 - UMC::SpazCtrl[PerBankRefEn] = 0
 - UMC::SpazCtrl[AutoRefFineGranMode] = "1h Fixed 2X"
 - DRAM MR4[Refresh tRFC Mode] = "1B: Fine Granularity Refresh Mode (tRFC2)"
- Mixed Refresh Mode (REFab, REFsb)
 - UMC::SpazCtrl[AllBankRefEn] = 1
 - UMC::SpazCtrl[PerBankRefEn] = 1
 - UMC::SpazCtrl[AutoRefFineGranMode] = "1h Fixed 2X"
 - DRAM MR4[Refresh tRFC Mode] = "1B: Fine Granularity Refresh Mode (tRFC2)"

AMD recommends Mixed Mode for 1-rank and 2-rank RDIMM channel populations. AMD recommends Normal Mode for greater-than-2-rank RDIMM channel populations. For all 3DS channel populations, AMD recommends Mixed Mode.

12.2.11 DDR5 UMC RAS

This section describes the Reliability, Availability, and Serviceability (RAS) Features for DDR5 DRAM controlled by the UMC. For SOC-level descriptions of RAS features, see 3.7 [DRAM RAS Features].

12.2.11.1 DDR5 DRAM ECC

In DDR5 mode, the UMC supports x8 ECC, and x16 ECC. The steps to initialize DRAM ECC as part of DDR initialization are as follows. After DDR training and before [DE](#) Startup P-State:

- Program UMC::EccCtrl[EccSymbolSize16], UMC::EccCtrl[EccSymbolSize]. DDR5 mode: x8 symbol not supported.
 - AMD recommends x16 symbol: Program EccCtrl[EccSymbolSize16,EccSymbolSize] = {1, 0}.
- If RDIMM/LRDIMM 72b then UMC::EccCtrl[PinReducedEcc] = 1. If RDIMM/LRDIMM 80b then UMC::EccCtrl[PinReducedEcc] = 0.

3. Program UMC::EccCtrl[EccBitInterleaving] = (EccSymbolSize||EccSymbolSize16 ? 1 : 0).
4. Program UMC::EccCtrl[EccHardwareHistoryEn] = (UMC::EccCtrl[EccBadDramSymEn] = 0).
5. UMC::EccCtrl[WREccEn] = 1.
6. Perform Memory Clear. See 12.5 [UMC Memory Clear].
7. UMC::EccCtrl[RdEccEn] = 1.
8. Program UMC::EccCtrl[EccHardwareHistoryEn] = (UMC::EccCtrl[EccBadDramSymEn] = (EccSymbolSize||EccSymbolSize16 ? 1 : 0)).
9. Program ECC error counter (AMD recommended).
 1. UMC::EccErrCntCtrl[EccErrCntEn] = 1.
 2. If interrupt needs to be generated on error counter overflow, program UMC::EccErrCntCtrl[EccErrInt].
 3. If using the ECC Leaky Bucket Counter feature:
 1. Program UMC::EccErrCntCtrl[EccErrLeakEn] = 1.
 2. Program UMC::EccErrCntCtrl[EccErrLeakRate] to the desired rate.
 3. Program UMC::EccErrCnt[EccErrThresh] as the desired threshold where the leaky EccErrCnt stops decrementing.

12.2.11.2 DDR5 Command and Address Parity

Command Address Parity is supported only by the RCD on RDIMMs. Additional control word settings are required to enable parity before DramReady. PHY configuration for Alert_n requires additional termination. See 12.2 [DDR5 UMC Initialization].

Initialize CA parity as follows before DDR Training:

1. Program RCD parity. (Recommended default enabled.)
2. Program UMC::UmcConfig[BurstCtrl[1:0]] = 1 Fixed. Note: Firmware should manage MR0.BurstMode to match UMC BurstMode/BurstCtrl configuration.
3. BIOS disables 32Byte SDP transactions: (DF:CS:DisDramAcc32ByteMode = 1). See 12.2.11.5 [DDR5 RAS Recovery].

12.2.11.3 DDR5 RAS Write CRC

CRC Modes must be disabled for DDR Training and Post Package Repair (PPR). CRC Modes increase the BurstLength to 18, even for BurstChop8 commands.

Write CRC should not be configured on ECC DIMMs if ECC is not enabled. If DIMM is ECC-capable (UMC::UmcConfig[DimmEccEn] == 1) and Write ECC not enabled (UMC::EccCtrl[WREccEn] == 0), then do not enable Write CRC.

Write CRC requires additional DRAM per-rank per-SubChannel MR settings before DramReady.

Initialize Write CRC as follows before DDR Training:

1. Program UMC::BeqCtrl1[WRCrcEn] = 1.

See 12.2.11.5 [DDR5 RAS Recovery].

12.2.11.4 DDR5 RAS Read CRC

CRC Modes must be disabled for DDR Training and Post Package Repair. CRC Modes increase the BurstLength to 18, even for BurstChop8 and MRR commands.

Read CRC requires additional per-rank per-SubChannel DRAM MR settings before DramReady.

CRC should be enabled if Controller ECC is disabled on ECC-Capable DIMM.

Initialize Read CRC as follows before DDR Training:

1. Program UMC::BeqCtrl1[RdCrcEn].
2. Program UMC::BeqCtrl1[RdCrcFatalEn].

See 12.2.11.5 [DDR5 RAS Recovery].

12.2.11.5 DDR5 RAS Recovery

In this section, UclkDiv1Mode refers to the value of

UMC::ClkDivCtrl[UclkDiv1Mode_M0,UclkDiv1Mode_M1,UclkDiv1Mode_M2,UclkDiv1Mode_M3] for the current memory P-State context.

Write CRC recovery may only be configured if parity recovery is also enabled.

Initialize the following recovery controls after DDR training and before [DF](#) Startup P-State Change. Additional settings are required before DramReady:

- UMC::RecTiming[RcvrWaitBus] = 0
- UMC::RecTiming[CmdStgFrc] = 1
- UMC::DramTiming19_mp[3:0]
 - UMC::DramTiming19[AlertCrcPulse] = (UclkDiv1Mode ? 1 : 2)
- UMC::RecCtrl1[CtrlUpdHold] = 28
- UMC::DramTiming15_mp[3:0]
 - UMC::DramTiming15[AlertParDly] = (UclkDiv1Mode ? 6 : 8) + (2DPC ? 3.0ns,2.0ns) + RCD
Max(tSTAOFF+ tALERT_HL) + 3 + 12
 - UMC::DramTiming15[AlertCrcDly] = UMC::ChanPipeDly[TxCtrlChnDly]+
UMC::ChanPipeDly[TxDatChnDly]+ 4 + UMC::DramTiming5[Tcwl] + (DRAM tCRC_ALERT max) +
Rcd [MAX](#)(tSTAOFF+tALERT) + (2DPC ? : 2.5ns,3.5ns) + 9 + 8 + 4. Constraint:
UMC::DramTiming15[AlertCrcDly] >= Tcwl + 40 + ((Memclk-800)/100).
- UMC::DramTiming35[Twrmp] = MAX(0, JEDEC tWR_MPR – UMC::DramTiming13[Tmod]).
- UMC::DramTiming35[CmdStgCnt] = 5
- UMC::DramTiming35[RcvrWait] = MAX(UMC::DramTiming11[Tzqcs],
IF(UMC::SpazCtrl[AutoRefFineGranMode] == 1) THEN (UMC::DramTiming14[Tdllk]-
UMC::DramTiming14[Txs])).

Adjust the following configuration settings based on recovery mode:

- UMC::DramTiming15[RankBusyDly] = If (UMC::RecCtrl[RecEn] > 0) THEN (MAX(EccRdLatency,
UMC::DramTiming15[AlertParDly], UMC::DramTiming15[AlertCrcDly]) + 1) ELSE 0 ENDIF. EccRdLatency =
UMC::DramTiming22[Trddata_en] + Tphy_rldat + (16 * ((UclkDiv1Mode==1) ? 1 : 2)).
- UMC::RecCtrl2[MaxEccRply] = Default 8. User controlled.
- UMC::RecCtrl2[MaxRdCrcRply] = Default 8. User controlled.
- UMC::RecCtrl[MaxCrcRply] = Default 8. User controlled Num Wrcrc Retries.
- UMC::RecCtrl[MaxParRply] = Default 8 – user controlled Num Parity Retries.
- UMC::RecCtrl[EccOptEn] = 0.
- UMC::RecCtrl[RplySpazReq] = 1.
- UMC::RecCtrl[BlockZqRply] = 0.
- UMC::RecCtrl[RcvrMode[0]] = 0. Pulse.
- UMC::RecCtrl[RcvrMode[2]] = ~UMC::BeqCtrl1[WrcrcEn].
- UMC::RecCtrl[BusyChk] = 0
- UMC::RecCtrl[RecEn[1]] = UMC::BeqCtrl1[WrcrcEn].
- UMC::RecCtrl[RecEn[2]] = UMC::EccCtrl[WrcrcEn].
- UMC::RecCtrl[RecEn[3]] = UMC::BeqCtrl1[RdCrcEn].

- IF UMC::RecCtrl[RecEn] > 0 THEN UMC::DramScrubCtrl[DisScrubUcRply] = 1

Before DramReady, perform final steps to enable Recovery as needed.

12.2.12 DDR5 Additional UMC Configuration Dependencies

12.2.12.1 DDR5 UMC and DDR Phy Configuration

The following subsections provide UMC programming guidance for parameters that are dependent on the configuration of the DDR Phy or the DFI channel.

12.2.12.1.1 DDR5 DFI Channel Stages

The DFI channel stage settings are programmed as follows:

- UMC::ChanPipeDly[RxDatChnDly, TxDatChnDly, TxCtrlChnDly] = {2, 3, 1}.

12.2.12.1.2 DDR5 MRL

The DDR PHY Interface Max Read Latency (DFIMRL) is the value returned by the PMU firmware after memory training for each memory P-State.

After training, for each memory P-State:

- Set the PHY CSR DFIMRL for all DBYTE instances (UMC::Phy::DFIMRL) to the maximum trained DFIMRL value across all DBYTEs connected to a UMC.

The UMC expects a common DFIMRL across both channels of a DDR5 module when UMC::BeqCtrl0[VcmEn] == 1.

12.2.13 DDR5 DDR Phy Initialization

The following are the DDR phy operational clocking requirements:

- DFICLK frequency = 1/2 MEMCLK frequency

12.2.13.1 DDR5 DDR PMU

The processor includes a Phy Micro-controller Unit (PMU) that trains the DDR interface during boot.

BIOS communicates with the PMU using the following methods:

- Mailbox: BIOS waits for upstream messages from the PMU for action synchronization or for status messages.
- SRAM Message Block: BIOS may read or write the memory used by the PMU to send or receive complex message data.

12.2.14 DDR5 Device Initialization and Training

The processor includes a Phy Micro-controller Unit (PMU) that initializes and trains DDR5 devices.

12.2.14.1 DRAM Initialization

DRAM, RCD, and DB initialization is performed by PMU firmware.

- If 1R RDIMM configuration, program RCD RW09[2] == 1.

12.2.14.2 DRAM Training

DRAM and DB data bus training is performed by PMU firmware.

12.2.15 DDR5 Memory Channel Disable

UMC supports disabling of a memory channel that is unpopulated or has a failure.

If `UMC::UmcCap[MemChanDis] == 1`, disable unused UMC channel:

1. Program `UMC::SdpCtrl[SdpInit] = 0` // disables the SDP port for unused UMC channels
2. Program DDR phy to LP3 mode:
 1. Program `UMC::Phy::MicroContMuxSel = 1`
 2. Initiate the phy state change. Perform one write to `UMC::Phy::SequencerOverride` as follows:
 1. [4:0] `ForceSeq0BDfiFreq = 1Fh`
 2. [5] `ForceSeq0BStart = 1`
 3. [6] `ForceSeq0BStop = 0`
 4. [7] `BlockSeq0BRequests = 1`
 5. [8] `BlockSeq0BAck = 1`
 6. [9] `DisableTerminateFlag = 0`
 7. [10] `SelectDFIFreqToGPRMux = 1`
 3. Wait 10us.
 1. Program `UMC::Phy::MicroContMuxSel = 0`
 2. Program `UMC::Phy::SequencerOverride[5] = 0` // `ForceSeq0BStart`
3. Program `UMC::ClkDivCtrl[DFICLK_SCGEEn,UCLK_SCGEEn,FCLK_SCGEEn] = {1, 1, 1}`.

Table 218: Encoding Of `UMC::Phy::SequencerOverride[10:0]`

Bits	Description
[4:0]	ForceSeq0BDfiFreq
[5]	ForceSeq0BStart
[6]	ForceSeq0BStop
[7]	BlockSeq0BRequests
[8]	BlockSeq0BAck
[9]	DisableTerminateFlag
[10]	SelectDFIFreqToGPRMux

12.2.16 DDR5 DRAM MBIST PPR

DDR5 DRAM devices can support optional Memory Built-In Self Test (MBIST) and Memory Built-In Self Test-Post Package Repair (mPPR) to help with hard failures such as single-bit or multi-bit failures in a single device so that weak cells can be scanned and repaired during the initialization phase.

This SOC does not support the DRAM MBIST PPR (mPPR) feature.

12.2.17 DDR5 UMC Post Package Repair

The DDR5 DRAM post package repair (PPR) feature provides a way to repair a failing DDR DRAM device row by swapping the failed row with a functioning reserved row.

DRAM devices recognize two methods:

- Hard Post Package Repair (hPPR) for a permanent Row repair, and
- Soft Post Package Repair (sPPR) for a temporary Row repair. If DRAM power is removed or the DRAM is RESET, the soft repair will revert to the un-repaired state.

hPPR and sPPR may not be enabled at the same time.

12.2.17.1 DDR5 UMC Boot Time PPR

The UMC supports static DDR5 post-package repair (PPR). This occurs at boot time. Static mode requires a system re-boot in order to replace the "new" device row.

Note: If 3DS, program DRAM device CID field (MR14:OP[3:0]) = 0.

The sequence for Post Package Repair (PPR) is as follows:

1. Configure prerequisites for DRAM PPR Mode.
 - a) Disable Write CRC.
2. Write a known data pattern to the full page of the repair target rank, bank, and row.
 - a) Save bit-level mis-compares for later use.
3. Program SWCLIMSTR to configure DRAM MR for PPR Mode:
 - a) Enable either Hard (hPPR) or Soft (sPPR) Post Package Repair by programming appropriate JEDEC MR Settings Key and MR Guard Key sequence.
 1. MR23 – PPR Settings
 - OP[0] = 1 // sets hPPR enabled
 - OP[1] = 1 // sets sPPR enabled
 2. Wait tMRD
 3. MR24 OP[7:0] – PPR Guard Key. Separate each MRW command by tMRD.
 - Seq1: CFh
 - Seq2: 73h
 - Seq3: BBh
 - Seq4: 3Bh
 - b) Issue ACT command: Row and Bank of failed row to be replaced.
 - c) Wait tRCD.
 - d) Issue WR command: Column and DQ of device to be selected for repair.
4. Wait for SWCLIMSTR to complete.
5. Configure the Post Package Repair
 - a) Program UMC::UmcBistCmdCntl[CmdTestEnable] = 1.
 - b) Send ACT command with target rank, bank, and row address.
 - c) Send WR command. Write data is used to select the individual DRAM in the rank for repair as follows:
 1. Set UMC::UmcBistCmdCntl[DataPatGenSel] = 10b. // User [Configurable](#) Data Pattern
 2. Set all instances of UMC::UserDataPattern0 = UMC::UserDataPattern1 = UMC::UserDataPattern2 = UMC::UserDataPattern3 = FFFF_FFFFh // all ones
 3. Set UMC::DQPatternOverride2[PatOvrVal] = 0.
 4. Set UMC::DQPatternOverride1[DQPatOvrEn] = 1 for all DQ lanes mapping to the desired repair device.
 - d) Send PRE command.
 - e) Program UMC::UmcBistCmdCntl[CmdTestEnable] = 0.
6. Disable Post Package Repair mode.
7. Read and validate the repair excluding saved bad bits (from step 2).
8. Once PPR mode is exited, to confirm if target row is repaired correctly, the host can verify the repair by writing data into the target row and reading it back.
9. Re-enable Write CRC if requested for mission-mode.

12.2.18 DDR5 UMC Interface Testing

The UMC contains logic to support DDR interface testing. BIOS may generate patterns in addition to those normally used for DRAM training, leveraging hardware and architectural awareness to efficiently interrogate the memory interface, detect errors, and potentially identify failing components.

The AMD BIOS Memory Test (DDR Interface MBIST) provides the following routines:

- Interface Connectivity Test:
 - Checks basic DDR address and data bus connectivity.
- Interface Data Bus Test:
 - Exercises the DDR data bus with a rich set of data patterns and traffic scenarios.

12.2.19 DDR5 Periodic Phase Training (PPT)

After DramReady:

- Send [MP1](#) message to enable periodic retrain only mode.
- Send MP1 message to program periodic interval (default 1,000ms).

12.2.20 DDR5 DRAM On-die ECC Error Check and Scrub (ECS)

The purpose of on-die ECC is to improve data integrity by detecting and correcting single-bit errors: Single Error Correction (SEC) ECC. The ECC Error Check and Scrub ([ECS](#)) feature on DRAM devices (refer to the JEDEC specification) allows the UMC to initiate, control, and provide status of the errors detected within the DRAM devices.

The UMC:

- Writes to the DRAM device.
 - The DRAM will correct any single-bit errors that result from the internal read before merging the incoming write data and then re-compute 8 ECC On-die Check bits. DDR5 devices that are x8 and x16 do not require internal read.
- Reads from the DRAM device.
- Performs internal "scrub" function where the device is internally performing periodic row "checks".

If on-die ECS is used, firmware resets the ECS Error counts through SWCLIMSTR during initialization.

It is important to note that for normal DRAM device write and read accesses the DRAM device does NOT log detected and corrected errors. This means that writes with single-bit errors will be detected and corrected and written into the array without logging for the SOC status. Reads with detected single-bit errors will be corrected and returned to the SOC but do NOT log the detection and correction nor write the corrected value back to the array.

12.2.20.1 ECS Modes

The UMC supports both Manual and Automatic mode [ECS](#) in DDR5: UMC::PETCtrl[ECSMode].

In ECS manual mode, the controller sends the Multi-Purpose Command (MPC) for ECS periodically (based on ECS scrub interval UMC::PETCtrl[tECSint] – varies by DRAM density) to each rank and SubChannel which requires a delay of UMC::DramTiming42[tECSc] MEMCLKs (~110ns) for each MPC.

In ECS Automatic mode, the controller must send periodic REFab commands or periodically enter Self Refresh mode. At the completion of all Row ECS scrub the UMC does a MRR MR16-20 from each Dram Device, calculates the DeviceID with the worst-case Error Count, and logs the failing Device ID and Failing ROW/BankGroup/BankAddress into the ECSStatus array.

ECS specifically refers to the scrub function and the resulting detection logging. It is only during ECS where on-die device errors are logged and MR16-20 updated for the DRAM devices on-die error transparency. The UMC will issue "manual" ECS commands via MPC commands (operation code: 0000_1100b) when the DRAM device is configured in "manual" mode.

The default is "automatic" mode. When "manual" mode is disabled, the UMC issues "automatic" ECS commands.

ECS is enabled in automatic mode by default when MR14 OP[7] = 0b and in manual mode when MR14 OP[7] = 1b. The entire DRAM array is initialized before ECS execution. If UMC ECC is enabled then ECS can rely on the fact that BIOS has initialized DRAM for "parallel" ECC purposes. However, if for some reason "parallel" ECC is not enabled in the SOC then BIOS is expected to initialize all DRAM locations purely for the purpose of ECS – if it is to be enabled.

In ECS auto mode the UMC will not issue MPC scrub commands but will still monitor TermCnt and pull ECS information from the devices when terminal count is reached.

The entire ECS RMW (Read-Modify-Write) operation takes tESCc. The tESCc time starts from the time the MPC manual ECS command is sent and no further commands can be issued after this MPC command is sent for the duration of tESCc. All rows must be in the idle state so a precharge precedes the ECS command.

12.2.20.2 ECS Counters

For transparency (observability), DRAM devices support 2 registers per die:

- EC (Error Count). Based on CFG setting MR14 OP[5], specifies either:
 - the number of rows with at least 1 error, or
 - the total number of code word errors.
- EpRC (Errors Per Row Count). Specifies the maximum error count of all the rows (Row address is provided).

When the controller reaches the 24 hours point, it performs an MRR MR16-20 from each DRAM Device, calculates the DeviceID with the worst-case Error Count, and logs the failing Device ID and Failing ROW/BankGroup/BankAddress into the [ECS](#) Status array.

The above counts are subject to the "thresholding" filter configuration as specified in MR15.

12.2.20.3 ECS Status Array

UMC::ECSSStatusSel – [ECS](#) Status Select register

- AutoInc[31] - Index read/write auto increment enable. (0=Disable. 1=Enable.)
- [30:6] - Reserved
- Index[5:0] - Index for accessing ECS status registers.
 - [1:0] Selects the register type:
 - 0x0 – Error Count Status.
 - 0x1 – Error Row Address.
 - 0x2 – ECS Transaction Count (TransCnt).
 - 0x3 – Reserved. Note: If ECSSStatusSel.Index[1:0] is set to 0x3, ECSSStatus returns 32'h0000_00000.
 - [3:2] Selects chip select (3:0).
 - [4] Selects SubChannel (0=A. 1=B.).
 - [5] Reserved. Note: When AutoInc enabled, ECSSStatusSel.Index automatically increments after the ECSSStatus register is read. If ECSSStatusSel.Index[5] is set to 0x1, ECSSStatus returns 32'h0000_00000.

UMC::ECSSStatus – ECS Status register

- When Index[1:0] = 00b (Error Count Status), ECS Status data per-subchannel and per-rank:
 - [7:0] – Errors Per Row Count (EpRC) from MR19 [5:0].
 - [11:8] – Device ID of x4 DRAM Device with highest EpRC error count. Lowest Device ID in case of a tie.
 - [11:9] – Device ID of x8 DRAM Device with highest EpRC error count. Lowest Device ID in case of a tie. Bit 8 not used.
 - [11:10] – Device ID of x16 DRAM device with highest EpRC error count. Lowest Device ID in case of a tie. Bits 9,8 not used.
 - [15:12] – CID (Only applies to 3DS). CID of highest error count.
 - [23:16] – MR20 [7:0] Error count
 - [27:24] – Device ID of Dram device with highest MR20 Error count.
 - [27:24] – Device ID of x4 DRAM device with highest MR20 Error count.
 - [27:25] – Device ID of x8 DRAM device with highest MR20 Error count.
 - [27:26] – Device ID of x16 DRAM device with highest MR20 Error count.
- When Index[1:0] = 01b (Error Row Address), ECS Status data per-subchannel and per-rank:
 - [17:0] – Row address (derived from MR16-18).
 - [19:18] – BankAddress (derived from MR16-18).
 - [22:20] – BankGroup (derived from MR16-18).
- When Index[1:0] = 10b (ECS TransCnt), ECS Status data per-subchannel and per-rank:
 - ECSTransCnt[31:0] – Only used for Z-states save/restore. The number of MPCs since the last termCnt expiration.

12.2.20.4 ECS Registers

See [ECS](#) registers UMC::PETCtrl, UMC::ECSTermCnt, UMC::DramTiming42.

The above registers reflect the worse case of the EC and EpRC (plus ROW) for each chip select. The UMC does not provide individual thresholding configurations but rather relies on the DRAM devices thresholding configuration. When the UMC "pulls" the status from all devices the worse non-zero EC and/or EpRC value from the log would initiate an [MCA](#) error. There is a separate MCA error source for ECS that signals as correctable.

The MCA error reporting occurs at a chip-select granularity, reporting the chip select and chip I.D. when a 3DS device detected. The UMC still supports "parallel" ECC – which is the separate 4/8-bit parallel ECC data store – but in general because on-die ECC corrects single-bit errors in the device, parallel ECC single-bit errors do not appear. This implies that parallel ECC logs multi-bit errors only (the exception being transient errors). In the event of Replay, the UMC does not "back-count" the ECS counter due to replay. In other words, the UMC counts only ECS commands it issues, and not updated counts that include replayed ECS commands.

The ECS address and counters are reset to a known value upon device reset of the writing of MR14 OP[6] = 1b (which is a rising edge detect – thus this bit would need to be written to a 0b afterwards before a following MR14 OP[6] write to a 1 to cause another reset.)

The UMC tracks the number of manual ECS operations to complete a full scrub of a device. The total number of manual ECS commands issued is device density specific. The UMC is aware of the array boundary – thus it is aware when a complete array is scrubbed. The UMC will manage an individual ECS scrub terminal count for each DRAM rank – which is inclusive of multiple chip selects and 3DS.

It is only upon the scrub completion of the array that the EC and EPRC counter values are transferred to the MR16-MR20 registers for UMC to query.

ECS supports 3DS stacking by utilizing the CID (Chip I.D.) field of MR14 to specify the die being accessed during the MRR command. During MRW commands the CID field will be ignored. The CID[3:0] bits must be set for MRR commands to MR14-MR20 to read out the data from the target die in the 3DS stack. These same CID[3:0] bits will also

be used by the Manual ECS MPC command. For single-die packages, the CID[3:0] bits should all be set to '0'. Mode register configuration and readout of mode register data requires PDA mode. Broadcasting of the Manual ECS MPC command to all die in the stack is not supported. The Manual ECS MPC command to command spacing requires waiting tECSc, even to different die in the stack. No other commands should be entered during the ECS operation time tECSc.

12.2.21 DDR5 Thermal Management and Bandwidth Capping

The UMC can be configured to increase the DRAM refresh rate and/or command throttle to reduce the command and data bus bandwidth as part of the DDR data integrity and thermal solution.

The UMC logic supports the following event types:

- Thermal Management Event:
 - UMC Periodic Thermal Polling of DRAM MR4
 - Sideband DIMM Temperature Sensor Polling
- Bandwidth Capping Event:
 - BIOS or [APML](#) based configuration of UMC::SwCmdThrottle

System thermal solutions should be designed per the DRAM device operating temperature limits to properly maintain data integrity. See DDR5 JEDEC SPD Byte 233: Operating Temperature Range.

UMC Periodic Thermal Polling:

The recommended BIOS configuration for the thermal polling feature is as follows:

- UMC::DramTiming12[OdsReadInterval] = 32 ms.
- UMC::MiscCfg[SpazMrrReadEn] = 0.
- UMC::MiscCfg[ThrottleThresh] = {3..5}. // option based on platform
- UMC::ThrottleCtrl[OdsEn] = 1.

Refresh Rate Configuration:

- UMC::ThrottleCtrl[OdsIncRefEn] programming:
 - Set UMC::ThrottleCtrl[OdsIncRefEn] = 1 to allow a dynamic change from 1x to 2x refresh rate during thermal events.
 - Set UMC::ThrottleCtrl[OdsIncRefEn] = 0 if UMC::DramTiming12[Tref] is programmed for a static JEDEC 2x refresh rate.
- Refer to DDR5 JEDEC defined temperature range options:
 - 1x Refresh Rate - temperature <= 85 °C.
 - 2x Refresh Rate - temperature > 85 °C.

Command Throttle Configuration:

- Program UMC::ThrottleCtrl[OdsCmdThrotCyc] based on the desired throttling rate to employ when a thermal event occurs. The actual throttling rate is workload dependent.
- Recommended settings are chosen as follows based on a desired integer ThrottlePercent rate reduction from 1 to 95:
 - Program UMC::ThrottleCtrl[RollWindowDepth] = 0x1FF.
 - Calculate: CmdThrotCyc = INT(((100-ThrottlePercent)*RollWindowDepth*2)/100).
 - Program UMC::ThrottleCtrl[OdsCmdThrotEn,OdsCmdThrotCyc] = {1, CmdThrotCyc}.
- UMC::ThrottleCtrl[OdsThrottleStatus] indicates active command throttling. Allow the DRAM devices to cool to return to full DDR bandwidth.

Sideband DIMM Temperature Sensor Polling:

In addition to the UMC Periodic Thermal Polling, platforms may choose to implement monitoring of the DIMM module temperature sensors by using the sideband I2C/I3C bus. The relationship between the DRAM case temperature, trip point, and the thermal event sampling interval is outlined as follows:

- The trip point for each DIMM is ordinarily configured to the case temperature specification minus a guard band

temperature for the DIMM.

- The temperature guard band is platform and vendor defined and is used to account for sensor inaccuracy, thermal event sample interval, and platform thermal design.
- The sampling interval is platform defined, expected to be in milliseconds range.
- The system temperature gradient, read interval, response delay, and hysteresis are platform defined .
- The thermal policy action is platform dependent. For example: an adjustment of the system fan policy in an attempt to prevent a thermal event.

Bandwidth Capping:

BIOS may enable bandwidth capping on a DRAM controller.

- Program UMC::ThrottleCtrl[RollWindowDepth] per guidance above.
- Program UMC::SwCmdThrottle[SwCmdThrotEn,SwCmdThrotCyc] = {1, CmdThrotCyc}.
 - See CmdThrotCyc calculation guidance above.
- Program UMC::SwCmdThrottle[SwCmdThrotEn] = 0 to return to full DDR bandwidth.
- The UMC employs the [MIN](#)(UMC::SwCmdThrottle[SwCmdThrotCyc], UMC::ThrottleCtrl[OdtsCmdThrotCyc]) cycle value if a DRAM thermal event is asserted when both modes are enabled.

DQ Byte Masking:

- For x16 devices only - Specifies the DQ byte for thermal polling threshold comparison.
- The DQ byte mapping for the platform would determine which DQ byte masks should be set.

Configuring DQ Byte Masking:

The UMC::ThermalPollByteMask register has eight 4-bit fields. Each bit in the field corresponds to one of the four DQ bytes.

- For x16 devices:
 - Byte-0 and Byte-2 are typically enabled (have valid MRR data) and Byte-1 and Byte-3 are masked off. This is done with a ThermByteMsk setting 0xA (1010).
 - If DQ bytes are swapped, Byte-1 and Byte-3 are enabled (have valid MRR data) and Byte-0 and Byte-2 would be masked off. This is done with a ThermByteMsk setting 0x5 (0101).
- For x4 and x8 devices, each DQ byte should have valid data and the ThermByteMsk fields are expected to be set to zero.

Table 219: UMC::ThermalPollByteMask Register - DQ Byte Mapping

Channel 0		Channel 1	
Field	Bits	Field	Bits
ThermByteMskCh0CS3	[15:12]	ThermByteMskCh1CS3	[31:28]
ThermByteMskCh0CS2	[11:8]	ThermByteMskCh1CS2	[27:24]
ThermByteMskCh0CS1	[7:4]	ThermByteMskCh1CS1	[23:20]
ThermByteMskCh0CS0	[3:0]	ThermByteMskCh1CS0	[19:16]

12.3 UMC Configuration Register Access

Each UMC contains an 8KB range of configuration registers accessed by SMN. The UMC configuration registers also provide an indirect index/data port access mechanism to the DDR PHY configuration space.

12.3.1 UMC Configuration Register Security

UMC configuration space is protected by a combination of RSMU security and local write-protect configuration bits.

RSMU security:

- Refer to individual registers for AccessType information based on InitiatorTrust levels.

The following bits disable write accesses to specified registers for all agents:

- UMC::MiscCfg[DisErrInj]
- UMC::MiscCfg[DisDebugBusSel]

12.3.2 DDR PHY Configuration Access

The following registers provide indirect index/data port access mechanism to the DDR PHY configuration space.

- UMC::ApbCmd
- UMC::ApbDat

The following paragraphs describe the supported access sequences.

12.3.2.1 Legacy DDR phy Configuration Access

Reads (without auto-increment):

1. Write the register number to ApbCmd[Address] with ApbCmd[Write, AutoInc] = {0,0}.
2. Wait until ApbStatus[Done] == 1.
3. Read the register contents from ApbDat[Data].

Reads (with auto-increment): Not supported.

Writes (without auto-increment):

1. Write all 32 bits of register data to ApbDat[Data].
2. Write the register number to ApbCmd[Address] with ApbCmd[Write, AutoInc] = {1,0}.
3. If (DisFastApbAccess == 1) wait until ApbStatus[Done] == 1.

Writes (with auto-increment):

1. Write all 32 bits of register data to ApbDat[Data].
2. Write the register number to ApbCmd[Address] with ApbCmd[Write, AutoInc] = {1,1}.
3. If (DisFastApbAccess == 1) wait until ApbStatus[Done] == 1.
4. Write all 32 bits of register data to ApbDat[Data]. Repeat steps 3 and 4 as needed for each additional register write.
5. Program ApbCmd[Write, AutoInc] = {0,0} to end the sequence.

Software should note the state of the DDR PHY MicroContMuxSel configuration bit which designates control of the phy internal configuration register bus.

12.4 UMC Data Encryption and Data Scrambling

12.4.1 UMC Data Encryption

The UMC data encryption feature is used as part of overall SOC solutions for main memory encryption. The data is encrypted in the UMC prior to being written to DRAM. The data is decrypted in the UMC when the data is returned on a DRAM read access.

BIOS recommended configuration is to enable the UMC data encryption hardware feature when available.

This feature is configured by a service call to the [ASP](#) during initialization.

BIOS configuration of the feature varies by BIOS options for:

- Transparent Secure Memory Encryption (TSME)
- AES-128 or AES-256 Mode

BIOS programs the following UMC registers:

- UMC::DataCtrl[DataEncrEn]
- UMC::DataCtrl[AES256En,AESXTSEn] for AES-256 mode
- UMC::DataCtrl[DisAddrTweak] for AES-128 mode
- UMC::EncrArrData per AES-128 or AES-256 mode
- UMC::TwkArrData per AES-128 mode
- UMC::XtsArrData per AES-256 mode

12.4.2 UMC Data Scrambling

The UMC data scramble feature reduces the likelihood of pathological bad signal integrity problems on the DDR interface. The data is scrambled in the UMC prior to being written to DRAM. The data is unscrambled in the UMC when the data is returned on a DRAM read access.

BIOS recommended configuration is to enable UMC data scramble.

BIOS programs the feature as follows:

- UMC::DataCtrl[DataScrambleEn] = 1.
- UMC::DataScrambleKey[DataScrambleKey]
- UMC::DataCtrl[DataScrambleEn] = (CbsCmnMemCtrllerDataScramble == "Enable") ? 1 : 0.

12.5 UMC Memory Clear

The UMC Memory Clear feature may be used to accelerate the initialization of DRAM contents during initial boot. The amount of time memory clear takes to complete depends on the DRAM capacity and the DRAM bandwidth of a UMC.

BIOS programs the region for the memory clear operation to the total DRAM memory size enabled on the UMC. BIOS executes the memory clear operation if ECC memory is populated or per BIOS option. It is recommended that BIOS accelerate the memory clear operation across all DDR channels of the SOC by initiating it on multiple UMCs in parallel, then polling for completion.

BIOS programs the following sequence to execute the feature:

1. Program the base and limit addresses into UMC::DramScrubBaseAddr and UMC::DramScrubLimitAddr.
2. Set UMC::DramScrubCtrl[MemClrEn] = 1.
3. Poll for UMC::DramScrubCtrl[MemClrDone] == 1.
4. Set UMC::DramScrubCtrl[MemClrEn,DisMemClr] = {0,1}.

The Memory Clear operation should not be executed during a UMC resume from a state such as ACPI S3 where the DRAM contents are expected to be preserved. If memory clear is not to be executed, firmware must set UMC::DramScrubCtrl[MemClrEn,DisMemClr] = {0,1}.

The Memory Clear operation cannot be aborted once initiated. When completed it cannot be re-triggered again without Reset.

12.6 UMC Memory Scrubber

The UMC memory scrubber feature can be divided into the following general categories of memory scrubbing:

- Patrol Scrubber (also known as a Periodic Scrubber) – Scrubs DRAM data at regular intervals.
- Demand Scrubber (also known as a Redirect Scrubber) – Scrubs DRAM data when [DF](#) makes a read request. UMC performs ECC calculations and sends corrected data to DF. In parallel, on an ECC error the UMC writes the corrected data to DRAM (sends an internally generated write command) to the address of the location of the error.
- Memory Clear – See 12.5 [UMC Memory Clear].

The AMD [RAS](#) team recommends that memory be scrubbed approximately once every 24 hours (+/- 10%). This value should be the default value for the [AGESA™](#) Platform Control Board (APCB) DRAM Scrub Time. Most power-of-two memory sizes will be scrubbed within 24.4 hours if the APCB DRAM Scrub Time is set to 24.

BIOS performs the following steps to calculate the UMC::DramScrubCtrl[DramScrubRate]:

1. Determine the desired DRAM Scrub Time (the number of desired hours).
2. Add 10% to this number. This is to accommodate some amount of slack. The scrub rate is selectable only in powers of 2. This does not necessarily translate perfectly to a number of hours.
3. Multiply the number by 3600 to convert to seconds.
4. Determine the amount of memory to scrub by calculating: (UMC::DramScrubLimitAddr[ScrubLimitAddr]+1) - UMC::DramScrubBaseAddr[ScrubBaseAddr].
5. Divide the amount of memory by 64 bytes to convert to the number of cache lines.
6. Divide the scrub time in seconds by the number of cache lines. This determines the rate at which to scrub.
7. Look up valid values for UMC::DramScrubCtrl[DramScrubRate]. Find the closest value that is faster than the scrub rate. If no value is faster, select the fastest rate (640 ns).
8. Program the value into UMC::DramScrubCtrl[DramScrubRate]. See below.

BIOS should separate updates to DramScrubCtrl[DramScrubEn] by at least 5 REFCLKs. Count starts from first posedge of [REFCLK](#) after DramScrubCtrl[DramScrubEn] has changed.

BIOS programs the following to enable the feature:

- UMC::SdpCtrl[DramScrubCrdt] = 1 (FEI reserves a credit for Scrub requests)
- Disable Memory Clear.
 - If DramScrubCtrl[MemClrDone] = 1, BIOS programs DramScrubCtrl[MemClrEn,DisMemClr]={0,1}.
- Program scrub rate:
 - DramScrubCtrl[DramScrubEn] = 0
 - Wait 6 REFCLKs
 - Update UMC::DramScrubCtrl[DramScrubRate] to the desired value
- DramScrubCtrl[DramScrubEn] = 1

To enable the Demand Scrubber, program:

- DramScrubCtrl[RedirectScrubEn] = 1

Firmware reads the UMC::DramScrubAddrLo[SeqScrubAddrLo] and UMC::DramScrubAddrHi[SeqScrubAddrHi] before entering S3/Z-state and restores the value after DramReady (exit S3/Z-State).

The SeqScrubAddrHi and SeqScrubAddrLo fields hold the last address that was scrubbed. It is updated with UMC::DramScrubBaseAddr on DramReady assertion.

If Post Package Repair (PPR) is in progress:

- Reads of DramScrubAddrLo[SeqScrubAddrLo] and DramScrubAddrHi[SeqScrubAddrHi] during Post Package Repair (PPR) return the address to be sent by the Patrol Scrubber after PPR ends.
- Writing to DramScrubAddrLo[SeqScrubAddrLo] and DramScrubAddrHi[SeqScrubAddrHi] during PPR is allowed.
- When the Patrol Scrubber operation resumes, the address to be sent is the last write performed while PPR was in progress.

12.7 UMC Refresh Management (RFM)

BIOS enables the UMC refresh management feature to support DRAMs that require the JEDEC Refresh Management (RFM) command and the Rolling Accumulated ACT (RAA) protocol. The SPD data structure identifies DRAM RFM requirements. In this mode of operation, UMC logic optimizes between issuing REF and RFM commands per the RAA threshold settings.

UMC RFM command shares scheduling timing configuration registers with the corresponding REF command timing register field. If RFM is enabled, BIOS programs the timing configuration per the maximum value required by the REF or RFM command. See UMC::TrfcTiming and UMC::RefTiming.

BIOS programs the following to enable the feature:

- Program UMC::RfmCtrl[RfmCsEn] per the populated chipselects which require RFM RAA tracking.
- Program UMC::RfmCtrl[RAADecRfm] per RFM command decrement requirements.
- Program UMC::RfmCtrl[RAADecRef] per REF command decrement requirements.
- Program UMC::RfmCtrl[RAAMult] per maximum management threshold requirements.
- Program UMC::RfmCtrl[RAAIMT] per initial management threshold requirements.
- Program UMC::RfmCtrl[RFMEn] = 1.

12.8 DDR Traffic Generation

The UMC supports test modes during initialization while system memory requests are quiesced.

DDR traffic pattern generation uses pattern generators in the UMC to generate controlled read and write traffic streams. During write pattern generation, data values based on a deterministic pattern are burst to the DRAM interface. Conversely for reads, data bursts from the DRAM interface are compared against expected data values on a per DQ lane basis.

Two address modes are available for DRAM traffic pattern generation, as configured by UMC::UmcBistCmdCntl[CmdTgt].

- To generate a stream of reads or writes to the same rank, address target A mode is used.
- To generate a stream of accesses to up to two different ranks, address target A and B mode is used.

An overview of the firmware sequence to generate training patterns is as follows:

- Configure training-specific register values. See 12.8.1 [Training-Specific Register Values].
- Configure the UMC for pattern generation. See 12.8.4 [Configurable Data Patterns].
- Ensure DIMMs are configured to support 16-beat burst(BL16).
 - If DDR5 mode and Write CRC is enabled, it takes 16 beats for data + 2 beats for CRC.
- Program UMC::UmcBistCmdCntl[CmdTestEnable] = 1.
- Send activate commands as appropriate. See 12.8.2 [Activate and Precharge Command Generation].
- Send read or write commands as desired. See 12.8.3 [Read and Write Command Generation].
- Send precharge commands as appropriate. See 12.8.2 [Activate and Precharge Command Generation].
- Program UMC::UmcBistCmdCntl[CmdTestEnable] = 0.

12.8.1 Training-Specific Register Values

The UMC requires certain features to be disabled during DRAM device initialization and training. BIOS should program the registers below before DRAM device initialization and training. BIOS must quiesce all other forms of DRAM traffic on the channel being trained.

Before DRAM device initialization and training, BIOS programs:

- UMC::SpazCtrl[DisAutoRefresh,PwrDownEn] = {1,0}

- UMC::ThrottleCtrl[OdtsEn] = 0
- UMC::DramTiming1[ZqcsInterval] = 0
- UMC::BeqCtrl1[DisPhyUpdRollOrderRst] = 1

12.8.2 Activate and Precharge Command Generation

Prior to sending read or write commands, BIOS must send an activate command to a row in a particular bank of the DRAM devices for access. To send an activate command, BIOS performs the following steps:

Program UMC::Command2 to the desired address as follows:

- CmdChipSelect = CS[7:0].
- CmdBank = BA[2:0].
- CmdAddress = A[17:0] .
- Program UMC::UmcBistCmdCntl[SendActCmd] = 1.
- Wait until UMC::UmcBistCmdCntl[SendActCmd] == 0.
- Wait 75 MEMCLKs.

After completing its accesses, BIOS must deactivate open rows in the DRAM devices. To send a precharge or precharge all command to deactivate open rows in a bank or in all banks, BIOS performs the following steps:

- Wait at least UMC::DramTiming3[Trtp]
- Program UMC::Command2 to the desired address as follows:
 - CmdChipSelect = CS[7:0]
 - Precharge all command:
 - CmdAddress[10] = 1
 - Precharge command:
 - CmdAddress[10] = 0
 - CmdBank = BA[2:0]
- Program UMC::UmcBistCmdCntl[SendPchgCmd] = 1.
- Wait until UMC::UmcBistCmdCntl[SendPchgCmd] == 0.
- Wait 25 MEMCLKs. UMC::DramTiming2[TrpPb,TrpAb]

12.8.3 Read and Write Command Generation

BIOS performs the following steps for read pattern generation:

- Program UMC::DQMaskEcc[EccMask], UMC::DQMaskLo[DQMask], and UMC::DQMaskHi[DQMask] with the data comparison masks for bit lanes of interest.
 - If ECC is disabled, BIOS must program (DQMaskEcc[EccMask]=FF) to disable comparison.
- Program UMC::Prbs[DataPrbsSeed] with the seed for the desired [PRBS](#).
- Program UMC::Command1[CmdCount] equal to the number of cache line commands.
- Program UMC::Command0[BubbleCnt,CmdStreamLen].
- Program UMC::TgtBase to the initial address.
- Program UMC::UmcBistCmdCntl as follows:
 - ResetAllErr as desired. See 12.8.5 [Data Comparison].
 - CmdTgt corresponding to UMC::TgtBase.
 - CmdType = 000b.
 - SendCmd = 1.
- If UMC::Command1[CmdCount] != 0 then
 - Wait for UMC::UmcBistCmdCntl[TestStatus] == 1 and UMC::UmcBistCmdCntl[CmdSendInProg] == 0.
- If UMC::Command1[CmdCount] = 0 then
 - Wait the desired amount of time.
 - Program UMC::Command1[CmdCount] = 1.

- Wait for UMC::UmcBistCmdCntl[TestStatus] == 1 and UMC::UmcBistCmdCntl[CmdSendInProg] == 0.
- Program UMC::UmcBistCmdCntl[SendCmd] = 0.
- Read UMC::Status0, UMC::Status1, and UMC::Status2 if applicable.

BIOS performs the following steps for write pattern generation:

- Program UMC::Prbs[DataPrbsSeed] the seed for the desired PRBS.
- Program UMC::Command1[CmdCount] equal to the number of cache line commands desired.
- Program UMC::Command0[BubbleCnt,CmdStreamLen].
- Program UMC::TgtBase[TgtAddress] to the initial address.
- Program UMC::UmcBistCmdCntl as follows:
 - CmdTgt corresponding to UMC::TgtBase.
 - CmdType = 001b.
 - SendCmd = 1.
- If UMC::Command1[CmdCount] != 0 then
 - Wait for UMC::UmcBistCmdCntl[TestStatus] == 1 and UMC::UmcBistCmdCntl[CmdSendInProg] == 0.
- If UMC::Command1[CmdCount] = 0 then
 - Wait the desired amount of time.
 - Program UMC::Command1[CmdCount] = 1.
 - Wait for UMC::UmcBistCmdCntl[TestStatus] == 1 and UMC::UmcBistCmdCntl[CmdSendInProg] == 0.
- Program UMC::UmcBistCmdCntl[SendCmd] = 0.

BIOS combines the two sets of steps listed above for alternating write and read pattern generation.

12.8.3.1 Infinite Write Mode

When defining a test subchannel and an aggressor subchannel (Infinite Write Mode), BIOS must program:

- If testing subchannel 0 and if subchannel 1 in Infinite Write Mode, then program:
 - UMC::UmcBistCmdCntl[CmdTgt[1]] = 0
 - UMC::Command0[InfWrMode_CmdTgt[1:0]] = 2
- If testing subchannel 1 and if subchannel 0 in Infinite Write Mode, then program:
 - UMC::UmcBistCmdCntl[CmdTgt[1]] = 1
 - UMC::Command0[InfWrMode_CmdTgt[1:0]] = 1

The UMC uses the same bubble count registers for both the test channel and the Infinite Write Mode channel. BIOS must program UMC::Command0[BubbleCnt2,BubbleCnt] values to satisfy traffic on both subchannels.

12.8.4 Configurable Data Patterns

In addition to [PRBS](#) mode, UMC::UmcBistCmdCntl[DataPatGenSel], UMC::UserDataPattern0, UMC::UserDataPattern1, UMC::UserDataPattern2, and UMC::UserDataPattern3 allow configurable data pattern generation. For example: Simultaneous Switching Output (SSO) and Inversion Patterns.

12.8.4.1 Configurable Data Pattern Example with 1 Address Target

Example data pattern below:

- [DataPatGenSel] = 10b // [Configurable](#) data pattern
- [UserDataPattern[3,2,1,0]] = {FFEE_DDCCh, BBAA_9988h, 7766_5544h, 3322_1100h}.

DQ/ Beat	Write TgtA Cmd 0							Write TgtA Cmd 1							Write TgtA Cmd 2							...			
	Beat 0	Beat 1	Beat 2	Beat 3	Beat 4	Beat 5	Beat 6	Beat 7	Beat 8	Beat 9	Beat 10	Beat 11	Beat 12	Beat 13	Beat 14	Beat 15	Beat 16	Beat 17	Beat 18	Beat 19	Beat 20	Beat 21	Beat 22	Beat 23	...
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
1	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	
2	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	
3	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	
4	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	
5	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	
6	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	
7	1	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	
8	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	
9	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	
10	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	
11	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1	
12	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	
13	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1	1	
14	0	1	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1	1	1	
15	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
...																									
ECC0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
ECC1	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	
ECC2	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	
ECC3	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	
ECC4	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	
ECC5	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	
ECC6	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	
ECC7	1	1	1	0	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	0	1	1	1	1	

Figure 50: Configurable Data Pattern Example with 1 Address Target

12.8.4.2 Configurable Data Pattern Circular Shift Example with 1 Address Target

Example data pattern below:

- [DataPatGenSel] = 11b. //User data pattern with circular lane shift
- [UserDataPattern[3,2,1,0]] = {FFEE_DDCCh, BBAA_9988h, 7766_5544h, 3322_1100h}.

DQ/ Beat	Write TgtA Cmd 0							Write TgtA Cmd 1							Write TgtA Cmd 2							...			
	Beat 0	Beat 1	Beat 2	Beat 3	Beat 4	Beat 5	Beat 6	Beat 7	Beat 8	Beat 9	Beat 10	Beat 11	Beat 12	Beat 13	Beat 14	Beat 15	Beat 16	Beat 17	Beat 18	Beat 19	Beat 20	Beat 21	Beat 22	Beat 23	...
0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	1	1	1	0	1	1	1	
1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	
2	0	1	0	0	0	1	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	
3	1	1	0	0	1	1	0	0	0	1	0	0	0	1	0	0	1	0	0	0	1	0	0	0	
4	0	0	1	0	0	0	1	0	1	1	0	0	1	1	0	0	0	1	0	0	0	1	0	0	
5	1	0	1	0	1	0	1	0	0	0	1	0	0	0	1	0	1	1	0	0	1	1	0	0	
6	0	1	1	0	0	1	1	0	1	0	1	0	1	0	1	0	0	0	1	0	0	0	1	0	
7	1	1	1	0	1	1	1	0	0	1	1	0	0	1	1	0	1	0	1	0	1	0	1	0	
8	0	0	0	1	0	0	0	1	1	1	1	0	1	1	1	0	0	1	1	0	0	1	1	0	
9	1	0	0	1	1	0	0	1	0	0	0	1	0	0	0	1	1	1	1	0	1	1	1	0	
10	0	1	0	1	0	1	0	1	1	0	0	1	1	0	0	1	0	0	0	1	0	0	0	1	
11	1	1	0	1	1	1	0	1	0	1	0	1	0	1	0	1	1	0	0	1	1	0	0	1	
12	0	0	1	1	0	0	1	1	1	1	0	1	1	1	0	1	0	1	0	1	0	1	0	1	
13	1	0	1	1	1	0	1	1	0	0	1	1	0	0	1	1	1	1	0	1	1	1	0	1	
14	0	1	1	1	0	1	1	1	1	0	1	1	1	0	1	1	0	0	1	1	0	0	1	1	
15	1	1	1	1	1	1	1	1	0	1	1	1	0	1	1	1	1	0	1	1	1	0	1	1	
...																									
ECC0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	1	1	1	0	1	1	1	
ECC1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	
ECC2	0	1	0	0	0	1	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	
ECC3	1	1	0	0	1	1	0	0	0	1	0	0	0	1	0	0	1	0	0	0	1	0	0	0	
ECC4	0	0	1	0	0	0	1	0	1	1	0	0	1	1	0	0	0	1	0	0	0	1	0	0	
ECC5	1	0	1	0	1	0	1	0	0	0	1	0	0	0	1	0	1	1	0	0	1	1	0	0	
ECC6	0	1	1	0	0	1	1	0	1	0	1	0	1	0	1	0	0	0	1	0	0	0	1	0	
ECC7	1	1	1	0	1	1	1	1	0	1	1	0	0	1	1	0	1	0	1	0	1	0	1	0	

Figure 51: Configurable Data Pattern Circular Shift Example with 1 Address Target

12.8.4.3 Static Data Pattern Override

Software is also able to use the bits of UMC::DQPatternOverride0, UMC::DQPatternOverride1, and UMC::DQPatternOverride2 to override the pattern generator on a DQ basis. This can only be used in the UMC::UmcBistCmdCntl[DataPatGenSel] == 10b mode. Software programs the following to enable the mode:

- Program UMC::DQPatternOverride2[PatOvrVal] to the data value desired.
- Program UMC::DQPatternOverride0[DQPatOvrEn], UMC::DQPatternOverride1[DQPatOvrEn], and UMC::DQPatternOverride2[EccPatOvrEn] to enable the override on the desired bit lanes.

12.8.4.4 Xor Data Pattern Override

Software can also use the bits of UMC::DQPatternOverride2[XorPatOvr] to override the pattern generator with the same

XOR function for all byte lanes. This feature can only be used with select UMC::UmcBistCmdCntl[DataPatGenSel] [PRBS](#) modes. Software programs the following to enable the mode:

- Program UMC::DQPatternOverride2[XorPatOvr] to the data value of the XOR function desired.
 - Output data = IF (DQPatOvrEn) THEN PatOvrVal. ELSE (DataPatGen ^ XorPatOvr). ENDIF.

12.8.5 Data Comparison

The UMC compares the incoming read data against the expected pattern sequence during pattern generation.

Data capture modes are specified by programming UMC::DQCapture2[SelMode,SelUI]. In parallel mode, the DQCapture* fields are mapped as follows:

Table 220: DQCapture* Mappings In Parallel Mode

Mode	DQCapture* Mapping
SelMode = 0 , SelUI = 0	DQCapture2 = ECC_CHNL_A [7:0] DQCapture1 = CHNL_A[63:32] DQCapture0 = CHNL_A [31:0]
SelMode = 0 , SelUI = 1	DQCapture2 = ECC_CHNL_A [15:8] DQCapture1 = CHNL_A[127:96] DQCapture0 = CHNL_A [95:64]
SelMode = 0 , SelUI = 2	DQCapture2 = ECC_CHNL_A [23:16] DQCapture1 = CHNL_B[63:32] DQCapture0 = CHNL_B [31:0]
SelMode = 0 , SelUI = 3	DQCapture2 = ECC_CHNL_A [31:24] DQCapture1 = CHNL_B[127:96] DQCapture0 = CHNL_B [95:64]

The UMC reports error information in status registers: UMC::Status0, UMC::Status1, UMC::Status2, UMC::Status4, UMC::Status5 and UMC::Status6. Error information can be masked on a per-bit basis by programming UMC::DQMaskLo, UMC::DQMaskHi, and UMC::DQMaskEcc.

DDR5 mode supports a 32-bit subchannel + ECC interface. Error status registers are mapped to errors as follows:

- UMC::Status4[DQErr] – [31:0] maps to DQ errors for the single subchannel (32 bits) on even [UI](#) data.
- UMC::Status5[DQErr] – [31:0] maps to DQ errors for the single subchannel (32 bits) on odd UI data.
- UMC::Status6[EccErr] – [7:0] maps to ECC errors for the single subchannel (8 bits).

BIOS resets the error information by programming UMC::UmcBistCmdCntl[ResetAllErr] = 1.

Error information is only valid in certain modes of UMC::UmcBistCmdCntl[CmdType,CmdTgt] and UMC::Command1[CmdCount] and when using 64-byte aligned addresses in UMC::TgtBase[TgtAddress]. Some modes require a series of writes to setup a DRAM data pattern. See the following table.

Table 221: Command Generation and Data Comparison

Commands	CmdType	CmdTgt	Maximum CmdCount ¹
Read	000b	00b ²	128
		01b ²	256 ³
Write-Read	010b	00b	Infinite

		01b ⁴	256 ⁵
Note: 1. [LsfrRollOver] = 0. The maximum CmdCount is infinite for all modes listed if [LsfrRollOver] == 1. 2. Requires setup writes to store a data pattern in DRAM. The write commands are generated using the same CmdTgt, CmdCount, and Data-PrbsSeed settings. 3. ([TgtAddress]TargetA) != ([TgtAddress]TargetB) 4. Requires setup writes to store a data pattern in DRAM. The write commands are generated programming [TgtAddress] to the intended Target B, CmdTgt = 00b, CmdCount to 1/2 of the intended command count, and the same DataPrbsSeed setting. 5. ([TgtAddress]TargetA) != ([TgtAddress]TargetB).			

12.8.5.1 Activate and Precharge Traffic Generation

The UMC generates ACT and PRE traffic in the available command bandwidth during a WR or RD sequence generated by UMC::UmcBistCmdCntl[SendCmd] when UMC::UmcBistCmdCntl[ActPchgGenEn] == 1. This 16 command sequence repeats until the WR or RD initiated by SendCmd hits a stopping condition.

Software is able to adjust the command spacing and address sequence by programming:

- UMC::Command2[CmdChipSelect,CmdAddress[17:4]]
- UMC::Command1[ActPchgCmdMin]

Software is responsible for the ACT/PRE protocol of the WR or RD targets initiated by SendCmd. On a stopping condition for the WR or RD traffic stream, software is responsible for returning all of the possibly open banks to the idle state.

Example configuration for DRAM Training:

- UMC::UmcBistCmdCntl[ActPchgGenEn,CmdTgt,CmdType] = {1b, 1b, 0b}. // Tgt A&B, Reads
- UMC::Command2[CmdChipSelect] = CS of TgtA.
- UMC::Command2[CmdAddress[17:4]] = desired upper row address bits.
 - Hardware places the sequence number in row address [3:0].
- UMC::Command1[ActPchgCmdMin] = Fh

12.8.6 BubbleCnt and CmdStreamLen Programming

BIOS programs UMC::Command0[BubbleCnt2,BubbleCnt,CmdStreamLen] to ensure proper channel command spacing in command generation mode.

For continuous pattern generation it is expected that BubbleCnt = 0.

In other modes, BIOS programs BubbleCnt, BubbleCnt2, and CmdStreamLen greater than or equal to the relevant DRAM timing parameters shown below to prevent contention on the DRAM bus.

In all cases, if (minimum BubbleCnt > 0) or (UmcBistCmdCntl[CmdType] == 010b) then BIOS programs CmdStreamLen = 1.

- In DDR5Mode, if WriteCRC is enabled then effective BubbleCnt is increased by 1 between read commands.
- Software should also ensure that bubble programming satisfies:
 - UMC::DramTiming8[TrdrdScL,TrdrdSc,TrdrdSd],
 - UMC::DramTiming9[TwrwrScL,TwrwrSc,TwrwrSd]
 - UMC::DramTiming56[TwrwrScL2]

In DDR5Mode, TwrwrSc <= TwrwrScL2 <= TwrwrScL.

Table 222: Command Generation and BubbleCnt Programming

Commands	CmdType	CmdTgt	BubbleCnt	BubbleCnt2
Read-Read same CS	000b	0Xb	UMC::DramTiming8[TrdrdScL] - 1; Exclude banned spacing: UMC::DramTiming8[TrdrdBan]	Xh ¹
Write-Write same CS	001b	00b	IF (BeqCtrl2[DisDatMsk]==1) && (DimmCfg[X4Dram]==!1) THEN (TwrwrScL2 -1) ELSE (TwrwrScL -1).	Xh ¹
		01b	UMC::DramTiming9[TwrwrSc] - 1	
Write-Read same CS	010b	00b	UMC::DramTiming5[TwtrL] + UMC::DramTiming5[Tcwl] + 3 IF (BubbleCnt < TwrwrScL/2) THEN BubbleCnt = TwrwrScL/2. IF (BubbleCnt < TrdrdScL/2) THEN BubbleCnt = tTrdrdScL/2.	[Trdwr]-1 IF ([Trdwr] < [TwrwrScL]/2) THEN BubbleCnt2 = [TwrwrScL]/2 IF (([TwrwrScL] < [TrdrdScL]) && ([Trdwr] < [TrdrdScL]/2))) THEN BubbleCnt2 = [TrdrdScL]/2 ¹
Read-Read different CS	000b	01b	UMC::DramTiming8[TrdrdSd] - 1; Exclude banned spacing: UMC::DramTiming8[TrdrdBan]	Xh ¹
Write-Write different CS	001b	01b	UMC::DramTiming9[TwrwrSd] - 1	Xh ¹
Write-Read different CS	010b	01b	UMC::DramTiming10[Twrrd] - 1	[Trdwr]-1 ¹
Read-Read different DIMM	000b	01b	UMC::DramTiming8[TrdrdDd] - 1; Exclude banned spacing: UMC::DramTiming8[TrdrdBan]	Xh ¹
Write-Write different DIMM	001b	01b	UMC::DramTiming9[TwrwrDd] - 1	Xh ¹
Write-Read different DIMM	010b	01b	UMC::DramTiming10[Twrrd] - 1 ²	[Trdwr]-1 ¹
Note: 1. IF DDR5Mode && (uclkdiv1mode == 0) THEN (IF (BubbleCnt%2 != 0) THEN BubbleCnt2 = (BubbleCnt2 + 1)) 2. IF DDR5Mode && (uclkdiv1mode == 0) THEN (IF (BubbleCnt%2 != 0) THEN BubbleCnt = (BubbleCnt + 1))				

12.9 DDR5 SRSM MOP Array Encodings

The Self Refresh State Machine (SRSM) Memory Operation (MOP) array is a mechanism for configuring DRAM during frequency changes (memory [P-state](#) change requests). These include the DRAM Commands prior to entering Self Refresh (SR) and the DRAM Commands after exiting Self Refresh for P-states Device/Memory states: (D1/M1), C-state/Stutter (D2/M2), and S-state (D3/M3) transitions.

The MOP Array is divided into two parallel sections, each having the same two, but independent MOP states: SrEnterMOP and SrExitMOP.

- SrEnterMOP is processed before entering self-refresh for a Memory P-state request.
- SrExitMOP is processed after exiting self-refresh for a Memory P-state request.

After training and before startup, the software populates the list of commands in each MOP section.

Each MOP Array section:

- Has up to 512 index pairs of UMC SRSR Array values.
- Contains multiple command lists located at different offset locations as described below.
- Any spare SRSR indexes within the section may be used by software as scratch registers.

Each Command List:

- Has the intended purpose of transitioning DRAM states which require enabling and disabling features as the DRAM transitions P-states.
- Is contiguous Memory P-state (MP-state) specific list of 40-bit Memory Operations (MOPs).
- Is a valid combination of sub-channel, rank, timer and command encodings and command pointers, and concluded with a null (all zeros) entry.
- Is responsible for enabling and disabling features as necessary for stable P-state transitions.
- Is terminated by a NULL entry, i.e., all 40 bits = 0.

Each Command:

- Is one of MR, MR w/ PDA, RCW, BCW commands, or Null (all zeros).

At the time of the startup, the P-state changes are executed by the Command List Initiator (CLIM). The CLIM issues commands starting at index zero within a list and continues sequentially until no valid entries remain in that list, i.e., null entries. Looping and branching operations are not supported.

SRSR Index and SrsrCmdOffset:

- SRSR Index[0] = 0 selects bits 31:0 (least significant word) of the MOP command. SRSR Index[0] = 1 selects bits 39:32 (most significant byte) of the MOP command.
- SrsrCmdOffset[SrsrEnterOffset] and [SrsrExitOffset] points to the SrsrIndex[9:1] of the first MOP command in the M-state specific enter and exit sequences.

For PDA/PBA mode, the OP[7:0] MRW OP data values are stored in the data array section of the MOP array in 80-bit patterns. Use the Send PDA to All EncDq with the OP[7:0] equal to the MOP Data Index 80-bit pointer.

Table 223: DDR5 SRSR MOP Array Encodings

Bits	Field Name	Description for Normal Behavior	Bits	Field Name	Description for Bonus MOP
0	MOP_CH0	1 = send command to sub-channel 0 - Required for channel A.	0	MOP_CH0	1 = send command to sub-channel 0 - Required for channel A.
1	MOP_CH1	1 = send command to sub-channel 1 - Required for Channel B.	1	MOP_CH1	1 = send command to sub-channel 1 - Required for Channel B.
5:2	MOP_Rank	Positive polarity chipselect 3:0. More than one rank can be enabled at a time.	5:2	MOP_Rank	Positive polarity chipselect Only one rank per Bonus MOP command.
9:6	MOP_CID	Positive polarity CID 2:0	14:6	MOP_CID	Used to select multiplied rank on LRDIMM
17:10	MOP_OP	OP[7:0] to be sent with MRW or MPC command	19:15	BonusMop	BEQ Mop to send out 0 - NOP

					8 - REF
25:18	MOP_MA	MA[7:0] to be sent with MRW or MPC command	28:20	BonusTimer	BonusTimer[8:0] - lower 9 bits of 12-bit command delay timer in MemClk units.
26	MOP_CW	1 = send RCW/BCW instead of MRW. Goes in CA[10]			
28:27	MOP_CMD	0 - send MRW command. 1 - Send MPC command. 2 - Send Dram-side VrefCA or VrefCs command. 3 - RESERVED			
33:29	MOP_EncDQ	Encode DQ. PDA sequence. 0 - Non-PDA 1..10 - Send PDA to x4 Devices 0..9 11..15 - Send PDA to x8 devices 0..4 16..25 - Send PDA to buffer 0..9 26 - Send PDA to All x4 Devices 27 - Send PDA to All x8/x16 Devices 28 - Send PDA to all buffers 31 - Bonus Mop select	33:29	MOP_EncDQ	31 - Use Bonus MOP encoding.
36:34	MOP_Timer	Encoded Timer Select 0 – Tmrdr 1 – TmrdrL 2 – TmrdrL2 3 – Tmrdr_pda 4 – TvrefDq 5 – Tmod 6 – TmrdrPba	36:34	BonusTimer	BonusTimer[11:9]
37	MOP_Vld_d1	Array Entry valid D1 State transition (DF P-state change) 0 - Array entry skipped 1 - Array entry is valid	37	MOP_Vld_d1	Array Entry valid D1 State transition (DF P-state change) 0 - Array entry skipped 1 - Array entry is valid
38	MOP_Vld_d2	Array entry valid for D2 State (C-states /Stutter) 0 - Array entry skipped 1 - Array entry is valid	38	MOP_Vld_d2	Array entry valid for D2 State (C-states /Stutter) 0 - Array entry skipped 1 - Array entry is valid
39	MOP_Vld_d3	Array entry valid for D3 State (S-states) 0 - Array entry skipped 1 - Array entry is valid	39	MOP_Vld_d3	Array entry valid for D3 State (S-states) 0 - Array entry skipped 1 - Array entry is valid

12.10 UMC Performance Monitors

The UMC provides Performance Monitor counters that software can use to count specific events that occur within the DDR channel. Each UMC provides multiple 48-bit performance counters. The UMC provides one dedicated clock

counter. All UMC performance monitor events can be counted on the remaining counters.

UMC Performance Monitors support the following features:

- 48-bit MEMCLK counter
- 48-bit programmable performance counters

The UMC includes both subchannels in all events counted. The programmable counters can be configured to monitor a variety of performance parameters. The counters support masking functions.

12.10.1 Software considerations for UMC Performance Monitors

UMC performance counter registers are accessible as MSRs as well as over SMN.

Software can clear the counters individually by writing zeros, or in larger increments with the global reset and global reset mask bits. The counters can be set to any value by writing to the Core::X86::Msr::UMC_PerfMonCnt registers (alias of UMC::PerfMonCtrHi and UMC::PerfMonCtrLo combined). When a counter reaches 48'hffff_ffff_ffff it will remain at all 1's until software resets the counter.

When accessing the UMC counters directly (not using MSRs), the 48-bit performance monitor counters require two 32-bit read operations to obtain the entire value. To maintain coherency, the high bits are latched when the low bits are read. Software should read UMC::PerfMonCtrClkLo first, then read UMC::PerfMonCtrClkHi to ensure the proper value is read.

For register writes, the software should follow a similar procedure.

When accessing UMC::PerfMonCtrLo and UMC::PerfMonCtrHi instances follow the same procedure as above. The UMC performance counter registers are a shared resource. Software needs to prevent conflicting accesses from multiple agents.

Software enables UMC performance monitoring by configuring specific perfmon events in the Core::X86::Msr::UMC_PerfMonCtl registers (alias of UMC::PerfMonCtl). Programming UMC::PerfMonCtlClk is also required when using the SMN interface.

For example, the following configuration enables UMC::PMC::MemClk (event 0) and UMC::PMC::DataSlotClks (event 14).

1. Program instance_n1 of Core::X86::Msr::UMC_PerfMonCtl (UMC::PerfMonCtl) = 0x80000000 // maps event 0 to instance_n1
2. Program instance_n2 of Core::X86::Msr::UMC_PerfMonCtl (UMC::PerfMonCtl) = 0x80000014 // maps event 14 to instance_n2
3. If using the SMN interface, program UMC::PerfMonCtlClk = 0x83000000

12.10.2 UMC Performance Monitor Events**UMCPMCx0 [Memory Clock Cycles] (UMC::PMC::MemClk)**

Read-write. Reset: 0000_0000h.	
Number of memory clock cycles.	
UMCPMCx0	
Bits	Description
31	Enable. Read-write. Reset: 0.
30:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx1 [Incoming Page Hit] (UMC::PMC::IncomingPh)

Read-write. Reset: 0000_0000h.	
UMCPMCx1	
Bits	Description
31	Enable. Read-write. Reset: 0.
30:29	Reserved.
28:24	VCSel. Read-write. Reset: 00h.
23:16	Reserved.
15:14	ReqSizeMask. Read-write. Reset: 0h.
13:10	PriorityMask. Read-write. Reset: 0h.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx2 [Incoming Page Miss] (UMC::PMC::IncomingPm)

Read-write. Reset: 0000_0000h.	
UMCPMCx2	
Bits	Description
31	Enable. Read-write. Reset: 0.
30:29	Reserved.
28:24	VCSel. Read-write. Reset: 00h.
23:16	Reserved.
15:14	ReqSizeMask. Read-write. Reset: 0h.
13:10	PriorityMask. Read-write. Reset: 0h.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx3 [Incoming Page Conflict] (UMC::PMC::IncomingPc)

Read-write. Reset: 0000_0000h.	
UMCPMCx3	
Bits	Description
31	Enable. Read-write. Reset: 0.
30:29	Reserved.
28:24	VCSel. Read-write. Reset: 00h.
23:16	Reserved.
15:14	ReqSizeMask. Read-write. Reset: 0h.
13:10	PriorityMask. Read-write. Reset: 0h.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx4 [Incoming Page Table Miss] (UMC::PMC::IncomingPgtMiss)

Read-write. Reset: 0000_0000h.

UMCPMCx4

Bits	Description
31	Enable. Read-write. Reset: 0.
30:29	Reserved.
28:24	VCSEL. Read-write. Reset: 00h.
23:16	Reserved.
15:14	ReqSizeMask. Read-write. Reset: 0h.
13:10	PriorityMask. Read-write. Reset: 0h.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx5 [Activate Command] (UMC::PMC::ActCmd)

Read-write. Reset: 0000_0000h.

Number of ACTIVATE commands sent. Cannot be used with SPM.

UMCPMCx5

Bits	Description										
31	Enable. Read-write. Reset: 0.										
30:10	Reserved.										
9:8	RdWrMask. Read-write. Reset: 0h. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>None.</td></tr> <tr> <td>1h</td><td>Mask Write.</td></tr> <tr> <td>2h</td><td>Mask Read.</td></tr> <tr> <td>3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	None.	1h	Mask Write.	2h	Mask Read.	3h	Reserved.
Value	Description										
0h	None.										
1h	Mask Write.										
2h	Mask Read.										
3h	Reserved.										
7:0	EventSelect. Read-write. Reset: 00h.										

UMCPMCx6 [Precharge Command] (UMC::PMC::PchgCmd)

Read-write. Reset: 0000_0000h.

Number of PRECHARGE commands sent. Cannot be used with SPM.

UMCPMCx6

Bits	Description										
31	Enable. Read-write. Reset: 0.										
30:10	Reserved.										
9:8	RdWrMask. Read-write. Reset: 0h. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>None.</td></tr> <tr> <td>1h</td><td>Mask Write.</td></tr> <tr> <td>2h</td><td>Mask Read.</td></tr> <tr> <td>3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	None.	1h	Mask Write.	2h	Mask Read.	3h	Reserved.
Value	Description										
0h	None.										
1h	Mask Write.										
2h	Mask Read.										
3h	Reserved.										
7:0	EventSelect. Read-write. Reset: 00h.										

UMCPMCx7 [Speculative Precharge Command] (UMC::PMC::SpecPchgCmd)

Read-write. Reset: 0000_0000h.

UMCPMCx7

Bits	Description
31	Enable. Read-write. Reset: 0.
30:24	Reserved.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx8 [Precharge All Command] (UMC::PMC::PchgAllCmd)

Read-write. Reset: 0000_0000h.

UMCPMCx8

Bits	Description
31	Enable. Read-write. Reset: 0.
30:24	Reserved.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx9 [Other Precharge Command] (UMC::PMC::OtherPchgCmd)

Read-write. Reset: 0000_0000h.

UMCPMCx9

Bits	Description
31	Enable. Read-write. Reset: 0.
30:24	Reserved.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCxA [CAS Command] (UMC::PMC::CasCmd)

Read-write. Reset: 0000_0000h.

Number of CAS commands sent.

UMCPMCxA

Bits	Description										
31	Enable. Read-write. Reset: 0.										
30:10	Reserved.										
9:8	RdWrMask. Read-write. Reset: 0h. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>None.</td></tr> <tr> <td>1h</td><td>Mask Write.</td></tr> <tr> <td>2h</td><td>Mask Read.</td></tr> <tr> <td>3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	None.	1h	Mask Write.	2h	Mask Read.	3h	Reserved.
Value	Description										
0h	None.										
1h	Mask Write.										
2h	Mask Read.										
3h	Reserved.										
7:0	EventSelect. Read-write. Reset: 00h.										

UMCPMCxB [CAS with Auto Precharge Command] (UMC::PMC::CasWithAP)

Read-write. Reset: 0000_0000h.

UMCPMCxB

Bits	Description
31	Enable. Read-write. Reset: 0.
30:29	Reserved.
28:24	VCSEL. Read-write. Reset: 00h.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:14	ReqSizeMask. Read-write. Reset: 0h.
13:10	PriorityMask. Read-write. Reset: 0h.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCxC [CAS with Burst Chop Command] (UMC::PMC::CasWithBurstChop)

Read-write. Reset: 0000_0000h.

UMCPMCxC

Bits	Description
31	Enable. Read-write. Reset: 0.
30:29	Reserved.
28:24	VCSEL. Read-write. Reset: 00h.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:14	Reserved.
13:10	PriorityMask. Read-write. Reset: 0h.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx13 [Command Slot Clocks] (UMC::PMC::CmdSlotClks)

Read-write. Reset: 0000_0000h.

UMCPMCx13

Bits	Description
31	Enable. Read-write. Reset: 0.
30:24	Reserved.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx14 [Data Slot Clocks] (UMC::PMC::DataSlotClks)

Read-write. Reset: 0000_0000h.

Number of clocks used by the data bus. Cannot be used with SPM.

UMCPMCx14

Bits	Description										
31	Enable. Read-write. Reset: 0.										
30:10	Reserved.										
9:8	RdWrMask. Read-write. Reset: 0h. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>None.</td></tr> <tr> <td>1h</td><td>Mask Write.</td></tr> <tr> <td>2h</td><td>Mask Read.</td></tr> <tr> <td>3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	None.	1h	Mask Write.	2h	Mask Read.	3h	Reserved.
Value	Description										
0h	None.										
1h	Mask Write.										
2h	Mask Read.										
3h	Reserved.										
7:0	EventSelect. Read-write. Reset: 00h.										

UMCPMCx15 [Page Table Overflow] (UMC::PMC::PageTableOverflow)

Read-write. Reset: 0000_0000h.

UMCPMCx15

Bits	Description
31	Enable. Read-write. Reset: 0.
30:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx16 [Command Parity Errors] (UMC::PMC::CmdError)

Read-write. Reset: 0000_0000h.

UMCPMCx16

Bits	Description
31	Enable. Read-write. Reset: 0.
30:24	Reserved.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx17 [Data Errors] (UMC::PMC::DataError)

Read-write. Reset: 0000_0000h.

UMCPMCx17

Bits	Description
31	Enable. Read-write. Reset: 0.
30:24	Reserved.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:10	Reserved.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx18 [Retry Parity] (UMC::PMC::RetryParity)

Read-write. Reset: 0000_0000h.

UMCPMCx18

Bits	Description
31	Enable. Read-write. Reset: 0.
30:24	Reserved.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx19 [Retry CRC] (UMC::PMC::RetryCRC)

Read-write. Reset: 0000_0000h.

Number of attempted recovery commands due to an ECC/RdCRC error. Note: The recovery command initiates a recovery sequence that is followed by Replay. Errors can happen during a recovery sequence. In such cases, another recovery command is attempted. For that reason the number of recovery attempts is not necessarily the same as the number of replays.

UMCPMCx19

Bits	Description
31	Enable. Read-write. Reset: 0.
30:24	Reserved.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:10	Reserved.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx1A [Bypass Saturation] (UMC::PMC::BypassSat)

Read-write. Reset: 0000_0000h.

UMCPMCx1A

Bits	Description
31	Enable. Read-write. Reset: 0.
30:29	Reserved.
28:24	VCSel. Read-write. Reset: 00h.
23:14	Reserved.
13:10	PriorityMask. Read-write. Reset: 0h.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx1B [Precharge Power Down Clocks] (UMC::PMC::PrechargePwrDownClks)

Read-write. Reset: 0000_0000h.

UMCPMCx1B

Bits	Description
31	Enable. Read-write. Reset: 0.
30:20	Reserved.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx1C [Thermal Throttle Clocks] (UMC::PMC::ThermalThrottleClks)

Read-write. Reset: 0000_0000h.

UMCPMCx1C

Bits	Description
31	Enable. Read-write. Reset: 0.
30:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx1D [DCQ Occupancy Greater Than 0 Percent] (UMC::PMC::DcqOccupancy)

Read-write. Reset: 0000_0000h.

UMCPMCx1D

Bits	Description
31	Enable. Read-write. Reset: 0.
30:29	Reserved.
28:24	VCSel. Read-write. Reset: 00h.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:14	ReqSizeMask. Read-write. Reset: 0h.
13:10	PriorityMask. Read-write. Reset: 0h.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx1E [DCQ Occupancy Greater Than 25 Percent] (UMC::PMC::DcqOccupancy25)

Read-write. Reset: 0000_0000h.

UMCPMCx1E

Bits	Description
31	Enable. Read-write. Reset: 0.
30:29	Reserved.
28:24	VCSel. Read-write. Reset: 00h.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:14	ReqSizeMask. Read-write. Reset: 0h.
13:10	PriorityMask. Read-write. Reset: 0h.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx1F [DCQ Occupancy Greater Than 50 Percent] (UMC::PMC::DcqOccupancy50)

Read-write. Reset: 0000_0000h.

UMCPMCx1F

Bits	Description
31	Enable. Read-write. Reset: 0.
30:29	Reserved.
28:24	VCSel. Read-write. Reset: 00h.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:14	ReqSizeMask. Read-write. Reset: 0h.
13:10	PriorityMask. Read-write. Reset: 0h.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx20 [DCQ Occupancy Greater Than 75 Percent] (UMC::PMC::DcqOccupancy75)

Read-write. Reset: 0000_0000h.

UMCPMCx20

Bits	Description
31	Enable. Read-write. Reset: 0.
30:29	Reserved.
28:24	VCSel. Read-write. Reset: 00h.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:14	ReqSizeMask. Read-write. Reset: 0h.
13:10	PriorityMask. Read-write. Reset: 0h.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx21 [DCQ Occupancy Greater Than 90 Percent] (UMC::PMC::DcqOccupancy90)

Read-write. Reset: 0000_0000h.

UMCPMCx21

Bits	Description
31	Enable. Read-write. Reset: 0.
30:29	Reserved.
28:24	VCSel. Read-write. Reset: 00h.
23:20	ChipIDSel. Read-write. Reset: 0h.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:14	ReqSizeMask. Read-write. Reset: 0h.
13:10	PriorityMask. Read-write. Reset: 0h.
9:8	RdWrMask. Read-write. Reset: 0h.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx22 [Self Refresh Clocks] (UMC::PMC::SelfRefreshClks)

Read-write. Reset: 0000_0000h.

UMCPMCx22

Bits	Description
31	Enable. Read-write. Reset: 0.
30:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx23 [Precharge Power Down All Clocks] (UMC::PMC::PrechargePwrDownAllClks)

Read-write. Reset: 0000_0000h.

UMCPMCx23

Bits	Description
31	Enable. Read-write. Reset: 0.
30:20	Reserved.
19:16	ChipSelMask. Read-write. Reset: 0h.
15:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx25 [Auto Refresh Count] (UMC::PMC::AutoRefCnt)

Read-write. Reset: 0000_0000h.

Number of auto refresh commands sent.

UMCPMCx25

Bits	Description										
31	Enable. Read-write. Reset: 0.										
30:24	Reserved.										
23:20	ChipIDSel. Read-write. Reset: 0h.										
19:16	ChipSelMask. Read-write. Reset: 0h.										
15:10	Reserved.										
9:8	RdWrMask. Read-write. Reset: 0h.										
ValidValues:											
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Counts REFpb (per-bank), REFsb (same-bank), and REFab (all-bank).</td></tr> <tr> <td>1h</td><td>Counts only REFab.</td></tr> <tr> <td>2h</td><td>Counts only REFpb (per-bank) and REFsb (same-bank).</td></tr> <tr> <td>3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Counts REFpb (per-bank), REFsb (same-bank), and REFab (all-bank).	1h	Counts only REFab.	2h	Counts only REFpb (per-bank) and REFsb (same-bank).	3h	Reserved.
Value	Description										
0h	Counts REFpb (per-bank), REFsb (same-bank), and REFab (all-bank).										
1h	Counts only REFab.										
2h	Counts only REFpb (per-bank) and REFsb (same-bank).										
3h	Reserved.										
7:0	EventSelect. Read-write. Reset: 00h.										

UMCPMCx28 [Arb Bypass Command] (UMC::PMC::ArbBypassCmd)

Read-write. Reset: 0000_0000h.

UMCPMCx28

Bits	Description
31	Enable. Read-write. Reset: 0.
30:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx29 [EPA Command] (UMC::PMC::EpaCmd)

Read-write. Reset: 0000_0000h.

UMCPMCx29

Bits	Description
31	Enable. Read-write. Reset: 0.
30:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

UMCPMCx57 [Refresh Management Commands] (UMC::PMC::RfmCmd)

Read-write. Reset: 0000_0000h.

UMCPMCx57

Bits	Description
31	Enable. Read-write. Reset: 0.
30:8	Reserved.
7:0	EventSelect. Read-write. Reset: 00h.

12.11 Definitions

Table 224: Link Definitions

Term	Description
DFI	DDR PHY Interface; the UMC interfaces to the PHY through the DFI. The DFI specification defines the protocol between memory controller logic and PHY interface (ddr-phy.org).
Ddr5Mode	DDR5 Mode Select (UMC::UmcConfig[DdrType] == 1h).
SRS	Self Refresh State Machine; the SRS in the UMC responds to power management and frequency change requests from the DFI . Its responsibilities includes putting memory in self-refresh, putting DDR PHYs in various power states, configuring Mode Registers in DRAM, etc.
MOP	Memory Operation; the MOP is a subblock of the SRS and contains the binary configuration information to be transmitted to the DDR PHYs and DRAM.
MR	Mode Register (refer to JEDEC DDR5 SDRAM standard).
MRL	Maximum Read Latency; MRL is the measure in units of MCLK between DFI Read Data Enable and DFI Read Data Valid.
MRR	Mode Register Read command (refer to JEDEC DDR5 SDRAM standard).
MRS	Mode Register Set; set of commands to program the DRAM Mode Register (MR).
MRW	Mode Register Write command (refer to JEDEC DDR5 SDRAM standard).
PBA	Per Buffer Addressability; ability to target individual DIMMs using commands that include one or more MRS commands or BCW commands (refer to JEDEC DDR5 SDRAM standard).
PDA	Per DRAM Addressability; uses the strobes and DQ[0] of each DRAM which allows the host to uniquely select individual DRAM to be programmed (refer to JEDEC DDR5 SDRAM standard).
PMU	PHY Microprocessor Unit; used to initialize the DDR PHY and train the PHY for optimum timing margins. Includes dedicated instruction and data SRAMs that are used to store the training code and data structures.
PLR	Per Logic Rank; used for addressing along with CS, Bank, Row, Channel. Often contains Chip ID (CID) addresses.
PRA	Per Rank Addressability.

12.12 UMC Registers

12.12.1 Controller Registers

UMC00CHx00000000...UMC11CHx0000000C [DRAM CS Base Address] (UMC::BaseAddr)	
Read-write. Reset: 0000_0000h.	
The BaseAddr and AddrMask registers translate request addresses to DRAM chipselects. For each chipselect there is a BaseAddr register and an AddrMask register. For each request normalized address, a chipselect[i] is asserted if: CSEnable[i] & ((RequestAddr & ~AddrMask[i]) == (BaseAddr[i] & ~AddrMask[i])). For each even/odd chipselect-pair, an even CS must be populated if the odd CS is populated.	
_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000000; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000004; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000008; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000000C; UMC[11:00]CH=00[B:0]5_0000h	
Bits	Description
31:1	BaseAddr: Base Address [39:9]. Read-write. Reset: 0000_0000h. Base address for chipselect decoding.
0	CSEnable. Read-write. Reset: 0. 0=Disable. 1=Enable. Enables chipselect decoding.

UMC00CHx00000010...UMC11CHx0000001C [DRAM CS Base Secondary Address] (UMC::BaseAddrSec)

Read-write. Reset: 0000_0000h.

The BaseAddrSec and AddrMaskSec registers translate request addresses to DRAM chipselects.

`_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000010; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000014; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000018; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000001C; UMC[11:00]CH=00[B:0]5_0000h`**Bits Description****31:1 BaseAddr: Base Address [39:9].** Read-write. Reset: 0000_0000h. Base address for chipselect decoding.**0 CSEnable.** Read-write. Reset: 0. 0=Disable. 1=Enable. Enables chipselect decoding.**UMC00CHx00000020...UMC11CHx0000002C [DRAM CS Mask Address] (UMC::AddrMask)**

Read-write. Reset: 0000_0000h.

The BaseAddr and AddrMask registers translate request addresses to DRAM chipselects.

`_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000020; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000024; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000028; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000002C; UMC[11:00]CH=00[B:0]5_0000h`**Bits Description****31:1 AddrMask: Address Mask [39:9].** Read-write. Reset: 0000_0000h. Address mask for chipselect decoding.**0** Reserved.**UMC00CHx00000030...UMC11CHx0000003C [DRAM CS Mask Secondary Address] (UMC::AddrMaskSec)**

Read-write. Reset: 0000_0000h.

The BaseAddrSec and AddrMaskSec registers translate request addresses to DRAM chipselects.

`_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000030; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000034; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000038; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000003C; UMC[11:00]CH=00[B:0]5_0000h`**Bits Description****31:1 AddrMask: Address Mask [39:9].** Read-write. Reset: 0000_0000h. Address mask for chipselect decoding.**0** Reserved.

UMC00CHx00000040...UMC11CHx0000004C [DRAM Address Configuration] (UMC::AddrCfg)

Read-write. Reset: 0015_0508h.

Specifies the number of bits used for DRAM address generation.

_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000040; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000044; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000048; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000004C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description												
31:30	CSXor : CSXor[1:0] . Read-write. Reset: 0h. Specifies XOR function bits to remap the CS decoder.												
29:22	Reserved.												
21:20	NumBanks . Read-write. Reset: 1h. Specifies the number of total bank address bits, including NumBankGroups. ValidValues:												
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>8 Banks (3 bit)</td></tr> <tr> <td>1h</td><td>16 Banks (4 bit)</td></tr> <tr> <td>2h</td><td>32 Banks (5 bit)</td></tr> <tr> <td>3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	8 Banks (3 bit)	1h	16 Banks (4 bit)	2h	32 Banks (5 bit)	3h	Reserved.		
Value	Description												
0h	8 Banks (3 bit)												
1h	16 Banks (4 bit)												
2h	32 Banks (5 bit)												
3h	Reserved.												
19:16	NumCol . Read-write. Reset: 5h. Specifies the number of column address bits. ValidValues:												
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Bh-0h</td><td>5 + NumCol Bits</td></tr> <tr> <td>Fh-Ch</td><td>Reserved.</td></tr> </table>	Value	Description	Bh-0h	5 + NumCol Bits	Fh-Ch	Reserved.						
Value	Description												
Bh-0h	5 + NumCol Bits												
Fh-Ch	Reserved.												
15:12	Reserved.												
11:8	NumRow . Read-write. Reset: 5h. Specifies the number of row address bits. ValidValues:												
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>8h-0h</td><td>10 + NumRow Bits</td></tr> <tr> <td>Fh-9h</td><td>Reserved.</td></tr> </table>	Value	Description	8h-0h	10 + NumRow Bits	Fh-9h	Reserved.						
Value	Description												
8h-0h	10 + NumRow Bits												
Fh-9h	Reserved.												
7	Reserved.												
6:4	NumRM . Read-write. Reset: 0h. Specifies the number of RM bits. ValidValues:												
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>No RM (0 bit)</td></tr> <tr> <td>1h</td><td>2x RM (1 bit)</td></tr> <tr> <td>2h</td><td>4x RM (2 bit)</td></tr> <tr> <td>3h</td><td>8x RM (3 bit)</td></tr> <tr> <td>7h-4h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	No RM (0 bit)	1h	2x RM (1 bit)	2h	4x RM (2 bit)	3h	8x RM (3 bit)	7h-4h	Reserved.
Value	Description												
0h	No RM (0 bit)												
1h	2x RM (1 bit)												
2h	4x RM (2 bit)												
3h	8x RM (3 bit)												
7h-4h	Reserved.												
3:2	NumBankGroups . Read-write. Reset: 2h. Specifies the number of BG bits. ValidValues:												
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>No Bank Groups</td></tr> <tr> <td>1h</td><td>2 Bank Groups (1 bit)</td></tr> <tr> <td>2h</td><td>4 Bank Groups (2 bit)</td></tr> <tr> <td>3h</td><td>8 Bank Groups (3 bit)</td></tr> </table>	Value	Description	0h	No Bank Groups	1h	2 Bank Groups (1 bit)	2h	4 Bank Groups (2 bit)	3h	8 Bank Groups (3 bit)		
Value	Description												
0h	No Bank Groups												
1h	2 Bank Groups (1 bit)												
2h	4 Bank Groups (2 bit)												
3h	8 Bank Groups (3 bit)												
1:0	Reserved.												

UMC00CHx00000050...UMC11CHx0000005C [DRAM Bank Address Select] (UMC::AddrSel)

Read-write. Reset: 070C_BA98h.

Specifies the normalized address bit position used for DRAM address generation.

`_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000050; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000054; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000058; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000005C; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description				
31:28	Reserved.				
27:24	Row. Read-write. Reset: 7h. Specifies the normalized address bit used as the least-significant row address bit. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>12 + Row Bit</td></tr> </table>	Value	Description	Fh-0h	12 + Row Bit
Value	Description				
Fh-0h	12 + Row Bit				
23:20	Reserved.				
19:16	BankBit4. Read-write. Reset: Ch. Specifies the normalized address bit used as the bank address bit. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>5 + BankBit NormAddr Bit</td></tr> </table>	Value	Description	Fh-0h	5 + BankBit NormAddr Bit
Value	Description				
Fh-0h	5 + BankBit NormAddr Bit				
15:12	BankBit3. Read-write. Reset: Bh. Specifies the normalized address bit used as the bank address bit. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>5 + BankBit NormAddr Bit</td></tr> </table>	Value	Description	Fh-0h	5 + BankBit NormAddr Bit
Value	Description				
Fh-0h	5 + BankBit NormAddr Bit				
11:8	BankBit2. Read-write. Reset: Ah. Specifies the normalized address bit used as the bank address bit. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>5 + BankBit NormAddr Bit</td></tr> </table>	Value	Description	Fh-0h	5 + BankBit NormAddr Bit
Value	Description				
Fh-0h	5 + BankBit NormAddr Bit				
7:4	BankBit1. Read-write. Reset: 9h. Specifies the normalized address bit used as the bank address bit. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>5 + BankBit NormAddr Bit</td></tr> </table>	Value	Description	Fh-0h	5 + BankBit NormAddr Bit
Value	Description				
Fh-0h	5 + BankBit NormAddr Bit				
3:0	BankBit0. Read-write. Reset: 8h. Specifies the normalized address bit used as the bank address bit. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>5 + BankBit NormAddr Bit</td></tr> </table>	Value	Description	Fh-0h	5 + BankBit NormAddr Bit
Value	Description				
Fh-0h	5 + BankBit NormAddr Bit				

UMC00CHx00000060...UMC11CHx00000078 [DRAM Column Address Select Low] (UMC::ColSelLo)

Read-write. Reset: 8765_4321h.

Specifies the normalized address bit position used for DRAM address generation.

`_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000060; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000068; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000070; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000078; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description				
31:28	ColBit7. Read-write. Reset: 8h. Specifies the normalized address bit used for the column address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>2 + ColBit Bit</td></tr> </table>	Value	Description	Fh-0h	2 + ColBit Bit
Value	Description				
Fh-0h	2 + ColBit Bit				
27:24	ColBit6. Read-write. Reset: 7h. Specifies the normalized address bit used for the column address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>2 + ColBit Bit</td></tr> </table>	Value	Description	Fh-0h	2 + ColBit Bit
Value	Description				
Fh-0h	2 + ColBit Bit				
23:20	ColBit5. Read-write. Reset: 6h. Specifies the normalized address bit used for the column address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>2 + ColBit Bit</td></tr> </table>	Value	Description	Fh-0h	2 + ColBit Bit
Value	Description				
Fh-0h	2 + ColBit Bit				
19:16	ColBit4. Read-write. Reset: 5h. Specifies the normalized address bit used for the column address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>2 + ColBit Bit</td></tr> </table>	Value	Description	Fh-0h	2 + ColBit Bit
Value	Description				
Fh-0h	2 + ColBit Bit				
15:12	ColBit3. Read-write. Reset: 4h. Specifies the normalized address bit used for the column address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>2 + ColBit Bit</td></tr> </table>	Value	Description	Fh-0h	2 + ColBit Bit
Value	Description				
Fh-0h	2 + ColBit Bit				
11:8	ColBit2. Read-write. Reset: 3h. Specifies the normalized address bit used for the column address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>2 + ColBit Bit</td></tr> </table>	Value	Description	Fh-0h	2 + ColBit Bit
Value	Description				
Fh-0h	2 + ColBit Bit				
7:4	ColBit1. Read-write. Reset: 2h. Specifies the normalized address bit used for the column address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>2 + ColBit Bit</td></tr> </table>	Value	Description	Fh-0h	2 + ColBit Bit
Value	Description				
Fh-0h	2 + ColBit Bit				
3:0	ColBit0. Read-write. Reset: 1h. Specifies the normalized address bit used for the column address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>2 + ColBit Bit</td></tr> </table>	Value	Description	Fh-0h	2 + ColBit Bit
Value	Description				
Fh-0h	2 + ColBit Bit				

UMC00CHx00000064...UMC11CHx0000007C [DRAM Column Address Select High] (UMC::ColSelHi)

Read-write. Reset: 0000_0543h.

Specifies the normalized address bit position used for DRAM address generation.

`_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000064; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000006C; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000074; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000007C; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description				
31:12	Reserved.				
11:8	ColBit10. Read-write. Reset: 5h. Specifies the normalized address bit used for the column address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>8 + ColBit Bit</td></tr> </table>	Value	Description	Fh-0h	8 + ColBit Bit
Value	Description				
Fh-0h	8 + ColBit Bit				
7:4	ColBit9. Read-write. Reset: 4h. Specifies the normalized address bit used for the column address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>8 + ColBit Bit</td></tr> </table>	Value	Description	Fh-0h	8 + ColBit Bit
Value	Description				
Fh-0h	8 + ColBit Bit				
3:0	ColBit8. Read-write. Reset: 3h. Specifies the normalized address bit used for the column address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>8 + ColBit Bit</td></tr> </table>	Value	Description	Fh-0h	8 + ColBit Bit
Value	Description				
Fh-0h	8 + ColBit Bit				

UMC00CHx00000080...UMC11CHx0000008C [DRAM Rank Multiply Address Select] (UMC::RmSel)

Read-write. Reset: 0000_0000h.

Specifies the normalized address bit position used for DRAM address generation.

`_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000080; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000084; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000088; UMC[11:00]CH=00[B:0]5_0000h``_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000008C; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description				
31:30	InvertRowMsbsSec. Read-write. Reset: 0h. Specifies secondary chipselect decoder additional row address translation to achieve DRAM encodings. MSBs determined by Row and NumRow programming.				
29:28	InvertRowMsbs. Read-write. Reset: 0h. Specifies additional row address translation to achieve DRAM encodings. MSBs determined by Row and NumRow programming.				
27:20	Reserved.				
19:16	ChanBit. Read-write. Reset: 0h. Specifies the normalized address bit used for sub-channel selection address bit in virtual controller mode. See: VcmEn. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>5 + ChanBit Bit</td></tr> </table>	Value	Description	Fh-0h	5 + ChanBit Bit
Value	Description				
Fh-0h	5 + ChanBit Bit				
15:12	Reserved.				
11:8	Rm2. Read-write. Reset: 0h. Specifies the normalized address bit used for the rank multiplication address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>12 + Rm Bit</td></tr> </table>	Value	Description	Fh-0h	12 + Rm Bit
Value	Description				
Fh-0h	12 + Rm Bit				
7:4	Rm1. Read-write. Reset: 0h. Specifies the normalized address bit used for the rank multiplication address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>12 + Rm Bit</td></tr> </table>	Value	Description	Fh-0h	12 + Rm Bit
Value	Description				
Fh-0h	12 + Rm Bit				
3:0	Rm0. Read-write. Reset: 0h. Specifies the normalized address bit used for the rank multiplication address bit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>12 + Rm Bit</td></tr> </table>	Value	Description	Fh-0h	12 + Rm Bit
Value	Description				
Fh-0h	12 + Rm Bit				

UMC00CHx00000090...UMC11CHx00000094 [DIMM Configuration] (UMC::DimmCfg)

Read-write. Reset: 0000_0000h.	
_umc0_dimm0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000090; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_dimm1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000094; UMC[11:00]CH=00[B:0]5_0000h	
Bits	Description
31:11	Reserved.
10	PkgRnkTimingAlign. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Enable Package Rank Timing Alignment.
9	DimmRefDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Per-DIMM Refresh Disable. Ranks are refreshed if (CSEnable & ~(DimmRefDis DisAutoRefresh)).
8	DqMapSwapDis. Read-write. Reset: 0. 0=Enable. 1=Disable. Disables odd rank DQMap bit swapping behavior at the DDR device. The DQMap swap from even -> odd rank is as follows: DQ0 -> DQ1, DQ1 -> DQ0, DQ2 -> DQ3, DQ3 -> DQ2.
7	X16Dram. Read-write. Reset: 0. 0=Disable. 1=Enable. Specifies x16 DRAMs are populated.
6	X4Dram. Read-write. Reset: 0. 0=Disable. 1=Enable. Specifies x4 DRAMs are populated.
5	Reserved.
4	RDIMM. Read-write. Reset: 0. 0=Disable. 1=Enable. Specifies an RDIMM is populated.
3	CIsCS. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Specifies the C pins decode as CS for addressing. Reserved if NumRM != 1.
2	Dram3DS. Read-write. Reset: 0. 0=Disable. 1=Enable. Specifies 3DS DRAMs are populated.
1	OutputInvert. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Specifies the DIMM has A-side/B-side inverted addressing. Reserved if DDR5 mode.
0	OnDimmMirror. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Specifies the DIMM has mirrored addressing for MRS commands. Reserved if DDR5 mode.

UMC00CHx00000098...UMC11CHx000000A8 [Address Hash Bank] (UMC::AddrHashBank)

Read-write.	
_umc0_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CHx00000098; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CHx0000009C; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n2_aliasSMN; UMC[11:00]CHx000000A0; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n3_aliasSMN; UMC[11:00]CHx000000A4; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n4_aliasSMN; UMC[11:00]CHx000000A8; UMC[11:00]CH=00[B:0]5_0000h	
Bits	Description
31:14	RowXor: RowXor[17:0]. Read-write. Reset: 0_0000h. Row bits used in hash function computation of the bank address bit.
13:1	ColXor: ColXor[12:0]. Read-write. Reset: Fixed,0000h. Column bits used in hash function computation of the bank address bit. ColXor[12:11,3:0] must be zero if DDR5 mode.
0	XorEnable. Read-write. Reset: 0. 0=Disable. 1=Enable. Enables hash function computation of the bank address bit.

UMC00CHx000000B0...UMC11CHx000000BC [Address Hash RM] (UMC::AddrHashRM)

Read-write. Reset: 0000_0000h.	
Register instance 3 is reserved in DDR5 mode.	
_umc0_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CHx000000B0; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CHx000000B4; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n2_aliasSMN; UMC[11:00]CHx000000B8; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n3_aliasSMN; UMC[11:00]CHx000000BC; UMC[11:00]CH=00[B:0]5_0000h	
Bits	Description
31:1	NormAddrXor: NormAddrXor[39:9]. Read-write. Reset: 0000_0000h. Normalized address bits to use in hash function computation of the rank multiply bit.
0	XorEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enables hash function computation of the rank multiply bit.

UMC[00...11]CHx000000C0 [Address Hash PC] (UMC::AddrHashPC)

Read-write.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000C0; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description
31:14	RowXor: RowXor[17:0] . Read-write. Reset: 0_0000h. Row bits to use in hash function computation of the channel bit.
13:1	ColXor: Col[12:0] . Read-write. Reset: Fixed,0000h. Column bits to use in hash function computation of the channel bit. In DDR5 mode, ColXor[12:11,3:0] must be zero.
0	XorEnable . Read-write. Reset: 0. 0=Disable. 1=Enable. Enables hash function computation of the channel bit. See: ChanBit, VcmEn.

UMC[00...11]CHx000000C4 [Address Hash PC 2] (UMC::AddrHashPC2)

Read-write. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CHx000000C4; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description
31:5	Reserved.
4:0	BankXor: BankXor[4:0] . Read-write. Reset: 00h. Bank bits to use in hash function computation of the channel bit.

UMC00CHx000000C8...UMC11CHx000000CC [Address Hash CS] (UMC::AddrHashCS)

Read-write. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CHx000000C8; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CHx000000CC; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description
31:1	NormAddrXor: NormAddrXor[39:9] . Read-write. Reset: 0000_0000h. Normalized address bits to use in hash function computation of the CS address bit.
0	XorEn . Read-write. Reset: 0. 0=Disable. 1=Enable. Enables hash function computation of the CS address bit.

UMC00CHx000000D0...UMC11CHx000000E4 [DQ Map DW0] (UMC::DQMapDW0)

Read-write. Reset: 0000_0000h.

Specifies the DIMM DQ Mapping. See: DqMapSwapDis.

`_umc0_dimm0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000D0; UMC[11:00]CH=00[B:0]5_0000h``_umc0_dimm1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000E4; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description						
31:30	Reserved.						
29:24	DQMap3. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
23:22	Reserved.						
21:16	DQMap2. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
15:14	Reserved.						
13:8	DQMap1. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
7:6	Reserved.						
5:0	DQMap0. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						

UMC00CHx000000D4...UMC11CHx000000E8 [DQ Map DW1] (UMC::DQMapDW1)

Read-write. Reset: 0000_0000h.

Specifies the DIMM DQ Mapping. See: DqMapSwapDis.

`_umc0_dimm0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000D4; UMC[11:00]CH=00[B:0]5_0000h``_umc0_dimm1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000E8; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description						
31:30	Reserved.						
29:24	DQMap7. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
23:22	Reserved.						
21:16	DQMap6. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
15:14	Reserved.						
13:8	DQMap5. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
7:6	Reserved.						
5:0	DQMap4. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						

UMC00CHx000000D8...UMC11CHx000000EC [DQ Map DW2] (UMC::DQMapDW2)

Read-write. Reset: 0000_0000h.

Specifies the DIMM DQ Mapping. See: DqMapSwapDis.

_umc0_dimm0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000D8; UMC[11:00]CH=00[B:0]5_0000h

_umc0_dimm1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000EC; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description						
31:30	Reserved.						
29:24	DQMap11. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
23:22	Reserved.						
21:16	DQMap10. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
15:14	Reserved.						
13:8	DQMap9. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
7:6	Reserved.						
5:0	DQMap8. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						

UMC00CHx000000DC...UMC11CHx000000F0 [DQ Map DW3] (UMC::DQMapDW3)

Read-write. Reset: 0000_0000h.

Specifies the DIMM DQ Mapping. See: DqMapSwapDis.

_umc0_dimm0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000DC; UMC[11:00]CH=00[B:0]5_0000h_umc0_dimm1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000F0; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description						
31:30	Reserved.						
29:24	DQMap15. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
23:22	Reserved.						
21:16	DQMap14. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
15:14	Reserved.						
13:8	DQMap13. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
7:6	Reserved.						
5:0	DQMap12. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						

UMC00CHx000000E0...UMC11CHx000000F4 [DQ Map DW4] (UMC::DQMapDW4)

Read-write. Reset: 0000_0000h.

_umc0_dimm0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000E0; UMC[11:00]CH=00[B:0]5_0000h_umc0_dimm1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000F4; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description						
31:30	Reserved.						
29:24	DQMap19. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
23:22	Reserved.						
21:16	DQMap18. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
15:14	Reserved.						
13:8	DQMap17. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						
7:6	Reserved.						
5:0	DQMap16. Read-write. Reset: 00h. Specifies the DIMM DQ Mapping. DQ[0,1,2,3] (DQ[4,5,6,7]) to DRAM[w,x,y,z]. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						

UMC[00...11]CHx000000F8 [DRAM CS to ZQ Map] (UMC::CSMapZQ)

Read-write. Reset: 0000_0000h.

Specifies the group of chipselects associated with a ZQ calibration interval.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000F8; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:29	Reserved.
28:24	CSMapZQ3. Read-write. Reset: 00h. CSMapZQ[4:0] mapped as {Rm[0], CS[3:0]}.
23:21	Reserved.
20:16	CSMapZQ2. Read-write. Reset: 00h. CSMapZQ[4:0] mapped as {Rm[0], CS[3:0]}.
15:13	Reserved.
12:8	CSMapZQ1. Read-write. Reset: 00h. CSMapZQ[4:0] mapped as {Rm[0], CS[3:0]}.
7:5	Reserved.
4:0	CSMapZQ0. Read-write. Reset: 00h. CSMapZQ[4:0] mapped as {Rm[0], CS[3:0]}.

UMC[00...11]CHx000000FC [DRAM CS to CKE Map] (UMC::CMapCKE)

Read-write. Reset: 0000_0000h.

Specifies the chipselects associated with a CKE.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000000FC; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:29	Reserved.
28:24	CMapCKE3 . Read-write. Reset: 00h. CMapCKE[4:0] mapped as {C[0], CS[3:0]}.
23:21	Reserved.
20:16	CMapCKE2 . Read-write. Reset: 00h. CMapCKE[4:0] mapped as {C[0], CS[3:0]}.
15:13	Reserved.
12:8	CMapCKE1 . Read-write. Reset: 00h. CMapCKE[4:0] mapped as {C[0], CS[3:0]}.
7:5	Reserved.
4:0	CMapCKE0 . Read-write. Reset: 00h. CMapCKE[4:0] mapped as {C[0], CS[3:0]}.

UMC[00...11]CHx00000100 [UMC Configuration] (UMC::UmcConfig)

Read-write. Reset: 0000_0200h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000100; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description										
31	DramReady . Read-write. Reset: 0. 0=Disable. 1=Enable. Specifies the memory sub-system is initialized and ready for system traffic.										
30:13	Reserved.										
12	DimmEccEn . Read-write. Reset: 0. 0=Disable. 1=Enable. Indicates all populated DIMMs on the channel support ECC check bits.										
11:10	BurstCtrl . Read-write. Reset: 0h. Specifies the burst control for BurstLength. Fixed BL mode must be set in conjunction with DF configuration bit for 64B accesses. Dynamic BL16/BC8 in DDR5 mode. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Dynamic BL/BC</td></tr> <tr> <td>1h</td><td>Fixed BL</td></tr> <tr> <td>3h-2h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Dynamic BL/BC	1h	Fixed BL	3h-2h	Reserved.		
Value	Description										
0h	Dynamic BL/BC										
1h	Fixed BL										
3h-2h	Reserved.										
9:8	BurstLength . Read-write. Reset: 2h. Init: 3h. Specifies the DDR burst length. Constraint: BL2, BL4, and BL8 modes are reserved in DDR5 mode. See: BurstCtrl. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>BL2</td></tr> <tr> <td>1h</td><td>BL4</td></tr> <tr> <td>2h</td><td>BL8</td></tr> <tr> <td>3h</td><td>BL16</td></tr> </table>	Value	Description	0h	BL2	1h	BL4	2h	BL8	3h	BL16
Value	Description										
0h	BL2										
1h	BL4										
2h	BL8										
3h	BL16										
7:3	Reserved.										
2:0	DdrType . Read-write. Reset: 0h. Init: 1h. Specifies the DRAM type. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Reserved.</td></tr> <tr> <td>1h</td><td>DDR5</td></tr> <tr> <td>7h-2h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Reserved.	1h	DDR5	7h-2h	Reserved.		
Value	Description										
0h	Reserved.										
1h	DDR5										
7h-2h	Reserved.										

UMC[00...11]CHx00000104 [SDP Control] (UMC::SdpCtrl)

Read-write. Reset: 3040_0008h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000104; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description								
31	SdpInit. Read-write. Reset: 0. 0=Disable. 1=Enable. Initializes the SDP port. BIOS should disable the SDP port for unused UMC channels.								
30:24	DatBufferCount. Read-write. Reset: 30h. Init: 40h. Specifies the number of SDP data buffer credits.								
	ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>40h-01h</td><td>DatBufferCount Credits</td></tr> <tr> <td>7Fh-41h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Reserved.	40h-01h	DatBufferCount Credits	7Fh-41h	Reserved.
Value	Description								
00h	Reserved.								
40h-01h	DatBufferCount Credits								
7Fh-41h	Reserved.								
23:16	CmdBufferCount. Read-write. Reset: 40h. Init: 7Fh. Specifies the number of SDP command buffer credits.								
	ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>80h-01h</td><td>CmdBufferCount Credits</td></tr> <tr> <td>FFh-81h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Reserved.	80h-01h	CmdBufferCount Credits	FFh-81h	Reserved.
Value	Description								
00h	Reserved.								
80h-01h	CmdBufferCount Credits								
FFh-81h	Reserved.								
15:10	Reserved.								
9	DramScrubCrdt. Read-write. Reset: 0. 0=No credit will be reserved for Scrub. And Scrub is not expected to be enabled. 1=Credit will be reserved for Scrub. Scrub can be enabled/disabled any time. Enable reserving credit for scrubs in the credit control logic in FEI. Needs to be set prior to MemClrDone assertion if MemClrEn=1 or prior to DramReady assertion if MemClrEn=0. See: DramScrubCtrl for scrub modes.								
8:4	Reserved.								
3	SdpCancelEn. Read-write. Reset: 1. Check: 0. 0=Disable. 1=Enable. Specifies SDP CANCEL command processing is enabled.								
2	Reserved.								
1	SdpParityEn. Read-write. Reset: 0. Check: 1. 0=Disable. 1=Enable. Enables SDP originator data parity checking.								
0	SdpFatalDatErr. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Specifies the SDP DATERR indication mode is enabled for all read responses after any FATAL error.								

UMC[00...11]CHx00000108 [PGT Control] (UMC::PgtCtrl)

Read-write. Reset: 400C_1012h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000108; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description										
31	DynIdleCycleEn. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Specifies operation of dynamic idle cycle precharge logic. Hardware adjusts the effective IdleCycleLimit settings based on access history. See: DynIdleHopClks, DynIdleStopLim.										
30	SpecPchgEn. Read-write. Reset: 1. 0=Disable. 1=Enable. Specifies speculative precharge logic is enabled.										
29	DynIdleStopLim. Read-write. Reset: 0. Check: 0. 0=No Limit. 1=Up to 4 stops. Specifies the hardware logic stop limit in dynamic idle cycle mode.										
28	Reserved.										
27:26	DynIdleHopClks. Read-write. Reset: 0h. Check: 0h. Specifies the number of clocks to increment or decrement the idle cycle limit for each stop in dynamic idle cycle mode. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>8 MEMCLKs</td></tr> <tr> <td>1h</td><td>16 MEMCLKs</td></tr> <tr> <td>2h</td><td>32 MEMCLKs</td></tr> <tr> <td>3h</td><td>64 MEMCLKs</td></tr> </table>	Value	Description	0h	8 MEMCLKs	1h	16 MEMCLKs	2h	32 MEMCLKs	3h	64 MEMCLKs
Value	Description										
0h	8 MEMCLKs										
1h	16 MEMCLKs										
2h	32 MEMCLKs										
3h	64 MEMCLKs										
25:24	IdleCycMult. Read-write. Reset: 0h. Check: 0h. Specifies the multiplication factor for IdleCycLimit. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>8 x IdleCycLimit MEMCLKs</td></tr> <tr> <td>1h</td><td>16 x IdleCycLimit MEMCLKs</td></tr> <tr> <td>3h-2h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	8 x IdleCycLimit MEMCLKs	1h	16 x IdleCycLimit MEMCLKs	3h-2h	Reserved.		
Value	Description										
0h	8 x IdleCycLimit MEMCLKs										
1h	16 x IdleCycLimit MEMCLKs										
3h-2h	Reserved.										
23:16	IdleCycLimit. Read-write. Reset: 0Ch. Init: 04h. Specifies the number of MEMCLKs an idle page remains open before it is closed. See: IdleCycMult. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>FFh-01h</td><td>IdleCycMult x IdleCycLimit MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	FFh-01h	IdleCycMult x IdleCycLimit MEMCLKs				
Value	Description										
00h	Reserved.										
FFh-01h	IdleCycMult x IdleCycLimit MEMCLKs										
15	EpaFarPort01. Read-write. Reset: 0. Init: 0. 0=EPA ports {2,3} are far ports. 1=EPA ports {0,1} are far ports. Specifies the EPA ports which are connected to CCMs on the far side of the SOC half. Only applicable on projects with four EPA ports.										
14	EpaClkGateDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disables medium grain clock gating in EPA block.										
13:10	EpaExpiryCnt. Read-write. Reset: 4h. Check: Fh. Expiration counter for pending EPA requests. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>EpaExpiryCnt MEMCLKs</td></tr> </table>	Value	Description	Fh-0h	EpaExpiryCnt MEMCLKs						
Value	Description										
Fh-0h	EpaExpiryCnt MEMCLKs										
9	EpaPchgEn. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Enable PRE commands for EPA request page conflicts. Constraint: EpaEn >= EpaPchgEn.										
8	EpaEn. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Enable ACT commands for Early Page Activate (EPA) requests.										
7:6	Reserved.										
5:2	SpecPchgCmdSep. Read-write. Reset: 4h. Check: 4h. Specifies the minimum number of memory clock cycles between speculative precharge commands.										

	ValidValues:	
	Value	Description
	3h-0h	Reserved.
	Fh-4h	SpecPchgCmdSep MEMCLKs
1	ForcePchgPgtFull. Read-write. Reset: 1. Check: 1. 0=Disable. 1=Enable. Enable forcing precharge when page table is full in associative page table mode. Constraint: ($\sim$ DDR5En (DDR5En && (ForcePchgPgtFull SpecPchgEn))).	
0	PgtClkGateDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disables medium grain clock gating in PGT block.	

UMC[00...11]CHx0000010C [AE Control 0] (UMC::AeCtrl0)

Read-write. Reset: 0000_0020h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000010C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description						
31:27	Reserved.						
26	RefMaskForceEntry. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Prevent a force entry request from blocking other DCQ entries if the rank of the force entry request is under refresh period.						
25	BlockRdAPTwrOk. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Block sending auto-precharge with read CAS commands based on Twr timing from the last write to the same page.						
24	DisplOptEn. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Enables a display optimization mode tracking the number of display urgent requests in the DCQ. If the count of display urgent requests is greater than DisplUrgReqCntThresh, and until expiration of the DisplUrgReqHyst hysteresis, the force entry logic is disabled.						
23:22	Reserved.						
21:16	PageSwitchThresh. Read-write. Reset: 00h. Check: 00h. Specifies the maximum number of CAS commands scheduled to a page in the presence of a page conflict. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>PageSwitchThresh Commands</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	PageSwitchThresh Commands
Value	Description						
00h	Disable						
3Fh-01h	PageSwitchThresh Commands						
15:14	Reserved.						
13:8	RankSwitchThresh. Read-write. Reset: 00h. Check: 00h. Specifies the maximum number of CAS commands scheduled to a rank in the presence of a different rank page hit. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>RankSwitchThresh Commands</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	RankSwitchThresh Commands
Value	Description						
00h	Disable						
3Fh-01h	RankSwitchThresh Commands						
7	ArbBypassEn. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Enables DCQ insertion bypass arbitration for new requests when the queue is empty.						
6	TwoActEn. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Enables arbitration of two ACTs per UCLK cycle when !UclkDiv1Mode.						
5	Arb1Cmd. Read-write. Reset: 1. Init: 0. 0=Enable. 1=Disable. Disable arbitration of two commands per UCLK cycle when !UclkDiv1Mode.						
4	DisPageHistPred. Read-write. Reset: 0. Init: 1. 0=Enable. 1=Disable. Disable use of page history information for auto-precharge prediction.						
3	DisReqChain. Read-write. Reset: 0. Init: 1. 0=Enable. 1=Disable. Disable use of SDP ReqChain indicator for auto-precharge prediction.						
2	ClosedPagePolicy. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Close page with the last CAS command if no other requests to that page are in the DCQ.						
1	DisAutoPchg. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disables sending any CAS command with auto-precharge.						
0	ForceAutoPchg. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Specifies a force of auto-precharge with every CAS command.						

UMC[00...11]CHx00000110 [AE Control 1] (UMC::AeCtrl1)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000110; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description						
31	Reserved.						
30	DisWrSameBGMask. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disable blocking ACTs to writes of the same bank group when there are page-miss writes to other bank-groups present in the queue. Used to optimize arbitration around DDR5 On-die ECC (Tccd_l_wr) limitations.						
29:24	MinWrThreshHiPriRd. Read-write. Reset: 00h. Minimum threshold count of writes before a switch to urgent or high priority reads, overrides MinWrThresh. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>3Fh-00h</td><td>MinWrThreshHiPriRd Writes</td></tr> </table>	Value	Description	3Fh-00h	MinWrThreshHiPriRd Writes		
Value	Description						
3Fh-00h	MinWrThreshHiPriRd Writes						
23	Reserved.						
22	DisHiPriWinOverPh. Read-write. Reset: 0. Init: 1. 0=Enable. 1=Disable. Specifies arbitration mode such that high priority requests win arbitration over lower priority requests with better page states.						
21:16	Reserved.						
15	UseRefMaskRdWrCnt. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Enables use of a masked version of outstanding reads/writes that considers pending urgent refreshes and the Trfc period after a refresh.						
14	DisWriteThresh. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disable write thresholding logic which bursts out reads and writes per priority thresholds. Constraint: IF (DisWriteThresh == 1) THEN RankSwitchThresh = 0 ENDIF. Constraint: IF (DisWriteThresh == 1) THEN MaxBypass.BypassPriEscEn != 0 ENDIF.						
13	DisSubUrgRefMask. Read-write. Reset: 0. Check: 1. 0=Enable. 1=Disable. Disable Sub Urgent Refresh Mask logic in DCQARB. SubUrgRefMsk is a performance feature that selectively blocks some DCQ requests when SPAZ indicates a rank is in SubUrgent Mode and SpazCtrl.PerBankRefEn is enabled.						
12	DisUrgRefSameBankDone. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disable Urgent Refresh Same Bank Done logic in DCQARB. UrgRefSameBankDone is a performance feature that selectively allows some DCQ requests to make progress even when SPAZ indicates a rank is in Urgent Mode and SpazCtrl.PerBankRefEn is enabled.						
11:9	Reserved.						
8:6	DisplUrgReqHyst. Read-write. Reset: 0h. Check: 0h. Hysteresis after DisplUrgReqCntThresh has been met. See: DisplOptEn. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>7h-0h</td><td>(DisplUrgReqHyst + 1) x 64 MEMCLKs</td></tr> </table>	Value	Description	7h-0h	(DisplUrgReqHyst + 1) x 64 MEMCLKs		
Value	Description						
7h-0h	(DisplUrgReqHyst + 1) x 64 MEMCLKs						
5:0	DisplUrgReqCntThresh. Read-write. Reset: 00h. Check: 00h. Threshold count of display urgent requests. See: DisplOptEn. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disable</td></tr> <tr> <td>3Fh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disable	3Fh-01h	Reserved.
Value	Description						
00h	Disable						
3Fh-01h	Reserved.						

UMC[00...11]CHx00000114 [AE Control 2] (UMC::AeCtrl2)

Read-write. Reset: 1818_2000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000114; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description				
31:24	MinRdThresh. Read-write. Reset: 18h. Init: 20h. Minimum threshold count of reads before a switch to writes. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>FFh-00h</td><td>MinRdThresh Reads</td></tr> </table>	Value	Description	FFh-00h	MinRdThresh Reads
Value	Description				
FFh-00h	MinRdThresh Reads				
23:16	MinWrThresh. Read-write. Reset: 18h. Init: 01h. Minimum threshold count of writes before a switch to reads. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>FFh-00h</td><td>MinWrThresh Writes</td></tr> </table>	Value	Description	FFh-00h	MinWrThresh Writes
Value	Description				
FFh-00h	MinWrThresh Writes				
15:8	TotWrThresh. Read-write. Reset: 20h. Check: 20h. Threshold count of total DCQ weighted writes to collect before switching to writes. See: LoPriWrWgt, MedPriWrWgt, HiPriWrWgt, UrgPriWrWgt. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>FFh-00h</td><td>TotWrThresh Writes</td></tr> </table>	Value	Description	FFh-00h	TotWrThresh Writes
Value	Description				
FFh-00h	TotWrThresh Writes				
7:0	TotRdThresh. Read-write. Reset: 00h. Check: 00h. Threshold count of total DCQ weighted reads to collect before switching to reads. See: LoPriRdWgt, MedPriRdWgt, HiPriRdWgt, UrgPriRdWgt. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>FFh-00h</td><td>TotRdThresh Reads</td></tr> </table>	Value	Description	FFh-00h	TotRdThresh Reads
Value	Description				
FFh-00h	TotRdThresh Reads				

UMC[00...11]CHx00000118 [UMC DCQ Control] (UMC::UmcDcqCntl)

Read-write. Reset: 0000_0047h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000118; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:7	Reserved.
6	CohTagFirst. Read-write. Reset: 1. Check: 1. 0=Disable. 1=Enable. Propagate Tag based priority escalation first, ahead of VC based priority escalation, if both are requested.
5:3	Reserved.
2	CohNewdepElevEn. Read-write. Reset: 1. Check: 1. 0=Disable. 1=Enable. Enable priority elevation of preceding requests per new request dependency with higher priority.
1	CohTagElevEn. Read-write. Reset: 1. Check: 1. 0=Disable. 1=Enable. Enable SDP Tag based priority elevation.
0	CohVcElevEn. Read-write. Reset: 1. Check: 1. 0=Disable. 1=Enable. Enable SDP VC based priority elevation.

UMC[00...11]CHx0000011C [BEQ Control 0] (UMC::BeqCtrl0)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000011C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	Reserved.
30	DramRst. Read-write. Reset: 0. Check: 0. 0=Deassert. 1=Assert. Direct control of dfi_reset_n pin via inverter. This bit is used for software initialization only and must be reset for normal operation. This bit must be reset prior to connecting UMC to PHY when in retention (S3).
29	Reserved.
28	DfiInitStart. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Direct control of dfi_init_start signal. This bit is used for software initialization only, is logically OR'd with hardware control of the same pin, and must be reset for normal operation.
27	Reserved.
26	RDimmSdrMode. Read-write. Reset: 0. 0=Disable. 1=Enable. RDIMM/LRDIMM of DDR5 is in SDR Mode.
25	RDimmParEn. Read-write. Reset: 0. Enable RCD command and address parity for RDIMM/LRDIMM. Constraint: !(RDimmParEn && CmdParEn).
24:20	Reserved.
19	CSOvrVal. Read-write. Reset: 0. Specifies override of the BEQ CS value.
18	CSOvrEn. Read-write. Reset: 0. Specifies override of the BEQ CS source is enabled. See: CsOvrEn.
17:4	Reserved.
3	VcmEn. Read-write. Reset: 0. Init: 1. Check: 1. 0=Disable. 1=Enable. Enables virtual controller mode.
2:0	Reserved.

UMC[00...11]CHx00000120 [BEQ Control 1] (UMC::BeqCtrl1)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000120; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:14	Reserved.
13	DisCtrlUpdRollOrderRst. Read-write. Reset: 0. Check: 0. Disable resetting of data rolling order pointer on CtrlUpd Request
12	DisPhyUpdRollOrderRst. Read-write. Reset: 0. Check: 0. Disable resetting of data rolling order pointer on PhyUpd Request
11	DisDramClkOnLpReq. Read-write. Reset: 0. Check: 0. 0=Do not disable the DRAM clock. 1=Disable the DRAM clock. Specifies behavior to disable the DRAM clock by dfi_dram_clk_disable when dfi_lp_ctrl_req and dfi_lp_ctrl_ack are asserted.
10	PdeOdtMode. Read-write. Reset: 0. 0=Disable. 1=Enable. Specifies DDR5 PDE with NT ODT mode.
9	DfiLpDatEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable DFI data low power requests.
8	RdCrcEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable read CRC generation.
7	RdCrcFatalEn. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Promote read CRC errors from deferred to fatal when not replayed or the replay threshold is met.
6:1	Reserved.
0	WrCrcEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable write CRC generation.

UMC[00...11]CHx00000124 [BEQ Control 2] (UMC::BeqCtrl2)

Read-write. Reset: 0080_0001h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000124; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description								
31:28	DfiDramClkDis: DfiDramClkDis[3:0] . Read-write. Reset: 0h. Specifies the value of dfi_dram_clk_disable. Note that value seen on DFI will be overridden if DisDramClkOnLpReq is set depending on lp_ctrl_req behavior.								
27:26	Reserved.								
25:24	PhyRdLatMode . Read-write. Reset: 0h. Init: 1h. Specifies the relation between Tphy_rdlat_even (Tphy_rdlat for dfi_rddata_en on even DFI PHY clock) and Tphy_rdlat_odd when dfi_freq_ratio is 1:2. Valid Values:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Tphy_rdlat_odd equals to Tphy_rdlat_even</td></tr> <tr> <td>1h</td><td>Tphy_rdlat_odd equals to Tphy_rdlat_even minus one</td></tr> <tr> <td>3h-2h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Tphy_rdlat_odd equals to Tphy_rdlat_even	1h	Tphy_rdlat_odd equals to Tphy_rdlat_even minus one	3h-2h	Reserved.
Value	Description								
0h	Tphy_rdlat_odd equals to Tphy_rdlat_even								
1h	Tphy_rdlat_odd equals to Tphy_rdlat_even minus one								
3h-2h	Reserved.								
23:20	Tvrefdur . Read-write. Reset: 8h. Init: 8h. Multi cycle CS assertion duration for VREFCA, VREFCS commands in cycles.								
19:16	Reserved.								
15	RdDbiEn . Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Enable read DBI.								
14:13	Reserved.								
12	WrDbiEn . Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Enable write DBI.								
11	DisDatMsk . Read-write. Reset: 0. Init: 1. 0=Enable. 1=Disable. Disable (DM) Data Masking feature. Must be set in conjunction with DF configuration bit.								
10:1	Reserved.								
0	AdrDf . Read-write. Reset: 1. Check: 1. 0=Low. 1=High. Specifies the default value for address.								

UMC[00...11]CHx00000128 [BEQ Control 3] (UMC::BeqCtrl3)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000128; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:4	Reserved.
3:0	Dfi1DramClkDis: Dfi1DramClkDis[3:0] . Read-write. Reset: 0h. Specifies the value of dfi_dram_clk_disable on sub channel 1. Note that value seen on DFI will be overridden if DisDramClkOnLpReq is set depending on lp_ctrl_req behavior.

UMC[00...11]CHx0000012C [SPAZ Control] (UMC::SpazCtrl)

Read-write. Reset: 0118_0C69h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000012C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	Reserved.
30	AggrPwrDownEn. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Enable Aggressive Power Down Mode.
29	PwrDownMode. Read-write. Reset: 0. Init: 1. 0=Full Channel Power Down. 1=Partial Channel Power Down. Specifies the DRAM Power Down Mode scheme.
28	PwrDownEn. Read-write. Reset: 0. Init: 0. 0=Disable. 1=Enable. Enable DRAM Power Down Mode.
27:24	AutoRefCmdSep. Read-write. Reset: 1h. Init: 5h. Specifies the minimum command separation between REF commands.
ValidValues:	
Value	Description
0h	Reserved.
Fh-1h	AutoRefCmdSep MEMCLKs
23:20	PchgCmdSep. Read-write. Reset: 1h. Init: 3h. Specifies the minimum command separation between PRE commands.
ValidValues:	
Value	Description
0h	Reserved.
Fh-1h	PchgCmdSep MEMCLKs
19	AllBankRefEn. Read-write. Reset: 1. Check: 1. 0=Disable. 1=Enable. Enable all-bank refresh. REFab is enabled if (AllBankRefEn !PerBankRefEn). A setting of AllBankRefEn & PerBankRefEn specifies mixed-mode operation. Constraint:1.
18:16	AutoRefFineGranMode. Read-write. Reset: 0h. Specifies the fine granularity refresh mode of operation. Fixed 4X, OTF 2X, and OTF 4X modes reserved if DDR5 mode.
ValidValues:	
Value	Description
0h	Fixed 1X
1h	Fixed 2X
2h	Fixed 4X
4h-3h	Reserved.
5h	OTF 2X
6h	OTF 4X
7h	Reserved.
15:11	Tlp_wakeup. Read-write. Reset: 01h. Check: 01h. Specifies the DFI lp_ctrl_wakeup and lp_data_wakeup values. This field is transmitted to the phy on dfi_lp_ctrl_wakeup and dfi_lp_data_wakeup on the asserting edge of the corresponding low power request.
10:8	SubUrgRefLowerBound. Read-write. Reset: 4h. Init: 1h. Specifies the stored refresh limit required to enter sub-urgent refresh mode. Constraint: SubUrgRefLowerBound <= UrgRefLimit.
ValidValues:	
Value	Description
0h	Reserved.
6h-1h	SubUrgRefLowerBound Refreshes
7h	Reserved.
7	Reserved.
6:4	UrgRefLimit. Read-write. Reset: 6h. Init: 4h. Specifies the stored refresh limit required to enter urgent refresh mode.

ValidValues:	
Value	Description
0h	Reserved.
6h-1h	UrgRefLimit Refreshes
7h	Reserved.
3	LpDis. Read-write. Reset: 1. Init: 0. 0=Enable. 1=Disable. Disable DFI low power requests.
2	ZqStartSpazRply. Read-write. Reset: 0. Check: 0. 0=REC to Replay ZqStart. 1=SPAZ to Replay ZqStart. When BlockZqRply is not asserted, this bit enables SPAZ to replay ZqStart instead of REC
1	PerBankRefEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable per-bank or same-bank refresh. Enables same-bank refresh (REFsb) if DDR5 mode.
0	DisAutoRefresh. Read-write. Reset: 1. Init: 0. 0=Enable. 1=Disable. Disables periodic refresh.

UMC[00...11]CHx00000130 [Throttle and Thermal Control] (UMC::ThrottleCtrl)

Constraint: IF (AutoRefFineGranMode > 0 && Ranks Populated >= 8) THEN SwCmdThrotCyc/RollWindowDepth >= 0.5 ENDIF.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000130; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	OdsThrottleStatus. Read-only, Volatile. Check: 0. Indicates when ODTS is throttling commands.
30:29	Reserved.
28	OdsIncRefEn. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Enable refresh rate increase based on thermal polling results.
27	OdsEn. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Enable thermal refresh rate polling MR4 for DDR5 thermal status. If this bit is not set, refresh rate will not be changed based on thermal status. When set, and when thermal status reaches the thermal threshold, refresh rate will be increased
26	ForcePwrDownThrotEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Force ranks into power down during throttling. Constraint: !ForcePwrDownThrotEn & (CMapCKE.CMapCKE0 == 0) (CMapCKE.CMapCKE1 == 0) (CMapCKE.CMapCKE2 == 0) (CMapCKE.CMapCKE3 == 0).
25	OdsCmdThrotEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable OdsCmdThrot based throttling.
24	I2CThermEvent. Read-write. Reset: 0. 0=Disable. 1=Enable. Indicates I2C thermal event
23:20	Reserved.
19:10	OdsCmdThrotCyc. Read-write. Reset: 000h. Specifies the weighted allowed cycles per rolling window depth. Constraint: 8 <= OdsCmdThrotCyc <= RollWindowDepth.
ValidValues:	
Value	Description
000h	Reserved.
3FFh-001h	OdsCmdThrotCyc MEMCLKs
9:0	RollWindowDepth. Read-write. Reset: 000h. Init: 1FFh. Specifies the throttling rolling window depth.
ValidValues:	
Value	Description
000h	Reserved.
3FFh-001h	RollWindowDepth MEMCLKs

UMC[00...11]CHx00000138 [Maximum Bypass Control] (UMC::MaxBypass)

Read-write. Reset: 0720_4060h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000138; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description										
31:29	Reserved.										
28:27	MaxBypassMult. Read-write. Reset: 0h. Check: 0h. Specifies the multiplier for the MaxBypassLo, MaxBypassMed, and MaxBypassHigh field encoding. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>1x</td></tr> <tr> <td>1h</td><td>2x</td></tr> <tr> <td>2h</td><td>4x</td></tr> <tr> <td>3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	1x	1h	2x	2h	4x	3h	Reserved.
Value	Description										
0h	1x										
1h	2x										
2h	4x										
3h	Reserved.										
26	BypassPriEscMode. Read-write. Reset: 1. Check: 1. 0=Urgent. 1=Next Higher Priority. Specifies the bypass priority escalation behavior when a request priority max bypass threshold is met.										
25	BypassPriEscEn. Read-write. Reset: 1. Check: 1. 0=Disable. 1=Enable. Specifies DCQ bypassing is enabled. See: ForceEntryEn and BypassPriEscMode. Constraint: IF (BypassPriEscEn == 0) THEN AeCtrl1.DisWriteThresh != 1 ENDIF.										
24	ForceEntryEn. Read-write. Reset: 1. Check: 1. 0=Disable. 1=Enable. Specifies DCQ ForceEntry logic is enabled when the urgent priority request bypass threshold is met.										
23:16	MaxBypassHi. Read-write. Reset: 20h. Init: 40h. Specifies high and urgent priority request bypass threshold. See: MaxBypassMult. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>FFh-00h</td><td>(MaxBypassHi << MaxBypassMult) Commands</td></tr> </table>	Value	Description	FFh-00h	(MaxBypassHi << MaxBypassMult) Commands						
Value	Description										
FFh-00h	(MaxBypassHi << MaxBypassMult) Commands										
15:8	MaxBypassMed. Read-write. Reset: 40h. Init: C0h. Specifies the medium priority request bypass threshold. See: MaxBypassMult. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>FFh-00h</td><td>(MaxBypassMed << MaxBypassMult) Commands</td></tr> </table>	Value	Description	FFh-00h	(MaxBypassMed << MaxBypassMult) Commands						
Value	Description										
FFh-00h	(MaxBypassMed << MaxBypassMult) Commands										
7:0	MaxBypassLo. Read-write. Reset: 60h. Init: C0h. Specifies the low priority request bypass threshold. See: MaxBypassMult. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>FFh-00h</td><td>(MaxBypassLo << MaxBypassMult) Commands</td></tr> </table>	Value	Description	FFh-00h	(MaxBypassLo << MaxBypassMult) Commands						
Value	Description										
FFh-00h	(MaxBypassLo << MaxBypassMult) Commands										

UMC[00...11]CHx00000144 [Data Control] (UMC::DataCtrl)

Read-write. Reset: 0000_2000h.

Configuration enables for UMC data encryption and data scrambling. See: EncrArrData, TwkArrData, XtsArrData. See: DataScrambleKey.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000144; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:17	Reserved.
16	DisAddrTweak: DisAddrTweak. Read-write. Reset: 0. Check: 0. 0=Address based tweak enabled. 1=Address based tweak disabled. Disable address run time variable tweaking in AES-128 Mode. Applies to all regions. Unused when AES-128 is disabled
15	Reserved.
14	AESXTSEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable AES-XTS Mode. Only valid when AES-256 mode is enabled. See: AES256En.
13	AES256En. Read-write. Reset: 1. 0=Disable. 1=Enable. Enable AES-256 mode. Encrypt memory with AES 256b encryption keys.
12:9	Reserved.
8	DataEncrEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enables data encryption.
7:1	Reserved.
0	DataScrambleEn. Read-write. Reset: 0. 0=Disable. 1=Enable.

UMC[00...11]CHx00000148 [Data Scramble Key] (UMC::DataScrambleKey)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000148; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:0	DataScrambleKey. Read-write. Reset: 0000_0000h. Specifies the data scrambling key. See: DataScrambleEn.

UMC[00...11]CHx0000014C [DRAM ECC Control] (UMC::EccCtrl)

Read-write. Reset: 0000_4000h.

Configuration bits for DRAM ECC.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000014C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:16	Reserved.
15	AddrXorEn. Read-write. Reset: 0. Init: 1. 0=Disable: Write/Read data will not be XORed with NormAddr. 1=Enable : Write/Read data will be XORed with NormAddr. XOR Normalized address with Write/Read data
14	PinReducedEcc. Read-write. Reset: 1. 0=Disable. 1=Enable. Reduce ECC to 8-bits per 64-bits of data for DDR5.
13:11	Reserved.
10	RdEccEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable DRAM data ECC checking and correction.
9	EccSymbolSize16. Read-write. Reset: 0. 0=Use EccSymbolSize (x4/x8 symbol). 1=x16 symbol.
8	UCFatalEn. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Promote uncorrectable errors from deferred to fatal.
7	EccSymbolSize. Read-write. Reset: 0. Check: 0. 0=x4 symbol. 1=x8 symbol. Specifies ECC symbol size. Reserved if EccSymbolSize16 = 1.
6	EccBitInterleaving. Read-write. Reset: 0. 0=Disable. 1=Enable. Enables data burst bit interleaving for ECC.
5	EccHardwareHistoryEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enables the hardware managed ECC history mechanism for x8 and x16 symbol size.
4	EccBadDramSymEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enables the software managed ECC history mechanism for x8 and x16 symbol size.
3:1	Reserved.
0	WrEccEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enables ECC generation for DRAM data. Also enables BEQ transmit of ECC data.

UMC[00...11]CHx00000150 [Recovery Control] (UMC::RecCtrl)

Reset: 0000_6000h.

Configuration for recovery logic.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000150; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	ClrFatal. Read-write, Volatile. Reset: 0. Check: 0. Cleared-by-hardware. Clear fatal status flag.
30	ClrError. Read-write, Volatile. Reset: 0. Cleared-by-hardware. Clear error status flag.
29	Reserved.
28	BlockOscStartRply. Read-write. Reset: 0. 0=Allow MPC OscStart to go through replay. 1=block MPC OscStart for replay.
27:22	MaxCrcRply. Read-write. Reset: 00h. Max allowed recursive Write CRC error replay.
ValidValues:	
Value	Description
00h	Infinite replay
3Fh-01h	MaxCrcRply replay
21:16	MaxParRply. Read-write. Reset: 00h. Max allowed recursive parity error replay attempts.
ValidValues:	
Value	Description
00h	Infinite replay
3Fh-01h	MaxParRply replay
15	BlockZqRply. Read-write. Reset: 0. Check: 0. 0=Allow ZqStart/Latch to go through replay. 1=block ZqStart/Latch for replay.
14	RplySpazReq. Read-write. Reset: 1. Init: 0. 0=Ignore SPAZ urgent request when REC controls the bus. 1=Accept SPAZ urgent request when REC controls the bus.
13	EccOptEn. Read-write. Reset: 1. Init: 0. 0=disable ECC optimization feature. 1=enable ECC optimization feature.
12:9	Reserved.
8:5	RcvrMode. Read-write. Reset: 0h. Recovery mode select.
4	BusyChk. Read-write. Reset: 0. 0=Recover all ranks. 1=Recover only ranks with activity. Specifies recovery behavior regarding rank activity in the protected command window.
3:0	RecEn. Read-write. Reset: 0h. Enables recovery for the specified error types.
ValidValues:	
Bit	Description
[0]	Parity.
[1]	Write CRC.
[2]	UECC. Reserved when ECC replay is disabled.
[3]	DDR5 Read CRC. Reserved when Read CRC replay is disabled.

UMC[00...11]CHx00000154 [Recovery Timing] (UMC::RecTiming)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000154; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description						
31:24	RcvrWaitBus. Read-write. Reset: 00h. Wait time to trigger dfi_ctrlupd_req signal. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Use RcvrWait counter.</td></tr> <tr> <td>FFh-01h</td><td>RcvrWaitBus clocks</td></tr> </table>	Value	Description	00h	Use RcvrWait counter.	FFh-01h	RcvrWaitBus clocks
Value	Description						
00h	Use RcvrWait counter.						
FFh-01h	RcvrWaitBus clocks						
23:11	Reserved.						
10	CmdStgFrc. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable recovery command stagger in recovery phase per CmdStgCnt.						
9:0	Reserved.						

UMC[00...11]CHx00000158 [Recovery Control 1] (UMC::RecCtrl1)

Read-write. Reset: 0010_8000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000158; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description				
31:29	Reserved.				
28:23	RecWtdg. Read-write. Reset: 00h. Recovery watchdog timer. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>3Fh-00h</td><td>RecWtdg clocks</td></tr> </table>	Value	Description	3Fh-00h	RecWtdg clocks
Value	Description				
3Fh-00h	RecWtdg clocks				
22	RcvrOrder. Read-write. Reset: 0. 0=Sequential. 1=Broadcast. Specifies rank ordering during recovery sequence				
21:16	CtrlUpdHold. Read-write. Reset: 10h. Check: 1Ch. DFI dfi_ctrlupd_reqctrl_upd hold time after deassertion.				
15:10	CtrlUpdPulse. Read-write. Reset: 20h. Check: 20h. DFI dfi_ctrlupd_reqctrl_upd pulse width.				
9:0	Reserved.				

UMC[00...11]CHx00000168 [AE Control 3] (UMC::AeCtrl3)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000168; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	Reserved.
30	DisEarlyRdActs. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disables early read ACTs based on read/write switching counts.
29	DynSnapRdThreshEn. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Dynamic snapshot based read threshold enable.
28	Reserved.
27:16	MaxRdCycleCnt. Read-write. Reset: 000h. Check: 000h. Maximum number of cycles to be in read mode before switching to writes if the MinRdThresh is not met. [0x000: Disabled, 0xFFFF .. 0x001: <MaxRdCycleCnt> MEMCLKs]
15	Reserved.
14	DisEarlyWrActs. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disable early write ACTs based on read/write switching counts.
13	DynSnapWrThreshEn. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Dynamic snapshot based write threshold enable.
12	Reserved.
11:0	MaxWrCycleCnt. Read-write. Reset: 000h. Init: 100h. Maximum number of cycles to be in write mode before switching to reads if the MinWrThresh is not met. [0x000: Disabled, 0xFFFF .. 0x001: <MaxWrCycleCnt> MEMCLKs]

UMC[00...11]CHx0000016C [AE Control 4] (UMC::AeCtrl4)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000016C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description				
31	Reserved.				
30:28	UrgPriRdWgt. Read-write. Reset: 0h. Check: 0h. Weight to use for counting Urgent Priority Reads to compare with configured thresholds. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>7h-0h</td><td>(UrgPriRdWgt)+1 Weighting</td></tr> </table>	Value	Description	7h-0h	(UrgPriRdWgt)+1 Weighting
Value	Description				
7h-0h	(UrgPriRdWgt)+1 Weighting				
27	Reserved.				
26:24	HiPriRdWgt. Read-write. Reset: 0h. Check: 0h. Weight to use for counting High Priority Reads to compare with configured thresholds. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>7h-0h</td><td>(HiPriRdWgt)+1 Weighting</td></tr> </table>	Value	Description	7h-0h	(HiPriRdWgt)+1 Weighting
Value	Description				
7h-0h	(HiPriRdWgt)+1 Weighting				
23	Reserved.				
22:20	MedPriRdWgt. Read-write. Reset: 0h. Check: 0h. Weight to use for counting Medium Priority Reads to compare with configured thresholds. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>7h-0h</td><td>(MedPriRdWgt)+1 Weighting</td></tr> </table>	Value	Description	7h-0h	(MedPriRdWgt)+1 Weighting
Value	Description				
7h-0h	(MedPriRdWgt)+1 Weighting				
19	Reserved.				
18:16	LoPriRdWgt. Read-write. Reset: 0h. Check: 0h. Weight to use for counting Low Priority Reads to compare with configured thresholds. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>7h-0h</td><td>(LoPriRdWgt)+1 Weighting</td></tr> </table>	Value	Description	7h-0h	(LoPriRdWgt)+1 Weighting
Value	Description				
7h-0h	(LoPriRdWgt)+1 Weighting				
15	Reserved.				
14:12	UrgPriWrWgt. Read-write. Reset: 0h. Check: 0h. Weight to use for counting Urgent Priority Writes to compare with configured thresholds. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>7h-0h</td><td>(UrgPriWrWgt)+1 Weighting</td></tr> </table>	Value	Description	7h-0h	(UrgPriWrWgt)+1 Weighting
Value	Description				
7h-0h	(UrgPriWrWgt)+1 Weighting				
11	Reserved.				
10:8	HiPriWrWgt. Read-write. Reset: 0h. Check: 0h. Weight to use for counting High Priority Writes to compare with configured thresholds. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>7h-0h</td><td>(HiPriWrWgt)+1 Weighting</td></tr> </table>	Value	Description	7h-0h	(HiPriWrWgt)+1 Weighting
Value	Description				
7h-0h	(HiPriWrWgt)+1 Weighting				
7	Reserved.				
6:4	MedPriWrWgt. Read-write. Reset: 0h. Check: 0h. Weight to use for counting Medium Priority Writes to compare with configured thresholds. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>7h-0h</td><td>(MedPriWrWgt)+1 Weighting</td></tr> </table>	Value	Description	7h-0h	(MedPriWrWgt)+1 Weighting
Value	Description				
7h-0h	(MedPriWrWgt)+1 Weighting				
3	Reserved.				
2:0	LoPriWrWgt. Read-write. Reset: 0h. Check: 0h. Weight to use for counting Low Priority Writes to compare with configured thresholds.				

	ValidValues:	
	Value	Description
	7h-0h	(LoPriWrWgt)+1 Weighting

UMC[00...11]CHx00000174 [DRAM Scrub Control] (UMC::DramScrubCtrl)

Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000174; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description																																		
31:20	Reserved.																																		
19:16	DramScrubRate. Read-write. Reset: 0h. Patrol Scrub Rate assuming 100MHz REFCLK . 64B is scrubbed every period. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr><td>0h</td><td>640 ns (64 REFCLKs)</td></tr> <tr><td>1h</td><td>1.28 us (128 REFCLKs)</td></tr> <tr><td>2h</td><td>2.56 us (256 REFCLKs)</td></tr> <tr><td>3h</td><td>5.12 us (512 REFCLKs)</td></tr> <tr><td>4h</td><td>10.24 us (1K REFCLKs)</td></tr> <tr><td>5h</td><td>20.48 us (2K REFCLKs)</td></tr> <tr><td>6h</td><td>40.96 us (4K REFCLKs)</td></tr> <tr><td>7h</td><td>81.92 us (8K REFCLKs)</td></tr> <tr><td>8h</td><td>163.84 us (16K REFCLKs)</td></tr> <tr><td>9h</td><td>327.68 us (32K REFCLKs)</td></tr> <tr><td>Ah</td><td>655.36 us (64K REFCLKs)</td></tr> <tr><td>Bh</td><td>1.31 ms (128K REFCLKs)</td></tr> <tr><td>Ch</td><td>2.62 ms (256K REFCLKs)</td></tr> <tr><td>Dh</td><td>5.24 ms (512K REFCLKs)</td></tr> <tr><td>Eh</td><td>10.48 ms (1M REFCLKs)</td></tr> <tr><td>Fh</td><td>20.96 ms (2M REFCLKs)</td></tr> </table>	Value	Description	0h	640 ns (64 REFCLKs)	1h	1.28 us (128 REFCLKs)	2h	2.56 us (256 REFCLKs)	3h	5.12 us (512 REFCLKs)	4h	10.24 us (1K REFCLKs)	5h	20.48 us (2K REFCLKs)	6h	40.96 us (4K REFCLKs)	7h	81.92 us (8K REFCLKs)	8h	163.84 us (16K REFCLKs)	9h	327.68 us (32K REFCLKs)	Ah	655.36 us (64K REFCLKs)	Bh	1.31 ms (128K REFCLKs)	Ch	2.62 ms (256K REFCLKs)	Dh	5.24 ms (512K REFCLKs)	Eh	10.48 ms (1M REFCLKs)	Fh	20.96 ms (2M REFCLKs)
Value	Description																																		
0h	640 ns (64 REFCLKs)																																		
1h	1.28 us (128 REFCLKs)																																		
2h	2.56 us (256 REFCLKs)																																		
3h	5.12 us (512 REFCLKs)																																		
4h	10.24 us (1K REFCLKs)																																		
5h	20.48 us (2K REFCLKs)																																		
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9h	327.68 us (32K REFCLKs)																																		
Ah	655.36 us (64K REFCLKs)																																		
Bh	1.31 ms (128K REFCLKs)																																		
Ch	2.62 ms (256K REFCLKs)																																		
Dh	5.24 ms (512K REFCLKs)																																		
Eh	10.48 ms (1M REFCLKs)																																		
Fh	20.96 ms (2M REFCLKs)																																		
15	RstWrBufPstMemClr. Read-write. Reset: 0. 0=Don't reset the buffers post MemClr operation. 1=Reset the buffers post MemClr operation. Reset buffers dependent on WrDeAlloc(WDB,BEQ,FEI ...) post MemClr operation																																		
14:12	Reserved.																																		
11	DisScrubUcRply. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disable replay of a patrol scrub read which results in an uncorrectable error. Applicable only when ECC Replay is enabled.																																		
10	DropScrubWrColl. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Configure logic to drop the scrub write for a patrol scrub read when there is already a demand write to same address in the DCQ.																																		
9	DropScrubWrNoErr. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Configure logic to drop the scrub write for a patrol scrub read without an ECC error.																																		
8	DramScrubEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable patrol scrubber to periodically read and write corrected data back to DRAM.																																		
7:6	ScrubReqLim. Read-write. Reset: 0h. Init: 0h. Specifies a limit on the number of redirect scrubs that can be handled simultaneously. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr><td>0h</td><td>8 Scrubs</td></tr> <tr><td>1h</td><td>4 Scrubs</td></tr> <tr><td>2h</td><td>2 Scrubs</td></tr> <tr><td>3h</td><td>1 Scrub</td></tr> </table>	Value	Description	0h	8 Scrubs	1h	4 Scrubs	2h	2 Scrubs	3h	1 Scrub																								
Value	Description																																		
0h	8 Scrubs																																		
1h	4 Scrubs																																		
2h	2 Scrubs																																		
3h	1 Scrub																																		
5	PoisonScrubEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable redirect scrubber to write back uncorrectable error as poison into DRAM.																																		
4	RedirectScrubEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable redirect scrubber to write back corrected data into DRAM.																																		

3	Reserved.
2	DisMemClr. Reset: 0. 0=Enable. 1=Disable. Disable Memory Clear. Expected to be set once memory initialization is complete. Once the bit is set it cannot be cleared until a Reset. AccessType: (UMC::DramScrubCtrl[DisMemClr] == 0) ? Read-write : Read-only.
1	MemClrDone. Read-only, Volatile. Reset: 0. Indicates the memory clear operation is done.
0	MemClrEn. Reset: 0. 0=Disable. 1=Enable. Enable memory clear of a region of DRAM by writing all zeroes to data and appropriate ECC. The region to be cleared is specified by DramScrubBaseAddr and DramScrubLimitAddr. AccessType: (UMC::DramScrubCtrl[DisMemClr] == 0) ? Read-write : Read-only.

UMC[00...11]CHx00000178 [DRAM Scrub Base Address] (UMC::DramScrubBaseAddr)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000178; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:4	ScrubBaseAddr: Scrub Base Address [47:20]. Read-write. Reset: 000_0000h. Check: 000_0000h. Specifies the normalized base address of the DRAM region for memory clear and periodic scrub.
3:0	Reserved.

UMC[00...11]CHx0000017C [DRAM Scrub Limit Address] (UMC::DramScrubLimitAddr)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000017C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:4	ScrubLimitAddr: Scrub Limit Address [47:20]. Read-write. Reset: 000_0000h. Specifies the normalized limit address of the DRAM region for memory clear and periodic scrub.
3:0	Reserved.

UMC[00...11]CHx00000180 [DRAM Scrub Addr Low] (UMC::DramScrubAddrLo)

Read-write, Volatile. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000180; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:6	SeqScrubAddrLo. Read-write, Volatile. Reset: 000_0000h. Patrol Scrubber current position address low. Specifies Addr[31:6] of the current position of the scrubber. This normalized address can be written by software to re-direct the patrol scrubber. Address is incremented by hardware. Note: Patrol scrub should be disabled by software before writing to this register to change the current address and AddrLo should be written first and then AddrHi.
5:0	Reserved.

UMC[00...11]CHx00000184 [DRAM Scrub Addr High] (UMC::DramScrubAddrHi)

Read-write, Volatile. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000184; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:20	Reserved.
19:4	SeqScrubAddrHi. Read-write, Volatile. Reset: 0000h. Patrol Scrubber current position address high. Specifies Addr[47:32] of the current position of the scrubber. This normalized address can be written by software to re-direct the patrol scrubber. Address is incremented by hardware.
3:0	Reserved.

UMC[00...11]CHx00000188 [Recovery Control 2] (UMC::RecCtrl2)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000188; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description								
31:15	Reserved.								
14	ForceRdAp. Read-write. Reset: 0. 0=during RPLY phase, no change of AP bit of RDs. 1=during RPLY phase, force AP bit to 1 for RDs. 0=during RPLY phase, no change of AP bit of RDs. 1=during RPLY phase, force AP bit to 1 for RDs. Constraint: IF 3DS or 2DPC configuration THEN (ForceRdAp = 1).								
13:12	Reserved.								
11:6	MaxRdCrcRply. Read-write. Reset: 00h. Max allowed recursive Read CRC error replay. ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Infinite replay</td></tr> <tr> <td>3Fh-01h</td><td>MaxRdCrcRply replay</td></tr> </table>	Value	Description	00h	Infinite replay	3Fh-01h	MaxRdCrcRply replay		
Value	Description								
00h	Infinite replay								
3Fh-01h	MaxRdCrcRply replay								
5:0	MaxEccRply. Read-write. Reset: 00h. Max allowed recursive ECC uncorrectable error replay. ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Infinite replay.</td></tr> <tr> <td>01h</td><td>Force Fatal UECC Error.</td></tr> <tr> <td>3Fh-02h</td><td>Maximum ECC Retry Replay count.</td></tr> </table>	Value	Description	00h	Infinite replay.	01h	Force Fatal UECC Error.	3Fh-02h	Maximum ECC Retry Replay count.
Value	Description								
00h	Infinite replay.								
01h	Force Fatal UECC Error.								
3Fh-02h	Maximum ECC Retry Replay count.								

UMC[00...11]CHx0000018C [Software Command Throttle] (UMC::SwCmdThrottle)

Read-write. Reset: 0000_0000h.

Constraints: IF (AutoRefFineGranMode > 0 && Ranks Populated >= 8) THEN SwCmdThrotCyc/RollWindowDepth >= 0.5 ENDIF.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx0000018C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description						
31:17	Reserved.						
16	SwCmdThrotEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable SwCmdThrotCyc based throttling.						
15:10	Reserved.						
9:0	SwCmdThrotCyc. Read-write. Reset: 000h. Specifies the weighted allowed cycles per rolling window depth. Constraint: 8 <= SwCmdThrotCyc <= RollWindowDepth. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>000h</td><td>Reserved.</td></tr> <tr> <td>3FFh-001h</td><td>SwCmdThrotCyc MEMCLKs</td></tr> </table>	Value	Description	000h	Reserved.	3FFh-001h	SwCmdThrotCyc MEMCLKs
Value	Description						
000h	Reserved.						
3FFh-001h	SwCmdThrotCyc MEMCLKs						

UMC[00...11]CHx00000190 [Dynamic PPR Scrub Control0] (UMC::DynPprCtrl0)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000190; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:0	DynPprBaseAddr: Dynamic PPR Scrub Base Address [39:8]. Read-write. Reset: 0000_0000h. Dynamic PPR Scrub Base Address [39:8].

UMC[00...11]CHx00000194 [Dynamic PPR Scrub Control1] (UMC::DynPprCtrl1)

Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000194; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	DynPprScrubVld. Read-write, Volatile. Reset: 0. 0=Disable. 1=Enable. Cleared-by-hardware. Setting this bit triggers scrubber operation starting from the DynPprBaseAddr incrementing until the DynPprScrubCnt is reached.
30	Reserved.
29:27	DynPprScrubPlr. Read-write. Reset: 0h. Plr of Dynamic PPR.
26:25	DynPprScrubCs. Read-write. Reset: 0h. CS of Dynamic PPR.
24	DynPprScrubChan. Read-write. Reset: 0. Sub-channel of Dynamic PPR.
23:16	DynPprBaseAddr: Dynamic PPR Scrub Base Address [47:40]. Read-write. Reset: 00h. Dynamic PPR Scrub Base Address [47:40].
15:0	DynPprScrubCnt. Read-write. Reset: 0000h. Specifies the Dynamic PPR Scrub Count.

UMC[00...11]CHx00000198 [AE Control 5] (UMC::AeCtrl5)

Read-write. Reset: 2004_0820h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000198; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description				
31:30	Reserved.				
29:24	WrActThresh. Read-write. Reset: 20h. Init: 1Bh. Number of writes to gather before allowing write ACTs when doing a read burst. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>3Fh-00h</td><td>WrActThresh pending writes</td></tr> </table>	Value	Description	3Fh-00h	WrActThresh pending writes
Value	Description				
3Fh-00h	WrActThresh pending writes				
23:22	Reserved.				
21:16	RdActThresh. Read-write. Reset: 04h. Init: 01h. Number of reads to gather before allowing read ACTs when doing a write burst. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>3Fh-00h</td><td>RdActThresh pending reads</td></tr> </table>	Value	Description	3Fh-00h	RdActThresh pending reads
Value	Description				
3Fh-00h	RdActThresh pending reads				
15:14	Reserved.				
13:9	WrBrstGap. Read-write. Reset: 04h. Specifies the number of memory clock cycles to wait before terminating write burst if no suitable write page hit is found. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>1Fh-00h</td><td>WrBrstGap MEMCLKs</td></tr> </table>	Value	Description	1Fh-00h	WrBrstGap MEMCLKs
Value	Description				
1Fh-00h	WrBrstGap MEMCLKs				
8	WrPhChk. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Check if a write page hit exists before terminating read burst and turning direction to writes.				
7:3	RdBrstGap. Read-write. Reset: 04h. Specifies the number of memory clock cycles to wait before terminating a read burst if no suitable read page hit is found. ValidValues:				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>1Fh-00h</td><td>RdBrstGap MEMCLKs</td></tr> </table>	Value	Description	1Fh-00h	RdBrstGap MEMCLKs
Value	Description				
1Fh-00h	RdBrstGap MEMCLKs				
2	RdPhChk. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Check if a read page hit exists before terminating a write burst and turning bus direction to reads.				
1	Reserved.				
0	PageHitBrstEn. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Enable Page Hit Burst Mode for Rd/Wr turnarounds.				

UMC00CHx000001B0...UMC11CHx000001BC [ODT Pattern Control] (UMC::OdtPat)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CHx000001B0; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CHx000001B4; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_n2_aliasSMN; UMC[11:00]CHx000001B8; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_n3_aliasSMN; UMC[11:00]CHx000001BC; UMC[11:00]CH=00[B:0]5_0000h**Bits Description**

31:12 Reserved.

11:8 **WrOdtPat**. Read-write. Reset: 0h. Specifies the state of ODT[3:0] pins when a write occurs to the specified chipselect.

7:4 Reserved.

3:0 **RdOdtPat**. Read-write. Reset: 0h.**Description:** Read-write. Reset: 0h. Specifies the ODT behavior when a read occurs to the specified chipselect.

Specifies the state of NT ODT to CS[3:0].

UMC00CHx000001C0...UMC11CHx000001CC [ECS Termination Count] (UMC::ECSTermCnt)

Read-write. Reset: 0000_0000h.

Configuration of the [ECS](#) termination count._umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000001C0; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000001C4; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000001C8; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000001CC; UMC[11:00]CH=00[B:0]5_0000h**Bits Description**31:0 **TermCnt**. Read-write. Reset: 0000_0000h. The total number of MPC transactions to complete an [ECS](#) scrub for all locations per die array. For 3DS, this corresponds to number of scrubs for each CID.

UMC[00...11]CHx000001D0 [Thermal Polling Byte Mask] (UMC::ThermalPollByteMask)

Read-write. Reset: 0000_0000h.

Thermal polling byte mask.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000001D0; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description										
31:16	Reserved.										
15:12	ThermByteMskCS3. Read-write. Reset: 0h. Each bit enables the respective DQ byte for the thermal polling comparator. 0 = Disable. 1 = Enable. ValidValues: <table> <tr> <th>Bit</th><th>Description</th></tr> <tr> <td>[0]</td><td>Byte0</td></tr> <tr> <td>[1]</td><td>Byte1</td></tr> <tr> <td>[2]</td><td>Byte2</td></tr> <tr> <td>[3]</td><td>Byte3</td></tr> </table>	Bit	Description	[0]	Byte0	[1]	Byte1	[2]	Byte2	[3]	Byte3
Bit	Description										
[0]	Byte0										
[1]	Byte1										
[2]	Byte2										
[3]	Byte3										
11:8	ThermByteMskCS2. Read-write. Reset: 0h. Each bit enables the respective DQ byte for the thermal polling comparator. 0 = Disable. 1 = Enable. ValidValues: <table> <tr> <th>Bit</th><th>Description</th></tr> <tr> <td>[0]</td><td>Byte0</td></tr> <tr> <td>[1]</td><td>Byte1</td></tr> <tr> <td>[2]</td><td>Byte2</td></tr> <tr> <td>[3]</td><td>Byte3</td></tr> </table>	Bit	Description	[0]	Byte0	[1]	Byte1	[2]	Byte2	[3]	Byte3
Bit	Description										
[0]	Byte0										
[1]	Byte1										
[2]	Byte2										
[3]	Byte3										
7:4	ThermByteMskCS1. Read-write. Reset: 0h. Each bit enables the respective DQ byte for the thermal polling comparator. 0 = Disable. 1 = Enable. ValidValues: <table> <tr> <th>Bit</th><th>Description</th></tr> <tr> <td>[0]</td><td>Byte0</td></tr> <tr> <td>[1]</td><td>Byte1</td></tr> <tr> <td>[2]</td><td>Byte2</td></tr> <tr> <td>[3]</td><td>Byte3</td></tr> </table>	Bit	Description	[0]	Byte0	[1]	Byte1	[2]	Byte2	[3]	Byte3
Bit	Description										
[0]	Byte0										
[1]	Byte1										
[2]	Byte2										
[3]	Byte3										
3:0	ThermByteMskCS0. Read-write. Reset: 0h. Each bit enables the respective DQ byte for the thermal polling comparator. 0 = Disable. 1 = Enable. ValidValues: <table> <tr> <th>Bit</th><th>Description</th></tr> <tr> <td>[0]</td><td>Byte0</td></tr> <tr> <td>[1]</td><td>Byte1</td></tr> <tr> <td>[2]</td><td>Byte2</td></tr> <tr> <td>[3]</td><td>Byte3</td></tr> </table>	Bit	Description	[0]	Byte0	[1]	Byte1	[2]	Byte2	[3]	Byte3
Bit	Description										
[0]	Byte0										
[1]	Byte1										
[2]	Byte2										
[3]	Byte3										

UMC[00...11]CHx000001D4 (UMC::PETCtrl)

Read-write. Reset: 0000_0001h.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000001D4; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description
31:26	Reserved.
25:16	tPPTInt . Read-write. Reset: 000h. PPT interval. Must be set to value > 0 to enable PPT. [0x000: Disabled, 0x3FF..0x001: <tPPTInt> * 2^(ShortInit ? 10 : 20) REFCLKs]
15:14	Reserved.
13:4	tECSInt . Read-write. Reset: 000h. ECS scrub interval. Must be set to value > 0 to enable ECS scrub. [0x000: Disabled, 0x3FF..0x001: <tECSInt> * 2^10 REFCLKs]
3	DisPPTAbort . Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. This bits allows the PPT Abort feature to be disabled. The PPT Abort feature is used to gracefully stop a PPT sequence when UMC is ready to enter SR.
2	PETIgnForceEntry . Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. This bit controls arbitration between conflicting page miss transactions and PET transactions. When set, the ARB will ignore ForceEntry status and PET transactions will get priority over conflicting page misses.
1	CodeWordMode . Read-write. Reset: 0. 0=Row Count. 1=Code Word. Specifies the ECS counter is in Code Word or Row Count error mode.
0	ECSMode . Read-write. Reset: 1. 0=Automatic Mode. 1=Manual Mode. Specifies the devices are in automatic or manual mode.

UMC[00...11]CHx000001D8 (UMC::LoadStepCtrl)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000001D8; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description																																		
31	LSEnabled. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Enable loadstep bandwidth management.																																		
30:20	Reserved.																																		
19:16	LSMaxDelta. Read-write. Reset: 0h. Init: 7h. The maximum additional bandwidth allowed compared to bandwidth of the previous window. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr><td>0h</td><td>6.25%</td></tr> <tr><td>1h</td><td>12.50%</td></tr> <tr><td>2h</td><td>18.75%</td></tr> <tr><td>3h</td><td>25.00%</td></tr> <tr><td>4h</td><td>31.25%</td></tr> <tr><td>5h</td><td>37.50%</td></tr> <tr><td>6h</td><td>43.75%</td></tr> <tr><td>7h</td><td>50.00%</td></tr> <tr><td>8h</td><td>56.25%</td></tr> <tr><td>9h</td><td>62.50%</td></tr> <tr><td>Ah</td><td>68.75%</td></tr> <tr><td>Bh</td><td>75.00%</td></tr> <tr><td>Ch</td><td>81.25%</td></tr> <tr><td>Dh</td><td>87.50%</td></tr> <tr><td>Eh</td><td>93.75%</td></tr> <tr><td>Fh</td><td>100.00%</td></tr> </table>	Value	Description	0h	6.25%	1h	12.50%	2h	18.75%	3h	25.00%	4h	31.25%	5h	37.50%	6h	43.75%	7h	50.00%	8h	56.25%	9h	62.50%	Ah	68.75%	Bh	75.00%	Ch	81.25%	Dh	87.50%	Eh	93.75%	Fh	100.00%
Value	Description																																		
0h	6.25%																																		
1h	12.50%																																		
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8h	56.25%																																		
9h	62.50%																																		
Ah	68.75%																																		
Bh	75.00%																																		
Ch	81.25%																																		
Dh	87.50%																																		
Eh	93.75%																																		
Fh	100.00%																																		
15:12	Reserved.																																		
11:8	LSInit. Read-write. Reset: 0h. Init: 7h. The initial max bandwidth after SRX. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr><td>0h</td><td>6.25%</td></tr> <tr><td>1h</td><td>12.50%</td></tr> <tr><td>2h</td><td>18.75%</td></tr> <tr><td>3h</td><td>25.00%</td></tr> <tr><td>4h</td><td>31.25%</td></tr> <tr><td>5h</td><td>37.50%</td></tr> <tr><td>6h</td><td>43.75%</td></tr> <tr><td>7h</td><td>50.00%</td></tr> <tr><td>8h</td><td>56.25%</td></tr> <tr><td>9h</td><td>62.50%</td></tr> <tr><td>Ah</td><td>68.75%</td></tr> <tr><td>Bh</td><td>75.00%</td></tr> <tr><td>Ch</td><td>81.25%</td></tr> <tr><td>Dh</td><td>87.50%</td></tr> <tr><td>Eh</td><td>93.75%</td></tr> <tr><td>Fh</td><td>100.00%</td></tr> </table>	Value	Description	0h	6.25%	1h	12.50%	2h	18.75%	3h	25.00%	4h	31.25%	5h	37.50%	6h	43.75%	7h	50.00%	8h	56.25%	9h	62.50%	Ah	68.75%	Bh	75.00%	Ch	81.25%	Dh	87.50%	Eh	93.75%	Fh	100.00%
Value	Description																																		
0h	6.25%																																		
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Ah	68.75%																																		
Bh	75.00%																																		
Ch	81.25%																																		
Dh	87.50%																																		
Eh	93.75%																																		
Fh	100.00%																																		
7:4	Reserved.																																		
3:0	LSWindow. Read-write. Reset: 0h. Init: 0h. The load step sample window.																																		

	ValidValues:	
	Value	Description
	Fh-0h	(LSWindow+1) * 512 MEMCLKs

UMC[00...11]CHx000001E0 [UMC Miscellaneous Configuration] (UMC::MiscCfg)

Read-write. Reset: 0000_6140h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000001E0; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description				
31:25	Reserved.				
24	TimClkGateDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disables medium grain clock gating in TIM block of ARB				
23	DcqClkGateDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disables medium grain clock gating in DCQARB block of ARB				
22	TstagSplitSc. Read-write. Reset: 0. 0=REF commands among both sub-channels when VCM is supported,. 1=REF commands between ranks of the same sub-channels when in virtual controller mode. Specifies the Tstag and Ttagsb timing behavior mode with respect to sub-channels (DDR5)				
21	TcshMult. Read-write. Reset: 0. Check: 0. 0=1x. 1=2x.				
20	Reserved.				
19	TstagSplit. Read-write. Reset: 0. Init: 1. 0=REF commands among all chipselects,. 1=REF commands between the chipselects of the same DIMM. Specifies the Tstag and Ttagsb timing behavior mode with respect to chipselects.				
18:17	Reserved.				
16	SpazMrrReadEn. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Specifies periodic MRR are issued by SPAZ.				
15:11	ThrottleThresh. Read-write. Reset: 0Ch. Init: 12.2.21 [DDR5 Thermal Management and Bandwidth Capping]. Specifies the minimum value to activate thermal throttle per the DRAM MR readout. Throttle will be activated for the value specified and greater. Constraint: Value inside {3:5} in DDR5 mode. ValidValues:				
	<table><tr><th>Value</th><th>Description</th></tr><tr><td>1Fh-00h</td><td><ThrottleThresh></td></tr></table>	Value	Description	1Fh-00h	<ThrottleThresh>
Value	Description				
1Fh-00h	<ThrottleThresh>				
10	Reserved.				
9	RecClkGateDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disables medium grain clock gating in REC.				
8	TrefShift. Read-write. Reset: 1. Check: 1. 0=Disable. 1=Enable. Specifies shifted Tref timeout rank grouping.				
7	SpecPchgClkGateDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disables medium grain clock gating in SPECPCHG block of ARB				
6	BlockAbHalfRefreshed. Read-write. Reset: 1. Check: 1. 0=Do not block REFab. 1=Block REFab. Specifies if SPAZ blocks REFab after half of the banks are refreshed by REFpb. Ignored if !PerBankRefEn				
5:4	Reserved.				
3	MCAGateDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disables medium grain clock gating for MCA logic.				
2	DisErrInj. Read-write. Reset: 0. Init: 1. 0=Enable. 1=Disable. Specifies UMC error injection configuration writes are disabled.				
1	DisDebugBusSel. Read-write. Reset: 0. Init: 1. 0=Enable. 1=Disable. Specifies UMC debug bus configuration writes are disabled.				
0	RegchClkGateEn. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Enable configuration register clock gating.				

UMC[00...11]CHx000001E4 [UMC RFM Control] (UMC::RfmCtrl)

Read-write. Reset: 0111_0010h.

UMC RFM control configuration. Programmed per 'Normal Refresh Mode' and adjusted by internal logic per AutoRefFineGranMode if DDR5 mode.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000001E4; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description																
31:28	RfmCsEn. Read-write. Reset: 0h. Specifies the chipselects which require RFM RAA tracking when RFMEn=1. ValidValues: <table> <tr> <th>Bit</th><th>Description</th></tr> <tr> <td>[0]</td><td>CS0. 0=Disable. 1=Enable.</td></tr> <tr> <td>[1]</td><td>CS1. 0=Disable. 1=Enable.</td></tr> <tr> <td>[2]</td><td>CS2. 0=Disable. 1=Enable.</td></tr> <tr> <td>[3]</td><td>CS3. 0=Disable. 1=Enable.</td></tr> </table>	Bit	Description	[0]	CS0. 0=Disable. 1=Enable.	[1]	CS1. 0=Disable. 1=Enable.	[2]	CS2. 0=Disable. 1=Enable.	[3]	CS3. 0=Disable. 1=Enable.						
Bit	Description																
[0]	CS0. 0=Disable. 1=Enable.																
[1]	CS1. 0=Disable. 1=Enable.																
[2]	CS2. 0=Disable. 1=Enable.																
[3]	CS3. 0=Disable. 1=Enable.																
27	Reserved.																
26:24	RAADecRfm. Read-write. Reset: 1h. Specifies the RAAIMT multiplier for the RFM decrement. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Reserved.</td></tr> <tr> <td>1h</td><td>1x</td></tr> <tr> <td>7h-2h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Reserved.	1h	1x	7h-2h	Reserved.								
Value	Description																
0h	Reserved.																
1h	1x																
7h-2h	Reserved.																
23	Reserved.																
22:20	RAADecRef. Read-write. Reset: 1h. Specifies the RAAIMT multiplier for the REF decrement. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>0.5x</td></tr> <tr> <td>1h</td><td>1x</td></tr> <tr> <td>7h-2h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	0.5x	1h	1x	7h-2h	Reserved.								
Value	Description																
0h	0.5x																
1h	1x																
7h-2h	Reserved.																
19	Reserved.																
18:16	RAAMult. Read-write. Reset: 1h. Specifies the RAAIMT multiplier for the maximum management threshold. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Reserved.</td></tr> <tr> <td>1h</td><td>2x</td></tr> <tr> <td>2h</td><td>3x</td></tr> <tr> <td>3h</td><td>4x</td></tr> <tr> <td>4h</td><td>5x</td></tr> <tr> <td>5h</td><td>6x</td></tr> <tr> <td>7h-6h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Reserved.	1h	2x	2h	3x	3h	4x	4h	5x	5h	6x	7h-6h	Reserved.
Value	Description																
0h	Reserved.																
1h	2x																
2h	3x																
3h	4x																
4h	5x																
5h	6x																
7h-6h	Reserved.																
15:12	Reserved.																
11:4	RAAIMT. Read-write. Reset: 01h. Specifies the accumulated activate initial management threshold. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>7Fh-01h</td><td>RAAIMT ACTs</td></tr> <tr> <td>FFh-80h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Reserved.	7Fh-01h	RAAIMT ACTs	FFh-80h	Reserved.								
Value	Description																
00h	Reserved.																
7Fh-01h	RAAIMT ACTs																
FFh-80h	Reserved.																
3:2	Reserved.																
1:0	RFMEn. Read-write. Reset: 0h. Specifies RFM RAA tracking mode is enabled.																

	ValidValues:	
	Value	Description
	0h	RFM Disabled
	1h	RFM Enabled: REF and RFM commands.
	3h-2h	Reserved.

UMC[00...11]CHx000001F0 [VDCI Status] (UMC::VdciStatus)

Read-only, Volatile. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000001F0; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:22	Reserved.
21:11	VdciStatusEpa. Read-only, Volatile. Reset: 000h. VDCI Epa Status
10:0	VdciStatusSdp. Read-only, Volatile. Reset: 000h. VDCI Sdp Status

UMC[00...11]CHx000001F4 [Phantom Write Data] (UMC::PhantomWrData)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000001F4; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:0	WrData. Read-write. Reset: 0000_0000h. Specifies the write data pattern associated with Phantom writes

UMC[00...11]CHx000001F8 [Phantom Write Control] (UMC::PhantomWrCtrl)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx000001F8; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:13	Reserved.
12:8	PhWrByteMask1. Read-write. Reset: 00h. Byte Masks for phantom writes on Sub-Channel 1
7:5	Reserved.
4:0	PhWrByteMask0. Read-write. Reset: 00h. Byte Masks for phantom writes on Sub-Channel 0

UMC00CHx00000200...UMC11CHx00000500 [DRAM Configuration] (UMC::DramConfiguration)

Read-write. Reset: 0001_0000h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000200; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000300; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000400; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000500; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description																																				
31	UclkGtFclk. Read-write. Reset: 0. 0=UCLK is less than or equal to FCLK. 1=UCLK is greater than FCLK. 0=UCLK is less than or equal to FCLK. 1=UCLK is greater than FCLK. Indicates UCLK frequency is greater than FCLK frequency. Constraint: IF (UclkGtFclk == 1) THEN UclkDiv1Mode != 0 ENDIF																																				
30:19	Reserved.																																				
18	GearDownEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Specifies GearDown mode. Programmed per RDimmSdrMode if DDR5 mode.																																				
17	Cmd2t. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Specifies 2T DDR address bus timing mode. Reserved if DDR5 mode.																																				
16	BankGroupEn. Read-write. Reset: 1. Check: 1. 0=Disable. 1=Enable. Specifies BankGroup timing parameters are enabled. Constraint: BankGroupEn == 1 in DDR5 mode.																																				
15:0	MemClkFreq. Read-write. Reset: 0000h. Specifies the memory clock frequency in MHz. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>03E7h-0000h</td><td>Reserved.</td></tr> <tr> <td>03E8h</td><td>1000 MHz (2000 MT/s)</td></tr> <tr> <td>063Fh-03E9h</td><td>Reserved.</td></tr> <tr> <td>0640h</td><td>1600 MHz (3200 MT/s)</td></tr> <tr> <td>0707h-0641h</td><td>Reserved.</td></tr> <tr> <td>0708h</td><td>1800 MHz (3600 MT/s)</td></tr> <tr> <td>07CFh-0709h</td><td>Reserved.</td></tr> <tr> <td>07D0h</td><td>2000 MHz (4000 MT/s)</td></tr> <tr> <td>0897h-07D1h</td><td>Reserved.</td></tr> <tr> <td>0898h</td><td>2200 MHz (4400 MT/s)</td></tr> <tr> <td>095Fh-0899h</td><td>Reserved.</td></tr> <tr> <td>0960h</td><td>2400 MHz (4800 MT/s)</td></tr> <tr> <td>0A27h-0961h</td><td>Reserved.</td></tr> <tr> <td>0A28h</td><td>2600 MHz (5200 MT/s)</td></tr> <tr> <td>0AEFh-0A29h</td><td>Reserved.</td></tr> <tr> <td>0AF0h</td><td>2800 MHz (5600 MT/s)</td></tr> <tr> <td>FFFFh-0AF1h</td><td>Reserved.</td></tr> </table>	Value	Description	03E7h-0000h	Reserved.	03E8h	1000 MHz (2000 MT/s)	063Fh-03E9h	Reserved.	0640h	1600 MHz (3200 MT/s)	0707h-0641h	Reserved.	0708h	1800 MHz (3600 MT/s)	07CFh-0709h	Reserved.	07D0h	2000 MHz (4000 MT/s)	0897h-07D1h	Reserved.	0898h	2200 MHz (4400 MT/s)	095Fh-0899h	Reserved.	0960h	2400 MHz (4800 MT/s)	0A27h-0961h	Reserved.	0A28h	2600 MHz (5200 MT/s)	0AEFh-0A29h	Reserved.	0AF0h	2800 MHz (5600 MT/s)	FFFFh-0AF1h	Reserved.
Value	Description																																				
03E7h-0000h	Reserved.																																				
03E8h	1000 MHz (2000 MT/s)																																				
063Fh-03E9h	Reserved.																																				
0640h	1600 MHz (3200 MT/s)																																				
0707h-0641h	Reserved.																																				
0708h	1800 MHz (3600 MT/s)																																				
07CFh-0709h	Reserved.																																				
07D0h	2000 MHz (4000 MT/s)																																				
0897h-07D1h	Reserved.																																				
0898h	2200 MHz (4400 MT/s)																																				
095Fh-0899h	Reserved.																																				
0960h	2400 MHz (4800 MT/s)																																				
0A27h-0961h	Reserved.																																				
0A28h	2600 MHz (5200 MT/s)																																				
0AEFh-0A29h	Reserved.																																				
0AF0h	2800 MHz (5600 MT/s)																																				
FFFFh-0AF1h	Reserved.																																				

UMC00CHx00000204...UMC11CHx00000504 [DRAM Timing 1] (UMC::DramTiming1)

Read-write. Reset: 1212_2712h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000204; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000304; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000404; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000504; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description								
31:30	Reserved.								
29:24	Trcdwr. Read-write. Reset: 12h. Specifies the minimum time in memory clock cycles from an activate command to a write command, both to the same bank. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>07h-00h</td><td>Reserved.</td></tr> <tr> <td>1Bh-08h</td><td>Trcdwr MEMCLKs</td></tr> <tr> <td>3Fh-1Ch</td><td>Reserved.</td></tr> </table>	Value	Description	07h-00h	Reserved.	1Bh-08h	Trcdwr MEMCLKs	3Fh-1Ch	Reserved.
Value	Description								
07h-00h	Reserved.								
1Bh-08h	Trcdwr MEMCLKs								
3Fh-1Ch	Reserved.								
23:22	Reserved.								
21:16	Trcdrd. Read-write. Reset: 12h. Specifies the minimum time in memory clock cycles from an activate command to a read command, both to the same bank. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>07h-00h</td><td>Reserved.</td></tr> <tr> <td>1Bh-08h</td><td>Trcdrd MEMCLKs</td></tr> <tr> <td>3Fh-1Ch</td><td>Reserved.</td></tr> </table>	Value	Description	07h-00h	Reserved.	1Bh-08h	Trcdrd MEMCLKs	3Fh-1Ch	Reserved.
Value	Description								
07h-00h	Reserved.								
1Bh-08h	Trcdrd MEMCLKs								
3Fh-1Ch	Reserved.								
15	Reserved.								
14:8	Tras. Read-write. Reset: 27h. Specifies the minimum time in memory clock cycles from an activate command to a precharge command, both to the same bank. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>14h-00h</td><td>Reserved.</td></tr> <tr> <td>4Dh-15h</td><td>Tras MEMCLKs.</td></tr> </table>	Value	Description	14h-00h	Reserved.	4Dh-15h	Tras MEMCLKs.		
Value	Description								
14h-00h	Reserved.								
4Dh-15h	Tras MEMCLKs.								
7:6	Reserved.								
5:0	Tcl. Read-write. Reset: 12h. Specifies the time in memory clock cycles from the CAS assertion for a read cycle until data return. Programmed per DDR5 DRAM CL + RCD nLadd. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>08h-00h</td><td>Reserved.</td></tr> <tr> <td>32h-09h</td><td>Number of MEMCLKs.</td></tr> <tr> <td>3Fh-33h</td><td>Reserved.</td></tr> </table>	Value	Description	08h-00h	Reserved.	32h-09h	Number of MEMCLKs.	3Fh-33h	Reserved.
Value	Description								
08h-00h	Reserved.								
32h-09h	Number of MEMCLKs.								
3Fh-33h	Reserved.								

UMC00CHx00000208...UMC11CHx00000508 [DRAM Timing 2] (UMC::DramTiming2)

Read-write. Reset: 0012_0039h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000208; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000308; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000408; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000508; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description								
31:30	Reserved.								
29:24	TrpPb. Read-write. Reset: 00h.								
23:22	Reserved.								
21:16	TrpAb. Read-write. Reset: 12h. Specifies the minimum time in memory clock cycles from a precharge command to an activate command or auto refresh command, both to the same bank. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>07h-00h</td><td>Reserved.</td></tr> <tr> <td>1Bh-08h</td><td>Trp MEMCLKs</td></tr> <tr> <td>3Fh-1Ch</td><td>Reserved.</td></tr> </table>	Value	Description	07h-00h	Reserved.	1Bh-08h	Trp MEMCLKs	3Fh-1Ch	Reserved.
Value	Description								
07h-00h	Reserved.								
1Bh-08h	Trp MEMCLKs								
3Fh-1Ch	Reserved.								
15:8	TrcPb. Read-write. Reset: 00h.								
7:0	TrcAb. Read-write. Reset: 39h. Specifies the minimum time in memory clock cycles from and activate command to another activate command or an auto refresh command, all to the same chipselect bank. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>1Ch-00h</td><td>Reserved.</td></tr> <tr> <td>87h-1Dh</td><td>Trc MEMCLKs</td></tr> <tr> <td>FFh-88h</td><td>Reserved.</td></tr> </table>	Value	Description	1Ch-00h	Reserved.	87h-1Dh	Trc MEMCLKs	FFh-88h	Reserved.
Value	Description								
1Ch-00h	Reserved.								
87h-1Dh	Trc MEMCLKs								
FFh-88h	Reserved.								

UMC00CHx0000020C...UMC11CHx0000050C [DRAM Timing 3] (UMC::DramTiming3)

Read-write. Reset: 0900_0404h.		
_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx0000020C; UMC[11:00]CH=00[B:0]5_0000h		
_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx0000030C; UMC[11:00]CH=00[B:0]5_0000h		
_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx0000040C; UMC[11:00]CH=00[B:0]5_0000h		
_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx0000050C; UMC[11:00]CH=00[B:0]5_0000h		
Bits	Description	
31:29	Reserved.	
28:24	Trtp. Read-write. Reset: 09h. Specifies the minimum time in memory clock cycles a page can be closed after having been read.	
	ValidValues:	
	Value	Description
	04h-00h	Reserved.
	12h-05h	Trtp MEMCLKs.
	1Fh-13h	Reserved.
23:21	Reserved.	
20:16	TrrdDlr. Read-write. Reset: 00h. Specifies the minimum time in memory clock cycles between activate commands to a different logical rank.	
	ValidValues:	
	Value	Description
	00h	Disabled
	03h-01h	Reserved.
	0Ch-04h	TrrdDlr MEMCLKs
	1Fh-0Dh	Reserved.
15:13	Reserved.	
12:8	TrrdL. Read-write. Reset: 04h. Specifies the minimum time in memory clock cycles between activate commands to different chipselect banks in the same bank group.	
	ValidValues:	
	Value	Description
	03h-00h	Reserved.
	0Ch-04h	TrrdL MEMCLKs
	1Fh-0Dh	Reserved.
7:5	Reserved.	
4:0	TrrdS. Read-write. Reset: 04h. Specifies the minimum time in memory clock cycles between activate commands to different chipselect banks. In a different bank group tRRD_S if BankGroupEn.	
	ValidValues:	
	Value	Description
	03h-00h	Reserved.
	0Ch-04h	TrrdS MEMCLKs
	1Fh-0Dh	Reserved.

UMC00CHx00000210...UMC11CHx00000510 [DRAM Timing 4] (UMC::DramTiming4)

Read-write. Reset: 0000_001Ah.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000210; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000310; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000410; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000510; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description										
31	Reserved.										
30:25	TfawDlr. Read-write. Reset: 00h. Specifies the rolling window in memory clock cycles during which no more than 4 banks may be activated in different logical ranks. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disabled</td></tr> <tr> <td>05h-01h</td><td>Reserved.</td></tr> <tr> <td>36h-06h</td><td>TfawDlr MEMCLKs</td></tr> <tr> <td>3Fh-37h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disabled	05h-01h	Reserved.	36h-06h	TfawDlr MEMCLKs	3Fh-37h	Reserved.
Value	Description										
00h	Disabled										
05h-01h	Reserved.										
36h-06h	TfawDlr MEMCLKs										
3Fh-37h	Reserved.										
24	Reserved.										
23:18	TfawSlr. Read-write. Reset: 00h. Specifies the rolling window in memory clock cycles during which no more than 4 banks may be activated in the same logical rank. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disabled</td></tr> <tr> <td>05h-01h</td><td>Reserved.</td></tr> <tr> <td>36h-06h</td><td>TfawSlr MEMCLKs</td></tr> <tr> <td>3Fh-37h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disabled	05h-01h	Reserved.	36h-06h	TfawSlr MEMCLKs	3Fh-37h	Reserved.
Value	Description										
00h	Disabled										
05h-01h	Reserved.										
36h-06h	TfawSlr MEMCLKs										
3Fh-37h	Reserved.										
17:7	Reserved.										
6:0	Tfaw. Read-write. Reset: 1Ah. Specifies the rolling window in memory clock cycles during which no more than 4 banks may be activated. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>05h-00h</td><td>Reserved.</td></tr> <tr> <td>36h-06h</td><td>Tfaw MEMCLKs</td></tr> <tr> <td>7Fh-37h</td><td>Reserved.</td></tr> </table>	Value	Description	05h-00h	Reserved.	36h-06h	Tfaw MEMCLKs	7Fh-37h	Reserved.		
Value	Description										
05h-00h	Reserved.										
36h-06h	Tfaw MEMCLKs										
7Fh-37h	Reserved.										

UMC00CHx00000214...UMC11CHx00000514 [DRAM Timing 5] (UMC::DramTiming5)

Read-write. Reset: 0003_030Ch.									
<code>_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000214; UMC[11:00]CH=00[B:0]5_0000h</code>									
<code>_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000314; UMC[11:00]CH=00[B:0]5_0000h</code>									
<code>_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000414; UMC[11:00]CH=00[B:0]5_0000h</code>									
<code>_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000514; UMC[11:00]CH=00[B:0]5_0000h</code>									
Bits	Description								
31:23	Reserved.								
22:16	TwtrL. Read-write. Reset: 03h. Specifies the minimum number of memory clock cycles from a write operation to a read operation, both to the same chipselect in the same bank group. This is measured from the rising clock edge following last non-masked data strobe of the write to the rising clock edge of the next read command.								
	ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>01h-00h</td><td>Reserved.</td></tr> <tr> <td>1Fh-02h</td><td>TwtrL MEMCLKs</td></tr> <tr> <td>7Fh-20h</td><td>Reserved.</td></tr> </table>	Value	Description	01h-00h	Reserved.	1Fh-02h	TwtrL MEMCLKs	7Fh-20h	Reserved.
Value	Description								
01h-00h	Reserved.								
1Fh-02h	TwtrL MEMCLKs								
7Fh-20h	Reserved.								
15:13	Reserved.								
12:8	TwtrS. Read-write. Reset: 03h. Specifies the minimum number of memory clock cycles from a write operation to a read operation, both to the same chipselect. This is measured from the rising clock edge following last non-masked data strobe of the write to the rising clock edge of the next read command. In a different bank group tWTR_S if BankGroupEn.								
	ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>01h-00h</td><td>Reserved.</td></tr> <tr> <td>0Eh-02h</td><td>TwtrS MEMCLKs</td></tr> <tr> <td>1Fh-0Fh</td><td>Reserved.</td></tr> </table>	Value	Description	01h-00h	Reserved.	0Eh-02h	TwtrS MEMCLKs	1Fh-0Fh	Reserved.
Value	Description								
01h-00h	Reserved.								
0Eh-02h	TwtrS MEMCLKs								
1Fh-0Fh	Reserved.								
7:6	Reserved.								
5:0	Tcwl. Read-write. Reset: 0Ch. Specifies the number of memory clock cycles from internal write command to first write data in at the DRAM. $Tcwl = DRAM\ CWL + DRAM\ PL + RCD\ nLadd$.								
	ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>08h-00h</td><td>Reserved.</td></tr> <tr> <td>16h-09h</td><td>Tcwl MEMCLKs</td></tr> <tr> <td>3Fh-17h</td><td>Reserved.</td></tr> </table>	Value	Description	08h-00h	Reserved.	16h-09h	Tcwl MEMCLKs	3Fh-17h	Reserved.
Value	Description								
08h-00h	Reserved.								
16h-09h	Tcwl MEMCLKs								
3Fh-17h	Reserved.								

UMC00CHx00000218...UMC11CHx00000518 [DRAM Timing 6] (UMC::DramTiming6)

Read-write. Reset: 0000_0012h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000218; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000318; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000418; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000518; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description								
31:8	Reserved.								
7:0	Twr. Read-write. Reset: 12h. Specifies the minimum time in memory clock cycles from the last data write until the chipselect bank precharge.								
	ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>09h-00h</td><td>Reserved.</td></tr> <tr> <td>64h-0Ah</td><td>Twr MEMCLKs</td></tr> <tr> <td>FFh-65h</td><td>Reserved.</td></tr> </table>	Value	Description	09h-00h	Reserved.	64h-0Ah	Twr MEMCLKs	FFh-65h	Reserved.
Value	Description								
09h-00h	Reserved.								
64h-0Ah	Twr MEMCLKs								
FFh-65h	Reserved.								

UMC00CHx0000021C...UMC11CHx0000051C [DRAM Timing 7] (UMC::DramTiming7)

Read-write. Reset: 0000_0004h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx0000021C; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx0000031C; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx0000041C; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx0000051C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:3	Reserved.
2:0	Tppd. Read-write. Reset: 4h.

UMC00CHx00000220...UMC11CHx00000520 [DRAM Timing 8] (UMC::DramTiming8)

Read-write. Reset: 0101_0303h.

Specifies CAS to CAS command spacing. The last clock of virtual CAS is defined as BL/2-1. Constraint: TrdrdSc <= TrdrdDc <= TrdrdDd. Constraint: TrdrdSc <= TrdrdScL.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000220; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000320; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000420; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000520; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description										
31:30	TrdrdBan. Read-write. Reset: 0h. Bans the traffic for the specified cases where the number of cycles from the last clock of virtual CAS of a first read-burst operation to the clock in which CAS is asserted for a following read-burst operation. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Disable</td></tr> <tr> <td>1h</td><td>Ban Trdrd traffic at 2 MEMCLKs. (e.g. BL=8 tCCD=5)</td></tr> <tr> <td>2h</td><td>Ban Trdrd traffic at 2 and 3 MEMCLKs. (e.g. BL=8 tCCD=5,6)</td></tr> <tr> <td>3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Disable	1h	Ban Trdrd traffic at 2 MEMCLKs. (e.g. BL=8 tCCD=5)	2h	Ban Trdrd traffic at 2 and 3 MEMCLKs. (e.g. BL=8 tCCD=5,6)	3h	Reserved.
Value	Description										
0h	Disable										
1h	Ban Trdrd traffic at 2 MEMCLKs. (e.g. BL=8 tCCD=5)										
2h	Ban Trdrd traffic at 2 and 3 MEMCLKs. (e.g. BL=8 tCCD=5,6)										
3h	Reserved.										
29:28	Reserved.										
27:24	TrdrdScL. Read-write. Reset: 1h. Specifies the minimum number of cycles from the last clock of virtual CAS of the first read-burst operation to the clock in which CAS is asserted for a following read-burst operation in the same chipselect in the same bank group. Used to satisfy JEDEC tCCD_L. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Reserved.</td></tr> <tr> <td>Fh-1h</td><td>TrdrdScL MEMCLKs</td></tr> </table>	Value	Description	0h	Reserved.	Fh-1h	TrdrdScL MEMCLKs				
Value	Description										
0h	Reserved.										
Fh-1h	TrdrdScL MEMCLKs										
23:20	TrdrdScDlr. Read-write. Reset: 0h. Specifies the minimum number of cycles from the last clock of virtual CAS of the first read-burst operation to the clock in which CAS is asserted for a following read-burst operation in a different logical rank. Used to satisfy JEDEC tCCD. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Disabled</td></tr> <tr> <td>Fh-1h</td><td>TrdrdScDlr MEMCLKs</td></tr> </table>	Value	Description	0h	Disabled	Fh-1h	TrdrdScDlr MEMCLKs				
Value	Description										
0h	Disabled										
Fh-1h	TrdrdScDlr MEMCLKs										
19:16	TrdrdSc. Read-write. Reset: 1h. Specifies the minimum number of cycles from the last clock of virtual CAS of the first read-burst operation to the clock in which CAS is asserted for a following read-burst operation in the same chipselect. Used to satisfy JEDEC tCCD/tCCD_S. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Reserved.</td></tr> <tr> <td>Fh-1h</td><td>TrdrdSc MEMCLKs</td></tr> </table>	Value	Description	0h	Reserved.	Fh-1h	TrdrdSc MEMCLKs				
Value	Description										
0h	Reserved.										
Fh-1h	TrdrdSc MEMCLKs										
15:12	Reserved.										
11:8	TrdrdSd. Read-write. Reset: 3h. Specifies the minimum number of cycles from the last clock of virtual CAS of the first read-burst operation to the clock in which CAS is asserted for a following read-burst operation in the same DIMM. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Reserved.</td></tr> <tr> <td>Fh-1h</td><td>TrdrdSd MEMCLKs</td></tr> </table>	Value	Description	0h	Reserved.	Fh-1h	TrdrdSd MEMCLKs				
Value	Description										
0h	Reserved.										
Fh-1h	TrdrdSd MEMCLKs										
7:4	Reserved.										

3:0	TrdrdDd . Read-write. Reset: 3h. Specifies the minimum number of cycles from the last clock of virtual CAS of the first read-burst operation to the clock in which CAS is asserted for a following read-burst operation in a different DIMM.	
	Valid Values:	
	Value	Description
	0h	Reserved.
	Fh-1h	TrdrdDd MEMCLKs

UMC00CHx00000224...UMC11CHx00000524 [DRAM Timing 9] (UMC::DramTiming9)

Read-write. Reset: 0101_0303h.

Specifies CAS to CAS command spacing. The last clock of virtual CAS is defined as BL/2-1. Constraint: TwrwrSc <= TwrwrDc <= TwrwrDd. Constraint: TwrwrSc <= TwrwrScL.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000224; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000324; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000424; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000524; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description										
31:30	TwrwrBan. Read-write. Reset: 0h. Bans the traffic for the specified cases where the number of cycles from the last clock of virtual CAS of a first write-burst operation to the clock in which CAS is asserted for a following write-burst operation. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Disable</td></tr> <tr> <td>1h</td><td>Ban Twrwr traffic at 2 MEMCLKs (e.g. BL=8 tCCD=5)</td></tr> <tr> <td>2h</td><td>Ban Twrwr traffic at 2 and 3 MEMCLKs (e.g. BL=8 tCCD=5,6)</td></tr> <tr> <td>3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Disable	1h	Ban Twrwr traffic at 2 MEMCLKs (e.g. BL=8 tCCD=5)	2h	Ban Twrwr traffic at 2 and 3 MEMCLKs (e.g. BL=8 tCCD=5,6)	3h	Reserved.
Value	Description										
0h	Disable										
1h	Ban Twrwr traffic at 2 MEMCLKs (e.g. BL=8 tCCD=5)										
2h	Ban Twrwr traffic at 2 and 3 MEMCLKs (e.g. BL=8 tCCD=5,6)										
3h	Reserved.										
29:24	TwrwrScL. Read-write. Reset: 01h. Specifies the minimum number of cycles from the last clock of virtual CAS of a first write-burst operation to the clock in which CAS is asserted for a following write-burst operation in the same bank group. Used to satisfy JEDEC tCCD_L. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>TwrwrScL MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	TwrwrScL MEMCLKs				
Value	Description										
00h	Reserved.										
3Fh-01h	TwrwrScL MEMCLKs										
23:20	TwrwrScDlr. Read-write. Reset: 0h. Specifies the minimum number of cycles from the last clock of virtual CAS of a first write-burst operation to the clock in which CAS is asserted for a following write-burst operation in a different logical rank. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Disabled</td></tr> <tr> <td>Fh-1h</td><td>TwrwrScDlr MEMCLKs</td></tr> </table>	Value	Description	0h	Disabled	Fh-1h	TwrwrScDlr MEMCLKs				
Value	Description										
0h	Disabled										
Fh-1h	TwrwrScDlr MEMCLKs										
19:16	TwrwrSc. Read-write. Reset: 1h. Specifies the minimum number of cycles from the last clock of virtual CAS of a first write-burst operation to the clock in which CAS is asserted for a following write-burst operation in the same chipselect. Used to satisfy JEDEC tCCD/tCCD_S. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Reserved.</td></tr> <tr> <td>Fh-1h</td><td>TwrwrSc MEMCLKs</td></tr> </table>	Value	Description	0h	Reserved.	Fh-1h	TwrwrSc MEMCLKs				
Value	Description										
0h	Reserved.										
Fh-1h	TwrwrSc MEMCLKs										
15:14	Reserved.										
13:8	TwrwrSd. Read-write. Reset: 03h. Specifies the minimum number of cycles from the last clock of virtual CAS of the first write-burst operation to the clock in which CAS is asserted for a following write-burst operation in the same DIMM. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>0Fh-01h</td><td>TwrwrSd MEMCLKs</td></tr> <tr> <td>3Fh-10h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Reserved.	0Fh-01h	TwrwrSd MEMCLKs	3Fh-10h	Reserved.		
Value	Description										
00h	Reserved.										
0Fh-01h	TwrwrSd MEMCLKs										
3Fh-10h	Reserved.										
7:4	Reserved.										

3:0	TwrrwrDd. Read-write. Reset: 3h. Specifies the minimum number of cycles from the last clock of virtual CAS of the first write-burst operation to the clock in which CAS is asserted for a following write-burst operation in a different DIMM.
ValidValues:	
Value	Description
0h	Reserved.
Fh-1h	TwrrwrDd MEMCLKs

UMC00CHx00000228...UMC11CHx00000528 [DRAM Timing 10] (UMC::DramTiming10)

Read-write. Reset: 0000_0501h.

Specifies CAS to CAS command spacing. The last clock of virtual CAS is defined as BL/2-1.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000228; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000328; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000428; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000528; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:22	Reserved.
21:16	TwrrdScDlr. Read-write. Reset: 00h. Specifies the minimum number of clock cycles from the last clock of virtual CAS of a first write-burst operation to the clock in which CAS is asserted for a following read-burst operation to a different logical rank.
ValidValues:	
Value	Description
00h	Disabled
3Fh-01h	TwrrdScDlr MEMCLKs
15:14	Reserved.
13:8	Trdwr. Read-write. Reset: 05h. Specifies the minimum number of clock cycles from the last clock of virtual CAS of a first read-burst operation to the clock in which CAS is asserted for a following write-burst operation.
ValidValues:	
Value	Description
00h	Reserved.
1Fh-01h	Trdwr MEMCLKs
3Fh-20h	Reserved.
7:4	Reserved.
3:0	Twrrd. Read-write. Reset: 1h. Specifies the minimum number of clock cycles from the last clock of virtual CAS of a first write-burst operation to the clock in which CAS is asserted for a following read-burst operation.
ValidValues:	
Value	Description
0h	Reserved.
Fh-1h	Twrrd MEMCLKs

UMC00CHx0000022C...UMC11CHx0000052C [DRAM Timing 11] (UMC::DramTiming11)

Read-write. Reset: 0000_4080h.		
_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx0000022C; UMC[11:00]CH=00[B:0]5_0000h		
_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx0000032C; UMC[11:00]CH=00[B:0]5_0000h		
_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx0000042C; UMC[11:00]CH=00[B:0]5_0000h		
_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx0000052C; UMC[11:00]CH=00[B:0]5_0000h		
Bits	Description	
31	ShortInit. Read-write. Reset: 0. 0=Disable. 1=Enable. 1=Interval by 2 ¹⁰ clocks. 0=Interval by 2 ²⁰ clocks.]	
30	Reserved.	
29:20	ZqcsInterval. Read-write. Reset: 000h. Specifies the interval in REFCLK cycles for the periodic ZQCS command. See: ShortInit. [0x000: Disabled, 0x3FF..0x001: ZqcsInterval * 2 ^{^(ShortInit ? 10 : 20)} REFCLKs]	
19:8	TzqOperCal. Read-write. Reset: 040h. Specifies the minimum time in memory clock cycles from a ZQ command. [0x000: Reserved, 0xFFF..0x001: <TzqOperCal> * 8 MEMCLKs]. Represents tZQCAL if DDR5 mode.	
7:0	Tzqcs. Read-write. Reset: 80h. Specifies the minimum time in memory clock cycles from a ZQ command to any other command on the channel. Represents tZQLATCH post ZqLatch if DDR5 mode.	
	ValidValues:	
	Value	Description
	00h	Reserved.
	FFh-01h	Tzqcs MEMCLKs

UMC00CHx00000230...UMC11CHx00000530 [DRAM Timing 12] (UMC::DramTiming12)

Read-write. Reset: 0000_2490h.		
Programming constraints:		
For 1X mode:		
Tref > ((MAX[Txs+Tzqoper, Trfc, Tzqcs] + MAX[32 * Tstag, 8 * TstagPlr, 32*AutoRefCmdSep, Trfc]) * (2 ^ IsThermalEventEnabled)).		
Here 32 = MAX_RANKS * MAX_PLRS_PER_RANK		
For 2X mode:		
Tref > ((MAX[Txs+Tzqoper, 2*Trfc2,2*Tzqcs] + MAX[32 * Tstag * 2, 8 * Tstag2plr * 2, 32*AutoRefCmdSep*2, 2 * Trfc2]) * (2 ^ IsThermalEventEnabled)).		
For 4X mode:		
Tref > ((MAX[Txs+Tzqoper, 4*Trfc2,4*Tzqcs]r + MAX[32 * Tstag * 4, 8 * Tstag4plr * 4, 32*AutoRefCmdSep*4, 4 * Trfc4]) * (2 ^ IsThermalEventEnabled)).		
For dynamic mode:		
Tref > ((MAX[Txs+Tzqoper, 4*Trfc4,4*Tzqcs] + MAX[32 * Tstag * 4, 32*AutoRefCmdSep*4, 8 * Tstagplr, 8 * Tstag2plr * 2, 8 * Tstag4plr * 4, Trfc, 2 * Trfc2, 4 * Trfc4]) * (2 ^ IsThermalEventEnabled)).		
_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000230; UMC[11:00]CH=00[B:0]5_0000h		
_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000330; UMC[11:00]CH=00[B:0]5_0000h		
_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000430; UMC[11:00]CH=00[B:0]5_0000h		
_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000530; UMC[11:00]CH=00[B:0]5_0000h		
Bits	Description	
31:30	Reserved.	
29:20	OdtsReadInterval. Read-write. Reset: 000h. Specifies the interval in REFCLK clock cycles for the periodic DRAM thermal sensor read. See ShortInit. [0x000: Disabled, 0x3FF..0x001: OdtsReadInterval * 2 ^{^(ShortInit ? 10 : 20)} REFCLKs]	
19:16	Reserved.	
15:0	Tref. Read-write. Reset: 2490h. Specifies the average time between refresh requests. See: DisAutoRefresh. [0x003B..0x000: Reserved, 0x36D9 .. 0x003C: <Tref> MEMCLKs, 0xFFFF..0x36D9: Reserved]	

UMC00CHx00000234...UMC11CHx00000534 [DRAM Timing 13] (UMC::DramTiming13)

Read-write. Reset: 2020_1808h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000234; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000334; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000434; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000534; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description						
31:30	Reserved.						
29:24	TmodPda. Read-write. Reset: 20h. Specifies the minimum time in memory clock cycles from an MRS PDA command to another non-MRS command, all to the same chipselect. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>TmodPda MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	TmodPda MEMCLKs
Value	Description						
00h	Reserved.						
3Fh-01h	TmodPda MEMCLKs						
23:22	Reserved.						
21:16	TmrdPda. Read-write. Reset: 20h. Specifies the minimum time in memory clock cycles from an MRS PDA command to another MRS command, all to the same chipselect. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>TmrdPda MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	TmrdPda MEMCLKs
Value	Description						
00h	Reserved.						
3Fh-01h	TmrdPda MEMCLKs						
15:14	Reserved.						
13:8	Tmod. Read-write. Reset: 18h. Specifies the minimum time in memory clock cycles from an MRS command to another non-MRS command, all to the same chipselect. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>Tmod MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	Tmod MEMCLKs
Value	Description						
00h	Reserved.						
3Fh-01h	Tmod MEMCLKs						
7:6	Reserved.						
5:0	Tmrd. Read-write. Reset: 08h. Specifies the minimum time in memory clock cycles from an MRS command to another MRS command, all to the same chipselect. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>Tmrd MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	Tmrd MEMCLKs
Value	Description						
00h	Reserved.						
3Fh-01h	Tmrd MEMCLKs						

UMC00CHx00000238...UMC11CHx00000538 [DRAM Timing 14] (UMC::DramTiming14)

Read-write. Reset: 0300_0144h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000238; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000338; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000438; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000538; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description
31:28	Reserved.
27:16	Tdllk. Read-write. Reset: 300h. Specifies the time to exit self refresh to commands requiring a locked DLL. [0x254..0x000: Reserved, 0x400..0x255 <Tdllk> MEMCLKs, 0x7FF..0x401: Reserved]
15:12	Reserved.
11:0	Txs. Read-write. Reset: 144h. Specifies the time to exit self refresh to commands not requiring a locked DLL. [0x070..0x000: Reserved, 0x30F..0x71: <Txs> MEMCLKs, 0x3FF..0x310: Reserved]

UMC00CHx0000023C...UMC11CHx0000053C [DRAM Timing 15] (UMC::DramTiming15)

Read-write. Reset: 2400_2024h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx0000023C; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx0000033C; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx0000043C; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx0000053C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description								
31:24	RankBusyDly. Read-write. Reset: 24h. Specifies the rank busy delay. Programmed to the max of (AlertCrcDly, AlertWcrcDly, or ECC return delay) if any recovery feature is enabled.								
23:20	Reserved.								
19:16	CmdParLatency. Read-write. Reset: 0h. Cmd/Addr Parity Latency. For RCD parity, represents the DDR5 nLadd value. ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>8h-0h</td><td>CmdParLatency MEMCLKs</td></tr> <tr> <td>Fh-9h</td><td>Reserved.</td></tr> </table>	Value	Description	8h-0h	CmdParLatency MEMCLKs	Fh-9h	Reserved.		
Value	Description								
8h-0h	CmdParLatency MEMCLKs								
Fh-9h	Reserved.								
15	Reserved.								
14:8	AlertParDly. Read-write. Reset: 20h. Expected ALERT parity error delay. Maximum supported value: 'd88 ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>58h-01h</td><td>AlertParDelay MEMCLKs</td></tr> <tr> <td>7Fh-59h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Reserved.	58h-01h	AlertParDelay MEMCLKs	7Fh-59h	Reserved.
Value	Description								
00h	Reserved.								
58h-01h	AlertParDelay MEMCLKs								
7Fh-59h	Reserved.								
7:0	AlertCrcDly. Read-write. Reset: 24h. Expected ALERT CRC error delay. Maximum supported value: 'd137 ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>89h-01h</td><td>AlertCrcDelay MEMCLKs</td></tr> <tr> <td>FFh-8Ah</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Reserved.	89h-01h	AlertCrcDelay MEMCLKs	FFh-8Ah	Reserved.
Value	Description								
00h	Reserved.								
89h-01h	AlertCrcDelay MEMCLKs								
FFh-8Ah	Reserved.								

UMC00CHx00000240...UMC11CHx00000540 [DRAM Timing 16] (UMC::DramTiming16)

Read-write. Reset: 0308_0801h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000240; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000340; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000440; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000540; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description						
31:27	Reserved.						
26:24	Tmrri. Read-write. Reset: 3h. Reserved if DDR5 mode. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Reserved.</td></tr> <tr> <td>7h-1h</td><td>Tmrri MEMCLKs</td></tr> </table>	Value	Description	0h	Reserved.	7h-1h	Tmrri MEMCLKs
Value	Description						
0h	Reserved.						
7h-1h	Tmrri MEMCLKs						
23:22	Reserved.						
21:16	Tmrr. Read-write. Reset: 08h. Specifies the minimum time in memory clock cycles for mode register read command period. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>Tmrr MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	Tmrr MEMCLKs
Value	Description						
00h	Reserved.						
3Fh-01h	Tmrr MEMCLKs						
15:14	Reserved.						
13:8	Tmrw. Read-write. Reset: 08h. Specifies the minimum time in memory clock cycles for mode register write command period. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>Tmrw MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	Tmrw MEMCLKs
Value	Description						
00h	Reserved.						
3Fh-01h	Tmrw MEMCLKs						
7:0	CtrlSwitchClks. Read-write. Reset: 01h. Description: Read-write. Reset: 01h. Specifies the minimum additional time in memory clock cycles for ARB control to UMCCTRL switch. Reserved. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>FFh-01h</td><td>CtrlSwitchClks MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	FFh-01h	CtrlSwitchClks MEMCLKs
Value	Description						
00h	Reserved.						
FFh-01h	CtrlSwitchClks MEMCLKs						

UMC00CHx00000244...UMC11CHx00000544 [DRAM Timing 17] (UMC::DramTiming17)

Read-write. Reset: FCC0_FF06h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000244; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000344; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000444; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000544; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description						
31:26	AggrPwrDownDly. Read-write. Reset: 3Fh. Specifies a count from the last DRAM activity for the UMC to close pages for precharge power down entry. The counter runs orthogonal to the page policy specified by IdleCycleLimit. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>AggrPwrDownDly MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	AggrPwrDownDly MEMCLKs
Value	Description						
00h	Reserved.						
3Fh-01h	AggrPwrDownDly MEMCLKs						
25:20	Tcsh. Read-write. Reset: 0Ch. Specifies the DRAM tCSH value. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>(Tcsh << TcshMult) MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	(Tcsh << TcshMult) MEMCLKs
Value	Description						
00h	Reserved.						
3Fh-01h	(Tcsh << TcshMult) MEMCLKs						
19:8	PwrDownDly. Read-write. Reset: 0FFh. Specifies the power down entry delay as the number of memory clock cycles from the last command. [0xFFF..0x1: <PwrDownDly> MEMCLKs].						
7:5	Reserved.						
4:0	Tpd. Read-write. Reset: 06h. Specifies the minimum duration for powerdown if not in low power state, as well as gap between exiting and entering powerdown in memory clock cycles. Also specifies the minimum time in memory clock cycles for powerdown entry to exit timing. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>1Fh-01h</td><td>Tpd MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	1Fh-01h	Tpd MEMCLKs
Value	Description						
00h	Reserved.						
1Fh-01h	Tpd MEMCLKs						

UMC00CHx00000248...UMC11CHx00000548 [DRAM Timing 18] (UMC::DramTiming18)

Read-write. Reset: 0010_0DACH.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000248; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000348; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000448; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000548; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description				
31:22	Reserved.				
21:16	Tsrx2srx. Read-write. Reset: 10h. Specifies the time delay in MEMCLKs between the first NOP for SRX to 3 consecutive NOPs in DDR5 mode. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>3Fh-00h</td><td>Tsrx2srx MEMCLKs</td></tr> </table>	Value	Description	3Fh-00h	Tsrx2srx MEMCLKs
Value	Description				
3Fh-00h	Tsrx2srx MEMCLKs				
15:14	Reserved.				
13:0	Tstab. Read-write. Reset: 0DACH. Specifies the time delay in MEMCLKs between the CS low to high transition to NOP command in DDR5 mode. Programmed per Tstab = tCKACT+tSTAB01 in DDR5 mode. [0x000: Reserved, 0x3FFF..0x1: <Tstab> MEMCLKs]				

UMC00CHx0000024C...UMC11CHx0000054C [DRAM Timing 19] (UMC::DramTiming19)

Read-write. Reset: 0000_0101h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx0000024C; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx0000034C; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx0000044C; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx0000054C; UMC[11:00]CH=00[B:0]5_0000h`**Bits Description**

31:11 Reserved.

10:8 **AlertCrcPulse**. Read-write. Reset: 1h. Expected CRC error pulse threshold.

7 Reserved.

6:0 **AlertParPulse**. Read-write. Reset: 01h. Expected parity error pulse threshold.**UMC00CHx00000250...UMC11CHx00000550 [DRAM Timing 20] (UMC::DramTiming20)**

Read-write. Reset: 0007_0000h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000250; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000350; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000450; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000550; UMC[11:00]CH=00[B:0]5_0000h`**Bits Description**

31:27 Reserved.

26:16 **Tsttag**. Read-write. Reset: 007h. Specifies the minimum time in memory clocks from a REFab/RFMab command for any chipselect to a REFab/RFMab/REFsb/RFMsb/REFpb/RFMpb command for a different chipselect. [0x7FF..0x01 <Tsttag> MEMCLKs] Programmed per DDR5 tRFC1_dpr if AutoRefFineGranMode is 'Fixed 1X'. Programmed per DDR5 tRFC2_dpr if AutoRefFineGranMode is 'Fixed 2X'.

15:9 Reserved.

8:0 **Tstagsb**. Read-write. Reset: 000h. Specifies the minimum time in memory clock cycles from a REFsb/RFMsb/REFpb/RFMpb command for any chipselect to a REFab/RFMab/REFsb/RFMsb/REFpb/RFMpb command for the same or a different chipselect. [0x1FF..0x01 <Tstags> MEMCLKs] Programmed per DDR5 tRFC1sb_dlr.

UMC00CHx00000254...UMC11CHx00000554 [DRAM Timing 21] (UMC::DramTiming21)

Read-write. Reset: 0604_0208h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000254; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000354; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000454; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000554; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description								
31:29	Reserved.								
28:24	Tcke. Read-write. Reset: 06h. Specifies the CKE minimum high and low pulse width in memory clock cycles. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>1Fh-01h</td><td>Tcke MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	1Fh-01h	Tcke MEMCLKs		
Value	Description								
00h	Reserved.								
1Fh-01h	Tcke MEMCLKs								
23:21	Reserved.								
20:16	Tcpded. Read-write. Reset: 04h. Specifies the DRAM tCPDED in memory clock cycles. Maximum supported value: 'd29 ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>03h-00h</td><td>Reserved.</td></tr> <tr> <td>1Dh-04h</td><td>Tcpded MEMCLKs</td></tr> <tr> <td>1Fh-1Eh</td><td>Reserved.</td></tr> </table>	Value	Description	03h-00h	Reserved.	1Dh-04h	Tcpded MEMCLKs	1Fh-1Eh	Reserved.
Value	Description								
03h-00h	Reserved.								
1Dh-04h	Tcpded MEMCLKs								
1Fh-1Eh	Reserved.								
15:14	Reserved.								
13:8	Tcacsh. Read-write. Reset: 02h. Specifies tCACSH value in memory clock cycles. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>Tcacsh MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	Tcacsh MEMCLKs		
Value	Description								
00h	Reserved.								
3Fh-01h	Tcacsh MEMCLKs								
7:6	Reserved.								
5:0	Txp. Read-write. Reset: 08h. Specifies the minimum time in memory clock cycles to issue a command after exiting precharge powerdown mode. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>03h-00h</td><td>Reserved.</td></tr> <tr> <td>0Bh-04h</td><td>Txp MEMCLKs</td></tr> <tr> <td>3Fh-0Ch</td><td>Reserved.</td></tr> </table>	Value	Description	03h-00h	Reserved.	0Bh-04h	Txp MEMCLKs	3Fh-0Ch	Reserved.
Value	Description								
03h-00h	Reserved.								
0Bh-04h	Txp MEMCLKs								
3Fh-0Ch	Reserved.								

UMC00CHx00000258...UMC11CHx00000558 [DRAM Timing 22] (UMC::DramTiming22)

Read-write. Reset: 0004_0420h.		
_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000258; UMC[11:00]CH=00[B:0]5_0000h		
_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000358; UMC[11:00]CH=00[B:0]5_0000h		
_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000458; UMC[11:00]CH=00[B:0]5_0000h		
_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000558; UMC[11:00]CH=00[B:0]5_0000h		
Bits	Description	
31:30	Reserved.	
29:28	Tparin_lat . Read-write. Reset: 0h. Specifies DFI tparin_lat (ctrl signals to parity delay) in memory clock cycles.	
	ValidValues:	
	Value	Description
	3h-0h	Tparin_lat MEMCLKs
27	Reserved.	
26:24	Tphy_wrdata . Read-write. Reset: 0h. Specifies DFI tphy_wrdata (dfi_wrdata_en to dfi_wrdata delay) in memory clock cycles.	
	ValidValues:	
	Value	Description
	7h-0h	Tphy_wrdata MEMCLKs
23:16	Tphy_rldat . Read-write. Reset: 04h. Specifies DFI tphy_rldat (dfi_rddata_en to dfi_rddata_vld delay) in memory clock cycles. See PhyRdLatMode.	
	ValidValues:	
	Value	Description
	FFh-00h	Tphy_rldat MEMCLKs
15:8	Tphy_wrlat . Read-write. Reset: 04h. Specifies DFI tphy_wrlat (write command to dfi_wrdata_en delay) in memory clock cycles. Maximum supported value: 'd56 when WrCrcEn=0, 'd47 when WrCrcEn=1	
	ValidValues:	
	Value	Description
	38h-00h	Tphy_wrlat MEMCLKs
	FFh-39h	Reserved.
7	Reserved.	
6:0	Trddata_en . Read-write. Reset: 20h. Specifies DFI trddat_en (read command to dfi_rddata_en delay) in memory clock cycles. Maximum supported value: 'd50	
	ValidValues:	
	Value	Description
	32h-00h	Trddata_en MEMCLKs
	7Fh-33h	Reserved.

UMC00CHx0000025C...UMC11CHx0000055C [DRAM Timing 23] (UMC::DramTiming23)

Read-write. Reset: 2200_2A2Ah.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx0000025C; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx0000035C; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx0000045C; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx0000055C; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description								
31	Reserved.								
30:28	TgearHold. Read-write. Reset: 2h. Specifies the tGEAR_hold in memory clock cycles.								
	ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>7h-0h</td><td>TgearHold MEMCLKs</td></tr> </table>	Value	Description	7h-0h	TgearHold MEMCLKs				
Value	Description								
7h-0h	TgearHold MEMCLKs								
27	Reserved.								
26:24	TgearSetup. Read-write. Reset: 2h. Specifies the tGEAR_setup in memory clock cycles.								
	ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>7h-0h</td><td>TgearSetup MEMCLKs</td></tr> </table>	Value	Description	7h-0h	TgearSetup MEMCLKs				
Value	Description								
7h-0h	TgearSetup MEMCLKs								
23:14	Reserved.								
13:8	LpExitDly. Read-write. Reset: 2Ah. Minimum number of MEMCLKs to count before taking a rank out of power down after the device gets into low power state.								
	ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>LpExitDly MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	LpExitDly MEMCLKs		
Value	Description								
00h	Reserved.								
3Fh-01h	LpExitDly MEMCLKs								
7:6	Reserved.								
5:0	LpDly. Read-write. Reset: 2Ah. Specifies the hysteresis delay prior to placing the phy in low power mode in memory clock cycles.								
	ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>1Fh-01h</td><td>LpDly MEMCLKs</td></tr> <tr> <td>3Fh-20h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Reserved.	1Fh-01h	LpDly MEMCLKs	3Fh-20h	Reserved.
Value	Description								
00h	Reserved.								
1Fh-01h	LpDly MEMCLKs								
3Fh-20h	Reserved.								

UMC00CHx00000260...UMC11CHx0000056C [DRAM Trfc Timing] (UMC::TrfcTiming)

Read-write. Reset: 00C0_0138h.

Constraint: Trfc >= TstagLr.Constraint: Trfc >= Trfc2. Constraint: Trfc2 >= TstagLr2.

_umc0_cs0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000260; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs1_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000264; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs2_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000268; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs3_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx0000026C; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000360; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs1_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000364; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs2_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000368; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs3_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx0000036C; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000460; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs1_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000464; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs2_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000468; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs3_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx0000046C; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000560; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs1_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000564; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs2_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000568; UMC[11:00]CH=00[B:0]5_0000h

_umc0_cs3_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx0000056C; UMC[11:00]CH=00[B:0]5_0000h

Bits Description

31:28 Reserved.

27:16 **Trfc2.** Read-write. Reset: 0C0h. Specifies the Fine Granularity Refresh mode REF command period. [0xFFF..0x1: <Trfc> MEMCLKs] In DDR5 mode, programmed per DDR5 tRFC2/tRFC2_slr.

15:12 Reserved.

11:0 **Trfc.** Read-write. Reset: 138h. Specifies the REF command period. [0xFFF..0x1: <Trfc> MEMCLKs] Programmed per DDR5 tRFC1, tRFC1_slr.**UMC00CHx00000274...UMC11CHx00000574 [Sync FIFO Margin] (UMC::SyncFifoMargin)**

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000274; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000374; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000474; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000574; UMC[11:00]CH=00[B:0]5_0000h

Bits Description

31:15 Reserved.

14:8 **S2M_Margin_Phases.** Read-write. Reset: 00h. Additional DFS phases of read margin in the S2M direction.**ValidValues:****Value Description**

00h Disabled.

7Fh-01h DFS Clock Phases.

7 Reserved.

6:0 **M2S_Margin_Phases.** Read-write. Reset: 00h. Additional DFS phases of read margin in the M2S direction.**ValidValues:****Value Description**

00h Disabled.

7Fh-01h DFS Clock Phases.

UMC00CHx00000278...UMC11CHx00000578 [Master VDCI Control] (UMC::MasterVdciCtl)

Read-write. Reset: A000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000278; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000378; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000478; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000578; UMC[11:00]CH=00[B:0]5_0000h`**Bits Description**31 **CombinationalBypass.** Read-write. Reset: 1.30 **SyncFifoMode.** Read-write. Reset: 0.29 **SkipToFreqAdjAckForBypass.** Read-write. Reset: 1.

28:18 Reserved.

17:11 **ReadLookAhead.** Read-write. Reset: 00h.10:9 **ReadSync.** Read-write. Reset: 0h.8:2 **WriteLookAhead.** Read-write. Reset: 00h.1:0 **WriteSync.** Read-write. Reset: 0h.**UMC00CHx0000027C...UMC11CHx0000057C [Slave VDCI Control] (UMC::SlaveVdciCtl)**

Read-write. Reset: A000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx0000027C; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx0000037C; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx0000047C; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx0000057C; UMC[11:00]CH=00[B:0]5_0000h`**Bits Description**31 **CombinationalBypass.** Read-write. Reset: 1.30 **SyncFifoMode.** Read-write. Reset: 0.29 **SkipToFreqAdjAckForBypass.** Read-write. Reset: 1.

28:18 Reserved.

17:11 **ReadLookAhead.** Read-write. Reset: 00h.10:9 **ReadSync.** Read-write. Reset: 0h.8:2 **WriteLookAhead.** Read-write. Reset: 00h.1:0 **WriteSync.** Read-write. Reset: 0h.

UMC00CHx00000280...UMC11CHx00000580 [DRAM ODT Timing] (UMC::OdtTiming)

Read-write. Reset: 0000_0000h.

RdOdtOnDur and WrOdtOnDur are configured relative to BurstLength setting.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000280; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000380; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000480; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000580; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description								
31:16	Reserved.								
15:12	WrOdtOnDur . Read-write. Reset: 0h. Specifies the number of memory clock cycles that ODT is asserted. ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Disabled</td></tr> <tr> <td>2h-1h</td><td>Reserved.</td></tr> <tr> <td>Fh-3h</td><td>WrOdtOnDur MEMCLKs</td></tr> </table>	Value	Description	0h	Disabled	2h-1h	Reserved.	Fh-3h	WrOdtOnDur MEMCLKs
Value	Description								
0h	Disabled								
2h-1h	Reserved.								
Fh-3h	WrOdtOnDur MEMCLKs								
11:8	WrOdtTrnOnDly . Read-write. Reset: 0h. Specifies the number of memory clock cycles that ODT assertion is delayed relative to write CAS. Constraint: (WrOdtTrnOnDly == 0 WrOdtOnDur > 0). ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>WrOdtTrnOnDly MEMCLKs</td></tr> </table>	Value	Description	Fh-0h	WrOdtTrnOnDly MEMCLKs				
Value	Description								
Fh-0h	WrOdtTrnOnDly MEMCLKs								
7:4	RdOdtOnDur . Read-write. Reset: 0h. Specifies the number of memory clock cycles that ODT is asserted. ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Disabled</td></tr> <tr> <td>2h-1h</td><td>Reserved.</td></tr> <tr> <td>Fh-3h</td><td>RdOdtOnDur MEMCLKs</td></tr> </table>	Value	Description	0h	Disabled	2h-1h	Reserved.	Fh-3h	RdOdtOnDur MEMCLKs
Value	Description								
0h	Disabled								
2h-1h	Reserved.								
Fh-3h	RdOdtOnDur MEMCLKs								
3:0	RdOdtTrnOnDly . Read-write. Reset: 0h. Specifies the number of memory clock cycles that ODT assertion is delayed relative to read CAS. Constraint: (RdOdtTrnOnDly == 0 RdOdtOnDur > 0). ValidValues:								
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>RdOdtTrnOnDly MEMCLKs</td></tr> </table>	Value	Description	Fh-0h	RdOdtTrnOnDly MEMCLKs				
Value	Description								
Fh-0h	RdOdtTrnOnDly MEMCLKs								

UMC00CHx00000284...UMC11CHx00000584 [Channel Pipe Delay] (UMC::ChanPipeDly)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000284; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000384; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000484; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000584; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description						
31:11	Reserved.						
10:8	RxDatChnDly. Read-write. Reset: 0h. Init: 2h. Number of delay stages on DFI read data from PHY to UMC. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>4h-0h</td><td>RxDatChnDly stages</td></tr> <tr> <td>7h-5h</td><td>Reserved.</td></tr> </table>	Value	Description	4h-0h	RxDatChnDly stages	7h-5h	Reserved.
Value	Description						
4h-0h	RxDatChnDly stages						
7h-5h	Reserved.						
7	Reserved.						
6:4	TxDatChnDly. Read-write. Reset: 0h. Init: 3h. Number of delay stages on DFI write data from UMC to PHY. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>4h-0h</td><td>TxDatChnDly stages</td></tr> <tr> <td>7h-5h</td><td>Reserved.</td></tr> </table>	Value	Description	4h-0h	TxDatChnDly stages	7h-5h	Reserved.
Value	Description						
4h-0h	TxDatChnDly stages						
7h-5h	Reserved.						
3	Reserved.						
2:0	TxCtrlChnDly. Read-write. Reset: 0h. Init: 1h. Number of delay stages on DFI control signals from UMC to PHY. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>4h-0h</td><td>TxCtrlChnDly stages</td></tr> <tr> <td>7h-5h</td><td>Reserved.</td></tr> </table>	Value	Description	4h-0h	TxCtrlChnDly stages	7h-5h	Reserved.
Value	Description						
4h-0h	TxCtrlChnDly stages						
7h-5h	Reserved.						

UMC00CHx00000288...UMC11CHx00000588 [DRAM Timing 34] (UMC::DramTiming34)

Read-write. Reset: 0000_0552h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx00000288; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx00000388; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx00000488; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx00000588; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description								
31:12	Reserved.								
11:8	Tphyupd_CmdDly. Read-write. Reset: 5h. Specifies the minimum number of memory clock cycles to delay dfi_phyupd_ack after any valid command. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Fh-0h</td><td>Tphyupd_CmdDly MEMCLKs</td></tr> </table>	Value	Description	Fh-0h	Tphyupd_CmdDly MEMCLKs				
Value	Description								
Fh-0h	Tphyupd_CmdDly MEMCLKs								
7:4	Tphyupd_WrDatDly. Read-write. Reset: 5h. Specifies the minimum number of memory clock cycles to delay dfi_phyupd_ack after write data movement. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>4h-0h</td><td>Reserved.</td></tr> <tr> <td>Fh-5h</td><td>Tphyupd_WrDatDly MEMCLKs</td></tr> </table>	Value	Description	4h-0h	Reserved.	Fh-5h	Tphyupd_WrDatDly MEMCLKs		
Value	Description								
4h-0h	Reserved.								
Fh-5h	Tphyupd_WrDatDly MEMCLKs								
3:0	Tphyupd_resp. Read-write. Reset: 2h. Check: 2h. Specifies the maximum number of memory clock cycles to acknowledge assertion of DFI phyupd_req. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Disabled</td></tr> <tr> <td>8h-1h</td><td>512 x Tphyupd_resp MEMCLKs</td></tr> <tr> <td>Fh-9h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Disabled	8h-1h	512 x Tphyupd_resp MEMCLKs	Fh-9h	Reserved.
Value	Description								
0h	Disabled								
8h-1h	512 x Tphyupd_resp MEMCLKs								
Fh-9h	Reserved.								

UMC00CHx0000028C...UMC11CHx0000058C [DRAM Timing 35] (UMC::DramTiming35)

Read-write. Reset: 1800_2980h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx0000028C; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx0000038C; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx0000048C; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx0000058C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description				
31:30	Reserved.				
29:24	Twrmpr. Read-write. Reset: 18h. Specifies the number of tWR_MPR clocks greater than Tmod. Program Twrmpr = MAX (0, JEDEC tWR_MPR - DramTiming13.Tmod). ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>3Fh-00h</td><td>Additional Twrmpr MEMCLKs</td></tr> </table>	Value	Description	3Fh-00h	Additional Twrmpr MEMCLKs
Value	Description				
3Fh-00h	Additional Twrmpr MEMCLKs				
23:22	Reserved.				
21:11	CmdStgCnt. Read-write. Reset: 005h. Recovery sequence command stagger counter. See CmdStgFrc. [0x5 .. 0x7FF: CmdStgCnt MEMCLKs]				
10:0	RcvrWait. Read-write. Reset: 180h. Wait time to start recovery sequence. [0x1 .. 0x7FF: RcvrWait MEMCLKs]				

UMC00CHx000002A0...UMC11CHx000005A0 [Clk_disable_timing] (UMC::ClkDisableTiming)

Read-write. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002A0; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003A0; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004A0; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005A0; UMC[11:00]CH=00[B:0]5_0000h`**Bits Description**

31:13 Reserved.

12:8 **Tdram_clk_enable**. Read-write. Reset: 00h. Specifies DFI tdram_clk_enable (deassertion of dfi_dram_clk_disable to CK toggling delay) in memclock cycles.

7:5 Reserved.

4:0 **Tdram_clk_disable**. Read-write. Reset: 00h. Specifies DFI tdram_clk_disable (assertion of dfi_dram_clk_disable to CK low delay) in memclock cycles.**UMC00CHx000002A4...UMC11CHx000005A4 [Dram Timing 34] (UMC::DramTiming41)**

Read-write. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002A4; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003A4; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004A4; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005A4; UMC[11:00]CH=00[B:0]5_0000h`**Bits Description**

31:15 Reserved.

14:12 **WrPost**. Read-write. Reset: 0h. Specifies the Write Postamble mode.

11 Reserved.

10:8 **WrPre**. Read-write. Reset: 0h. Specifies the Write Preamble mode.

7 Reserved.

6:4 **RdPost**. Read-write. Reset: 0h. Specifies the Read Postamble mode.

3 Reserved.

2:0 **RdPre**. Read-write. Reset: 0h. Specifies the Read Preamble mode.**UMC00CHx000002A8...UMC11CHx000005A8 [Dram Timing 42] (UMC::DramTiming42)**

Read-write. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002A8; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003A8; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004A8; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005A8; UMC[11:00]CH=00[B:0]5_0000h`**Bits Description**

31:9 Reserved.

8:0 **tECS**. Read-write. Reset: 000h. [ECS](#) RMW operation time in MEMCLKs. Programmed per JEDEC tECS.

UMC00CHx000002B0...UMC11CHx000005BC [DRAM Tstag Logical Rank Timing] (UMC::TstagLrTiming)

Read-write. Reset: 0000_0000h.

_umc0_cs0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002B0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs1_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002B4; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs2_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002B8; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs3_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002BC; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003B0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs1_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003B4; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs2_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003B8; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs3_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003BC; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004B0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs1_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004B4; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs2_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004B8; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs3_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004BC; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005B0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs1_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005B4; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs2_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005B8; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs3_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005BC; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:27	Reserved.
26:16	Tstag2Lr. Read-write. Reset: 000h. Specifies the minimum number of clocks between REF commands to different logical ranks during Fine Granularity Refresh mode. [0x000: Disabled, 0x7FF..0x001: <Tstag2Lr> MEMCLKs] Programmed per DDR5 tRFC2_dlr.
15:11	Reserved.
10:0	TstagLr. Read-write. Reset: 000h. Specifies the minimum number of clocks between REF commands to different logical ranks. [0x000: Disabled, 0x7FF..0x001: <TstagLr> MEMCLKs] Programmed per DDR5 tRFC1_dlr.

UMC00CHx000002C0...UMC11CHx000005CC [DRAM Refresh Timing] (UMC::RefTiming)

Read-write. Reset: 0000_0000h.

_umc0_cs0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002C0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs1_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002C4; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs2_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002C8; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs3_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002CC; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003C0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs1_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003C4; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs2_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003C8; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs3_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003CC; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004C0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs1_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004C4; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs2_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004C8; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs3_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004CC; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005C0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs1_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005C4; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs2_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005C8; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs3_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005CC; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:24	Reserved.
23:16	Trefsbrd. Read-write. Reset: 00h. Reserved if !PerBankRefEn. Programmed per DDR5 tREFSBRD/tREFSBRD_slr.
15:11	Reserved.
10:0	Trfcsb. Read-write. Reset: 000h. Reserved if !PerBankRefEn. Programmed per DDR5 tRFCsb/tRFCsb_slr.

UMC00CHx000002D0...UMC11CHx000005DC [DRAM Refresh Logical Rank Timing] (UMC::RefLrTiming)

Read-write. Reset: 0000_0000h.

_umc0_cs0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002D0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs1_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002D4; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs2_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002D8; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs3_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002DC; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003D0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs1_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003D4; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs2_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003D8; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs3_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003DC; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004D0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs1_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004D4; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs2_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004D8; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs3_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004DC; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005D0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs1_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005D4; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs2_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005D8; UMC[11:00]CH=00[B:0]5_0000h
_umc0_cs3_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005DC; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:24	Reserved.
23:16	TrfcsbDlR . Read-write. Reset: 00h. tREFSBRD_dlr in DDR5.
15:9	Reserved.
8:0	TrfcsbDlR . Read-write. Reset: 000h. tREFSBRD_dlr in DDR5.

UMC00CHx000002E0...UMC11CHx000005E0 [DRAM Timing 10] (UMC::DramTiming56)

Read-write. Reset: 0000_0001h.

Specifies CAS to CAS command spacing. The last clock of virtual CAS is defined as BL/2-1.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002E0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003E0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004E0; UMC[11:00]CH=00[B:0]5_0000h
_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005E0; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description						
31:6	Reserved.						
5:0	TwrrwrScL2 . Read-write. Reset: 01h. Description: Read-write. Reset: 01h. Specifies the minimum number of cycles from the last clock of virtual CAS of a first write-burst operation to the clock in which CAS is asserted for a following write-burst operation, where all bytes are enabled, in the same bank group. Used to satisfy JEDEC tCCD_L_WR2.						
	ValidValues: <table border="1"> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>TwrrwrScL2 MEMCLKs</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	TwrrwrScL2 MEMCLKs
Value	Description						
00h	Reserved.						
3Fh-01h	TwrrwrScL2 MEMCLKs						

UMC00CHx000002E4...UMC11CHx000005E4 [Sub Channel Throttle Ctrl] (UMC::SubChanThrotCtrl)

Read-write. Reset: 0000_0000h.

Features to control sub-channel CAS throttling to reduce DDRPHY VDD Droop

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CHx000002E4; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CHx000003E4; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CHx000004E4; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CHx000005E4; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description						
31:28	Twprhwr. Read-write. Reset: 0h. Specifies the minimum number of cycles from the last clock of virtual CAS of a regular write to the clock in which CAS is asserted for a phantom write. Should be programmed based on DramTiming41 register WrPre+WrPost values ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Disabled.</td></tr> <tr> <td>Fh-1h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Disabled.	Fh-1h	Reserved.
Value	Description						
0h	Disabled.						
Fh-1h	Reserved.						
27	PhWrWrRdEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable sending phantom writes on Wr->Rd turnarounds						
26	PhWrRdWrEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable sending phantom writes on Rd->Wr turnaround						
25	PhWrLdRelEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable sending phantom writes on load release						
24	PhWrLdStpEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable sending phantom writes on load step						
23:22	Reserved.						
21:16	SubChanThrotClk. Read-write. Reset: 00h. Number of MEMCLKs to throttle sending CAS on both sub-channels simultaneously if idle is detected ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disabled</td></tr> <tr> <td>3Fh-01h</td><td>Sub Channel Throttling Clks</td></tr> </table>	Value	Description	00h	Disabled	3Fh-01h	Sub Channel Throttling Clks
Value	Description						
00h	Disabled						
3Fh-01h	Sub Channel Throttling Clks						
15:8	SubChanThrotIdleClk. Read-write. Reset: 00h. Number of MEMCLKs to look in the past to determine idle status along with SubChanThrotIdleBW ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disabled.</td></tr> <tr> <td>FFh-01h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disabled.	FFh-01h	Reserved.
Value	Description						
00h	Disabled.						
FFh-01h	Reserved.						
7:4	SubChanThrotIdleBW. Read-write. Reset: 0h. Bandwidth limit under which sub-channel throttling is enabled. See also SubChanThrotIdleClk ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>fully idle</td></tr> <tr> <td>Fh-1h</td><td>N*6.25% BW</td></tr> </table>	Value	Description	0h	fully idle	Fh-1h	N*6.25% BW
Value	Description						
0h	fully idle						
Fh-1h	N*6.25% BW						
3:1	Reserved.						
0	SubChanThrotEn. Read-write. Reset: 0. 0=Disabled. 1=Enabled. Enables Sub-channel Throttling between CAS						

UMC[00...11]CHx00000A00 [Encryption Key Array Index] (UMC::EncrArrIndex)

Read-write. Reset: 0000_0000h.

Encryption Key array indexed 32b access port. The key index is Index[12:3] irrespective of AES-128 or AES-256 mode. Software must access in order from low to high incrementing Index[1:0] for each 128b key. Software must access in order from low to high incrementing Index[2:0] for each 256b key. Software using AutoInc for block key writes in AES-128 mode should zero-fill the upper 128b of the key index.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000A00; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	AutoInc. Read-write. Reset: 0. 0=Disable. 1=Enable. Index read/write auto increment enable.
30:13	Reserved.
12:0	Index. Read-write. Reset: 0000h. Encryption Key Array Index.

UMC[00...11]CHx00000A04 [Encryption Key Array Data.] (UMC::EncrArrData)

Read-write. Reset: 0000_0000h.

Software should configure EncrArrIndex and then read or write EncrArrData. EncrArrData accesses increment EncrArrIndex.Index if EncrArrIndex.AutoInc == 1.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000A04; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:0	Data. Read-write. Reset: 0000_0000h.

UMC[00...11]CHx00000A08 [Tweak Array Index] (UMC::TwkArrIndex)

Read-write. Reset: 0000_0000h.

Key array indexed 32b access port. Software must access in order from low to high incrementing Index[1:0] for each 128b key.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000A08; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	AutoInc. Read-write. Reset: 0. 0=Disable. 1=Enable. Index read/write auto increment enable.
30:2	Reserved.
1:0	Index. Read-write. Reset: 0h. Tweak array index

UMC[00...11]CHx00000A0C [Tweak Array Data] (UMC::TwkArrData)

Read-write. Reset: 0000_0000h.

Software should configure TwkArrIndex and then read or write TwkArrData. TwkArrData accesses increment TwkArrIndex.Index if TwkArrIndex.AutoInc == 1.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000A0C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:0	Data. Read-write. Reset: 0000_0000h.

UMC[00...11]CHx00000A18 [XTS Key Array Index] (UMC::XtsArrIndex)

Read-write. Reset: 0000_0000h.

XTS Key array indexed 32b access port. The key index is Index[12:3]. Software must access in order from low to high incrementing Index[2:0] for each 256b key.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000A18; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	AutoInc. Read-write. Reset: 0. 0=Disable. 1=Enable. Index read/write auto increment enable.
30:13	Reserved.
12:0	Index. Read-write. Reset: 0000h. XTS Key array index

UMC[00...11]CHx00000A1C [XTS Key Array Data] (UMC::XtsArrData)

Read-write. Reset: 0000_0000h.

Software should configure XtsArrIndex and then read or write XtsArrData. XtsArrData accesses increment XtsArrIndex.Index if XtsArrIndex.AutoInc == 1.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000A1C; UMC[11:00]CH=00[B:0]5_0000h**Bits Description**31:0 **Data.** Read-write. Reset: 0000_0000h.**UMC00CHx00000B00...UMC11CHx00000B0C [DRAM CS Base Address Extended] (UMC::BaseAddrExt)**

Read-write. Reset: 0000_0000h.

The BaseAddr and AddrMask registers translate request addresses to DRAM chipselects. A chipselect[i] is asserted if: CSEnable[i] & ((RequestAddr & ~AddrMask[i]) == (BaseAddr[i] & ~AddrMask[i])).

_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B00; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B04; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B08; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B0C; UMC[11:00]CH=00[B:0]5_0000h**Bits Description**

31:8 Reserved.

7:0 **BaseAddr: Base Address [47:40].** Read-write. Reset: 00h. Base address for chipselect decoding.**UMC00CHx00000B10...UMC11CHx00000B1C [DRAM CS Base Secondary Address Extended] (UMC::BaseAddrSecExt)**

Read-write. Reset: 0000_0000h.

The BaseAddr and AddrMask registers translate request addresses to DRAM chipselects. A chipselect[i] is asserted if: CSEnable[i] & ((RequestAddr & ~AddrMask[i]) == (BaseAddr[i] & ~AddrMask[i])).

_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B10; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B14; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B18; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B1C; UMC[11:00]CH=00[B:0]5_0000h**Bits Description**

31:8 Reserved.

7:0 **BaseAddr: Base Address [47:40].** Read-write. Reset: 00h. Base address for chipselect decoding.**UMC00CHx00000B20...UMC11CHx00000B2C [DRAM CS Mask Address Extended] (UMC::AddrMaskExt)**

Read-write. Reset: 0000_0000h.

_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B20; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B24; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B28; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B2C; UMC[11:00]CH=00[B:0]5_0000h**Bits Description**

31:8 Reserved.

7:0 **AddrMask: Address Mask [47:40].** Read-write. Reset: 00h. Address mask for chipselect decoding.**UMC00CHx00000B30...UMC11CHx00000B3C [DRAM CS Mask Secondary Address Extended] (UMC::AddrMaskSecExt)**

Read-write. Reset: 0000_0000h.

_umc0_cs0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B30; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B34; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B38; UMC[11:00]CH=00[B:0]5_0000h_umc0_cs3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000B3C; UMC[11:00]CH=00[B:0]5_0000h**Bits Description**

31:8 Reserved.

7:0 **AddrMask: Address Mask [47:40].** Read-write. Reset: 00h. Address mask for chipselect decoding.

UMC00CHx00000BB0...UMC11CHx00000BBC [Address Hash RM Extended] (UMC::AddrHashRMExt)

Read-write. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CHx00000BB0; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CHx00000BB4; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_n2_aliasSMN; UMC[11:00]CHx00000BB8; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_n3_aliasSMN; UMC[11:00]CHx00000BBC; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description
31:8	Reserved.
7:0	NormAddrXor: NormAddrXor [47:40] . Read-write. Reset: 00h. Check: 00h. Normalized address bits to use in hash function computation of the rank multiply bit.

UMC00CHx00000BC8...UMC11CHx00000BCC [Address Hash CS Extended] (UMC::AddrHashExtCS)

Read-write. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CHx00000BC8; UMC[11:00]CH=00[B:0]5_0000h``_umc0_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CHx00000BCC; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description
31:8	Reserved.
7:0	NormAddrXor: NormAddrXor[47:40] . Read-write. Reset: 00h. Check: 00h. Normalized address bits to use in hash function computation of the CS address bit.

UMC[00...11]CHx00000D00 [Performance Monitor Control] (UMC::PerfMonCtlClk)

Read-write. Reset: 0200_0000h.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000D00; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description
31	CtrClkEn . Read-write. Reset: 0. 0=Disable. 1=Enable. Individual enable bit for the clock cycle counter.
30:26	Reserved.
25	GlblMonEn . Read-write. Reset: 1. Check: 1. Global counter enable. Start any counter whose individual enable bit is active.
24	GlblReset . Read-write. Reset: 0. Check: 0. Reset each counter not being masked by GlblResetMsk.
23:6	Reserved.
5:0	GlblResetMsk . Read-write. Reset: 00h. Mask each counter from being reset by GlblReset.

UMC[00...11]CHx00000D04 [Performance Monitor Data] (UMC::PerfMonCtrClkLo)

Read-write. Reset: 0000_0000h.

Access low 32b of counter data. Software must always access in order from low to high.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000D04; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description
31:0	Data: Data[31:0] . Read-write. Reset: 0000_0000h.

UMC[00...11]CHx00000D08 [Performance Monitor Data] (UMC::PerfMonCtrClkHi)

Read-write. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000D08; UMC[11:00]CH=00[B:0]5_0000h`

Bits	Description
31:17	Reserved.
16	Overflow . Read-write. Reset: 0. Indicates the overflow of the 48b counter.
15:0	Data: Data[47:32] . Read-write. Reset: 0000h. Access high 16 bits of clock cycle counter. Software must always access in order from low to high.

UMC00CHx00000D10...UMC11CHx00000D40 [Performance Monitor Control] (UMC::PerfMonCtl)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CHx00000D10; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_n2_aliasSMN; UMC[11:00]CHx00000D1C; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_n3_aliasSMN; UMC[11:00]CHx00000D28; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_n4_aliasSMN; UMC[11:00]CHx00000D34; UMC[11:00]CH=00[B:0]5_0000h_umc0_inst[UMCWPHY[11:0]]_n5_aliasSMN; UMC[11:00]CHx00000D40; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	Enable. Read-write. Reset: 0. 0=Disable. 1=Enable. Counter enable.
30:29	SubChanMask. Read-write. Reset: 0h. Mask on UMC virtual controller mode subchannel.
ValidValues:	
Bit	Description
[0]	SubChan0
[1]	SubChan1
28:24	VCSel. Read-write. Reset: 00h. Log only events matching SDP Virtual Channel specified by VCSel[3:0] when enabled.
ValidValues:	
Bit	Description
[0]	VCSel0
[1]	VCSel1
[2]	VCSel2
[3]	VCSel3
[4]	Enable
23:20	ChipIDSel. Read-write. Reset: 0h. Log only events matching C[2:0] when enabled.
ValidValues:	
Bit	Description
[0]	C0
[1]	C1
[2]	C2
[3]	Enable
19:16	ChipSelMask. Read-write. Reset: 0h. Mask based on chipselect.
ValidValues:	
Bit	Description
[0]	CS0
[1]	CS1
[2]	CS2
[3]	CS3
15:14	ReqSizeMask. Read-write. Reset: 0h. Mask transactions based on request size.
ValidValues:	
Value	Description
0h	No masking
1h	Mask 32B transactions
2h	Mask 64B transactions
3h	Reserved.
13:10	PriorityMask. Read-write. Reset: 0h. Mask transactions based on low, medium, high, and urgent priority.

	ValidValues:	
	Bit	Description
	[0]	Mask Low Priority
	[1]	Mask Medium Priority
	[2]	Mask High Priority
	[3]	Mask Urgent Priority
9:8	RdWrMask. Read-write. Reset: 0h. Mask transactions based on read or write.	
	ValidValues:	
	Value	Description
	0h	No masking
	1h	Mask writes
	2h	Mask reads
	3h	Reserved.
7:0	EventSelect. Read-write. Reset: 00h. Select the performance monitor event.	

UMC00CHx00000D14...UMC11CHx00000D44 [Performance Monitor Data] (UMC::PerfMonCtrLo)

Read-write. Reset: 0000_0000h.	
_umc0_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CHx00000D14; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n2_aliasSMN; UMC[11:00]CHx00000D20; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n3_aliasSMN; UMC[11:00]CHx00000D2C; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n4_aliasSMN; UMC[11:00]CHx00000D38; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n5_aliasSMN; UMC[11:00]CHx00000D44; UMC[11:00]CH=00[B:0]5_0000h	
Bits	Description
31:0	Data: Data[31:0]. Read-write. Reset: 0000_0000h. Access low 32b of counter data. Software must always access in order from low to high.

UMC00CHx00000D18...UMC11CHx00000D48 [Performance Monitor Data] (UMC::PerfMonCtrHi)

Read-write. Reset: 0000_0000h.	
_umc0_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CHx00000D18; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n2_aliasSMN; UMC[11:00]CHx00000D24; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n3_aliasSMN; UMC[11:00]CHx00000D30; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n4_aliasSMN; UMC[11:00]CHx00000D3C; UMC[11:00]CH=00[B:0]5_0000h	
_umc0_inst[UMCWPHY[11:0]]_n5_aliasSMN; UMC[11:00]CHx00000D48; UMC[11:00]CH=00[B:0]5_0000h	
Bits	Description
31:17	Reserved.
16	Overflow. Read-write. Reset: 0. Overflow bit of ther performance counter.
15:0	Data: Data[47:32]. Read-write. Reset: 0000h. Access high 16b of counter data. Software must always access in order from low to high.

UMC[00...11]CHx00000D6C (UMC::DebugMisc)

Read-only, Volatile.

Indicates DFI initialization has completed.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000D6C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:17	Reserved.
16	InDstate . Read-only, Volatile. Indicates the current memory InDstate status.
15:13	DstateSel . Read-only, Volatile. Indicates the current memory Dstate select.
12:11	CurrentMemoryFSP . Read-only, Volatile. Indicates the current memory frequency set point.
10:6	Reserved.
5	UmcReady . Read-only, Volatile. Indicates UMC initialization internal logic is ready.
4	DfiInitComplete . Read-only, Volatile.
3	UclkDiv1Mode . Read-only, Volatile. Indicates the current UCLK divisor mode. 1=UCLK:MEMCLK is 1:1. 0=UCLK:MEMCLK is 1:2 (i.e. UCLK frequency = MEMCLK/2 frequency).
2:0	CurrentMemoryPstate . Read-only, Volatile. Indicates the current memory pstate.

UMC[00...11]CHx00000D80 [DRAM ECC Error Count Select] (UMC::EccErrCntSel)

Reset: 0000_0000h.

Specifies the index used by the EccErrCnt register.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000D80; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	AutoInc . Read-write. Reset: 0. 0=Disable. 1=Enable. Index read/write auto increment. When enabled, the index register will auto increment in hardware after a RD or WR access.
30:6	Reserved.
5:0	Index . Read-write, Volatile. Reset: 00h. ECC Error Counter Select Index. Bits [5:0] map to subchannel, chipselect, Plr as follows: {bit [5] - DDR5 DIMM Subchannel (0=A,1=B), bits [4:3] - Chipselect 3:0, bits [2:0] - RankMul 7:0 (LRDIMM/3DS)}.

UMC[00...11]CHx00000D84 [DRAM ECC Error Count] (UMC::EccErrCnt)

Reset: 0000_0000h.

The ECC error count value as selected by EccErrCntSel which maps to SubCh/CS/Plr.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000D84; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:16	EccErrThresh . Read-write. Reset: 0000h. ECC Threshold value, selected by EccErrCntSel[Index] which maps to SubCh/CS/Plr.
15:0	EccErrCnt . Read-write, Volatile. Reset: 0000h. ECC error counter value, selected by EccErrCntSel[Index] which maps to SubCh/CS/Plr. If EccErrCntSel[AutoInc] == 1, accesses incremented EccErrCntSel[Index].

UMC[00...11]CHx00000D88 [DRAM ECC Error Count Control] (UMC::EccErrCntCtrl)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000D88; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description										
31:13	Reserved.										
12:8	EccErrLeakRate. Read-write. Reset: 00h. Minimum Leak Rate assuming 100MHz REFCLK and InDstate0. (Times in parentheses are computed for 100MHz REFCLK.) EccErrLeakRate counter is started at the end of DDR Initialization when BIOS sets (DramReady = 1) and restarts at expiration of the Leak rate counter. LeakRate is paused during clock stop power management modes.										
	ValidValues: See: UMC::EccErrCntCtrl[EccErrLeakRate] Valid Values.										
7:6	Reserved.										
5:4	EccErrInt. Read-write. Reset: 0h. Specifies the type of interrupt generated when any EccErrCnt field transitions to FFFFh.										
	ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>No Interrupt</td></tr> <tr> <td>1h</td><td>LVT (Interrupt Request 2)</td></tr> <tr> <td>2h</td><td>SMI (Interrupt Request 0)</td></tr> <tr> <td>3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	No Interrupt	1h	LVT (Interrupt Request 2)	2h	SMI (Interrupt Request 0)	3h	Reserved.
Value	Description										
0h	No Interrupt										
1h	LVT (Interrupt Request 2)										
2h	SMI (Interrupt Request 0)										
3h	Reserved.										
3:2	Reserved.										
1	EccErrLeakEn. Read-write. Reset: 0. Enable Leaky Bucket mode for ECC Error Counters										
0	EccErrCntEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enables the per-chipselect ECC error counters. See EccErrCntSel and EccErrCnt.										

UMC::EccErrCntCtrl[EccErrLeakRate] Valid Values

Value	Description
00h	8 REFCLKs (80ns)
01h	16 REFCLKs (160ns)
02h	32 REFCLKs (320ns)
03h	64 REFCLKs (640ns)
04h	128 REFCLKs (1.28us)
05h	256 REFCLKs (2.56us)
06h	512 REFCLKs (5.12us)
07h	1K REFCLKs (10.24us)
08h	2K REFCLKs (20.48us)
09h	4K REFCLKs (40.96us)
0Ah	8K REFCLKs (81.92us)
0Bh	16K REFCLKs (163.84us)
0Ch	32K REFCLKs (327.68us)
0Dh	64K REFCLKs (655.36us)
0Eh	128K REFCLKs (1.31ms)
0Fh	256K REFCLKs (2.62ms)
10h	512K REFCLKs (5.24ms)
11h	2 ²⁰ REFCLKs (10.49ms)
12h	2 ²¹ REFCLKs (20.97ms)
13h	2 ²² REFCLKs (41.94ms)
14h	2 ²³ REFCLKs (83.88ms)
15h	2 ²⁴ REFCLKs (167.77ms)
16h	2 ²⁵ REFCLKs (335.54ms)

17h	64M REFCLKs (671.77ms)
18h	128M REFCLKs (1.34s)
19h	256M REFCLKs (2.68s)
1Ah	2^29 REFCLKs (5.37s)
1Bh	2^30 REFCLKs (10.73s)
1Ch	2^31 REFCLKs (21.47s)
1Dh	2^32 REFCLKs (42.95s)
1Eh	2^33 REFCLKs (85.90s)
1Fh	2^34 REFCLKs (171.80s)

UMC[00...11]CHx00000D8C [DRAM ECC Error Inject Control] (UMC::EccErrInjCtrl)

Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000D8C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:3	Reserved.
2	EccErrAddrEn. Reset: 0. 0=Disable. 1=Enable. Allow address-based injection from the DF CS. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
1	EccErrOneShotEn. Reset: 0. 0=Disable. 1=Enable. Cleared-by-hardware. Enable a single error injection. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
0	EccErrPersistentEn. Reset: 0. 0=Disable. 1=Enable. Enable continuous error injection. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.

UMC[00...11]CHx00000D90 [DRAM ECC Bad Symbol Index] (UMC::EccBadSymbolIndex)

Reset: 0000_0000h.

Specifies DRAM ECC bad symbol. It is per SubChannel, Plr, CS. In PinReducedEcc DDR5 mode, valid values: 0-18. In FullECC DDR5 mode: 0-7 for Data, 8-15 Reserved, and 16-17 for ECC.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000D90; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	AutoInc. Read-write. Reset: 0. 0=Disable. 1=Enable. Index read/write auto increment enable.
30:6	Reserved.
5:0	Index. Read-write, Volatile. Reset: 00h. Index for accessing DRAM ECC bad symbol registers.

UMC[00...11]CHx00000D94 [DRAM ECC Bad Symbol Data] (UMC::EccBadSymbolData)

Read-write, Volatile. Reset: 0000_0000h.

Software should configure EccBadSymbolIndex and then read or write EccBadSymbolData. EccBadSymbolData accesses increment EccBadSymbolIndex.Index if EccBadSymbolIndex.AutoInc == 1. BadSymbolData[0]=BadSymbolEn BadSymbol[8:4]= 5bit Bad symbol value

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000D94; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:0	Data. Read-write, Volatile. Reset: 0000_0000h.

UMC[00...11]CHx00000D98 [ECS Status Select] (UMC::ECSSStatusSel)

Reset: 0000_0000h.

Specifies the index into the [ECS](#) status registers. See 12.2.20.3 [ECS Status Array]

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000D98; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31	AutoInc. Read-write. Reset: 0. 0=Disable. 1=Enable. Index read/write auto increment enable.
30:6	Reserved.
5:0	Index. Read-write, Volatile. Reset: 00h. Index for accessing ECS status registers.

UMC[00...11]CHx00000D9C [ECS Status Data] (UMC::ECSSStatus)

Read-write, Volatile. Reset: 0000_0000h.

Software should configure ECSSStatusSel and then read or write ECSSStatus. ECSSStatusData accesses increment ECSSStatusSel.Index if ECSSStatusSel.AutoInc == 1.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000D9C; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:0	Data. Read-write, Volatile. Reset: 0000_0000h. Description: ECS Status per-subchannel and per-rank for Index[1:0] = 00 (Error Count Status) [7:0] - Errors Per Row Count (EpRC) from MR19 [5:0] [11:8] - Device ID of x4 Dram Device with highest EpRC error count. Lowest Device ID in case of a tie. [11:9] - Device ID of x8 Dram Device with highest EpRC error count. Lowest Device ID in case of a tie. [11:10] - Device ID of x16 Dram device with highest EpRC error count. Lowest Device ID in case of a tie. Bits 9,8 not used. [15:12] - CID (Only applies to 3DS). CID of highest error count [23:16] - MR20 [7:0] Error count [27:24] - Device ID of x4 Dram device with highest MR20 Error count [27:25] - Device ID of x8 Dram device with highest MR20 Error count [27:26] - Device ID of x16 Dram device with highest MR20 Error count See 12.2.20.3 [ECS Status Array].

UMC00CHx00000DB0...UMC11CHx00000DBC [DRAM Error Inject] (UMC::EccErrInj)

Reset: 0000_0000h.

Specifies error injection on DRAM data. There is one register per ECC code word, corresponding to sequential DDR bus data beats.

_umc0_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CHx00000DB0; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CHx00000DB4; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_n2_aliasSMN; UMC[11:00]CHx00000DB8; UMC[11:00]CH=00[B:0]5_0000h

_umc0_inst[UMCWPHY[11:0]]_n3_aliasSMN; UMC[11:00]CHx00000DBC; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description																						
31:26	Reserved.																						
25:16	EccInjEn. Reset: 000h. Specifies the double byte of data that the pattern in EccInjVector should be XORed to AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only. ValidValues: <table> <tr> <th>Bit</th><th>Description</th></tr> <tr> <td>[0]</td><td>Data[15:0]</td></tr> <tr> <td>[1]</td><td>Data[31:16]</td></tr> <tr> <td>[2]</td><td>Data[47:32]</td></tr> <tr> <td>[3]</td><td>Data[63:48]</td></tr> <tr> <td>[4]</td><td>Data[79:64]</td></tr> <tr> <td>[5]</td><td>Data[95:80]</td></tr> <tr> <td>[6]</td><td>Data[111:96]</td></tr> <tr> <td>[7]</td><td>Data[127:112]</td></tr> <tr> <td>[8]</td><td>ECC[15:0]</td></tr> <tr> <td>[9]</td><td>ECC[31:16] : Applicable only for DDR5 FullEcc mode</td></tr> </table>	Bit	Description	[0]	Data[15:0]	[1]	Data[31:16]	[2]	Data[47:32]	[3]	Data[63:48]	[4]	Data[79:64]	[5]	Data[95:80]	[6]	Data[111:96]	[7]	Data[127:112]	[8]	ECC[15:0]	[9]	ECC[31:16] : Applicable only for DDR5 FullEcc mode
Bit	Description																						
[0]	Data[15:0]																						
[1]	Data[31:16]																						
[2]	Data[47:32]																						
[3]	Data[63:48]																						
[4]	Data[79:64]																						
[5]	Data[95:80]																						
[6]	Data[111:96]																						
[7]	Data[127:112]																						
[8]	ECC[15:0]																						
[9]	ECC[31:16] : Applicable only for DDR5 FullEcc mode																						
15:0	EccInjVector. Reset: 0000h. Specifies the pattern XORed into each double byte of data specified by the EccInjEn field. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.																						

UMC[00...11]CHx00000DC0 [BEQ Error Inject] (UMC::BeqErrInj)

Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000DC0; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:24	InjRxDatPeriod. Reset: 00h. Inject frequency, by 32B transaction, multiplied by 32. Zero means one error per transaction. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
23:16	InjTxDatPeriod. Reset: 00h. Inject frequency, by 32B transaction, multiplied by 32. Zero means one error per transaction. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
15:8	InjTxCmdPeriod. Reset: 00h. Injection frequency, by clock or command per InjTxCmdByClk, multiplied by 32. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
7	InjTxCmdAddr3. Reset: 0. For command injection, 1 means flip CA[3] bit. 0 means CA[3] not flipped AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
6	InjErrEn. Reset: 0. continuously Inject error. Use with InjTxCmdEn/InjTxDatEn/InjRxDatEn AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
5	InjTxCmdParBit. Reset: 0. For command injection, 1 means flip parity bit. 0 means default bit AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
4	InjErrOneShotEn. Reset: 0. 0=Disable. 1=Enable. Cleared-by-hardware. Inject error for one shot. Write this bit to 1 to trigger one shot error injection. Use with InjTxCmdEn/InjTxDatEn/InjRxDatEn. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
3	InjTxCmdByClk. Reset: 0. 0=Commands. 1=Clocks. Inject error on command/address periodically by memory clocks versus commands. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
2	InjRxDatEn. Reset: 0. 0=Disable. 1=Enable. Enables Rx data error injection. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
1	InjTxDatEn. Reset: 0. 0=Disable. 1=Enable. Enables Tx data error injection. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
0	InjTxCmdEn. Reset: 0. 0=Disable. 1=Enable. Enables Tx command error injection. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.

UMC[00...11]CHx00000DC4 [BEQ Error Injection 2] (UMC::BeqErrInj2)

Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000DC4; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description																								
31:30	InjPeriodMult. Reset: 0h. Injection period is multiplied by 2 ^{InjPeriodMult} . AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.																								
29:12	InjTxDatVec. Reset: 0_0000h. Specify which nibbles to inject wrdat errors (ECC4, ECC0, DQ60,DQ56...DQ0) AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.																								
11	Reserved.																								
10:0	InjTxCmdMsk. Reset: 000h. Mask command from Error Injection when InjTxCmdByClk=0. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only. ValidValues: <table> <tr> <th>Bit</th><th>Description</th></tr> <tr><td>[0]</td><td>RD</td></tr> <tr><td>[1]</td><td>WR</td></tr> <tr><td>[2]</td><td>DDR5 PDE</td></tr> <tr><td>[3]</td><td>SRE</td></tr> <tr><td>[4]</td><td>DDR5 REF/RFM</td></tr> <tr><td>[5]</td><td>DDR5 MPC (includes ZQ)</td></tr> <tr><td>[6]</td><td>MRR</td></tr> <tr><td>[7]</td><td>MRW</td></tr> <tr><td>[8]</td><td>Reserved.</td></tr> <tr><td>[9]</td><td>ACT</td></tr> <tr><td>[10]</td><td>PRE</td></tr> </table>	Bit	Description	[0]	RD	[1]	WR	[2]	DDR5 PDE	[3]	SRE	[4]	DDR5 REF/RFM	[5]	DDR5 MPC (includes ZQ)	[6]	MRR	[7]	MRW	[8]	Reserved.	[9]	ACT	[10]	PRE
Bit	Description																								
[0]	RD																								
[1]	WR																								
[2]	DDR5 PDE																								
[3]	SRE																								
[4]	DDR5 REF/RFM																								
[5]	DDR5 MPC (includes ZQ)																								
[6]	MRR																								
[7]	MRW																								
[8]	Reserved.																								
[9]	ACT																								
[10]	PRE																								

UMC[00...11]CHx00000DC8 [BEQ Error Inject3] (UMC::BeqErrInj3)

Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000DC8; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description
31:6	Reserved.
5:2	InjTxCmdCyc. Reset: 0h. Cycle of the command to which error is injected. This is expected to be configured based on whether 1N/2N/Rdimmsdr mode is enabled. AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.
1:0	InjTxCmdChn. Reset: 0h. Subchannel to which error is injected. Only one Subchannel is supported at any time. Applicable for all InjTxCmd/InjTxDat/InjRxDat AccessType: (UMC::MiscCfg[DisErrInj] == 0) ? Read-write : Read-only.

UMC[00...11]CHx00000DF0 [UMC Capabilities] (UMC::UmcCap)

Read-only.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000DF0; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description				
31:20	Reserved.				
19	MemChanDis. Read-only. Disable this memory channel and UMC.				
18	EncryptionDis. Read-only. Disable support for encryption/decryption of DRAM data.				
17	EccDis. Read-only. Disable support for DRAM ECC.				
16	RegDimmDis. Read-only. Disable support for registered DIMMs (RDIMM/LRDIMM).				
15:8	DramTypeDis. Read-only. Disable different DDR types if multiple DRAM types are supported in a SOC.				
7:0	DdrMaxRate. Read-only. Description: In this field bits [3:0] encode the DDR5 limit. DDR5 encoded values: 0: Unlimited 1: DDR-2000 2: DDR-3200 3: DDR-3600 4: DDR-4000 5: DDR-4400 6: DDR-4800 7: DDR-5200 ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>FFh-00h</td><td>Encoded DDR Max Rate.</td></tr> </table>	Value	Description	FFh-00h	Encoded DDR Max Rate.
Value	Description				
FFh-00h	Encoded DDR Max Rate.				

UMC[00...11]CHx00000DF4 [UMC Capabilities High] (UMC::UmcCapHi)

Read-only.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CHx00000DF4; UMC[11:00]CH=00[B:0]5_0000h

Bits	Description				
31	EccChipKillCap. Read-only. Indicates ECC chipkill configuration is enabled.				
30	EccEnabled. Read-only. Indicates that ECC logic configuration is enabled.				
29:8	Reserved.				
7:0	DdrMaxRateEnf. Read-only. Description: In this field bits [3:0] encode the DDR5 limit. DDR5 encoded values: 0: Unlimited 1: DDR-2000 2: DDR-3200 3: DDR-3600 4: DDR-4000 5: DDR-4400 6: DDR-4800 7: DDR-5200 ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>FFh-00h</td><td>Encoded DDR Max Rate Enforced.</td></tr> </table>	Value	Description	FFh-00h	Encoded DDR Max Rate Enforced.
Value	Description				
FFh-00h	Encoded DDR Max Rate Enforced.				

UMC[00...11]CTLx00000000 [UMC BIST Command Control] (UMC::UmcBistCmdCntl)

Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000000; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description														
31	SendActCmd. Read-write, Volatile. Reset: 0. Cleared-by-hardware. Sends an ACT command per Command2 target.														
30	SendPchgCmd. Read-write, Volatile. Reset: 0. Cleared-by-hardware. Sends a PRE/PREA command per Command2 target.														
29	RrwCtrlUpdEn. Read-write. Reset: 0. Enables RRW to send a CtrlUpd to the PHY when SendActCmd is set.														
28	RrwClkGateEn. Read-write. Reset: 0. Check: 0. Turns on the clock inside the RRW when set to one.														
27:25	Reserved.														
24	ActPchgGenEn. Read-write. Reset: 0. Indicates UMC should manage the DRAM page specifies by the Command2 target. SendCmd triggers ACT, command sequence, and PRE. Constraint: (CmdTgt == 'Target A').														
23:21	Reserved.														
20:18	DataPatGenSel. Read-write. Reset: 0h. Specifies the data pattern generation mode.														
ValidValues:															
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>PRBS23 I</td></tr> <tr> <td>1h</td><td>PRBS23 II. Within a byte lane, the DQ[n] value is offset 8n bit times from DQ[0].</td></tr> <tr> <td>2h</td><td>Configurable data pattern. See UserDataPattern.</td></tr> <tr> <td>3h</td><td>Configurable data pattern with circular lane shift. See UserDataPattern.</td></tr> <tr> <td>4h</td><td>PRBS23 III. Within a byte lane, each DQ[n] value is equal to DQ[0].</td></tr> <tr> <td>7h-5h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	PRBS23 I	1h	PRBS23 II. Within a byte lane, the DQ[n] value is offset 8n bit times from DQ[0].	2h	Configurable data pattern. See UserDataPattern.	3h	Configurable data pattern with circular lane shift. See UserDataPattern.	4h	PRBS23 III. Within a byte lane, each DQ[n] value is equal to DQ[0].	7h-5h	Reserved.
Value	Description														
0h	PRBS23 I														
1h	PRBS23 II. Within a byte lane, the DQ[n] value is offset 8n bit times from DQ[0].														
2h	Configurable data pattern. See UserDataPattern.														
3h	Configurable data pattern with circular lane shift. See UserDataPattern.														
4h	PRBS23 III. Within a byte lane, each DQ[n] value is equal to DQ[0].														
7h-5h	Reserved.														
17:14	Reserved.														
13	LfsrRollOver. Read-write. Reset: 0. 0=Disable. 1=Enable. Specifies if DataPrbsSeed is used to re-seed data comparison logic if the generated address wraps around to equal TgtAddress.														
12	CmdSendInProg. Read-only, Volatile. Reset: 0. Indicates UMCCTRL logic is in control of BEQ.														
11	SendCmd. Read-write. Reset: 0. 0=Disable. 1=Enable. Initiate command generation. BIOS must clear this field after command sequence completion.														
10	TestStatus. Read-only, Volatile. Reset: 0. 0=In progress. 1=Completed. Indicates completion of the SendCmd sequence.														
9:8	CmdTgt. Read-write. Reset: 0h. Specifies the SendCmd target address mode.														
ValidValues:															
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Target A mode, Subchannel A</td></tr> <tr> <td>1h</td><td>Alternating Target A and Target B, Subchannel A</td></tr> <tr> <td>2h</td><td>Target A mode, Subchannel B</td></tr> <tr> <td>3h</td><td>Alternating Target A and Target B, Subchannel B</td></tr> </table>	Value	Description	0h	Target A mode, Subchannel A	1h	Alternating Target A and Target B, Subchannel A	2h	Target A mode, Subchannel B	3h	Alternating Target A and Target B, Subchannel B				
Value	Description														
0h	Target A mode, Subchannel A														
1h	Alternating Target A and Target B, Subchannel A														
2h	Target A mode, Subchannel B														
3h	Alternating Target A and Target B, Subchannel B														
7:5	CmdType. Read-write. Reset: 0h. Specifies the SendCmd command type.														
ValidValues:															
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Read</td></tr> <tr> <td>1h</td><td>Write</td></tr> <tr> <td>2h</td><td>Alternating write and read</td></tr> <tr> <td>7h-3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Read	1h	Write	2h	Alternating write and read	7h-3h	Reserved.				
Value	Description														
0h	Read														
1h	Write														
2h	Alternating write and read														
7h-3h	Reserved.														
4	Reserved.														
3	ResetAllErr. Read-write, Volatile. Reset: 0. 0=Disable. 1=Enable. Cleared-by-hardware. Set to clear error status bits and error counters. This bit may be written simultaneously with SendCmd.														
2	CmdTestEnable. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable command generation mode.														

1:0	Reserved.
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UMC00CTLx00000004...UMC11CTLx00000008 [DRAM Target Base] (UMC::TgtBase)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CTLx00000004; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CTLx00000008; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description						
31	Reserved.						
30:28	TgtCid. Read-write. Reset: 0h. Specifies the chipselect.						
27:24	TgtChipSelect. Read-write. Reset: 0h. Specifies the chipselect. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>3h-0h</td><td>CS TgtChipSelect</td></tr> <tr> <td>Fh-4h</td><td>Reserved.</td></tr> </table>	Value	Description	3h-0h	CS TgtChipSelect	Fh-4h	Reserved.
Value	Description						
3h-0h	CS TgtChipSelect						
Fh-4h	Reserved.						
23:19	TgtBank. Read-write. Reset: 00h. Specifies the bank address bits.						
18	Reserved.						
17:0	TgtAddress. Read-write. Reset: 0_0000h. Specifies the column address bits.						

UMC[00...11]CTLx0000000C (UMC::Command0)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx0000000C; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description						
31:22	BubbleCnt2. Read-write. Reset: 000h. Specifies the number of command slots inserted after the last clock of virtual CAS of each read-burst operation in alternating write and read mode.						
21:12	BubbleCnt. Read-write. Reset: 000h. Specifies the number of command slots inserted after the last clock of virtual CAS of each read or write command.						
11:10	Reserved.						
9:8	InfWrMode_CmdTgt. Read-write. Reset: 0h. Targets specific subchannel to run continuous writes.						
7:0	CmdStreamLen. Read-write. Reset: 00h. Specifies the number of CAS commands generated before BubbleCnt bubbles are inserted. Constraint: CmdStreamLen == 1 ~(CmdType==2). ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>FFh-01h</td><td>CmdStreamLen commands</td></tr> </table>	Value	Description	00h	Reserved.	FFh-01h	CmdStreamLen commands
Value	Description						
00h	Reserved.						
FFh-01h	CmdStreamLen commands						

UMC[00...11]CTLx00000010 (UMC::Command1)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000010; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description						
31:29	Reserved.						
28:21	ActPchgCmdMin. Read-write. Reset: 00h. Specifies the minimum time in memory clock cycles from an activate or precharge command to the next command. ValidValues:						
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disabled.</td></tr> <tr> <td>FFh-01h</td><td>ActPchgCmdMin MEMCLKs.</td></tr> </table>	Value	Description	00h	Disabled.	FFh-01h	ActPchgCmdMin MEMCLKs.
Value	Description						
00h	Disabled.						
FFh-01h	ActPchgCmdMin MEMCLKs.						
20:0	CmdCount. Read-write. Reset: 00_0000h. Specifies the maximum number of CAS commands generated by SendCmd sequence. [0: Infinite, 0x1..0x1FFFFFF: CmdCountCommands]						

UMC[00...11]CTLx00000014 (UMC::Command2)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000014; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31	Reserved.
30:28	CmdCid. Read-write. Reset: 0h. Specifies the Cid bits.
27:24	CmdChipSelect. Read-write. Reset: 0h. Specifies the command chipselect as CS_L[3:0].
23	Reserved.
22:18	CmdBank. Read-write. Reset: 00h. Specifies the bank address bits.
17:0	CmdAddress. Read-write. Reset: 0_0000h. Specifies the row address bits.

UMC[00...11]CTLx00000018 (UMC::Status0)

Read-only, Volatile. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000018; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description						
31:25	ErrDqNum. Read-only, Volatile. Reset: 00h. Indicates the DQ bit of the first error occurrence. Valid when ErrCnt > 0. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>47h-00h</td><td>ErrDqNum</td></tr> <tr> <td>7Fh-48h</td><td>Reserved.</td></tr> </table>	Value	Description	47h-00h	ErrDqNum	7Fh-48h	Reserved.
Value	Description						
47h-00h	ErrDqNum						
7Fh-48h	Reserved.						
24:0	ErrCnt. Read-only, Volatile. Reset: 000_0000h. Specifies a saturating counter indicating the number of bit errors detected. Counts a maximum of 1 error per-bit lane per bit-time. Status is accumulated until cleared by ResetAllErr. Errors can an be masked on per-bit basis by programming DQMask.						

UMC[00...11]CTLx0000001C (UMC::Status1)

Read-only, Volatile. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx0000001C; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:23	Reserved.
22:0	NibbleErrSts. Read-only, Volatile. Reset: 00_0000h. Indicates error detection status on a per nibble basis. Status is accumulated until cleared by ResetAllErr.

UMC[00...11]CTLx00000020 (UMC::Status2)

Read-only, Volatile. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000020; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:21	Reserved.
20:0	NibbleErr180Sts. Read-only, Volatile. Reset: 00_0000h. Indicates error detection status on a per nibble basis, comparing received data against data shifted 1-bit time earlier. Status is accumulated until cleared by ResetAllErr.

UMC[00...11]CTLx00000024 (UMC::Prbs)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000024; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:19	Reserved.
18:0	DataPrbsSeed. Read-write. Reset: 0_0000h. Specifies the seed value used for creating pseudo random traffic on the data bus. This register must be written with a non-zero seed value.

UMC[00...11]CTLx00000028 (UMC::DQMaskLo)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000028; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:0	DQMask : DQMask[31:0] . Read-write. Reset: 0000_0000h. 1=Disable comparison. 0=Enable comparison.

UMC[00...11]CTLx0000002C (UMC::DQMaskHi)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx0000002C; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:0	DQMask : DQMask[63:32] . Read-write. Reset: 0000_0000h. 1=Disable comparison. 0=Enable comparison.

UMC[00...11]CTLx00000030 (UMC::DQMaskEcc)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000030; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:8	Reserved.
7:0	EccMask . Read-write. Reset: 00h. 1=Disable comparison. 0=Enable comparison.

UMC[00...11]CTLx00000034 (UMC::DQPatternOverride0)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000034; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:0	DQPatOvrEn : DQPatOvrEn[31:0] . Read-write. Reset: 0000_0000h. The pattern on this bit lane is overridden with a static value specified by PatOvrVal. Reserved if CmdTestEnable && DataPatGenSel != 10b.

UMC[00...11]CTLx00000038 (UMC::DQPatternOverride1)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000038; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:0	DQPatOvrEn : DQPatOvrEn[63:32] . Read-write. Reset: 0000_0000h. The pattern on this bit lane is overridden with a static value specified by PatOvrVal. Reserved if CmdTestEnable && DataPatGenSel != 10b.

UMC[00...11]CTLx0000003C (UMC::DQPatternOverride2)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx0000003C; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:24	XorPatOvr . Read-write. Reset: 00h. Specifies the override data pattern used for creating traffic on the data bus. The output data for each DQ of each byte lane will be XOR with the specified values. Reserved if DataPatGenSel != 100b.
23:16	DbiPatOvrEn . Read-write. Reset: 00h. The pattern on this bit lane is overridden with a static value specified by PatOvrVal. Reserved if CmdTestEnable && DataPatGenSel != 10b.
15:9	Reserved.
8	PatOvrVal . Read-write. Reset: 0. 0=Static 0 s. 1=Static 1 s. Specifies the override data value used for creating traffic on the data bus.
7:0	EccPatOvrEn . Read-write. Reset: 00h. The pattern on this bit lane is overridden with a static value specified by PatOvrVal. Reserved if CmdTestEnable && DataPatGenSel != 10b.

UMC[00...11]CTLx00000040 (UMC::Status3)

Read-only, Volatile. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000040; UMC[11:00]CTL=00[B:0]5_1000h`

Bits	Description
31:28	Reserved.
27:24	ErrBeatNum. Read-only, Volatile. Reset: 0h. Indicates the data beat of the first error occurrence in the command reported by ErrCmdNum. Valid when ErrCnt > 0 CmdCount != 'Infinite'. Cleared by ResetAllErr.
23:21	Reserved.
20:0	ErrCmdNum. Read-only, Volatile. Reset: 00_0000h. Indicates the CAS command number of the first error occurrence in the command reported by ErrCmdNum. Valid when ErrCnt > 0 CmdCount != 'Infinite'. Cleared by ResetAllErr.

UMC[00...11]CTLx00000044 (UMC::Status4)

Read-only, Volatile. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000044; UMC[11:00]CTL=00[B:0]5_1000h`

Bits	Description
31:0	DQErr: DQErr[31:0]. Read-only, Volatile. Reset: 0000_0000h. Indicates error detection status on a per bit-lane basis. Status is accumulated until cleared by ResetAllErr.

UMC[00...11]CTLx00000048 (UMC::Status5)

Read-only, Volatile. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000048; UMC[11:00]CTL=00[B:0]5_1000h`

Bits	Description
31:0	DQErr: DQErr[63:32]. Read-only, Volatile. Reset: 0000_0000h. Indicates error detection status on a per bit-lane basis. Status is accumulated until cleared by ResetAllErr.

UMC[00...11]CTLx0000004C (UMC::Status6)

Read-only, Volatile. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx0000004C; UMC[11:00]CTL=00[B:0]5_1000h`

Bits	Description
31:8	Reserved.
7:0	EccErr. Read-only, Volatile. Reset: 00h. Indicates error detection status on a per bit-lane basis. Status is accumulated until cleared by ResetAllErr.

UMC[00...11]CTLx00000050 (UMC::UserDataPattern0)

Read-write. Reset: 0000_0000h.

Four registers specify the user data pattern for DQ[15:0]. This pattern is repeated across the entire data bus width.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000050; UMC[11:00]CTL=00[B:0]5_1000h`

Bits	Description
31:24	DataPatternDQ3. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].
23:16	DataPatternDQ2. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].
15:8	DataPatternDQ1. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].
7:0	DataPatternDQ0. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].

UMC[00...11]CTLx00000054 (UMC::UserDataPattern1)

Read-write. Reset: 0000_0000h.

Four registers specify the user data pattern for DQ[15:0]. This pattern is repeated across the entire data bus width.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000054; UMC[11:00]CTL=00[B:0]5_1000h`

Bits	Description
31:24	DataPatternDQ7. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].
23:16	DataPatternDQ6. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].
15:8	DataPatternDQ5. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].
7:0	DataPatternDQ4. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].

UMC[00...11]CTLx00000058 (UMC::UserDataPattern2)

Read-write. Reset: 0000_0000h.

Four registers specify the user data pattern for DQ[15:0]. This pattern is repeated across the entire data bus width.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000058; UMC[11:00]CTL=00[B:0]5_1000h`

Bits	Description
31:24	DataPatternDQ11. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].
23:16	DataPatternDQ10. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].
15:8	DataPatternDQ9. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].
7:0	DataPatternDQ8. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].

UMC[00...11]CTLx0000005C (UMC::UserDataPattern3)

Read-write. Reset: 0000_0000h.

Four registers specify the user data pattern for DQ[15:0]. This pattern is repeated across the entire data bus width.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx0000005C; UMC[11:00]CTL=00[B:0]5_1000h`

Bits	Description
31:24	DataPatternDQ15. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].
23:16	DataPatternDQ14. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].
15:8	DataPatternDQ13. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].
7:0	DataPatternDQ12. Read-write. Reset: 00h. Specifies the DQ lane bit value per 8 data beats. DataPattern[7:0] corresponds to data beat[7:0].

UMC[00...11]CTLx00000060 (UMC::DQCapture0)

Read-only, Volatile. Reset: 0000_0000h.

Contains the low 32 bits of MPR read data

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000060; UMC[11:00]CTL=00[B:0]5_1000h`

Bits	Description
31:0	DQ: DQ[31:0]. Read-only, Volatile. Reset: 0000_0000h. Lower 32 bits of MPR read data.

UMC[00...11]CTLx00000064 (UMC::DQCapture1)

Read-only, Volatile. Reset: 0000_0000h.

Contains the high 32 bits of MPR read data

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000064; UMC[11:00]CTL=00[B:0]5_1000h`**Bits Description**31:0 **DQ: DQ[63:32]**. Read-only, Volatile. Reset: 0000_0000h. Upper 32 bits of MPR read data.**UMC[00...11]CTLx00000068 (UMC::DQCapture2)**

Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000068; UMC[11:00]CTL=00[B:0]5_1000h`**Bits Description**31:29 **SelUI**. Read-write. Reset: 0h. Selects the [UI](#) or bit-lane to be captured in the internal buffer. See SelMode.28 **SelMode**. Read-write. Reset: 0. 0=Single data beat specified by SelUI (entire width of the data bus for 1 [UI](#)). 1=Single lane (entire lane for 8 UI). Selects the capture mode for the internal data buffer selected by SelUI.

27:8 Reserved.

7:0 **DQ: ECC[7:0]**. Read-only, Volatile. Reset: 00h. Contains the MPR read data from the ECC DRAM**UMC[00...11]CTLx0000006C [RRWArray Index Register] (UMC::RRWArrayIndex)**`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx0000006C; UMC[11:00]CTL=00[B:0]5_1000h`**Bits Description**31 **AutoInc**. Read-write. Reset: 0. 0=Disable. 1=Enable. RRW Array auto increment enable.

30:10 Reserved.

9:0 **Index**. Read-write, Volatile. RRW array index. Updated-by-hardware if AutoInc = 1.**UMC[00...11]CTLx00000070 [RRW Array Data] (UMC::RRWArrayData)**

Read-write, Volatile.

Software should configure RRW Array Index and then read or write RRWArrayData. Data accesses increment RRW ArrayIndex.Index if SrsIndex.AutoInc == 1.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000070; UMC[11:00]CTL=00[B:0]5_1000h`**Bits Description**31:0 **RRWArrayData**. Read-write, Volatile.

UMC[00...11]CTLx00000080 (UMC::Command3)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000080; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description										
31:22	BubbleCnt2InfCh. Read-write. Reset: 000h. Specifies the number of command slots inserted on irritator subchannel after the last clock of virtual CAS of each read-burst operation in alternating write and read mode.										
21:12	BubbleCntInfCh. Read-write. Reset: 000h. Specifies the number of command slots inserted on irritator subchannel for continuous reads/writes after the last clock of virtual CAS of each read or write command.										
11	Reserved.										
10:8	CmdTypeInfCh. Read-write. Reset: 0h. Determines to run continuous reads or writes on irritator subchannel specified by InfWrMode_CmdTgt[1:0]. ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Read</td></tr> <tr> <td>1h</td><td>Write</td></tr> <tr> <td>2h</td><td>Alternating write and read</td></tr> <tr> <td>7h-3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Read	1h	Write	2h	Alternating write and read	7h-3h	Reserved.
Value	Description										
0h	Read										
1h	Write										
2h	Alternating write and read										
7h-3h	Reserved.										
7:0	CmdStreamLenInfCh. Read-write. Reset: 00h. Specifies the number of CAS commands generated before BubbleCnt bubbles are inserted on the irritator subchannel for infinite wr/rd (for di/dt). Constraint: CmdStreamLen == 1 ~(CmdType==2). ValidValues:										
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disabled</td></tr> <tr> <td>FFh-01h</td><td>CmdStreamLenInfCh commands.</td></tr> </table>	Value	Description	00h	Disabled	FFh-01h	CmdStreamLenInfCh commands.				
Value	Description										
00h	Disabled										
FFh-01h	CmdStreamLenInfCh commands.										

UMC[00...11]CTLx00000084 (UMC::Command4)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000084; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description														
31:3	Reserved.														
2:0	DataPatGenSelIrrCh. Read-write. Reset: 0h. Specifies the data pattern generation mode. ValidValues:														
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>PRBS23 I</td></tr> <tr> <td>1h</td><td>PRBS23 II. Within a byte lane, the DQ[n] value is offset 8n bit times from DQ[0].</td></tr> <tr> <td>2h</td><td>Configurable data pattern. See UserDataPattern.</td></tr> <tr> <td>3h</td><td>Configurable data pattern with circular lane shift. See UserDataPattern.</td></tr> <tr> <td>4h</td><td>PRBS23 III. Within a byte lane, each DQ[n] value is equal to DQ[0].</td></tr> <tr> <td>7h-5h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	PRBS23 I	1h	PRBS23 II. Within a byte lane, the DQ[n] value is offset 8n bit times from DQ[0].	2h	Configurable data pattern. See UserDataPattern.	3h	Configurable data pattern with circular lane shift. See UserDataPattern.	4h	PRBS23 III. Within a byte lane, each DQ[n] value is equal to DQ[0].	7h-5h	Reserved.
Value	Description														
0h	PRBS23 I														
1h	PRBS23 II. Within a byte lane, the DQ[n] value is offset 8n bit times from DQ[0].														
2h	Configurable data pattern. See UserDataPattern.														
3h	Configurable data pattern with circular lane shift. See UserDataPattern.														
4h	PRBS23 III. Within a byte lane, each DQ[n] value is equal to DQ[0].														
7h-5h	Reserved.														

UMC[00...11]CTLx000000A0 [SRSM MOP Array Index] (UMC::SrsmIndex)

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx000000A0; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31	AutoInc. Read-write. Reset: 0. 0=Disable. 1=Enable. SRSM auto increment enable.
30:10	Reserved.
9:0	SrsmIndex. Read-write, Volatile. SRSM array index. Updated-by-hardware if AutoInc = 1.

UMC[00...11]CTLx000000A4 [SRSM MOP Array Data] (UMC::SrsmData)

Read-write, Volatile.

Software should configure SrsmIndex and then read or write SrsmData. SrsmData accesses increment SrsmIndex.Index if SrsmIndex.AutoInc == 1.

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx000000A4; UMC[11:00]CTL=00[B:0]5_1000h`**Bits Description**31:0 **SrsmData**. Read-write, Volatile.**UMC[00...11]CTLx000000A8 [Register to hold 4bit device ids for devices 0 to 7] (UMC::PdaDevIdList0)**

Read-write. Reset: 7654_3210h.

Register to hold 4bit device ids for devices 0 to 7

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx000000A8; UMC[11:00]CTL=00[B:0]5_1000h`**Bits Description**31:0 **DevId70**. Read-write. Reset: 7654_3210h. Register to hold 4bit device ids for devices 0 to 7**UMC[00...11]CTLx000000AC [Register to hold 4bit device ids for devices 8 to 9] (UMC::PdaDevIdList1)**

Read-write. Reset: 0000_0098h.

Register to hold 4bit device ids for devices 8 to 9

`_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx000000AC; UMC[11:00]CTL=00[B:0]5_1000h`**Bits Description**

31:8 Reserved.

7:0 **DevId98**. Read-write. Reset: 98h. Register to hold 4bit device ids for devices 8 to 9

UMC00CTLx000000B0...UMC11CTLx000000BC [SRSM Command Offset] (UMC::SrsmCmdOffset)

Read-write.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx000000B0; UMC[11:00]CTL=00[B:0]5_1000h_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx000000B4; UMC[11:00]CTL=00[B:0]5_1000h_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx000000B8; UMC[11:00]CTL=00[B:0]5_1000h_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx000000BC; UMC[11:00]CTL=00[B:0]5_1000h**Bits Description**

31:25 Reserved.

24:16 **SrsmExitOffset**. Read-write. Array offset for M-state exit.Reset: _umc0_instUMCWPHY0_mp0_aliasSMN: 040hReset: _umc0_instUMCWPHY0_mp1_aliasSMN: 0C0hReset: _umc0_instUMCWPHY0_mp2_aliasSMN: 140hReset: _umc0_instUMCWPHY0_mp3_aliasSMN: 1C0hReset: _umc0_instUMCWPHY1_mp0_aliasSMN: 040hReset: _umc0_instUMCWPHY1_mp1_aliasSMN: 0C0hReset: _umc0_instUMCWPHY1_mp2_aliasSMN: 140hReset: _umc0_instUMCWPHY1_mp3_aliasSMN: 1C0hReset: _umc0_instUMCWPHY2_mp0_aliasSMN: 040hReset: _umc0_instUMCWPHY2_mp1_aliasSMN: 0C0hReset: _umc0_instUMCWPHY2_mp2_aliasSMN: 140hReset: _umc0_instUMCWPHY2_mp3_aliasSMN: 1C0hReset: _umc0_instUMCWPHY3_mp0_aliasSMN: 040hReset: _umc0_instUMCWPHY3_mp1_aliasSMN: 0C0hReset: _umc0_instUMCWPHY3_mp2_aliasSMN: 140hReset: _umc0_instUMCWPHY3_mp3_aliasSMN: 1C0hReset: _umc0_instUMCWPHY4_mp0_aliasSMN: 040hReset: _umc0_instUMCWPHY4_mp1_aliasSMN: 0C0hReset: _umc0_instUMCWPHY4_mp2_aliasSMN: 140hReset: _umc0_instUMCWPHY4_mp3_aliasSMN: 1C0hReset: _umc0_instUMCWPHY5_mp0_aliasSMN: 040hReset: _umc0_instUMCWPHY5_mp1_aliasSMN: 0C0hReset: _umc0_instUMCWPHY5_mp2_aliasSMN: 140hReset: _umc0_instUMCWPHY5_mp3_aliasSMN: 1C0hReset: _umc0_instUMCWPHY6_mp0_aliasSMN: 040hReset: _umc0_instUMCWPHY6_mp1_aliasSMN: 0C0hReset: _umc0_instUMCWPHY6_mp2_aliasSMN: 140hReset: _umc0_instUMCWPHY6_mp3_aliasSMN: 1C0hReset: _umc0_instUMCWPHY7_mp0_aliasSMN: 040hReset: _umc0_instUMCWPHY7_mp1_aliasSMN: 0C0hReset: _umc0_instUMCWPHY7_mp2_aliasSMN: 140hReset: _umc0_instUMCWPHY7_mp3_aliasSMN: 1C0hReset: _umc0_instUMCWPHY8_mp0_aliasSMN: 040hReset: _umc0_instUMCWPHY8_mp1_aliasSMN: 0C0hReset: _umc0_instUMCWPHY8_mp2_aliasSMN: 140hReset: _umc0_instUMCWPHY8_mp3_aliasSMN: 1C0hReset: _umc0_instUMCWPHY9_mp0_aliasSMN: 040hReset: _umc0_instUMCWPHY9_mp1_aliasSMN: 0C0hReset: _umc0_instUMCWPHY9_mp2_aliasSMN: 140hReset: _umc0_instUMCWPHY9_mp3_aliasSMN: 1C0hReset: _umc0_instUMCWPHY10_mp0_aliasSMN: 040hReset: _umc0_instUMCWPHY10_mp1_aliasSMN: 0C0hReset: _umc0_instUMCWPHY10_mp2_aliasSMN: 140hReset: _umc0_instUMCWPHY10_mp3_aliasSMN: 1C0hReset: _umc0_instUMCWPHY11_mp0_aliasSMN: 040hReset: _umc0_instUMCWPHY11_mp1_aliasSMN: 0C0hReset: _umc0_instUMCWPHY11_mp2_aliasSMN: 140hReset: _umc0_instUMCWPHY11_mp3_aliasSMN: 1C0h

15:9 Reserved.

8:0 **SrsmEnterOffset**. Read-write. Array offset for M-state entry.Reset: _umc0_instUMCWPHY0_mp0_aliasSMN: 000hReset: _umc0_instUMCWPHY0_mp1_aliasSMN: 080hReset: _umc0_instUMCWPHY0_mp2_aliasSMN: 100h

Reset: _umc0_instUMCWPHY0_mp3_aliasSMN: 180h
Reset: _umc0_instUMCWPHY1_mp0_aliasSMN: 000h
Reset: _umc0_instUMCWPHY1_mp1_aliasSMN: 080h
Reset: _umc0_instUMCWPHY1_mp2_aliasSMN: 100h
Reset: _umc0_instUMCWPHY1_mp3_aliasSMN: 180h
Reset: _umc0_instUMCWPHY2_mp0_aliasSMN: 000h
Reset: _umc0_instUMCWPHY2_mp1_aliasSMN: 080h
Reset: _umc0_instUMCWPHY2_mp2_aliasSMN: 100h
Reset: _umc0_instUMCWPHY2_mp3_aliasSMN: 180h
Reset: _umc0_instUMCWPHY3_mp0_aliasSMN: 000h
Reset: _umc0_instUMCWPHY3_mp1_aliasSMN: 080h
Reset: _umc0_instUMCWPHY3_mp2_aliasSMN: 100h
Reset: _umc0_instUMCWPHY3_mp3_aliasSMN: 180h
Reset: _umc0_instUMCWPHY4_mp0_aliasSMN: 000h
Reset: _umc0_instUMCWPHY4_mp1_aliasSMN: 080h
Reset: _umc0_instUMCWPHY4_mp2_aliasSMN: 100h
Reset: _umc0_instUMCWPHY4_mp3_aliasSMN: 180h
Reset: _umc0_instUMCWPHY5_mp0_aliasSMN: 000h
Reset: _umc0_instUMCWPHY5_mp1_aliasSMN: 080h
Reset: _umc0_instUMCWPHY5_mp2_aliasSMN: 100h
Reset: _umc0_instUMCWPHY5_mp3_aliasSMN: 180h
Reset: _umc0_instUMCWPHY6_mp0_aliasSMN: 000h
Reset: _umc0_instUMCWPHY6_mp1_aliasSMN: 080h
Reset: _umc0_instUMCWPHY6_mp2_aliasSMN: 100h
Reset: _umc0_instUMCWPHY6_mp3_aliasSMN: 180h
Reset: _umc0_instUMCWPHY7_mp0_aliasSMN: 000h
Reset: _umc0_instUMCWPHY7_mp1_aliasSMN: 080h
Reset: _umc0_instUMCWPHY7_mp2_aliasSMN: 100h
Reset: _umc0_instUMCWPHY7_mp3_aliasSMN: 180h
Reset: _umc0_instUMCWPHY8_mp0_aliasSMN: 000h
Reset: _umc0_instUMCWPHY8_mp1_aliasSMN: 080h
Reset: _umc0_instUMCWPHY8_mp2_aliasSMN: 100h
Reset: _umc0_instUMCWPHY8_mp3_aliasSMN: 180h
Reset: _umc0_instUMCWPHY9_mp0_aliasSMN: 000h
Reset: _umc0_instUMCWPHY9_mp1_aliasSMN: 080h
Reset: _umc0_instUMCWPHY9_mp2_aliasSMN: 100h
Reset: _umc0_instUMCWPHY9_mp3_aliasSMN: 180h
Reset: _umc0_instUMCWPHY10_mp0_aliasSMN: 000h
Reset: _umc0_instUMCWPHY10_mp1_aliasSMN: 080h
Reset: _umc0_instUMCWPHY10_mp2_aliasSMN: 100h
Reset: _umc0_instUMCWPHY10_mp3_aliasSMN: 180h
Reset: _umc0_instUMCWPHY11_mp0_aliasSMN: 000h
Reset: _umc0_instUMCWPHY11_mp1_aliasSMN: 080h
Reset: _umc0_instUMCWPHY11_mp2_aliasSMN: 100h
Reset: _umc0_instUMCWPHY11_mp3_aliasSMN: 180h

UMC[00...11]CTLx000001AC [Clock Divider Control] (UMC::ClkDivCtrl)

Read-write. Reset: 0000_0F00h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx000001AC; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31	Reserved.
30	DFICLK_CGDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disable dynamic DFICLK coarse gater in DFI channel. This bit should be reset to zero for maximum power savings.
29	UCLK_CGDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disable dynamic UCLK coarse gater in UMC during D2, D3, and D7 requests. This bit should be reset to zero for maximum power savings.
28	FCLK_CGDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disable dynamic FCLK coarse gater in UMC during D2, D3, and D7 requests. This bit should be reset to zero for maximum power savings.
27	APBCLK_CGDis. Read-write. Reset: 0. Init: 1. 0=Enable. 1=Disable. Disable dynamic APBCLK coarse gater in DFI channel. This bit should be reset to zero for maximum power savings.
26	DFICLK_SCGEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable static DFICLK coarse gater in DFI channel. This bit should be set to one for an unused UMC.
25	UCLK_SCGEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable static UCLK coarse gater in UMC. This bit should be set to one for an unused UMC.
24	FCLK_SCGEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable static FCLK coarse gater in UMC. This bit should be set to one for an unused UMC.
23	SyncFifoMode_M3. Read-write. Reset: 0. SyncFifo Mode for M-state 3.
22	SyncFifoMode_M2. Read-write. Reset: 0. SyncFifo Mode for M-state 2.
21	SyncFifoMode_M1. Read-write. Reset: 0. SyncFifo Mode for M-state 1.
20	SyncFifoMode_M0. Read-write. Reset: 0. SyncFifo Mode for M-state 0.
19:18	Reserved.
17	UCLK_CGMCDis. Read-write. Reset: 0. Init: 0. 1=Disable dynamic UCLK Coarse Gater in UMC during Mstate changes.
16	FCLK_CGMCDis. Read-write. Reset: 0. Check: 0. 1=Disable dynamic FCLK Coarse Gater in UMC during Mstate changes.
15	DisjointClkBypass_M3. Read-write. Reset: 0. Disjoint clock bypass mode for M-state 3.
14	DisjointClkBypass_M2. Read-write. Reset: 0. Disjoint clock bypass mode for M-state 2.
13	DisjointClkBypass_M1. Read-write. Reset: 0. Disjoint clock bypass mode for M-state 1.
12	DisjointClkBypass_M0. Read-write. Reset: 0. Disjoint clock bypass mode for M-state 0.
11	FclkDiv1Mode_M3. Read-write. Reset: 1. Check: 1. FCLK Div1Mode for M-state 3.
10	FclkDiv1Mode_M2. Read-write. Reset: 1. Check: 1. FCLK Div1Mode for M-state 2.
9	FclkDiv1Mode_M1. Read-write. Reset: 1. Check: 1. FCLK Div1Mode for M-state 1.
8	FclkDiv1Mode_M0. Read-write. Reset: 1. Check: 1. FCLK Div1Mode for M-state 0.
7	DfiClkDiv1Mode_M3. Read-write. Reset: 0. Check: 0. DFICLK Div1Mode for M-state 3.
6	DfiClkDiv1Mode_M2. Read-write. Reset: 0. Check: 0. DFICLK Div1Mode for M-state 2.
5	DfiClkDiv1Mode_M1. Read-write. Reset: 0. Check: 0. DFICLK Div1Mode for M-state 1.
4	DfiClkDiv1Mode_M0. Read-write. Reset: 0. Check: 0. DFICLK Div1Mode for M-state 0.
3	UclkDiv1Mode_M3. Read-write. Reset: 0. UCLK Div1Mode for M-state 3.
2	UclkDiv1Mode_M2. Read-write. Reset: 0. UCLK Div1Mode for M-state 2.
1	UclkDiv1Mode_M1. Read-write. Reset: 0. UCLK Div1Mode for M-state 1.
0	UclkDiv1Mode_M0. Read-write. Reset: 0. UCLK Div1Mode for M-state 0.

UMC[00...11]CTLx000001B4 [Data Fabric Power Management Control] (UMC::DFPM)

Read-write. Reset: 0003_0022h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx000001B4; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:30	Reserved.
29	FakeAckEn. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Enable fake acknowledge for Dstate requests if DFPMEn is disabled.
28:21	Reserved.
20	EnableDllReset. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Enable sending MPC command after SRX to reset DLL.
19	PairedLP2. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Enable putting UMC0 and UMC1 into LP2 simultaneously.
18	FspEncodeEnable. Read-write. Reset: 0. 0=Disable Encoded FSP. FSP toggles with every P-state. 1=Enable Encoded FSP. Enable sending encoded FSP values to phy. If cleared FSP values to phy will simply toggle for every P-state change. See dfi_freq_fsp_M0 to dfi_freq_fsp_M3 .
17	EnablePDX. Read-write. Reset: 1. Init: 0. 0=Disable. 1=Enable. Enable PDX after exiting self-refresh. Applies only to LP variants.
16	EnablePDE. Read-write. Reset: 1. Init: 0. 0=Disable. 1=Enable. Enable PDE before entering self-refresh. Applies only to LP variants.
15:14	dfi_freq_fsp_M3. Read-write. Reset: 0h. Check: 0h. DFI frequency set point for M-state 3.
13:12	dfi_freq_fsp_M2. Read-write. Reset: 0h. Check: 0h. DFI frequency set point for M-state 2.
11:10	dfi_freq_fsp_M1. Read-write. Reset: 0h. Check: 0h. DFI frequency set point for M-state 1.
9:8	dfi_freq_fsp_M0. Read-write. Reset: 0h. Check: 0h. DFI frequency set point for M-state 0.
7	ConvertD2toPhyLP3. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Convert a D2-state request from DF to an LP3 request encoding at the phy dfi_frequency interface. Used for long idle power savings.
6	Reserved.
5	SRSMEEn. Read-write. Reset: 1. 0=Disable. 1=Enable. Enable SRSME for CLKARB requests. CLKARB requests could be DFPM Dstate requests or Local Dstate requests. If SRSMEEn = 0 then CLKARB requests are bypassed and acknowledged.
4	S3ResumeEnEMops. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. Enable SRSME MOP Array Entry commands when S3ResumeMode = 1.
3	SendCmd2. Read-write. Reset: 0. Check: 0. 0=Disable. 1=Enable. 1=Enable UMC to mirror each command in the sequence of commands to a second set of chipselect and cid pins. Must be used for QR encoded RDIMM. The chipselect and cid for the first-of-two command is specified in Ch0Cmd1CsEnMap and Ch0Cmd1CidEnMap. The chipselect and cid for the second-of-two command is specified in Ch0Cmd2CsEnMap and Ch0Cmd2CidEnMap.
2	S3ResumeMode. Read-write. Reset: 0. 0=Disable. 1=Enable. Enable S3 resume mode, the SRE command is skipped during a DFPM Dstate request.
1	DfiClkStpEn. Read-write. Reset: 1. Check: 1. 0=Disable. 1=Enable. Enable DFI stopclock during DFPM Dstate requests. DFI stopclock typically puts the phy into a LP2 state.
0	DFPMEn. Read-write. Reset: 0. 0=Disable. 1=Enable. Specifies UMC handling of DFPM Dstate requests. See FakeAckEn.

UMC[00...11]CTLx000001B8 (UMC::DFPM2)

Read-write. Reset: 0000_0000h.

Data Fabric Power Management Control 2

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx000001B8; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:27	Reserved.
26:24	Cmd2CidEnMap . Read-write. Reset: 0h. Cid value for second-of-two commands on even memclk phase.
23:20	Reserved.
19:16	Cmd2CsEnMap . Read-write. Reset: 0h. Chipselect enable map for second-of-two commands on even memclk phase or even sub-channel.
15:11	Reserved.
10:8	Cmd1CidEnMap . Read-write. Reset: 0h. Cid value for first-of-two commands on even memclk phase.
7:4	Reserved.
3:0	Cmd1CsEnMap . Read-write. Reset: 0h. Chipselect enable map for first-of-two commands on even memclk phase or even sub-channel.

UMC[00...11]CTLx000001BC (UMC::DFPM3)

Read-write. Reset: 0F0F_1F78h.

Data Fabric Power Management Control 3

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx000001BC; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description								
31:28	Reserved.								
27:24	Cmd2DramClkDis: Cmd2DramClkDis[3:0]. Read-write. Reset: Fh. DFI DRAM clock disable pattern for second (staggered) group powering down. Corresponds to dfi_dram_clk_disable[3:0].								
23:20	Reserved.								
19:16	Cmd1DramClkDis: Cmd1DramClkDis[3:0]. Read-write. Reset: Fh. DFI DRAM clock disable pattern for first group powering down. Corresponds to dfi_dram_clk_disable[3:0].								
15	Reserved.								
14:12	PeCtrlTypeD7. Read-write. Reset: 1h. Check: 1h. Specifies configurable actions per Dstate request and system policy. ValidValues:								
	<table> <tr> <th>Bit</th><th>Description</th></tr> <tr> <td>[0]</td><td>Flush WR</td></tr> <tr> <td>[1]</td><td>Flush RD</td></tr> <tr> <td>[2]</td><td>Payback REF</td></tr> </table>	Bit	Description	[0]	Flush WR	[1]	Flush RD	[2]	Payback REF
Bit	Description								
[0]	Flush WR								
[1]	Flush RD								
[2]	Payback REF								
11:9	PeCtrlTypeD4. Read-write. Reset: 7h. Init: 0h. Specifies configurable actions per Dstate request and system policy. ValidValues:								
	<table> <tr> <th>Bit</th><th>Description</th></tr> <tr> <td>[0]</td><td>Flush WR</td></tr> <tr> <td>[1]</td><td>Flush RD</td></tr> <tr> <td>[2]</td><td>Payback REF</td></tr> </table>	Bit	Description	[0]	Flush WR	[1]	Flush RD	[2]	Payback REF
Bit	Description								
[0]	Flush WR								
[1]	Flush RD								
[2]	Payback REF								
8:6	PeCtrlTypeD3. Read-write. Reset: 5h. Init: 7h. Specifies configurable actions per Dstate request and system policy. ValidValues:								
	<table> <tr> <th>Bit</th><th>Description</th></tr> <tr> <td>[0]</td><td>Flush WR</td></tr> <tr> <td>[1]</td><td>Flush RD</td></tr> <tr> <td>[2]</td><td>Payback REF</td></tr> </table>	Bit	Description	[0]	Flush WR	[1]	Flush RD	[2]	Payback REF
Bit	Description								
[0]	Flush WR								
[1]	Flush RD								
[2]	Payback REF								
5:3	PeCtrlTypeD2. Read-write. Reset: 7h. Check: 7h. Specifies configurable actions per Dstate request and system policy. ValidValues:								
	<table> <tr> <th>Bit</th><th>Description</th></tr> <tr> <td>[0]</td><td>Flush WR</td></tr> <tr> <td>[1]</td><td>Flush RD</td></tr> <tr> <td>[2]</td><td>Payback REF</td></tr> </table>	Bit	Description	[0]	Flush WR	[1]	Flush RD	[2]	Payback REF
Bit	Description								
[0]	Flush WR								
[1]	Flush RD								
[2]	Payback REF								
2:0	PeCtrlTypeD1. Read-write. Reset: 0h. Init: 4h. Check: 4h. Specifies configurable actions per Dstate request and system policy. ValidValues:								
	<table> <tr> <th>Bit</th><th>Description</th></tr> <tr> <td>[0]</td><td>Flush WR</td></tr> <tr> <td>[1]</td><td>Flush RD</td></tr> <tr> <td>[2]</td><td>Payback REF</td></tr> </table>	Bit	Description	[0]	Flush WR	[1]	Flush RD	[2]	Payback REF
Bit	Description								
[0]	Flush WR								
[1]	Flush RD								
[2]	Payback REF								

UMC[00...11]CTLx000001C0 (UMC::DFPM4)

Reset: 3E31_0400h.

Data Fabric Power Management Control 4

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx000001C0; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description																
31:30	Reserved.																
29:25	Freq_D3 . Read-write. Reset: 1Fh. Check: 1Fh. Specifies the frequency value sent to phy for D3 state.																
24:20	Freq_M3 . Read-write. Reset: 03h. Check: 03h. Specifies the frequency value sent to phy for M3 state.																
19:15	Freq_M2 . Read-write. Reset: 02h. Check: 02h. Specifies the frequency value sent to phy for M2 state.																
14:10	Freq_M1 . Read-write. Reset: 01h. Check: 01h. Specifies the frequency value sent to phy for M1 state.																
9:5	Freq_M0 . Read-write. Reset: 00h. Check: 00h. Specifies the frequency value sent to phy for M0 state.																
4	LocalInDstate . Read-only, Volatile. Reset: 0. 0=Not in local Dstate. 1=In local Dstate. Indicates the UMC is in a local Dstate. See LocalDstateSel.																
3	Reserved.																
2:0	LocalDstateSel . Read-write. Reset: 0h. Specifies a software-based local Dstate request. Should only be used as part of a controlled initialization sequence. ValidValues:																
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>D0</td></tr> <tr> <td>1h</td><td>D1</td></tr> <tr> <td>2h</td><td>D2</td></tr> <tr> <td>3h</td><td>D3</td></tr> <tr> <td>4h</td><td>D4</td></tr> <tr> <td>6h-5h</td><td>Reserved.</td></tr> <tr> <td>7h</td><td>D7</td></tr> </table>	Value	Description	0h	D0	1h	D1	2h	D2	3h	D3	4h	D4	6h-5h	Reserved.	7h	D7
Value	Description																
0h	D0																
1h	D1																
2h	D2																
3h	D3																
4h	D4																
6h-5h	Reserved.																
7h	D7																

UMC[00...11]CTLx000001C4 [Clock Divider Control 2] (UMC::ClkDivCtrl2)

Read-write. Reset: 0000_000Fh.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx000001C4; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:4	Reserved.
3	MclkDiv1Mode_M3 . Read-write. Reset: 1. Memclk Div1Mode for M-state 3.
2	MclkDiv1Mode_M2 . Read-write. Reset: 1. Memclk Div1Mode for M-state 2.
1	MclkDiv1Mode_M1 . Read-write. Reset: 1. Memclk Div1Mode for M-state 1.
0	MclkDiv1Mode_M0 . Read-write. Reset: 1. Memclk Div1Mode for M-state 0.

UMC[00...11]CTLx000001E4 [UMCCTRL Miscellaneous Configuration] (UMC::UmcCtrlMiscCfg)

Read-write. Reset: 0000_2000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx000001E4; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:17	Reserved.
16	EnableDynamicZq. Read-write. Reset: 0. Init: 1. 0=Dynamic ZQ commands disabled. 1=Dynamic ZQ commands enabled. Enable ZQ commands dynamically from umcctrl when SPAZ issues ZQ requests during DstateSel!=0
15	StaggerEnable. Read-write. Reset: 0. Init: 1. 0=Commands generated by SRSM will be broadcasted to both subchannels. MOP commands broadcasted if MOP_CH0 and MOP_CH1 are set. 1=Commands on SubChan1 are staggered by StaggerDelay with respect to SubChan0. MOP commands follow same behavior if MOP_CH0 and MOP_CH1 are set. Enable staggering of SRSM commands. Only relevant when in VCM mode.
14	InDstateOnLP3. Read-write. Reset: 0. Check: 0. 0=InDstate will be set on LP2 entry. 1=InDstate will be set on LP3 entry. Sets InDstate on entering LP3
13	SRin2N. Read-write. Reset: 1. 0=UMC goes into SR in 1N mode. 1=UMC goes into SR in 2N mode. Sets the DRAM timing used by the UMC to go into SR. Constraint: IF (DimmCfgCS01[RDIMM] == 1 DimmCfgCS23[RDIMM] == 1 DimmCfgCS01[LRDIMM] == 1 DimmCfgCS23[LRDIMM] == 1) THEN (UmcCtrlMiscCfg[SRin2N] = BeqCtrl0[RDimmSdrMode]) ELSE (UmcCtrlMiscCfg[SRin2N] = 1).
12:6	Reserved.
5	CliArbClkGateDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disables medium-grain clock gating of CliArb block.
4	Reserved.
3	SrefEn. Read-write. Reset: 0. Init: 0. 0=Disable. 1=Enable. Enables SREF for SRE associated with frequency change in DDR5.
2	SwClimClkGateDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disables medium-grain clock gating of SWCLIM block.
1	SrsmClkGateDis. Read-write. Reset: 0. Check: 0. 0=Enable. 1=Disable. Disables medium-grain clock gating.
0	RegClkGateEn. Read-write. Reset: 0. Init: 1. 0=Disable. 1=Enable. Enables medium-grain configuration register clock gating.

UMC00CTLx00000200...UMC11CTLx00000260 [SRSM Timing 0] (UMC::SrsmTiming0)

Read-write. Reset: 0000_3000h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx00000200; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx00000220; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx00000240; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx00000260; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:30	Reserved.
29:16	Tcksrx. Read-write. Reset: 0000h. Specifies the minimum time in memory clock cycles to assert CKE after clock frequency stable to CKE assertion.
15:8	TcksrxS. Read-write. Reset: 30h. Specifies the minimum time in memory clock cycles to assert CKE after clock frequency stable to CKE assertion. Applies only to C-states and DfiClkStpEn == 0.
7:0	Tcksre. Read-write. Reset: 00h. Specifies the minimum time in memory clock cycles to remove external clocks after entering self refresh.

UMC00CTLx00000204...UMC11CTLx00000264 [SRSM Timing 1] (UMC::SrsmTiming1)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx00000204; UMC[11:00]CTL=00[B:0]5_1000h_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx00000224; UMC[11:00]CTL=00[B:0]5_1000h_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx00000244; UMC[11:00]CTL=00[B:0]5_1000h_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx00000264; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:28	Reserved.
27:16	Txsddl. Read-write. Reset: 000h. Specifies the exit self refresh to commands requiring a locked DLL.
15:12	Reserved.
11:0	Txs. Read-write. Reset: 000h. Specifies the time to exit self refresh to commands not requiring a locked DLL.

UMC00CTLx00000208...UMC11CTLx00000268 [SRSM Timing 2] (UMC::SrsmTiming2)

Read-write. Reset: 0000_0000h.

Constraint: SrsmTiming6.TmrdL2 >= SrsmTiming6.TmrdL1 >= SrsmTiming2.Tmrd. IF (RDIMM | LRDIMM) THEN
 Constraint: DramTiming13.Tmrd >= max(SrsmTiming6.TmrdL2, SrsmTiming6.TmrdL1, SrsmTiming2.Tmrd) ELSE
 Constraint: DramTiming13.Tmrd == SrsmTiming2.Tmrd ENDIF.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx00000208; UMC[11:00]CTL=00[B:0]5_1000h_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx00000228; UMC[11:00]CTL=00[B:0]5_1000h_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx00000248; UMC[11:00]CTL=00[B:0]5_1000h_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx00000268; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description								
31:24	TmodPda. Read-write. Reset: 00h. Specifies the minimum time in memory clock cycles from an MRS PDA command to another non-MRS command, all to the same chip select. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Disabled.</td></tr> <tr> <td>3Fh-01h</td><td>TmodPda MEMCLKs.</td></tr> <tr> <td>FFh-40h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Disabled.	3Fh-01h	TmodPda MEMCLKs.	FFh-40h	Reserved.
Value	Description								
00h	Disabled.								
3Fh-01h	TmodPda MEMCLKs.								
FFh-40h	Reserved.								
23:16	Reserved.								
15:8	Tmod. Read-write. Reset: 00h. Specifies the minimum time in memory clock cycles from an MRS command to another non-MRS command, all to the same chip select. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>Tmod MEMCLKs</td></tr> <tr> <td>FFh-40h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	Tmod MEMCLKs	FFh-40h	Reserved.
Value	Description								
00h	Reserved.								
3Fh-01h	Tmod MEMCLKs								
FFh-40h	Reserved.								
7:0	Tmrd. Read-write. Reset: 00h. Specifies the minimum time in memory clock cycles from an MRS command to another MRS command, all to the same chip select. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>Reserved.</td></tr> <tr> <td>3Fh-01h</td><td>Tmrd MEMCLKs</td></tr> <tr> <td>FFh-40h</td><td>Reserved.</td></tr> </table>	Value	Description	00h	Reserved.	3Fh-01h	Tmrd MEMCLKs	FFh-40h	Reserved.
Value	Description								
00h	Reserved.								
3Fh-01h	Tmrd MEMCLKs								
FFh-40h	Reserved.								

UMC00CTLx0000020C...UMC11CTLx0000026C [SRSM Timing 3] (UMC::SrsmTiming3)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx0000020C; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx0000022C; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx0000024C; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx0000026C; UMC[11:00]CTL=00[B:0]5_1000h

Bits Description31:24 **TmrdPda.** Read-write. Reset: 00h. Specifies the minimum time in memory clock cycles from an MRS PDA command to another MRS command, all to the same chip select.**ValidValues:**

Value	Description
00h	Disabled.
3Fh-01h	TmrdPda MEMCLKs.
FFh-40h	Reserved.

23:20 Reserved.

19 **EnableSRXZqLatch.** Read-write. Reset: 0. Enables issuing of ZQ Latch command(after ZQ Start command) upon self refresh exit. Don't care if ExitSRZqCalDis is set18 **ExitSRZqCalDis.** Read-write. Reset: 0. Init: 1. 0=Enable. 1=Disable. Reserved if DDR5 mode.17 **Cmd2t.** Read-write. Reset: 0. 0=Disable. 1=Enable. Specifies 2T DDR address bus timing mode.16 **GearDownMode.** Read-write. Reset: 0. 0=Disable. 1=Enable. Programmed per RDimmSdrMode if DDR5 mode.

15:4 Reserved.

3:0 **Tescke.** Read-write. Reset: 0h. Reserved in DDR5 mode.**UMC00CTLx00000210...UMC11CTLx00000270 [SRSM Timing 4] (UMC::SrsmTiming4)**

Read-write. Reset: 0000_0040h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx00000210; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx00000230; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx00000250; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx00000270; UMC[11:00]CTL=00[B:0]5_1000h

Bits Description

31:12 Reserved.

11:0 **TzqOperCal.** Read-write. Reset: 040h. Specifies the minimum time in memory clock cycles from a ZQ command. [0x000: Reserved, 0xFFFF.0x001: <TzqOperCal> * 8 MEMCLKs] Programmed per tZQCAL if DDR5 mode.**UMC00CTLx00000214...UMC11CTLx00000274 [SRSM Timing 5] (UMC::SrsmTiming5)**

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx00000214; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx00000234; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx00000254; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx00000274; UMC[11:00]CTL=00[B:0]5_1000h

Bits Description

31:24 Reserved.

23:16 **Tzqlat.** Read-write. Reset: 00h. Specifies the minimum number of memory clock cycles from the ZQ Latch command and any other command.

15:0 Reserved.

UMC00CTLx00000218...UMC11CTLx00000278 [SRSM Timing 6] (UMC::SrsTiming6)

Read-write. Reset: 0000_2010h.

Constraint: TmrdPba >= TmrdL2.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx00000218; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx00000238; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx00000258; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx00000278; UMC[11:00]CTL=00[B:0]5_1000h

Bits Description

31:24 Reserved.

23:16 **TmrdPba**. Read-write. Reset: 00h. Specifies the minimum number of memory clock cycles from a PBA MR to another MR.15:8 **TmrdL2**. Read-write. Reset: 20h. Constraint: TmrdL2%2==0.7:0 **TmrdL**. Read-write. Reset: 10h. Specifies the minimum number of memory clock cycles from a MR, RCW, or BCW command to another MR, RCW, or BCW command in the MOP array. This is the medium-delay and is specified with MOP_Timer=1 in a MOP array entry.**UMC00CTLx0000021C...UMC11CTLx0000027C [SRSM Timing 7] (UMC::SrsTiming7)**

Read-write. Reset: 207C_7D80h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx0000021C; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx0000023C; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx0000025C; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx0000027C; UMC[11:00]CTL=00[B:0]5_1000h

Bits Description31:26 **StaggerDelay**. Read-write. Reset: 08h. Check: 08h. Time in memclks state machine staggers commands to SubChan1 w.r.t SubChan0. Only relevant when StaggerEnable is set25:18 **MopExpTimer**. Read-write. Reset: 1Fh. Time in memclks state machine waits before two expanded MOPs17:10 **Tpdx**. Read-write. Reset: 1Fh. Specifies the number of memory clock cycles SRSM waits after issuing PDX command9:0 **Tvref**. Read-write. Reset: 180h. Specifies the minimum number of memory clock cycles from an MR command which modifies DRAM Vref to another command. [0x0: Reserved, 0x3FF .. 0x1: <Tvref> MEMCLKs]**UMC[00...11]CTLx00000330 [LP2 timing parameters] (UMC::Lp2Timing)**

Read-write. Reset: 0010_2010h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000330; UMC[11:00]CTL=00[B:0]5_1000h

Bits Description

31:24 Reserved.

23:16 **Lp2DfiClkOnTimer**. Read-write. Reset: 10h. Check: 10h. Upon LP2 exit request DFICLK is turned on and SRSM waits for this timer to expire before initiating LP2 exit handshake15:8 **Lp2ResidencyTimer**. Read-write. Reset: 20h. Check: 20h. If entered into LP2 mode phy will not be instructed to get out of LP2 until this timer is expired. Only used when PairedLP2 is set7:0 **Lp2DfiClkOffTimer**. Read-write. Reset: 10h. Check: 10h. DFICLK will be turned off only after this timer is expired after phy enters LP2

UMC[00...11]CTLx00000400 [CLI Master control] (UMC::ClimCtrl)

Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000400; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:14	Reserved.
13	EnableRestart. Read-write. Reset: 0. Enable restart of SWCLIM sequence on RestartReq input assertion.
12	EnableMrrSnoop. Read-write. Reset: 0. Enable snooping MRR data for SW initiated PPT.
11	EnableAtomic. Read-write. Reset: 0. This bit will used to drive the Atomic bit of CLI Request interface.
10:6	RdWrMux. Read-write. Reset: 00h. Mux used to read received data or configure write data to be written.
5:1	CmdCount. Read-write. Reset: 00h. Number of consecutive slots with valid commands to send. Valid values are 0x0 to 0xF indicating 1 to 16 commands sent
0	SendCmd. Read-write, Volatile. Reset: 0. Cleared-by-hardware. Start sending CLI cmds from the cmd slots. Cleared by HW once SWLCIM transactions are complete.

UMC[00...11]CTLx00000404 [CLIM Read Data] (UMC::ClimRdData)

Read-only.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000404; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:0	Data. Read-only. Read Data demuxed from 512 bit CLI read data word using RdWrMux.

UMC[00...11]CTLx00000408 [CLIM Write Data] (UMC::ClimWrData)

Read-write.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000408; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:0	Data. Read-write. Write Data muxed to compose 512 bit CLI write data word using RdWrMux

UMC[00...11]CTLx0000040C [CLIM Write Pattern] (UMC::ClimWrPat)

Read-write.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx0000040C; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:0	Data. Read-write. If enabled using WrDataSel, this register will be expanded into a nibble pattern and used for generating write data

UMC00CTLx00000410...UMC11CTLx00000498 [CLIM Cmd Low word] (UMC::ClimCmdLo)

Read-write.

_umc0_instUMCWPHY0_n[17:0]_aliasSMN; UMC00CTLx0000_04[98,90,88,80,78,70,68,60,58,50,48,40,38,30,28,20,18,10]; UMC00CTL=0005_1000h
 _umc0_instUMCWPHY1_n[17:0]_aliasSMN; UMC01CTLx0000_04[98,90,88,80,78,70,68,60,58,50,48,40,38,30,28,20,18,10]; UMC01CTL=0015_1000h
 _umc0_instUMCWPHY2_n[17:0]_aliasSMN; UMC02CTLx0000_04[98,90,88,80,78,70,68,60,58,50,48,40,38,30,28,20,18,10]; UMC02CTL=0025_1000h
 _umc0_instUMCWPHY3_n[17:0]_aliasSMN; UMC03CTLx0000_04[98,90,88,80,78,70,68,60,58,50,48,40,38,30,28,20,18,10]; UMC03CTL=0035_1000h
 _umc0_instUMCWPHY4_n[17:0]_aliasSMN; UMC04CTLx0000_04[98,90,88,80,78,70,68,60,58,50,48,40,38,30,28,20,18,10]; UMC04CTL=0045_1000h
 _umc0_instUMCWPHY5_n[17:0]_aliasSMN; UMC05CTLx0000_04[98,90,88,80,78,70,68,60,58,50,48,40,38,30,28,20,18,10]; UMC05CTL=0055_1000h
 _umc0_instUMCWPHY6_n[17:0]_aliasSMN; UMC06CTLx0000_04[98,90,88,80,78,70,68,60,58,50,48,40,38,30,28,20,18,10]; UMC06CTL=0065_1000h
 _umc0_instUMCWPHY7_n[17:0]_aliasSMN; UMC07CTLx0000_04[98,90,88,80,78,70,68,60,58,50,48,40,38,30,28,20,18,10]; UMC07CTL=0075_1000h
 _umc0_instUMCWPHY8_n[17:0]_aliasSMN; UMC08CTLx0000_04[98,90,88,80,78,70,68,60,58,50,48,40,38,30,28,20,18,10]; UMC08CTL=0085_1000h
 _umc0_instUMCWPHY9_n[17:0]_aliasSMN; UMC09CTLx0000_04[98,90,88,80,78,70,68,60,58,50,48,40,38,30,28,20,18,10]; UMC09CTL=0095_1000h
 _umc0_instUMCWPHY10_n[17:0]_aliasSMN; UMC10CTLx0000_04[98,90,88,80,78,70,68,60,58,50,48,40,38,30,28,20,18,10]; UMC10CTL=00A5_1000h
 _umc0_instUMCWPHY11_n[17:0]_aliasSMN; UMC11CTLx0000_04[98,90,88,80,78,70,68,60,58,50,48,40,38,30,28,20,18,10]; UMC11CTL=00B5_1000h

Bits	Description
31:29	CmdWaitTmrSel. Read-write. Select one of 8 post command timer values to wait after a cmd is sent before sending next command. Timers 0-3 are generic, 4-7 are special
28	NoWait. Read-write. If set after sending current command state machine will proceed immediately to next command ignoring any interruptions
27	WrDataSel. Read-write. Selects either ClimWrData or ClimWrPat expanded into a nibble pattern for sending out write data
26	WaitForData. Read-write. Set if there is a data transaction on CLI Rd/WrData Bus with BEQ involved with the command.
25:22	Reserved.
21	CliBchop. Read-write. Sets the Burst Chop bit which is sent by this command
20	CliAp. Read-write. Sets the Ap which is sent by this command
19:15	CliBank. Read-write. Sets the Bank which is sent by this command
14:11	CliCs. Read-write. Sets the Cs which is sent by this command
10	Reserved.
9:7	CliPlr. Read-write. Sets the Plr which is sent by this command
6:2	CliMop. Read-write. Sets the Mop which is sent by this command
1:0	CliChannel. Read-write. Selects the channel on which command is sent. If both channels are set same command will be sent to both the channels

UMC00CTLx00000414...UMC11CTLx0000049C [CLIM Cmd High word] (UMC::ClimCmdHi)

Read-write.

_umc0_instUMCWPHY0_n[17:0]_aliasSMN; UMC00CTLx0000_04[9C,94,8C,84,7C,74,6C,64,5C,54,4C,44,3C,34,2C,24,1C,14]; UMC00CTL=0005_1000h
 _umc0_instUMCWPHY1_n[17:0]_aliasSMN; UMC01CTLx0000_04[9C,94,8C,84,7C,74,6C,64,5C,54,4C,44,3C,34,2C,24,1C,14]; UMC01CTL=0015_1000h
 _umc0_instUMCWPHY2_n[17:0]_aliasSMN; UMC02CTLx0000_04[9C,94,8C,84,7C,74,6C,64,5C,54,4C,44,3C,34,2C,24,1C,14]; UMC02CTL=0025_1000h
 _umc0_instUMCWPHY3_n[17:0]_aliasSMN; UMC03CTLx0000_04[9C,94,8C,84,7C,74,6C,64,5C,54,4C,44,3C,34,2C,24,1C,14]; UMC03CTL=0035_1000h
 _umc0_instUMCWPHY4_n[17:0]_aliasSMN; UMC04CTLx0000_04[9C,94,8C,84,7C,74,6C,64,5C,54,4C,44,3C,34,2C,24,1C,14]; UMC04CTL=0045_1000h
 _umc0_instUMCWPHY5_n[17:0]_aliasSMN; UMC05CTLx0000_04[9C,94,8C,84,7C,74,6C,64,5C,54,4C,44,3C,34,2C,24,1C,14]; UMC05CTL=0055_1000h
 _umc0_instUMCWPHY6_n[17:0]_aliasSMN; UMC06CTLx0000_04[9C,94,8C,84,7C,74,6C,64,5C,54,4C,44,3C,34,2C,24,1C,14]; UMC06CTL=0065_1000h
 _umc0_instUMCWPHY7_n[17:0]_aliasSMN; UMC07CTLx0000_04[9C,94,8C,84,7C,74,6C,64,5C,54,4C,44,3C,34,2C,24,1C,14]; UMC07CTL=0075_1000h
 _umc0_instUMCWPHY8_n[17:0]_aliasSMN; UMC08CTLx0000_04[9C,94,8C,84,7C,74,6C,64,5C,54,4C,44,3C,34,2C,24,1C,14]; UMC08CTL=0085_1000h
 _umc0_instUMCWPHY9_n[17:0]_aliasSMN; UMC09CTLx0000_04[9C,94,8C,84,7C,74,6C,64,5C,54,4C,44,3C,34,2C,24,1C,14]; UMC09CTL=0095_1000h
 _umc0_instUMCWPHY10_n[17:0]_aliasSMN; UMC10CTLx0000_04[9C,94,8C,84,7C,74,6C,64,5C,54,4C,44,3C,34,2C,24,1C,14]; UMC10CTL=00A5_1000h
 _umc0_instUMCWPHY11_n[17:0]_aliasSMN; UMC11CTLx0000_04[9C,94,8C,84,7C,74,6C,64,5C,54,4C,44,3C,34,2C,24,1C,14]; UMC11CTL=00B5_1000h

Bits	Description
31	Reserved.
30:20	CliCol. Read-write. Sets the Col which is sent by this command
19:18	Reserved.
17:0	CliRow. Read-write. Sets the Row which is sent by this command

UMC00CTLx00000500...UMC11CTLx0000050C [CLIM Timer0 and Timer1] (UMC::ClimTimer01)

Read-write.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx00000500; UMC[11:00]CTL=00[B:0]5_1000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx00000504; UMC[11:00]CTL=00[B:0]5_1000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx00000508; UMC[11:00]CTL=00[B:0]5_1000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx0000050C; UMC[11:00]CTL=00[B:0]5_1000h`**Bits Description**31:16 **Timer1.** Read-write. Post command timer 115:0 **Timer0.** Read-write. Post command timer 0**UMC00CTLx00000510...UMC11CTLx0000051C [CLIM Timer2 and Timer3] (UMC::ClimTimer23)**

Read-write.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx00000510; UMC[11:00]CTL=00[B:0]5_1000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx00000514; UMC[11:00]CTL=00[B:0]5_1000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx00000518; UMC[11:00]CTL=00[B:0]5_1000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx0000051C; UMC[11:00]CTL=00[B:0]5_1000h`**Bits Description**31:16 **Timer3.** Read-write. Post command timer 315:0 **Timer2.** Read-write. Post command timer 2**UMC00CTLx00000520...UMC11CTLx0000052C [CLIM Miscellaneous timers] (UMC::ClimTimerMisc)**

Read-write. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx00000520; UMC[11:00]CTL=00[B:0]5_1000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx00000524; UMC[11:00]CTL=00[B:0]5_1000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx00000528; UMC[11:00]CTL=00[B:0]5_1000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx0000052C; UMC[11:00]CTL=00[B:0]5_1000h`**Bits Description**

31:24 Reserved.

23:16 **PostSyncTimer.** Read-write. Reset: 00h. Number of uclks state machine waits after a Cmd17(Usually Sync command) to PhyAck assertion15:8 **PostDeSyncTimer.** Read-write. Reset: 00h. Number of uclks state machine waits after a Cmd16(Usually De-Sync command) to PhyAck assertion

7:0 Reserved.

UMC00CTLx00000530...UMC11CTLx0000053C [ARB Training Timers] (UMC::ArbTimerTraining)

Read-write. Reset: 0000_0000h.

`_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx00000530; UMC[11:00]CTL=00[B:0]5_1000h``_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx00000534; UMC[11:00]CTL=00[B:0]5_1000h``_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx00000538; UMC[11:00]CTL=00[B:0]5_1000h``_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx0000053C; UMC[11:00]CTL=00[B:0]5_1000h`**Bits Description**

31:25 Reserved.

24 **DisDQoscillatorStopMPC.** Read-write. Reset: 0. 0=Enable. 1=Disable. Disables an explicit MPC command to stop the DQ oscillator at the end of the run time interval. When disabled the DRAM timer should be configured to stop automatically.23:16 **DQoscillatorOffTime.** Read-write. Reset: 00h. Specifies the minimum time interval in MEMCLKs from DQ oscillator stop to the first oscillator read MRR. Programmed per DDR5 tOSCO.15:0 **DQoscillatorOnTime.** Read-write. Reset: 0000h. Specifies the DQ oscillator run time interval in MEMCLKs.

UMC00CTLx00000540...UMC11CTLx0000054C [CLIM Advanced Control Register] (UMC::ClimAdvCtrl)

Read-write. Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx00000540; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx00000544; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx00000548; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx0000054C; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:3	Reserved.
2	EnIntSlotsSW . Read-write. Reset: 0. If enabled interrupt slots will be enabled for software request generated commands
1:0	Reserved.

UMC00CTLx00000550...UMC11CTLx0000055C [CLI Arb Config register] (UMC::CliArbCfg)

Read-write. Reset: 0000_0004h.

_umc0_inst[UMCWPHY[11:0]]_mp0_aliasSMN; UMC[11:00]CTLx00000550; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp1_aliasSMN; UMC[11:00]CTLx00000554; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp2_aliasSMN; UMC[11:00]CTLx00000558; UMC[11:00]CTL=00[B:0]5_1000h

_umc0_inst[UMCWPHY[11:0]]_mp3_aliasSMN; UMC[11:00]CTLx0000055C; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:8	Reserved.
7:0	CtrlSwitchDelay . Read-write. Reset: 04h. Specifies an additional offset in memory clock cycles to Tphyupd_CmdDly and Tphyupd_WrDatDly for CLIARB logic. Set CtrlSwitchDelay = 4 + 6 + (UMC::DramTiming17[Tcsh] << UMC::MiscCfg[TcshMult]).

UMC[00...11]CTLx00000800 [APB Command] (UMC::ApbCmd)

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000800; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31	Write . Read-write. Reset: 0. 0=APB Read. 1=APB Write. APB command select.
30	AutoInc . Read-write. Reset: 0. 0=Disable. 1=Enable. APB auto increment enable.
29:0	Address . Read-write, Volatile. APB address index. Updated-by-hardware if AutoInc = 1.

UMC[00...11]CTLx00000804 [APB Data] (UMC::ApbDat)

Read-write, Volatile.

Read data port controlled by ApbCmd.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:0	Data . Read-write, Volatile. APB data register.

UMC[00...11]CTLx00000808 [APB Status] (UMC::ApbStatus)

APB status register.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000808; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:15	Reserved.
14:8	LoopCnt . Read-write. Reset: 00h. Loop count for APB looping mode.
7:2	Reserved.
1	PslvErr . Read-only, Volatile. Indicates APB PSLVERR.
0	Done . Read-only, Volatile. Indicates APB command has finished.

UMC[00...11]CTLx0000080C [APB Increment] (UMC::ApbIncrement)

Reset: 0000_0000h.

_umc0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx0000080C; UMC[11:00]CTL=00[B:0]5_1000h

Bits	Description
31:0	Reserved.

12.12.2 PHY Registers

**UMC00CTLx00000804_address00000008...UMC11CTLx00000804_address0000B008
(UMC::Phy::ANIBReserved)**

Read-write. Reset: 0000h.	
Spares for the ANIB Hard Macro	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]008; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]008; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]008; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]008; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]008; UMC04CTL=0045_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]008; UMC05CTL=0055_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]008; UMC06CTL=0065_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]008; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]008; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]008; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]008; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]008; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	ANIBReserved. Read-write. Reset: 000h. Spares for the ANIB Hard Macro

UMC00CTLx00000804_address00000020...UMC11CTLx00000804_address0000B020 (UMC::Phy::AsyncAnibMode)

Read-write. Reset: 0000h.

Enables Async Flyover for all lanes of the ANIB

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]020; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]020; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]020; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]020; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]020; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]020; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]020; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]020; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]020; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]020; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]020; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]020; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	AsyncAnibMode. Read-write. Reset: 0h. Description: Used for giving firmware direct access to ANIB pins. Each bit controls the corresponding lane

UMC00CTLx00000804_address00000021...UMC11CTLx00000804_address0000B021 (UMC::Phy::AsyncAnibTxEn)

Read-write. Reset: 0000h.

TxEnable bits for Async Flyover

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]021; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]021; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]021; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]021; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]021; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]021; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]021; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]021; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]021; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]021; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]021; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]021; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	AsyncAnibTxEn. Read-write. Reset: 0h. Description: Used for giving firmware direct access to ANIB pins. Each bit controls the corresponding lane

UMC00CTLx00000804_address00000022...UMC11CTLx00000804_address0000B022 (UMC::Phy::AsyncAnibTxData)

Read-write. Reset: 0000h.

TxData bits for Async Flyover

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]022; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]022; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]022; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]022; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]022; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]022; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]022; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]022; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]022; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]022; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]022; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]022; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	AsyncAnibTxData. Read-write. Reset: 0h.
	Description: Used for giving firmware direct access to ANIB pins. Each bit controls the corresponding lane

UMC00CTLx00000804_address00000023...UMC11CTLx00000804_address0000B023 (UMC::Phy::AsyncAnibRxEn)

Read-write. Reset: 0000h.

TxEnable bits for Async Flyover

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]023; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]023; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]023; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]023; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]023; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]023; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]023; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]023; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]023; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]023; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]023; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]023; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	AsyncAnibRxEn. Read-write. Reset: 0h. Description: Used for giving firmware direct access to ANIB pins. Each bit controls the corresponding lane

UMC00CTLx00000804_address00000024...UMC11CTLx00000804_address0000B024 (UMC::Phy::AsyncAnibRxData)

Read-only, Volatile.

RxData bits for Async Flyover

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]024; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]024; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]024; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]024; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]024; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]024; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]024; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]024; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]024; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]024; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]024; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]024; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	AsyncAnibRxData. Read-only, Volatile.
	Description: Used for giving firmware direct access to ANIB pins. Each bit controls the corresponding lane

UMC00CTLx00000804_address00000027...UMC11CTLx00000804_address0000B027 (UMC::Phy::AForceDrvCont)

Read-write. Reset: 0000h.

Force Address/Command Driven (Lanes A3-A0)

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]027; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]027; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]027; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]027; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]027; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]027; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]027; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]027; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]027; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]027; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]027; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]027; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	AForceDrvCont. Read-write. Reset: 0h. Description: Force continuous drive, per-lane, of the ACX4 instance controlled by this register. Setting this register will cause the PHY to drive the target lane when dfi_init_complete==1 - Bit [0] = controls lane 0 of the target ACX4 block - Bit [1] = controls lane 1 of the target ACX4 block - Bit [2] = controls lane 2 of the target ACX4 block - Bit [3] = controls lane 3 of the target ACX4 block

UMC00CTLx00000804_address00000028...UMC11CTLx00000804_address0000B028 (UMC::Phy::AForceTriCont)

Read-write. Reset: 0000h.

Force Address/Command Tristate (Lanes A3-A0)

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]028; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]028; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]028; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]028; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]028; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]028; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]028; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]028; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]028; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]028; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]028; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]028; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	AForceTriCont. Read-write. Reset: 0h. Description: Force tristate control, per-lane, of the ACX4 instance controlled by this register. Setting this register will cause the PHY to tristate the target lane when dfi_init_complete==1 - Bit [0] = controls lane 0 of the target ACX4 block - Bit [1] = controls lane 1 of the target ACX4 block - Bit [2] = controls lane 2 of the target ACX4 block - Bit [3] = controls lane 3 of the target ACX4 block

UMC00CTLx00000804_address00000029...UMC11CTLx00000804_address00320029 [Voltage Regulator Control1] (UMC::Phy::VREGCtrl1)

Read-write. Reset: 00ACh.

Controls the voltage regulator generation of VSH, VDS3 and VshAnalogTestOut.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p0_aliasSMN; UMC00CTLx00000804_address0000 [D:0]029; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY0_p0_aliasSMN; UMC00CTLx00000804_address0001 [9:0]029; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020029; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p1_aliasSMN; UMC00CTLx00000804_address0010 [D:0]029; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY0_p1_aliasSMN; UMC00CTLx00000804_address0011 [9:0]029; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120029; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p2_aliasSMN; UMC00CTLx00000804_address0020 [D:0]029; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY0_p2_aliasSMN; UMC00CTLx00000804_address0021 [9:0]029; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220029; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p3_aliasSMN; UMC00CTLx00000804_address0030 [D:0]029; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY0_p3_aliasSMN; UMC00CTLx00000804_address0031 [9:0]029; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320029; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p0_aliasSMN; UMC01CTLx00000804_address0000 [B:0]029; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY1_p0_aliasSMN; UMC01CTLx00000804_address0001 [9:0]029; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p1_aliasSMN; UMC01CTLx00000804_address0010 [B:0]029; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY1_p1_aliasSMN; UMC01CTLx00000804_address0011 [9:0]029; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p2_aliasSMN; UMC01CTLx00000804_address0020 [B:0]029; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY1_p2_aliasSMN; UMC01CTLx00000804_address0021 [9:0]029; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p3_aliasSMN; UMC01CTLx00000804_address0030 [B:0]029; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY1_p3_aliasSMN; UMC01CTLx00000804_address0031 [9:0]029; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p0_aliasSMN; UMC02CTLx00000804_address0000 [B:0]029; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY2_p0_aliasSMN; UMC02CTLx00000804_address0001 [9:0]029; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p1_aliasSMN; UMC02CTLx00000804_address0010 [B:0]029; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY2_p1_aliasSMN; UMC02CTLx00000804_address0011 [9:0]029; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p2_aliasSMN; UMC02CTLx00000804_address0020 [B:0]029; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY2_p2_aliasSMN; UMC02CTLx00000804_address0021 [9:0]029; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p3_aliasSMN; UMC02CTLx00000804_address0030 [B:0]029; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY2_p3_aliasSMN; UMC02CTLx00000804_address0031 [9:0]029; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p0_aliasSMN; UMC03CTLx00000804_address0000 [D:0]029; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY3_p0_aliasSMN; UMC03CTLx00000804_address0001 [9:0]029; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p1_aliasSMN; UMC03CTLx00000804_address0010 [D:0]029; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

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[illegible]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY11_p3_aliasSMN; UMC11CTLx00000804_address0031_[9:0]029; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:9	Reserved.
8:2	VshDAC. Read-write. Reset: 2Bh. Description: DAC control bits. - D5 (VDDnom=0.75): 0100010 - D5 (VDDnom=0.85): 0010010
1:0	VshCurrentLoad. Read-write. Reset: 0h. Future feature to adjust regulator for lower current load. Set to 0.

UMC00CTLx00000804_address0000002B...UMC11CTLx00000804_address0002002B [Voltage Regulator Control2] (UMC::Phy::VREGCtrl2)

Read-write. Reset: 0020h.

Controls the voltage regulator generation of VSH, VDS3 and VshAnalogTestOut

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]02B; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0001_[9:0]02B; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002002B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]02B; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0001_[9:0]02B; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]02B; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0001_[9:0]02B; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]02B; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0001_[9:0]02B; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]02B; UMC04CTL=0045_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0001_[9:0]02B; UMC04CTL=0045_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]02B; UMC05CTL=0055_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0001_[9:0]02B; UMC05CTL=0055_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]02B; UMC06CTL=0065_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0001_[9:0]02B; UMC06CTL=0065_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]02B; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0001_[9:0]02B; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]02B; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0001_[9:0]02B; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]02B; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0001_[9:0]02B; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]02B; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0001_[9:0]02B; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]02B; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0001_[9:0]02B; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15	VshMissionMode. Read-write. Reset: 0. When asserted, switches VSH from fail-safe default to mission mode after coming back from S3
14:13	VshBypassModes. Read-write. Reset: 0h. Bypass regulator output from DAC. Debug only. Set to 0.
12:11	VshPowerModes. Read-write. Reset: 0h.

	Description: Used to enable/control bleeder currents to VSH during calibration for TX - 00 No bleeder (recommended mission mode default) - 01 1x unit bleeder - 10 2x unit bleeder - 11 4x unit bleeder (TX calibration default)
10:8	VshAnalogOutCtrl. Read-write. Reset: 0h. Description: Selection control for analog test output - 0XX Hi-Z (recommended default) - 100 select VSH to test output - 101 select VDS3 to test output - 110 select internal bias voltage to test output - 111 select internal vref to test output
7:0	VshAnalog. Read-write. Reset: 20h. Description: Allows gain/current of OpAmp to be adjusted. - VshAnalog[7] is used to enable optional VSH bleeder current. - VshAnalog[6] is used to enable VSH bleeder current as load for testing. Recommended mission mode default = 0x20

UMC00CTLx00000804_address00000040...UMC11CTLx00000804_address0000B040 (UMC::Phy::ATxImpedanceCtrl1)

Read-write. Reset: 0FFFh.

Reserved for future use.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]040; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]040; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]040; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]040; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]040; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]040; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]040; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]040; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]040; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]040; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]040; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]040; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	Reserved.
11:0	RFU. Read-write. Reset: FFFh.

UMC00CTLx00000804_address00000041...UMC11CTLx00000804_address0000B041 (UMC::Phy::ATxImpedanceCtrl2)

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]041; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]041; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]041; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]041; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]041; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]041; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]041; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]041; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]041; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]041; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]041; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]041; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	RFU. Read-write. Reset: 000h.
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UMC00CTLx00000804_address00000042...UMC11CTLx00000804_address0000B042 [Address TX ODT driver strength control] (UMC::Phy::ATxOdtDrvStren)

Read-write. Reset: 0000h.

Selects desired impedance value for Host ODT

Must be set by user before PHY initialization.

Setting for each protocol:

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]042; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]042; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]042; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]042; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]042; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]042; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]042; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]042; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]042; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]042; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]042; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]042; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:12 Reserved.

11:6 **AODTStrenN.** Read-write. Reset: 00h.**Description:** Selects the ODT pull-down impedance.

Coding for [ODTStren120N[3:0] , ODTStren240N , ODTStren480N[0:0]] is:

- 0000_0_0 = High Impedance
- 0000_0_1 = 480 ohms
- 0000_1_0 = 240 ohms
- 0000_1_1 = 160 ohms
- 0001_0_0 = 120 ohms
- 0001_0_1 = 96.0 ohms
- 0001_1_0 = 80.0 ohms
- 0001_1_1 = 68.6 ohms
- 0011_0_0 = 60.0 ohms
- 0011_0_1 = 53.3 ohms
- 0011_1_0 = 48.0 ohms
- 0011_1_1 = 43.6 ohms
- 0111_0_0 = 40.0 ohms
- 0111_0_1 = 36.9 ohms
- 0111_1_0 = 34.3 ohms
- 0111_1_1 = 32.0 ohms
- 1111_0_0 = 30.0 ohms
- 1111_0_1 = 28.2 ohms
- 1111_1_0 = 26.7 ohms
- 1111_1_1 = 25.3 ohms

5:0 **AODTStrenP.** Read-write. Reset: 00h.

Description: Selects the ODT pull-up impedance.

Coding for [ODTStren120P[3:0] , ODTStren240P , ODTStren480P[0:0]] is:

- 0000_0_0 = High Impedance
- 0000_0_1 = 480 ohms
- 0000_1_0 = 240 ohms
- 0000_1_1 = 160 ohms
- 0001_0_0 = 120 ohms
- 0001_0_1 = 96.0 ohms
- 0001_1_0 = 80.0 ohms
- 0001_1_1 = 68.6 ohms
- 0011_0_0 = 60.0 ohms
- 0011_0_1 = 53.3 ohms
- 0011_1_0 = 48.0 ohms
- 0011_1_1 = 43.6 ohms
- 0111_0_0 = 40.0 ohms
- 0111_0_1 = 36.9 ohms
- 0111_1_0 = 34.3 ohms
- 0111_1_1 = 32.0 ohms
- 1111_0_0 = 30.0 ohms
- 1111_0_1 = 28.2 ohms
- 1111_1_0 = 26.7 ohms
- 1111_1_1 = 25.3 ohms

UMC00CTLx00000804_address00000043...UMC11CTLx00000804_address0000B043 [Address TX impedance controls] (UMC::Phy::ATxImpedance)

Read-write. Reset: CFFFh.

Tx impedance of Address driver cells.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]043; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]043; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]043; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]043; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]043; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]043; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]043; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]043; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]043; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]043; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]043; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]043; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	ATxCalBaseP. Read-write. Reset: 1. Enables the base transistor leg of each output driver section.
14	ATxCalBaseN. Read-write. Reset: 1. Enables the base transistor leg of each output driver section.
13:12	ATxReserved13x12. Read-write. Reset: 0h. Reserved for future use in TXFE.
11:6	ADrvStrenN. Read-write. Reset: 3Fh. Description: This 6-bit bus is used to select the target pull-down output impedance. Connects to the DrvStren pins of the driver as follows: [Csr_DrvStrenFS120N[3:0], Csr_DrvStrenFS240N,Csr_DrvStrenFS480N[0:0]] = [csrADrvStrenN[5:0]] See table of csr ADrvStrenN for list of possible impedances. Setting ADrvStrenN[11:6] to '1' is the same as HiZ as 480 segments are disconnected from the PAD.
5:0	ADrvStrenP. Read-write. Reset: 3Fh. Description: This 6-bit bus used to select the target pull-up output impedance. Connects to the DrvStren pins of the driver as follows: [Csr_DrvStrenFS120P[3:0], Csr_DrvStrenFS240P,Csr_DrvStrenFS480P[0:0]] = [csrADrvStrenP[5:0]] - X0001X - 120.0 - X0011X - 60.0 - X0111X - 40.0 - X1111X - 30.0

UMC00CTLx00000804_address00000055...UMC11CTLx00000804_address0030B055 [Address TX slew rate and predriver controls] (UMC::Phy::ATxSlewRate)

Read-write. Reset: 0700h.

Controls slew rates of address/command/memclk drivers.

Tx predriver cal trims driver timing to account for clock duty cycle variation.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]055; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p1_aliasSMN; UMC00CTLx00000804_address0010_[D:0]055; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p2_aliasSMN; UMC00CTLx00000804_address0020_[D:0]055; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p3_aliasSMN; UMC00CTLx00000804_address0030_[D:0]055; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p0_aliasSMN; UMC01CTLx00000804_address0000_[B:0]055; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p1_aliasSMN; UMC01CTLx00000804_address0010_[B:0]055; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p2_aliasSMN; UMC01CTLx00000804_address0020_[B:0]055; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p3_aliasSMN; UMC01CTLx00000804_address0030_[B:0]055; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p0_aliasSMN; UMC02CTLx00000804_address0000_[B:0]055; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p1_aliasSMN; UMC02CTLx00000804_address0010_[B:0]055; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p2_aliasSMN; UMC02CTLx00000804_address0020_[B:0]055; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p3_aliasSMN; UMC02CTLx00000804_address0030_[B:0]055; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p0_aliasSMN; UMC03CTLx00000804_address0000_[D:0]055; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p1_aliasSMN; UMC03CTLx00000804_address0010_[D:0]055; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p2_aliasSMN; UMC03CTLx00000804_address0020_[D:0]055; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p3_aliasSMN; UMC03CTLx00000804_address0030_[D:0]055; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p0_aliasSMN; UMC04CTLx00000804_address0000_[B:0]055; UMC04CTL=0045_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p1_aliasSMN; UMC04CTLx00000804_address0010_[B:0]055; UMC04CTL=0045_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p2_aliasSMN; UMC04CTLx00000804_address0020_[B:0]055; UMC04CTL=0045_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p3_aliasSMN; UMC04CTLx00000804_address0030_[B:0]055; UMC04CTL=0045_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p0_aliasSMN; UMC05CTLx00000804_address0000_[B:0]055; UMC05CTL=0055_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p1_aliasSMN; UMC05CTLx00000804_address0010_[B:0]055; UMC05CTL=0055_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p2_aliasSMN; UMC05CTLx00000804_address0020_[B:0]055; UMC05CTL=0055_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p3_aliasSMN; UMC05CTLx00000804_address0030_[B:0]055; UMC05CTL=0055_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p0_aliasSMN; UMC06CTLx00000804_address0000_[D:0]055; UMC06CTL=0065_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p1_aliasSMN; UMC06CTLx00000804_address0010_[D:0]055; UMC06CTL=0065_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p2_aliasSMN; UMC06CTLx00000804_address0020_[D:0]055; UMC06CTL=0065_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p3_aliasSMN; UMC06CTLx00000804_address0030_[D:0]055; UMC06CTL=0065_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p0_aliasSMN; UMC07CTLx00000804_address0000_[B:0]055; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p1_aliasSMN; UMC07CTLx00000804_address0010_[B:0]055; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p2_aliasSMN; UMC07CTLx00000804_address0020_[B:0]055; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p3_aliasSMN; UMC07CTLx00000804_address0030_[B:0]055; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p0_aliasSMN; UMC08CTLx00000804_address0000_[B:0]055; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p1_aliasSMN; UMC08CTLx00000804_address0010_[B:0]055; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p2_aliasSMN; UMC08CTLx00000804_address0020_[B:0]055; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p3_aliasSMN; UMC08CTLx00000804_address0030_[B:0]055; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p0_aliasSMN; UMC09CTLx00000804_address0000_[D:0]055; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p1_aliasSMN; UMC09CTLx00000804_address0010_[D:0]055; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p2_aliasSMN; UMC09CTLx00000804_address0020_[D:0]055; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p3_aliasSMN; UMC09CTLx00000804_address0030_[D:0]055; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p0_aliasSMN; UMC10CTLx00000804_address0000_[B:0]055; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p1_aliasSMN; UMC10CTLx00000804_address0010_[B:0]055; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p2_aliasSMN; UMC10CTLx00000804_address0020_[B:0]055; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p3_aliasSMN; UMC10CTLx00000804_address0030_[B:0]055; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p0_aliasSMN; UMC11CTLx00000804_address0000_[B:0]055; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p1_aliasSMN; UMC11CTLx00000804_address0010_[B:0]055; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p2_aliasSMN; UMC11CTLx00000804_address0020_[B:0]055; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p3_aliasSMN; UMC11CTLx00000804_address0030_[B:0]055; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15	Reserved.
14:11	ATxChargeCancel. Read-write. Reset: 0h. Controls strength of discharge function. 4'h0 no discharge. 4'hF max discharge.
10:8	ATxPreDrvMode. Read-write. Reset: 7h. Description: Controls VDD high function. - TxPreDrvMode[8] is high VDD. When VDD=0.9V nominal, set this to 1'b1, otherwise set to 1'b0. - TxPreDrvMode[10:9] reserved for future use.
7:0	CsrATxSrc. Read-write. Reset: 00h. Description: 8-bit binary trim for the driver pull-up/down slew rate. The 4 upper bits don't care if the 4 lowest bits are set to 4'b0000. Bit 7 reserved for future use Bit 6 Low VDD(0.75V nominal)=1'b0; High VDD(0.85V nominal)=1'b1 Bit 5 D5=1'b1 Bit 4 AC=1'b0, CK=1'b1 Bits[3:2] Slew rate control rise edge: 2'b00 = slew rate control off 2'b01 = slew rate control on, slower edge 2'b10 = slew rate control on, slow edge 2'b11 = reserved for future use Bits[1:0] Slew rate control fall edge: 2'b00 = slew rate control off 2'b01 = slew rate control on, slower edge 2'b10 = slew rate control on, slow edge 2'b11 = reserved for future use

UMC00CTLx00000804_address00000056...UMC11CTLx00000804_address0000B056 (UMC::Phy::ATestPrbsErrCntln0)

Read-only, Volatile.

Address Loopback Test Result register

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]056; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]056; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]056; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]056; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]056; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]056; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]056; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]056; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]056; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]056; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]056; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]056; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	ATestPrbsErrCntln0. Read-only, Volatile.
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Description: Overall error indicator for each prbs bump checker.

Non-zero value indicating the number of errors seen by self-seeding prbs checker.

UMC00CTLx00000804_address00000066...UMC11CTLx00000804_address00020066 [Voltage Regulator Control3] (UMC::Phy::VREGCtrl3)

Read-write. Reset: 0000h.

Controls the voltage regulator generation of VSH, VDS3 and VshAnalogTestOut

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]066; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0001_[9:0]066; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_instUMCWPHY[11:0]_aliasSMN; UMC[11:0]CTLx00000804_address00020066; UMC[11:0]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]066; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0001_[9:0]066; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]066; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0001_[9:0]066; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]066; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0001_[9:0]066; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]066; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0001_[9:0]066; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]066; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0001_[9:0]066; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]066; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0001_[9:0]066; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]066; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0001_[9:0]066; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]066; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0001_[9:0]066; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]066; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0001_[9:0]066; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]066; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0001_[9:0]066; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]066; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[9:0]_chiplettype1_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0001_[9:0]066; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:1 Reserved.

0 **VshCtrlUpdate.** Read-write. Reset: 0. Strobe for VshDAC and VshAnalog

UMC00CTLx00000804_address000007F..UMC11CTLx00000804_address0030B07F (UMC::Phy::ATuDly)

Read-write. Reset: 0000h.

Address/Command Delay, per pstate.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]07F; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p1_aliasSMN; UMC00CTLx00000804_address0010_[D:0]07F; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p2_aliasSMN; UMC00CTLx00000804_address0020_[D:0]07F; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p3_aliasSMN; UMC00CTLx00000804_address0030_[D:0]07F; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p0_aliasSMN; UMC01CTLx00000804_address0000_[B:0]07F; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p1_aliasSMN; UMC01CTLx00000804_address0010_[B:0]07F; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p2_aliasSMN; UMC01CTLx00000804_address0020_[B:0]07F; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p3_aliasSMN; UMC01CTLx00000804_address0030_[B:0]07F; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p0_aliasSMN; UMC02CTLx00000804_address0000_[B:0]07F; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p1_aliasSMN; UMC02CTLx00000804_address0010_[B:0]07F; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p2_aliasSMN; UMC02CTLx00000804_address0020_[B:0]07F; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p3_aliasSMN; UMC02CTLx00000804_address0030_[B:0]07F; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p0_aliasSMN; UMC03CTLx00000804_address0000_[D:0]07F; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p1_aliasSMN; UMC03CTLx00000804_address0010_[D:0]07F; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p2_aliasSMN; UMC03CTLx00000804_address0020_[D:0]07F; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p3_aliasSMN; UMC03CTLx00000804_address0030_[D:0]07F; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p0_aliasSMN; UMC04CTLx00000804_address0000_[B:0]07F; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p1_aliasSMN; UMC04CTLx00000804_address0010_[B:0]07F; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p2_aliasSMN; UMC04CTLx00000804_address0020_[B:0]07F; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p3_aliasSMN; UMC04CTLx00000804_address0030_[B:0]07F; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p0_aliasSMN; UMC05CTLx00000804_address0000_[B:0]07F; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p1_aliasSMN; UMC05CTLx00000804_address0010_[B:0]07F; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p2_aliasSMN; UMC05CTLx00000804_address0020_[B:0]07F; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p3_aliasSMN; UMC05CTLx00000804_address0030_[B:0]07F; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p0_aliasSMN; UMC06CTLx00000804_address0000_[D:0]07F; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p1_aliasSMN; UMC06CTLx00000804_address0010_[D:0]07F; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p2_aliasSMN; UMC06CTLx00000804_address0020_[D:0]07F; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p3_aliasSMN; UMC06CTLx00000804_address0030_[D:0]07F; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p0_aliasSMN; UMC07CTLx00000804_address0000_[B:0]07F; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p1_aliasSMN; UMC07CTLx00000804_address0010_[B:0]07F; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p2_aliasSMN; UMC07CTLx00000804_address0020_[B:0]07F; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p3_aliasSMN; UMC07CTLx00000804_address0030_[B:0]07F; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p0_aliasSMN; UMC08CTLx00000804_address0000_[B:0]07F; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p1_aliasSMN; UMC08CTLx00000804_address0010_[B:0]07F; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p2_aliasSMN; UMC08CTLx00000804_address0020_[B:0]07F; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p3_aliasSMN; UMC08CTLx00000804_address0030_[B:0]07F; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p0_aliasSMN; UMC09CTLx00000804_address0000_[D:0]07F; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p1_aliasSMN; UMC09CTLx00000804_address0010_[D:0]07F; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p2_aliasSMN; UMC09CTLx00000804_address0020_[D:0]07F; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p3_aliasSMN; UMC09CTLx00000804_address0030_[D:0]07F; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p0_aliasSMN; UMC10CTLx00000804_address0000_[B:0]07F; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p1_aliasSMN; UMC10CTLx00000804_address0010_[B:0]07F; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p2_aliasSMN; UMC10CTLx00000804_address0020_[B:0]07F; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p3_aliasSMN; UMC10CTLx00000804_address0030_[B:0]07F; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p0_aliasSMN; UMC11CTLx00000804_address0000_[B:0]07F; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p1_aliasSMN; UMC11CTLx00000804_address0010_[B:0]07F; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p2_aliasSMN; UMC11CTLx00000804_address0020_[B:0]07F; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p3_aliasSMN; UMC11CTLx00000804_address0030_[B:0]07F; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:8	Reserved.
7:0	ATuDly. Read-write. Reset: 00h. Description: Trained to generate timed address and command signals to the DRAMs, per ACX4, (upper 2 bits). The upper two signals generated by an ACX4 will have common timing. This value is applied as discussed in the csrAtxDlySelect Register. - ATuDly[7:6] is the coarse delay, ie one unit of delay is 1 UI. - ATuDly[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64;

UMC00CTLx00000804_address0000080...UMC11CTLx00000804_address0030B080 (UMC::Phy::ATxDly)

Read-write. Reset: 0000h.

Address/Command Delay, per pstate.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]080; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p1_aliasSMN; UMC00CTLx00000804_address0010_[D:0]080; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p2_aliasSMN; UMC00CTLx00000804_address0020_[D:0]080; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p3_aliasSMN; UMC00CTLx00000804_address0030_[D:0]080; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p0_aliasSMN; UMC01CTLx00000804_address0000_[B:0]080; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p1_aliasSMN; UMC01CTLx00000804_address0010_[B:0]080; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p2_aliasSMN; UMC01CTLx00000804_address0020_[B:0]080; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p3_aliasSMN; UMC01CTLx00000804_address0030_[B:0]080; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p0_aliasSMN; UMC02CTLx00000804_address0000_[B:0]080; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p1_aliasSMN; UMC02CTLx00000804_address0010_[B:0]080; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p2_aliasSMN; UMC02CTLx00000804_address0020_[B:0]080; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p3_aliasSMN; UMC02CTLx00000804_address0030_[B:0]080; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p0_aliasSMN; UMC03CTLx00000804_address0000_[D:0]080; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p1_aliasSMN; UMC03CTLx00000804_address0010_[D:0]080; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p2_aliasSMN; UMC03CTLx00000804_address0020_[D:0]080; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p3_aliasSMN; UMC03CTLx00000804_address0030_[D:0]080; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p0_aliasSMN; UMC04CTLx00000804_address0000_[B:0]080; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p1_aliasSMN; UMC04CTLx00000804_address0010_[B:0]080; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p2_aliasSMN; UMC04CTLx00000804_address0020_[B:0]080; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p3_aliasSMN; UMC04CTLx00000804_address0030_[B:0]080; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p0_aliasSMN; UMC05CTLx00000804_address0000_[B:0]080; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p1_aliasSMN; UMC05CTLx00000804_address0010_[B:0]080; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p2_aliasSMN; UMC05CTLx00000804_address0020_[B:0]080; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p3_aliasSMN; UMC05CTLx00000804_address0030_[B:0]080; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p0_aliasSMN; UMC06CTLx00000804_address0000_[D:0]080; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p1_aliasSMN; UMC06CTLx00000804_address0010_[D:0]080; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p2_aliasSMN; UMC06CTLx00000804_address0020_[D:0]080; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p3_aliasSMN; UMC06CTLx00000804_address0030_[D:0]080; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p0_aliasSMN; UMC07CTLx00000804_address0000_[B:0]080; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p1_aliasSMN; UMC07CTLx00000804_address0010_[B:0]080; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p2_aliasSMN; UMC07CTLx00000804_address0020_[B:0]080; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p3_aliasSMN; UMC07CTLx00000804_address0030_[B:0]080; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p0_aliasSMN; UMC08CTLx00000804_address0000_[B:0]080; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p1_aliasSMN; UMC08CTLx00000804_address0010_[B:0]080; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p2_aliasSMN; UMC08CTLx00000804_address0020_[B:0]080; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p3_aliasSMN; UMC08CTLx00000804_address0030_[B:0]080; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p0_aliasSMN; UMC09CTLx00000804_address0000_[D:0]080; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p1_aliasSMN; UMC09CTLx00000804_address0010_[D:0]080; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p2_aliasSMN; UMC09CTLx00000804_address0020_[D:0]080; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p3_aliasSMN; UMC09CTLx00000804_address0030_[D:0]080; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p0_aliasSMN; UMC10CTLx00000804_address0000_[B:0]080; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p1_aliasSMN; UMC10CTLx00000804_address0010_[B:0]080; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p2_aliasSMN; UMC10CTLx00000804_address0020_[B:0]080; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p3_aliasSMN; UMC10CTLx00000804_address0030_[B:0]080; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p0_aliasSMN; UMC11CTLx00000804_address0000_[B:0]080; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p1_aliasSMN; UMC11CTLx00000804_address0010_[B:0]080; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p2_aliasSMN; UMC11CTLx00000804_address0020_[B:0]080; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p3_aliasSMN; UMC11CTLx00000804_address0030_[B:0]080; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:8	Reserved.
7:0	ATxDly. Read-write. Reset: 00h. Description: Trained to generate timed address and command signals to the DRAMs, per ACX4, (lower 2 bits) The lower two signals generated by an ACX4 will have common timing. This value is applied as discussed in the csrAtxDlySelect Register. - ATxDly[7:6] is the coarse delay, ie one unit of delay is 1 UI. - ATxDly[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64;

UMC00CTLx00000804_address0000081...UMC11CTLx00000804_address0000B081 [Controls the static PowerDown of ANIB drivers and receivers] (UMC::Phy::AcPowerDownStatic)

Read-write. Reset: 0000h.

Forces power down ANIB per lane regardless of low-power state.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]081; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]081; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]081; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]081; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]081; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]081; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]081; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]081; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]081; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]081; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]081; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]081; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:4	AcRxPowerDownStatic. Read-write. Reset: 0h. AcRxPowerDownStatic[n]=1 will statically power down BP_ACn receiver
3:0	AcTxPowerDownStatic. Read-write. Reset: 0h. AcTxPowerDownStatic[n]=1 will statically power down BP_ACn driver

UMC00CTLx00000804_address0000082...UMC11CTLx00000804_address0030B082 [Controls the PowerDown of ANIB drivers] (UMC::Phy::AcPowerDownDynEn)

Read-write. Reset: 0000h.

Enable power down ANIB per lane when PowerDnDynAC=1.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]082; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p1_aliasSMN; UMC00CTLx00000804_address0010_[D:0]082; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p2_aliasSMN; UMC00CTLx00000804_address0020_[D:0]082; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p3_aliasSMN; UMC00CTLx00000804_address0030_[D:0]082; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p0_aliasSMN; UMC01CTLx00000804_address0000_[B:0]082; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p1_aliasSMN; UMC01CTLx00000804_address0010_[B:0]082; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p2_aliasSMN; UMC01CTLx00000804_address0020_[B:0]082; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p3_aliasSMN; UMC01CTLx00000804_address0030_[B:0]082; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p0_aliasSMN; UMC02CTLx00000804_address0000_[B:0]082; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p1_aliasSMN; UMC02CTLx00000804_address0010_[B:0]082; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p2_aliasSMN; UMC02CTLx00000804_address0020_[B:0]082; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p3_aliasSMN; UMC02CTLx00000804_address0030_[B:0]082; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p0_aliasSMN; UMC03CTLx00000804_address0000_[D:0]082; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p1_aliasSMN; UMC03CTLx00000804_address0010_[D:0]082; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p2_aliasSMN; UMC03CTLx00000804_address0020_[D:0]082; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p3_aliasSMN; UMC03CTLx00000804_address0030_[D:0]082; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p0_aliasSMN; UMC04CTLx00000804_address0000_[B:0]082; UMC04CTL=0045_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p1_aliasSMN; UMC04CTLx00000804_address0010_[B:0]082; UMC04CTL=0045_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p2_aliasSMN; UMC04CTLx00000804_address0020_[B:0]082; UMC04CTL=0045_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p3_aliasSMN; UMC04CTLx00000804_address0030_[B:0]082; UMC04CTL=0045_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p0_aliasSMN; UMC05CTLx00000804_address0000_[B:0]082; UMC05CTL=0055_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p1_aliasSMN; UMC05CTLx00000804_address0010_[B:0]082; UMC05CTL=0055_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p2_aliasSMN; UMC05CTLx00000804_address0020_[B:0]082; UMC05CTL=0055_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p3_aliasSMN; UMC05CTLx00000804_address0030_[B:0]082; UMC05CTL=0055_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p0_aliasSMN; UMC06CTLx00000804_address0000_[D:0]082; UMC06CTL=0065_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p1_aliasSMN; UMC06CTLx00000804_address0010_[D:0]082; UMC06CTL=0065_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p2_aliasSMN; UMC06CTLx00000804_address0020_[D:0]082; UMC06CTL=0065_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p3_aliasSMN; UMC06CTLx00000804_address0030_[D:0]082; UMC06CTL=0065_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p0_aliasSMN; UMC07CTLx00000804_address0000_[B:0]082; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p1_aliasSMN; UMC07CTLx00000804_address0010_[B:0]082; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p2_aliasSMN; UMC07CTLx00000804_address0020_[B:0]082; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p3_aliasSMN; UMC07CTLx00000804_address0030_[B:0]082; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p0_aliasSMN; UMC08CTLx00000804_address0000_[B:0]082; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p1_aliasSMN; UMC08CTLx00000804_address0010_[B:0]082; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p2_aliasSMN; UMC08CTLx00000804_address0020_[B:0]082; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p3_aliasSMN; UMC08CTLx00000804_address0030_[B:0]082; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p0_aliasSMN; UMC09CTLx00000804_address0000_[D:0]082; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p1_aliasSMN; UMC09CTLx00000804_address0010_[D:0]082; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p2_aliasSMN; UMC09CTLx00000804_address0020_[D:0]082; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p3_aliasSMN; UMC09CTLx00000804_address0030_[D:0]082; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p0_aliasSMN; UMC10CTLx00000804_address0000_[B:0]082; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p1_aliasSMN; UMC10CTLx00000804_address0010_[B:0]082; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p2_aliasSMN; UMC10CTLx00000804_address0020_[B:0]082; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p3_aliasSMN; UMC10CTLx00000804_address0030_[B:0]082; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p0_aliasSMN; UMC11CTLx00000804_address0000_[B:0]082; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p1_aliasSMN; UMC11CTLx00000804_address0010_[B:0]082; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p2_aliasSMN; UMC11CTLx00000804_address0020_[B:0]082; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p3_aliasSMN; UMC11CTLx00000804_address0030_[B:0]082; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:8	Reserved.
7:4	AcRxPowerDownDynEn. Read-write. Reset: 0h. Per lane, AcTxPowerDownDynEn=1 enables power down of ANIB receivers when csrStandbyDynAC =1
3:0	AcTxPowerDownDynEn. Read-write. Reset: 0h. Per lane, AcTxPowerDownDynEn=1 enables power down of ANIB drivers when csrStandbyDynAC =1

UMC00CTLx00000804_address0000083...UMC11CTLx00000804_address0030B083 (UMC::Phy::ATxDly2nMode)

Read-write. Reset: 0000h.

Address/Command Delay, 2nMode Asserted, per pstate.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]083; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p1_aliasSMN; UMC00CTLx00000804_address0010_[D:0]083; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p2_aliasSMN; UMC00CTLx00000804_address0020_[D:0]083; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p3_aliasSMN; UMC00CTLx00000804_address0030_[D:0]083; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p0_aliasSMN; UMC01CTLx00000804_address0000_[B:0]083; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p1_aliasSMN; UMC01CTLx00000804_address0010_[B:0]083; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p2_aliasSMN; UMC01CTLx00000804_address0020_[B:0]083; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p3_aliasSMN; UMC01CTLx00000804_address0030_[B:0]083; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p0_aliasSMN; UMC02CTLx00000804_address0000_[B:0]083; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p1_aliasSMN; UMC02CTLx00000804_address0010_[B:0]083; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p2_aliasSMN; UMC02CTLx00000804_address0020_[B:0]083; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p3_aliasSMN; UMC02CTLx00000804_address0030_[B:0]083; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p0_aliasSMN; UMC03CTLx00000804_address0000_[D:0]083; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p1_aliasSMN; UMC03CTLx00000804_address0010_[D:0]083; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p2_aliasSMN; UMC03CTLx00000804_address0020_[D:0]083; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p3_aliasSMN; UMC03CTLx00000804_address0030_[D:0]083; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p0_aliasSMN; UMC04CTLx00000804_address0000_[B:0]083; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p1_aliasSMN; UMC04CTLx00000804_address0010_[B:0]083; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p2_aliasSMN; UMC04CTLx00000804_address0020_[B:0]083; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p3_aliasSMN; UMC04CTLx00000804_address0030_[B:0]083; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p0_aliasSMN; UMC05CTLx00000804_address0000_[B:0]083; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p1_aliasSMN; UMC05CTLx00000804_address0010_[B:0]083; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p2_aliasSMN; UMC05CTLx00000804_address0020_[B:0]083; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p3_aliasSMN; UMC05CTLx00000804_address0030_[B:0]083; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p0_aliasSMN; UMC06CTLx00000804_address0000_[D:0]083; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p1_aliasSMN; UMC06CTLx00000804_address0010_[D:0]083; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p2_aliasSMN; UMC06CTLx00000804_address0020_[D:0]083; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p3_aliasSMN; UMC06CTLx00000804_address0030_[D:0]083; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p0_aliasSMN; UMC07CTLx00000804_address0000_[B:0]083; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p1_aliasSMN; UMC07CTLx00000804_address0010_[B:0]083; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p2_aliasSMN; UMC07CTLx00000804_address0020_[B:0]083; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p3_aliasSMN; UMC07CTLx00000804_address0030_[B:0]083; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p0_aliasSMN; UMC08CTLx00000804_address0000_[B:0]083; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p1_aliasSMN; UMC08CTLx00000804_address0010_[B:0]083; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p2_aliasSMN; UMC08CTLx00000804_address0020_[B:0]083; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p3_aliasSMN; UMC08CTLx00000804_address0030_[B:0]083; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p0_aliasSMN; UMC09CTLx00000804_address0000_[D:0]083; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p1_aliasSMN; UMC09CTLx00000804_address0010_[D:0]083; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p2_aliasSMN; UMC09CTLx00000804_address0020_[D:0]083; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p3_aliasSMN; UMC09CTLx00000804_address0030_[D:0]083; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p0_aliasSMN; UMC10CTLx00000804_address0000_[B:0]083; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p1_aliasSMN; UMC10CTLx00000804_address0010_[B:0]083; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p2_aliasSMN; UMC10CTLx00000804_address0020_[B:0]083; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p3_aliasSMN; UMC10CTLx00000804_address0030_[B:0]083; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p0_aliasSMN; UMC11CTLx00000804_address0000_[B:0]083; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p1_aliasSMN; UMC11CTLx00000804_address0010_[B:0]083; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p2_aliasSMN; UMC11CTLx00000804_address0020_[B:0]083; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p3_aliasSMN; UMC11CTLx00000804_address0030_[B:0]083; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:8	Reserved.
7:0	ATxDly2nMode. Read-write. Reset: 00h. Description: Trained to generate timed address and command signals to the DRAMs, per ACX4, (lower 2 bits). The two lower signals generated by an ACX4 will have common timing. This value is applied as discussed in the csrAtxDlySelect Register. - ATxDly2nMode[7:6] is the coarse delay, ie one unit of delay is 1 UI . - ATxDly2nMode[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64;

UMC00CTLx00000804_address00000804...UMC11CTLx00000804_address0030B084 (UMC::Phy::ATuDly2nMode)

Read-write. Reset: 0000h.

Address/Command Delay, 2nMode Asserted, per pstate.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]084; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p1_aliasSMN; UMC00CTLx00000804_address0010_[D:0]084; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p2_aliasSMN; UMC00CTLx00000804_address0020_[D:0]084; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p3_aliasSMN; UMC00CTLx00000804_address0030_[D:0]084; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p0_aliasSMN; UMC01CTLx00000804_address0000_[B:0]084; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p1_aliasSMN; UMC01CTLx00000804_address0010_[B:0]084; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p2_aliasSMN; UMC01CTLx00000804_address0020_[B:0]084; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p3_aliasSMN; UMC01CTLx00000804_address0030_[B:0]084; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p0_aliasSMN; UMC02CTLx00000804_address0000_[B:0]084; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p1_aliasSMN; UMC02CTLx00000804_address0010_[B:0]084; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p2_aliasSMN; UMC02CTLx00000804_address0020_[B:0]084; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p3_aliasSMN; UMC02CTLx00000804_address0030_[B:0]084; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p0_aliasSMN; UMC03CTLx00000804_address0000_[D:0]084; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p1_aliasSMN; UMC03CTLx00000804_address0010_[D:0]084; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p2_aliasSMN; UMC03CTLx00000804_address0020_[D:0]084; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p3_aliasSMN; UMC03CTLx00000804_address0030_[D:0]084; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p0_aliasSMN; UMC04CTLx00000804_address0000_[B:0]084; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p1_aliasSMN; UMC04CTLx00000804_address0010_[B:0]084; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p2_aliasSMN; UMC04CTLx00000804_address0020_[B:0]084; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p3_aliasSMN; UMC04CTLx00000804_address0030_[B:0]084; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p0_aliasSMN; UMC05CTLx00000804_address0000_[B:0]084; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p1_aliasSMN; UMC05CTLx00000804_address0010_[B:0]084; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p2_aliasSMN; UMC05CTLx00000804_address0020_[B:0]084; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p3_aliasSMN; UMC05CTLx00000804_address0030_[B:0]084; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p0_aliasSMN; UMC06CTLx00000804_address0000_[D:0]084; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p1_aliasSMN; UMC06CTLx00000804_address0010_[D:0]084; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p2_aliasSMN; UMC06CTLx00000804_address0020_[D:0]084; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p3_aliasSMN; UMC06CTLx00000804_address0030_[D:0]084; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p0_aliasSMN; UMC07CTLx00000804_address0000_[B:0]084; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p1_aliasSMN; UMC07CTLx00000804_address0010_[B:0]084; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p2_aliasSMN; UMC07CTLx00000804_address0020_[B:0]084; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p3_aliasSMN; UMC07CTLx00000804_address0030_[B:0]084; UMC07CTL=0075_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p0_aliasSMN; UMC08CTLx00000804_address0000_[B:0]084; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p1_aliasSMN; UMC08CTLx00000804_address0010_[B:0]084; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p2_aliasSMN; UMC08CTLx00000804_address0020_[B:0]084; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p3_aliasSMN; UMC08CTLx00000804_address0030_[B:0]084; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p0_aliasSMN; UMC09CTLx00000804_address0000_[D:0]084; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p1_aliasSMN; UMC09CTLx00000804_address0010_[D:0]084; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p2_aliasSMN; UMC09CTLx00000804_address0020_[D:0]084; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p3_aliasSMN; UMC09CTLx00000804_address0030_[D:0]084; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p0_aliasSMN; UMC10CTLx00000804_address0000_[B:0]084; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p1_aliasSMN; UMC10CTLx00000804_address0010_[B:0]084; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p2_aliasSMN; UMC10CTLx00000804_address0020_[B:0]084; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p3_aliasSMN; UMC10CTLx00000804_address0030_[B:0]084; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p0_aliasSMN; UMC11CTLx00000804_address0000_[B:0]084; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p1_aliasSMN; UMC11CTLx00000804_address0010_[B:0]084; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p2_aliasSMN; UMC11CTLx00000804_address0020_[B:0]084; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p3_aliasSMN; UMC11CTLx00000804_address0030_[B:0]084; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:8	Reserved.
7:0	ATuDly2nMode. Read-write. Reset: 00h. Description: Trained to generate timed address and command signals to the DRAMs, per ACX4, (upper 2 bits). The two upper signals generated by an ACX4 will have common timing. This value is applied as discussed in the csrAtxDlySelect Register. - ATuDly2nMode[7:6] is the coarse delay, ie one unit of delay is 1 UI. - ATuDly2nMode[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64;

UMC00CTLx00000804_address00000805...UMC11CTLx00000804_address0030B085 (UMC::Phy::ATxDlySelect)

Read-write. Reset: 0000h.

Address/Command Delay Select

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]085; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p1_aliasSMN; UMC00CTLx00000804_address0010_[D:0]085; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p2_aliasSMN; UMC00CTLx00000804_address0020_[D:0]085; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_p3_aliasSMN; UMC00CTLx00000804_address0030_[D:0]085; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p0_aliasSMN; UMC01CTLx00000804_address0000_[B:0]085; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p1_aliasSMN; UMC01CTLx00000804_address0010_[B:0]085; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p2_aliasSMN; UMC01CTLx00000804_address0020_[B:0]085; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_p3_aliasSMN; UMC01CTLx00000804_address0030_[B:0]085; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p0_aliasSMN; UMC02CTLx00000804_address0000_[B:0]085; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p1_aliasSMN; UMC02CTLx00000804_address0010_[B:0]085; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p2_aliasSMN; UMC02CTLx00000804_address0020_[B:0]085; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_p3_aliasSMN; UMC02CTLx00000804_address0030_[B:0]085; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p0_aliasSMN; UMC03CTLx00000804_address0000_[D:0]085; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p1_aliasSMN; UMC03CTLx00000804_address0010_[D:0]085; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p2_aliasSMN; UMC03CTLx00000804_address0020_[D:0]085; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_p3_aliasSMN; UMC03CTLx00000804_address0030_[D:0]085; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p0_aliasSMN; UMC04CTLx00000804_address0000_[B:0]085; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p1_aliasSMN; UMC04CTLx00000804_address0010_[B:0]085; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p2_aliasSMN; UMC04CTLx00000804_address0020_[B:0]085; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_p3_aliasSMN; UMC04CTLx00000804_address0030_[B:0]085; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p0_aliasSMN; UMC05CTLx00000804_address0000_[B:0]085; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p1_aliasSMN; UMC05CTLx00000804_address0010_[B:0]085; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p2_aliasSMN; UMC05CTLx00000804_address0020_[B:0]085; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_p3_aliasSMN; UMC05CTLx00000804_address0030_[B:0]085; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p0_aliasSMN; UMC06CTLx00000804_address0000_[D:0]085; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p1_aliasSMN; UMC06CTLx00000804_address0010_[D:0]085; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p2_aliasSMN; UMC06CTLx00000804_address0020_[D:0]085; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_p3_aliasSMN; UMC06CTLx00000804_address0030_[D:0]085; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p0_aliasSMN; UMC07CTLx00000804_address0000_[B:0]085; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p1_aliasSMN; UMC07CTLx00000804_address0010_[B:0]085; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p2_aliasSMN; UMC07CTLx00000804_address0020_[B:0]085; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_p3_aliasSMN; UMC07CTLx00000804_address0030_[B:0]085; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p0_aliasSMN; UMC08CTLx00000804_address0000_[B:0]085; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p1_aliasSMN; UMC08CTLx00000804_address0010_[B:0]085; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p2_aliasSMN; UMC08CTLx00000804_address0020_[B:0]085; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_p3_aliasSMN; UMC08CTLx00000804_address0030_[B:0]085; UMC08CTL=0085_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p0_aliasSMN; UMC09CTLx00000804_address0000_[D:0]085; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p1_aliasSMN; UMC09CTLx00000804_address0010_[D:0]085; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p2_aliasSMN; UMC09CTLx00000804_address0020_[D:0]085; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_p3_aliasSMN; UMC09CTLx00000804_address0030_[D:0]085; UMC09CTL=0095_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p0_aliasSMN; UMC10CTLx00000804_address0000_[B:0]085; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p1_aliasSMN; UMC10CTLx00000804_address0010_[B:0]085; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p2_aliasSMN; UMC10CTLx00000804_address0020_[B:0]085; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_p3_aliasSMN; UMC10CTLx00000804_address0030_[B:0]085; UMC10CTL=00A5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p0_aliasSMN; UMC11CTLx00000804_address0000_[B:0]085; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p1_aliasSMN; UMC11CTLx00000804_address0010_[B:0]085; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p2_aliasSMN; UMC11CTLx00000804_address0020_[B:0]085; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_p3_aliasSMN; UMC11CTLx00000804_address0030_[B:0]085; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:2	Reserved.
1:0	ATxDlySelect. Read-write. Reset: 0h. Description: Selects between csrAt[x u]Dly and csrAt[x u]Dly2nMode as determined by dfi_2n_mode. - 0x: Select CsrAt[x u]Dly when !dfi_2n_mode. Otherwise select CsrAt[x u]Dly2nMode - 10: Select CsrAt[x u]Dly - 11: Select CsrAt[x u]Dly2nMode

UMC00CTLx00000804_address0000086...UMC11CTLx00000804_address0000B086 (UMC::Phy::TxAcPhase1UI)

Read-only, Volatile.

Tx AC DLLPhase1UI Code

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]086; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]086; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]086; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]086; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]086; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]086; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]086; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]086; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]086; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]086; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]086; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]086; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:10	Reserved.
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9:0	TxAcPhase1UI . Read-only, Volatile. The raw AC DLL Phase1UI code, used for debug only.
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UMC00CTLx00000804_address0000087...UMC11CTLx00000804_address0000B087 (UMC::Phy::TuAcPhase1UI)

Read-only, Volatile.

Tu AC DLLPhase1UI Code

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]087; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]087; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]087; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]087; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]087; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]087; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]087; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]087; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]087; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]087; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]087; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]087; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:10	Reserved.
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9:0	TuAcPhase1UI . Read-only, Volatile. The raw AC DLL Phase1UI code, used for debug only.
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UMC00CTLx00000804_address00000E0...UMC11CTLx00000804_address0000B0E0 (UMC::Phy::ATestMode2)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]0E0; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]0E0; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]0E0; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]0E0; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]0E0; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]0E0; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]0E0; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]0E0; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]0E0; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]0E0; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]0E0; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]0E0; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15 Reserved.

14:0 RFU. Read-write. Reset: 0000h.

UMC00CTLx00000804_address00000E1...UMC11CTLx00000804_address0000B0E1 [ATestMode3 control] (UMC::Phy::ATestMode3)

Read-write. Reset: 0000h.

Controls for Address Command Loopback. This CSR is written by ATE firmware.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]0E1; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]0E1; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]0E1; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]0E1; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]0E1; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]0E1; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]0E1; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]0E1; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]0E1; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]0E1; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]0E1; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]0E1; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:4 Reserved.

3:0 **PrbsDdrModeToggle**. Read-write. Reset: 0h. Sets the prbs mode to DDR. Each bit controls the corresponding lane (0 -> 0, etc) in an ANIB. When 0, SDR mode is selected.

UMC00CTLx00000804_address000000E2...UMC11CTLx00000804_address0000B0E2 (UMC::Phy::APrbsOverride)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]0E2; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]0E2; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]0E2; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]0E2; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]0E2; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]0E2; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]0E2; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]0E2; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]0E2; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]0E2; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]0E2; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]0E2; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	APrbsOverride. Read-write. Reset: 0h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address00000156...UMC11CTLx00000804_address0000B156 (UMC::Phy::ATestPrbsErrCntln1)

Read-only, Volatile.

Address Loopback Test Result register

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]156; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]156; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]156; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]156; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]156; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]156; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]156; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]156; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]156; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]156; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]156; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]156; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	ATestPrbsErrCntln1. Read-only, Volatile.
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Description: Overall error indicator for each prbs bump checker.

Non-zero value indicating the number of errors seen by self-seeding prbs checker.

UMC00CTLx00000804_address00000190...UMC11CTLx00000804_address0000B190 (UMC::Phy::DlyTestCntRingOscAcldl0)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]190; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]190; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]190; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]190; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]190; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]190; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]190; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]190; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]190; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]190; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]190; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]190; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscAcldl0. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address00000191...UMC11CTLx00000804_address0000B191 (UMC::Phy::DlyTestCntRingOscAcldl1)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]191; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]191; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]191; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]191; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]191; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]191; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]191; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]191; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]191; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]191; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]191; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]191; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	DlyTestCntRingOscAcldl1. Read-only, Volatile. Reserved for PHY training firmware use.

UMC00CTLx00000804_address00000192...UMC11CTLx00000804_address0000B192 (UMC::Phy::DlyTestCntRingOscAcldl2)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]192; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]192; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]192; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]192; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]192; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]192; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]192; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]192; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]192; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]192; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]192; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]192; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	DlyTestCntRingOscAcldl2. Read-only, Volatile. Reserved for PHY training firmware use.

UMC00CTLx00000804_address00000256...UMC11CTLx00000804_address0000B256 (UMC::Phy::ATestPrbsErrCntln2)

Read-only, Volatile.

Address Loopback Test Result register

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]256; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]256; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]256; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]256; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]256; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]256; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]256; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]256; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]256; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]256; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]256; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]256; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	ATestPrbsErrCntln2. Read-only, Volatile.
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Description: Overall error indicator for each prbs bump checker.

Non-zero value indicating the number of errors seen by self-seeding prbs checker.

UMC00CTLx00000804_address00000356...UMC11CTLx00000804_address0000B356 (UMC::Phy::ATestPrbsErrCntln3)

Read-only, Volatile.

Address Loopback Test Result register

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY0_aliasSMN; UMC00CTLx00000804_address0000_[D:0]356; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY1_aliasSMN; UMC01CTLx00000804_address0000_[B:0]356; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY2_aliasSMN; UMC02CTLx00000804_address0000_[B:0]356; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY3_aliasSMN; UMC03CTLx00000804_address0000_[D:0]356; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY4_aliasSMN; UMC04CTLx00000804_address0000_[B:0]356; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY5_aliasSMN; UMC05CTLx00000804_address0000_[B:0]356; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY6_aliasSMN; UMC06CTLx00000804_address0000_[D:0]356; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY7_aliasSMN; UMC07CTLx00000804_address0000_[B:0]356; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY8_aliasSMN; UMC08CTLx00000804_address0000_[B:0]356; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[13:0]_chiplettype0_instUMCWPHY9_aliasSMN; UMC09CTLx00000804_address0000_[D:0]356; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY10_aliasSMN; UMC10CTLx00000804_address0000_[B:0]356; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet[11:0]_chiplettype0_instUMCWPHY11_aliasSMN; UMC11CTLx00000804_address0000_[B:0]356; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	ATestPrbsErrCntln3. Read-only, Volatile.
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Description: Overall error indicator for each prbs bump checker.

Non-zero value indicating the number of errors seen by self-seeding prbs checker.

UMC00CTLx00000804_address00010000...UMC11CTLx00000804_address00019000 [DBYTE Module Disable] (UMC::Phy::DbyteMiscMode)

Reset: 0000h.

Controls the enable/disable state of the DBYTE module.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010000; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011000; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012000; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013000; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014000; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015000; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016000; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017000; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018000; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019000; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2	DBYTEDisable. Read-write. Reset: 0. Description: Controls whether this DBYTE module is disabled. If this DBYTE module is not enabled, it receives no clocks and remains in reset. - 0 - Enable this DBYTE module - 1 - Disable this DBYTE module This field should only be changed PHY initialization step C.
1:0	Reserved.

[illegible]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00311003; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312003; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313003; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314003; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315003; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316003; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317003; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318003; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319003; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:1	Reserved.
0	UseDqsEnReplica . Read-write. Reset: 0. Reserved for PHY training firmware use.

UMC00CTLx00000804_address00010004...UMC11CTLx00000804_address00019004 (UMC::Phy::TrainingParam1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010004; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011004; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012004; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013004; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014004; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015004; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016004; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017004; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018004; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019004; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:2	Reserved.
1:0	RFU . Read-write. Reset: 0h.

UMC00CTLx00000804_address00010005...UMC11CTLx00000804_address00019005 [Controls the static PowerDown of DBYTE drivers] (UMC::Phy::DbPowerDownStatic)

Read-write. Reset: 0000h.

Forces power down

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010005; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011005; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012005; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013005; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014005; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015005; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016005; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017005; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018005; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019005; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:10	Reserved.
9:0	DbTxPowerDownStatic. Read-write. Reset: 000h. Description: Per-lane power down of DBYTE drivers. DbTxPowerDownStatic[9:0] (BP_DAT9,10),(BP_DAT8,11),BP_DAT7..BP_DAT0

[illegible]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00311006; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312006; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313006; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314006; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315006; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316006; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317006; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318006; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319006; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:2	Reserved.
1	DbRxPowerDownDynEn. Read-write. Reset: 0. DbRxPowerDownDynEn=1 enables power down of DBYTE receivers when PowerDownDynDB=1
0	DbTxPowerDownDynEn. Read-write. Reset: 0. DbTxPowerDownDynEn=1 enables power down of DBYTE drivers when PowerDownDynDB=1

UMC00CTLx00000804_address00010008...UMC11CTLx00000804_address00019008 (UMC::Phy::DBYTEReserved)

Read-write. Reset: 0000h.	
Spares for the DBYTE Hard Macro	
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010008; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011008; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012008; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013008; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014008; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015008; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016008; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017008; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018008; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019008; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	DBYTEReserved. Read-write. Reset: 000h. Spares for the DBYTE Hard Macro

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312020; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313020; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314020; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315020; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316020; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317020; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318020; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319020; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:5	Reserved.
4:0	DFIMRL. Read-write. Reset: 06h. Description: This Max Read Latency CSR is to be trained to ensure the rx-data fifo is not read until after all DBYTEs have their read data valid. This is the value, in units of two mem clocks, between dfi_rddata_en and dfi_rddata_valid A unit change in the LSB is a change in MRL of two mem clocks. The maximum value allowed is 28, 0x1C.

[illegible]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00311021; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312021; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313021; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314021; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315021; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316021; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317021; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318021; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319021; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:1	Reserved.
0	DFIMRLmCK . Read-write. Reset: 0. Reserved for future use.

UMC00CTLx00000804_address00010024...UMC11CTLx00000804_address00019024 (UMC::Phy::AsyncDbyteMode)

Read-write. Reset: 0000h.	
Enables Async Flyover for all lanes of the DBYTE	
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:9	Reserved.
8:0	AsyncDbyteMode . Read-write. Reset: 000h.
	Description: Used for giving firmware direct access to data and DBI pins - Bit 7:0 controls Dq lanes for both RX and TX - Bit 8 controls Dqs lane for both RX and TX

UMC00CTLx00000804_address00010026...UMC11CTLx00000804_address00019026 (UMC::Phy::AsyncDbyteTxEn)

Read-write. Reset: 0000h.

TxEnable bits for Async Flyover

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010026; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011026; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012026; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013026; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014026; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015026; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016026; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017026; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018026; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019026; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AsyncDbyteTxEn. Read-write. Reset: 000h.
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Description: Used for giving firmware direct access to data and DBI pins.

Each bit controls one I/O:

Bit[7:0] = DQ[7:0]

- Bit[8] = DBI

- Bit[9] = DQS_H

- Bit[10] = DQS_L

- Bit[11] = DQSU_L

UMC00CTLx00000804_address00010028...UMC11CTLx00000804_address00019028 (UMC::Phy::AsyncDbyteTxData)

Read-write. Reset: 0000h.

TxData bits for Async Flyover

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AsyncDbyteTxData. Read-write. Reset: 000h.
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Description: Used for giving firmware direct access to data and DBI pins.

Each bit controls one I/O:

- Bit[7:0] = DQ[7:0]
- Bit[8] = DBI
- Bit[9] = DQS_H
- Bit[10] = DQS_L
- Bit[11] = DQSU_L

UMC00CTLx00000804_address0001002A...UMC11CTLx00000804_address0001902A (UMC::Phy::AsyncDbyteRxData)

Read-only, Volatile.

RxData bits for Async Flyover

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001002A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001102A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001202A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001302A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001402A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001502A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001602A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001702A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001802A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001902A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AsyncDbyteRxData. Read-only, Volatile.
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Description: Used for giving firmware direct access to data and DBI pins.

Each bit controls one I/O:

- Bit[7:0] = DQ[7:0]

- Bit[8] = DBI

- Bit[9] = DQS_H

- Bit[10] = DQS_L

- Bit[11] = DQSU_L

[illegible]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031102C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031202C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031302C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031402C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031502C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031602C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031702C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031802C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031902C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:1	Reserved.
0	RFU_TxStandbyControl. Read-write. Reset: 0.

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address00317530; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address00317630; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address00317730; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address00317830; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address00318030; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address00318130; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address00318230; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address00318330; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address00318430; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address00318530; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address00318630; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address00318730; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address00318830; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address00319030; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address00319130; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address00319230; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address00319330; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address00319430; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address00319530; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address00319630; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address00319730; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address00319830; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:7	Reserved.
6:0	VrefDAC1. Read-write. Reset: 3Fh. Description: VrefDAC1 control for DQ Receiver. Used only when DFE is enabled for DDR5. DAC control for rxdq cell internal VREF, trained by Firmware. The VREF generators have different ranges, depending on the Mission Mode settings for [DqDqsRcvCntrl:ExtVrefRange] - [0] : VREF = VDDQ*(0.46 + VrefDAC1[6:0]*0.00379) - [1] : VREF = VDDQ*(0.302 + VrefDAC1[6:0]*0.0049) - [non-enumerated encodings are reserved] - Register Block Offset Address 0x030 is the VrefDAC1 for lane0. - Register Block Offset Address 0x130 is the VrefDAC1 for lane1. - Register Block Offset Address 0x230 is the VrefDAC1 for lane2. - Register Block Offset Address 0x330 is the VrefDAC1 for lane3. - Register Block Offset Address 0x430 is the VrefDAC1 for lane4. - Register Block Offset Address 0x530 is the VrefDAC1 for lane5. - Register Block Offset Address 0x630 is the VrefDAC1 for lane6. - Register Block Offset Address 0x730 is the VrefDAC1 for lane7. - Register Block Offset Address 0x830 is the VrefDAC1 for lane[DBI].

Read-write. Reset: 003Fh.

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_r0_aliasSMN; UMC[11:0]CTLx00000804_address00010040; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_r1_aliasSMN; UMC[11:0]CTLx00000804_address00010140; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_r2_aliasSMN; UMC[11:0]CTLx00000804_address00010240; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_r3_aliasSMN; UMC[11:00]CTLx00000804_address00010340; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_r4_aliasSMN; UMC[11:0]CTLx00000804_address00010440; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_r5_aliasSMN; UMC[11:0]CTLx00000804_address00010540; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_r6_aliasSMN; UMC[11:0]CTLx00000804_address00010640; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_r7_aliasSMN; UMC[11:0]CTLx00000804_address00010740; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_r8_aliasSMN; UMC[11:0]CTLx00000804_address00010840; UMC[11:0]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::AnbCmd[Address]
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_r0_aliasSMN; UMC[11:00]CTLx00000804_address00011040; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::AnbCmd[Address]
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_r1_aliasSMN; UMC[11:0]CTLx00000804_address00011140; UMC[11:0]CTL=00[B:0]5_1000h;
DataPortWrite=[UMC::AnbCmd[Address]
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_r2_aliasSMN; UMC[11:00]CTLx00000804_address00011240; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_r3_aliasSMN; UMC[11:00]CTLx00000804_address00011340; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_r4_aliasSMN; UMC[11:00]CTLx00000804_address00011440; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_r5_aliasSMN; UMC[11:00]CTLx00000804_address00011540; UMC[11:00]CTL=00[B:0]5_1000h;
```

```

_dataPortWrite=UMC...ArbCmd[Address]
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_r6_aliasSMN; UMC[11:00]CTLx00000804_address00011640; UMC[11:00]CTL=00[B:0]5_1000h;
_dataPortWrite=UMC...ArbCmd[Address]

```

```

_dataPortWrite=UMC::AnbCmd[Address]
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_r7_aliasSMN; UMC[11:00]CTLx00000804_address00011740; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::AnbCmd[Address]

```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_r8_aliasSMN; UMC[11:00]CTLx00000804_address00011840; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_r0_aliasSMN; UMC[11:00]CTLx00000804_address00012040; UMC[11:00]CTL=00[B:0]5_1000h;
```

```

DataPortWrite=UMC...ApbCmId[Address]
_umc0_chiplet2_chiplettype1_inst1UMCWPHY[11:0]_p0_r1_aliasSMN; UMC[11:00]CTLx00000804_address00012140; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC...ApbCmId[Address]

```

```
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_r2_aliasSMN; UMC[11:00]CTLx00000804_address00012244; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
DataPortWrite=UMC::ApbCmd[Address]
```

```

DataPortWrite=UMC...ApbCmId[Address]
...umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_r4_aliasSMN; UMC[11:00]CTLx00000804_address00012440; UMC[11:00]CTL=00[B:0]5_1000h;
...B:0]5_1000h; UMC[11:00]CTL=00[B:0]5_1000h;

```

```
DataPortWrite=UMC::ApbCmd[Address]
```

```
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_r5_aliasSMN; UMC[11:00]CTLx00000804_address00012540; UMC[11:00]CTL=00[B:0]5_1000h;
```

```

DataPortWrite=UMC::ApbCmd[Address]
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_r6_aliasSMN; UMC[11:00]CTLx00000804_address00012640; UMC[11:00]CTL=00[B:0]5_1000h;
_p0_r6_aliasSMN; UMC[11:00]CTL=00[B:0]5_1000h;

```

```
_umc0_chiplet2_chiplettetype1_inst[UMCWPHY[11:0]]_p0_r7_aliasSMN; UMC[11:00]CTLx00000804_address00012740; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_r8_aliasSMN; UMC[11:00]CTLx00000804_address00012840; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_r0_aliasSMN; UMC[11:00]CTLx00000804_address00013040; UMC[11:00]CTL=00[B:0]5_1000h;
```

DataPortWrite=UMC::ApbCmd[Address]
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_r1_aliasSMN; UMC[11:00]CTLx00000804_address00013140; UMC[11:00]CTL=00[B:0]5_1000h;

DataPortWrite=UMC::ApbCmd[Address]
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_r2_aliasSMN; UMC[11:00]CTLx00000804_address00013240; UMC[11:00]CTL=00[B:0]5_1000h;

```
DataPortWrite=UMC::ApbCmd[Address]
```

```
DataPortWrite=UMC::ApbCmd[Address]
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address00317540; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address00317640; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address00317740; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address00317840; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address00318040; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address00318140; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address00318240; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address00318340; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address00318440; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address00318540; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address00318640; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address00318740; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address00318840; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address00319040; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address00319140; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address00319240; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address00319340; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address00319440; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address00319540; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address00319640; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address00319740; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address00319840; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:7	Reserved.
6:0	VrefDAC0. Read-write. Reset: 3Fh.

Description: VrefDAC0 control for DQ Receiver. Can be used either when DFE not enabled or when DFE is enabled.

DAC control for rxdq cell internal VREF, trained by Firmware.

The VREF generators have different ranges, depending on the Mission Mode settings for [DqDqsRcvCntrl:ExtVrefRange]

- [0] : $VREF = VDDQ * (0.46 + VrefDAC0[6:0] * 0.00379)$
- [1] : $VREF = VDDQ * (0.302 + VrefDAC0[6:0] * 0.0049)$
- [non-enumerated encodings are reserved]
- Register Block Offset Address 0x040 is the VrefDAC0 for lane0.
- Register Block Offset Address 0x140 is the VrefDAC0 for lane1.
- Register Block Offset Address 0x240 is the VrefDAC0 for lane2.
- Register Block Offset Address 0x340 is the VrefDAC0 for lane3.
- Register Block Offset Address 0x440 is the VrefDAC0 for lane4.
- Register Block Offset Address 0x540 is the VrefDAC0 for lane5.
- Register Block Offset Address 0x640 is the VrefDAC0 for lane6.
- Register Block Offset Address 0x740 is the VrefDAC0 for lane7.
- Register Block Offset Address 0x840 is the VrefDAC0 for lane[DBI].

UMC00CTLx00000804_address00010041...UMC11CTLx00000804_address00319141 [Data TX impedance controls] (UMC::Phy::TxImpedanceCtrl0)

Read-write. Reset: 079Eh.

This field is programmed as follows:

- Tx DQ driver impedance when equalization is enabled per nibble.
- Register Block Offset Address 0x041 controls the lower DQ nibble.
- Register Block Offset Address 0x141 controls the upper DQ nibble.

_umc0_b0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00010041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00010141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00011041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00011141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00012041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00012141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00013041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00013141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00014041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00014141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00015041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00015141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00016041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00016141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00017041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00017141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00018041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00018141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00019041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00019141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00110041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00110141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00111041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00111141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00112041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00112141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00113041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00113141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00114041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

[illegible]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00311141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319041; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319141; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:6	TxStrenEqLoPd. Read-write. Reset: 1Eh. Description: 6-bit bus used to select the target pull-down output impedance. Connects to the DrvStren pins of the driver as follows: [Csr_DrvStren120N[3:0],Csr_DrvStren240N,Csr_DrvStren480N[0:0]] = [csrDrvStrenDqN] See table of csr DrvStrenDqP for list of possible impedances.
5:0	TxStrenEqHiPu. Read-write. Reset: 1Eh. Description: This 6-bit bus used to select the target pull-down output impedance. Connects to the DrvStren pins of the driver as follows: [Csr_DrvStren120P[3:0], Csr_DrvStren240P,Csr_DrvStren480P[0:0]] = [csrDrvStrenDqP] - 000000 - HiZ. - X0111X - 48.0 - X1110X - 40.0 - X1111X - 34.3

UMC00CTLx00000804_address00010042...UMC11CTLx00000804_address00019042 [Dq/Dqs receiver control] (UMC::Phy::DqDqsRcvCntrl3)

Read-write. Reset: 0000h.

Connected to DBYTE for VIL/VIH testing.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010042; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011042; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012042; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013042; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014042; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015042; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016042; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017042; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018042; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019042; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	VilVihTestAmp. Read-write. Reset: 0h. Description: This field is programmed as follows: - [0]: Reserved/unavailable - should not be set. - [1]: Set to 1 to enable VIL/VIH testing of the DFE0 or DFE1 Amp. - [2]: Set to 1 to enable VIL/VIH testing of the DFE2 or DFE3 Amp. Note: VilVihTestAmp[2:1] should not be asserted simultaneously.

UMC00CTLx00000804_address00010043...UMC11CTLx00000804_address00319143 [Dq/Dqs receiver control] (UMC::Phy::DqDqsRcvCtrl)

Read-write. Reset: 1500h.

Connected to the rxdqs and rxdq Mode selects and trim controls, per nibble.

- Register Block Offset Address 0x043 controls the lower DQ nibble

- Register Block Offset Address 0x143 controls the upper DQ nibble

_umc0_b0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00010043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00010143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00011043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00011143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00012043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00012143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00013043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00013143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00014043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00014143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00015043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00015143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00016043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00016143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00017043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00017143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00018043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00018143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00019043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00019143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00110043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00110143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00111043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00111143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00112043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00112143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00113043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00113143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00114043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00114143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

[illegible]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319043; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:8	GainCurrAdj. Read-write. Reset: 15h. Description: Adjust gain current of RX amplifier stage. It is recommended to use default values for this CSR.
7	RxEnLEDly. Read-write. Reset: 0. Description: This field is programmed as follows: - 1 delays the leading edge of RxEn by 0.5UI to accommodate shortened DQS preamble; reduces drift tolerance. - 0 uses standard RxEn
6:5	DqDqsReserved65. Read-write. Reset: 0h. Reserved for future usage.
4:2	DfeCtrl. Read-write. Reset: 0h. This field is programmed as follows:- 000 - no DFE fxn- 001 - DFE 1 tap- 010 - DFE train channel 0- 011 - DFE train channel 1- 100 - Reserved- 101 - DFE 2 tap- 110 - 3 tap DFE- 111 - 4 tap DFEThese settings are determined by PHY Training FW and should not be overridden.
1	ExtVrefRange. Read-write. Reset: 0. Description: Extends the range available in the local per-bit VREF generator. - 0 : VREF = VDDQ*(0.46 + VrefDAC0[6:0]*0.00379) - 1 : VREF = VDDQ*(0.302 + VrefDAC0[6:0]*0.0049)
0	SelAnalogVref. Read-write. Reset: 0. Description: Setting this signal high will force the local per-bit VREF generator to pass the global VREFA to the samplers.

UMC00CTLx00000804_address00010046...UMC11CTLx00000804_address00019046 [TX base transistor leg enables] (UMC::Phy::TxCtrl0)

Read-write. Reset: 0003h.

Tx driver base transistor leg enables.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010046; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011046; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012046; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013046; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014046; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015046; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016046; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017046; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018046; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019046; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15:2 Reserved.

1 **DbTxCalBaseP.** Read-write. Reset: 1. Enables the base transistor leg of each output driver section.0 **DbTxCalBaseN.** Read-write. Reset: 1. Enables the base transistor leg of each output driver section.

UMC00CTLx00000804_address00010048...UMC11CTLx00000804_address00319048 [Tx DQ driver equalization mode controls. FFE, de-emphasis] (UMC::Phy::TxEqualizationMode)

Read-write. Reset: 0000h.

Tx DQ driver equalization is a technique for improving the write-data eye at the DRAMs.

The driver can drive with full strength (configured with DrvStrenFSHiDqP and DrvStrenFSLoDqN0 or with weak strength (configured with DrvStrenEQHiDqP and DrvStrenEQLoDqN) depending on the DQ value sequence.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00010048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00011048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00012048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00013048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00014048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00015048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00016048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00017048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00018048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00019048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00110048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00111048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00112048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00113048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00114048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00115048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00116048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00117048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00118048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00119048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00210048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00211048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00212048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00213048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00214048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00215048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00216048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00217048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00218048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00219048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00310048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00311048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319048; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:2	Reserved.
1:0	TxEqMode. Read-write. Reset: 0h. Description: This field is programmed as follows: - TxEqMode = 00 Off; drive with full strength only - TxEqMode = 01 Mode0 drive with full strength.or weak strength depending on present logical value and the previous logical value. - TxEqMode = 10 Mode1 drive with full strength.or weak strength depending on present logical value and the next logical value.Mode1 is a lower power mode. - TxEqMode = 11 reserved.

UMC00CTLx00000804_address00010049...UMC11CTLx00000804_address00319149 [TX impedance controls] (UMC::Phy::TxImpedanceCtrl1)

Read-write. Reset: 079Eh.

Tx impedance of DQ driver cells when equalization is disabled.

- Register Block Offset Address 0x049 controls the lower DQ nibble.

- Register Block Offset Address 0x149 controls the upper DQ nibble.

_umc0_b0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00010049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00010149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00011049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00011149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00012049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00012149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00013049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00013149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00014049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00014149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00015049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00015149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00016049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00016149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00017049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00017149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00018049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00018149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00019049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00019149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00110049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00110149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00111049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00111149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00112049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00112149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00113049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00113149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00114049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00114149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

[illegible]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319049; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319149; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:6	TxStrenPd. Read-write. Reset: 1Eh. Description: 6-bit bus used to select the target pull-up output impedance. Connects to the DrvStren pins of the driver as follows: [Csr_DrvStrenFS120N[3:0], Csr_DrvStrenFS240N, Csr_DrvStrenFS480N[0:0]] = [csrDrvStrenDqN] See table of csr TxStrenPu for list of possible impedances.
5:0	TxStrenPu. Read-write. Reset: 1Eh. Description: The 6-bit bus used to select the target pull-up output impedance. Connects to the DrvStren pins of the driver as follows: [Csr_DrvStrenFS120P[3:0], Csr_DrvStrenFS240P, Csr_DrvStrenFS480P[0]] = [csrFSDrvStrenDq] - 000000 - HiZ. - X0001X - 240.0 - X0010X - 120.0 - X0011X - 80.0 - X0110X - 60.0 - X0111X - 48.0 - X1110X - 40.0 - X1111X - 34.3

UMC00CTLx00000804_address0001004A...UMC11CTLx00000804_address0001904A [Dq/Dqs receiver control] (UMC::Phy::DqDqsRcvCntrl1)

Read-write. Reset: 0000h.

Misc control of rxdq cell and rxdqs cell

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001004A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001104A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001204A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001304A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001404A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001504A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001604A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001704A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001804A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001904A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	Reserved.
11	EnLPReqPDR . Read-write. Reset: 0. Reserved for future use
10	RxPadStandbyEn . Read-write. Reset: 0. Enables the rxdq/rxdqs StandBy power savings.
9	PowerDownRcvrDqs . Read-write. Reset: 0. Description: Active high signal which powers down the receiver. After this pin is de-asserted the receiver cannot be used for a minimum of 100 ns. This control powers down both the upper and lower DQS receivers. If Register X4TG[3:0]=0000, then the upper DQS receiver is powered down.
8:0	PowerDownRcvr . Read-write. Reset: 000h. Description: Active high signal which powers down the receiver. - 7:0 -> dq[7:0] - 8 -> dbi After this pin is de-asserted the receiver cannot be used for a minimum of 100 ns.

UMC00CTLx00000804_address0001004B...UMC11CTLx00000804_address0031914B [TX equalization impedance controls] (UMC::Phy::TxImpedanceCtrl2)

Read-write. Reset: 0861h.

Tx DQ driver opposition impedance when equalization is enabled.

- Register Block Offset Address 0x04b controls the lower DQ nibble.

- Register Block Offset Address 0x14b controls the upper DQ nibble.

_umc0_b0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001004B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001014B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001104B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001114B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001204B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001214B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001304B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001314B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001404B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001414B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001504B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001514B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001604B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001614B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001704B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001714B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001804B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001814B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001904B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001914B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011004B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011014B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011104B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011114B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011204B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011214B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011304B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011314B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011404B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011414B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

[illegible]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031204B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031214B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031304B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031314B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031404B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031414B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031504B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031514B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031604B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031614B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031704B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031714B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031804B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031814B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031904B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031914B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:6	TxStrenEqHiPd. Read-write. Reset: 21h. Description: The 6-bit bus used to select the target pull-down output impedance used in equalization. Connects to the DrvStren pins of the driver as follows: [Csr_DrvStrenEQLo120N[3:0], Csr_DrvStrenEQLo240N, Csr_DrvStrenEQLo480N] = [csrDrvStrenEQLoDq]
5:0	TxStrenEqLoPu. Read-write. Reset: 21h. Description: The 6-bit bus used to select the target pull-up output impedance used in equalization. Connects to the DrvStren pins of the driver as follows: [Csr_DrvStrenEQHi120P[3:0], Csr_DrvStrenEQHi240P, Csr_DrvStrenEQHi480P] = [csrDrvStrenEQHiDq]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031104C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031204C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031304C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031404C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031504C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031604C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031704C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031804C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031904C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:5	Reserved.
4:1	RxChPowerDown. Read-write. Reset: Eh. Description: These pins are for internal use only. Please set to all '0's for all other purposes.
0	EnRxAggressivePDR. Read-write. Reset: 0. Reserved for future use.

UMC00CTLx00000804_address0001004D...UMC11CTLx00000804_address0031914D [TX ODT driver strength control] (UMC::Phy::TxOdtDrvStren)

Read-write. Reset: 0861h.

Selects desired impedance value for Host ODT.

Must be set by user before PHY initialization.

Setting for each protocol:

- Register Block Offset Address 0x04d controls the lower DQ nibble

- Register Block Offset Address 0x14d controls the upper DQ nibble

_umc0_b0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001004D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001014D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001104D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001114D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001204D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001214D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001304D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001314D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001404D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001414D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001504D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001514D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001604D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001614D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001704D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001714D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001804D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001814D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001904D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001914D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011004D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011014D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011104D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011114D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011204D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011214D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011304D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

[illegible]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031014D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031104D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031114D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031204D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031214D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031304D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031314D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031404D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031414D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031504D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031514D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031604D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031614D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031704D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031714D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031804D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031814D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031904D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031914D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:6	TxOdtStrenPd. Read-write. Reset: 21h.

	Description: Selects the ODT pull-down impedance. [ODTStren120N[3:0], ODTStren240N,ODTStren480N[0:0]] - 0000_0_0 = High Impedance - 0000_0_1 = 480 ohms - 0000_1_0 = 240 ohms - 0000_1_1 = 160 ohms - 0001_0_0 = 120 ohms - 0001_0_1 = 96.0 ohms - 0001_1_0 = 80.0 ohms - 0001_1_1 = 68.6 ohms - 0011_0_0 = 60.0 ohms - 0011_0_1 = 53.3 ohms - 0011_1_0 = 48.0 ohms - 0011_1_1 = 43.6 ohms - 0111_0_0 = 40.0 ohms - 0111_0_1 = 36.9 ohms - 0111_1_0 = 34.3 ohms - 0111_1_1 = 32.0 ohms - 1111_0_0 = 30.0 ohms - 1111_0_1 = 28.2 ohms - 1111_1_0 = 26.7 ohms - 1111_1_1 = 25.3 ohms
5:0	TxOdtStrenPu. Read-write. Reset: 21h. Description: Selects the ODT pull-up impedance. [ODTStren120P[3:0], ODTStren240P,ODTStren480P[0:0]] - 0000_0_0 = High Impedance - 0000_0_1 = 480 ohms - 0000_1_0 = 240 ohms - 0000_1_1 = 160 ohms - 0001_0_0 = 120 ohms - 0001_0_1 = 96.0 ohms - 0001_1_0 = 80.0 ohms - 0001_1_1 = 68.6 ohms - 0011_0_0 = 60.0 ohms - 0011_0_1 = 53.3 ohms - 0011_1_0 = 48.0 ohms - 0011_1_1 = 43.6 ohms - 0111_0_0 = 40.0 ohms - 0111_0_1 = 36.9 ohms - 0111_1_0 = 34.3 ohms - 0111_1_1 = 32.0 ohms - 1111_0_0 = 30.0 ohms - 1111_0_1 = 28.2 ohms - 1111_1_0 = 26.7 ohms - 1111_1_1 = 25.3 ohms

[illegible]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00017350; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00017450; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00017550; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00017650; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00017750; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00017850; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00018050; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00018150; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00018250; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00018350; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00018450; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00018550; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00018650; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00018750; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00018850; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00019050; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00019150; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00019250; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00019350; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00019450; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00019550; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00019650; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00019750; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00019850; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:7	Reserved.
6:0	VrefDAC2 . Read-write. Reset: 3Fh. - Reserved; deprecated.

UMC00CTLx00000804_address00010056...UMC11CTLx00000804_address00019056 [Status of RX FIFO Consistency Checks] (UMC::Phy::RxFifoCheckStatus)

Read-only, Volatile.

When a PhyRxFifoCheck interrupt is posted. These bits indicate which pair of fifo pointers did not match as expected. The non-zero pointer values can be read from the RxFifoCheckStatusValues register.

Setting the PhyRxFifoCheckClr in the PhyInterruptClear register clears these bits.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010056; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011056; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012056; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013056; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014056; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015056; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016056; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017056; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018056; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019056; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11	RxFifoLocDBErr. Read-only, Volatile.
----	---

Description: If set, there is a mismatch between the pointers of lanes 4 and 0.

For example, the write pointer of lane 4 differs from that of lane 0.

10	RxFifoLocUErr. Read-only, Volatile.
----	--

Description: If set, there is a mismatch between the pointers of lanes 4,5,6,7,8.

For example, the read pointer of lane 4 differs from that of lane 7.

9	RxFifoLocLErr. Read-only, Volatile.
---	--

Description: If set, there is a mismatch between the pointers of lanes 0,1,2,3.

For example, the read pointer of lane 1 differs from that of lane 2.

8	RxFifoLocErr8. Read-only, Volatile. See RxFifoLocErr0
---	--

7	RxFifoLocErr7. Read-only, Volatile. See RxFifoLocErr0
---	--

6	RxFifoLocErr6. Read-only, Volatile. See RxFifoLocErr0
---	--

5	RxFifoLocErr5. Read-only, Volatile. See RxFifoLocErr0
---	--

4	RxFifoLocErr4. Read-only, Volatile. See RxFifoLocErr0
---	--

3	RxFifoLocErr3. Read-only, Volatile. See RxFifoLocErr0
---	--

2	RxFifoLocErr2. Read-only, Volatile. See RxFifoLocErr0
---	--

1	RxFifoLocErr1. Read-only, Volatile. See RxFifoLocErr0
---	--

0	RxFifoLocErr0. Read-only, Volatile.
---	--

Description: Per DQ lane.

If set, the read pointer (DFI side) on the read FIFO mismatches the write pointer (DQS side) on the read data FIFO of the dq lane when the PHY is idle.

UMC00CTLx00000804_address00010057...UMC11CTLx00000804_address00019857 [Contains the captured values associated with an Rx Fifo consistency error] (UMC::Phy::Rx Fifo Check Err Values)

Read-only, Volatile.

Contains the value of the fifo pointer for the first mismatch captured in the Rx Fifo Check Status register. Setting the PhyRx Fifo Check Clr in the Phy Interrupt Clear register clears these bits.

Compare to Rx Fifo Info

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00010057; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00010157; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00010257; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00010357; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00010457; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00010557; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00010657; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00010757; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00010857; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00011057; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00011157; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00011257; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00011357; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00011457; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00011557; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00011657; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00011757; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00011857; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00012057; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00012157; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00012257; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00012357; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00012457; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00012557; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00012657; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00012757; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00012857; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00013057; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00013157; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00013257; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

[illegible]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00017157; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00017257; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00017357; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00017457; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00017557; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00017657; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00017757; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00017857; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00018057; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00018157; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00018257; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00018357; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00018457; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00018557; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00018657; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00018757; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00018857; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00019057; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00019157; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00019257; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00019357; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00019457; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00019557; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00019657; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00019757; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00019857; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15	Reserved.
14:10	RxFifoRdLocERR. Read-only, Volatile. The Mission mode read pointer.
9:5	RxFifoWrLocOddERR. Read-only, Volatile. The Mission mode write pointer odd-phase UI .
4:0	RxFifoWrLocEvnERR. Read-only, Volatile. The Mission mode write pointer even-phase UI .

UMC00CTLx00000804_address0001005F...UMC11CTLx00000804_address0031915F [TX slew rate controls] (UMC::Phy::TxSlewRate)

Read-write. Reset: 0700h.

Controls slew rates of the DQ/DQS drivers per nibble.

- Tx predriver cal trims driver timing to account for clock duty cycle variation.
- Register Block Offset Address 0x05f is the TxSlewRate for the lower nibble.
- Register Block Offset Address 0x15f is the TxSlewRate for the upper nibble.

_umc0_b0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001005F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001015F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001105F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001115F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001205F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001215F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001305F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001315F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001405F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001415F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001505F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001515F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001605F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001615F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001705F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001715F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001805F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001815F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001905F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0001915F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011005F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011015F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011105F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011115F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011205F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011215F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011305F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011315F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0011405F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

[illegible]

_umc0_b1_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031115F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031205F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031215F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031305F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031315F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031405F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031415F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031505F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031515F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031605F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031615F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031705F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031715F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031805F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031815F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031905F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_b1_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031915F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15	Reserved.
14:11	TxChargeCancel. Read-write. Reset: 0h. Controls strength of discharge function. 4'h0 no discharge. 4'hF max discharge.
10:8	TxPreDrvMode. Read-write. Reset: 7h. Description: Controls VDD high function. - TxPreDrvMode[8] is highVDD. When VDD=0.85V nominal, set this to 1'b1, otherwise set to 1'b0. - TxPreDrvMode[10:9] reserved for future use.
7:0	CsrTxSrc. Read-write. Reset: 00h. Description: 8-bit binary trim for the driver pull-up/down slew rate. The 4 upper bits don't care if the 4 lowest bits are set to 4'b0000. Bit 7 reserved for future use Bit 6 Low VDD(0.75V nominal)=1'b0; High VDD(0.85V nominal)=1'b1 Bit 5 D5=1'b1 Bit 4 DQ/DQS=1'b0 Bits[3:2] Slew rate control rise edge: 2'b00 = slew rate control off 2'b01 = slew rate control on, slower edge 2'b10 = slew rate control on, slow edge 2'b11 = reserved for future use Bits[1:0] Slew rate control fall edge: 2'b00 = slew rate control off 2'b01 = slew rate control on, slower edge 2'b10 = slew rate control on, slow edge 2'b11 = reserved for future use

[illegible]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00017360; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00017460; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00017560; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00017660; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00017760; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00017860; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00018060; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00018160; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00018260; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00018360; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00018460; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00018560; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00018660; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00018760; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00018860; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00019060; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00019160; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00019260; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00019360; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00019460; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00019560; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00019660; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00019760; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00019860; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:7	Reserved.
6:0	VrefDAC3 . Read-write. Reset: 3Fh. Reserved; deprecated.

UMC00CTLx00000804_address00010065...UMC11CTLx00000804_address00019065 [DFE miscellaneous control] (UMC::Phy::RxDFEMisc)

Read-write. Reset: 0000h.

Contains miscellaneous control signals for DFE.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010065; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011065; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012065; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013065; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014065; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015065; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016065; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017065; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018065; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019065; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:1	DFEbiassel. Read-write. Reset: 0h. DAC control for DFE bias.
0	SDen. Read-write. Reset: 0.
	Description: Used to enable source degeneration in analog slices. 1 - enable source degeneration. 0 - disable source degeneration.

Contains the control signals and sign bits for DFE tap

666

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address00317467; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address00317567; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address00317667; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address00317767; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address00317867; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address00318067; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address00318167; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address00318267; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address00318367; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address00318467; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address00318567; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address00318667; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address00318767; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address00318867; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address00319067; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address00319167; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address00319267; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address00319367; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address00319467; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address00319567; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address00319667; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address00319767; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address00319867; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:9	DFEtap4sel. Read-write. Reset: 0h. Description: MSB, bit 11, is Sign bit for DFE tap 4. Bits [11:9] are DAC control for DFE tap4
8:5	DFEtap3sel. Read-write. Reset: 0h. Description: MSB, bit 8, is Sign bit for DFE tap 3. Bits [8:5] are DAC control for DFE tap3
4:0	DFEtap2sel. Read-write. Reset: 00h. Description: MSB is Sign bit for DFE tap 2. Bits [4:0] are DAC control for DFE tap2

[illegible]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00017368; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00017468; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00017568; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00017668; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00017768; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00017868; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00018068; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00018168; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00018268; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00018368; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00018468; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00018568; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00018668; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00018768; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00018868; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00019068; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00019168; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00019268; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00019368; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00019468; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00019568; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00019668; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00019768; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00019868; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:7	Reserved.
6:0	RxPBDlyTg0 . Read-write. Reset: 00h.

Description: Read DQ per-bit BDL delay (Timing Group 0).

Trained to deskew the read DQ data that will be sampled by the incoming per-nibble read DQS, to improve the composite eye.

The unit of change is a LSB in the control of the delay circuit, and is not denominated in units of [UI](#).

- Register Block Offset Address 0x068 controls the lane0 read deskew timing. (For Timing Group 0)
- Register Block Offset Address 0x168 controls the lane1 read deskew timing. (For Timing Group 0)
- Register Block Offset Address 0x268 controls the lane2 read deskew timing. (For Timing Group 0)
- Register Block Offset Address 0x368 controls the lane3 read deskew timing. (For Timing Group 0)
- Register Block Offset Address 0x468 controls the lane4 read deskew timing. (For Timing Group 0)
- Register Block Offset Address 0x568 controls the lane5 read deskew timing. (For Timing Group 0)
- Register Block Offset Address 0x668 controls the lane6 read deskew timing. (For Timing Group 0)
- Register Block Offset Address 0x768 controls the lane7 read deskew timing. (For Timing Group 0)
- Register Block Offset Address 0x868 controls the lane[DBI] read deskew timing. (For Timing Group 0)

Note: This register is not per-pstate. The same deskew (for channel-routing differences) is used for all pstates.

Note: Timing Group 0 = RANK0 for all systems except 3DS/LRDIMM.

[illegible]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00017369; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00017469; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00017569; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00017669; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00017769; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00017869; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00018069; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00018169; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00018269; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00018369; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00018469; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00018569; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00018669; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00018769; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00018869; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address00019069; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address00019169; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address00019269; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address00019369; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address00019469; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address00019569; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address00019669; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address00019769; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address00019869; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:7	Reserved.
6:0	RxPBDlyTg1. Read-write. Reset: 00h.

Description: Read DQ per-bit BDL delay (Timing Group 1).

Trained to deskew the read DQ data that will be sampled by the incoming per-nibble read DQS, to improve the composite eye.

The unit of change is a LSB in the control of the delay circuit, and is not denominated in units of [UI](#).

- Register Block Offset Address 0x069 controls the lane0 read deskew timing. (For Timing Group 1)
- Register Block Offset Address 0x169 controls the lane1 read deskew timing. (For Timing Group 1)
- Register Block Offset Address 0x269 controls the lane2 read deskew timing. (For Timing Group 1)
- Register Block Offset Address 0x369 controls the lane3 read deskew timing. (For Timing Group 1)
- Register Block Offset Address 0x469 controls the lane4 read deskew timing. (For Timing Group 1)
- Register Block Offset Address 0x569 controls the lane5 read deskew timing. (For Timing Group 1)
- Register Block Offset Address 0x669 controls the lane6 read deskew timing. (For Timing Group 1)
- Register Block Offset Address 0x769 controls the lane7 read deskew timing. (For Timing Group 1)
- Register Block Offset Address 0x869 controls the lane[DBI] read deskew timing. (For Timing Group 1)

Note: This register is not per-pstate. The same deskew (for channel-routing differences) is used for all pstates.

Note: Timing Group 1 = RANK1 for all systems except 3DS/LRDIMM.

[illegible]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address0001736A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address0001746A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address0001756A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address0001766A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address0001776A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address0001786A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address0001806A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address0001816A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address0001826A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address0001836A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address0001846A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address0001856A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address0001866A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address0001876A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address0001886A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address0001906A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address0001916A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address0001926A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address0001936A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address0001946A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address0001956A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address0001966A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address0001976A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address0001986A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:7	Reserved.
6:0	RxPBDlyTg2 . Read-write. Reset: 00h.

Description: Read DQ per-bit BDL delay (Timing Group 2).

Trained to deskew the read DQ data that will be sampled by the incoming per-nibble read DQS, to improve the composite eye.

The unit of change is a LSB in the control of the delay circuit, and is not denominated in units of [UI](#).

- Register Block Offset Address 0x06a controls the lane0 read deskew timing. (For Timing Group 2)
- Register Block Offset Address 0x16a controls the lane1 read deskew timing. (For Timing Group 2)
- Register Block Offset Address 0x26a controls the lane2 read deskew timing. (For Timing Group 2)
- Register Block Offset Address 0x36a controls the lane3 read deskew timing. (For Timing Group 2)
- Register Block Offset Address 0x46a controls the lane4 read deskew timing. (For Timing Group 2)
- Register Block Offset Address 0x56a controls the lane5 read deskew timing. (For Timing Group 2)
- Register Block Offset Address 0x66a controls the lane6 read deskew timing. (For Timing Group 2)
- Register Block Offset Address 0x76a controls the lane7 read deskew timing. (For Timing Group 2)
- Register Block Offset Address 0x86a controls the lane[DBI] read deskew timing. (For Timing Group 2)

Note: This register is not per-pstate. The same deskew (for channel-routing differences) is used for all pstates.

Note: Timing Group 2 = RANK2 for all systems except 3DS/LRDIMM.

[illegible]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address0001736B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address0001746B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address0001756B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address0001766B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address0001776B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address0001786B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address0001806B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address0001816B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address0001826B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address0001836B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address0001846B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address0001856B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address0001866B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address0001876B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address0001886B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address0001906B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address0001916B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address0001926B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address0001936B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address0001946B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address0001956B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address0001966B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address0001976B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address0001986B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:7	Reserved.
6:0	RxPBDlyTg3 . Read-write. Reset: 00h.

Description: Read DQ per-bit BDL delay (Timing Group 3).

Trained to deskew the read DQ data that will be sampled by the incoming per-nibble read DQS, to improve the composite eye.

The unit of change is a LSB in the control of the delay circuit, and is not denominated in units of [UI](#).

- Register Block Offset Address 0x06b controls the lane0 read deskew timing. (For Timing Group 3)

- Register Block Offset Address 0x16b controls the lane1 read deskew timing. (For Timing Group 3)

- Register Block Offset Address 0x26b controls the lane2 read deskew timing. (For Timing Group 3)

- Register Block Offset Address 0x36b controls the lane3 read deskew timing. (For Timing Group 3)

- Register Block Offset Address 0x46b controls the lane4 read deskew timing. (For Timing Group 3)

- Register Block Offset Address 0x56b controls the lane5 read deskew timing. (For Timing Group 3)

- Register Block Offset Address 0x66b controls the lane6 read deskew timing. (For Timing Group 3)

- Register Block Offset Address 0x76b controls the lane7 read deskew timing. (For Timing Group 3)

- Register Block Offset Address 0x86b controls the lane[DBI] read deskew timing. (For Timing Group 3)

Note: This register is not per-pstate. The same deskew (for channel-routing differences) is used for all pstates.

Note: Timing Group 3 = RANK3 for all systems except 3DS/LRDIMM.

[illegible]

[illegible]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address0001726C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address0001736C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address0001746C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address0001756C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address0001766C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address0001776C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address0001786C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address0001806C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address0001816C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address0001826C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address0001836C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address0001846C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address0001856C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address0001866C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address0001876C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address0001886C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r0_aliasSMN; UMC[11:00]CTLx00000804_address0001906C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r1_aliasSMN; UMC[11:00]CTLx00000804_address0001916C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r2_aliasSMN; UMC[11:00]CTLx00000804_address0001926C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r3_aliasSMN; UMC[11:00]CTLx00000804_address0001936C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r4_aliasSMN; UMC[11:00]CTLx00000804_address0001946C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r5_aliasSMN; UMC[11:00]CTLx00000804_address0001956C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r6_aliasSMN; UMC[11:00]CTLx00000804_address0001966C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r7_aliasSMN; UMC[11:00]CTLx00000804_address0001976C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_r8_aliasSMN; UMC[11:00]CTLx00000804_address0001986C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Offselodd1. Read-write. Reset: 0h. Description: MSB, bit 12, is offset calibration sign bit for slice 1 odd summer. Bits [15:12] are offset calibration sel bits for slice 1 odd summer.
11:8	Offselodd0. Read-write. Reset: 0h. Description: MSB, bit 11, is offset calibration sign bit for slice 0 odd summer. Bits [11:8] are offset calibration sel bits for slice 0 odd summer.
7:4	Offseleven1. Read-write. Reset: 0h. Description: MSB, bit 7, is offset calibration sign bit for slice 1 even summer. Bits [7:4] are offset calibration sel bits for slice 1 even summer.
3:0	Offseleven0. Read-write. Reset: 0h. Description: MSB, bit 3, is offset calibration sign bit for slice 0 even summer. Bits [3:0] are offset calibration sel bits for slice 0 even summer.

[illegible]

[illegible]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00312170; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00313070; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00313170; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00314070; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00314170; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00315070; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00315170; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00316070; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00316170; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00317070; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00317170; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00318070; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00318170; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00319070; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00319170; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:0	PptDqsCntInvTrnTg0. Read-write. Reset: 0000h. Programmed by PHY training firmware to support DDR5 DRAM drift compensation.

[illegible]

[illegible]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00312171; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00313071; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00313171; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00314071; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00314171; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00315071; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00315171; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00316071; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00316171; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00317071; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00317171; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00318071; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00318171; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00319071; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00319171; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:0	PptDqsCntInvTrnTg1. Read-write. Reset: 0000h. Programmed by PHY training firmware to support DDR5 DRAM drift compensation.

[illegible]

[illegible]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00312172; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00313072; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00313172; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00314072; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00314172; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00315072; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00315172; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00316072; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00316172; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00317072; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00317172; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00318072; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00318172; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00319072; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00319172; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:0	PptDqsCntInvTrnTg2. Read-write. Reset: 0000h. Programmed by PHY training firmware to support DDR5 DRAM drift compensation.

[illegible]

[illegible]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00312173; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00313073; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00313173; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00314073; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00314173; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00315073; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00315173; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00316073; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00316173; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00317073; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00317173; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00318073; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00318173; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00319073; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00319173; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:0	PptDqsCntInvTrnTg3. Read-write. Reset: 0000h. Programmed by PHY training firmware to support DDR5 DRAM drift compensation.

[illegible]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00313080; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00313180; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00314080; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00314180; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00315080; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00315180; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00316080; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00316180; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00317080; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00317180; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00318080; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00318180; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00319080; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00319180; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	RxEnDlyTg0. Read-write. Reset: 100h. Description: Trained Receive Enable Delay (For Timing Group 0) Trained to set the delay from the memory-read command to the signal enabling the read DQS to generate read-data strobes. The trained read DQS enable asserts during the read DQS preamble and deasserts at the end of the read burst. RxEnDlyTg0[11:6] is the coarse delay, for example one unit of delay is 1 UI and has been changed to max of 12 bits from 11. The maximum value allowed for RxEnDlyTg0[11:6] is 63, 0x3F, - If Dly64Prec=0, RxEnDlyTg0[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32; - RxEnDlyTg0[5] is reserved. - If Dly64Prec=1, RxEnDlyTg0[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64; - Register Block Offset Address 0x080 controls the lower DQ nibble DqsEn. (For Timing Group 0) - Register Block Offset Address 0x180 controls the upper DQ nibble DqsEn. (For Timing Group 0) Note: These Registers are replicated per pstate. See Description of csr TwoTckRxDqsPre for description of expected trained values of RxEnDly, and drift tolerance of the read DQS for the various supported DDR5 read preamble modes. Note: Timing Group 0 = RANK0 for all systems except 3DS/LRDIMM.

[illegible]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00313081; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00313181; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00314081; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00314181; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00315081; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00315181; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00316081; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00316181; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00317081; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00317181; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00318081; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00318181; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00319081; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00319181; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	RxEnDlyTg1. Read-write. Reset: 100h. Description: Trained Receive Enable Delay (For Timing Group 1) Trained to set the delay from the memory-read command to the signal enabling the read DQS to generate read-data strobes. The trained read DQS enable asserts during the read DQS preamble and deasserts at the end of the read burst. RxEnDlyTg1[11:6] is the coarse delay, for example one unit of delay is 1 UI and has been changed to max of 12 bits from 11. The maximum value allowed for RxEnDlyTg1[11:6] is 63, 0x3F, - If Dly64Prec=0, RxEnDlyTg1[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32; - RxEnDlyTg1[5] is reserved. - If Dly64Prec=1, RxEnDlyTg1[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64; - Register Block Offset Address 0x081 controls the lower DQ nibble DqsEn. (For Timing Group 1) - Register Block Offset Address 0x181 controls the upper DQ nibble DqsEn. (For Timing Group 1) Note: These Registers are replicated per pstate. See Description of csr TwoTckRxDqsPre for description of expected trained values of RxEnDly, and drift tolerance of the read DQS for the various supported DDR5 read preamble modes. Note: Timing Group 1 = RANK1 for all systems except 3DS/LRDIMM.

Read-write. Reset: 0100h.

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:00]CTLx00000804_address00010082; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:00]CTLx00000804_address00010182; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:00]CTLx00000804_address00011082; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:00]CTLx00000804_address00011182; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:0]CTLx00000804_address00012082; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:00]CTLx00000804_address00012182; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:0]CTLx00000804_address00013082; UMC[11:0]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:00]CTLx00000804_address00013182; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=[UMC::AnbCmd[Address]
```

```
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:00]CTLx00000804_address00014082; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=[UMC::ApbCmd[Address]
```

```
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:00]CTLx00000804_address00014182; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:00]CTLx00000804_address00015082; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=[UMC::AnbCmd[Address]
```

```
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:00]CTLx00000804_address00015182; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:00]CTLx00000804_address00016082; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:00]CTLx00000804_address00016182; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:00]CTLx00000804_address00017082; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11h:00]CTLx00000804_address00017182; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:00]CTLx00000804_address00018082; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:00]CTLx00000804_address00018182; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:00]CTLx00000804_address00019082; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:00]CTLx00000804_address00019182; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_u0_aliasSMN; UMC[11:0]CTLx00000804_address00110082; UMC[11:0]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet0_chiplettype1_inst1_umc0phy[11:0]_p1_u1_aliasSMN; UMC[11:00]CTLx00000804_address00110182; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_u0_aliasSMN; UMC[11:00]CTLx00000804_address00111082; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_u1_aliasSMN; UMC[11:00]CTLx00000804_address00111182; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet2_chiplettype1_inst1UMCWPHY[11:0]]_p1_u0_aliasSMN[UMC[11:00]CTLx00000804_address00112082;UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_u1_aliasSMN; UMC[11:00]CTLx00000804_address00112182; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_u0_aliasSMN; UMC[11:00]CTLx00000804_address00113082; UMC[11:00]CTLx00[B:0]5_1000h;
```

```
DataPortWrite=UMC::ApbCmd[Address]
    _umc_0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_u1_aliasSMN; UMC[11:00]CTLx00000804_address00113182; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p1_u0_aliasSMN; UMC[11:00]CTLx00000804_address00114082; UMC[11:00]CTL=00[B:0]5_1000h;
```

```

DataPortWrite=UMC::ApbCmqlAddress]
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_lp1_u1_aliasSMN; UMC[11:00]CTLx00000804_address00114182; UMC[11:00]CTL=00[B:0]5_1000h;

```

```
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_lp1_u0_aliasSMN; UMC[11:00]CTLx00000804_address00115082; UMC[11:00]CTL=00[B:0]5_1000h;
```

```

DataPortWrite=UMC::ApbCmd[Address]
_umco_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p1_u1_aliasSMN; UMC[11:00]CTLx00000804_address00115182; UMC[11:00]CTL=00[B:0]5_1000h;
B; B=UMC[11:00]CTL; UMC[11:00]CTL=00[B:0]5_1000h;

```

[illegible]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00313082; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00313182; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00314082; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00314182; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00315082; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00315182; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00316082; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00316182; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00317082; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00317182; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00318082; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00318182; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00319082; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00319182; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	RxEnDlyTg2. Read-write. Reset: 100h. Description: Trained Receive Enable Delay (For Timing Group 2) Trained to set the delay from the memory-read command to the signal enabling the read DQS to generate read-data strobes. The trained read DQS enable asserts during the read DQS preamble and deasserts at the end of the read burst. RxEnDlyTg2[11:6] is the coarse delay, for example one unit of delay is 1 UI and has been changed to max of 12 bits from 11. The maximum value allowed for RxEnDlyTg2[11:6] is 63, 0x3F, - If Dly64Prec=0, RxEnDlyTg2[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32; - RxEnDlyTg2[5] is reserved. - If Dly64Prec=1, RxEnDlyTg2[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64; - Register Block Offset Address 0x082 controls the lower DQ nibble DqsEn. (For Timing Group 2) - Register Block Offset Address 0x182 controls the upper DQ nibble DqsEn. (For Timing Group 2) Note: These Registers are replicated per pstate. See Description of csr TwoTckRxDqsPre for description of expected trained values of RxEnDly, and drift tolerance of the read DQS for the various supported DDR5 read preamble modes. Note: Timing Group 2 = RANK2 for all systems except 3DS/LRDIMM.

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_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00313083; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00313183; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00314083; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00314183; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00315083; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00315183; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00316083; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00316183; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00317083; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00317183; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00318083; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00318183; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00319083; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00319183; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	RxEnDlyTg3. Read-write. Reset: 100h. Description: Trained Receive Enable Delay (For Timing Group 3) Trained to set the delay from the memory-read command to the signal enabling the read DQS to generate read-data strobes. The trained read DQS enable asserts during the read DQS preamble and deasserts at the end of the read burst. RxEnDlyTg3[11:6] is the coarse delay, for example one unit of delay is 1 UI and has been changed to max of 12 bits from 11. The maximum value allowed for RxEnDlyTg3[11:6] is 63, 0x3F, - If Dly64Prec=0, RxEnDlyTg3[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32; - RxEnDlyTg3[5] is reserved. - If Dly64Prec=1, RxEnDlyTg3[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64; - Register Block Offset Address 0x083 controls the lower DQ nibble DqsEn. (For Timing Group 3) - Register Block Offset Address 0x183 controls the upper DQ nibble DqsEn. (For Timing Group 3) Note: These Registers are replicated per pstate. See Description of csr TwoTckRxDqsPre for description of expected trained values of RxEnDly, and drift tolerance of the read DQS for the various supported DDR5 read preamble modes. Note: Timing Group 3 = RANK3 for all systems except 3DS/LRDIMM.

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_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031218C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031308C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031318C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031408C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031418C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031508C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031518C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031608C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031618C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031708C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031718C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031808C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031818C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031908C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031918C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:6	Reserved.
5:0	RxClkDlyTg0. Read-write. Reset: 10h. Description: Trained Read DQS_t to RxClk Delay (Timing Group DEST=0). Trained to generate read even-phase DQ receive clock from the received read DQS_t. - RxClk Delay (6-bits, fine delay only); (For Timing Group 0) - If Dly64Prec=0, RxClkDlyTg0[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a $UI = UI/32$; - RxClkDlyTg0[5] is reserved. - If Dly64Prec=1, RxClkDlyTg0[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a $UI = UI/64$; - Register Block Offset Address 0x08c controls the lower DQ nibble RxClk timing. (For Timing Group 0) - Register Block Offset Address 0x18c controls the upper DQ nibble RxClk timing. (For Timing Group 0) Note: These Registers are replicated per pstate. Note: Timing Group 0 = RANK0 for all systems except 3DS/LRDIMM.

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_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031218D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031308D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031318D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031408D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031418D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031508D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031518D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031608D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031618D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031708D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031718D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031808D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031818D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031908D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031918D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:6	Reserved.
5:0	RxClkDlyTg1. Read-write. Reset: 10h. Description: Trained Read DQS_t to RxClk Delay (Timing Group DEST=1). Trained to generate read even-phase DQ receive clock from the received read DQS_t. - RxClk Delay (6-bits, fine delay only); (For Timing Group 1) - If Dly64Prec=0, RxClkDlyTg1[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a $UI = UI/32$; - RxClkDlyTg1[5] is reserved. - If Dly64Prec=1, RxClkDlyTg1[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a $UI = UI/64$; - Register Block Offset Address 0x08d controls the lower DQ nibble RxClk timing. (For Timing Group 1) - Register Block Offset Address 0x18d controls the upper DQ nibble RxClk timing. (For Timing Group 1) Note: These Registers are replicated per pstate. Note: Timing Group 1 = RANK1 for all systems except 3DS/LRDIMM.

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_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031218E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031308E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031318E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031408E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031418E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031508E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031518E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031608E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031618E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031708E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031718E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031808E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031818E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031908E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031918E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:6	Reserved.
5:0	RxClkDlyTg2. Read-write. Reset: 10h. Description: Trained Read DQS_t to RxClk Delay (Timing Group DEST=2). Trained to generate read even-phase DQ receive clock from the received read DQS_t. - RxClk Delay (6-bits, fine delay only); (For Timing Group 2) - If Dly64Prec=0, RxClkDlyTg2[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a $UI = UI/32$; - RxClkDlyTg2[5] is reserved. - If Dly64Prec=1, RxClkDlyTg2[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a $UI = UI/64$; - Register Block Offset Address 0x08e controls the lower DQ nibble RxClk timing. (For Timing Group 2) - Register Block Offset Address 0x18e controls the upper DQ nibble RxClk timing. (For Timing Group 2) Note: These Registers are replicated per pstate. Note: Timing Group 2 = RANK2 for all systems except 3DS/LRDIMM.

[illegible]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031308F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031318F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031408F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031418F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031508F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031518F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031608F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031618F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031708F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031718F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031808F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031818F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address0031908F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address0031918F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:6	Reserved.
5:0	RxClkDlyTg3. Read-write. Reset: 10h. Description: Trained Read DQS_t to RxClk Delay (Timing Group DEST=3). Trained to generate read even-phase DQ receive clock from the received read DQS_t. - RxClk Delay (6-bits, fine delay only); (For Timing Group 3) - If Dly64Prec=0, RxClkDlyTg3[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32; - RxClkDlyTg3[5] is reserved. - If Dly64Prec=1, RxClkDlyTg3[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64; - Register Block Offset Address 0x08f controls the lower DQ nibble RxClk timing. (For Timing Group 3) - Register Block Offset Address 0x18f controls the upper DQ nibble RxClk timing. (For Timing Group 3) Note: These Registers are replicated per pstate. Note: Timing Group 3 = RANK3 for all systems except 3DS/LRDIMM.

[illegible]

[illegible]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00312190; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00313090; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00313190; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00314090; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00314190; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00315090; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00315190; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00316090; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00316190; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00317090; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00317190; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00318090; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00318190; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00319090; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00319190; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:6	Reserved.
5:0	RxClkcDlyTg0. Read-write. Reset: 10h. Description: Trained Read DQS_c to RxClkc Delay (Timing Group DEST=0). Trained to generate read odd-phase DQ receive clock from the received read DQS_c. - RxClkc Delay (6-bits, fine delay only); (For Timing Group 0) - If Dly64Prec=0, RxClkcDlyTg0[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32; - RxClkcDlyTg0[5] is reserved. - If Dly64Prec=1, RxClkcDlyTg0[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64; - Register Block Offset Address 0x090 controls the lower DQ nibble RxClkc timing. (For Timing Group 0) - Register Block Offset Address 0x190 controls the upper DQ nibble RxClkc timing. (For Timing Group 0) Note: These Registers are replicated per pstate. Note: Timing Group 0 = RANK0 for all systems except 3DS/LRDIMM.

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_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00312191; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00313091; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00313191; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00314091; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00314191; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00315091; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00315191; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00316091; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00316191; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00317091; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00317191; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00318091; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00318191; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00319091; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00319191; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:6	Reserved.
5:0	RxClkcDlyTg1. Read-write. Reset: 10h. Description: Trained Read DQS_c to RxClkc Delay (Timing Group DEST=1). Trained to generate read odd-phase DQ receive clock from the received read DQS_c. - RxClkc Delay (6-bits, fine delay only); (For Timing Group 1) - If Dly64Prec=0, RxClkcDlyTg1[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32; - RxClkcDlyTg1[5] is reserved. - If Dly64Prec=1, RxClkcDlyTg1[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64; - Register Block Offset Address 0x091 controls the lower DQ nibble RxClkc timing. (For Timing Group 1) - Register Block Offset Address 0x191 controls the upper DQ nibble RxClkc timing. (For Timing Group 1) Note: These Registers are replicated per pstate. Note: Timing Group 1 = RANK1 for all systems except 3DS/LRDIMM.

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_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00312192; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00313092; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00313192; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00314092; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00314192; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00315092; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00315192; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00316092; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00316192; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00317092; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00317192; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00318092; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00318192; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00319092; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00319192; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:6	Reserved.
5:0	RxClkcDlyTg2. Read-write. Reset: 10h. Description: Trained Read DQS_c to RxClkc Delay (Timing Group DEST=2). Trained to generate read odd-phase DQ receive clock from the received read DQS_c. - RxClkc Delay (6-bits, fine delay only); (For Timing Group 2) - If Dly64Prec=0, RxClkcDlyTg2[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32; - RxClkcDlyTg2[5] is reserved. - If Dly64Prec=1, RxClkcDlyTg2[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64; - Register Block Offset Address 0x092 controls the lower DQ nibble RxClkc timing. (For Timing Group 2) - Register Block Offset Address 0x192 controls the upper DQ nibble RxClkc timing. (For Timing Group 2) Note: These Registers are replicated per pstate. Note: Timing Group 2 = RANK2 for all systems except 3DS/LRDIMM.

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_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00312193; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00313093; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00313193; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00314093; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00314193; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00315093; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00315193; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00316093; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00316193; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00317093; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00317193; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00318093; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00318193; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address00319093; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address00319193; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:6	Reserved.
5:0	RxClkcDlyTg3. Read-write. Reset: 10h. Description: Trained Read DQS_c to RxClkc Delay (Timing Group DEST=3). Trained to generate read odd-phase DQ receive clock from the received read DQS_c. - RxClkc Delay (6-bits, fine delay only); (For Timing Group 3) - If Dly64Prec=0, RxClkcDlyTg3[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32; - RxClkcDlyTg3[5] is reserved. - If Dly64Prec=1, RxClkcDlyTg3[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64; - Register Block Offset Address 0x093 controls the lower DQ nibble RxClkc timing. (For Timing Group 3) - Register Block Offset Address 0x193 controls the upper DQ nibble RxClkc timing. (For Timing Group 3) Note: These Registers are replicated per pstate. Note: Timing Group 3 = RANK3 for all systems except 3DS/LRDIMM.

UMC00CTLx00000804_address000100A0...UMC11CTLx00000804_address000190A0 (UMC::Phy::Dq0LnSelTg0)

Read-write. Reset: 0000h.

Maps PHY DQ lane to memory DQ0

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100A0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110A0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120A0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130A0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140A0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150A0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160A0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170A0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180A0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190A0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:3	Reserved.
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2:0	Dq0LnSelTg0. Read-write. Reset: 0h.
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Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle).
Intended to undo the swizzle of board-level phy-to-dram connections.
such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE.
Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to
memory dq0 (on this DBYTE), then Register Dq0LnSelTg0 for this DBYTE should be 3.

UMC00CTLx00000804_address000100A1...UMC11CTLx00000804_address000190A1 (UMC::Phy::Dq1LnSelTg0)

Read-write. Reset: 0001h.

Maps PHY DQ lane to memory DQ1

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:3	Reserved.
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2:0	Dq1LnSelTg0. Read-write. Reset: 1h.
-----	--

Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle).
Intended to undo the swizzle of board-level phy-to-dram connections.
such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE.
Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to
memory dq1 (on this DBYTE), then Register Dq1LnSelTg0 for this DBYTE should be 3.

UMC00CTLx00000804_address000100A2...UMC11CTLx00000804_address000190A2 (UMC::Phy::Dq2LnSelTg0)

Read-write. Reset: 0002h.

Maps PHY DQ lane to memory DQ2

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:3	Reserved.
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2:0	Dq2LnSelTg0. Read-write. Reset: 2h.
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Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle).
Intended to undo the swizzle of board-level phy-to-dram connections.
such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE.
Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to
memory dq2 (on this DBYTE), then Register Dq2LnSelTg0 for this DBYTE should be 3.

UMC00CTLx00000804_address000100A3...UMC11CTLx00000804_address000190A3 (UMC::Phy::Dq3LnSelTg0)

Read-write. Reset: 0003h.

Maps PHY DQ lane to memory DQ3

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100A3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110A3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120A3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130A3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140A3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150A3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160A3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170A3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180A3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190A3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:3	Reserved.
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2:0	Dq3LnSelTg0. Read-write. Reset: 3h.
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Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle).
Intended to undo the swizzle of board-level phy-to-dram connections.
such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE.
Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to
memory dq3 (on this DBYTE), then Register Dq3LnSelTg0 for this DBYTE should be 3.

UMC00CTLx00000804_address000100A4...UMC11CTLx00000804_address000190A4 (UMC::Phy::Dq4LnSelTg0)

Read-write. Reset: 0004h.

Maps PHY DQ lane to memory DQ4

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:3	Reserved.
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2:0	Dq4LnSelTg0. Read-write. Reset: 4h.
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Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle).
Intended to undo the swizzle of board-level phy-to-dram connections.
such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE.
Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to
memory dq4 (on this DBYTE), then Register Dq4LnSelTg0 for this DBYTE should be 3.

UMC00CTLx00000804_address000100A5...UMC11CTLx00000804_address000190A5 (UMC::Phy::Dq5LnSelTg0)

Read-write. Reset: 0005h.

Maps PHY DQ lane to memory DQ5

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:3	Reserved.
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2:0	Dq5LnSelTg0. Read-write. Reset: 5h.
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Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle).

Intended to undo the swizzle of board-level phy-to-dram connections.

such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE.

Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to

memory dq5 (on this DBYTE), then Register Dq5LnSelTg0 for this DBYTE should be 3.

UMC00CTLx00000804_address000100A6...UMC11CTLx00000804_address000190A6 (UMC::Phy::Dq6LnSelTg0)

Read-write. Reset: 0006h.

Maps PHY DQ lane to memory DQ6

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15:3 Reserved.

2:0 **Dq6LnSelTg0.** Read-write. Reset: 6h.

Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle).
Intended to undo the swizzle of board-level phy-to-dram connections.
such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE.
Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to
memory dq6 (on this DBYTE), then Register Dq6LnSelTg0 for this DBYTE should be 3.

UMC00CTLx00000804_address000100A7...UMC11CTLx00000804_address000190A7 (UMC::Phy::Dq7LnSelTg0)

Read-write. Reset: 0007h.

Maps PHY DQ lane to memory DQ7

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15:3 Reserved.

2:0 **Dq7LnSelTg0.** Read-write. Reset: 7h.

Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle).
Intended to undo the swizzle of board-level phy-to-dram connections.
such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE.
Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to
memory dq7 (on this DBYTE), then Register Dq7LnSelTg0 for this DBYTE should be 3.

UMC00CTLx00000804_address000100AA...UMC11CTLx00000804_address000190AA (UMC::Phy::PptCtlStatic)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100AA; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110AA; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120AA; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130AA; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140AA; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150AA; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160AA; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170AA; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180AA; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190AA; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15 Reserved.

14:0 RFU. Read-write. Reset: 0000h.

UMC00CTLx00000804_address000100AB...UMC11CTLx00000804_address000190AB (UMC::Phy::PptCtlDyn)

Read-write. Reset: 0000h.

This register is dynamically written by PHY Initialization Engine during frequency changes and should not be written by the user.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100AB; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110AB; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120AB; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130AB; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140AB; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150AB; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160AB; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170AB; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180AB; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190AB; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15:2 Reserved.

1:0 RFU. Read-write. Reset: 0h.

UMC00CTLx00000804_address000100AC...UMC11CTLx00000804_address000190AC (UMC::Phy::PptInfo)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100AC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110AC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120AC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130AC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140AC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150AC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160AC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170AC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180AC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190AC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits Description**15:0 PptInfo.** Read-only, Volatile. Reserved for PHY training firmware use.**UMC00CTLx00000804_address000100AD...UMC11CTLx00000804_address000190AD (UMC::Phy::RxDqsTrackInfo)**

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100AD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110AD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120AD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130AD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140AD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150AD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160AD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170AD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180AD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190AD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits Description**15:0 RxDqsTrackInfo.** Read-only, Volatile. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000100AE...UMC11CTLx00000804_address000190AE [Rx DQS tracking information] (UMC::Phy::RxDqsTrackInfoSelect)

Read-write. Reset: 0000h.

Information for debug of tracking read DQS drift with respect to RxEn

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100AE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110AE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120AE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130AE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140AE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150AE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160AE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170AE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180AE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190AE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15:5 Reserved.

4:1 **RxDqsTrackInfoSel.** Read-write. Reset: 0h.**Description:** Select the information from RxDqsTracking logic that is muxed into RxDqsTrackInfo.

- 0 0000000,RxEnDlyOffset[8:0] , lower nibble, timing-group 0
- 1 IncDecHistory, nDqsSampNegRxEn[5:0], lower nibble, timing-group 0
- 2 0000000,RxEnDlyOffset[8:0] , lower nibble, timing-group 1
- 3 IncDecHistory, nDqsSampNegRxEn[5:0], lower nibble, timing-group 1
- 4 0000000,RxEnDlyOffset[8:0] , lower nibble, timing-group 2
- 5 IncDecHistory, nDqsSampNegRxEn[5:0], lower nibble, timing-group 2
- 6 0000000,RxEnDlyOffset[8:0] , lower nibble, timing-group 3
- 7 IncDecHistory, nDqsSampNegRxEn[5:0], lower nibble, timing-group 3
- 8 0000000,RxEnDlyOffset[8:0] , upper nibble, timing-group 0
- 9 IncDecHistory, nDqsSampNegRxEn[5:0], upper nibble, timing-group 0
- 10 0000000,RxEnDlyOffset[8:0] , upper nibble, timing-group 1
- 11 IncDecHistory, nDqsSampNegRxEn[5:0], upper nibble, timing-group 1
- 12 0000000,RxEnDlyOffset[8:0] , upper nibble, timing-group 2
- 13 IncDecHistory, nDqsSampNegRxEn[5:0], upper nibble, timing-group 2
- 14 0000000,RxEnDlyOffset[8:0] , upper nibble, timing-group 3
- 15 IncDecHistory, nDqsSampNegRxEn[5:0], upper nibble, timing-group 3

The RxEnDlyOffset are the two's complement adjustment to the RxEnDly.

The nDqsSampNegRxEn are the two-complement number of samples used to make the determination of RxEnDlyOffset at the time of the last update event.

This count is limited by the csrRxDqsTrackingThreshold[2:0]; see that description.

- IncDecHistory is the history of [inc,dec] that was applied to determine nDqsSampNegRxEn.

- IncDecHistory[9:8] is the most recent value of [inc,dec]

- IncDecHistory[7:6] is the 2nd most recent value of [inc,dec], etc.

- IncDecHistory[1:0] is the 5th most recent value of [inc,dec]

- A pair of IncDecHistory bits will be 2'b00 if there have not been that many sampling events.

For example, IncDecHistory[3:0] will be 4'b00_00 if there are 3 or fewer sampling events.

IncDecHistory is not cleared by update events.

0	SnapRxDqsTrackInfo. Read-write. Reset: 0. De-asserting edge snapshots information for reading in RxDqsTrackInfo
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UMC00CTLx00000804_address000100AF...UMC11CTLx00000804_address000190AF (UMC::Phy::DbCurrentDlyTimingInfoTg0)

Read-only, Volatile.

Reserved for PHY training firmware use and for debugging

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	DbCurrentDlyTimingInfoTg0. Read-only, Volatile. Reserved for PHY training firmware use and for debugging

UMC00CTLx00000804_address000100BF...UMC11CTLx00000804_address000190BF [Selects the delay information readable from DbCurrentDlyTimingInfo.] (UMC::Phy::SelDbCurDlyTmngInfo)

Read-write. Reset: 0000h.

Controls what information is read from DbCurrentDlyTimingInfo.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100BF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110BF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120BF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130BF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140BF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150BF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160BF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170BF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180BF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190BF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7	FreezeAllDbCDTI. Read-write. Reset: 0. This field is reserved for PHY training firmware use and for debugging
6	FreezeSelDbCDTI. Read-write. Reset: 0. Description: This field is programmed as follows: - 1 freezes the delay selected by SelDbCDTI so that it may be read - 1 has the side-effect of also freezing the delay selected by SelDbCDTI going to the pipes - 0 passes the delay, unless FreezeAllDbCDTI=1
5:0	SelDbCDTI. Read-write. Reset: 00h.

Description: Selects the delay information readable from CurrentDlyTimingInfo.

- 6'd00 [7'd0, useTxDqDlyTg*_r0[8:6],useDlITxDqPhaseD*_r0[5:0]]
- 6'd01 [7'd0, useTxDqDlyTg*_r1[8:6],useDlITxDqPhaseD*_r1[5:0]]
- 6'd02 [7'd0, useTxDqDlyTg*_r2[8:6],useDlITxDqPhaseD*_r2[5:0]]
- 6'd03 [7'd0, useTxDqDlyTg*_r3[8:6],useDlITxDqPhaseD*_r3[5:0]]
- 6'd04 [7'd0, useTxDqDlyTg*_r4[8:6],useDlITxDqPhaseD*_r4[5:0]]
- 6'd05 [7'd0, useTxDqDlyTg*_r5[8:6],useDlITxDqPhaseD*_r5[5:0]]
- 6'd06 [7'd0, useTxDqDlyTg*_r6[8:6],useDlITxDqPhaseD*_r6[5:0]]
- 6'd07 [7'd0, useTxDqDlyTg*_r7[8:6],useDlITxDqPhaseD*_r7[5:0]]
- 6'd08 [7'd0, useTxDqDlyTg*_r8[8:6],useDlITxDqPhaseD*_r8[5:0]]
- 6'd09 [5'd0, useTxDqsDlyTg*_u0[10:6],useDlITxDqsPhaseD*_u0[5:0]]
- 6'd10 [5'd0, useTxDqsDlyTg*_u1[10:6],useDlITxDqsPhaseD*_u1[5:0]]
- 6'd11 [10'd0, useDlIRxClkPhaseD*_u0[5:0]]
- 6'd12 [10'd0, useDlIRxClkPhaseD*_u0[5:0]]
- 6'd13 [4'd0, useRxEnDlyTg*_u0[11:6],useDlIRxEnPhaseD*_u0[5:0]]
- 6'd14 [10'd0, useDlIRxClkPhaseD*_u1[5:0]]
- 6'd15 [10'd0, useDlIRxClkPhaseD*_u1[5:0]]
- 6'd16 [4'd0, useRxEnDlyTg*_u1[11:6],useDlIRxEnPhaseD*_u1[5:0]]
- 6'd17 [9'd0, useRxPBDlyTg*_r0[6:0]]
- 6'd18 [9'd0, useRxPBDlyTg*_r1[6:0]]
- 6'd19 [9'd0, useRxPBDlyTg*_r2[6:0]]
- 6'd20 [9'd0, useRxPBDlyTg*_r3[6:0]]
- 6'd21 [9'd0, useRxPBDlyTg*_r4[6:0]]
- 6'd22 [9'd0, useRxPBDlyTg*_r5[6:0]]
- 6'd23 [9'd0, useRxPBDlyTg*_r6[6:0]]
- 6'd24 [9'd0, useRxPBDlyTg*_r7[6:0]]
- 6'd25 [9'd0, useRxPBDlyTg*_r8[6:0]]
- >=6'd26 [16'd0] where * is the timing-group index 0..3

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[illegible]

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_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address003175C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address003176C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address003177C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address003178C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address003180C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address003181C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address003182C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address003183C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address003184C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address003185C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address003186C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address003187C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address003188C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address003190C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address003191C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address003192C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address003193C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address003194C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address003195C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address003196C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address003197C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address003198C0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:9	Reserved.
8:0	TxDqDlyTg0. Read-write. Reset: 010h.

Description: Write DQ Delay (Timing Group 0).

Trained to center the delay between the write DQ to the write DQS.

- Tx DQ Delay (9 bits, includes 3b coarse and 6b fine delays); (For Timing Group 0)

- TxDqDlyTg0[8:6] is the coarse delay, ie one unit of delay is 1 [UI](#).

- If Dly64Prec=0,

- TxDqDlyTg0[4:0] is the fine (fractional UI) delay, ie one unit of delay is one-thirtysecond of a UI = $UI/32$.

- TxDqDlyTg0[5] is reserved.

Note: The reset value is $1/2$ UI.

- If Dly64Prec=1,

- TxDqDlyTg0[5:0] is the fine (fractional UI) delay, ie one unit of delay is one-sixtyfourth of a UI = $UI/64$.

Note: The reset value is $1/4$ UI.

- Register Block Offset Address 0x0C0 controls the lane0 write DQ timing. (For Timing Group 0)

- Register Block Offset Address 0x1C0 controls the lane1 write DQ timing. (For Timing Group 0)

- Register Block Offset Address 0x2C0 controls the lane2 write DQ timing. (For Timing Group 0)

- Register Block Offset Address 0x3C0 controls the lane3 write DQ timing. (For Timing Group 0)

- Register Block Offset Address 0x4C0 controls the lane4 write DQ timing. (For Timing Group 0)

- Register Block Offset Address 0x5C0 controls the lane5 write DQ timing. (For Timing Group 0)

- Register Block Offset Address 0x6C0 controls the lane6 write DQ timing. (For Timing Group 0)

- Register Block Offset Address 0x7C0 controls the lane7 write DQ timing. (For Timing Group 0)

- Register Block Offset Address 0x8C0 controls the lane[DMDBI] write DQ timing. (For Timing Group 0)

Note: These Registers are replicated per pstate.

Note: Timing Group 0 = RANK0 for all systems except 3DS/LRDIMM.

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_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address003175C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address003176C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address003177C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address003178C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address003180C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address003181C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address003182C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address003183C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address003184C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address003185C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address003186C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address003187C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address003188C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address003190C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address003191C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address003192C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address003193C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address003194C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address003195C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address003196C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address003197C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address003198C1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:9	Reserved.
8:0	TxDqDlyTg1. Read-write. Reset: 010h.

Description: Write DQ Delay (Timing Group 1).

Trained to center the delay between the write DQ to the write DQS.

- Tx DQ Delay (9 bits, includes 3b coarse and 6b fine delays); (For Timing Group 1)

- Tx DqDlyTg1[8:6] is the coarse delay, ie one unit of delay is 1 [UI](#).

- If Dly64Prec=0,

- Tx DqDlyTg1[4:0] is the fine (fractional UI) delay, ie one unit of delay is one-thirtysecond of a UI = $UI/32$.

- Tx DqDlyTg1[5] is reserved.

Note: The reset value is $1/2$ UI.

- If Dly64Prec=1,

- Tx DqDlyTg1[5:0] is the fine (fractional UI) delay, ie one unit of delay is one-sixtyfourth of a UI = $UI/64$.

Note: The reset value is $1/4$ UI.

- Register Block Offset Address 0x0C1 controls the lane0 write DQ timing. (For Timing Group 1)

- Register Block Offset Address 0x1C1 controls the lane1 write DQ timing. (For Timing Group 1)

- Register Block Offset Address 0x2C1 controls the lane2 write DQ timing. (For Timing Group 1)

- Register Block Offset Address 0x3C1 controls the lane3 write DQ timing. (For Timing Group 1)

- Register Block Offset Address 0x4C1 controls the lane4 write DQ timing. (For Timing Group 1)

- Register Block Offset Address 0x5C1 controls the lane5 write DQ timing. (For Timing Group 1)

- Register Block Offset Address 0x6C1 controls the lane6 write DQ timing. (For Timing Group 1)

- Register Block Offset Address 0x7C1 controls the lane7 write DQ timing. (For Timing Group 1)

- Register Block Offset Address 0x8C1 controls the lane[DIMM] write DQ timing. (For Timing Group 1)

Note: These Registers are replicated per pstate.

Note: Timing Group 1 = RANK1 for all systems except 3DS/LRDIMM.

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_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address003175C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address003176C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address003177C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address003178C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address003180C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address003181C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address003182C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address003183C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address003184C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address003185C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address003186C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address003187C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address003188C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address003190C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address003191C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address003192C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address003193C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address003194C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address003195C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address003196C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address003197C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address003198C2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:9	Reserved.
8:0	TxDqDlyTg2. Read-write. Reset: 010h.

Description: Write DQ Delay (Timing Group 2).

Trained to center the delay between the write DQ to the write DQS.

- Tx DQ Delay (9 bits, includes 3b coarse and 6b fine delays); (For Timing Group 2)

- TxDqDlyTg2[8:6] is the coarse delay, ie one unit of delay is 1 [UI](#).

- If Dly64Prec=0,

- TxDqDlyTg2[4:0] is the fine (fractional UI) delay, ie one unit of delay is one-thirtysecond of a UI = $UI/32$.

- TxDqDlyTg2[5] is reserved.

Note: The reset value is $1/2$ UI.

- If Dly64Prec=1,

- TxDqDlyTg2[5:0] is the fine (fractional UI) delay, ie one unit of delay is one-sixtyfourth of a UI = $UI/64$.

Note: The reset value is $1/4$ UI.

- Register Block Offset Address 0x0C2 controls the lane0 write DQ timing. (For Timing Group 2)

- Register Block Offset Address 0x1C2 controls the lane1 write DQ timing. (For Timing Group 2)

- Register Block Offset Address 0x2C2 controls the lane2 write DQ timing. (For Timing Group 2)

- Register Block Offset Address 0x3C2 controls the lane3 write DQ timing. (For Timing Group 2)

- Register Block Offset Address 0x4C2 controls the lane4 write DQ timing. (For Timing Group 2)

- Register Block Offset Address 0x5C2 controls the lane5 write DQ timing. (For Timing Group 2)

- Register Block Offset Address 0x6C2 controls the lane6 write DQ timing. (For Timing Group 2)

- Register Block Offset Address 0x7C2 controls the lane7 write DQ timing. (For Timing Group 2)

- Register Block Offset Address 0x8C2 controls the lane[DMDBI] write DQ timing. (For Timing Group 2)

Note: These Registers are replicated per pstate.

Note: Timing Group 2 = RANK2 for all systems except 3DS/LRDIMM.

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address003175C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address003176C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address003177C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address003178C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address003180C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address003181C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address003182C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address003183C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address003184C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address003185C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address003186C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address003187C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address003188C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r0_aliasSMN; UMC[11:00]CTLx00000804_address003190C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r1_aliasSMN; UMC[11:00]CTLx00000804_address003191C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r2_aliasSMN; UMC[11:00]CTLx00000804_address003192C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r3_aliasSMN; UMC[11:00]CTLx00000804_address003193C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r4_aliasSMN; UMC[11:00]CTLx00000804_address003194C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r5_aliasSMN; UMC[11:00]CTLx00000804_address003195C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r6_aliasSMN; UMC[11:00]CTLx00000804_address003196C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r7_aliasSMN; UMC[11:00]CTLx00000804_address003197C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_r8_aliasSMN; UMC[11:00]CTLx00000804_address003198C3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:9	Reserved.
8:0	TxDqDlyTg3. Read-write. Reset: 010h.

Description: Write DQ Delay (Timing Group 3).

Trained to center the delay between the write DQ to the write DQS.

- Tx DQ Delay (9 bits, includes 3b coarse and 6b fine delays); (For Timing Group 3)

- Tx DqDlyTg3[8:6] is the coarse delay, ie one unit of delay is 1 [UI](#).

- If Dly64Prec=0,

- Tx DqDlyTg3[4:0] is the fine (fractional UI) delay, ie one unit of delay is one-thirtysecond of a UI = $UI/32$.

- Tx DqDlyTg3[5] is reserved.

Note: The reset value is $1/2$ UI.

- If Dly64Prec=1,

- Tx DqDlyTg3[5:0] is the fine (fractional UI) delay, ie one unit of delay is one-sixtyfourth of a UI = $UI/64$.

Note: The reset value is $1/4$ UI.

- Register Block Offset Address 0x0C3 controls the lane0 write DQ timing. (For Timing Group 3)

- Register Block Offset Address 0x1C3 controls the lane1 write DQ timing. (For Timing Group 3)

- Register Block Offset Address 0x2C3 controls the lane2 write DQ timing. (For Timing Group 3)

- Register Block Offset Address 0x3C3 controls the lane3 write DQ timing. (For Timing Group 3)

- Register Block Offset Address 0x4C3 controls the lane4 write DQ timing. (For Timing Group 3)

- Register Block Offset Address 0x5C3 controls the lane5 write DQ timing. (For Timing Group 3)

- Register Block Offset Address 0x6C3 controls the lane6 write DQ timing. (For Timing Group 3)

- Register Block Offset Address 0x7C3 controls the lane7 write DQ timing. (For Timing Group 3)

- Register Block Offset Address 0x8C3 controls the lane[DIMM] write DQ timing. (For Timing Group 3)

Note: These Registers are replicated per pstate.

Note: Timing Group 3 = RANK3 for all systems except 3DS/LRDIMM.

Read-write. Reset: 0100h.

Write DQS Delay (Timing Group DEST=0).

```
_umc0_chiplet0_chiplettype1_inst[UMCWHY[11:0]]_p0_u0_aliasSMN; UMC[11:0]CTLx00000804_address000100D0; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:0]CTLx00000804_address000101D0; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:0]CTLx00000804_address000110D0; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:0]CTLx00000804_address000111D0; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:00]CTLx00000804_address000120D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:0]CTLx00000804_address000121D0; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:0]CTLx00000804_address000130D0; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:0]CTLx00000804_address000131D0; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]
```

```
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:0]CTLx00000804_address000140D0; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:0]CTLx00000804_address000141D0; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:00]CTLx00000804_address000150D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:0]CTLx00000804_address000151D0; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]
```

```
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:00]CTLx00000804_address000160D0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:0]CTLx00000804_address000161D0; UMC[11:0]CTL=00[B:0]5_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:00]CTLx00000804_address000170D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]
```

```
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:00]CTLx00000804_address000171D0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:0]CTLx00000804_address000180D0; UMC[11:0]CTL=00[B:0]5_1000h;
DataPortWrite=[UMC::ApbCmd[Address]
```

```
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:00]CTLx00000804_address000181D0; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_u0_aliasSMN; UMC[11:0]CTLx00000804_address000190D0; UMC[11:0]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]
```

```
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p0_u1_aliasSMN; UMC[11:00]CTLx00000804_address000191D0; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_u0_aliasSMN; UMC[11:00]CTLx00000804_address001100D0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::AnhCmd[Address]
```

```
_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_p1_u1_aliasSMN; UMC[11:00]CTLx00000804_address001101D0; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_u0_aliasSMN; UMC[11:00]CTLx00000804_address001110D0; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p1_u1_aliasSMN; UMC[11:00]CTLx00000804_address001111D0; UMC[11:00]CTL=00[B:0]5_1000h;
```

```
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_u0_aliasSMN; UMC[11:00]CTLx00000804_address001120D0; UMC[11:00]CTL=00[B:0]5_1000h;
```

```

_dataPortWrite=UMC...ApbCmd[Address]
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p1_u1_aliasSMN; UMC[11:00]CTLx00000804_address001121D0; UMC[11:00]CTL=00[B:0]5_1000h;
_dataPortWrite=UMC...ApbCmd[Address]

```

```

_dataPortWrite=UMC::ApbCmd[Address]
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_u0_aliasSMN; UMC[11:00]CTLx00000804_address001130D0; UMC[11:00]CTL=00[B:0]5_1000h;
_dataPortWrite=UMC::ApbCmd[Address]

```

```
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p1_u1_aliasSMN; UMC[11:00]CTLx00000804_addradd001131D0; UMC[11:00]CTL=00[B:0]5_1000h;
```

```

_dataPortWrite=UMC...ApbCmnd[Address]
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p1_u0_aliasSMN; UMC[11:00]CTLx00000804_address001140D0; UMC[11:00]CTL=00[B:0]5_1000h;
_dataPortWrite=UMC...ApbCmnd[Address]

```

```

DataPortWrite=UMC::ApbCmnd[Address]
_umc0_chiplet4_chipletpe1_inst[UMCWPHY[11:0]]_p1_u1_aliasSMN; UMC[11:00]CTLx00000804_address001141D0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmnd[Address]

```

```
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p1_u0_aliasSMN; UMC[11:00]CTLx00000804_address001150D0; UMC[11:00]CTL=00[B:0]5_1000h;
```

[illegible]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003121D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003130D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003131D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003140D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003141D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003150D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003151D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003160D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003161D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003170D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003171D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003180D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003181D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003190D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003191D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:11	Reserved.
10:0	TxDqsDlyTg0. Read-write. Reset: 100h. Description: Write DQS Delay (Timing Group DEST=0). Trained to set the delay from the memory-write command to the signal driving the write DQS - Tx DQS Delay (11 bits, includes coarse and fine delays).(For Timing Group 0) - TxDqsDlyTg0[10:6] is the coarse delay, ie one unit of delay is 1 UI. The maximum value allowed is 26, 0x1A. - If Dly64Prec=0, - TxDqsDlyTg0[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32. - TxDqsDlyTg0[5] is reserved. - If Dly64Prec=1, - TxDqsDlyTg0[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64. Register is per nibble, per timing-group, per pstate. - Register Block Offset Address 0x0D0 controls the lower DQ nibble write DQS timing. (For Timing Group 0) - Register Block Offset Address 0x1D0 controls the upper DQ nibble write DQS timing. (For Timing Group 0) Note: Timing Group 0 = RANK0 for all systems except 3DS/LRDIMM.

[illegible]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003121D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003130D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003131D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003140D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003141D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003150D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003151D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003160D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003161D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003170D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003171D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003180D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003181D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003190D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003191D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:11	Reserved.
10:0	TxDqsDlyTg1. Read-write. Reset: 100h. Description: Write DQS Delay (Timing Group DEST=1). Trained to set the delay from the memory-write command to the signal driving the write DQS - Tx DQS Delay (11 bits, includes coarse and fine delays).(For Timing Group 1) - TxDqsDlyTg1[10:6] is the coarse delay, ie one unit of delay is 1 UI. The maximum value allowed is 26, 0x1A. - If Dly64Prec=0, - TxDqsDlyTg1[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32. - TxDqsDlyTg1[5] is reserved. - If Dly64Prec=1, - TxDqsDlyTg1[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64. Register is per nibble, per timing-group, per pstate. - Register Block Offset Address 0x0D1 controls the lower DQ nibble write DQS timing. (For Timing Group 1) - Register Block Offset Address 0x1D1 controls the upper DQ nibble write DQS timing. (For Timing Group 1) Note: Timing Group 1 = RANK1 for all systems except 3DS/LRDIMM.

[illegible]

[illegible]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003121D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003130D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003131D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003140D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003141D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003150D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003151D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003160D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003161D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003170D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003171D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003180D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003181D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003190D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003191D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:11	Reserved.
10:0	TxDqsDlyTg2. Read-write. Reset: 100h. Description: Write DQS Delay (Timing Group DEST=2). Trained to set the delay from the memory-write command to the signal driving the write DQS - Tx DQS Delay (11 bits, includes coarse and fine delays).(For Timing Group 2) - TxDqsDlyTg2[10:6] is the coarse delay, ie one unit of delay is 1 UI. The maximum value allowed is 26, 0x1A. - If Dly64Prec=0, - TxDqsDlyTg2[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32. - TxDqsDlyTg2[5] is reserved. - If Dly64Prec=1, - TxDqsDlyTg2[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64. Register is per nibble, per timing-group, per pstate. - Register Block Offset Address 0x0D2 controls the lower DQ nibble write DQS timing. (For Timing Group 2) - Register Block Offset Address 0x1D2 controls the upper DQ nibble write DQS timing. (For Timing Group 2) Note: Timing Group 2 = RANK2 for all systems except 3DS/LRDIMM.

[illegible]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003121D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003130D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003131D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003140D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003141D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003150D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003151D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003160D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003161D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003170D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003171D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003180D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003181D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u0_aliasSMN; UMC[11:00]CTLx00000804_address003190D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_u1_aliasSMN; UMC[11:00]CTLx00000804_address003191D3; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:11	Reserved.
10:0	TxDqsDlyTg3. Read-write. Reset: 100h. Description: Write DQS Delay (Timing Group DEST=3). Trained to set the delay from the memory-write command to the signal driving the write DQS - Tx DQS Delay (11 bits, includes coarse and fine delays).(For Timing Group 3) - TxDqsDlyTg3[10:6] is the coarse delay, ie one unit of delay is 1 UI. The maximum value allowed is 26, 0x1A. - If Dly64Prec=0, - TxDqsDlyTg3[4:0] is the fine delay, ie one unit of delay is one-thirtysecond of a UI = UI/32. - TxDqsDlyTg3[5] is reserved. - If Dly64Prec=1, - TxDqsDlyTg3[5:0] is the fine delay, ie one unit of delay is one-sixtyfourth of a UI = UI/64. Register is per nibble, per timing-group, per pstate. - Register Block Offset Address 0x0D3 controls the lower DQ nibble write DQS timing. (For Timing Group 3) - Register Block Offset Address 0x1D3 controls the upper DQ nibble write DQS timing. (For Timing Group 3) Note: Timing Group 3 = RANK3 for all systems except 3DS/LRDIMM.

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_umc0_chiplet7_chiplettype1_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000172E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000173E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000174E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000175E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000176E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000177E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000178E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000181E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000182E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000183E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000184E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000185E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000186E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000187E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000188E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000191E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000192E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000193E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000194E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000195E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000196E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000197E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000198E1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:6	Reserved.
5:0	RFU_Dtsmprbserresetup. Read-write. Reset: 00h.

UMC00CTLx00000804_address000100E3...UMC11CTLx00000804_address000190E3 (UMC::Phy::DxPhase1UI)

Read-only, Volatile.

DBYTE DLLPhase1UI Code_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100E3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110E3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120E3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130E3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140E3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150E3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160E3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170E3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180E3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190E3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:10	Reserved.
9:0	DxPhase1UI . Read-only, Volatile. The raw DBYTE DLL Phase1UI code, used for debug only, selected by Register LcdlStatusSel

UMC00CTLx00000804_address000100E4...UMC11CTLx00000804_address000190E4 [Debug status of the DBYTE LCDL] (UMC::Phy::DxLcdlStatus)

Read-only, Volatile.

Status of DBYTE macro LCDL selected by Register LcdlStatusSel.

Note: This register is per DBYTE.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100E4; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110E4; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120E4; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130E4; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140E4; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150E4; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160E4; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170E4; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180E4; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190E4; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13	DxLcdlLiveLock. Read-only, Volatile. Present value of whether the LCDL is locked, valid when LcdlTstEnable=1.
12	DxLcdlStickyUnlock. Read-only, Volatile. Latched value of whether the LCDL ever lost lock after the assertion of LcdlTstEnable.
11	DxLcdlStickyLock. Read-only, Volatile. Latched value of whether the LCDL ever achieved lock after the assertion of LcdlTstEnable.
10	DxLcdlPhdSnapVal. Read-only, Volatile. Value of the LCDL phase-detector output, latched by pulse on csr LcdlFineSnap while csr LcdlTstEnable=1.
9:0	DxLcdlFineSnapVal. Read-only, Volatile. Description: Value of the LCDL 1UI estimate code, latched by pulse on csr LcdlFineSnap while csr LcdlTstEnable=1. Index 9 is reserved for growth.

UMC00CTLx00000804_address000100E5...UMC11CTLx00000804_address000190E5 (UMC::Phy::DlyTestCntRingOscDbldcl0)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100E5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110E5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120E5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130E5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140E5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150E5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160E5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170E5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180E5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190E5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits **Description**

15:0 **DlyTestCntRingOscDbldcl0.** Read-only, Volatile. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000100E6...UMC11CTLx00000804_address000190E6 (UMC::Phy::DlyTestCntRingOscDbldcl1)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100E6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110E6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120E6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130E6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140E6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150E6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160E6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170E6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180E6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190E6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits **Description**

15:0 **DlyTestCntRingOscDbldcl1.** Read-only, Volatile. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000100E7...UMC11CTLx00000804_address000190E7 (UMC::Phy::DlyTestCntRingOscDbldcl2)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcl2. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100E8...UMC11CTLx00000804_address000190E8 (UMC::Phy::DlyTestCntRingOscDbldcl3)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100E8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110E8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120E8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130E8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140E8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150E8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160E8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170E8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180E8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190E8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcl3. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100E9...UMC11CTLx00000804_address000190E9 (UMC::Phy::DlyTestCntRingOscDbldcl4)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100E9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110E9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120E9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130E9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140E9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150E9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160E9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170E9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180E9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190E9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcl4. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100EA...UMC11CTLx00000804_address000190EA (UMC::Phy::DlyTestCntRingOscDbldcl5)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcl5. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100EB...UMC11CTLx00000804_address000190EB (UMC::Phy::DlyTestCntRingOscDbldcl6)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcl6. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100EC...UMC11CTLx00000804_address000190EC (UMC::Phy::DlyTestCntRingOscDbldcl7)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100EC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110EC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120EC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130EC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140EC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150EC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160EC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170EC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180EC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190EC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcl7. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100ED...UMC11CTLx00000804_address000190ED (UMC::Phy::DlyTestCntRingOscDbldcl8)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100ED; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110ED; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120ED; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130ED; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140ED; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150ED; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160ED; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170ED; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180ED; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190ED; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcl8. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100EE...UMC11CTLx00000804_address000190EE (UMC::Phy::DlyTestCntRingOscDbldcl9)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100EE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110EE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120EE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130EE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140EE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150EE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160EE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170EE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180EE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190EE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcl9. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100EF...UMC11CTLx00000804_address000190EF (UMC::Phy::DlyTestCntRingOscDbldcdl10)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100EF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110EF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120EF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130EF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140EF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150EF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160EF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170EF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180EF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190EF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl10. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100F0...UMC11CTLx00000804_address000190F0 (UMC::Phy::DlyTestCntRingOscDbldcdl11)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100F0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110F0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120F0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130F0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140F0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150F0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160F0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170F0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180F0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190F0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl11. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100F1...UMC11CTLx00000804_address000190F1 (UMC::Phy::DlyTestCntRingOscDbldcdl12)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100F1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110F1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120F1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130F1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140F1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150F1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160F1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170F1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180F1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190F1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl12. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100F2...UMC11CTLx00000804_address000190F2 (UMC::Phy::DlyTestCntRingOscDbldcdl13)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100F2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110F2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120F2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130F2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140F2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150F2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160F2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170F2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180F2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190F2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl13. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100F3...UMC11CTLx00000804_address000190F3 (UMC::Phy::DlyTestCntRingOscDbldcdl14)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100F3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110F3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120F3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130F3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140F3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150F3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160F3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170F3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180F3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190F3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl14. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100F4...UMC11CTLx00000804_address000190F4 (UMC::Phy::DlyTestCntRingOscDbldcdl15)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100F4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110F4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120F4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130F4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140F4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150F4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160F4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170F4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180F4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190F4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl15. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100F5...UMC11CTLx00000804_address000190F5 (UMC::Phy::DlyTestCntRingOscDbldcdl16)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100F5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110F5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120F5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130F5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140F5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150F5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160F5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170F5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180F5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190F5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl16. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100F6...UMC11CTLx00000804_address000190F6 (UMC::Phy::DlyTestCntRingOscDbldcdl17)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100F6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110F6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120F6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130F6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140F6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150F6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160F6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170F6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180F6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190F6; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl17. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100F7...UMC11CTLx00000804_address000190F7 (UMC::Phy::DlyTestCntRingOscDbldcdl18)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100F7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110F7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120F7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130F7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140F7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150F7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160F7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170F7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180F7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190F7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl18. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100F8...UMC11CTLx00000804_address000190F8 (UMC::Phy::DlyTestCntRingOscDbldcdl19)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100F8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110F8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120F8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130F8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140F8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150F8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160F8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170F8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180F8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190F8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl19. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100F9...UMC11CTLx00000804_address000190F9 (UMC::Phy::DlyTestCntRingOscDbldcdl20)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100F9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110F9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120F9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130F9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140F9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150F9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160F9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170F9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180F9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190F9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl20. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100FA...UMC11CTLx00000804_address000190FA (UMC::Phy::DlyTestCntRingOscDbldcdl21)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl21. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100FB...UMC11CTLx00000804_address000190FB (UMC::Phy::DlyTestCntRingOscDbldcdl22)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100FB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110FB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120FB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130FB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140FB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150FB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160FB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170FB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180FB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190FB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl22. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100FC...UMC11CTLx00000804_address000190FC (UMC::Phy::DlyTestCntRingOscDbldcdl23)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100FC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110FC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120FC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130FC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140FC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150FC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160FC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170FC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180FC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190FC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl23. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000100FD...UMC11CTLx00000804_address000190FD (UMC::Phy::DlyTestCntRingOscDbldcdl24)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000100FD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000110FD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000120FD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000130FD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000140FD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000150FD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000160FD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000170FD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000180FD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000190FD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DlyTestCntRingOscDbldcdl24. Read-only, Volatile. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000101AF...UMC11CTLx00000804_address000191AF (UMC::Phy::DbCurrentDlyTimingInfoTg1)

Read-only, Volatile.

Reserved for PHY training firmware use and for debugging

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000101AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000111AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000121AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000131AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000141AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000151AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000161AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000171AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000181AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000191AF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DbCurrentDlyTimingInfoTg1. Read-only, Volatile. Reserved for PHY training firmware use and for debugging
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UMC00CTLx00000804_address000102AF...UMC11CTLx00000804_address000192AF (UMC::Phy::DbCurrentDlyTimingInfoTg2)

Read-only, Volatile.

Reserved for PHY training firmware use and for debugging

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000102AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000112AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000122AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000132AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000142AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000152AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000162AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000172AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000182AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000192AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DbCurrentDlyTimingInfoTg2. Read-only, Volatile. Reserved for PHY training firmware use and for debugging
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UMC00CTLx00000804_address000103AF...UMC11CTLx00000804_address000193AF (UMC::Phy::DbCurrentDlyTimingInfoTg3)

Read-only, Volatile.

Reserved for PHY training firmware use and for debugging

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000103AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000113AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000123AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000133AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000143AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000153AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000163AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000173AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000183AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000193AF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	DbCurrentDlyTimingInfoTg3. Read-only, Volatile. Reserved for PHY training firmware use and for debugging
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UMC00CTLx00000804_address000103F0...UMC11CTLx00000804_address000193F0 **(UMC::Phy::LnTgSelForceTg0)**

Read-write. Reset: 0000h.

Force all Timing Groups to use csrDq<n>LnSelTg0

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000103F0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000113F0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000123F0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000133F0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000143F0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000153F0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000163F0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000173F0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000183F0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000193F0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:1	Reserved.
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0	LnTgSelForceTg0. Read-write. Reset: 0.
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Description: When set, the values of csrDq<0..7>LnSel<1..3> are overridden by csrDq<0,,7>LnSelTg0

- 0 -- LnSelTg<0..3> are all individually assigned.

- 1 -- LnSelTg<0..3> are all assigned to LnSelTg0.

UMC00CTLx00000804_address00010500...UMC11CTLx00000804_address00019500 (UMC::Phy::Dq0LnSelTg1)

Read-write. Reset: 0000h.

Maps PHY DQ lane to memory DQ0_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010500; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011500; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012500; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013500; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014500; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015500; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016500; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017500; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018500; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019500; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq0LnSelTg1. Read-write. Reset: 0h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq0 (on this DBYTE), then Register Dq0LnSelTg1 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010501...UMC11CTLx00000804_address00019501 (UMC::Phy::Dq1LnSelTg1)

Read-write. Reset: 0001h.

Maps PHY DQ lane to memory DQ1_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010501; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011501; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012501; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013501; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014501; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015501; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016501; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017501; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018501; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019501; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq1LnSelTg1. Read-write. Reset: 1h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq1 (on this DBYTE), then Register Dq1LnSelTg1 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0</br>

UMC00CTLx00000804_address00010502...UMC11CTLx00000804_address00019502 (UMC::Phy::Dq2LnSelTg1)

Read-write. Reset: 0002h.

Maps PHY DQ lane to memory DQ2_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010502; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011502; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012502; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013502; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014502; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015502; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016502; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017502; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018502; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019502; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq2LnSelTg1. Read-write. Reset: 2h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq2 (on this DBYTE), then Register Dq2LnSelTg1 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010503...UMC11CTLx00000804_address00019503 (UMC::Phy::Dq3LnSelTg1)

Read-write. Reset: 0003h.

Maps PHY DQ lane to memory DQ3_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010503; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011503; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012503; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013503; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014503; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015503; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016503; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017503; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018503; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019503; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq3LnSelTg1. Read-write. Reset: 3h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq3 (on this DBYTE), then Register Dq3LnSelTg1 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010504...UMC11CTLx00000804_address00019504 (UMC::Phy::Dq4LnSelTg1)

Read-write. Reset: 0004h.

Maps PHY DQ lane to memory DQ4_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010504; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011504; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012504; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013504; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014504; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015504; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016504; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017504; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018504; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019504; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq4LnSelTg1. Read-write. Reset: 4h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq4 (on this DBYTE), then Register Dq4LnSelTg1 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010505...UMC11CTLx00000804_address00019505 (UMC::Phy::Dq5LnSelTg1)

Read-write. Reset: 0005h.

Maps PHY DQ lane to memory DQ5

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010505; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011505; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012505; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013505; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014505; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015505; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016505; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017505; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018505; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019505; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq5LnSelTg1. Read-write. Reset: 5h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq5 (on this DBYTE), then Register Dq5LnSelTg1 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010506...UMC11CTLx00000804_address00019506 (UMC::Phy::Dq6LnSelTg1)

Read-write. Reset: 0006h.

Maps PHY DQ lane to memory DQ6_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010506; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011506; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012506; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013506; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014506; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015506; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016506; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017506; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018506; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019506; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq6LnSelTg1. Read-write. Reset: 6h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq6 (on this DBYTE), then Register Dq6LnSelTg1 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010507...UMC11CTLx00000804_address00019507 (UMC::Phy::Dq7LnSelTg1)

Read-write. Reset: 0007h.

Maps PHY DQ lane to memory DQ7

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010507; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011507; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012507; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013507; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014507; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015507; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016507; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017507; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018507; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019507; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq7LnSelTg1. Read-write. Reset: 7h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq7 (on this DBYTE), then Register Dq7LnSelTg1 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010508...UMC11CTLx00000804_address00019508 (UMC::Phy::Dq0LnSelTg2)

Read-write. Reset: 0000h.

Maps PHY DQ lane to memory DQ0_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010508; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011508; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012508; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013508; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014508; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015508; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016508; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017508; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018508; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019508; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq0LnSelTg2. Read-write. Reset: 0h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq0 (on this DBYTE), then Register Dq0LnSelTg2 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010509...UMC11CTLx00000804_address00019509 (UMC::Phy::Dq1LnSelTg2)

Read-write. Reset: 0001h.

Maps PHY DQ lane to memory DQ1

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010509; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011509; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012509; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013509; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014509; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015509; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016509; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017509; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018509; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019509; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq1LnSelTg2. Read-write. Reset: 1h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq1 (on this DBYTE), then Register Dq1LnSelTg2 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0</br>

UMC00CTLx00000804_address0001050A...UMC11CTLx00000804_address0001950A **(UMC::Phy::Dq2LnSelTg2)**

Read-write. Reset: 0002h.

Maps PHY DQ lane to memory DQ2

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001050A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001150A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001250A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001350A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001450A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001550A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001650A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001750A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001850A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001950A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq2LnSelTg2. Read-write. Reset: 2h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq2 (on this DBYTE), then Register Dq2LnSelTg2 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0</br>

UMC00CTLx00000804_address0001050B...UMC11CTLx00000804_address0001950B (UMC::Phy::Dq3LnSelTg2)

Read-write. Reset: 0003h.

Maps PHY DQ lane to memory DQ3

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001050B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001150B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001250B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001350B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001450B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001550B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001650B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001750B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001850B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001950B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:3	Reserved.
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2:0	Dq3LnSelTg2. Read-write. Reset: 3h.
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Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle).
Intended to undo the swizzle of board-level phy-to-dram connections.
such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE.
Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to
memory dq3 (on this DBYTE), then Register Dq3LnSelTg2 for this DBYTE should be 3.

Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by
csrDq<0..7>LnSelTg0</br>

UMC00CTLx00000804_address0001050C...UMC11CTLx00000804_address0001950C **(UMC::Phy::Dq4LnSelTg2)**

Read-write. Reset: 0004h.

Maps PHY DQ lane to memory DQ4

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001050C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001150C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001250C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001350C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001450C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001550C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001650C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001750C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001850C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001950C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15:3 Reserved.

2:0 **Dq4LnSelTg2.** Read-write. Reset: 4h.

Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle).

Intended to undo the swizzle of board-level phy-to-dram connections.

such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE.

Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to memory dq4 (on this DBYTE), then Register Dq4LnSelTg2 for this DBYTE should be 3.

Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0</br>

UMC00CTLx00000804_address0001050D...UMC11CTLx00000804_address0001950D (UMC::Phy::Dq5LnSelTg2)

Read-write. Reset: 0005h.

Maps PHY DQ lane to memory DQ5

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001050D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001150D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001250D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001350D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001450D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001550D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001650D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001750D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001850D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001950D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15:3 Reserved.

2:0 **Dq5LnSelTg2.** Read-write. Reset: 5h.

Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle).
Intended to undo the swizzle of board-level phy-to-dram connections.
such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE.
Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to
memory dq5 (on this DBYTE), then Register Dq5LnSelTg2 for this DBYTE should be 3.

Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by
csrDq<0..7>LnSelTg0</br>

UMC00CTLx00000804_address0001050E...UMC11CTLx00000804_address0001950E (UMC::Phy::Dq6LnSelTg2)

Read-write. Reset: 0006h.

Maps PHY DQ lane to memory DQ6

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001050E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001150E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001250E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001350E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001450E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001550E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001650E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001750E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001850E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001950E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15:3 Reserved.

2:0 **Dq6LnSelTg2.** Read-write. Reset: 6h.

Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle).
Intended to undo the swizzle of board-level phy-to-dram connections.
such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE.
Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to
memory dq6 (on this DBYTE), then Register Dq6LnSelTg2 for this DBYTE should be 3.

Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by
csrDq<0..7>LnSelTg0</br>

UMC00CTLx00000804_address0001050F..UMC11CTLx00000804_address0001950F (UMC::Phy::Dq7LnSelTg2)

Read-write. Reset: 0007h.

Maps PHY DQ lane to memory DQ7

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001050F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001150F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001250F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001350F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001450F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001550F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001650F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001750F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001850F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0001950F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15:3 Reserved.

2:0 **Dq7LnSelTg2.** Read-write. Reset: 7h.

Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values.

Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set.

For example, if, on this DBYTE, PHY lane 3 is connected to memory dq7 (on this DBYTE), then Register Dq7LnSelTg2 for this DBYTE should be 3.

Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0</br>

UMC00CTLx00000804_address00010510...UMC11CTLx00000804_address00019510 (UMC::Phy::Dq0LnSelTg3)

Read-write. Reset: 0000h.

Maps PHY DQ lane to memory DQ0

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010510; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011510; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012510; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013510; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014510; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015510; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016510; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017510; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018510; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019510; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq0LnSelTg3. Read-write. Reset: 0h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq0 (on this DBYTE), then Register Dq0LnSelTg3 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010511...UMC11CTLx00000804_address00019511 (UMC::Phy::Dq1LnSelTg3)

Read-write. Reset: 0001h.

Maps PHY DQ lane to memory DQ1

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010511; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011511; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012511; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013511; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014511; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015511; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016511; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017511; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018511; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019511; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq1LnSelTg3. Read-write. Reset: 1h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq1 (on this DBYTE), then Register Dq1LnSelTg3 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010512...UMC11CTLx00000804_address00019512 (UMC::Phy::Dq2LnSelTg3)

Read-write. Reset: 0002h.

Maps PHY DQ lane to memory DQ2_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010512; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011512; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012512; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013512; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014512; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015512; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016512; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017512; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018512; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019512; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq2LnSelTg3. Read-write. Reset: 2h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq2 (on this DBYTE), then Register Dq2LnSelTg3 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010513...UMC11CTLx00000804_address00019513 (UMC::Phy::Dq3LnSelTg3)

Read-write. Reset: 0003h.

Maps PHY DQ lane to memory DQ3_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010513; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011513; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012513; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013513; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014513; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015513; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016513; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017513; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018513; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019513; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq3LnSelTg3. Read-write. Reset: 3h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq3 (on this DBYTE), then Register Dq3LnSelTg3 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010514...UMC11CTLx00000804_address00019514 (UMC::Phy::Dq4LnSelTg3)

Read-write. Reset: 0004h.

Maps PHY DQ lane to memory DQ4

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010514; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011514; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012514; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013514; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014514; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015514; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016514; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017514; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018514; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019514; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq4LnSelTg3. Read-write. Reset: 4h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq4 (on this DBYTE), then Register Dq4LnSelTg3 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010515...UMC11CTLx00000804_address00019515 (UMC::Phy::Dq5LnSelTg3)

Read-write. Reset: 0005h.

Maps PHY DQ lane to memory DQ5

_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010515; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011515; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012515; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013515; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014515; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015515; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016515; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017515; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018515; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019515; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq5LnSelTg3. Read-write. Reset: 5h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq5 (on this DBYTE), then Register Dq5LnSelTg3 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010516...UMC11CTLx00000804_address00019516 (UMC::Phy::Dq6LnSelTg3)

Read-write. Reset: 0006h.

Maps PHY DQ lane to memory DQ6_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010516; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011516; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012516; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013516; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014516; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015516; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016516; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017516; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018516; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019516; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq6LnSelTg3. Read-write. Reset: 6h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq6 (on this DBYTE), then Register Dq6LnSelTg3 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0

UMC00CTLx00000804_address00010517...UMC11CTLx00000804_address00019517 (UMC::Phy::Dq7LnSelTg3)

Read-write. Reset: 0007h.

Maps PHY DQ lane to memory DQ7_umc0_chiplet0_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00010517; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00011517; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00012517; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00013517; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00014517; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00015517; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00016517; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00017517; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00018517; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00019517; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	Dq7LnSelTg3. Read-write. Reset: 7h. Description: Supports mapping of PHY DQ to DRAM DQ within a byte (swizzle). Intended to undo the swizzle of board-level phy-to-dram connections. such that MRR operations of binary counters may be return correct values. Note: This register is per-DBYTE such that the swizzle may be different per DBYTE. Each register in a byte's set of DqLnSelTg must have a unique value within the set. For example, if, on this DBYTE, PHY lane 3 is connected to memory dq7 (on this DBYTE), then Register Dq7LnSelTg3 for this DBYTE should be 3. Note: When LnTgSelForceTg0 is set, csrDq<0..7>LnSelTg<1..3> value is overridden by csrDq<0..7>LnSelTg0</br>

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312520; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313520; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314520; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315520; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316520; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317520; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318520; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319520; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:10	Reserved.
9:0	TxTrkStateS0 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312521; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313521; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314521; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315521; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316521; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317521; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318521; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319521; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:10	Reserved.
9:0	TxTrkStateS1 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312522; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313522; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314522; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315522; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316522; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317522; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318522; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319522; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:10	Reserved.
9:0	TxTrkStateS2 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312523; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313523; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314523; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315523; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316523; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317523; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318523; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319523; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:10	Reserved.
9:0	TxTrkStateS3 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312524; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313524; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314524; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315524; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316524; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317524; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318524; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319524; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:10	Reserved.
9:0	TxTrkStateS4 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312525; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313525; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314525; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315525; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316525; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317525; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318525; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319525; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:10	Reserved.
9:0	TxTrkStateS5 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312526; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313526; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314526; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315526; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316526; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317526; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318526; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319526; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:10	Reserved.
9:0	TxTrkStateS6 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312527; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313527; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314527; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315527; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316527; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317527; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318527; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319527; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:10	Reserved.
9:0	TxTrkStateS7 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312528; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313528; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314528; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315528; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316528; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317528; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318528; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319528; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	RxTrkStateS0 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00312529; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00313529; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00314529; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00315529; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00316529; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00317529; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00318529; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00319529; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	RxTrkStateS1 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

[illegible]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031152A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031252A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031352A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031452A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031552A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031652A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031752A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031852A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031952A; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	RxTrkStateS2. Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

[illegible]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031152B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031252B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031352B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031452B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031552B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031652B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031752B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031852B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031952B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	RxTrkStateS3 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

[illegible]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031152C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031252C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031352C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031452C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031552C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031652C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031752C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031852C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031952C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	RxTrkStateS4. Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

[illegible]

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031152D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031252D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031352D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031452D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031552D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031652D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031752D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031852D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031952D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	RxTrkStateS5 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

_umc0_chiplet1_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031152E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031252E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031352E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031452E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031552E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031652E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031752E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031852E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031952E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	RxTrkStateS6. Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

_umc0_chiplet2_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031252F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet3_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031352F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet4_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031452F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet5_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031552F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet6_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031652F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet7_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031752F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet8_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031852F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chiplet9_chiplettype1_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0031952F; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:12	Reserved.
11:0	RxTrkStateS7 . Read-write. Reset: 000h. Reserved for bridging tracking results across S3.

UMC[00...11]CTLx00000804_address00020001 (UMC::Phy::ForceClkDisable)

Read-write. Reset: 0000h.

Clock gating control

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020001; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	ForceClkDisable . Read-write. Reset: 0h. Description: This CSR forces the gating of MEMCLKs driven from the PHY - ForceClkDisable[0] - controls CLK_[H/L]0. - ForceClkDisable[1] - controls CLK_[H/L]1.(if present) - ForceClkDisable[2] - controls CLK_[H/L]2.(if present) - ForceClkDisable[3] - controls CLK_[H/L]3.(if present)

UMC[00...11]CTLx00000804_address00020003 (UMC::Phy::ForceInternalUpdate)

Read-write. Reset: 0000h.

This Register used by Training Firmware to force an internal PHY Update Event.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020003; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	ForceInternalUpdate . Read-write. Reset: 0. Description: This Register is used by Training Firmware to force an internal PHY Update Event. The register may be written to 1'b1 only when all of the following are true: - The DFI interface is idle - The Training Hardware is sending only DES and all transactions have retired. Additionally, this Register must stay asserted for at least 32 DFICLKs before clearing. Prematurely clearing this register may result in an incomplete update.

UMC[00...11]CTLx00000804_address00020004 [Read Only displays PHY Configuration.]
(UMC::Phy::PhyConfig)

Read-only, Volatile.

Displays PHY Configuration

```
_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020004; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]
```

Bits	Description
15:10	Reserved.
9:8	PhyConfigDfi. Read-only, Volatile. Description: Returns the following value... depending on the define - 0x1 if DWC_DDRPHY_DFI1_EXISTS is not defined - 0x2 if DWC_DDRPHY_DFI1_EXISTS is defined
7:4	PhyConfigDbytes. Read-only, Volatile. Description: Returns the following value... depending on the chosen define - 0x1 if DWC_DDRPHY_NUM_DBYTES_1 - 0x2 if DWC_DDRPHY_NUM_DBYTES_2 - 0x3 if DWC_DDRPHY_NUM_DBYTES_3 - 0x4 if DWC_DDRPHY_NUM_DBYTES_4 - 0x5 if DWC_DDRPHY_NUM_DBYTES_5 - 0x6 if DWC_DDRPHY_NUM_DBYTES_6 - 0x7 if DWC_DDRPHY_NUM_DBYTES_7 - 0x8 if DWC_DDRPHY_NUM_DBYTES_8 - 0x9 if DWC_DDRPHY_NUM_DBYTES_9 - 0xA if DWC_DDRPHY_NUM_DBYTES_10
3:0	PhyConfigAnibs. Read-only, Volatile. Description: Returns the following value... depending on the chosen define - 0x6 if DWC_DDRPHY_NUM_ANIBS_6 - 0xA if DWC_DDRPHY_NUM_ANIBS_10 - 0xC if DWC_DDRPHY_NUM_ANIBS_12 - 0xE if DWC_DDRPHY_NUM_ANIBS_14

**UMC[00...11]CTLx00000804_address00020005 [PHY General Configuration Register(PGCR).]
(UMC::Phy::PGCR)**

Read-write. Reset: 0000h.

PHY General Configuration Register(PGCR)

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020005; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:9	Reserved.
8	LPGaterDis. Read-write. Reset: 0. Disables selective clock gating during LP mode. Should always be 0.
7	BsiInterfaceEnable. Read-write. Reset: 0. For saving power, set this bit to enable the Bsi Interface
6	PubPreCfgClkGateDisable. Read-write. Reset: 0. The clock to the pub piping of the Cfg*signals to the CSR blocks is normally gated. Setting this bit disables the gating.
5	PubCfgClkGateDisable. Read-write. Reset: 0. The clock to the pub piping of the CSRs is normally gated. Setting this bit disables the gating.
4:3	HardMacroOverride. Read-write. Reset: 0h. Description: The PUB determines the revision of Hard Macros from the PHYREV value. This field is an emergency override and should always be 0. 0x = disabled. 10 = HMC, 11 = HMD or later
2	MAAlertAsync. Read-write. Reset: 0. Description: To provide an async path from the MAAlert receiver to the dfi_alert_n pins, set this bit, and put the MAAlert Bump into Bypass Mode by setting the appropriate bit of csrAsyncAnibMode
1	DfiCtrlRxFifoRst. Read-write. Reset: 0. Description: When set, a phyupd or ctrlupd transaction resets the RxFifos. This will reset the rolling order when BL18 transactions are enabled in DDR5
0	RxClkRiseFallMode. Read-write. Reset: 0. Description: This register field controls independent training for RxClk_c and RxClk_t. - 0: Independent training for RxClk_c and RxClk_t is disabled. - 1: Independent training for RxClk_c and RxClk_t is enabled. Default Value is 0.

UMC[00...11]CTLx00000804_address00020006 [Value of dfi_freq_fsp pins] (UMC::Phy::FreqFsp)

Read-only, Volatile.

Value of dfi_freq_fsp pins - captured on rising edge of dfi[0,1]_init_start

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020006; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:2	FreqFsp1. Read-only, Volatile. Value of dfi1_freq_fsp[1:0] (or 0 in single-channel system)
1:0	FreqFsp0. Read-only, Volatile. Value of dfi0_freq_fsp[1:0]

UMC00CTLx00000804_address00020008...UMC11CTLx00000804_address00320008 [Impedance Calibration Clock Ratio] (UMC::Phy::CalUclkInfo)

Read-write. Reset: 0320h.

Informs the impedance calibration engine about frequency of DfiClk

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020008; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120008; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220008; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320008; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:11	Reserved.
10:0	CalUclkTicksPer1uS. Read-write. Reset: 320h. Description: Must be programmed to the number of DfiClks in 1us (rounded up), with minimum value of 24. - if (DfiClk < 24MHz) CalUclkInfo = 24 - else CalUclkInfo = (number of DfiClks in 1us)

UMC[00...11]CTLx00000804_address00020009 [Enables MTestCombo output on core-side copy] (UMC::Phy::MTestDtoCtrl)

Read-write. Reset: 0000h.

If set, allows the MTestCombo observability signal to be routed out core-side through the dwc_ddrphy_dto pin

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020009; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	MTESTdtoEn. Read-write. Reset: 0. Description: This field is programmed as follows: - MTESTdtoEn==[0], dwc_ddrphy_dto will be squelched (0) - MTESTdtoEn==[1], dwc_ddrphy_dto will reflect the observability signal multiplexed on MTestCombo

UMC00CTLx00000804_address0002000B...UMC11CTLx00000804_address0032000B (UMC::Phy::Seq0BDLY0)

Read-write. Reset: 0000h.

PHY Initialization Engine (PIE) Delay Register 0_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002000B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012000B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022000B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032000B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Seq0BDLY0. Read-write. Reset: 0000h. Description: PHY Initialization Engine (PIE) Delay Register 0 This register is available for selection by the NOP and WAIT instructions in the PIE for the delay value. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes. Programming Seq0BDLY[3,2,1,0] Registers: - These registers are used to control various delays during PLL, DLL initialization. - These registers should be programmed with the corresponding number of MEMCLK cycles representing a fixed amount of delay irrespective of DfiClk Frequency. - When calculating the number of cycles to be programmed for each PState, it should be noted that the sequencer multiplies programmed values by a factor of 8. -- Register: Seq0BDLY0 -- Required Delay: 0.5us -- Programmed Value: (0.5us / MEMCLK period) / 8

UMC00CTLx00000804_address0002000C...UMC11CTLx00000804_address0032000C (UMC::Phy::Seq0BDLY1)

Read-write. Reset: 0000h.

PHY Initialization Engine (PIE) Delay Register 1_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002000C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012000C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022000C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032000C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Seq0BDLY1. Read-write. Reset: 0000h. Description: PHY Initialization Engine (PIE) Delay Register 1 This register is available for selection by the NOP and WAIT instructions in the PIE for the delay value. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes. Programming Seq0BDLY[3,2,1,0] Registers: - These registers are used to control various delays during PLL, DLL initialization. - These registers should be programmed with the corresponding number of MEMCLK cycles representing a fixed amount of delay irrespective of DfiClk Frequency. - When calculating the number of cycles to be programmed for each PState, it should be noted that the sequencer multiplies programmed values by a factor of 8. -- Register: Seq0BDLY1 -- Required Delay: 1.0us -- Programmed Value: (1.0us / MEMCLK period) / 8

UMC00CTLx00000804_address0002000D...UMC11CTLx00000804_address0032000D (UMC::Phy::Seq0BDLY2)

Read-write. Reset: 0000h.

PHY Initialization Engine (PIE) Delay Register 2_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002000D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012000D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022000D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032000D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Seq0BDLY2. Read-write. Reset: 0000h. Description: PHY Initialization Engine (PIE) Delay Register 2 This register is available for selection by the NOP and WAIT instructions in the PIE for the delay value. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes. Programming Seq0BDLY[3,2,1,0] Registers: - These registers are used to control various delays during PLL, DLL initialization. - These registers should be programmed with the corresponding number of MEMCLK cycles representing a fixed amount of delay irrespective of DfiClk Frequency. - When calculating the number of cycles to be programmed for each PState, it should be noted that the sequencer multiplies programmed values by a factor of 8. -- Register: Seq0BDLY2 -- Required Delay: 10.0us. -- Programmed Value: (10.0us / MEMCLK period) / 8

UMC00CTLx00000804_address0002000E...UMC11CTLx00000804_address0032000E (UMC::Phy::Seq0BDLY3)

Read-write. Reset: 0000h.

PHY Initialization Engine (PIE) Delay Register 3_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002000E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012000E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022000E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032000E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description****15:0** **Seq0BDLY3.** Read-write. Reset: 0000h.**Description:** PHY Initialization Engine (PIE) Delay Register 3

This register is available for selection by the NOP and WAIT instructions in the PIE for the delay value.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

Programming Seq0BDLY[3,2,1,0] Registers:

- These registers are used to control various delays during PLL, DLL initialization.
- These registers should be programmed with the corresponding number of MEMCLK cycles representing a fixed amount of delay irrespective of DfiClk Frequency.
- When calculating the number of cycles to be programmed for each PState, it should be noted that the sequencer multiplies programmed values by a factor of 8.

-- Register: Seq0BDLY3

-- Required Delay: --

-- Programmed Value:

64/4 if (MEMCLK Freq <= 400MHz),

132/4 if (400MHz Freq < MEMCLK Freq <= 533MHz),

176/4 if (MEMCLK Freq > 533MHz)

UMC[00...11]CTLx00000804_address0002000F [PHY Alert status bit] (UMC::Phy::PhyAlertStatus)

Read-only, Volatile.

This Read Only CSR represents the status of the ALERT_N pin to the PHY

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002000F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description****15:1** Reserved.**0** **PhyAlert.** Read-only, Volatile. Current state of ALERT_N.

UMC00CTLx00000804_address00020010...UMC11CTLx00000804_address00320010 [Setup Intervals for DFI PHY Master operations] (UMC::Phy::PPTTrainSetup)

Read-write. Reset: 0000h.

Some tracking is required periodically using the DFI PHY MASTER interface.

This register control the interval.

Setting the register to 0 completely disables the PHY Master Interface.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support DRAM drift compensation.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020010; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120010; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220010; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320010; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15:8 Reserved.

7 **PhyMstrCtrlMode.** Read-write. Reset: 0.

Description: When this bit is 1, a PHY Master transaction is initiated only by a dfi_ctrlmsg transaction. When this bit is 0, a PHY Master transaction is initiated only by timer function.

6:4 **PhyMstrMaxReqToAck.** Read-write. Reset: 0h.

Description: Bits 6:4 of this register specify the max time from tdfi_phymstr_req asserted to tdfi_phymstr_ack asserted.

- 3'b000 Disable PHY Master Interface
- 3'b001 set tPHYMSTR_resp. = 512 MEMCLKs
- 3'b010 set tPHYMSTR_resp. = 1024 MEMCLKs
- 3'b011 set tPHYMSTR_resp. = 2048 MEMCLKs
- 3'b100 set tPHYMSTR_resp. = 4096 MEMCLKs
- 3'b101 set tPHYMSTR_resp. = 8192 MEMCLKs
- 3'b110 set tPHYMSTR_resp. = 32768 MEMCLKs
- 3'b111 set tPHYMSTR_resp. = undefined

Note 1: The above are nominal values. The actual values are calculated as:

(nominal_value - (8*(2^PhyMstrMaxReqToAck)) - (2*DWC_DDRPHY_PIPE_DFI_MISC))

Note 2: This signal interacts with dfi_phyupd_req.

3:0 **PhyMstrTrainInterval.** Read-write. Reset: 0h.

Description: Bits 3:0 of this register specifies the time between the end of one training and the start of the next. As per The DFI Specification, it is the max expected time from dfi_init_complete asserted to tdfi_phymstr_ack asserted.

- 4'b0000 Disable PHY Master Interface unless csrPhyMstrCtrlMode=1
- 4'b0001 PHY MASTER Request Interval = 524288 MEMCLKs
- 4'b0010 PHY MASTER Request Interval = 1048576 MEMCLKs
- 4'b0011 PHY MASTER Request Interval = 2097152 MEMCLKs
- 4'b0100 PHY MASTER Request Interval = 4194304 MEMCLKs
- 4'b0101 PHY MASTER Request Interval = 8388608 MEMCLKs
- 4'b0110 PHY MASTER Request Interval = 16777216 MEMCLKs
- 4'b0111 PHY MASTER Request Interval = 33554432 MEMCLKs
- 4'b1000 PHY MASTER Request Interval = 67108864 MEMCLKs
- 4'b1001 PHY MASTER Request Interval = 134217728 MEMCLKs
- 4'b1010 PHY MASTER Request Interval = 268435456 MEMCLKs
- 4'b1011 - 4'b1111 PHY MASTER Request Interval = undefined

UMC00CTLx00000804_address00020011...UMC11CTLx00000804_address00320011 (UMC::Phy::PPTTrainSetup2)

Read-write. Reset: 0002h.

Programmed by PHY training firmware to support DRAM drift compensation.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020011; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120011; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220011; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320011; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	RFU_PPTTrainSetup2. Read-write. Reset: 2h.

UMC[00...11]CTLx00000804_address00020013 (UMC::Phy::DfiCtrlMsgCode)

Read-only, Volatile.

Last captured value on dfi_ctrlmsg

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020013; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	DfiCtrlMsgCode. Read-only, Volatile. Description: Last captured value on dfi_ctrlmsg - Bits 7:0 reflect the last captured value on channel 0, and in a 2 channel system, - Bits 15:8 reflect the last captured value on channel 1.

UMC[00...11]CTLx00000804_address00020018 [Enables for update and low-power interfaces for DFI0 and DFI1.] (UMC::Phy::DfiMode)

Read-write. Reset: 0001h.

Each bit enables the PHY Update, PHY Master Update, and low power logic for the respective interfaces.

When DWC_DDRPHY_DFI1_EXISTS is defined:

- 'b001: DFI0 controls 1/2 the PHY, external logic ties off inputs to DFI1.
- 'b011: DFI0 controls 1/2 PHY (CH. A), DFI1 controls 1/2 PHY (CH. B).

- All other combinations are illegal.

When DWC_DDRPHY_DFI1_EXISTS is not defined:

- 'b001: DFI0 controls entire PHY.
- All other combinations are illegal.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020018; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2	Dfi0Override. Read-write. Reset: 0. DFI0 is used to control the PHY logic associated with both DFI0 and DFI1.
1	Dfi1Enable. Read-write. Reset: 0. Enables operation for the PHY logic associated with DFI1.
0	Dfi0Enable. Read-write. Reset: 1. Enables operation for the PHY logic associated with DFI0.

UMC00CTLx00000804_address00020019...UMC11CTLx00000804_address00320019 [Mode select register for MEMCLK/Address/Command Tristates] (UMC::Phy::TristateModeCA)

Read-write. Reset: 0000h.

This register is used to control the PHY Dynamic Address/Command Tristate Feature, and the stopped MEMCLK state.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020019; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120019; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220019; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320019; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:2	CkDisVal. Read-write. Reset: 0h. Description: The PHY provides 4 memory clocks, n=0..3. - When the toggling of memory clock CK_t[n] is disabled with dfi_clk_disable[n]=1, the memory clock CK_t[n] is driven with Register CkDisVal[1]. - When the toggling of memory clock CK_c[n] is disabled with dfi_clk_disable[n]=1, the memory clock CK_c[n] is driven with ~(Register CkDisVal[0]) (ie inverted). Note: The non-toggling differential memory clock [CK_t[n],CK_c[n]] may be driven with any combination, LL,LH,HL,HH; the default [CK_t[n],CK_c[n]]=LH.
1	DDR2TMode. Read-write. Reset: 0. Description: Must be set to 1 for Dynamic Tristate to work when CA bus is 2T mode. This CSR bit has no effect when DisDynAdrTri==1
0	DisDynAdrTri. Read-write. Reset: 0. Description: Dynamic Tristating is on by default. - When DisDynAdrTri=1, Dynamic Tristating is disabled. Dynamic Tristating should be disabled (DisDynAdrTri=1) for these modes: -- DDR5 - When DisDynAdrTri=0 (default). -- In 1T mode, the PHY will tristate the relevant pins when all ranks are deselected

UMC[00...11]CTLx00000804_address0002001D (UMC::Phy::PMIEnable)

Read-write. Reset: 0000h.

This register is dynamically written by PHY Initialization Engine during frequency changes and should not be written by the user.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002001D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	PMIEnable. Read-write. Reset: 0. This register is dynamically written by PHY Initialization Engine during frequency changes and should not be written by the user.

UMC[00...11]CTLx00000804_address0002001E (UMC::Phy::PhyTID)

Read-write. Reset: 0000h.

PHY Technology ID Register

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002001E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	PhyTID. Read-write. Reset: 0000h. This register is a spare to store technology-specific information.

UMC00CTLx00000804_address0002001F...UMC11CTLx00000804_address0032001F (UMC::Phy::D5RdimmSDRmode)

Read-write. Reset: 0000h.

SDR Mode enable for DDR5 RDIMM

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002001F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012001F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022001F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032001F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:1	Reserved.
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0	D5RdimmSDRmode. Read-write. Reset: 0.
---	--

Description: SDR Mode enable for DDR5 RDIMM.

- 1 = SDR Mode

- 0 = DDR Mode

UMC00CTLx00000804_address00020020...UMC11CTLx00000804_address00320020 (UMC::Phy::HwtMRL)

Read-write. Reset: 0006h.

HWT MaxReadLatency.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020020; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120020; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220020; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320020; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:5	Reserved.
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4:0	HwtMRL. Read-write. Reset: 06h.
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Description: This Max Read Latency CSR is to be trained to ensure the rx-data fifo is not read until after all DBYTES have their read data valid.

The CSR is in units of two mem clocks; that is, a unit change in the LSB is a change in MRL of two mem clocks. This MASTER copy of MRL is used by the PHY training hardware only.

The maximum value allowed is 28, 0x1C.

UMC[00...11]CTLx00000804_address00020021 [DFI PhyUpdate Request time counter (in MEMCLKs)] (UMC::Phy::DFIPHYUPD)

Read-write. Reset: 0007h.

This Register is used to determine the DFI Update request interval and the maximum update response time.

Note: DFIPHYUPDCNT and DFIPHYUPDRESP must be chosen such that DFIPHYUPDCNT > DFIPHYUPDRESP

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020021; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:7	Reserved.
6:4	DFIPHYUPDRESP. Read-write. Reset: 0h. Description: Enforces the t_phyupd_resp time, the maximum time that is allowed to controller to respond to the request for a PHY update. A dfi_error will be signaled if there is no acknowledgement of the update request within nMEMCLKs_phyupd_resp. - 3'b000 nMEMCLKs_phyupd_resp = 1024 MEMCLKs, default value - 3'b001 nMEMCLKs_phyupd_resp = 2048 MEMCLKs - 3'b010 nMEMCLKs_phyupd_resp = 4096 MEMCLKs - 3'b011 nMEMCLKs_phyupd_resp = 8192 MEMCLKs - 3'b100 nMEMCLKs_phyupd_resp = 16384 MEMCLKs Note 1: The above are nominal values. The actual values are calculated as $(\text{nominal_value} - (8 * (2^{\text{DFIPHYUPDRESP}})) - (2 * \text{DWC_DDRPHY_PIPE_DFI_MISC}))$ Note 2: Any code not enumerated above is RSVD and should not be set.
3:0	DFIPHYUPDCNT. Read-write. Reset: 7h. Description: This controls the interval between the end of a phyupdate transaction and a subsequent request. - 4'b0000 - Disable timer-based PHY Update - 4'b1001 - 2K MEMCLKs minus nMEMCLKs_phyupd_resp - 4'b1010 - 4K MEMCLKs minus nMEMCLKs_phyupd_resp - 4'b1011 - 8K MEMCLKs minus nMEMCLKs_phyupd_resp - 4'b0001 - 16K MEMCLKs minus nMEMCLKs_phyupd_resp - 4'b0011 - 32K MEMCLKs minus nMEMCLKs_phyupd_resp - 4'b0111 - 64K MEMCLKs minus nMEMCLKs_phyupd_resp - 4'b1111 - 128K MEMCLKs minus nMEMCLKs_phyupd_resp Any code not enumerated above is RSVD and should not be set.

UMC[00...11]CTLx00000804_address00020023 (UMC::Phy::Dfi2nModeCtl)

Read-write. Reset: 0000h.

Controls whether dfi_2n_mode will cause CS and CKE timing to change.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020023; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1:0	Dfi2nModeCtl. Read-write. Reset: 0h. Description: This field is programmed as follows: - Dfi2nModeCtl[0] controls whether dfi_2n_mode will cause chip-select (CS) timing to change. -- 0: CS timing will not be dynamically modified with dfi_2n_mode. -- 1: CS will be delayed by an additional 1UI dynamically when dfi_2n_mode==1. - Dfi2nModeCtl[1] controls whether dfi_2n_mode will cause clock-enable (CKE) timing to change. -- 0: CKE timing will not be dynamically modified with dfi_2n_mode. -- 1: CKE will be delayed by an additional 1UI dynamically when dfi_2n_mode==1.

UMC00CTLx00000804_address00020024...UMC11CTLx00000804_address00320024 [Control the PHY logic related to the read and write DQS preamble] (UMC::Phy::DqsPreambleControl)

Read-write. Reset: 0008h.

For improved robustness in operating with DRAM timing drift, this register provides controls for the width of the write DQS preamble and postamble, and controls for the anticipated width of DQS preamble for reads.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320024; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits Description15:14 **PositionRxPhaseUpdate.** Read-write. Reset: 0h.

Description: Control for setup of PhaseUpdate to ClkIn for RxEn and RxClk LCDL.
For firmware use only.

13:12 **PositionTxPhaseUpdate.** Read-write. Reset: 0h.

Description: Control for setup of PhaseUpdate to ClkIn for Tx LCDL
For firmware use only.

11 **DDR5RxPostamble.** Read-write. Reset: 0.

Description: DDR5 Read Postamble Mode must have the same value as DDR5 MR8 OP[6]
- 0 0.5 tCK postamble
- 1 1.5 tCK postamble.
For normal operation, this should always be set to 1.

10:8 **DDR5RxPreamble.** Read-write. Reset: 0h.

Description: DDR5 Read Preamble Mode
- 000 1 tCK Preamble, DDR5 MR8 OP[2:0]==0
-- Position the nominal RxEn 1UI before the first strobing edge.
-- This mode has +/-63/64UI max tolerance of read DQS drift.

- 001 2 tCK Preamble(toggling), DDR5 MR8 OP[2:0]==1
-- Position the nominal RxEn 1UI before the first strobing edge.
-- This mode has +/-63/64UI max tolerance of read DQS drift.

- 010 2 tCK Preamble(static), DDR5 MR8 OP[2:0]==2
-- Position the nominal RxEn 2UI before the first strobing edge.
-- This mode has +/-127/64UI max tolerance of read DQS drift.

- 011 3 tCK Preamble(toggling), DDR5 MR8 OP[2:0]==3
-- Position the nominal RxEn 4UI before the first preamble posedge
-- This mode has +/-127/64UI max tolerance of read DQS drift.
- 100 4 tCK Preamble(toggling), DDR5 MR8 OP[2:0]==4
-- Position the nominal RxEn 6UI before the first preamble posedge
-- This mode has +/-127/64UI max tolerance of read DQS drift.
- - 101..111 reserved.

7 **DDR5RxPreambleEn.** Read-write. Reset: 0.

Description: Use the csrDDR5RxPreamble and csrDDR5RxPostamble to control the read-state machine, RSM,
If set the csrTwoTckRxDqsPre will not be used.

6 **WDQSEXTENSION.** Read-write. Reset: 0. Reserved. This bit must always be 0.

5	DisDlyAdjPosDI. Read-write. Reset: 0. Disable adjustment of the DfeInit coarse delay that is based on RxEnDly, zerodelay and RxClkDly
4:2	PositionDfeInit. Read-write. Reset: 2h. Description: - PositionDfeInit[2]=1 causes DfeInit to be asserted with 1UI less setup to RxEn - PositionDfeInit[1]=1 causes DfeInit to be asserted at the nominal time - PositionDfeInit[0]=1 causes DfeInit to be asserted with 1UI more setup to RxEn PositionDfeInit=3'b010 is the nominal, recommended value for leading edge used by receiver. 2 is the supported value; 4 & 1 are for debug only. others are reserved. Value 1 is not supported for DDR5RxPreambleEn=1,DDR5RxPreamble=4
1	TwoTckTxDqsPre. Read-write. Reset: 0. - 0: Standard 1tck TxDqs Preamble
0	TwoTckRxDqsPre. Read-write. Reset: 0. Description: Used only when DDR5RxPreambleEn=0. Widens the RxDqsEn window to allow larger drift in the incoming read DQS to take advantage of the larger/wider preamble generated by the DRAMs for static 2tCK preamble modes.

UMC[00...11]CTLx00000804_address00020025 (UMC::Phy::MasterX4Config)

Read-write. Reset: 0000h.

DBYTE module controls to select X4 DRAM device mode

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020025; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	X4TG. Read-write. Reset: 0h. Description: Set to 1 if this Timing Group/Rank is x4 (as opposed to x8) memory. - XXX[1 0] : TxReqDest/RxReqDest 0 X4/X8. - XX[1 0]X : TxReqDest/RxReqDest 1 X4/X8. - X[1 0]XX : TxReqDest/RxReqDest 2 X4/X8. - [1 0]XXX : TxReqDest/RxReqDest 3 X4/X8. This CSR must be written before dfi_init_start is asserted.

UMC00CTLx00000804_address00020028...UMC11CTLx00000804_address00320028 (UMC::Phy::RxDqsTrkLimit)

Read-write. Reset: 0000h.

Limits the excursion of effective RxEnDly when tracking read DQS

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:6	Reserved.
5:3	RxEnDlyTrkLimitHigh. Read-write. Reset: 0h. Description: Controls the upper limit to the effective RxEnDly when tracking read DQS (EnRxDqsTracking=1) If 0, then the upper limit on the effective RxEnDly is (39+63/64) UI else it is the lesser of (39+63/64) UI or (csrRxEnDly + (RxEnDlyTrkHighLimit+0/64) UI.
2:0	RxEnDlyTrkLimitLow. Read-write. Reset: 0h. Description: Controls the lower limit to the effective RxEnDly when tracking read DQS (EnRxDqsTracking=1) If 0, then the lower limit on the effective RxEnDly is (2+0/64) UI else it is the greater of (2+0/64) UI or (csrRxEnDly - (RxEnDlyTrkLowLimit+0/64) UI.

UMC[00...11]CTLx00000804_address0002002A [Test Bump Control2] (UMC::Phy::TestBumpCntrl2)

Read-write. Reset: 0000h.

Control for Test Bumps

PHY RX VREF DAC control for rxdq cell internal VREF.

DAC control for rxdq cell internal VREF.

The VREF generators have different ranges, depending on the Mission Mode settings for

UMC::Phy::DqDqsRcvCntrl[ExtVrefRange]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002002A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	TestRxStandby. Read-write. Reset: 0. Test Bump Control2
14	TestRxPowerdown. Read-write. Reset: 0. Test Bump Control2
13:7	TestVrefDAC1. Read-write. Reset: 00h. Test Bump Control2
6:0	TestVrefDAC0. Read-write. Reset: 00h. Test Bump Control2

UMC[00...11]CTLx00000804_address0002002C (UMC::Phy::Acx4AnibDis)

Read-write. Reset: 0000h.

Disable for unused ACX Nibbles

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002002C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:0	Acx4AnibDis. Read-write. Reset: 0000h. When a bit is set, the Dfi-side FIFO clock to the corresponding ANIB is disabled.

UMC00CTLx00000804_address0002002D...UMC11CTLx00000804_address0032002D [This Register is used to enable the Read-DBI function in each DBYTE] (UMC::Phy::DMIPinPresent)

Read-write. Reset: 0000h.

This Register is used to instruct the PHY whether to enable the Read-DBI function on Lane8 of each DBYTE.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002002D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012002D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022002D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032002D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	RdDbiEnabled. Read-write. Reset: 0. Description: This bit must be set to 1'b1 if Read-DBI is enabled in a connected DRAM. If set, the following DRAM MR should also be set [DDR5.MR5.A7=1]

UMC00CTLx00000804_address0002002E...UMC11CTLx00000804_address0032002E (UMC::Phy::ARdPtrInitVal)

Read-write. Reset: 0000h.

Address/Command FIFO ReadPointer Initial Value

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002002E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012002E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022002E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032002E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	ARdPtrInitVal. Read-write. Reset: 0h. Description: This is the initial Pointer Offset for the free-running FIFOs in the DBYTE and ACX4 hardIPs. The units of this register are in UI . The pointer separation should be chosen to compensate for all sources of skew and drift of the PHY DFICLK and PCLK networks. This CSR must be programmed in Step C of the PHY Initialization sequence. Supported values for this register are 0-10.

UMC[00...11]CTLx00000804_address0002002F [Test Bump Control3] (UMC::Phy::TestBumpCntrl3)

Read-write. Reset: 0000h.

Control for Test Bumps

Reserved; deprecated.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002002F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:7	TestVrefDAC3. Read-write. Reset: 00h. Test Bump Control3
6:0	TestVrefDAC2. Read-write. Reset: 00h. Test Bump Control3

UMC[00...11]CTLx00000804_address0002003A [DLL Mode control CSR for DBYTEs] (UMC::Phy::DbyteDllModeCntrl)

Reset: 0000h.

DLL Mode control Register in DBYTE modules

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002003A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2	DllRxBurstLengthMode. Read-write. Reset: 0. Description: Controls the DBYTE macro DQS read-pipeline burst-length mode - 0 for BC4,BL8,BL16 (non-CRC) reads - 1 for BL18 (CRC) reads (DDR5 only)
1	DllRxBurstPreambleMode. Read-write. Reset: 0. Obsolete. Has no function.
0	Reserved.

UMC[00...11]CTLx00000804_address0002003F (UMC::Phy::AnTxLcdlCalPhDetOut)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002003F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:14 Reserved.

13:0 **AnTxLcdlCalPhDetOut**. Read-only, Volatile. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address00020040 (UMC::Phy::TimingModeCtrl)**

Read-write. Reset: 0000h.

Mode controls for fine timing.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020040; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:1 Reserved.

0 **Dly64Prec**. Read-write. Reset: 0. Use 1/64 [UI](#) precision mode for TxDqDly, TxDqsDly, RxEnDly, RxClkDly, ATxDly

UMC00CTLx00000804_address00020041...UMC11CTLx00000804_address00320041 (UMC::Phy::TrackingModeCtrl)

Read-write. Reset: 1000h.

Mode controls for fine timing adjustment during mission-mode operation.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020041; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120041; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220041; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320041; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	Reserved.
14:12	RxDqsTrackingThreshold. Read-write. Reset: 1h. Description: When EnRxDqsTracking=1, this csr is used to set the max value of the signed count of the sense of the count. - 0 3 - 1 5, default value - 2 9 - 3 15 - 4 31 - 5-7 reserved
11:8	DqsOscRunTimeSel. Read-write. Reset: 0h. Description: Informs the PHY of the DQS oscillator runtime configured in the DRAMs. or the runtime used by the controller. That time must agree with the configuration of this CSR. - 0 for a runtime of 256 memCK, legacy, for PHY-initiated tracking - 1 for a runtime of 512 memCK - 2 for a runtime of 1024 memCK - 3 for a runtime of 2048 memCK - 4 for a runtime of 4096 memCK - 5 for a runtime of 8192 memCK - 6..15 reserved
7:5	Tdqs2dqTrackingLimit. Read-write. Reset: 0h. Description: When tracking the DRAM tdqs2dq, in either dfi-controlled mode or phy-controlled mode, limit the change to the Tx DQ delay that may be applied at each update. This does not limit the total excursion from the trained value of TxDqDly. - 0 no limit - 1 1/64 UI - 2 2/64 UI - 3 4/64 UI - 4 8/64 UI - 5 16/64 UI - 6,7 reserved
4	EnRxDqsTracking. Read-write. Reset: 0. Description: Enable tracking of the incoming read DQS to modify the RxEn timing. In normal use, the value in this field should be equal to the value in csrEnDqsSampNegRxEn
3	EnDfiTdqs2dqTrackingTg3. Read-write. Reset: 0.

	Description: Enable tracking of tDQS2DQ tracking for timing group 3 under control of DFI initiated reads tagged with assertion of dwc_ddrphy_snoop_en. When training writes PptDqsCntInvTrnTg3, if this csr is written 0-to-1, then the prior-history delta-TxDqDly will be initialized to zero. See csr DisPtrInitClrTxTracking.
2	EnDfiTdqs2dqTrackingTg2. Read-write. Reset: 0. Description: Enable tracking of tDQS2DQ tracking for timing group 2 under control of DFI initiated reads tagged with assertion of dwc_ddrphy_snoop_en. When training writes PptDqsCntInvTrnTg2, if this csr is written 0-to-1, then the prior-history delta-TxDqDly will be initialized to zero. See csr DisPtrInitClrTxTracking.
1	EnDfiTdqs2dqTrackingTg1. Read-write. Reset: 0. Description: Enable tracking of tDQS2DQ tracking for timing group 1 under control of DFI initiated reads tagged with assertion of dwc_ddrphy_snoop_en. When training writes PptDqsCntInvTrnTg1, if this csr is written 0-to-1, then the prior-history delta-TxDqDly will be initialized to zero. See csr DisPtrInitClrTxTracking.
0	EnDfiTdqs2dqTrackingTg0. Read-write. Reset: 0. Description: Enable tracking of tDQS2DQ tracking for timing group 0 under control of DFI initiated reads tagged with assertion of dwc_ddrphy_snoop_en. When training writes PptDqsCntInvTrnTg0, if this csr is written 0-to-1, then the prior-history delta-TxDqDly will be initialized to zero. See csr DisPtrInitClrTxTracking.

UMC[00...11]CTLx00000804_address00020042 (UMC::Phy::AnTuLcdlCalPhDetOut)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020042; UMC[11:00]CTL=00[B:0]5_1000h;
 DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:0	AnTuLcdlCalPhDetOut. Read-only, Volatile. Reserved for PHY training firmware use.

UMC00CTLx00000804_address00020051...UMC11CTLx00000804_address00320051 [Controls PtrInit clear of timing correction determined by previous tracking] (UMC::Phy::PtrInitTrackingModeCtrl)

Read-write. Reset: 0000h.

Controls whether PtrInit clears the timing correction determined by previous tracking

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020051; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120051; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220051; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320051; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:2 Reserved.

1 **DisPtrInitClrRxTracking.** Read-write. Reset: 0. Reserved. Must be zero.0 **DisPtrInitClrTxTracking.** Read-write. Reset: 0.**Description:** Disables PtrInit from clearing the results of previous tDQS2SQ tracking.

0 PtrInit causes the original firmware trained values of csrTxDqDly to be used immediately after PtrInit. Subsequent tracking events will update the values used for Tx DQ timing if enabled with csrEnDfiTdqs2dqTrackingTg[i]

1 PtrInit will not affect the values used for TxDqDly as previously determined by tracking events if enabled with csrEnDfiTdqs2dqTrackingTg[i].

default 0

UMC[00...11]CTLx00000804_address00020055 [ODT Disable Control] (UMC::Phy::OdtDebugDisable)

Read-write. Reset: 0000h.

Used for debugging without ODT mode, allowing to turn it on/off in both the DBYTE and acx4.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020055; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:2 Reserved.

1 **OdtAnibDebugDisable.** Read-write. Reset: 0.**Description:** Disables ODT mode for the ANIB.

- Setting this bit will Disable ODT mode for Debugging.

- Clearing this bit will leave ODT mode on.

For a description of ODT mode behavior please refer to the Ctlpipe_ODT pin description.

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0 **OdtDbyteDebugDisable.** Read-write. Reset: 0.**Description:** Disables ODT mode for the DBYTE.

- Setting this bit will Disable ODT mode for Debugging.

- Clearing this bit will leave ODT mode on.

For a description of ODT mode behavior please refer to the Ctlpipe_ODT pin description.

UMC[00...11]CTLx00000804_address00020057 (UMC::Phy::RemoteImpCalCtl)

Read-write. Reset: 0000h.

Remote Impedance Calibration Control

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020057; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1:0	RemoteImpCalCtl. Read-write. Reset: 0h. Description: Controls the mode for Remote Impedance Calibration - 00: Use dwc_ddrphy_CalibSlave pin (hi = slave) - 10: Force Master Mode - 01: Force Slave Mode - 11: Disabled (lower power)

UMC[00...11]CTLx00000804_address00020058 (UMC::Phy::RemoteImpCalStatus)

Read-only, Volatile.

Remote Impedance Calibration Status

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020058; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1:0	RemoteImpCalStatus. Read-only, Volatile. Description: Status indicators for Remote Impedance Calibration - Bit 0: The state of the dwc_ddrphy_CalibSlave pin - Bit 1: This bit is set after the first set of codes is received/sent. Changing mode between master and slave clears the bit.

UMC[00...11]CTLx00000804_address0002005B [This Register is used to configure the Mtest Receiver] (UMC::Phy::MAlerTControl)

Read-write. Reset: 4000h.

Controls for the receiver for MemTest

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002005B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	MALERTForceError. Read-write. Reset: 0. When MALERTForceError is set, this CSR state is used to force parity error to memory.
14	MALERTDisableVal. Read-write. Reset: 1. When MALERTRxEn is not set, this CSR state is used to drive dfi_alert_n.
13	MALERTRxEn. Read-write. Reset: 0. Description: This field is programmed as follows: - 1 - Received data is forwarded to dfi_alert_n. - 0 - Send MALERTDisableVal to dfi_alert_n
12	MalertControlReserved12. Read-write. Reset: 0. Reserved - Unused
11:8	MTESTPuStrenReserved118. Read-write. Reset: 0h. Reserved - Unused
7:0	MalertControlReserved70. Read-write. Reset: 00h. Reserved - Unused

UMC[00...11]CTLx00000804_address00020060 [Protection and control of BP_MemReset_L] (UMC::Phy::MemResetL)

Read-write. Reset: 0000h.

Control the MemResetL output of the PHY. Intended to be used to protect/force the BP_MemReset_L output.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020060; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1	ProtectMemReset. Read-write. Reset: 0. Description: Control the MemResetL output of the PHY. The state of the MemResetL field is transferred to the PHY BP_MemReset_L output when ProtectMemReset is set.
0	MemResetLValue. Read-write. Reset: 0. Description: Control the MemResetL output of the PHY. The state of this field is transferred to the PHY BP_MemReset_L output when ProtectMemReset is set.

UMC[00...11]CTLx00000804_address00020068 (UMC::Phy::MASTERReserved)

Read-write. Reset: 0000h.

Spares for the MASTER Hard Macro

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020068; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	MASTERReserved. Read-write. Reset: 0000h. Spares for the MASTER Hard Macro

UMC[00...11]CTLx00000804_address0002006E [PUBMODE - HWT Mux Select] (UMC::Phy::PUBMODE)

Read-write. Reset: 0000h.

Selects between HWT and DCT as the source of memory transactions.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002006E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	HwtMemSrc. Read-write. Reset: 0. Description: When this is set to a 1, the mux that switches between DCT and HWT for the source of memory transactions is switched to HWT. - Setting this bit relinquishes control to the HWT. - Clearing this bit relinquishes control to the DCT. The transition can only be made when the FIFOs are all in PtrInit (for example, all PtrInit signals are high).

UMC[00...11]CTLx00000804_address00020072 (UMC::Phy::HwtLpCsEnA)

Read-write. Reset: 0003h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020072; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1:0	HwtLpCsEnA. Read-write. Reset: 3h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address00020073 (UMC::Phy::HwtLpCsEnB)

Read-write. Reset: 0003h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020073; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1:0	HwtLpCsEnB . Read-write. Reset: 3h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address00020074 (UMC::Phy::HwtLpCsEnBypass)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020074; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	HwtLpCsEnBypass . Read-write. Reset: 0. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address00020075 [Dfi Command/Address Mode] (UMC::Phy::DfiCAMode)

Read-write. Reset: 0000h.

This CSR controls the PHY signal-to-bump-mapping mode.

Setting bits fields to 1 changes the arch mode and are required to be 1-hot.

While all fields of this csr are present in all products, some will not always be relevant

as a particular DRAM type may not be supported. Irrelevant bits must be set to 0, always.

Programmed by dwc_ddrphy_phyinit_C_initPhyConfig() according to PHY configuration and DRAM type.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020075; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:6	Reserved.
5	DfiD5DCAMode . Read-write. Reset: 0. Description: Enable D5 DDR Address Mode - 0: D5 DDR Address mode disabled - 1: D5 DDR Address mode enabled
4	DfiD5CAMode . Read-write. Reset: 0. Description: Enable D5 Mode - 0: D5 mode disabled - 1: D5 mode enabled
3	DfiCAModeReserved3 . Read-write. Reset: 0. Reserved
2	DfiLp4CAMode . Read-write. Reset: 0. Reserved.
1	Reserved.
0	DfiCAModeReserved0 . Read-write. Reset: 0. Reserved

UMC[00...11]CTLx00000804_address00020076 (UMC::Phy::HwtCACtl)

Read-write. Reset: 0001h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020076; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	RFU . Read-write. Reset: 1.

UMC[00...11]CTLx00000804_address00020077 (UMC::Phy::HwtCAMode)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020077; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:8 Reserved.

7:0 **RFU**. Read-write. Reset: 00h.**UMC[00...11]CTLx00000804_address0002007A (UMC::Phy::HwtControlOvr0)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002007A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:13 Reserved.

12:0 **RFU**. Read-write. Reset: 0000h.**UMC[00...11]CTLx00000804_address0002007B (UMC::Phy::HwtControlOvr1)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002007B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:4 Reserved.

3:0 **RFU**. Read-write. Reset: 0h.**UMC00CTLx00000804_address0002007C...UMC11CTLx00000804_address0032007C [DLL gain control] (UMC::Phy::DllGainCtl)**

Read-write. Reset: 00A1h.

Controls for the DllGain sequencing for locking and maintaining lock of the LCDL delay cells.

The DllGain sequencer will start locking with the DllGainIV.

As the end of locking approaches, the DllGain will be incremented in stages

until, at the end of locking, the DllGain is the terminal values DllGainTV.

 $2^{(5-DllGain)}$ is fractional change in the DAC code MstPhase1UI[8:0] of Dll per pclk.

Firmware uses values for DllGainIV, DllGainTV, and LcdlSeed0 that are optimized

for minimum locking time; these controls are not to be configured by the user.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002007C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012007C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022007C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032007C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:8 Reserved.

7:4 **DllGainTV**. Read-write. Reset: Ah.**Description:** Terminal value of DllGain, ie the value in effect when locking is done and the value used for maintaining lock, ie tracking pclk variation.

DllGainTV must be greater than or equal to DllGainIV.

A value of 5 results in a unit lsb change in the Phase1UI[8:0]

3:0 **DllGainIV**. Read-write. Reset: 1h. Initial value of DllGain.

**UMC00CTLx00000804_address0002007D...UMC11CTLx00000804_address0032007D
(UMC::Phy::DllLockParam)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002007D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012007D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022007D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032007D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:10 Reserved.

9:0 **RFU_DllLockParam.** Read-write. Reset: 000h.**UMC[00...11]CTLx00000804_address0002007E (UMC::Phy::HwtControlVal0)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002007E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:13 Reserved.

12:0 **RFU.** Read-write. Reset: 0000h.**UMC[00...11]CTLx00000804_address0002007F (UMC::Phy::HwtControlVal1)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002007F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:4 Reserved.

3:0 **RFU.** Read-write. Reset: 0h.**UMC00CTLx00000804_address00020086...UMC11CTLx00000804_address00320086
(UMC::Phy::LcdlCharacterization)**

Read-write. Reset: 0008h.

Reserved for PHY test firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020086; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120086; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220086; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320086; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:6 Reserved.

5:0 **RFU_LcdlCharacterization.** Read-write. Reset: 08h.

UMC[00...11]CTLx00000804_address00020088 [Impedance Calibration Control] (UMC::Phy::CalRate)

Read-write. Reset: 0009h.

This CSR controls the impedance calibration engine in the PHY.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020088; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:7	Reserved.
6	DisableBackgroundZQUpdates. Read-write. Reset: 0. Description: Reserved for debug only. When this bit is set, ZQ compensation code updates will occur only during a DFI sideband [PHYUPDATE,CTRLUPDATE,PHYMSTR] handshake. This debug feature must be disabled if DFI1 is active. Since the accumulated update may be more than 1 LSB, the output may glitch in this mode. The user should mitigate this by calibrating only in an idle state or via some other method.
5	CalOnce. Read-write. Reset: 0. Description: The setting of this CSR defines how impedance calibration runs are triggered, for example, periodically or not. - 1: The 0->1 transition of CSR CalRun causes a single iteration of the calibration sequence to occur. Once the calibration is complete, the calibration engine will remain IDLE until another CalRun Trigger. This option is for cases where impedance calibration is to be run during DevInit only. - 0: Calibration will proceed at the rate determined by Register CalInterval. This field should only be changed while the calibrator is idle. For example before csr CalRun is set.
4	CalRun. Read-write. Reset: 0. Description: This field is programmed as follows: - 1: A calibration sequence will be triggered by the 0->1 transition of this bit, as determined by CSR CalOnce. - 0: Calibrator will not run. If already active then it will complete the current sequence before quiescing. The default value of this field is 0. This csr will not do anything while CalZap is asserted.
3:0	CalInterval. Read-write. Reset: 9h. Description: This CSR specifies the interval between successive calibrations, in mS. csrValue : Interval (mS) - 0 : 0 (continuous) - 1 : 0.013 - 2 : 0.10 - 3 : 1 - 4 : 2 - 5 : 3 - 6 : 4 - 7 : 8 - 8 : 10 - 9 : 20 - 10-15 : Reserved This field should only be changed while the calibrator is idle. For example before csr CalRun is set.

UMC[00...11]CTLx00000804_address0002008B (UMC::Phy::PState)

Read-write. Reset: 000Ch.

PSTATE Selection

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002008B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	PState. Read-write. Reset: Ch. Description: This CSR is written by PHY Initialization Engine (PIE) and the data in here will be overwritten. The current PState inferred by the PIE from handshake requests on DFI status interface. - 0000 - PState P0 - 0001 - PState P1 - 0010 - PState P2 - 0011 - PState P3 - 0100 - Reserved - 0101 - Reserved - 0110 - Reserved - 0111 - Reserved - 1000 - Reserved - 1001 - Reserved - 1010 - Reserved - 1011 - Reserved - 1100 - LP2 - 1101 - Reserved - 1110 - Reserved - 1111 - LP3 The lower two bits of this CSR are used to select the PHY's current Pstate for all Pstateable registers. Note: It is a don't care what value is selected in either LP2 or LP3 because no transactions will be taking place when in those states.

UMC[00...11]CTLx00000804_address00020097 (UMC::Phy::CalBusy)

Read-only, Volatile.

Impedance Calibration Busy Status

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020097; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	CalBusy. Read-only, Volatile. Description: Read 1 if the calibrator is actively calibrating. Any changes to calibrator-related CSRs may only be made if the calibrator is disabled (via CSR CalRun) and this CSR reads 0.

UMC[00...11]CTLx00000804_address00020098 [Miscellaneous impedance calibration controls.]
(UMC::Phy::CalMisc2)

Read-write. Reset: 2034h.

Second set of miscellaneous impedance calibrator controls.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020098; UMC[11:00]CTL=00[B:0]5_1000h;
 DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	CalMisc2Reserved15. Read-write. Reset: 0. Miscellaneous impedance calibration controls.
14	CalSlowCmpana. Read-write. Reset: 0. When set, this CSR increases the time allowed for the cmpana cell to settle, by 50%. It changes from 3 to 6 refClk ticks.
13	CalCancelRoundErrDis. Read-write. Reset: 1. Description: The PEXT calibration result and NINT calibration results naturally include a rounding error which manifests as a change of impedance at the pad. Where both of these errors are in the same direction the PEXT calibration result will be adjusted by 1 to offset NINT error. Setting this csr to 1 prevents this adjustment.
12	CalCmptrResTrim. Read-write. Reset: 0. Reserved for future use.
11	CalMisc2Reserved11. Read-write. Reset: 0. Description: Reserved for future use. Formerly used to Skip ODT calibration if required.
10:6	CalMisc2Reserved10x6. Read-write. Reset: 00h. Reserved for future use.
5	SarRoundUpPu. Read-write. Reset: 1. Enable the round-up of the SAR result, to improve PU calibration accuracy by making SAR output stronger by one extra unit
4	SarRoundUpPd. Read-write. Reset: 1. Enable the round-up of the SAR result, to improve PD calibration accuracy by making SAR output stronger by one extra unit
3	CalMisc2Reserved3x3. Read-write. Reset: 0. Reserved for future use.
2:0	CalNumVotes. Read-write. Reset: 4h. Description: This CSR controls the number of consecutive comparator output bits over which majority voting is done. The result of the voting is fed into the SAR. The voting is initiated after the time configured via csrs. CalSlowCmpana has elapsed. CSR Value : number of Votes - 0 : 1 - 1 : 9 - 2 : 17 - 3 : 33 - 4 : 65 (defaults) - 5 : 129 - 6 : 193 - 7 : 255

UMC00CTLx00000804_address000200A1...UMC11CTLx00000804_address003200A1 [Control write DQS preamble by specifying a pattern and an enable.] (UMC::Phy::DqsPreamblePattern)

Read-write. Reset: 0000h.

Control write DQS preamble by specifying a pattern and an enable.
Function available only when WLM9=1.

Note: Programming Limitations:

- Pre=1, TxDqsPreamblePattern = 8'h01, EnTxDqsPreamblePattern = 8'h01
- Pre=2, TxDqsPreamblePattern = 8'h01, EnTxDqsPreamblePattern = 8'h07
- Pre=3, TxDqsPreamblePattern = 8'h01, EnTxDqsPreamblePattern = 8'h1F
- Pre=4, Static, TxDqsPreamblePattern = 8'h01, EnTxDqsPreamblePattern = 8'h7F
- Pre=4 Toggling, TxDqsPreamblePattern = 8'h05, EnTxDqsPreamblePattern = 8'h7F

_umc0_chiplletype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address000200A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplletype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address001200A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplletype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address002200A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplletype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address003200A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	EnTxDqsPreamblePattern. Read-write. Reset: 00h. Description: When WLM9=1, csr EnTxDqsPreamblePattern must be non-zero, and csr TxDqsPreamblePattern must be valie. 0x00 enables the default 1 memclk toggling preamble, and the function of csrTwoTckTxDqsPre; WLM9 must be 0. - 0x01 Find the low level of the first strobing DQSt ,drive 1UI before it with TxDqsPreamblePattern[0] - 0x03 Find the low level of the first strobing DQSt ,drive 2UI before it with TxDqsPreamblePattern[1:0] - 0x07 Find the low level of the first strobing DQSt ,drive 3UI before it with TxDqsPreamblePattern[2:0] - 0x0f Find the low level of the first strobing DQSt ,drive 4UI before it with TxDqsPreamblePattern[3:0] - 0x1f Find the low level of the first strobing DQSt ,drive 5UI before it with TxDqsPreamblePattern[4:0] - 0x3f Find the low level of the first strobing DQSt ,drive 6UI before it with TxDqsPreamblePattern[5:0] - 0x7f Find the low level of the first strobing DQSt ,drive 7UI before it with TxDqsPreamblePattern[6:0] - 0xff Find the low level of the first strobing DQSt ,drive 8UI before it with TxDqsPreamblePattern[7:0] - other values are reserved.
7:0	TxDqsPreamblePattern. Read-write. Reset: 00h. Description: The DQSt 8 UI before the 0 before the first strobing edge will be driven with this pattern for the UI that have EnTxDqsPreamblePattern=1. Index 7 is the first UI, and index 0 will be the last UI. For example, for a BL8 write, DQSt =(tri),(Pre7),(Pre6),,(Pre5),(Pre4),(Pre3),(Pre2),(Pre1),(Pre0), 0,1,0,1,0,1,0,1,0, (Pst3),(Pst2),(Pst1),(Pst0),(tri), assuming that EnTxDqsPreamblePattern=0xff and assuming that EnTxDqsPostamblePattern=0xf. The DQSt,DQSc are always driven differentially.

UMC00CTLx00000804_address000200A2...UMC11CTLx00000804_address003200A2 [Control write DQS postamble by specifying a pattern and an enable] (UMC::Phy::DqsPostamblePattern)

Read-write. Reset: 0000h.

Control write DQS postamble by specifying a pattern and an enable.
Function available only when Wlm9=1.

Note: Programming Limitations:

- Post=0.5, TxDqsPostamblePattern = 4'h0, EnTxDqsPostamblePattern = 4'h0.

- Post=1.5, TxDqsPostamblePattern = 4'h0, EnTxDqsPostamblePattern = 4'hC

`_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address000200A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]``_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address001200A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]``_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address002200A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]``_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address003200A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]`**Bits Description**

15:8 Reserved.

7:4 **EnTxDqsPostamblePattern.** Read-write. Reset: 0h.**Description:** This field is programmed as follows:

- 0x8 Find the low level of the last DQSt ,drive 1 UI after it with TxDqPostamblePattern[3]
- 0xc Find the low level of the last DQSt ,drive 2 UI after it with TxDqPostamblePattern[3:2]
- 0xe Find the low level of the last DQSt ,drive 3 UI after it with TxDqPostamblePattern[3:1]
- 0xf Find the low level of the last DQSt ,drive 4 UI after it with TxDqPostamblePattern[3:0]
- other values are reserved.

3:0 **TxDqsPostamblePattern.** Read-write. Reset: 0h.

Description: The DQSt 4 UI after the 0 after the last strobing edge will be driven with this pattern for the UI that have EnTxDqsPostamblePattern=1.
Index 3 is the first UI, and index 0 will be the last UI.
See example in DqsPreamblePattern.
The DQSt,DQSc are always driven differentially.

UMC[00...11]CTLx00000804_address000200A6 (UMC::Phy::ForceClkGaterEnables)

Read-write. Reset: 0000h.

Force internal clock gaters to be on or off.

`_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]`**Bits Description**

15:8 Reserved.

7 **ForceRdDatClkEnLow.** Read-write. Reset: 0. Forces the RdDatClk gater to be 0.6 **ForceRdDatClkEnHigh.** Read-write. Reset: 0.

Description: Forces the RdDatClk gater to be 1.
Required to be set for RxFifoVisibility function.

5 **ForceACSMClkEnLow.** Read-write. Reset: 0. Forces the ACSM Clk gater to be 0.4 **ForceACSMClkEnHigh.** Read-write. Reset: 0. Forces the ACSM Clk gater to be 1.3 **ForcePIEClkEnLow.** Read-write. Reset: 0. Forces the PIE Clk gater to be 0.2 **ForcePIEClkEnHigh.** Read-write. Reset: 0. Forces the PIE Clk gater to be 1.1 **ForcePubDxCkEnLow.** Read-write. Reset: 0. Forces the gaters of DfiTxClkEn and DfiRxClkEn to be 0.0 **ForcePubDxCkEnHigh.** Read-write. Reset: 0. Forces the gaters of DfiTxClkEn and DfiRxClkEn to be 1.

UMC00CTLx00000804_address000200B2...UMC11CTLx00000804_address003200B2 [PHY Global Vref Controls] (UMC::Phy::VrefInGlobal)

Read-write. Reset: 0200h.

This Register controls the PHY global VREF generator, and the PHY analog pad BP_ATO.

The mission mode function of the PHY VREF generator is to generate VrefDQ for the PHY RX function.

This register is written and configured by the PHY training firmware for mission mode function.

`_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address000200B2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]``_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address001200B2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]``_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address002200B2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]``_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address003200B2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]`

Bits	Description
15	Reserved.
14	GlobalVrefInMode. Read-write. Reset: 0. RSVD
13:10	GlobalVrefInTrim. Read-write. Reset: 0h. RSVD
9:3	GlobalVrefInDAC. Read-write. Reset: 40h. Description: DAC code for internal Vref generation. The DAC has two ranges; the range is set by GlobalVrefInSel[2] - RANGE0 : DDR5 [GlobalVrefInSel[2] = 0] - DAC Output Voltage = GlobalVrefInDAC == 7'h00
2:0	GlobalVrefInSel. Read-write. Reset: 0h. Description: GlobalVrefInSel[1:0] controls the mode of the PHY VREF DAC and the BP_ATO pin - 2'b00 - PHY Vref DAC Range0 -- BP_ATO = Hi-Z - 2'b01 - Reserved Encoding - 2'b10 - PHY Vref DAC Range0 -- BP_ATO connected to PLL Analog Bus - 2'b11 - PHY Vref DAC Range0 -- BP_ATO connected to PHY Vref DAC GlobalVrefInSel[2] shall be set according to DRAM Protocol: Protocol GlobalVrefInSel[2] DDR5 1'b0

UMC[00...11]CTLx00000804_address000200B5 (UMC::Phy::MasUpdGoodCtr)

Read-only, Volatile.

Counts successful PHY Master Interface Updates (PPTs)

`_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200B5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]`

Bits	Description
15:0	MasUpdGoodCtr. Read-only, Volatile. Description: This register increments whenever the Memory Controller acknowledges a PHY Master Interface request (for example, a request for the PHY to take over the DFI for Periodic Phase Training (PPT)). In a 2-channel system, both MC channels must ack. The counter saturates at max value 65535.

UMC[00...11]CTLx00000804_address000200B6 (UMC::Phy::PhyUpd0GoodCtr)

Read-only, Volatile.

Counts successful PHY-initiated DFI0 Interface Updates

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200B6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	PhyUpd0GoodCtr. Read-only, Volatile. Description: This register increments whenever the Memory Controller acknowledges a PHY-initiated DFI0 interface update request. The counter saturates at max value 65535.

UMC[00...11]CTLx00000804_address000200B7 (UMC::Phy::PhyUpd1GoodCtr)

Read-only, Volatile.

Counts successful PHY-initiated DFI1 Interface Updates

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200B7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	PhyUpd1GoodCtr. Read-only, Volatile. Description: This register increments whenever the Memory Controller acknowledges a PHY-initiated DFI1 interface update request. The counter saturates at max value 65535.

UMC[00...11]CTLx00000804_address000200B8 (UMC::Phy::CtlUpd0GoodCtr)

Read-only, Volatile.

Counts successful Memory Controller DFI0 Interface Updates

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200B8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	CtlUpd0GoodCtr. Read-only, Volatile. Description: This register increments whenever the PHY acknowledges a Memory Controller-initiated DFI0 interface update request. The counter saturates at max value 65535.

UMC[00...11]CTLx00000804_address000200B9 (UMC::Phy::CtlUpd1GoodCtr)

Read-only, Volatile.

Counts successful Memory Controller DFI1 Interface Updates

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200B9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	CtlUpd1GoodCtr. Read-only, Volatile. Description: This register increments whenever the PHY acknowledges a Memory Controller-initiated DFI1 interface update request. The counter saturates at max value 65535.

UMC[00...11]CTLx00000804_address000200BA (UMC::Phy::MasUpdFailCtr)

Read-only, Volatile.

Counts unsuccessful PHY Master Interface Updates

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200BA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **MasUpdFailCtr.** Read-only, Volatile.

Description: This register increments whenever the PHY asserts a PHY Master Interface request, but the Memory Controller does not acknowledge the request within the allowed interval, such that dfi_error is asserted. The counter saturates at max value 65535. In a 2-channel system, a failure occurs if either MC channels fails to ack.

UMC[00...11]CTLx00000804_address000200BB (UMC::Phy::PhyUpd0FailCtr)

Read-only, Volatile.

Counts unsuccessful PHY-initiated DFI0 Interface Updates

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200BB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **PhyUpd0FailCtr.** Read-only, Volatile.

Description: This register increments whenever the PHY asserts a DFI0 Interface update request, but the Memory Controller does not acknowledge the request within the allowed interval, such that dfi_error is asserted. The counter saturates at max value 65535.

UMC[00...11]CTLx00000804_address000200BC (UMC::Phy::PhyUpd1FailCtr)

Read-only, Volatile.

Counts unsuccessful PHY-initiated DFI1 Interface Updates

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200BC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **PhyUpd1FailCtr.** Read-only, Volatile.

Description: This register increments whenever the PHY asserts a DFI1 Interface update request, but the Memory Controller does not acknowledge the request within the allowed interval, such that dfi_error is asserted. The counter saturates at max value 65535.

UMC[00...11]CTLx00000804_address000200BD [Enables for Performance Counters] (UMC::Phy::PhyPerfCtrEnable)

Read-write. Reset: 0000h.

Each bit corresponds to one performance counter. Set the bit to enable the counter.

Clearing the bit stops and clears the counter.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200BD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7	PhyUpd1FailCtl. Read-write. Reset: 0. Enables PhyUpd1FailCtr
6	PhyUpd0FailCtl. Read-write. Reset: 0. Enables PhyUpd0FailCtr
5	MasUpdFailCtl. Read-write. Reset: 0. Enables MasUpdFailCtr
4	CtlUpd1GoodCtl. Read-write. Reset: 0. Enables CtlUpd1GoodCtr
3	CtlUpd0GoodCtl. Read-write. Reset: 0. Enables CtlUpd0GoodCtr
2	PhyUpd1GoodCtl. Read-write. Reset: 0. Enables PhyUpd1GoodCtr
1	PhyUpd0GoodCtl. Read-write. Reset: 0. Enables PhyUpd0GoodCtr
0	MasUpdGoodCtl. Read-write. Reset: 0. Enables MasUpdGoodCtr

UMC00CTLx00000804_address000200C5...UMC11CTLx00000804_address003200C5 [PState dependent PLL Control Register 2] (UMC::Phy::PllCtrl2)

Read-write. Reset: 0019h.

This csr is to be programmed by bios or other means to set PLL charge-pump settings in various PStates.

- For DfiClk range 937.5 - 1200 MHz: PllFreqSel = 11000'b
- For DfiClk range 625 - 937.5 MHz: PllFreqSel = 11001'b
- For DfiClk range 468.75 - 625 MHz: PllFreqSel = 01010'b
- For DfiClk range 312.5 - 468.75 MHz: PllFreqSel = 01011'b
- For DfiClk range 234.4 - 312.5 MHz: PllFreqSel = 00110'b
- For DfiClk range 166 - 234.4 MHz: PllFreqSel = 00111'b

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address000200C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address001200C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address002200C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address003200C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:5	Reserved.
4:0	PllFreqSel. Read-write. Reset: 19h. Description: Adjusts the loop parameters to compensate for different VCO bias points, and input/output clock division ratios. Program based on reference clock frequency (DfiClk) with this table.

UMC[00...11]CTLx00000804_address000200C9 (UMC::Phy::PllLockStatus)

Read-only, Volatile.

PLL's pll_lock pin output

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200C9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits Description

15:1 Reserved.

0 **PllLockStatus.** Read-only, Volatile.**Description:** Directly connected to the pll_Lock output.

When asserted Pll macro has asserted the Lock status.

To avoid metastability issues, this register should perform a majority-vote after reading this register several times.

UMC00CTLx00000804_address000200CA...UMC11CTLx00000804_address003200CA (UMC::Phy::PllTestMode)

Read-write. Reset: 0124h.

Additional controls for PLL CP/VCO modes of operation

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address000200CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address001200CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address002200CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address003200CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits Description15:0 **PllTestMode.** Read-write. Reset: 0124h.**Description:** Directly connected to testmode[15:0] pin of PLL.

Note: This table might not be optimal for all technology process.

- Bits [8,5,2,1] shall be set according to table below. All other bits shall be set to 0.

- For DfIClk range 937.5 - 1200 MHz: PllTestMode[8,5,2,1] = 1110'b

- For DfIClk range 625 - 937.5 MHz: PllTestMode[8,5,2,1] = 1110'b

- For DfIClk range 468.75 - 625 MHz: PllTestMode[8,5,2,1] = 1110'b

- For DfIClk range 312.5 - 468.75 MHz: PllTestMode[8,5,2,1] = 1110'b

- For DfIClk range 234.4 - 312.5 MHz: PllTestMode[8,5,2,1] = 1110'b

- For DfIClk range 166 - 234.4 MHz: PllTestMode[8,5,2,1] = 1110'b

UMC[00...11]CTLx00000804_address000200CB [PLL Control Register 3] (UMC::Phy::PllCtrl3)

Read-write. Reset: 01F0h.

It is recommended to use default values for this CSR. Otherwise, it can be programmed by bios or other means.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200CB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	PllEnCal. Read-write. Reset: 0. Description: Connects directly to en_cal of PLL and controls behavior at both Preset assertion at PLL startup and Standby assertion to enter standby mode. At Preset assertion, if en_cal =1, force_cal controls the source of the DAC calibration value to be used during PLL startup; else if en_cal=0, the default hardwired value will be used. Entering standby mode, if en_cal=1, calibration will run at standby rising edge.
14	PllForceCal. Read-write. Reset: 0. Description: Connects directly to force_cal of PLL. If en_cal = 0, force_cal has no effect. If en_cal = 1, force_cal = 1 forces calibration code dacval_in<4:0> to be used at preset mode; else if force_cal=0, the last calibration result stored internal to the Pll will be used.
13:9	PllDacValIn. Read-write. Reset: 00h. Description: Connects directly to dacval_in<4:0> of PLL. Calibration DAC input current setting, can be forced at rising edge of preset if force_cal =1
8:4	PllMaxRange. Read-write. Reset: 1Fh. Description: Connects directly to maxrange of PLL. Setting for Saturation control of PLL.
3:0	Reserved.

UMC00CTLx00000804_address000200CC...UMC11CTLx00000804_address003200CC [PState dependent PLL Control Register 4] (UMC::Phy::PllCtrl4)

Read-write. Reset: 017Fh.

This csr is to be programmed by bios or other means to set PLL charge-pump settings in various PStates. See this table for programming based on reference clock frequency (DfiClk).

Note: This table might not be optimal for all technology process.

- For DfiClk range 937.5 - 1200 MHz: PllCpPropGsCtrl = 1011'b, PllCpIntGsCtrl = 11111'b
- For DfiClk range 625 - 937.5 MHz: PllCpPropGsCtrl = 1011'b, PllCpIntGsCtrl = 11111'b
- For DfiClk range 468.75 - 625 MHz: PllCpPropGsCtrl = 1011'b, PllCpIntGsCtrl = 11111'b
- For DfiClk range 312.5 - 468.75 MHz: PllCpPropGsCtrl = 1011'b, PllCpIntGsCtrl = 11111'b
- For DfiClk range 234.4 - 312.5 MHz: PllCpPropGsCtrl = 1011'b, PllCpIntGsCtrl = 11111'b
- For DfiClk range 166 - 234.4 MHz: PllCpPropGsCtrl = 1011'b, PllCpIntGsCtrl = 11111'b

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address000200CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address001200CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address002200CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address003200CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:9	Reserved.
8:5	PllCpPropGsCtrl. Read-write. Reset: Bh. Description: Connects directly to cp_prop_gs_cntrl<6:0> of the PLL. Charge pump proportional current control for fast relock and gearshift.
4:0	PllCpIntGsCtrl. Read-write. Reset: 1Fh. Description: Connects directly to cp_int_gs_cntrl<6:0> of the PLL. Charge pump integrating current control for fast relock and gearshift.

UMC[00...11]CTLx00000804_address000200CD (UMC::Phy::PllEndofCal)

Reset: 0000h.

Reserved

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200CD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Reserved.

UMC[00...11]CTLx00000804_address000200CE (UMC::Phy::PllStandbyEff)

Read-only, Volatile.

PLL's standby_eff (effective standby) output

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200CE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	PllStandbyEff. Read-only, Volatile. Returns state off PLL standby. PLL will be in standby in PHY LP2 states.

UMC[00...11]CTLx00000804_address000200CF (UMC::Phy::PllDacValOut)

Read-only, Volatile.

PLL's Dacval_out output

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200CF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:5	Reserved.
4:0	PllDacValOut . Read-only, Volatile. Directly connected to the pll's dacval_out output. Used for observation of PLL DAC code.

UMC[00...11]CTLx00000804_address000200D1 (UMC::Phy::DlyTestRingSel1)

Read-write. Reset: 0000h.

Reserved for PHY test firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200D1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	DlyTestRingSel1 . Read-write. Reset: 0000h. Reserved for PHY test firmware use.

UMC[00...11]CTLx00000804_address000200D2 (UMC::Phy::DlyTestRingSel2)

Read-write. Reset: 0000h.

Reserved for PHY test firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200D2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	Reserved.
11:0	DlyTestRingSel2 . Read-write. Reset: 000h. Reserved for PHY test firmware use.

UMC[00...11]CTLx00000804_address000200D5 (UMC::Phy::PllLockedStatus)

Read-only, Volatile.

Status capture of PLL's pll_lock pin output

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200D5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1	PllUnlocked . Read-only, Volatile. Description: When PllLocked is set, the first deassertion of PllLock sets this bit. It is cleared by csrPllLockedClear
0	PllLocked . Read-only, Volatile. Description: The first assertion of PllLock sets this bit. It is cleared by csrPllLockedClear

UMC[00...11]CTLx00000804_address000200D6 (UMC::Phy::PllLockedControl)

Read-write. Reset: 0000h.

Control for PllLock monitoring logic

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200D6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:5	Reserved.
4	PllLockedCtrsClear . Read-write. Reset: 0. When 1, csrPllLockHiCtr and csrPllLockTransCtr are forced to 0.
3	PllLockedCtrsEnable . Read-write. Reset: 0. When 1, (and csrPllCtrClr=0), csrPllLockHiCtr, and csrPllLockTransCtr count.
2	PllLockedTransCtrMode . Read-write. Reset: 0. When 0, csrPllLockTransCtr counts 0->1 transitions of PLLLock. When 1, it counts 1->0 transitions.
1	PllLockedHiCtrMode . Read-write. Reset: 0. Description: When 0, csrPllLockHiCtr counts (in DfiClk cycles) the length of the next assertion of PllLock. When 1, csrPllLockHiCtr counts (in DfiClk cycles) every time PllLock is 1.
0	PllLockedClear . Read-write. Reset: 0. When this register is a 1, csrPllLocked and csrPllUnlocked are cleared.

UMC[00...11]CTLx00000804_address000200D7 (UMC::Phy::PllLockedHiCtr)

Read-only, Volatile.

Counts high assertions of PLLLock

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200D7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	PllLockedHiCtr . Read-only, Volatile. Counts DfiClk cycles when PllLock is a 1 according to programming of PllLockControl. Saturates at max value.

UMC[00...11]CTLx00000804_address000200D8 (UMC::Phy::PllLockedTransCtr)

Read-only, Volatile.

Counts transitions of PLLLock

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200D8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	PllLockedTransCtr . Read-only, Volatile. Counts 0->1, 1->0 transitions of PllLock according to programming of PllLockControl. Saturates at max value.

UMC[00...11]CTLx00000804_address000200DC [Controls for DFE in MEMRESET macro] (UMC::Phy::Master4TapDfeCtrl)

Read-write. Reset: 0000h.

Contains controls for 4 tap DFE for MEMRESET

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200DC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:12	MasterDFEtap4sel . Read-write. Reset: 0h. DAC control for DFE tap4
11:9	MasterDFEtap3sel . Read-write. Reset: 0h. DAC control for DFE tap3
8:5	MasterDFEtap2sel . Read-write. Reset: 0h. DAC control for DFE tap2
4:3	MasterDfebiasssel . Read-write. Reset: 0h. DAC control for DFE bias.
2:0	MasterDfeCtrl . Read-write. Reset: 0h. Description: This field is programmed as follows:- 000 - no DFE fxn- 001 - DFE 1 tap- 010 - DFE train channel 0- 011 - DFE train channel 1- 100 - Reserved- 101 - DFE 2 tap- 110 - 3 tap DFE- 111 - 4 tap DFE These settings are determined by PHY Training FW and should not be overridden.

UMC[00...11]CTLx00000804_address000200E0 [Small delays on handshake signals] (UMC::Phy::DfiHandshakeDelays0)

Read-write. Reset: 0000h.

Add additional assertion/deassertion delays on handshake signals (for example, these delays add to existing delays)
Logic assumes that dfi signal assertions exceed the programmed delays.

IMPORTANT: Timing parameters for dfi_phyupd and dfi_ctrlupd interfaces are valid only when this register is 0.

IMPORTANT: Non-zero values of this register are Reserved for Debug.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200E0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	CtrlUpdReqDelay0. Read-write. Reset: 0h. Adds 0-15 DfiClks of additional delay after dfi0_ctrlupd_req asserts, before the PHY takes any action (such as starting DDL calibration).
11:8	CtrlUpdAckDelay0. Read-write. Reset: 0h. Adds 0-15 DfiClks of additional delay after the PHY completes all PHY update activities, before de-asserting dfi0_ctrlupd_ack.
7:4	PhyUpdReqDelay0. Read-write. Reset: 0h. Adds 0-15 DfiClks of additional delay after the PHY completes all PHY update activities, before de-asserting dfi0_phyupd_req.
3:0	PhyUpdAckDelay0. Read-write. Reset: 0h. Adds 0-15 DfiClks of additional delay after dfi0_phyupd_ack asserts, before the PHY takes any action (such as starting DDL calibration).

UMC[00...11]CTLx00000804_address000200E1 [Small delays on handshake signals] (UMC::Phy::DfiHandshakeDelays1)

Read-write. Reset: 0000h.

Add additional assertion/deassertion delays on handshake signals (for example, these delays add to existing delays)
Logic assumes that dfi signal assertions exceed the programmed delays.

IMPORTANT: Timing parameters for dfi_phyupd and dfi_ctrlupd interfaces are valid only when this register is 0.

IMPORTANT: Non-zero values of this register are Reserved for Debug.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200E1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	CtrlUpdReqDelay1. Read-write. Reset: 0h. Adds 0-15 DfiClks of additional delay after dfi1_ctrlupd_req asserts, before the PHY takes any action (such as starting DDL calibration).
11:8	CtrlUpdAckDelay1. Read-write. Reset: 0h. Adds 0-15 DfiClks of additional delay after the PHY completes all PHY update activities, before de-asserting dfi1_ctrlupd_ack.
7:4	PhyUpdReqDelay1. Read-write. Reset: 0h. Adds 0-15 DfiClks of additional delay after the PHY completes all PHY update activities, before de-asserting dfi1_phyupd_req.
3:0	PhyUpdAckDelay1. Read-write. Reset: 0h. Adds 0-15 DfiClks of additional delay after dfi1_phyupd_ack asserts, before the PHY takes any action (such as starting DDL calibration).

UMC[00...11]CTLx00000804_address000200E3 (UMC::Phy::LcdlDbgCntl)

Read-write. Reset: 0000h.

Controls for use in observing and testing the LCDLs.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200E3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	LcdlStatusSel. Read-write. Reset: 0h. Description: Selects the LCDL status, from among the status for the 16 LCDLs in the DBYTE, for reading via Register DxLcdlStatus and an LCDL from among the LCDLs in the ANIB for reading via Register AcLcdlStatus. LcdlStatusSel source for DxLcdlStatus source for AcLcdlStatus - 15: lcdl_rxclk1t reserved - 14: lcdl_rxclk1c reserved - 13: lcdl_rxclk0t reserved - 12: lcdl_rxclk0c reserved - 11: lcdl_rxen1 ANIB11-tx/tu* - 10: lcdl_rxen0 ANIB10-tx/tu* - 9: lcdl_txln9 (dqs-lower) ANIB9-tx/tu* - 8: lcdl_txln8 (dm/dqs-upper) ANIB8-tx/tu* - 7: lcdl_txln7 (dq7) ANIB7-tx/tu* - 6: lcdl_txln6 (dq6) ANIB6-tx/tu* - 5: lcdl_txln5 (dq5) ANIB5-tx/tu* - 4: lcdl_txln4 (dq4) ANIB4-tx/tu* - 3: lcdl_txln3 (dq3) ANIB3-tx/tu* - 2: lcdl_txln2 (dq2) ANIB2-tx/tu* - 1: lcdl_txln1 (dq1) ANIB1-tx/tu* - 0: lcdl_txln0 (dq0) ANIB0-tx/tu* There is only one set of calibration logic, and the registers reflect the state of that logic. For non-clock ANIBs, there is only one DLL to calibrate, so the registers always reflect the results of the last TxDLL calibration. For clock ANIBs, by default, the MsDLL is selected, so the registers will reflect the results of the last MsDLL calibration. However, if user set the appropriate bits in csrAcLcdlMasDis, the calibration logic can instead calibrate the TxDLL, and after the calibration the registers will reflect the results of that calibration. There is a pair of DLLs for Tx. When set, csrLcdlTuStatusSel selects the DLL associated with the DLL used for [3:2] (the "Tu" DLL)
11	LcdlTstEnable. Read-write. Reset: 0. Enables the debug/test operations and status Ovr,Snap,StickyLock,StickyUnlock, and LiveLock.
10	LcdlFineSnap. Read-write. Reset: 0. Description: Latch enable for reading the present LCDL 1UI estimate code in LcdlFineSnapVal and the present phase-detector value in LcdlPhdSnapVal
9	LcdlFineOvr. Read-write. Reset: 0. Description: Forces the value of the present LCDL 1UI estimate code to be LcdlFineOvrVal for all LCDLs, used only when LcdlPh10b=0. The correct way to use LcdlFineOvrVal and LcdlFineOvr is to write them before running locking or before update events. The values that are written will become applied to the LCDL when locking update events. Do not write it and expect it to be applied immediately, and be careful that this value is not otherwise overwritten. Also leave LcdlFineOvr set to 1, ie do not pulse it.

8:0	LcdlFineOvrVal. Read-write. Reset: 000h. Description: Value forced as the initial value while LcdlTstEnable=1 and LcdlFineOvr. After the deassertion of LcdlFineOvr, the locking state-machine will attempt locking. Used only when LcdlPh10b=0
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UMC[00...11]CTLx00000804_address000200E4 [Debug status of the ANIB LCDL] (UMC::Phy::AcLcdlStatus)

Read-only, Volatile.

Status of ANIB macro LCDL selected by Register LcdlStatusSel.

Note: There is just one copy of this register, with the information from the implemented ANIBs multiplexed per csr LcdlStatus Sel.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200E4; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13	AcLcdlLiveLock. Read-only, Volatile. Present value of whether the LCDL is locked, valid when LcdlTstEnable=1.
12	AcLcdlStickyUnlock. Read-only, Volatile. Latched value of whether the LCDL ever lost lock after the assertion of LcdlTstEnable.
11	AcLcdlStickyLock. Read-only, Volatile. Latched value of whether the LCDL ever achieved lock after the assertion of LcdlTstEnable.
10	AcLcdlPhdSnapVal. Read-only, Volatile. Value of the LCDL phase-detector output, latched by pulse on LcdlFineSnap while csr LcdlTstEnable=1.
9:0	AcLcdlFineSnapVal. Read-only, Volatile. Description: Value of the LCDL 1UI estimate code, latched by pulse on csrLcdlFineSnap while csr LcdlTstEnable=1. Index 9 is reserved for growth.

UMC[00...11]CTLx00000804_address000200E5 [Controls for use in observing and testing the LCDLs.] (UMC::Phy::LcdlDbgCntl2)

Read-write. Reset: 0000h.

Controls for use in observing and testing the LCDLs. See description of csrLcdlFineOvr for more info.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200E5; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:13	Reserved.
12	LcdlTuStatusSel. Read-write. Reset: 0. When set, selects TuDll for csrLcdlStatusSel.
11	LcdlFineSeed2OvrVal. Read-write. Reset: 0. Description: Selects the csr to use for fine override. - 1 use the per-pstate LcdlSeed0 as the override value - 0 use LcdlFineOvrVal or LcdlFineOvrVal10b as the override value
10	LcdlPh10b. Read-write. Reset: 0. Description: Used only for selecting the csr to use for fine override. - 1 use LcdlFineOvrVal10b - 0 use LcdlFineOvrVal
9:0	LcdlFineOvrVal10b. Read-write. Reset: 000h. Description: Value forced as the initial value while LcdlTstEnable=1 and LcdlFineOvr. After the deassertion of LcdlFineOvr, the locking state-machine will attempt locking. Used only when LcdlPh10b=1

UMC[00...11]CTLx00000804_address000200E7 (UMC::Phy::AnibClockAssignment)

Read-only, Volatile.

Bump Map derived Clock ANIB Assignment

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:0	AnibClockAssignment. Read-only, Volatile. Description: Bump Map derived Clock ANIB Assignment A one in bit N means that ANIB N is a Clock ANIB, which are the only ANIBs that use the MDLL.

UMC[00...11]CTLx00000804_address000200E8 (UMC::Phy::AcLcdlMasDis)

Read-write. Reset: 0000h.

Disable Master DLL for AC DLL calibration

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200E8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:0	AcLcdlMasDis. Read-write. Reset: 0000h. Description: In a clock ANIB, the MsDLL is calibrated. In a non-clock ANIB, the TxDLL is calibrated. The bits in this register are don't cares except for the bits that correspond to clock ANIBs. (see csrAnibClockAssignment) For those bits, a zero provides normal operation. A one converts the clock ANIB to operate as if it is a non-clock ANIB, permitting direct calibration of the clock-ANIB TxDLL. This register is used only for testing purposes.

UMC[00...11]CTLx00000804_address000200E9 (UMC::Phy::AcClkLcdlControl)

Read-write. Reset: 00A0h.

Various Controls for AC LCDL test and manual control

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200E9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:4	MDllGainTrk. Read-write. Reset: Ah. Description: Value of DllGain for MDLL when in tracking mode, ie after initial lock. Larger values will cause increased filtering before changing the 1UI phase value; that is, it will be more stable but have slower response to VT changes. Valid values are 5-10. Values over 5 low-pass-filter the dither
3	AcLcdlMDllGainTrk. Read-write. Reset: 0. Setting this bit causes the MDllGainTrk to be used for the MDLL when tracking, ie after initial lock.
2	AcLcdlMasDllGainTv6. Read-write. Reset: 0. Setting this bit causes the DllGainTV values on the Clock MDLLs to be forced to 6 during locking (so that the TXDLLs in the non-clock ANIBs can use a different calibration value)
1	AcLcdlPhaseOvr. Read-write. Reset: 0. Setting this bit causes the CalPhase values for the MDLLs (Which can be overridden by csrLcdlCalPhase) to be applied to the inputs of the rampers for all clock ANIBs.
0	AcLcdlRampByp. Read-write. Reset: 0. Setting this bit causes the rampers to be bypassed. Updates applied to a ramper input are immediately applied to the ramper output, and ramps in progress are terminated.

UMC00CTLx00000804_address000200EA...UMC11CTLx00000804_address003200EA (UMC::Phy::AcClkDLLControl)

Read-write. Reset: 1080h.

Controls for the clock ANIB DLLs

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address000200EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address001200EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address002200EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address003200EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:12	AcLcdlStepInterval. Read-write. Reset: 1h. Description: This selects the shortest interval between steps for the ramp mechanism for the phase code to AC TxDLLs. The three selections [1,2,3] correspond to delays of [8,16,32] DfiClk cycles respectively. A zero disables the step mechanism, no updates are applied to the DLL unless the ramper is bypassed (csr AcLcdlRampByp).
11:0	AcLcdlUpdInterval. Read-write. Reset: 080h. Description: This contains the interval in DfiClks between updates applied to the TxDLL in the Clock. ANIBs. It affects only DLLs in ANIBs where MDLL is selected. A zero disables updates. Minimum value = 64.

UMC00CTLx00000804_address000200EB...UMC11CTLx00000804_address003200EB (UMC::Phy::LcdlDbgCntl3)

Read-write. Reset: 0000h.

Threshold for the LcdlStickyUnlock detection.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address000200EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address001200EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address002200EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address003200EB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	StickyUnlockThreshold. Read-write. Reset: 0h. Description: This field sets a threshold for the assertion of DxLcdlStickyUnlock and AcLcdlStickyUnlock. If LCDL updates are proceeding at sufficient pace then the calibration phase-detector samples should dither. However, after LCDL locking is completed, if the number of calibration phase-detector samples have the same value for more than csrStickyUnlockThreshold samples, then the DxLcdlStickyUnlock or AcLcdlStickyUnlock bit will be set, and the StickyUnlock interrupt will be asserted. This may indicate that the VT drift is faster than the frequency of update events can track. - A different value of the calibration phase-detector sample will reset the count to zero. - If 0, then the LcdlStickyUnlock will not set. - The value 1 is reserved. Otherwise, the recommended minimum number is 3.

UMC[00...11]CTLx00000804_address000200EF [Start vector value to be used for LP3-exit or Init PIE Sequence] (UMC::Phy::LP3ExitSeq0BStartVector)

Read-write. Reset: 00DEh.

Programmed by dwc_ddrphy_phyinit_C_initPhyConfig() to support frequency changes.

Set the start vector to be used in LP3-exit or Init PIE sequence.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200EF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:4	LP3ExitSeq0BStartVecPllBypassed. Read-write. Reset: Dh. Description: PIE Start Vector value to be used for LP3-exit or Init and target P-state has PLL bypassed
3:0	LP3ExitSeq0BStartVecPllEnabled. Read-write. Reset: Eh. Description: PIE Start Vector value to be used for LP3-exit or Init and target P-state has PLL enabled

UMC00CTLx00000804_address000200F0...UMC11CTLx00000804_address000200F7 [DFI Frequency Translation Register 0..7] (UMC::Phy::DfiFreqXlat)

Read-write. Reset: 0000h.

DFI Frequency Translation Register

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CTLx00000804_address000200F0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CTLx00000804_address000200F1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_n2_aliasSMN; UMC[11:00]CTLx00000804_address000200F2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_n3_aliasSMN; UMC[11:00]CTLx00000804_address000200F3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_n4_aliasSMN; UMC[11:00]CTLx00000804_address000200F4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_n5_aliasSMN; UMC[11:00]CTLx00000804_address000200F5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_n6_aliasSMN; UMC[11:00]CTLx00000804_address000200F6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_n7_aliasSMN; UMC[11:00]CTLx00000804_address000200F7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	DfiFreqXlatVal3. Read-write. Reset: 0h. The sequencer start vector used when dfi_freq value is $(n * 4) + 3$. i.e. 3,7,11,15,19,23,27,31.
11:8	DfiFreqXlatVal2. Read-write. Reset: 0h. The sequencer start vector used when dfi_freq value is $(n * 4) + 2$. i.e. 2,6,10,14,18,22,26,30.
7:4	DfiFreqXlatVal1. Read-write. Reset: 0h. The sequencer start vector used when dfi_freq value is $(n * 4) + 1$. i.e. 1,5,9,13,17,21,25,29.
3:0	DfiFreqXlatVal0. Read-write. Reset: 0h. The sequencer start vector used when dfi_freq value is $(n * 4) + 0$. i.e. 0,4,8,12,16,20,24,28.

UMC[00...11]CTLx00000804_address000200F9 (UMC::Phy::DfiInitComplete)

Read-write. Reset: 0000h.

Dfi Init Complete control

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000200F9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	DfiInitComplete. Read-write. Reset: 0. Description: This register directly controls DfiInitComplete, and is meant to be written by the PState sequencer as part of the power state switching sequence. It should be written to a 1 once the transition to LP2 is complete. Likewise it should be written to 0 once the transition to an active and PState is complete. It should reset to a 0 on both warm and cold reset.

UMC00CTLx00000804_address000200FA...UMC11CTLx00000804_address003200FA (UMC::Phy::DfiFreqRatio)

Read-write. Reset: 0001h.

Dfi Frequency Ratio

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address000200FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address001200FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address002200FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address003200FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1:0	DfiFreqRatio. Read-write. Reset: 1h. Description: Specifies the ratio DfiCtlClk:MemClk. - 00 = 1:1 mode (unsupported) - 01 = 1:2 mode - 10 = 1:4 mode When DWC_DDRPHY_FIXED_DFICTLCLK_RATIO is defined, this register is ignored and 1:2 mode is selected.

UMC00CTLx00000804_address000200FC...UMC11CTLx00000804_address003200FC [Control write DQ preamble by specifying a pattern and an enable] (UMC::Phy::DqPreamblePatternU0)

Read-write. Reset: 0000h.

Generate signal-conditioning, non-payload write DQ preamble by specifying a pattern and an enable. Function available only when WLn9=1. Per nibble.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address000200FC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address001200FC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address002200FC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address003200FC; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:4	EnTxDqPreamblePatternU0. Read-write. Reset: 0h. Description: This field is programmed as follows: - 0x0 no DQ preamble except as determined by tx equalization logic mode 2 - 0x1 drive the 1 UI before the first payload UI with TxDqPreamblePatternU0[0] - 0x3 drive the 2 UI before the first payload UI with TxDqPreamblePatternU0[1:0] - 0x7 drive the 3 UI before the first payload UI with TxDqPreamblePatternU0[2:0] - 0xf drive the 4 UI before the first payload UI with TxDqPreamblePatternU0[3:0] - other values are reserved. PdaMrsWriteMode has priority over this mode.
3:0	TxDqPreamblePatternU0. Read-write. Reset: 0h. Description: The DQ 4 UI before the first payload UI. will be driven with this pattern for the UI that have EnTxDqPreamblePattern=1. Index 3 is the first UI, and index 0 will be the last UI.

UMC00CTLx00000804_address000200FD...UMC11CTLx00000804_address003200FD [Control write DQ preamble by specifying a pattern and an enable] (UMC::Phy::DqPreamblePatternU1)

Read-write. Reset: 0000h.

Generate signal-conditioning, non-payload write DQ preamble by specifying a pattern and an enable. Function available only when Wlm9=1. Per nibble.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address000200FD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address001200FD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address002200FD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address003200FD; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:4	EnTxDqPreamblePatternU1. Read-write. Reset: 0h. Description: This field is programmed as follows: - 0x0 no DQ preamble except as determined by tx equalization logic mode 2 - 0x1 drive the 1 UI before the first payload UI with TxDqPreamblePatternU1[0] - 0x3 drive the 2 UI before the first payload UI with TxDqPreamblePatternU1[1:0] - 0x7 drive the 3 UI before the first payload UI with TxDqPreamblePatternU1[2:0] - 0xf drive the 4 UI before the first payload UI with TxDqPreamblePatternU1[3:0] - other values are reserved. PdaMrsWriteMode has priority over this mode.
3:0	TxDqPreamblePatternU1. Read-write. Reset: 0h. Description: The DQ 4 UI before the first payload UI. will be driven with this pattern for the UI that have EnTxDqPreamblePattern=1. Index 3 is the first UI, and index 0 will be the last UI.

UMC00CTLx00000804_address000200FE...UMC11CTLx00000804_address003200FE [Control write DM preamble by specifying a pattern and an enable] (UMC::Phy::DmPreamblePattern)

Read-write. Reset: 0000h.

Generate signal-conditioning, non-payload write DM preamble by specifying a pattern and an enable. Function available only when WLM9=1. Per nibble.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address000200FE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address001200FE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address002200FE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address003200FE; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:4	EnTxDmPreamblePattern. Read-write. Reset: 0h. Description: This field is programmed as follows: - 0x0 no DQ preamble except as determined by tx equalization logic mode 2 - 0x1 drive the 1 UI before the first payload UI with TxDmPreamblePattern[0] - 0x3 drive the 2 UI before the first payload UI with TxDmPreamblePattern[1:0] - 0x7 drive the 3 UI before the first payload UI with TxDmPreamblePattern[2:0] - 0xf drive the 4 UI before the first payload UI with TxDmPreamblePattern[3:0] - other values are reserved. PdaMrsWriteMode has priority over this mode.
3:0	TxDmPreamblePattern. Read-write. Reset: 0h. Description: The DM 4 UI before the first payload UI, will be driven with this pattern for the UI that have EnTxDmPreamblePattern=1. Index 3 is the first UI, and index 0 will be the last UI.

UMC[00...11]CTLx00000804_address00020100 [Interrupt Enable Bits] (UMC::Phy::PhyInterruptEnable)

Read-write. Reset: 0000h.

Enables various interrupts to appear in the PhyInterruptStatus CSR. Only interrupts which are enabled by writing a 1 to the appropriate location in this CSR will be observable in the PhyInterruptStatus CSR. If a 0 is written in a particular location the corresponding interrupt will not be visible in the PhyInterruptStatus CSR.

See the description of PhyInterruptStatus for more information on specific interrupts.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020100; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	PhyStickyUnlockEn. Read-write. Reset: 0. Description: Enable for the StickyUnlock Interrupt - 0 : Interrupt not enabled - 1 : Interrupt enabled
14	PhyBsiEn. Read-write. Reset: 0. Description: Enable for the Bsi Interrupt - 0 : Interrupt not enabled - 1 : Interrupt enabled
13	PhyAnibRcvErrEn. Read-write. Reset: 0. Description: Enable for ANIB Receive Error (See csrAnibRcvErr) - 0 : Interrupt not enabled - 1 : Interrupt enabled
12	PhyD5ACSM1ParityEn. Read-write. Reset: 0. Description: Enable for D5ACSM Channel 1 Parity Error - 0 : Interrupt not enabled - 1 : Interrupt enabled
11	PhyD5ACSM0ParityEn. Read-write. Reset: 0. Description: Enable for D5ACSM Channel 0 Parity Error - 0 : Interrupt not enabled - 1 : Interrupt enabled
10	PhyRxFifoCheckEn. Read-write. Reset: 0. Description: Enable for the RxFifo Pointers Check Interrupt - 0 : Interrupt not enabled - 1 : Interrupt enabled
9	PhyTxPPTEn. Read-write. Reset: 0. Description: Enable for the TX PPT Interrupt - 0 : Interrupt not enabled - 1 : Interrupt enabled
8	PhyEccEn. Read-write. Reset: 0. Description: Enable for the ARC ECC Interrupt - 0 : Interrupt not enabled - 1 : Interrupt enabled
7:3	PhyFWReservedEn. Read-write. Reset: 00h. Description: Enable for the Reserved Interrupts - 0 : Interrupt not enabled - 1 : Interrupt enabled
2	PhyTrngFailEn. Read-write. Reset: 0. Description: Enable for the PHY Training Failure interrupt. - 0 : Interrupt not enabled - 1 : Interrupt enabled
1	PhyInitCmplEn. Read-write. Reset: 0.

	Description: Enable for the PHY Initialization Complete interrupt. - 0 : Interrupt not enabled - 1 : Interrupt enabled
0	PhyTrngCmplEn. Read-write. Reset: 0. Description: Enable for the PHY Training Complete interrupt. - 0 : Interrupt not enabled - 1 : Interrupt enabled

UMC[00...11]CTLx00000804_address00020101 [Interrupt Firmware Control Bits] (UMC::Phy::PhyInterruptFWControl)

Read-write. Reset: 0000h.

This register is written ONLY by the Firmware. A 0 to 1 transition for any of the interrupts will set the corresponding bit location in the PhyInterruptStatus CSR if the corresponding bit location in the PhyInterruptEnable CSR is set to 1. See the description of PhyInterruptStatus for more information on specific interrupts.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020101; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:3	PhyFWReservedFW. Read-write. Reset: 00h. Description: Implemented, reserved for future use. - 0 : Interrupt not asserted - 1 : Interrupt asserted - Reserved
2	PhyTrngFailFW. Read-write. Reset: 0. Description: PHY Training Failure Firmware interrupt. - 0 : Interrupt not asserted - 1 : Interrupt asserted
1	PhyInitCmplFW. Read-write. Reset: 0. Description: PHY Initialization Complete Firmware interrupt. Note: Use of this interrupt is optional. The PHY does not set this interrupt. In the event that there are multiple processors or threads in the SOC, the thread loading PHYINIT can write this bit as part of. PhyInit userCustomPost() or manually via APB after first dfi_init_start. - 0 : Interrupt not asserted - 1 : Interrupt asserted
0	PhyTrngCmplFW. Read-write. Reset: 0. Description: PHY Training Complete Firmware interrupt. - 0 : Interrupt not asserted - 1 : Interrupt asserted

UMC[00...11]CTLx00000804_address00020102 [Interrupt Mask Bits] (UMC::Phy::PhyInterruptMask)

Read-write. Reset: 0000h.

This register is used to mask various interrupts to prevent them from affecting the output pin dwc_ddsphy_int_n. Only interrupts which are not masked by clearing to 0 the corresponding location in this CSR will be able to affect the dwc_ddsphy_int_n output when asserted in the PhyInterruptStatus CSR. If a 1 is written in a particular location that interrupt will still be visible in the PhyInterruptStatus CSR but that interrupt will not affect the output pin dwc_ddsphy_int_n.

See the description of PhyInterruptStatus for more information on specific interrupts.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020102; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	PhyStickyUnlockMsk. Read-write. Reset: 0. Description: Mask for StickyUnlock Interrupt - 0 : Interrupt not masked - 1 : Interrupt masked
14	PhyBsiMsk. Read-write. Reset: 0. Description: Mask for Bsi Interrupt - 0 : Interrupt not masked - 1 : Interrupt masked
13	PhyAnibRcvErrMsk. Read-write. Reset: 0. Description: Mask for ANIB Receive Error (See csrAnibRcvErr) - 0 : Interrupt not masked - 1 : Interrupt masked
12	PhyD5ACSM1ParityMsk. Read-write. Reset: 0. Description: Mask for D5ACSM Channel 1 Parity Error - 0 : Interrupt not masked - 1 : Interrupt masked
11	PhyD5ACSM0ParityMsk. Read-write. Reset: 0. Description: Mask for D5ACSM Channel 0 Parity Error - 0 : Interrupt not masked - 1 : Interrupt masked
10	PhyRxFifoCheckMsk. Read-write. Reset: 0. Description: Mask for the RxFifo Pointers Check Interrupt - 0 : Interrupt not masked - 1 : Interrupt masked
9	PhyTxPPTMsk. Read-write. Reset: 0. Description: Enable for the TX PPT Interrupt - 0 : Interrupt not masked - 1 : Interrupt masked
8	PhyEccMsk. Read-write. Reset: 0. Description: Mask for the ARC ECC Interrupt - 0 : Interrupt not masked - 1 : Interrupt masked
7:3	PhyFWReservedMsk. Read-write. Reset: 00h. Description: Mask for the Reserved interrupts, - 0 : Interrupt not masked - 1 : Interrupt masked
2	PhyTrngFailMsk. Read-write. Reset: 0. Description: Mask for the PHY Training Failure interrupt. - 0 : Interrupt not masked - 1 : Interrupt masked

1	PhyInitCmpltMsk. Read-write. Reset: 0.
	Description: Mask for the PHY Initialization Complete interrupt. - 0 : Interrupt not masked - 1 : Interrupt masked
0	PhyTrngCmpltMsk. Read-write. Reset: 0.
	Description: Mask for the PHY Training Complete interrupt. - 0 : Interrupt not masked - 1 : Interrupt masked

UMC[00...11]CTLx00000804_address00020103 [Interrupt Clear Bits] (UMC::Phy::PhyInterruptClear)

Read-write. Reset: 0000h.

This register clears various interrupts and will negate the output pin dwc_ddrphy_int_n if the cleared interrupt was the cause

of the assertion of the output pin dwc_ddrphy_int_n. A 0 to 1 transition on a bit will clear the corresponding interrupt, and

cause the hardware to immediately clear the bit, causing the register to behave as a Write-Only register.

See the description of PhyInterruptStatus for more information on specific interrupts.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020103; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	PhyStickyUnlockClr. Read-write. Reset: 0. Description: Clear for StickyUnlock Interrupt - 0 : Interrupt not affected - 1 : Interrupt cleared
14	PhyBsiClr. Read-write. Reset: 0. Description: Clear Bsi Interrupt and csrBsiInterfaceErr register - 0 : Interrupt not affected - 1 : Interrupt cleared
13	PhyAnibRcvErrClr. Read-write. Reset: 0. Description: Clear for ANIB Receive Error and csrAnibRcvErr - 0 : Interrupt not affected - 1 : Interrupt cleared
12	PhyD5ACSM1ParityClr. Read-write. Reset: 0. Description: Clear for D5ACSM Channel 1 Parity Error - 0 : Interrupt not affected - 1 : Interrupt cleared
11	PhyD5ACSM0ParityClr. Read-write. Reset: 0. Description: Clear for D5ACSM Channel 0 Parity Error - 0 : Interrupt not affected - 1 : Interrupt cleared
10	PhyRxFifoCheckClr. Read-write. Reset: 0. Description: Clear for the RxFifo Pointers Check Interrupt - 0 : Interrupt not affected - 1 : Interrupt cleared
9	PhyTxPPTClr. Read-write. Reset: 0. Description: Clear for the TX PPT Interrupt - 0 : Interrupt not affected - 1 : Interrupt cleared
8	PhyEccClr. Read-write. Reset: 0. Description: Clear for the ARC ECC Interrupt - 0 : Interrupt not affected - 1 : Interrupt cleared
7:3	PhyFWReservedClr. Read-write. Reset: 00h. Description: Clear for the Reserved Interrupts - 0 : Interrupt not affected - 1 : Interrupt cleared
2	PhyTrngFailClr. Read-write. Reset: 0. Description: Clear for the PHY Training Failure interrupt. - 0 : Interrupt not affected - 1 : Interrupt cleared

1	PhyInitCmpltClr. Read-write. Reset: 0.
	Description: Clear for the PHY Initialization Complete interrupt. - 0 : Interrupt not affected - 1 : Interrupt cleared
0	PhyTrngCmpltClr. Read-write. Reset: 0.
	Description: Clear for the PHY Training Complete interrupt. - 0 : Interrupt not affected - 1 : Interrupt cleared

UMC[00...11]CTLx00000804_address00020104 [Interrupt Status Bits] (UMC::Phy::PhyInterruptStatus)

Read-only, Volatile.

This register displays the current status of various PHY interrupts. This is a read only CSR. The PHY is the only entity which can write to it.

In order for a bit to become set in the PhyInterruptStatus CSR the corresponding bit must be set in the PhyInterruptEnable CSR.

If an interrupt is set in the PhyInterruptStatus CSR and the corresponding bit in the PhyInterruptMask CSR is cleared the dwc_ddrphy_int_n output signal will be asserted. Once the dwc_ddrphy_int_n output signal is asserted it can only be negated when all unmasked interrupts in the PhyInterruptStatus CSR have been cleared by writing the corresponding bit in the PhyInterruptClear CSR to 1.

_umc0_chiplotype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020104; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	PhyStickyUnlockErr. Read-only, Volatile. Description: Indicates that csrDxLcdlStickyUnlock or csrAcLcdlStickyUnlock were set, possibly indicating that VT drift is faster than the phase update frequency.
14	PhyBsiInt. Read-only, Volatile. Description: Bsi Interrupt occurred (csrBsiInterfaceErr non-zero, or interrupt from BSI Interface occurred) - 0 : Interrupt not enabled or not asserted - 1 : Interrupt enabled and asserted
13	PhyAnibRcvErr. Read-only, Volatile. Description: ANIB Receive Error (See csrAnibRcvErr) - 0 : Interrupt not enabled or not asserted - 1 : Interrupt enabled and asserted
12	PhyD5ACSM1ParityErr. Read-only, Volatile. Description: Parity Error (even parity) for D5ACSM Channel 1 Parity Error - 0 : Interrupt not enabled or not asserted - 1 : Interrupt enabled and asserted
11	PhyD5ACSM0ParityErr. Read-only, Volatile. Description: Parity Error (even parity) for D5ACSM Channel 0 Parity Error - 0 : Interrupt not enabled or not asserted - 1 : Interrupt enabled and asserted
10	PhyRxFifoCheckErr. Read-only, Volatile. Description: A mechanism in the PHY checks the Read Fifo pointers for consistency at times they are idle. If a check fails, this interrupt is asserted, and detailed status is found in the RxFifoCheckStatus register. - 0 : Interrupt not enabled or not asserted - 1 : Interrupt enabled and asserted
9	PhyRxTxPPTErr. Read-only, Volatile. Description: TxPPT Error - Check the PPT-specific Error registers for more information - 0 : Interrupt not enabled or not asserted - 1 : Interrupt enabled and asserted
8	PhyEccErr. Read-only, Volatile. Description: ARC ECC Interrupt - see csrArcEccIndications for specifics - 0 : Interrupt not enabled or not asserted - 1 : Interrupt enabled and asserted
7:3	PhyFWReserved. Read-only, Volatile.

	Description: Reserved Firmware Interrupts. These are implemented and can be used by the Firmware (only) for any purpose. - 0 : Interrupt not enabled or not asserted - 1 : Interrupt enabled and asserted
2	PhyTrngFail. Read-only,Volatile. Description: PHY Training Failure interrupt. Assertion of this interrupt indicates there was an issue during training which did not allow the PHY to successfully complete training. - 0 : Interrupt not enabled or not asserted - 1 : Interrupt enabled and asserted
1	PhyInitCmpl. Read-only,Volatile. Description: PHY Initialization Complete interrupt. Assertion of this interrupt indicates that the PHY Initialization Engine has been fully loaded and the PHY is ready and able to assert dfi_init_complete following the assertion of dfi_init_start. Note: Use of this interrupt is optional. The PHY does not set this interrupt. In the event that there are multiple processors or threads in the SOC, the thread loading PHYINIT can write this bit as part of. PhyInit userCustomPost() or manually via APB after first dfi_init_start. - 0 : Interrupt not enabled or not asserted - 1 : Interrupt enabled and asserted
0	PhyTrngCmpl. Read-only,Volatile. Description: PHY Training Complete interrupt. Assertion of this interrupt indicates that the PHY has successfully trained the memory interface at all indicated frequencies and is ready for the PHY Initialization Engine to be loaded. - 0 : Interrupt not enabled or not asserted - 1 : Interrupt enabled and asserted

UMC[00...11]CTLx00000804_address00020107 (UMC::Phy::PhyInterruptOverride)

Read-write. Reset: 0000h.

Interrupt Override Debug Register

_umc0_chiplletype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020107; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	PhyInterruptOverride. Read-write. Reset: 0000h. Description: This register is used for debug. A 0 -> 1 transition on any bit causes the corresponding interrupt in csrPhyInterruptStatus to be triggered, if the interrupt is masked and enabled. This permits software emulation of HW interrupts.

UMC[00...11]CTLx00000804_address00020110 (UMC::Phy::BsiInterfaceAddr)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplletype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020110; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	Reserved.
11:0	BsiInterfaceAddr. Read-write. Reset: 000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address00020111 (UMC::Phy::BsiInterfaceCtl)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020111; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1:0	BsiInterfaceCtl . Read-write. Reset: 0h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address00020112 (UMC::Phy::BsiInterfaceWrDataLo)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020112; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	BsiInterfaceWrDataLo . Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address00020113 (UMC::Phy::BsiInterfaceWrDataHi)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020113; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	BsiInterfaceWrDataHi . Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address00020114 (UMC::Phy::BsiInterfaceRdDataLo)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020114; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	BsiInterfaceRdDataLo . Read-only, Volatile. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address00020115 (UMC::Phy::BsiInterfaceRdDataHi)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020115; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	BsiInterfaceRdDataHi . Read-only, Volatile. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address00020116 (UMC::Phy::BsiInterfaceErr)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020116; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	BsiInterfaceErr . Read-only, Volatile. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address00020117 (UMC::Phy::BsiInterfaceRdy)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020117; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	BsiInterfaceRdy . Read-only, Volatile. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address0002011C (UMC::Phy::D5ACSMInfiniteOLRC)

Read-write. Reset: 0000h.

Enables outer loop as infinite for D5ACSMs

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002011C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1:0	D5ACSMInfiniteOLRC . Read-write. Reset: 0h. Description: This field is programmed as follows: - When set, the outer loop runs forever if corresponding csrLP54ACSMRun is set. - When the bit is cleared, the loop counter is set to 0 and the control flow exits the loop. - Bit 0 represents channel 0 and Bit 1 represents channel 1.

UMC[00...11]CTLx00000804_address0002011D (UMC::Phy::D5ACSMRun)

Read-write. Reset: 0000h.

Enable for D5ACSMs

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002011D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1:0	D5ACSMRun . Read-write. Reset: 0h. Description: These are the start bits for the playback engines. Bit 0 governs channel 0 and Bit 1 governs channel 1. The D5ACSM starts when it detects a 0->1 transition on the corresponding bit. When the Run bit is set and the Done bit is not set, the D5ACSM is running.

UMC[00...11]CTLx00000804_address0002011E (UMC::Phy::D5ACSMDone)

Read-only, Volatile.

Done Status for D5ACSMs

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002011E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1:0	D5ACSMDone . Read-only, Volatile. Description: These are the done bits for the playback engines. Bit 0 represents channel 0 and Bit 1 represents channel 1. They are set when the corresponding D5ACSM has completed its sequence. When the Run bit is set and the Done bit is not set, the D5ACSM is running.

UMC[00...11]CTLx00000804_address0002011F (UMC::Phy::D5ACSMiscControl)

Read-write. Reset: 0000h.

Miscellaneous D5ACSM Control

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002011F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1	D5ACSMdtsmEnOnInactiveRank. Read-write. Reset: 0. Enables dtsm_rdata_en on inactive ranks when using csrD5ACSMMaskCs;
0	Reserved.

UMC00CTLx00000804_address00020120...UMC11CTLx00000804_address00320120 [Timing Parameters for RxEn Pulse for Channel 0] (UMC::Phy::D5ACSM0RxEnPulse)

Read-write. Reset: 0000h.

Contains the delay and pulse width for RxEn for Channel 0

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020120; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120120; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220120; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320120; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:8	D5ACSM0RxEnWidth. Read-write. Reset: 00h. The number of MEMCLKs the RxEn bit is asserted
7:6	D5ACSM0RxEnDelayReserved. Read-write. Reset: 0h. Reserved
5:0	D5ACSM0RxEnDelay. Read-write. Reset: 00h. Description: The number of MEMCLKs between the cycle that the RxEn bit is set in the D5ACSM and the cycle that RxEn is asserted

UMC00CTLx00000804_address00020121...UMC11CTLx00000804_address00320121 [Timing Parameters for RxVal Pulse for Channel 0] (UMC::Phy::D5ACSM0RxValPulse)

Read-write. Reset: 0000h.

Contains the delay and pulse width for RxVal for Channel 0

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020121; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120121; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220121; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320121; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:8	D5ACSM0RxValWidth. Read-write. Reset: 00h. The number of MEMCLKs the RxVal bit is asserted
7:6	D5ACSM0RxValDelayReserved. Read-write. Reset: 0h. Reserved
5:0	D5ACSM0RxValDelay. Read-write. Reset: 00h. Description: The number of MEMCLKs between the cycle that the RxVal bit is set in the D5ACSM and the cycle that RxVal is asserted

UMC00CTLx00000804_address00020122...UMC11CTLx00000804_address00320122 [Timing Parameters for TxEn Pulse for Channel 0] (UMC::Phy::D5ACSM0TxEnPulse)

Read-write. Reset: 0000h.

Contains the delay and pulse width for TxEn for Channel 0

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020122; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120122; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220122; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320122; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:8	D5ACSM0TxEnWidth. Read-write. Reset: 00h. The number of MEMCLKs the TxEn bit is asserted
7:6	D5ACSM0TxEnDelayReserved. Read-write. Reset: 0h. Reserved
5:0	D5ACSM0TxEnDelay. Read-write. Reset: 00h. Description: The number of MEMCLKs between the cycle that the TxEn bit is set in the. D5ACSM and the cycle that TxEn is asserted

UMC00CTLx00000804_address00020123...UMC11CTLx00000804_address00320123 [Timing Parameters for WrCs Pulse for Channel 0] (UMC::Phy::D5ACSM0WrCsPulse)

Read-write. Reset: 0000h.

Contains the delay and pulse width for WrCs for Channel 0

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020123; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120123; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220123; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320123; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:8	D5ACSM0WrCsWidth. Read-write. Reset: 00h. The number of MEMCLKs the WrCs bit is asserted
7:6	D5ACSM0WrCsDelayReserved. Read-write. Reset: 0h. Reserved
5:0	D5ACSM0WrCsDelay. Read-write. Reset: 00h. Description: The number of MEMCLKs between the cycle that the TxEn bit is set in the. D5ACSM and the cycle that wrdata_cs (as selected by the TG bits) is asserted

UMC00CTLx00000804_address00020124...UMC11CTLx00000804_address00320124 [Timing Parameters for Rdcs Pulse for Channel 0] (UMC::Phy::D5ACSM0RdcsPulse)

Read-write. Reset: 0000h.

Contains the delay and pulse width for Rdcs for Channel 0

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020124; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120124; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220124; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320124; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:8	D5ACSM0RdcsWidth. Read-write. Reset: 00h. The number of MEMCLKs the Rdcs bit is asserted
7:6	D5ACSM0RdcsDelayReserved. Read-write. Reset: 0h. Reserved
5:0	D5ACSM0RdcsDelay. Read-write. Reset: 00h. Description: The number of MEMCLKs between the cycle that the RxEn bit is set in the. D5ACSM and the cycle that rddata_cs (as selected by the TG bits) is asserted

UMC00CTLx00000804_address00020125...UMC11CTLx00000804_address00320125 [Timing Parameters for hwt_snoop assertion for Channel 0] (UMC::Phy::D5ACSM0SnoopPulse)

Read-write. Reset: 0000h.

Contains the delay and pulse width for hwt_snoop for Channel 0

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020125; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120125; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220125; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320125; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:8	D5ACSM0SnoopWidth. Read-write. Reset: 00h. The number of MEMCLKs hwt_snoop_en is asserted
7:6	D5ACSM0SnoopDelayReserved. Read-write. Reset: 0h. Reserved
5:0	D5ACSM0SnoopDelay. Read-write. Reset: 00h. Description: The number of MEMCLKs between the phase specified by the SPS bits (when SVS bits are set) in the. D5ACSM control word, and the cycle that hwt_snoop_en is asserted

UMC[00...11]CTLx00000804_address00020126 (UMC::Phy::D5ACSM0AddressMask)

Read-write. Reset: 0000h.

CA Address Mask for Channel 0

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020126; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:0	D5ACSM0AddressMask. Read-write. Reset: 0000h. A one indicates that the corresponding bit in CA[13:0] is not part of the address increment calculation.

UMC[00...11]CTLx00000804_address00020127 (UMC::Phy::D5ACSM0AlgaIncVal)

Read-write. Reset: 0000h.

Increment value for D5ACSM Algorithmic Address Generator for Channel 0

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020127; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:0	D5ACSM0AlgaIncVal. Read-write. Reset: 0000h. Description: This value is added to an internal counter which is then added to the CA Address generated by D5ACSM 0. When the D5ACSM0AddressMask is non-zero, bit locations set to one in that register should not be considered active in this register. So, for example, if the AddressMask is 0x2, and user want to specify a value of 0x3, user would specify 0x5 (ignoring bit 1).

UMC[00...11]CTLx00000804_address00020128 (UMC::Phy::D5ACSM0ParityCtlH)

Read-write. Reset: 3FFFh.

CA Parity Control for Channel 0, First [UI](#)_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020128; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	Reserved.
14	D5ACSM0ParityOverH. Read-write. Reset: 0. A one indicates that the parity bit field from the D5ACSM control word is driven directly during the first UI .
13:0	D5ACSM0ParityMaskH. Read-write. Reset: 3FFFh. A one indicates that the corresponding bit in CA[13:0] is considered in the parity calculation for the first UI .

UMC[00...11]CTLx00000804_address00020129 (UMC::Phy::D5ACSM0ParityCtlL)

Read-write. Reset: 3FFFh.

CA Parity Control for Channel 0, Second [UI](#)_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020129; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	Reserved.
14	D5ACSM0ParityOverL. Read-write. Reset: 0. A one indicates that the parity bit field from the D5ACSM control word is driven directly during the second UI .
13:0	D5ACSM0ParityMaskL. Read-write. Reset: 3FFFh. A one indicates that the corresponding bit in CA[13:0] is considered in the parity calculation for the second UI .

UMC[00...11]CTLx00000804_address0002012A (UMC::Phy::D5ACSM0OuterLoopRepeatCnt)

Read-write. Reset: 0000h.

Outer Loop Repeat for Channel 0

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002012A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:0	D5ACSM0OuterLoopRepeatCnt. Read-write. Reset: 00h. Description: One more than the repeat count for the outer loop for D5ACSM Channel 0. A value of 0 implies a repeat count of 1 which means the code is executed twice. - When csrD5ACSMInfiniteOLRC is set, the outer loop repeats forever if csrLP54ACSMRun is set. - When csrD5ACSMInfiniteOLRC is cleared, the loop counter is set to 0 and the control flow exits the loop.

UMC00CTLx00000804_address0002012B...UMC11CTLx00000804_address0032012B (UMC::Phy::D5ACSM0StartAddr)

Read-write. Reset: 0000h.

Start Address for D5ACSM Sequence 0

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002012B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012012B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022012B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032012B; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:0	D5ACSM0StartAddr. Read-write. Reset: 00h. Allows user to change the first executed address. Bit 0 is ignored and assumed to be 0.

UMC00CTLx00000804_address0002012C...UMC11CTLx00000804_address0032012C (UMC::Phy::D5ACSM0StopAddr)

Read-write. Reset: 0000h.

Last address of Playback Sequence

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002012C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012012C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022012C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032012C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:0	D5ACSM0StopAddr. Read-write. Reset: 00h. Indicates the last address to be executed by the corresponding D5ACSM

UMC[00...11]CTLx00000804_address0002012D (UMC::Phy::D5ACSM0LastAddr)

Read-only, Volatile.

Last executed address of Playback Sequence for Channel 0

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002012D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:0	D5ACSM0LastAddr. Read-only, Volatile. Indicates the last executed address by D5ACSM 0.

UMC[00...11]CTLx00000804_address0002012E [HWT Snoop Values] (UMC::Phy::D5ACSM0SnoopVal)

Read-write. Reset: 0000h.

the D5ACSM has 2 SVS bits that select one of 3 hwt_snoop_en values to be driven

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002012E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	Reserved.
11:8	D5ACSM0HwtSnoopEn3. Read-write. Reset: 0h. The hwt_snoop_en value selected when SVS = 3
7:4	D5ACSM0HwtSnoopEn2. Read-write. Reset: 0h. The hwt_snoop_en value selected when SVS = 2
3:0	D5ACSM0HwtSnoopEn1. Read-write. Reset: 0h. The hwt_snoop_en value selected when SVS = 1

UMC[00...11]CTLx00000804_address0002012F (UMC::Phy::D5ACSM0DefaultAddr)

Read-write. Reset: 0000h.

ca value driven by D5ACSM when idle

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002012F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:0	D5ACSM0DefaultAddr. Read-write. Reset: 0000h. ca value driven by D5ACSM when idle

UMC[00...11]CTLx00000804_address00020130 (UMC::Phy::D5ACSM0DefaultCs)

Read-write. Reset: 000Fh.

CS values driven by D5ACSM when idle

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020130; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	D5ACSM0DefaultCs. Read-write. Reset: Fh. CS value driven by D5ACSM when idle

UMC[00...11]CTLx00000804_address00020131 (UMC::Phy::D5ACSM0MaskCs)

Read-write. Reset: 0000h.

Mask of ACSM Control Word CS values

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020131; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	D5ACSM0MaskCs. Read-write. Reset: 0h. A set bit in this register forces the corresponding bit in the CS[3:0] field of the D5ACSM control word to the inactive state.

UMC00CTLx00000804_address00020140...UMC11CTLx00000804_address00320140 [Timing Parameters for RxEn Pulse for Channel 1] (UMC::Phy::D5ACSM1RxEnPulse)

Read-write. Reset: 0000h.

Contains the delay and pulse width for RxEn for Channel 1

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020140; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120140; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220140; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320140; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:8	D5ACSM1RxEnWidth. Read-write. Reset: 00h. The number of MEMCLKs the RxEn bit is asserted
7:6	D5ACSM1RxEnDelayReserved. Read-write. Reset: 0h. Reserved
5:0	D5ACSM1RxEnDelay. Read-write. Reset: 00h. Description: The number of MEMCLKs between the cycle that the RxEn bit is set in the D5ACSM and the cycle that RxEn is asserted

UMC00CTLx00000804_address00020141...UMC11CTLx00000804_address00320141 [Timing Parameters for RxVal Pulse for Channel 1] (UMC::Phy::D5ACSM1RxValPulse)

Read-write. Reset: 0000h.

Contains the delay and pulse width for RxVal for Channel 1

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020141; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120141; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220141; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320141; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:8	D5ACSM1RxValWidth. Read-write. Reset: 00h. The number of MEMCLKs the RxVal bit is asserted
7:6	D5ACSM1RxValDelayReserved. Read-write. Reset: 0h. Reserved
5:0	D5ACSM1RxValDelay. Read-write. Reset: 00h. Description: The number of MEMCLKs between the cycle that the RxVal bit is set in the. D5ACSM and the cycle that RxVal is asserted

UMC00CTLx00000804_address00020142...UMC11CTLx00000804_address00320142 [Timing Parameters for TxEn Pulse for Channel 1] (UMC::Phy::D5ACSM1TxEnPulse)

Read-write. Reset: 0000h.

Contains the delay and pulse width for TxEn for Channel 1

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020142; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120142; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220142; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320142; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:8	D5ACSM1TxEnWidth. Read-write. Reset: 00h. The number of MEMCLKs the TxEn bit is asserted
7:6	D5ACSM1TxEnDelayReserved. Read-write. Reset: 0h. Reserved
5:0	D5ACSM1TxEnDelay. Read-write. Reset: 00h. Description: The number of MEMCLKs between the cycle that the TxEn bit is set in the. D5ACSM and the cycle that TxEn is asserted

UMC00CTLx00000804_address00020143...UMC11CTLx00000804_address00320143 [Timing Parameters for WrCs Pulse for Channel 1] (UMC::Phy::D5ACSM1WrCsPulse)

Read-write. Reset: 0000h.

Contains the delay and pulse width for WrCs for Channel 1

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320143; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:8	D5ACSM1WrCsWidth. Read-write. Reset: 00h. The number of MEMCLKs the WrCs bit is asserted
7:6	D5ACSM1WrCsDelayReserved. Read-write. Reset: 0h. Reserved
5:0	D5ACSM1WrCsDelay. Read-write. Reset: 00h. Description: The number of MEMCLKs between the cycle that the TxEn bit is set in the. D5ACSM and the cycle that wrdata_cs (as selected by the TG bits) is asserted

UMC00CTLx00000804_address00020144...UMC11CTLx00000804_address00320144 [Timing Parameters for RdCs Pulse for Channel 1] (UMC::Phy::D5ACSM1RdCsPulse)

Read-write. Reset: 0000h.

Contains the delay and pulse width for RdCs for Channel 1

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020144; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120144; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220144; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320144; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:8	D5ACSM1RdCsWidth. Read-write. Reset: 00h. The number of MEMCLKs the RdCs bit is asserted
7:6	D5ACSM1RdCsDelayReserved. Read-write. Reset: 0h. Reserved
5:0	D5ACSM1RdCsDelay. Read-write. Reset: 00h. Description: The number of MEMCLKs between the cycle that the RxEn bit is set in the. D5ACSM and the cycle that rddata_cs (as selected by the TG bits) is asserted

UMC00CTLx00000804_address00020145...UMC11CTLx00000804_address00320145 [Timing Parameters for hwt_snoop assertion for Channel 1] (UMC::Phy::D5ACSM1SnoopPulse)

Read-write. Reset: 0000h.

Contains the delay and pulse width for hwt_snoop for Channel 1

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00020145; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00120145; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00220145; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00320145; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:8	D5ACSM1SnoopWidth. Read-write. Reset: 00h. The number of MEMCLKs hwt_snoop_en is asserted
7:6	D5ACSM1SnoopDelayReserved. Read-write. Reset: 0h. Reserved
5:0	D5ACSM1SnoopDelay. Read-write. Reset: 00h. Description: The number of MEMCLKs between the phase specified by the SPS bits (when SVS bits are set) in the D5ACSM control word, and the cycle that hwt_snoop_en is asserted

UMC[00...11]CTLx00000804_address00020146 (UMC::Phy::D5ACSM1AddressMask)

Read-write. Reset: 0000h.

CA Address Mask for Channel 1

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020146; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:0	D5ACSM1AddressMask. Read-write. Reset: 0000h. A one indicates that the corresponding bit in CA[13:0] is not part of the address increment calculation.

UMC[00...11]CTLx00000804_address00020147 (UMC::Phy::D5ACSM1AlgaIncVal)

Read-write. Reset: 0000h.

Increment value for D5ACSM Algorithmic Address Generator for Channel 1

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020147; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:0	D5ACSM1AlgaIncVal. Read-write. Reset: 0000h. Description: This value is added to an internal counter which is then added to the CA Address generated by D5ACSM 1. When the D5ACSM1AddressMask is non-zero, bit locations set to one in that register should not be considered active in this register. So, for example, if the AddressMask is 0x2, and user want to specify a value of 0x3, user would specify 0x5 (ignoring bit 1).

UMC[00...11]CTLx00000804_address00020148 (UMC::Phy::D5ACSM1ParityCtlH)

Read-write. Reset: 3FFFh.

CA Parity Control for Channel 1, First [UI](#)_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020148; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	Reserved.
14	D5ACSM1ParityOverH. Read-write. Reset: 0. A one indicates that the parity bit field from the D5ACSM control word is driven directly during the first UI .
13:0	D5ACSM1ParityMaskH. Read-write. Reset: 3FFFh. A one indicates that the corresponding bit in CA[13:0] is considered in the parity calculation for the first UI .

UMC[00...11]CTLx00000804_address00020149 (UMC::Phy::D5ACSM1ParityCtlL)

Read-write. Reset: 3FFFh.

CA Parity Control for Channel 1, Second [UI](#)_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020149; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15	Reserved.
14	D5ACSM1ParityOverL. Read-write. Reset: 0. A one indicates that the parity bit field from the D5ACSM control word is driven directly during the second UI .
13:0	D5ACSM1ParityMaskL. Read-write. Reset: 3FFFh. A one indicates that the corresponding bit in CA[13:0] is considered in the parity calculation for the second UI .

UMC[00...11]CTLx00000804_address0002014A (UMC::Phy::D5ACSM1OuterLoopRepeatCnt)

Read-write. Reset: 0000h.

Outer Loop Repeat for Channel 1

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002014A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:0	D5ACSM1OuterLoopRepeatCnt. Read-write. Reset: 00h. Description: One more than the repeat count for the outer loop for D5ACSM Channel 1. A value of 0 implies a repeat count of 1 which means the code is executed twice. - When csrD5ACSMInfiniteOLRC is set, the outer loop repeats forever if csrLP54ACSMRun is set. - When csrD5ACSMInfiniteOLRC is cleared, the loop counter is set to 0 and the control flow exits the loop.

UMC00CTLx00000804_address0002014B...UMC11CTLx00000804_address0032014B (UMC::Phy::D5ACSM1StartAddr)

Read-write. Reset: 0000h.

Start Address for D5ACSM Sequence 1

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002014B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012014B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022014B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032014B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:0	D5ACSM1StartAddr. Read-write. Reset: 00h. Allows user to change the first executed address. Bit 0 is ignored and assumed to be 0.

UMC00CTLx00000804_address0002014C...UMC11CTLx00000804_address0032014C (UMC::Phy::D5ACSM1StopAddr)

Read-write. Reset: 0000h.

Last address of Playback Sequence

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0002014C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0012014C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0022014C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0032014C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:0	D5ACSM1StopAddr. Read-write. Reset: 00h. Indicates the last address to be executed by the corresponding D5ACSM

UMC[00...11]CTLx00000804_address0002014D (UMC::Phy::D5ACSM1LastAddr)

Read-only, Volatile.

Last executed address of Playback Sequence for Channel 1

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002014D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:0	D5ACSM1LastAddr. Read-only, Volatile. Indicates the last executed address by D5ACSM 1.

UMC[00...11]CTLx00000804_address0002014E [HWT Snoop Values] (UMC::Phy::D5ACSM1SnoopVal)

Read-write. Reset: 0000h.

the D5ACSM has 2 SVS bits that select one of 3 hwt_snoop_en values to be driven

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002014E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	Reserved.
11:8	D5ACSM1HwtSnoopEn3. Read-write. Reset: 0h. The hwt_snoop_en value selected when SVS = 3
7:4	D5ACSM1HwtSnoopEn2. Read-write. Reset: 0h. The hwt_snoop_en value selected when SVS = 2
3:0	D5ACSM1HwtSnoopEn1. Read-write. Reset: 0h. The hwt_snoop_en value selected when SVS = 1

UMC[00...11]CTLx00000804_address0002014F (UMC::Phy::D5ACSM1DefaultAddr)

Read-write. Reset: 0000h.

ca value driven by D5ACSM when idle

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002014F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:14	Reserved.
13:0	D5ACSM1DefaultAddr. Read-write. Reset: 0000h. ca value driven by D5ACSM when idle

UMC[00...11]CTLx00000804_address00020150 (UMC::Phy::D5ACSM1DefaultCs)

Read-write. Reset: 000Fh.

CS values driven by D5ACSM when idle

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020150; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	D5ACSM1DefaultCs. Read-write. Reset: Fh. CS value driven by D5ACSM when idle

UMC[00...11]CTLx00000804_address00020151 (UMC::Phy::D5ACSM1MaskCs)

Read-write. Reset: 0000h.

Mask of ACSM Control Word CS values

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020151; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	D5ACSM1MaskCs . Read-write. Reset: 0h. A set bit in this register forces the corresponding bit in the CS[3:0] field of the D5ACSM control word to the inactive state.

UMC[00...11]CTLx00000804_address00020160 (UMC::Phy::AnibRcvControl0)

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020160; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:0	RFU . Read-write. Reset: 00h.

UMC[00...11]CTLx00000804_address00020161 (UMC::Phy::AnibRcvControl1)

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020161; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:0	RFU . Read-write. Reset: 00h.

UMC[00...11]CTLx00000804_address00020162 (UMC::Phy::AnibRcvControl2)

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020162; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:0	RFU . Read-write. Reset: 00h.

UMC[00...11]CTLx00000804_address00020163 (UMC::Phy::AnibRcvControl3)

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020163; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:8	Reserved.
7:0	RFU . Read-write. Reset: 00h.

UMC[00...11]CTLx00000804_address00020168 (UMC::Phy::AnibRcvErr)

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020168; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	AnibRcvErr . Read-only, Volatile. Reserved for future use.

UMC[00...11]CTLx00000804_address00020169 (UMC::Phy::AcReceiveFilters)

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020169; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1:0	RFU. Read-write. Reset: 0h.

UMC[00...11]CTLx00000804_address0002016A (UMC::Phy::AnibRcvPtrInitVal)

Reset: 0000h.

AnibRcv FIFO pointer init value

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002016A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Reserved.

UMC[00...11]CTLx00000804_address0002019A (UMC::Phy::DfiDataEnLatency)

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002019A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2:0	RFU. Read-write. Reset: 0h.

UMC[00...11]CTLx00000804_address000201A0 (UMC::Phy::PpgcFltCfg0)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201A0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:5	Reserved.
4:0	RFU. Read-write. Reset: 00h.

UMC[00...11]CTLx00000804_address000201A1 (UMC::Phy::PpgcFltCfg1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201A1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:5	Reserved.
4:0	RFU. Read-write. Reset: 00h.

UMC[00...11]CTLx00000804_address000201A2 (UMC::Phy::PpgcFltCfg2)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201A2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	RFU. Read-write. Reset: 0h.

UMC[00...11]CTLx00000804_address000201C0 (UMC::Phy::AnibRcvPerfCtr0C1)

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201C0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr0C1.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201C1 (UMC::Phy::AnibRcvPerfCtr0C2)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201C1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr0C2.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201C2 (UMC::Phy::AnibRcvPerfCtr0C3)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201C2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr0C3.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201C3 (UMC::Phy::AnibRcvPerfCtr0C4)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201C3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr0C4.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201C4 (UMC::Phy::AnibRcvPerfCtr0CE)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201C4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr0CE.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201C5 (UMC::Phy::AnibRcvPerfCtrEn0)**

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:5 Reserved.

4:0 **AnibRcvPerfCtrEn0.** Read-write. Reset: 00h. Reserved for future use.

UMC[00...11]CTLx00000804_address000201C8 (UMC::Phy::AnibRcvPerfCtr1C1)

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201C8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr1C1**. Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201C9 (UMC::Phy::AnibRcvPerfCtr1C2)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201C9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr1C2**. Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201CA (UMC::Phy::AnibRcvPerfCtr1C3)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr1C3**. Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201CB (UMC::Phy::AnibRcvPerfCtr1C4)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201CB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr1C4**. Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201CC (UMC::Phy::AnibRcvPerfCtr1CE)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr1CE**. Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201CD (UMC::Phy::AnibRcvPerfCtrEn1)**

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201CD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:5 Reserved.

4:0 **AnibRcvPerfCtrEn1**. Read-write. Reset: 00h. Reserved for future use.

UMC[00...11]CTLx00000804_address000201D0 (UMC::Phy::AnibRcvPerfCtr2C1)

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201D0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr2C1.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201D1 (UMC::Phy::AnibRcvPerfCtr2C2)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201D1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr2C2.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201D2 (UMC::Phy::AnibRcvPerfCtr2C3)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201D2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr2C3.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201D3 (UMC::Phy::AnibRcvPerfCtr2C4)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201D3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr2C4.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201D4 (UMC::Phy::AnibRcvPerfCtr2CE)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201D4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr2CE.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201D5 (UMC::Phy::AnibRcvPerfCtrEn2)**

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201D5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:5 Reserved.

4:0 **AnibRcvPerfCtrEn2.** Read-write. Reset: 00h. Reserved for future use.

UMC[00...11]CTLx00000804_address000201D8 (UMC::Phy::AnibRcvPerfCtr3C1)

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201D8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr3C1.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201D9 (UMC::Phy::AnibRcvPerfCtr3C2)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201D9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr3C2.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201DA (UMC::Phy::AnibRcvPerfCtr3C3)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201DA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr3C3.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201DB (UMC::Phy::AnibRcvPerfCtr3C4)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201DB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr3C4.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201DC (UMC::Phy::AnibRcvPerfCtr3CE)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201DC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **AnibRcvPerfCtr3CE.** Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000201DD (UMC::Phy::AnibRcvPerfCtrEn3)**

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201DD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:5 Reserved.

4:0 **AnibRcvPerfCtrEn3.** Read-write. Reset: 00h. Reserved for future use.

UMC[00...11]CTLx00000804_address000201FE (UMC::Phy::AlertPerfCtr)

Read-only, Volatile.

ALERT_N Error Performance Counter

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201FE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	AlertPerfCtr. Read-only, Volatile. Description: Counts the number of assertions of ALERT_N. Enabled by setting csrAlertPerfEn. Disabled/cleared by clearing csrAlertPerfEn.

UMC[00...11]CTLx00000804_address000201FF (UMC::Phy::AlertPerfCtrEn)

Read-write. Reset: 0000h.

ALERT_N Error Performance Counter Enable

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000201FF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	AlertPerfCtrEn. Read-write. Reset: 0. Setting the bit enables the csrAlertPerfCtr. Clearing the bit clears and disables the counter.

UMC[00...11]CTLx00000804_address00020200 (UMC::Phy::MapCAA0toDfi)

Read-write. Reset: 0000h.

Maps PHY CAA lane 0 from dfi0_address of the index of the register contents

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020200; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA0toDfi. Read-write. Reset: 0h. Description: These CSRs map a dfi0_address to CAA 0, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtoDfi must have a unique value within the set. If PHY CAA0 is mapped from dfi0_address[0], then Register MapCAA0toDfi should be 0.

UMC[00...11]CTLx00000804_address00020201 (UMC::Phy::MapCAA1toDfi)

Read-write. Reset: 0001h.

Maps PHY CAA lane 1 from dfi0_address of the index of the register contents

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020201; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA1toDfi. Read-write. Reset: 1h. Description: These CSRs map a dfi0_address to CAA 1, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtoDfi must have a unique value within the set. If PHY CAA1 is mapped from dfi0_address[1], then Register MapCAA1toDfi should be 1.

UMC[00...11]CTLx00000804_address00020202 (UMC::Phy::MapCAA2toDfi)

Read-write. Reset: 0002h.

Maps PHY CAA lane 2 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020202; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA2toDfi. Read-write. Reset: 2h. Description: These CSRs map a dfi0_address to CAA 2, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtoDfi must have a unique value within the set. If PHY CAA2 is mapped from dfi0_address[2], then Register MapCAA2toDfi should be 2.

UMC[00...11]CTLx00000804_address00020203 (UMC::Phy::MapCAA3toDfi)

Read-write. Reset: 0003h.

Maps PHY CAA lane 3 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020203; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA3toDfi. Read-write. Reset: 3h. Description: These CSRs map a dfi0_address to CAA 3, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtoDfi must have a unique value within the set. If PHY CAA3 is mapped from dfi0_address[3], then Register MapCAA3toDfi should be 3.

UMC[00...11]CTLx00000804_address00020204 (UMC::Phy::MapCAA4toDfi)

Read-write. Reset: 0004h.

Maps PHY CAA lane 4 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020204; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA4toDfi. Read-write. Reset: 4h. Description: These CSRs map a dfi0_address to CAA 4, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtoDfi must have a unique value within the set. If PHY CAA4 is mapped from dfi0_address[4], then Register MapCAA4toDfi should be 4.

UMC[00...11]CTLx00000804_address00020205 (UMC::Phy::MapCAA5toDfi)

Read-write. Reset: 0005h.

Maps PHY CAA lane 5 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020205; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA5toDfi. Read-write. Reset: 5h. Description: These CSRs map a dfi0_address to CAA 5, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtoDfi must have a unique value within the set. If PHY CAA5 is mapped from dfi0_address[5], then Register MapCAA5toDfi should be 5.

UMC[00...11]CTLx00000804_address00020206 (UMC::Phy::MapCAA6toDfi)

Read-write. Reset: 0006h.

Maps PHY CAA lane 6 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020206; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA6toDfi. Read-write. Reset: 6h. Description: These CSRs map a dfi0_address to CAA 6, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtoDfi must have a unique value within the set. If PHY CAA6 is mapped from dfi0_address[6], then Register MapCAA6toDfi should be 6.

UMC[00...11]CTLx00000804_address00020207 (UMC::Phy::MapCAA7toDfi)

Read-write. Reset: 0007h.

Maps PHY CAA lane 7 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020207; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA7toDfi. Read-write. Reset: 7h. Description: These CSRs map a dfi0_address to CAA 7, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtoDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAA7 is mapped from dfi0_address[0], then Register MapCAA7toDfi should be 7. Otherwise, If PHY CAA7 is mapped from dfi0_address[7], then Register MapCAA7toDfi should be 7.

UMC[00...11]CTLx00000804_address00020208 (UMC::Phy::MapCAA8toDfi)

Read-write. Reset: 0008h.

Maps PHY CAA lane 8 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020208; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA8toDfi. Read-write. Reset: 8h. Description: These CSRs map a dfi0_address to CAA 8, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtoDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAA8 is mapped from dfi0_address[1], then Register MapCAA8toDfi should be 8. Otherwise, If PHY CAA8 is mapped from dfi0_address[8], then Register MapCAA8toDfi should be 8.

UMC[00...11]CTLx00000804_address00020209 (UMC::Phy::MapCAA9toDfi)

Read-write. Reset: 0009h.

Maps PHY CAA lane 9 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020209; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA9toDfi. Read-write. Reset: 9h. Description: These CSRs map a dfi0_address to CAA 9, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtoDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAA9 is mapped from dfi0_address[2], then Register MapCAA9toDfi should be 9. Otherwise, If PHY CAA9 is mapped from dfi0_address[9], then Register MapCAA9toDfi should be 9.

UMC[00...11]CTLx00000804_address0002020A (UMC::Phy::MapCAA10toDfi)

Read-write. Reset: 000Ah.

Maps PHY CAA lane 10 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002020A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA10toDfi. Read-write. Reset: Ah. Description: These CSRs map a dfi0_address to CAA 10, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtoDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAA10 is mapped from dfi0_address[3], then Register MapCAA10toDfi should be 10. Otherwise, If PHY CAA10 is mapped from dfi0_address[10], then Register MapCAA10toDfi should be 10.

UMC[00...11]CTLx00000804_address0002020B (UMC::Phy::MapCAA11toDfi)

Read-write. Reset: 000Bh.

Maps PHY CAA lane 11 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002020B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA11toDfi. Read-write. Reset: Bh. Description: These CSRs map a dfi0_address to CAA 11, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtoDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAA11 is mapped from dfi0_address[4], then Register MapCAA11toDfi should be 11. Otherwise, If PHY CAA11 is mapped from dfi0_address[11], then Register MapCAA11toDfi should be 11.

UMC[00...11]CTLx00000804_address0002020C (UMC::Phy::MapCAA12toDfi)

Read-write. Reset: 000Ch.

Maps PHY CAA lane 12 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002020C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA12toDfi. Read-write. Reset: Ch. Description: These CSRs map a dfi0_address to CAA 12, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAA12 is mapped from dfi0_address[5], then Register MapCAA12toDfi should be 12. Otherwise, If PHY CAA12 is mapped from dfi0_address[12], then Register MapCAA12toDfi should be 12.

UMC[00...11]CTLx00000804_address0002020D (UMC::Phy::MapCAA13toDfi)

Read-write. Reset: 000Dh.

Maps PHY CAA lane 13 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002020D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAA13toDfi. Read-write. Reset: Dh. Description: These CSRs map a dfi0_address to CAA 13, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCAAtDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAA13 is mapped from dfi0_address[6], then Register MapCAA13toDfi should be 13. Otherwise, If PHY CAA13 is mapped from dfi0_address[13], then Register MapCAA13toDfi should be 13.

UMC[00...11]CTLx00000804_address00020210 (UMC::Phy::MapCAB0toDfi)

Read-write. Reset: 0000h.

Maps PHY CAB lane 0 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020210; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB0toDfi. Read-write. Reset: 0h. Description: These CSRs map a dfi1_address to CAB 0, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. If PHY CAB0 is mapped from dfi1_address[0], then Register MapCAB0toDfi should be 0.

UMC[00...11]CTLx00000804_address00020211 (UMC::Phy::MapCAB1toDfi)

Read-write. Reset: 0001h.

Maps PHY CAB lane 1 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020211; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB1toDfi. Read-write. Reset: 1h. Description: These CSRs map a dfi1_address to CAB 1, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. If PHY CAB1 is mapped from dfi1_address[1], then Register MapCAB1toDfi should be 1.

UMC[00...11]CTLx00000804_address00020212 (UMC::Phy::MapCAB2toDfi)

Read-write. Reset: 0002h.

Maps PHY CAB lane 2 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020212; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB2toDfi. Read-write. Reset: 2h. Description: These CSRs map a dfi1_address to CAB 2, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. If PHY CAB2 is mapped from dfi1_address[2], then Register MapCAB2toDfi should be 2.

UMC[00...11]CTLx00000804_address00020213 (UMC::Phy::MapCAB3toDfi)

Read-write. Reset: 0003h.

Maps PHY CAB lane 3 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020213; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB3toDfi. Read-write. Reset: 3h. Description: These CSRs map a dfi1_address to CAB 3, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. If PHY CAB3 is mapped from dfi1_address[3], then Register MapCAB3toDfi should be 3.

UMC[00...11]CTLx00000804_address00020214 (UMC::Phy::MapCAB4toDfi)

Read-write. Reset: 0004h.

Maps PHY CAB lane 4 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020214; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB4toDfi. Read-write. Reset: 4h. Description: These CSRs map a dfi1_address to CAB 4, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. If PHY CAB4 is mapped from dfi1_address[4], then Register MapCAB4toDfi should be 4.

UMC[00...11]CTLx00000804_address00020215 (UMC::Phy::MapCAB5toDfi)

Read-write. Reset: 0005h.

Maps PHY CAB lane 5 from dfi0_address of the index of the register contents

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020215; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB5toDfi. Read-write. Reset: 5h. Description: These CSRs map a dfi1_address to CAB 5, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. If PHY CAB5 is mapped from dfi1_address[5], then Register MapCAB5toDfi should be 5.

UMC[00...11]CTLx00000804_address00020216 (UMC::Phy::MapCAB6toDfi)

Read-write. Reset: 0006h.

Maps PHY CAB lane 6 from dfi0_address of the index of the register contents

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020216; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB6toDfi. Read-write. Reset: 6h. Description: These CSRs map a dfi1_address to CAB 6, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. If PHY CAB6 is mapped from dfi1_address[6], then Register MapCAB6toDfi should be 6.

UMC[00...11]CTLx00000804_address00020217 (UMC::Phy::MapCAB7toDfi)

Read-write. Reset: 0007h.

Maps PHY CAB lane 7 from dfi0_address of the index of the register contents

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020217; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB7toDfi. Read-write. Reset: 7h. Description: These CSRs map a dfi1_address to CAB 7, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAB7 is mapped from dfi1_address[0], then Register MapCAB7toDfi should be 7. Otherwise, If PHY CAB7 is mapped from dfi1_address[7], then Register MapCAB7toDfi should be 7.

UMC[00...11]CTLx00000804_address00020218 (UMC::Phy::MapCAB8toDfi)

Read-write. Reset: 0008h.

Maps PHY CAB lane 8 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020218; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB8toDfi. Read-write. Reset: 8h. Description: These CSRs map a dfi1_address to CAB 8, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAB8 is mapped from dfi1_address[1], then Register MapCAB8toDfi should be 8. Otherwise, If PHY CAB8 is mapped from dfi1_address[8], then Register MapCAB8toDfi should be 8.

UMC[00...11]CTLx00000804_address00020219 (UMC::Phy::MapCAB9toDfi)

Read-write. Reset: 0009h.

Maps PHY CAB lane 9 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020219; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB9toDfi. Read-write. Reset: 9h. Description: These CSRs map a dfi1_address to CAB 9, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAB9 is mapped from dfi1_address[2], then Register MapCAB9toDfi should be 9. Otherwise, If PHY CAB9 is mapped from dfi1_address[9], then Register MapCAB9toDfi should be 9.

UMC[00...11]CTLx00000804_address0002021A (UMC::Phy::MapCAB10toDfi)

Read-write. Reset: 000Ah.

Maps PHY CAB lane 10 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002021A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB10toDfi. Read-write. Reset: Ah. Description: These CSRs map a dfi1_address to CAB 10, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAB10 is mapped from dfi1_address[3], then Register MapCAB10toDfi should be 10. Otherwise, If PHY CAB10 is mapped from dfi1_address[10], then Register MapCAB10toDfi should be 10.

UMC[00...11]CTLx00000804_address0002021B (UMC::Phy::MapCAB11toDfi)

Read-write. Reset: 000Bh.

Maps PHY CAB lane 11 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002021B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB11toDfi. Read-write. Reset: Bh. Description: These CSRs map a dfi1_address to CAB 11, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAB11 is mapped from dfi1_address[4], then Register MapCAB11toDfi should be 11. Otherwise, If PHY CAB11 is mapped from dfi1_address[11], then Register MapCAB11toDfi should be 11.

UMC[00...11]CTLx00000804_address0002021C (UMC::Phy::MapCAB12toDfi)

Read-write. Reset: 000Ch.

Maps PHY CAB lane 12 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002021C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB12toDfi. Read-write. Reset: Ch. Description: These CSRs map a dfi1_address to CAB 12, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAB12 is mapped from dfi1_address[5], then Register MapCAB12toDfi should be 12. Otherwise, If PHY CAB12 is mapped from dfi1_address[12], then Register MapCAB12toDfi should be 12.

UMC[00...11]CTLx00000804_address0002021D (UMC::Phy::MapCAB13toDfi)

Read-write. Reset: 000Dh.

Maps PHY CAB lane 13 from dfi0_address of the index of the register contents

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002021D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	MapCAB13toDfi. Read-write. Reset: Dh. Description: These CSRs map a dfi1_address to CAB 13, used only for DDR5, and enabled only when the global define DWC_DDRPHY_CAXBAR is enabled. Each register in the set of MapCABtoDfi must have a unique value within the set. For RDIMM, bits[6:0] are for the rising edge of MEMCLK, and bits [13:7] are for the falling edge of MEMCLK. So mapping should be programmed within each group of 7 bits and if PHY CAB13 is mapped from dfi1_address[6], then Register MapCAB13toDfi should be 13. Otherwise, If PHY CAB13 is mapped from dfi1_address[13], then Register MapCAB13toDfi should be 13.

UMC[00...11]CTLx00000804_address00020240 (UMC::Phy::D5ACSMPtrXlat0)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020240; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020241 (UMC::Phy::D5ACSMPtrXlat1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020241; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020242 (UMC::Phy::D5ACSMPtrXlat2)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020242; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020243 (UMC::Phy::D5ACSMPtrXlat3)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020243; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020244 (UMC::Phy::D5ACSMPtrXlat4)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020244; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020245 (UMC::Phy::D5ACSMPtrXlat5)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020245; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020246 (UMC::Phy::D5ACSMPtrXlat6)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020246; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020247 (UMC::Phy::D5ACSMPtrXlat7)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020247; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020248 (UMC::Phy::D5ACSMPtrXlat8)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020248; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020249 (UMC::Phy::D5ACSMPtrXlat9)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020249; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002024A (UMC::Phy::D5ACSMPtrXlat10)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002024A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002024B (UMC::Phy::D5ACSMPtrXlat11)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002024B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002024C (UMC::Phy::D5ACSMPtrXlat12)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002024C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002024D (UMC::Phy::D5ACSMPtrXlat13)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002024D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002024E (UMC::Phy::D5ACSMPtrXlat14)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002024E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002024F (UMC::Phy::D5ACSMPtrXlat15)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002024F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020250 (UMC::Phy::D5ACSMPtrXlat16)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020250; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020251 (UMC::Phy::D5ACSMPtrXlat17)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020251; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020252 (UMC::Phy::D5ACSMPtrXlat18)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020252; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020253 (UMC::Phy::D5ACSMPtrXlat19)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020253; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020254 (UMC::Phy::D5ACSMPtrXlat20)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020254; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020255 (UMC::Phy::D5ACSMPtrXlat21)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020255; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020256 (UMC::Phy::D5ACSMPtrXlat22)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020256; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020257 (UMC::Phy::D5ACSMPtrXlat23)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020257; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020258 (UMC::Phy::D5ACSMPtrXlat24)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020258; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020259 (UMC::Phy::D5ACSMPtrXlat25)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020259; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002025A (UMC::Phy::D5ACSMPtrXlat26)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002025A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002025B (UMC::Phy::D5ACSMPtrXlat27)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002025B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002025C (UMC::Phy::D5ACSMPtrXlat28)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002025C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002025D (UMC::Phy::D5ACSMPtrXlat29)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002025D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002025E (UMC::Phy::D5ACSMPtrXlat30)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002025E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002025F (UMC::Phy::D5ACSMPtrXlat31)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002025F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020260 (UMC::Phy::D5ACSMPtrXlat32)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020260; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020261 (UMC::Phy::D5ACSMPtrXlat33)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020261; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020262 (UMC::Phy::D5ACSMPtrXlat34)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020262; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020263 (UMC::Phy::D5ACSMPtrXlat35)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020263; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020264 (UMC::Phy::D5ACSMPtrXlat36)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020264; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020265 (UMC::Phy::D5ACSMPtrXlat37)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020265; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020266 (UMC::Phy::D5ACSMPtrXlat38)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020266; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020267 (UMC::Phy::D5ACSMPtrXlat39)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020267; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020268 (UMC::Phy::D5ACSMPtrXlat40)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020268; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020269 (UMC::Phy::D5ACSMPtrXlat41)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020269; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002026A (UMC::Phy::D5ACSMPtrXlat42)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002026A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002026B (UMC::Phy::D5ACSMPtrXlat43)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002026B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002026C (UMC::Phy::D5ACSMPtrXlat44)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002026C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002026D (UMC::Phy::D5ACSMPtrXlat45)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002026D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002026E (UMC::Phy::D5ACSMPtrXlat46)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002026E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002026F (UMC::Phy::D5ACSMPtrXlat47)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002026F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020270 (UMC::Phy::D5ACSMPtrXlat48)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020270; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020271 (UMC::Phy::D5ACSMPtrXlat49)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020271; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020272 (UMC::Phy::D5ACSMPtrXlat50)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020272; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020273 (UMC::Phy::D5ACSMPtrXlat51)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020273; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020274 (UMC::Phy::D5ACSMPtrXlat52)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020274; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020275 (UMC::Phy::D5ACSMPtrXlat53)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020275; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020276 (UMC::Phy::D5ACSMPtrXlat54)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020276; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020277 (UMC::Phy::D5ACSMPtrXlat55)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020277; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020278 (UMC::Phy::D5ACSMPtrXlat56)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020278; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address00020279 (UMC::Phy::D5ACSMPtrXlat57)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020279; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002027A (UMC::Phy::D5ACSMPtrXlat58)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002027A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002027B (UMC::Phy::D5ACSMPtrXlat59)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002027B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:0 RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address0002027C (UMC::Phy::D5ACSMPtrXlat60)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002027C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **RFU.** Read-write. Reset: 0000h.**UMC[00...11]CTLx00000804_address0002027D (UMC::Phy::D5ACSMPtrXlat61)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002027D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **RFU.** Read-write. Reset: 0000h.**UMC[00...11]CTLx00000804_address0002027E (UMC::Phy::D5ACSMPtrXlat62)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002027E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **RFU.** Read-write. Reset: 0000h.**UMC[00...11]CTLx00000804_address0002027F (UMC::Phy::D5ACSMPtrXlat63)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address0002027F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **RFU.** Read-write. Reset: 0000h.**UMC[00...11]CTLx00000804_address00020280 (UMC::Phy::D5ACSMProgPtr)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020280; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:4 Reserved.

3:0 **D5ACSMProgPtr.** Read-write. Reset: 0h. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address00020281 (UMC::Phy::D5ACSMXlatSelect)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00020281; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:1 Reserved.

0 **D5ACSMXlatSelect.** Read-write. Reset: 0. Reserved for PHY training firmware use.

UMC00CTLx00000804_address00040080...UMC11CTLx00000804_address00340080 (UMC::Phy::AcsmpPlayback0x0)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00040080; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00140080; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00240080; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00340080; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AcsmpPlayback0x0. Read-write. Reset: 000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address00040081...UMC11CTLx00000804_address00340081 (UMC::Phy::AcsmpPlayback1x0)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00040081; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00140081; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00240081; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00340081; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AcsmpPlayback1x0. Read-write. Reset: 000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address00040082...UMC11CTLx00000804_address00340082 (UMC::Phy::AcsmpPlayback0x1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00040082; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00140082; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00240082; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00340082; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AcsmpPlayback0x1. Read-write. Reset: 000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address00040083...UMC11CTLx00000804_address00340083 (UMC::Phy::AcsmpPlayback1x1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00040083; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00140083; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00240083; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00340083; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AcsmpPlayback1x1. Read-write. Reset: 000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address00040084...UMC11CTLx00000804_address00340084 (UMC::Phy::AcsmpPlayback0x2)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00040084; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00140084; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00240084; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00340084; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AcsmpPlayback0x2. Read-write. Reset: 000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address00040085...UMC11CTLx00000804_address00340085 (UMC::Phy::AcsmpPlayback1x2)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00040085; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00140085; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00240085; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00340085; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AcsmpPlayback1x2. Read-write. Reset: 000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address00040086...UMC11CTLx00000804_address00340086 (UMC::Phy::AcsmPlayback0x3)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00040086; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00140086; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00240086; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00340086; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	Reserved.
11:0	AcsmPlayback0x3. Read-write. Reset: 000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address00040087...UMC11CTLx00000804_address00340087 (UMC::Phy::AcsmPlayback1x3)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00040087; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00140087; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00240087; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00340087; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	Reserved.
11:0	AcsmPlayback1x3. Read-write. Reset: 000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address00040088...UMC11CTLx00000804_address00340088 (UMC::Phy::AcsmPlayback0x4)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00040088; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00140088; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00240088; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00340088; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	Reserved.
11:0	AcsmPlayback0x4. Read-write. Reset: 000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address00040089...UMC11CTLx00000804_address00340089 (UMC::Phy::AcsmPlayback1x4)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00040089; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00140089; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00240089; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00340089; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AcsmPlayback1x4. Read-write. Reset: 000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address0004008A...UMC11CTLx00000804_address0034008A (UMC::Phy::AcsmPlayback0x5)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0004008A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0014008A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0024008A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0034008A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AcsmPlayback0x5. Read-write. Reset: 000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address0004008B...UMC11CTLx00000804_address0034008B (UMC::Phy::AcsmPlayback1x5)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0004008B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0014008B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0024008B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0034008B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AcsmPlayback1x5. Read-write. Reset: 000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address0004008C...UMC11CTLx00000804_address0034008C (UMC::Phy::AcsmpPlayback0x6)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0004008C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0014008C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0024008C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0034008C; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AcsmpPlayback0x6. Read-write. Reset: 000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address0004008D...UMC11CTLx00000804_address0034008D (UMC::Phy::AcsmpPlayback1x6)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0004008D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0014008D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0024008D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0034008D; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AcsmpPlayback1x6. Read-write. Reset: 000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address0004008E...UMC11CTLx00000804_address0034008E (UMC::Phy::AcsmpPlayback0x7)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0004008E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0014008E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0024008E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0034008E; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:12	Reserved.
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11:0	AcsmpPlayback0x7. Read-write. Reset: 000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address0004008F...UMC11CTLx00000804_address0034008F (UMC::Phy::AcsmpPlayback1x7)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address0004008F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address0014008F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address0024008F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype4_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address0034008F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	Reserved.
11:0	AcsmpPlayback1x7 . Read-write. Reset: 000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address00040090 (UMC::Phy::AcsmpPStateOvrEn)

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() for Controlling Playback Engine

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00040090; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	AcsmpPStateOvrEn . Read-write. Reset: 0. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() for Controlling Playback Engine

UMC[00...11]CTLx00000804_address00040091 (UMC::Phy::AcsmpPStateOvrVal)

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() for Controlling Playback Engine

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00040091; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	AcsmpPStateOvrVal . Read-write. Reset: 0h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() for Controlling Playback Engine

UMC[00...11]CTLx00000804_address000400C2 (UMC::Phy::AcsmpCkeVal)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000400C2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	AcsmpCkeVal . Read-write. Reset: 0h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000400C8 (UMC::Phy::LowSpeedClockDivider)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000400C8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:6	Reserved.
5:0	LowSpeedClockDivider . Read-write. Reset: 00h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000400EA (UMC::Phy::LowSpeedClockStopVal)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000400EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:1 Reserved.

0 **LowSpeedClockStopVal**. Read-write. Reset: 0. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address00070091 (UMC::Phy::PPGC2Ctrl1)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00070091; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:5 Reserved.

4:1 **RFU**. Read-write. Reset: 0h.

0 Reserved.

UMC[00...11]CTLx00000804_address00070095 (UMC::Phy::Ppgc2Lane2CrcInMap0)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00070095; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:12 Reserved.

11:0 **RFU**. Read-write. Reset: 000h.**UMC[00...11]CTLx00000804_address00070096 (UMC::Phy::Ppgc2Lane2CrcInMap1)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00070096; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:12 Reserved.

11:0 **RFU**. Read-write. Reset: 000h.

UMC00CTLx00000804_address000700A4...UMC11CTLx00000804_address000708A4 (UMC::Phy::Ppgc2PrbsTapDly0)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708A4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2PrbsTapDly0. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700A5...UMC11CTLx00000804_address000708A5 (UMC::Phy::Ppgc2PrbsTapDly1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708A5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2PrbsTapDly1. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700A6...UMC11CTLx00000804_address000708A6 (UMC::Phy::Ppgc2PrbsTapDly2)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708A6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2PrbsTapDly2. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700A7...UMC11CTLx00000804_address000708A7 (UMC::Phy::Ppgc2PrbsTapDly3)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708A7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2PrbsTapDly3. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700B0 (UMC::Phy::Ppgc2GenPrbsByte0)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700B0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2GenPrbsByte0. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700B1 (UMC::Phy::Ppgc2GenPrbsByte1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700B1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2GenPrbsByte1. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700B2 (UMC::Phy::Ppgc2GenPrbsByte2)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700B2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2GenPrbsByte2. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700B3 (UMC::Phy::Ppgc2GenPrbsByte3)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700B3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2GenPrbsByte3. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700B4 (UMC::Phy::Ppgc2GenPrbsByte4)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700B4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2GenPrbsByte4. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700B5 (UMC::Phy::Ppgc2GenPrbsByte5)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700B5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2GenPrbsByte5. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700B6 (UMC::Phy::Ppgc2GenPrbsByte6)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700B6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2GenPrbsByte6. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700B7 (UMC::Phy::Ppgc2GenPrbsByte7)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700B7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **Ppgc2GenPrbsByte7**. Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address000700B8 (UMC::Phy::Ppgc2GenPrbsByte8)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700B8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **Ppgc2GenPrbsByte8**. Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address000700B9 (UMC::Phy::Ppgc2GenPrbsByte9)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700B9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **Ppgc2GenPrbsByte9**. Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address000700BA (UMC::Phy::Ppgc2GenPrbsByte10)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700BA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **Ppgc2GenPrbsByte10**. Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address000700BB (UMC::Phy::Ppgc2GenPrbsByte11)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700BB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **Ppgc2GenPrbsByte11**. Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address000700BC (UMC::Phy::Ppgc2GenPrbsByte12)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700BC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **Ppgc2GenPrbsByte12**. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700BD (UMC::Phy::Ppgc2GenPrbsByte13)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700BD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2GenPrbsByte13 . Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700BE (UMC::Phy::Ppgc2GenPrbsByte14)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700BE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2GenPrbsByte14 . Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700BF (UMC::Phy::Ppgc2GenPrbsByte15)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700BF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2GenPrbsByte15 . Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700C0...UMC11CTLx00000804_address000708C0 (UMC::Phy::GenData0)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700C0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701C0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702C0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703C0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704C0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705C0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706C0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707C0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708C0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	GenData0 . Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700C1...UMC11CTLx00000804_address000708C1 (UMC::Phy::GenData1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700C1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701C1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702C1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703C1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704C1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705C1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706C1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707C1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708C1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **GenData1.** Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC00CTLx00000804_address000700C2...UMC11CTLx00000804_address000708C2 (UMC::Phy::GenData2)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700C2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701C2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702C2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703C2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704C2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705C2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706C2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707C2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708C2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **GenData2.** Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700C3...UMC11CTLx00000804_address000708C3 (UMC::Phy::GenData3)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700C3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701C3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702C3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703C3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704C3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705C3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706C3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707C3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708C3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **GenData3.** Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC00CTLx00000804_address000700C4...UMC11CTLx00000804_address000708C4 (UMC::Phy::ChkData0)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700C4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701C4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702C4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703C4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704C4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705C4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706C4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707C4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708C4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **ChkData0.** Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700C5...UMC11CTLx00000804_address000708C5 (UMC::Phy::ChkData1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708C5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **ChkData1.** Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC00CTLx00000804_address000700C6...UMC11CTLx00000804_address000708C6 (UMC::Phy::ChkData2)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700C6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701C6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702C6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703C6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704C6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705C6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706C6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707C6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708C6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **ChkData2.** Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700C7...UMC11CTLx00000804_address000708C7 (UMC::Phy::ChkData3)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700C7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701C7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702C7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703C7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704C7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705C7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706C7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707C7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708C7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **ChkData3.** Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC00CTLx00000804_address000700C8...UMC11CTLx00000804_address000708C8 (UMC::Phy::Ppgc2GenData0)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700C8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701C8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702C8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703C8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704C8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705C8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706C8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707C8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708C8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **Ppgc2GenData0.** Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700C9...UMC11CTLx00000804_address000708C9 (UMC::Phy::Ppgc2GenData1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700C9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701C9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702C9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703C9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704C9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705C9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706C9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707C9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708C9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	Ppgc2GenData1. Read-write. Reset: 0000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000700CA...UMC11CTLx00000804_address000708CA (UMC::Phy::Ppgc2GenData2)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708CA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	Ppgc2GenData2. Read-write. Reset: 0000h. Reserved for PHY training firmware use.
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UMC00CTLx00000804_address000700CB...UMC11CTLx00000804_address000708CB (UMC::Phy::Ppgc2GenData3)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700CB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701CB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702CB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703CB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704CB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705CB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706CB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707CB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708CB; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2GenData3. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700CC...UMC11CTLx00000804_address000708CC (UMC::Phy::Ppgc2ChkData0)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708CC; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2ChkData0. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700CD...UMC11CTLx00000804_address000708CD (UMC::Phy::Ppgc2ChkData1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700CD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701CD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702CD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703CD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704CD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705CD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706CD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707CD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708CD; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2ChkData1. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700CE...UMC11CTLx00000804_address000708CE (UMC::Phy::Ppgc2ChkData2)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700CE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701CE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702CE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703CE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704CE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705CE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706CE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707CE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708CE; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2ChkData2. Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC00CTLx00000804_address000700CF...UMC11CTLx00000804_address000708CF (UMC::Phy::Ppgc2ChkData3)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_i0_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700CF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i1_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000701CF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i2_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000702CF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i3_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000703CF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i4_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000704CF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i5_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000705CF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i6_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000706CF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000707CF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype7_i8_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000708CF; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2ChkData3 . Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700D0 (UMC::Phy::PattModeLocCfg)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700D0; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	RFU . Read-write. Reset: 0.

UMC[00...11]CTLx00000804_address000700D1 (UMC::Phy::Ppgc2PattModeLocCfg)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700D1; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	RFU . Read-write. Reset: 0.

UMC[00...11]CTLx00000804_address000700D2 (UMC::Phy::PpgcMskPat0)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700D2; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	PpgcMskPat0 . Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700D3 (UMC::Phy::PpgcMskPat1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700D3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **PpgcMskPat1**. Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address000700D4 (UMC::Phy::Ppgc2MskPat0)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700D4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **Ppgc2MskPat0**. Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address000700D5 (UMC::Phy::Ppgc2MskPat1)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700D5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **Ppgc2MskPat1**. Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address000700E0 (UMC::Phy::Ppgc2PrbsGenCtl)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700E0; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:7 Reserved.

6:0 **RFU**. Read-write. Reset: 00h.**UMC[00...11]CTLx00000804_address000700E1 (UMC::Phy::Ppgc2PrbsGenStateLo)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700E1; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **Ppgc2PrbsGenStateLo**. Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address000700E2 (UMC::Phy::Ppgc2PrbsGenStateHi)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiptype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700E2; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:7 Reserved.

6:0 **Ppgc2PrbsGenStateHi**. Read-write. Reset: 00h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700E3 (UMC::Phy::Ppgc2PrbsChkStateLo)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700E3; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Ppgc2PrbsChkStateLo . Read-write. Reset: 0000h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700E4 (UMC::Phy::Ppgc2PrbsChkStateHi)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700E4; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:7	Reserved.
6:0	Ppgc2PrbsChkStateHi . Read-write. Reset: 00h. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000700E5 (UMC::Phy::Ppgc2PrbsGenCtl1)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700E5; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:9	Reserved.
8:0	RFU . Read-write. Reset: 000h.

UMC[00...11]CTLx00000804_address000700E6 (UMC::Phy::Ppgc2PrbsGenCtl2)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700E6; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	RFU . Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address000700E7 (UMC::Phy::PpgcGenOutputInvertMode)

Read-write. Reset: 0000h.

PPGC mode control.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:9	Reserved.
8:0	PpgcGenOutputInvertMode . Read-write. Reset: 000h. Description: General PRBS Control0 (per-DBYTE module) Inverts the output of the Gen PPGC when set to 1.

UMC[00...11]CTLx00000804_address000700E8 (UMC::Phy::PpgcChkOutputInvertMode)

Read-write. Reset: 0000h.

PPGC mode control.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700E8; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:9	Reserved.
8:0	PpgcChkOutputInvertMode. Read-write. Reset: 000h. Description: General PRBS Control0 (per-DBYTE module) Inverts the output of the Check PPGC when set to 1.

UMC[00...11]CTLx00000804_address000700E9 (UMC::Phy::Ppgc2GenOutputInvertMode)

Read-write. Reset: 0000h.

PPGC2 mode control.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700E9; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:9	Reserved.
8:0	Ppgc2GenOutputInvertMode. Read-write. Reset: 000h. Description: General PRBS Control0 (per-DBYTE module) Inverts the output of the Gen PPGC2 when set to 1.

UMC[00...11]CTLx00000804_address000700EA (UMC::Phy::Ppgc2ChkOutputInvertMode)

Read-write. Reset: 0000h.

PPGC2 mode control.

_umc0_chiplettype7_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000700EA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:9	Reserved.
8:0	Ppgc2ChkOutputInvertMode. Read-write. Reset: 000h. Description: General PRBS Control0 (per-DBYTE module) Inverts the output of the Check PPGC2 when set to 1.

UMC00CTLx00000804_address00090000...UMC11CTLx00000804_address00090003 (UMC::Phy::PreSequenceReg0bs0)

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CTLx00000804_address00090000; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CTLx00000804_address00090003; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	PreSequenceReg0bs0. Read-write. Reset: 0000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

UMC00CTLx00000804_address00090001...UMC11CTLx00000804_address00090004 (UMC::Phy::PreSequenceReg0bs1)

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CTLx00000804_address00090001; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CTLx00000804_address00090004; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	PreSequenceReg0bs1. Read-write. Reset: 0000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.
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UMC00CTLx00000804_address00090002...UMC11CTLx00000804_address00090005 (UMC::Phy::PreSequenceReg0bs2)

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CTLx00000804_address00090002; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CTLx00000804_address00090005; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:9	Reserved.
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8:0	PreSequenceReg0bs2. Read-write. Reset: 000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.
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UMC00CTLx00000804_address00090006...UMC11CTLx00000804_address00090009 (UMC::Phy::PostSequenceReg0bs0)

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CTLx00000804_address00090006; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CTLx00000804_address00090009; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	PostSequenceReg0bs0. Read-write. Reset: 0000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.
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UMC00CTLx00000804_address00090007...UMC11CTLx00000804_address0009000A (UMC::Phy::PostSequenceReg0bs1)

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CTLx00000804_address00090007; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CTLx00000804_address0009000A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	PostSequenceReg0bs1. Read-write. Reset: 0000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.
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UMC00CTLx00000804_address00090008...UMC11CTLx00000804_address0009000B (UMC::Phy::PostSequenceReg0bs2)

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CTLx00000804_address00090008; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CTLx00000804_address0009000B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:9	Reserved.
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8:0	PostSequenceReg0bs2. Read-write. Reset: 000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.
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UMC00CTLx00000804_address0009000C...UMC11CTLx00000804_address00090013 (UMC::Phy::Seq0BDisableFlag)

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n0_aliasSMN; UMC[11:00]CTLx00000804_address0009000C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n1_aliasSMN; UMC[11:00]CTLx00000804_address0009000D; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n2_aliasSMN; UMC[11:00]CTLx00000804_address0009000E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n3_aliasSMN; UMC[11:00]CTLx00000804_address0009000F; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n4_aliasSMN; UMC[11:00]CTLx00000804_address00090010; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n5_aliasSMN; UMC[11:00]CTLx00000804_address00090011; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n6_aliasSMN; UMC[11:00]CTLx00000804_address00090012; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_n7_aliasSMN; UMC[11:00]CTLx00000804_address00090013; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
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15:0	Seq0BDisableFlag. Read-write. Reset: 0000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.
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UMC00CTLx00000804_address00090017...UMC11CTLx00000804_address00090026 (UMC::Phy::StartVector0b)

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

_umc0_chiplettype9_instUMCWPHY0_n[15:0]_aliasSMN; UMC00CTLx00000804_address0009_00[26:17]; UMC00CTL=0005_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_instUMCWPHY1_n[15:0]_aliasSMN; UMC01CTLx00000804_address0009_00[26:17]; UMC01CTL=0015_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_instUMCWPHY2_n[15:0]_aliasSMN; UMC02CTLx00000804_address0009_00[26:17]; UMC02CTL=0025_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_instUMCWPHY3_n[15:0]_aliasSMN; UMC03CTLx00000804_address0009_00[26:17]; UMC03CTL=0035_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_instUMCWPHY4_n[15:0]_aliasSMN; UMC04CTLx00000804_address0009_00[26:17]; UMC04CTL=0045_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_instUMCWPHY5_n[15:0]_aliasSMN; UMC05CTLx00000804_address0009_00[26:17]; UMC05CTL=0055_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_instUMCWPHY6_n[15:0]_aliasSMN; UMC06CTLx00000804_address0009_00[26:17]; UMC06CTL=0065_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_instUMCWPHY7_n[15:0]_aliasSMN; UMC07CTLx00000804_address0009_00[26:17]; UMC07CTL=0075_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_instUMCWPHY8_n[15:0]_aliasSMN; UMC08CTLx00000804_address0009_00[26:17]; UMC08CTL=0085_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_instUMCWPHY9_n[15:0]_aliasSMN; UMC09CTLx00000804_address0009_00[26:17]; UMC09CTL=0095_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_instUMCWPHY10_n[15:0]_aliasSMN; UMC10CTLx00000804_address0009_00[26:17]; UMC10CTL=00A5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_instUMCWPHY11_n[15:0]_aliasSMN; UMC11CTLx00000804_address0009_00[26:17]; UMC11CTL=00B5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:8 Reserved.

7:0 RFU. Read-write. Reset: 00h.

UMC[00...11]CTLx00000804_address00090028 (UMC::Phy::PhyInLP3)

Read-write. Reset: 0001h.

Indicator for PIE Lower Power 3 (LP3) Status

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address00090028; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:1 Reserved.

0 **PhyInLP3.** Read-write. Reset: 1.**Description:** Read Only. Set to 1 by the PIE once completed LP3 Entry sequence;
Cleared during LP3 Exit sequence. System software can
read this csr for the status of PIE engine.

Read-write. Reset: 0000h.

Programmed by dwc ddrphy phyinit I loadPIEImage() to support frequency changes.

```
_umc0_chiplettype9_instUMCWPHY0_n[143:0]_aliasSMN;  
UMC00C0TL000000804_address0009_0[D16,D3,D10,1CD,1CA,1C7,1C4,1C1,1BE,1BB,1B8,1B5,1B2,1AF,1AC,1A9,1A6,1A3,1A0,19D,19A,197,194,191,18E,18  
B,188,185,182,17F,17C,179,176,173,170,16D,16A,167,164,161,15E,15B,158,155,152,14F,14C,149,146,143,140,13D,13A,137,134,131,12E,12B,128,125,122,11F,11  
C,119,116,113,110,10D,10A,107,104,101,0FE,0FB,0F8,0F5,0F2,0EF,0EC,0E9,0E6,0E3,0E0,0DD,0DA,0D7,0D4,0D1,0CE,0CB,0C8,0C5,0C2,0BF,0BC,0B9,0B6,0B  
3,0B0,0AD,0AA,0A7,0A4,0A1,09E,09B,098,095,092,08F,08C,089,086,083,080,07D,07A,077,074,071,06E,06B,068,065,062,05F,05C,059,056,053,050,04D,04A,04  
7,004,041,03E,03B,038,035,032,02E,02C,029]; UMC00C0TL=0005 1000h: DataPortWrite=UMC0: AdbCamel Address1
```

```
_umc0_chiplettype9_instUMCWPHY1_n[143:0]_aliasSMN;  
UMC0C1CTLX00000804_addr0009_0[D16,D3,D10,1CD,1CA,1C7,1C4,1C1,1BE,1BB,1B8,1B5,1B2,1AF,1AC,1A9,1A6,1A3,1A0,19D,19A,197,194,191,18E,18  
B,188,185,182,17F,17C,179,176,173,170,16D,16A,167,164,161,15E,15B,158,155,152,14F,14C,149,146,143,140,13D,13A,137,134,131,12E,12B,128,125,122,11F,11  
C,119,116,113,110,10D,10A,107,104,101,0FE,0FB,0F8,0F5,0F2,0EF,0EC,0E9,0E6,0E3,0E0,0DD,0DA,0D7,0D4,0D1,0CE,0CB,0C8,0C5,0C2,0BF,0BC,0B9,0B6,0B  
3,0B0,0AD,0AA,0A7,0A4,0A1,09E,09B,098,095,092,08F,08C,089,086,083,080,07D,07A,077,074,071,06E,06B,068,065,062,05F,05C,059,056,053,050,04D,04A,04  
7,044,041,03E,03B,038,035,032,02F,02C,029]; UMC0C1CTLX0015_1000h; DataPortWrite=UMC0C1CTLX0015_1000h; AppCmId[Address]
```

```
_umc0_chiplettype9_instUMCWPHY2_n[143:0]_aliasSMN;  
UMC02CTLx00000804_address009_0[D16,D3,D10,1CD,1CA,1C7,1C4,1C1,1BE,1BB,1B8,1B5,1B2,1AF,1AC,1A9,1A6,1A3,1A0,19D,19A,197,194,191,18E,18  
B,188,185,182,17F,17C,179,176,173,170,16D,16A,167,164,161,15E,15B,158,155,152,14F,14C,149,146,143,140,13D,13A,137,134,131,12E,12B,128,125,122,11F,11  
C,119,116,113,110,10D,10A,107,104,101,0FE,0FB,0F8,0F5,0F2,0EF,0EC,0E9,0E6,0E3,0E0,0DD,0DA,0D7,0D4,0D1,0CE,0CB,0C8,0C5,0C2,0BF,0BC,0B9,0B6,0B  
3,0B0,0AD,0AA,0A7,0A4,0A1,09E,09B,098,095,092,08F,08C,089,086,083,080,07D,07A,077,074,071,06E,06B,068,065,062,05F,05C,059,056,053,050,04D,04A,04  
7,044,041,03E,03B,038,035,032,02F,02C,029]; UMC02CTLx00025_1000h; DataPortWrite=UMC025_1000h; AppCmb[Address]
```

```
_umc0_chiplettype9_instUMCWPHY3_n[143:0]_aliasSMN;  
UMC03CTLX00000804_address009_0[D16,D3,D10,1CD,1CA,1C7,1C4,1C1,1BE,1BB,1B8,1B5,1B2,1AF,1AC,1A9,1A6,1A3,1A0,19D,19A,197,194,191,18E,18B,188,185,182,17F,17C,179,176,173,170,16D,16A,167,164,161,15E,15B,158,155,152,14F,14C,149,146,143,140,13D,13A,137,134,131,12E,12B,128,125,122,11F,11C,119,116,113,110,10D,10A,107,104,101,0FE,0FB,0F8,0F5,0F2,0EF,0EC,0E9,0E6,0E3,0E0,0DD,0DA,0D7,0D4,0D1,0CE,0CB,0C8,0C5,0C2,0BF,0BC,0B9,0B6,0B3,0B0,0AD,0AA,0A7,0A4,0A1,09E,09B,098,095,092,08F,08C,089,086,083,080,07D,07A,077,074,071,06F,06B,068,065,062,05F,05C,059,056,053,050,04D,04A,047,044,041,03E,03B,038,035,032,02E,02C,029]; UMC03CTLX=0035 1000h: DataPortWrite=UMC03CTLX=0035,0036,0037,0038,0039,003A,003B,003C,003D,003E,003F,0040,0041,0042,0043,0044,0045,0046,0047,0048,0049,004A,004B,004C,004D,004E,004F,0050,0051,0052,0053,0054,0055,0056,0057,0058,0059,005A,005B,005C,005D,005E,005F,0060,0061,0062,0063,0064,0065,0066,0067,0068,0069,006A,006B,006C,006D,006E,006F,0070,0071,0072,0073,0074,0075,0076,0077,0078,0079,007A,007B,007C,007D,007E,007F,0080,0081,0082,0083,0084,0085,0086,0087,0088,0089,008A,008B,008C,008D,008E,008F,0090,0091,0092,0093,0094,0095,0096,0097,0098,0099,009A,009B,009C,009D,009E,009F,00A0,00A1,00A2,00A3,00A4,00A5,00A6,00A7,00A8,00A9,00AA,00AB,00AC,00AD,00AE,00AF,00B0,00B1,00B2,00B3,00B4,00B5,00B6,00B7,00B8,00B9,00BA,00BB,00BC,00BD,00BE,00BF,00C0,00C1,00C2,00C3,00C4,00C5,00C6,00C7,00C8,00C9,00CA,00CB,00CC,00CD,00CE,00CF,00D0,00D1,00D2,00D3,00D4,00D5,00D6,00D7,00D8,00D9,00DA,00DB,00DC,00DD,00DE,00DF,00E0,00E1,00E2,00E3,00E4,00E5,00E6,00E7,00E8,00E9,00EA,00EB,00EC,00ED,00EE,00EF,00F0,00F1,00F2,00F3,00F4,00F5,00F6,00F7,00F8,00F9,00FA,00FB,00FC,00FD,00FE,00FF,0100,0101,0102,0103,0104,0105,0106,0107,0108,0109,010A,010B,010C,010D,010E,010F,0110,0111,0112,0113,0114,0115,0116,0117,0118,0119,011A,011B,011C,011D,011E,011F,0120,0121,0122,0123,0124,0125,0126,0127,0128,0129,012A,012B,012C,012D,012E,012F,0130,0131,0132,0133,0134,0135,0136,0137,0138,0139,013A,013B,013C,013D,013E,013F,0140,0141,0142,0143,0144,0145,0146,0147,0148,0149,014A,014B,014C,014D,014E,014F,0150,0151,0152,0153,0154,0155,0156,0157,0158,0159,015A,015B,015C,015D,015E,015F,0160,0161,0162,0163,0164,0165,0166,0167,0168,0169,016A,016B,016C,016D,016E,016F,0170,0171,0172,0173,0174,0175,0176,0177,0178,0179,017A,017B,017C,017D,017E,017F,0180,0181,0182,0183,0184,0185,0186,0187,0188,0189,018A,018B,018C,018D,018E,018F,0190,0191,0192,0193,0194,0195,0196,0197,0198,0199,019A,019B,019C,019D,019E,019F,01A0,01A1,01A2,01A3,01A4,01A5,01A6,01A7,01A8,01A9,01AA,01AB,01AC,01AD,01AE,01AF,01B0,01B1,01B2,01B3,01B4,01B5,01B6,01B7,01B8,01B9,01BA,01BB,01BC,01BD,01BE,01BF,01C0,01C1,01C2,01C3,01C4,01C5,01C6,01C7,01C8,01C9,01CA,01CB,01CC,01CD,01CE,01CF,01D0,01D1,01D2,01D3,01D4,01D5,01D6,01D7,01D8,01D9,01DA,01DB,01DC,01DD,01DE,01DF,01E0,01E1,01E2,01E3,01E4,01E5,01E6,01E7,01E8,01E9,01EA,01EB,01EC,01ED,01EE,01EF,01F0,01F1,01F2,01F3,01F4,01F5,01F6,01F7,01F8,01F9,01FA,01FB,01FC,01FD,01FE,01FF,0200,0201,0202,0203,0204,0205,0206,0207,0208,0209,020A,020B,020C,020D,020E,020F,0210,0211,0212,0213,0214,0215,0216,0217,0218,0219,021A,021B,021C,021D,021E,021F,0220,0221,0222,0223,0224,0225,0226,0227,0228,0229,022A,022B,022C,022D,022E,022F,0230,0231,0232,0233,0234,0235,0236,0237,0238,0239,023A,023B,023C,023D,023E,023F,0240,0241,0242,0243,0244,0245,0246,0247,0248,0249,024A,024B,024C,024D,024E,024F,0250,0251,0252,0253,0254,0255,0256,0257,0258,0259,025A,025B,025C,025D,025E,025F,0260,0261,0262,0263,0264,0265,0266,0267,0268,0269,026A,026B,026C,026D,026E,026F,0270,0271,0272,0273,0274,0275,0276,0277,0278,0279,027A,027B,027C,027D,027E,027F,0280,0281,0282,0283,0284,0285,0286,0287,0288,0289,028A,028B,028C,028D,028E,028F,0290,0291,0292,0293,0294,0295,0296,0297,0298,0299,029A,029B,029C,029D,029E,029F,02A0,02A1,02A2,02A3,02A4,02A5,02A6,02A7,02A8,02A9,02AA,02AB,02AC,02AD,02AE,02AF,02B0,02B1,02B2,02B3,02B4,02B5,02B6,02B7,02B8,02B9,02BA,02BB,02BC,02BD,02BE,02BF,02C0,02C1,02C2,02C3,02C4,02C5,02C6,02C7,02C8,02C9,02CA,02CB,02CC,02CD,02CE,02CF,02D0,02D1,02D2,02D3,02D4,02D5,02D6,02D7,02D8,02D9,02DA,02DB,02DC,02DD,02DE,02DF,02E0,02E1,02E2,02E3,02E4,02E5,02E6,02E7,02E8,02E9,02EA,02EB,02EC,02ED,02EE,02EF,02F0,02F1,02F2,02F3,02F4,02F5,02F6,02F7,02F8,02F9,02FA,02FB,02FC,02FD,02FE,
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```
_umc0_chiplettype9_instUMCWPHY4_n[143:0]_aliasSMN;  
UMC04CTLSL0K00000804_address0009_0[D16,D13,D10,1CD,1CA,1C7,1C4,1C1,1BE,1BB,1B8,1B5,1B2,1AF,1AC,1A9,1A6,1A3,1A0,19D,19A,197,194,191,18E,18  
B,188,185,182,17F,17C,179,176,173,170,16D,16A,167,164,161,15E,15B,158,155,152,14F,14C,149,146,143,140,13D,13A,137,134,131,12E,12B,128,125,122,11F,11  
C,119,116,113,110,10D,10A,107,104,101,0FE,0FB,0F8,0F5,0F2,0EF,0EC,0E9,0E6,0E3,0E0,0DD,0DA,0D7,0D4,0D1,0CE,0CB,0C8,0C5,0C2,0BF,0BC,0B9,0B6,0B  
3,0B0,0AD,0AA,0A7,0A4,0A1,09E,09B,098,095,092,08F,08C,089,086,083,080,07D,07A,077,074,071,06E,06B,068,065,062,05F,05C,059,056,053,050,04D,04A,04  
7,044,041,03E,03B,038,035,032,02F,02C,029]; UMC04CTLSL0K00000804_n[143:0]_aliasSMN; UMC04CTLSL0K00000804_n[143:0]_aliasSMN;
```

```
_umc0_chiplettype9_instUMCWPHY5_n[143:0]_aliasSMN;  
UMC05CTL9x00000804_address0009_0[D16,D13,D10,1CD,1CA,1C7,1C4,1C1,1BE,1BB,1B8,1B5,1B2,1AF,1AC,1A9,1A6,1A3,1A0,19D,19A,197,194,191,18E,18  
B,188,185,182,17F,17C,179,176,173,170,16D,16A,167,164,161,15E,15B,158,155,152,14F,14C,149,146,143,140,13D,13A,137,134,131,12E,12B,128,125,122,11F,11  
C,119,116,113,110,10D,10A,107,104,101,0FE,0FB,0F8,0F5,0F2,0EF,0EC,0E9,0E6,0E3,0E0,0DD,0DA,0D7,0D4,0D1,0CE,0CB,0C8,0C5,0C2,0BF,0BC,0B9,0B6,0B  
3,0B0,0AD,0AA,0A7,0A4,0A1,09E,09B,098,095,092,08F,08C,089,086,083,080,07D,07A,077,074,071,06E,06B,066,065,062,05F,05C,059,056,053,050,04D,04A,04  
7,044,041,03E,03B,038,035,032,02F,02C,029]; UMC05CTL9=0055 1000h; DataPortWrite=UMC05:AppCmd/Address
```

```
_umc0_chiplettype9_instUMCWPHY6_n[143:0]_aliasSMN;  
UMC06CTLTLR00000804_address0009_0[D16,D13,D10,1CD,1CA,1C7,1C4,1C1,1BE,1BB,1B8,1B5,1B2,1AF,1AC,1A9,1A6,1A3,1A0,19D,19A,197,194,191,18E,18  
B,188,185,182,17F,17C,179,176,173,170,16D,16A,167,164,161,15E,15B,158,155,152,14F,14C,149,146,143,140,13D,13A,137,134,131,12E,12B,128,125,122,11F,11  
C,119,116,113,110,10D,10A,107,104,101,0FE,0FB,0F8,0F5,0F2,0EF,0EC,0E9,0E6,0E3,0E0,0DD,0DA,0D7,0D4,0D1,0CE,0CB,0C8,0C5,0C2,0BF,0BC,0B9,0B6,0B  
3,0B0,0AD,0AA,0A7,0A4,0A1,09E,09B,098,095,092,08F,08C,089,086,083,080,07D,07A,077,074,071,06E,06B,068,065,062,05F,05C,059,056,053,050,04D,04A,04  
7,004,041,03E,03B,038,035,032,02F,02C,029]; UMC06CTLTLR=0065 1000h; DataPortWrite=UMC::ApbCmd[Address]
```

```
umc0_chiplettype9_instUMCWPHY7_n[143:0]_aliasSMN;  
UMC0C7CTLX00000804_address0009_0[D16,D13,D10,1CD,1CA,1C7,1C4,1C1,1BE,1BB,1B8,1B5,1B2,1AF,1AC,1A9,1A6,1A3,1A0,19D,19A,197,194,191,18E,18  
B,188,185,182,17F,17C,179,176,173,170,16D,16A,167,164,161,15E,15B,158,155,152,14F,14C,149,146,143,140,13D,13A,137,134,131,12E,12B,128,125,122,11F,11  
C,119,116,113,110,10D,10A,107,104,101,0FE,0FB,0F8,0F5,0F2,0EF,0EC,0E9,0E6,0E3,0E0,0DD,0DA,0D7,0D4,0D1,0CE,0CB,0C8,0C5,0C2,0BF,0BC,0B9,0B6,0B  
3,0B0,0AD,0AA,0A7,0A4,0A1,09E,09B,098,095,092,08F,08C,089,086,083,080,07D,07A,077,074,071,06E,06B,068,065,062,05F,05C,059,056,053,050,04D,04A,04  
7,040,041,03E,03B,038,035,032,02F,02C,029]; UMC0C7CTLX=0075_1000h; DataPortWrite=UMC::AppCmId[Address]
```

```
_umc0_chiplttype9_instUMCWPHY8_n[143:0]_aliasSMN;  
UMC0885TLX00000804_addr0009_0[D16,D13,D10,1CD,1CA,1C7,1C4,1C1,1BE,1BB,1B8,1B5,1B2,1AF,1AC,1A9,1A6,1A3,1A0,19D,19A,197,194,191,18E,18  
B,188,185,182,17F,17C,179,176,173,170,16D,16A,167,164,161,15E,15B,158,155,152,14F,14C,149,146,143,140,13D,13A,137,134,131,12E,12B,128,125,122,11F,11  
C,119,116,113,110,10D,10A,107,104,101,0FE,0FB,0F8,0F5,0F2,0EF,0EC,0E9,0E6,0E3,0E0,0DD,0DA,0D7,0D4,0D1,0CE,0CB,0C8,0C5,0C2,0BF,0BC,0B9,0B6,0B  
3,0B0,0AD,0AA,0A7,0A4,0A1,09E,09B,098,095,092,08F,08C,089,086,083,080,07D,07A,077,074,071,06E,06B,068,065,062,05F,05C,059,056,053,050,04D,04A,04  
7,044,041,03E,03B,038,035,032,02F,02C,029]; UMC0885TLX00085_1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplettype9_instUMCWPHY9_n[143:0]_aliasSMN;  
UMC09C9TLX00000804 _0[D16,D3,D10,D1,1CD,1CA,1C7,1C4,1C1,1BE,1BB,1B8,1B5,1B2,1AF,1AC,1A9,1A6,1A3,1A0,19D,19A,197,194,191,18E,18  
B,188,185,182,17F,17C,179,176,173,170,16D,16A,167,164,161,15E,15B,158,155,152,14F,14C,149,146,143,140,13D,13A,137,134,131,12E,12B,128,125,122,11F,11  
C,119,116,113,110,10D,10A,107,104,101,0FE,0FB,0F8,0F5,0F2,0EF,0EC,0E9,0E6,0E3,0E0,0DD,0DA,0D7,0D4,0D1,0CE,0CB,0C8,0C5,0C2,0BF,0BC,0B9,0B6,0B  
3,0B0,0AD,0AA,0A7,0A4,0A1,09E,09B,098,095,092,08F,08C,089,086,083,080,07D,07A,077,074,071,06E,06B,068,065,062,05F,05C,059,056,053,050,04D,04A,04  
7,044,041,03E,03B,038,035,032,02F,02C,029]; UMC09C9TLX=0095 1000h; DataPortWrite=UMC::AppCmd[Address]
```

```
_umc0_chiplettype9_instUMCWPHY10_n[143:0]_aliasSMN;
UMC10CTLx00000804_address009_0[D16,D3,D10,D1C,D1CA,1C7,1C4,1C1,1BE,1BB,1B8,1B5,1B2,1AF,1AC,1A9,1A6,1A3,1A0,19D,19A,197,194,191,18E,18B,188,185,182,17F,17C,179,176,173,170,16D,16A,167,164,161,15E,15B,158,155,152,14F,14C,149,146,143,140,13D,13A,137,134,131,12E,12B,128,125,122,11F,11C,119,116,113,110,10D,10A,107,104,101,0FE,0FB,0F8,0F5,0F2,0EF,0EC,0E9,0E6,0E3,0E0,0DD,0DA,0D7,0D4,0D1,0CE,0CB,0C8,0C5,0C2,0BF,0BC,0B9,0B6,0B3,0B0,0AD,0AA,0A7,0A4,0A1,09E,09B,098,095,092,08F,08C,089,086,083,080,07D,07A,077,074,071,06E,06B,068,065,062,05F,05C,059,056,053,050,04D,04A,047,044,041,03E,03B,038,035,032,02F,02C,029]; UMC10CTLx00A5_1000h; DataPortWrite=0x0000:AppbCmd[Address]
```

```
_umc0_chiptype9_instUMCWPHY11_n[143:0]_aliasSMN;
UMC11CTLx00000804_address0009_0[1D6,1D3,1D0,1CD,1CA,1C7,1C4,1C1,1BE,1BB,1B8,1B5,1B2,1AF,1AC,1A9,1A6,1A3,1A0,19D,19A,197,194,191,18E,18
B,188,185,182,17F,17C,179,176,173,170,16D,16A,167,164,161,15E,15B,158,155,152,14F,14C,149,146,143,140,13D,13A,137,134,131,12E,12B,128,125,122,11F,11
C,119,116,113,110,10D,10A,107,104,101,0FE,0FB,0F8,0F5,0F2,0EF,0EC,0E9,0E6,0E3,0E0,0DD,0DA,0D7,0D4,0D1,0CE,0CB,0C8,0C5,0C2,0BF,0BC,0B9,0B6,0B
3,0B0,0AD,0AA,0A7,0A4,0A1,09E,09B,098,095,092,08F,08C,089,086,083,080,07D,07A,077,074,071,06E,06B,068,065,062,05F,05C,059,056,053,050,04D,04A,04
7,044,041,03E,03B,038,035,032,02F,02C,029]; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]
```

Bits	Description
15:0	SequenceReg0bs0. Read-write. Reset: 0000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

```
umc0_chiplettype9_instUMCWPHY0_n[143:0]_aliasSMN;
UMC00CTLX00000804_address0009_0[D17,D14,D11,1CE,1CB,1C8,1C5,1C2,1BF,1BC,1B9,1B6,1B3,1B0,1AD,1AA,1A7,1A4,1A1,19E,19B,198,195,192,18F,18C,
189,186,183,180,17D,17A,177,174,171,16E,16B,165,162,15F,15C,159,156,153,150,14D,14A,147,144,141,13E,13B,138,135,132,12F,12C,129,126,123,120,11D,
11A,117,114,111,10E,10B,108,105,102,0FF,0FC,0F9,0F6,0F3,0F0,0ED,0EA,0E7,0E4,0E1,0DE,0DB,0D8,0D5,0D2,0CF,0CC,0C9,0C6,0C3,0C0,0BD,0BA,0B7,0B4,0B1,
0B1,0AE,0AB,0A8,0A5,0A2,09F,09C,099,096,093,090,08D,08A,087,084,081,07E,07B,078,075,072,06F,06C,069,066,063,060,05D,05A,057,054,051,04E,04B,048,
045,042,03F,03C,039,036,033,030,02D,02A); UMC00CTLX00000804=0005 1000h; DataPortWrite=UMC=:AppbMcM[Address]
```

```
umc0_chiplettype9_instUMCWPHY1_n[143:0]_aliasSMN;
UMC0CTLX00000804_address0009_0[D17,D14,D11,1CE,1CB,1C8,1C5,1C2,1BF,1BC,1B9,1B6,1B3,1B0,1AD,1AA,1A7,1A4,1A1,19E,19B,198,195,192,18F,18C,
,189,186,183,180,17D,17A,177,174,171,16E,16B,165,162,15F,15C,159,156,153,150,14D,14A,147,144,141,13E,13B,138,135,132,12F,12C,129,126,123,120,11D,
,11A,117,114,111,10E,10B,108,105,102,0FF,0FC,0F9,0F6,0F3,0F0,0ED,0EA,0E7,0E4,0E1,0DE,0DB,0D8,0D5,0D2,0CF,0CC,0C9,0C6,0C3,0C0,0BD,0BA,0B7,0B4,
,0B1,0AE,0AB,0A8,0A5,0A2,09F,09C,099,096,093,090,08D,08A,087,084,081,07E,07B,078,075,072,06F,06C,069,066,063,060,05D,05A,057,054,051,04E,04B,048,
045,042,03C,039,036,033,030,02D,02A1; UMC0CTL=0015 1000h; DataPortWrite=UMC::AppbMcM[Address
```

```
umc0_chiplettype9_instUMCWPHY2_n[143:0]_aliasSMN;  
UMC0CTLX00000804_address0009_0[D17,D14,D11,1CE,1CB,1C8,1C5,1C2,1BF,1BC,1B9,1B6,1B3,1B0,1AD,1AA,1A7,1A4,1A1,19E,19B,198,195,192,18F,18C,  
189,186,183,180,17D,17A,177,174,171,16E,16B,165,162,15F,15C,159,156,153,150,14D,14A,147,144,141,13E,13B,138,135,132,12F,12C,129,126,123,120,11D,  
11A,117,114,111,10E,10B,108,105,102,0FF,0FC,0F9,0F6,0F3,0F0,0ED,0EA,0E7,0E4,0E1,0DE,0DB,0D8,0D5,0D2,0CF,0CC,0C9,0C6,0C3,0C0,0BD,0BA,0B7,0B4,  
0B1,0AE,0AB,0A8,0A5,0A2,09F,09C,099,096,093,090,08D,08A,087,084,081,07E,07B,078,075,072,06F,06C,069,066,063,060,05D,05A,057,054,051,04E,04B,048,  
045,042,03C,03C,039,036,033,030,02D,02A1; UMC0CTL=0025 1000h; DataPortWrite=UMC::AppbCMf[Address]
```

```
_umc0_chiplettype9_instUMCWPHY3_n[143:0]_aliasSMN;
UMC0CTL3L00000804_address0009_0[D17,D14,D11,1CE,1CB,1C8,1C5,1C2,1BF,1BC,1B9,1B6,1B3,1B0,1AD,1AA,1A7,1A4,1A1,19E,19B,198,195,192,18F,18C,
,189,186,183,180,17D,17A,177,174,171,16E,16B,165,162,15F,15C,159,156,153,150,14D,14A,147,144,141,13E,13B,138,135,132,12F,12C,129,126,123,120,11D,
,11A,117,114,111,10E,10B,108,105,102,0FF,0FC,0F9,0F6,0F3,0F0,0ED,0EA,0E7,0E4,0E1,0DE,0DB,0D8,0D5,0D2,0CF,0CC,0C9,0C6,0C3,0C0,0BD,0BA,0B7,0B4,0B1,0AE,0AB,0A8,0A5,0A2,09F,09C,099,096,093,090,08D,08A,087,084,081,07E,07B,078,075,072,06F,06C,069,066,063,060,05D,05A,057,054,051,04E,04B,048,
,045,042,03F,03C,039,036,033,030,02D,02A]; UMC0CTL=0035 1000h; DataPortWrite=UMC::AppbMcM[Address]
```

```
umc0_chiplettype9_instUMCWPHY4_n[143:0]_aliasSMN;
UMC04CTLX00000804_addr000009_0[D17,D14,D11,1CE,1CB,1C8,1C5,1C2,1BF,1BC,1B9,1B6,1B3,1B0,1AD,1AA,1A7,1A4,1A1,19E,19B,198,195,192,18F,18C,
,189,186,183,180,17D,17A,177,174,171,16E,16B,165,162,15F,15C,159,156,153,150,14D,14A,147,144,141,13E,13B,138,135,132,12F,12C,129,126,123,120,11D,
,11A,117,114,111,10E,10B,108,105,102,0FF,0FC,0F9,0F6,0F3,0F0,0ED,0EA,0E7,0E4,0E1,0DE,0DB,0D8,0D5,0D2,0CF,0CC,0C9,0C6,0C3,0C0,0BD,0BA,0B7,0B4,0B1,0AE,0AB,0A8,0A5,0A2,09F,09C,
,099,096,093,090,08D,08A,087,084,081,07E,07B,078,075,072,06F,06C,069,066,063,060,05D,05A,057,054,051,04E,04B,048,
,045,042,03F,03C,039,036,033,030,02D,02A]; UMC04CTL=0045 1000h; DataPortWrite=UMC::AppbMcM[Address]
```

```
_umc0_chiplettype9_instUMCWPHY5_n[143:0]_aliasSMN;  
UMC0CTLX00000804_address0009_0[D17,D14,D11,1CE,1CB,1C8,1C5,1C2,1BF,1BC,1B9,1B6,1B3,1B0,1AD,1AA,1A7,1A4,1A1,19E,19B,198,195,192,18F,18C,  
189,186,183,180,17D,17A,177,174,171,16E,16B,165,162,15F,15C,159,156,153,150,14D,14A,147,144,141,13E,13B,138,135,132,12F,12C,129,126,123,120,11D,  
11A,117,114,111,10E,10B,108,105,102,0FF,0FC,0F9,0F6,0F3,0F0,0ED,0EA,0E7,0E4,0E1,0DE,0DB,0D8,0D5,0D2,0CF,0CC,0C9,0C6,0C3,0C0,0BD,0BA,0B7,0B4,0B1,  
0B1,0AE,0AB,0A8,0A5,0A2,09F,09C,099,096,093,090,08D,08A,087,084,081,07E,07B,078,075,072,06F,06C,069,066,063,060,05D,05A,057,054,051,04E,04B,048,  
045,042,03FC,03C,039,036,033,030,02D,02A1]; UMC0CTL=0055 1000h; DataPortWrite=UMC::AppbMcM[Address]
```

```
_umc0_chiplettype9_instUMCWPHY6_n[143:0]_aliasSMN;  
UMC06CTLX00000804_address0009_0[D17,D14,D11,1CE,1CB,1C8,1C5,1C2,1BF,1BC,1B9,1B6,1B3,1B0,1AD,1AA,1A7,1A4,1A1,19E,19B,198,195,192,18F,18C,  
189,186,183,180,17D,17A,177,174,171,16E,16B,165,162,15F,15C,159,156,153,150,14D,14A,147,144,141,13E,13B,138,135,132,12F,12C,129,126,123,120,11D,  
11A,117,114,111,10E,10B,108,105,102,0FF,0FC,0F9,0F6,0F3,0F0,0ED,0EA,0E7,0E4,0E1,0DE,0DB,0D8,0D5,0D2,0CF,0CC,0C9,0C6,0C3,0C0,0BD,0BA,0B7,0B4,  
0B1,0AE,0AB,0A8,0A5,0A2,09F,09C,099,096,093,090,08D,08A,087,084,081,07E,07B,078,075,072,06F,06C,069,066,063,060,05D,05A,057,054,051,04E,04B,048,  
045,042,03C,039,036,033,030,02D,02A1: UMC06CTLX=0065 1000h: DataPortWrite=UMC5: AppbCM6[Address]
```

```
_umc0_chiplettype9_instUMCWPHY7_n[143:0]_aliasSMN;  
UMC07CTLx00000804_address0009_0[D17,D14,D11,D1,I1,C1B,1C8,1C5,1C2,1BF,1BC,1B9,1B6,1B3,1B0,1AD,1AA,1A7,1A4,1A1,19E,19B,198,195,192,18F,18C,  
189,186,183,180,17D,17A,177,174,171,16E,16B,165,162,15F,15C,159,156,153,150,14D,14A,147,144,141,13E,13B,138,135,132,12F,12C,129,126,123,120,11D,  
11A,117,114,111,10E,10B,108,105,102,0FF,0FC,0F9,0F6,0F3,0F0,0ED,0EA,0E7,0E4,0E1,0DE,0DB,0D8,0D5,0D2,0CF,0CC,0C9,0C6,0C3,0C0,0BD,0BA,0B7,0B4,  
0B1,0A6,0AB,0A8,0A5,0A2,09F,09C,099,096,093,090,08D,08A,087,084,081,07E,07B,078,075,072,06F,06C,069,066,063,060,05D,05A,057,054,051,04E,04B,048,  
045,042,03C,039,036,033,030,02D,02A1: UMC07CTL=0075 1000h: DataPortWrite=UMC::AppbCmf[Address]
```

```
_umc0_chiplettype9_instUMCWPHY8_n[143:0]_aliasSMN;  
UMC08CTLx00000804_address0009_0[D17,D14,D11,D1,I1,C1B,C1C,C1S,C1C2,1BF,1BC,1B9,1B6,1B3,1B0,1AD,1AA,1A7,1A4,1A1,19E,19B,198,195,192,18F,18C,  
189,186,183,180,17D,17A,177,174,171,16E,16B,165,162,15F,15C,159,156,153,150,14D,14A,147,144,141,13E,13B,138,135,132,12F,12C,129,126,123,120,11D,  
11A,117,114,111,10E,10B,108,105,102,0FF,0FC,0F9,0F6,0F3,0F0,0ED,0EA,0E7,0E4,0E1,0DE,0DB,0D8,0D5,0D2,0CF,0CC,0C9,0C6,0C3,0C0,0BD,0BA,0B7,0B4,  
0B1,0A4,0A8,0AB,0A5,0A0,09F,09C,099,096,093,090,08D,08A,087,084,081,07E,07B,078,075,072,06F,06C,069,066,063,060,05D,05A,057,054,051,04E,04B,048,  
045,042,03F,03C,039,036,033,030,02D,02A1: UMC08CTLx=0885 1000h: DataPortWrite=UMC::AppbCmf[Address]
```

```
_umc0_chiplettype9_instUMCWPWHY9_n[143:0]_aliasSMN;
UMC09C6T1X00000804_address0009_0[D17,D14,D11,D1,I,C1B,1C8,1C5,1C2,1BF,1BC,1B9,1B6,1B3,1B0,1AD,1AA,1A7,1A4,1A1,19E,19B,198,195,192,18F,18C,
189,186,183,180,17D,17A,177,174,171,16E,16B,165,162,15F,15C,159,156,153,150,14D,14A,147,144,141,13E,13B,138,135,132,12F,12C,129,126,123,120,11D,
11A,117,114,111,10E,10B,108,105,102,0FF,0FC,0F9,0F6,0F3,0F0,0ED,0EA,0E7,0E4,0E1,0DE,0DB,0D8,0D5,0D2,0CF,0CC,0C9,0C6,0C3,0C0,0BD,0BA,0B7,0B4,
0B1,0A6,0AB,0A8,0A5,0A2,09F,09C,099,096,093,090,08D,08A,087,084,081,07E,07B,078,075,072,06F,06C,069,066,063,060,05D,05A,057,054,051,04E,04B,048,
045,042,03F,03C,039,036,033,030,02D,02A1: UMC09C6T1X=0095 1000h: DataPortWrite=UMC::AppbCmf[Address]
```

```

_umc0_chiplettype9_instUMCWPHY10_n[143:0]_aliasSMN;
UMC10CTL=0x00000804_address0009_0[D17,D14,D11,1CE,1CB,1C8,1C5,1C2,1BF,1BC,1B9,1B6,1B3,1B0,1AD,1AA,1A7,1A4,1A1,19E,19B,198,195,192,18F,18C,
189,186,183,180,17D,17A,177,174,171,16E,16B,168,165,162,15F,15C,159,156,153,150,14D,14A,147,144,141,13E,13B,138,135,132,12F,12C,129,126,123,120,11D,
11A,117,114,111,10E,10B,108,105,102,0FF,0FC,0F9,0F6,0F3,0F0,0ED,0EA,0E7,0E4,0E1,0DE,0DB,0D8,0D5,0D2,0CF,0CC,0C9,0C6,0C3,0C0,0BD,0BA,0B7,0B4,0B1,0AE,0AB,0A8,0A5,0A2,09F,09C,099,096,093,090,08D,08A,087,084,081,07E,07B,078,075,072,06F,06C,069,066,063,060,05D,05A,057,054,051,04E,04B,048,
045,042,03F,03C,039,036,033,030,02D,02A1; UMC10CTL=000A5 1000h; DataPortWrite=UMC::AppBMC[Address]

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```
_umc0_chiptype9_instUMCWPHY11_n[143:0]_aliasSMN;
UMC11CTLx00000804_address0009_0[1D7,1D4,1D1,1CE,1CB,1C8,1C5,1C2,1BF,1BC,1B9,1B6,1B3,1B0,1AD,1AA,1A7,1A4,1A1,19E,19B,198,195,192,18F,18C,189,186,183,180,17D,17A,177,174,171,16E,16B,168,165,162,15F,15C,159,156,153,150,14D,14A,147,144,141,13E,13B,138,135,132,12F,12C,129,126,123,120,11D,11A,117,114,111,10E,10B,108,105,102,0FF,0FC,0F9,0F6,0F3,0F0,0ED,0EA,0E7,0E4,0E1,0DE,0DB,0D8,0D5,0D2,0CF,0CC,0C9,0C6,0C3,0C0,0BD,0BA,0B7,0B4,0B1,0AE,0AB,0A8,0A5,0A2,09F,09C,099,096,093,090,08D,08A,087,084,081,07E,07B,078,075,072,06F,06C,069,066,063,060,05D,05A,057,054,051,04E,04B,048,045,042,03F,03C,039,036,033,030,02D,02A]; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]
```

Bits	Description
15:0	SequenceReg0bs1. Read-write. Reset: 0000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

```

Read-write. Reset: 0000h.
Programmed by dwc_ddrphy_phyinit_i_loadPIEImage() to support frequency changes.
_umc0_chiplettype9_instUMCWPHY0_n[143:0]_aliasSMN;
UMC00CTLx00000804_address0009_0[1D8,1D5,1D2,1CF,1CC,1C9,1C6,1C3,1C0,1BD,1BA,1B7,1B4,1B1,1AE,1AB,1A8,1A5,1A2,19F,19C,199,196,193,190,18D,18A,187,184,181,17E,17B,178,175,172,16F,16C,169,166,163,160,15D,15A,157,154,151,14E,14B,148,145,142,13F,13C,139,136,133,130,12D,12A,127,124,121,11E,11B,118,115,112,10F,10C,109,106,103,100,0FD,0FA,0F7,0F4,0F1,0EE,0EB,0E8,0E5,0E2,0DF,0DC,0D9,0D6,0D3,0D0,0CD,0CA,0C7,0C4,0C1,0BE,0BB,0B8,0B5,0B2,0AF,0AC,0A9,0A6,0A3,0A0,09D,09A,097,094,091,08E,08B,088,085,082,07F,07C,079,076,073,070,06D,06A,067,064,061,05E,05B,058,055,052,04F,04C,049,046,043,040,03D,03A,037,034,031,02E,02B]; UMC00CTL=0005_1000h; DataPortWrite=UMC::ApbCmd[Address]
_umc0_chiplettype9_instUMCWPHY1_n[143:0]_aliasSMN;
UMC01CTLx00000804_address0009_0[1D8,1D5,1D2,1CF,1CC,1C9,1C6,1C3,1C0,1BD,1BA,1B7,1B4,1B1,1AE,1AB,1A8,1A5,1A2,19F,19C,199,196,193,190,18D,18A,187,184,181,17E,17B,178,175,172,16F,16C,169,166,163,160,15D,15A,157,154,151,14E,14B,148,145,142,13F,13C,139,136,133,130,12D,12A,127,124,121,11E,11B,118,115,112,10F,10C,109,106,103,100,0FD,0FA,0F7,0F4,0F1,0EE,0EB,0E8,0E5,0E2,0DF,0DC,0D9,0D6,0D3,0D0,0CD,0CA,0C7,0C4,0C1,0BE,0BB,0B8,0B5,0B2,0AF,0AC,0A9,0A6,0A3,0A0,09D,09A,097,094,091,08E,08B,088,085,082,07F,07C,079,076,073,070,06D,06A,067,064,061,05E,05B,058,055,052,04F,04C,049,046,043,040,03D,03A,037,034,031,02E,02B]; UMC01CTL=0015_1000h; DataPortWrite=UMC::ApbCmd[Address]
_umc0_chiplettype9_instUMCWPHY2_n[143:0]_aliasSMN;
UMC02CTLx00000804_address0009_0[1D8,1D5,1D2,1CF,1CC,1C9,1C6,1C3,1C0,1BD,1BA,1B7,1B4,1B1,1AE,1AB,1A8,1A5,1A2,19F,19C,199,196,193,190,18D,18A,187,184,181,17E,17B,178,175,172,16F,16C,169,166,163,160,15D,15A,157,154,151,14E,14B,148,145,142,13F,13C,139,136,133,130,12D,12A,127,124,121,11E,11B,118,115,112,10F,10C,109,106,103,100,0FD,0FA,0F7,0F4,0F1,0EE,0EB,0E8,0E5,0E2,0DF,0DC,0D9,0D6,0D3,0D0,0CD,0CA,0C7,0C4,0C1,0BE,0BB,0B8,0B5,0B2,0AF,0AC,0A9,0A6,0A3,0A0,09D,09A,097,094,091,08E,08B,088,085,082,07F,07C,079,076,073,070,06D,06A,067,064,061,05E,05B,058,055,052,04F,04C,049,046,043,040,03D,03A,037,034,031,02E,02B]; UMC02CTL=0025_1000h; DataPortWrite=UMC::ApbCmd[Address]
_umc0_chiplettype9_instUMCWPHY3_n[143:0]_aliasSMN;
UMC03CTLx00000804_address0009_0[1D8,1D5,1D2,1CF,1CC,1C9,1C6,1C3,1C0,1BD,1BA,1B7,1B4,1B1,1AE,1AB,1A8,1A5,1A2,19F,19C,199,196,193,190,18D,18A,187,184,181,17E,17B,178,175,172,16F,16C,169,166,163,160,15D,15A,157,154,151,14E,14B,148,145,142,13F,13C,139,136,133,130,12D,12A,127,124,121,11E,11B,118,115,112,10F,10C,109,106,103,100,0FD,0FA,0F7,0F4,0F1,0EE,0EB,0E8,0E5,0E2,0DF,0DC,0D9,0D6,0D3,0D0,0CD,0CA,0C7,0C4,0C1,0BE,0BB,0B8,0B5,0B2,0AF,0AC,0A9,0A6,0A3,0A0,09D,09A,097,094,091,08E,08B,088,085,082,07F,07C,079,076,073,070,06D,06A,067,064,061,05E,05B,058,055,052,04F,04C,049,046,043,040,03D,03A,037,034,031,02E,02B]; UMC03CTL=0035_1000h; DataPortWrite=UMC::ApbCmd[Address]
_umc0_chiplettype9_instUMCWPHY4_n[143:0]_aliasSMN;
UMC04CTLx00000804_address0009_0[1D8,1D5,1D2,1CF,1CC,1C9,1C6,1C3,1C0,1BD,1BA,1B7,1B4,1B1,1AE,1AB,1A8,1A5,1A2,19F,19C,199,196,193,190,18D,18A,187,184,181,17E,17B,178,175,172,16F,16C,169,166,163,160,15D,15A,157,154,151,14E,14B,148,145,142,13F,13C,139,136,133,130,12D,12A,127,124,121,11E,11B,118,115,112,10F,10C,109,106,103,100,0FD,0FA,0F7,0F4,0F1,0EE,0EB,0E8,0E5,0E2,0DF,0DC,0D9,0D6,0D3,0D0,0CD,0CA,0C7,0C4,0C1,0BE,0BB,0B8,0B5,0B2,0AF,0AC,0A9,0A6,0A3,0A0,09D,09A,097,094,091,08E,08B,088,085,082,07F,07C,079,076,073,070,06D,06A,067,064,061,05E,05B,058,055,052,04F,04C,049,046,043,040,03D,03A,037,034,031,02E,02B]; UMC04CTL=0045_1000h; DataPortWrite=UMC::ApbCmd[Address]
_umc0_chiplettype9_instUMCWPHY5_n[143:0]_aliasSMN;
UMC05CTLx00000804_address0009_0[1D8,1D5,1D2,1CF,1CC,1C9,1C6,1C3,1C0,1BD,1BA,1B7,1B4,1B1,1AE,1AB,1A8,1A5,1A2,19F,19C,199,196,193,190,18D,18A,187,184,181,17E,17B,178,175,172,16F,16C,169,166,163,160,15D,15A,157,154,151,14E,14B,148,145,142,13F,13C,139,136,133,130,12D,12A,127,124,121,11E,11B,118,115,112,10F,10C,109,106,103,100,0FD,0FA,0F7,0F4,0F1,0EE,0EB,0E8,0E5,0E2,0DF,0DC,0D9,0D6,0D3,0D0,0CD,0CA,0C7,0C4,0C1,0BE,0BB,0B8,0B5,0B2,0AF,0AC,0A9,0A6,0A3,0A0,09D,09A,097,094,091,08E,08B,088,085,082,07F,07C,079,076,073,070,06D,06A,067,064,061,05E,05B,058,055,052,04F,04C,049,046,043,040,03D,03A,037,034,031,02E,02B]; UMC05CTL=0055_1000h; DataPortWrite=UMC::ApbCmd[Address]
_umc0_chiplettype9_instUMCWPHY6_n[143:0]_aliasSMN;
UMC06CTLx00000804_address0009_0[1D8,1D5,1D2,1CF,1CC,1C9,1C6,1C3,1C0,1BD,1BA,1B7,1B4,1B1,1AE,1AB,1A8,1A5,1A2,19F,19C,199,196,193,190,18D,18A,187,184,181,17E,17B,178,175,172,16F,16C,169,166,163,
```


_umc0_chipllettype9_instUMCWPHY11_n[143:0]_aliasSMN; UMC11CTLx00000804_address0009_0[1D8,1D5,1D2,1CF,1CC,1C9,1C6,1C3,1C0,1BD,1BA,1B7,1B4,1B1,1AE,1AB,1A8,1A5,1A2,19F,19C,199,196,193,190,18D,18A,187,184,181,17E,17B,178,175,172,16F,16C,169,166,163,160,15D,15A,157,154,151,14E,14B,148,145,142,13F,13C,139,136,133,130,12D,12A,127,124,121,11E,11B,118,115,112,10F,10C,109,106,103,100,0FD,0FA,0F7,0F4,0F1,0EE,0EB,0E8,0E5,0E2,0DF,0DC,0D9,0D6,0D3,0D0,0CD,0CA,0C7,0C4,0C1,0BE,0BB,0B8,0B5,0B2,0AF,0AC,0A9,0A6,0A3,0A0,09D,09A,097,094,091,08E,08B,088,085,082,07F,07C,079,076,073,070,06D,06A,067,064,061,05E,05B,058,055,052,04F,04C,049,046,043,040,03D,03A,037,034,031,02E,02B]; UMC11CTL=00B5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:9	Reserved.
8:0	SequenceReg0bs2 . Read-write. Reset: 000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

UMC00CTLx00000804_address00090204...UMC11CTLx00000804_address00390204 (UMC::Phy::Seq0BGPR4)

Read-write. Reset: 0000h.	
Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.	
_umc0_chipllettype9_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00090204; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chipllettype9_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00190204; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chipllettype9_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00290204; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chipllettype9_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00390204; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:0	Seq0BGPR4 . Read-write. Reset: 0000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

UMC00CTLx00000804_address00090205...UMC11CTLx00000804_address00390205 (UMC::Phy::Seq0BGPR5)

Read-write. Reset: 0000h.	
Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.	
_umc0_chipllettype9_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00090205; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chipllettype9_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00190205; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chipllettype9_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00290205; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chipllettype9_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00390205; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:0	Seq0BGPR5 . Read-write. Reset: 0000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

UMC00CTLx00000804_address00090206...UMC11CTLx00000804_address00390206 (UMC::Phy::Seq0BGPR6)

Read-write. Reset: 0000h.	
Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.	
_umc0_chipllettype9_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00090206; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chipllettype9_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00190206; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chipllettype9_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00290206; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
_umc0_chipllettype9_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00390206; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]	
Bits	Description
15:0	Seq0BGPR6 . Read-write. Reset: 0000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

UMC00CTLx00000804_address00090207...UMC11CTLx00000804_address00390207 (UMC::Phy::Seq0BGPR7)

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00090207; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00190207; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00290207; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00390207; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Seq0BGPR7. Read-write. Reset: 0000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

UMC00CTLx00000804_address00090208...UMC11CTLx00000804_address00390208 (UMC::Phy::Seq0BGPR8)

Read-write. Reset: 0000h.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_p0_aliasSMN; UMC[11:00]CTLx00000804_address00090208; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_inst[UMCWPHY[11:0]]_p1_aliasSMN; UMC[11:00]CTLx00000804_address00190208; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_inst[UMCWPHY[11:0]]_p2_aliasSMN; UMC[11:00]CTLx00000804_address00290208; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]_umc0_chiplettype9_inst[UMCWPHY[11:0]]_p3_aliasSMN; UMC[11:00]CTLx00000804_address00390208; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Seq0BGPR8. Read-write. Reset: 0000h. Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

UMC[00...11]CTLx00000804_address000902FF (UMC::Phy::Seq0BFixedAddrBits)

Read-write. Reset: 000Fh.

Programmed by dwc_ddrphy_phyinit_I_loadPIEImage() to support frequency changes.

_umc0_chiplettype9_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000902FF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:7	Reserved.
6:0	RFU. Read-write. Reset: 0Fh.

UMC[00...11]CTLx00000804_address000C0004 (UMC::Phy::DctShadowRegs)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0004; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	RFU. Read-only, Volatile.

UMC[00...11]CTLx00000804_address000C0030 (UMC::Phy::DctWriteOnlyShadow)

Read-only, Volatile.

Reserved for PHY training firmware use.

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0030; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	DctWriteOnlyShadow. Read-only, Volatile. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000C0032 (UMC::Phy::UctWriteOnly)

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0032; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **UctWriteOnly**. Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address000C0033 (UMC::Phy::UctWriteProt)**

Read-write. Reset: 0001h.

Reserved for PHY training firmware use.

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0033; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:1 Reserved.

0 **UctWriteProt**. Read-write. Reset: 1. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address000C0034 (UMC::Phy::UctDatWriteOnly)**

Read-write. Reset: 0000h.

Reserved for PHY training firmware use.

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0034; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **UctDatWriteOnly**. Read-write. Reset: 0000h. Reserved for PHY training firmware use.**UMC[00...11]CTLx00000804_address000C0035 (UMC::Phy::UctDatWriteProt)**

Read-write. Reset: 0001h.

Reserved for PHY training firmware use.

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0035; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:1 Reserved.

0 **UctDatWriteProt**. Read-write. Reset: 1. Reserved for PHY training firmware use.

UMC[00...11]CTLx00000804_address000C007E [Controls reset and clock shutdown on the local microcontroller] (UMC::Phy::MicroResetPIE)

Read-write. Reset: 0001h.

When enabled by csrMicroResetPIEEn, controls reset and clock shutdown on the local microcontroller.

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C007E; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3	ResetToMicroPIE. Read-write. Reset: 0. Description: This field is programmed as follows: - Set this bit to apply synchronous reset to the microcontroller. - Clear this bit to release reset to the microcontroller. This reset initializes the program counter to begin execution from the boot vector. This reset also clears the interrupt sticky bits.
2:1	MicroResetPIEReserved. Read-write. Reset: 0h. Reserved.
0	StallToMicroPIE. Read-write. Reset: 1. Description: This field is programmed as follows: - Set this bit to stall the microcontroller by hardware. - Clear this bit to allow the microcontroller to continue executing from its current program counter location. Typically, this bit is used at power up to hold the program counter at the boot vector while BIOS loads the microcontroller program code. While stalled, the microcontroller clocks are gated off for power reduction.

UMC[00...11]CTLx00000804_address000C0080 [Uclk and Hclk enables] (UMC::Phy::UclkHclkEnables)

Read-write. Reset: 0003h.

UcClk/Hclk Enables. [UcClk and Hclk run at either DFICLK or are gated]

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0080; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:3	Reserved.
2	UcClkFull. Read-write. Reset: 0. When set, allows the ARC to run at full DfiClk Speed. Reserved for Debug. Non-zero values are not supported.
1	HclkEn. Read-write. Reset: 1. When training has completed (and assuming no further need for the training hardware), the enable should be set to 0 to reduce power.
0	UcclkEn. Read-write. Reset: 1. When training has completed (and assuming no further need for the microcontroller), the enable should be set to 0 to reduce power.

UMC[00...11]CTLx00000804_address000C0081 (UMC::Phy::CurPstate0b)

Read-write. Reset: 0000h.

PIE current Pstate value

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0081; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3:0	CurPstate0b. Read-write. Reset: 0h. Description: PIE current Pstate value. This register is used to select values for writing by the Pstate sequencer and is written in the beginning of the Pstate switch. It can also be used to reference which Pstate the sequencer thinks it's in.

UMC[00...11]CTLx00000804_address000C0082 [Indications of ARC double bit and exception errors] (UMC::Phy::ArcEccIndications)

Read-only, Volatile.

Indications of ARC double bit and exception errors.

Indications are cleared by resetting the ARC, or explicitly via ARC firmware.

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0082; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:5	Reserved.
4	ArcExceptionError . Read-only, Volatile. Error during software exception
3	ArcDccmAdrError . Read-only, Volatile. Error in the DCCM Addr
2	ArcDccmDbError . Read-only, Volatile. Double-bit error in the DCCM Data
1	ArcIccmAdrError . Read-only, Volatile. Error in the ICCM Addr
0	ArcIccmDbError . Read-only, Volatile. Double-bit error in the ICCM Data

UMC[00...11]CTLx00000804_address000C0083 (UMC::Phy::ArcIccmSbErrCtr)

Read-only, Volatile.

Count of ICCM Single-bit corrected ECC Errors

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0083; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	ArcIccmSbErrCtr . Read-only, Volatile. Count of ICCM Single-bit corrected ECC Errors. Controlled by csrArcIccmSbCtrEn.

UMC[00...11]CTLx00000804_address000C0084 (UMC::Phy::ArcDccmSbErrCtr)

Read-only, Volatile.

Count of DCCM Single-bit corrected ECC Errors

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0084; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	ArcDccmSbErrCtr . Read-only, Volatile. Count of DCCM Single-bit corrected ECC Errors. Controlled by csrArcDccmSbCtrEn.

UMC[00...11]CTLx00000804_address000C0085 [Enables for single-bit error counters] (UMC::Phy::ArcSbCtrEnables)

Read-write. Reset: 0000h.

Setting the bit enables the counter. Clearing the bit stops and clears the counter.

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0085; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:2	Reserved.
1	ArcDccmSbCtrEn . Read-write. Reset: 0. Enable for counter of single-bit data errors in the DCCM Data
0	ArcIccmSbCtrEn . Read-write. Reset: 0. Enable for counter of single-bit data errors in the ICCM Data

UMC[00...11]CTLx00000804_address000C0086 (UMC::Phy::ArcPmuEccCtl)

Read-write. Reset: 0000h.

ARC ECC Control

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0086; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:6	Reserved.
5:0	ArcPmuEccCtl. Read-write. Reset: 00h. Description: Overrides/Control for ARC Error Protection Hardware Control Register (ERP_CTRL) Modify this register only during initialization before any firmware is loaded. - Bits 1:0: 00 = ECC Enabled, 01 = ECC Disabled, 1x = Unsupported Debug mode (enables use of bits 5:2) - Bit 2: (Debug mode only) override value of ERP_CTRL IP field (1 Enable parity protection for ICCM, correct single-bit errors, and raise exception for double-bit, address-bus, or parity errors) - Bit 3: (Debug mode only) override value of ERP_CTRL IE field (0 enables protection exceptions for double-bit, address bus, and parity errors on ICCM) - Bit 4: (Debug mode only) override value of ERP_CTRL DP field (1 Enable parity protection for DCCM, correct single-bit errors, and raise exception for double-bit, address-bus, or parity errors) - Bit 5: (Debug mode only) override value of ERP_CTRL DE field (0 enables protection exceptions for double-bit, address bus, and parity errors on DCCM)

UMC[00...11]CTLx00000804_address000C0088 (UMC::Phy::StartDCCMClear)

Read-write. Reset: 0000h.

Starts DCCM Auto-clear

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0088; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	StartDCCMClear. Read-write. Reset: 0. Description: A 0->1 transition on this bit starts a clear of the DCCM. UcClk must be enabled, but the ARC should be disabled by setting csrStallToMicro when this function is operating. While the function is operating, APB writes to ICCM or DCCM will appear to complete but they will be ignored. APB accesses to internal PHY registers can occur while this function is running. This operation takes 98308 DfiClk cycles.

UMC[00...11]CTLx00000804_address000C0089 (UMC::Phy::DCCMClearRunning)

Read-only, Volatile.

DCCM Auto-clear Status

_umc0_chiplettype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C0089; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	DCCMClearRunning. Read-only, Volatile. When 1, indicates that DCCM Auto-clear is running.

UMC[00...11]CTLx00000804_address000C008A (UMC::Phy::APBMode16BitOverride)

Read-write. Reset: 0000h.

Override APB Data Width for ICCM/DCCM

_umc0_chiptype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C008A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	APBMode16BitOverride. Read-write. Reset: 0. Description: When set, overrides APB data width for ICCM/DCCM (which can be 32b mode based on `DWC_DDRPHY_APBMODE32BIT) to be 16b mode for TDR interface

UMC[00...11]CTLx00000804_address000C008B (UMC::Phy::APBMode32Bit)

Read-only, Volatile.

APB Data Width for ICCM/DCCM

_umc0_chiptype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C008B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	APBMode32Bit. Read-only, Volatile. 1 = 32b, 0 = 16b

UMC[00...11]CTLx00000804_address000C00EF (UMC::Phy::PUBVAR)

Read-only, Volatile.

The hardware variant of this PUB

_umc0_chiptype12_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000C00EF; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	PUBVAR. Read-only, Volatile. Description: The hardware variant of this PUB 0xFFFF indicates the mainline release. Variants are identified by specific hex value.

UMC[00...11]CTLx00000804_address000D0000 (UMC::Phy::MicroContMuxSel)

Read-write. Reset: 0000h.

PMU Config Mux Select

_umc0_chiptype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0000; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	MicroContMuxSel. Read-write. Reset: 0. Description: This register controls access to the PHY configuration registers. - 1 : MicroController/PHY Init Engine has control of csr bus. - 0 : MicroController/PHY Init Engine csr requests are ignored. Note: When writing host Firmware that interacts with the ARC processor using the message interface, and receiving a message from the ARC that it is time to clear csrMicroContMuxSel, Firmware should wait 40 DfiClk before writing the bit to allow the ARC to finish message processing.

UMC[00...11]CTLx00000804_address000D0004 [PMU/Controller Protocol - Controller Read-only Shadow] (UMC::Phy::UctShadowRegs)

Read-only, Volatile.

This is used for the mailbox protocol between the firmware and the system.

See section 3.4.1 of the training firmware AppNote for details on the protocol.

`_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0004; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]`

Bits	Description
15:2	Reserved.
1	UctDatWriteProtShadow . Read-only, Volatile. Reserved for future use.
0	UctWriteProtShadow . Read-only, Volatile. When set to 0, the PMU has a message for the user

UMC[00...11]CTLx00000804_address000D0030 (UMC::Phy::DctWriteOnly)

Read-write. Reset: 0000h.

Reserved for future use.

`_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0030; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]`

Bits	Description
15:0	DctWriteOnly . Read-write. Reset: 0000h. Reserved for future use.

UMC[00...11]CTLx00000804_address000D0031 (UMC::Phy::DctWriteProt)

Read-write. Reset: 0001h.

DCT downstream mailbox protocol CSR.

`_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0031; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]`

Bits	Description
15:1	Reserved.
0	DctWriteProt . Read-write. Reset: 1. By setting this register to 0, the user acknowledges the receipt of the message.

UMC[00...11]CTLx00000804_address000D0032 (UMC::Phy::UctWriteOnlyShadow)

Read-only, Volatile.

Read-only view of the csr UctDatWriteOnly

`_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0032; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]`

Bits	Description
15:0	UctWriteOnlyShadow . Read-only, Volatile. Used to pass the message ID for major messages. Also used to pass the lower 16-bits for streaming messages.

UMC[00...11]CTLx00000804_address000D0034 (UMC::Phy::UctDatWriteOnlyShadow)

Read-only, Volatile.

Read-only view of the csr UctDatWriteOnly

`_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0034; UMC[11:00]CTL=00[B:0]5_1000h; DataPortWrite=UMC::ApbCmd[Address]`

Bits	Description
15:0	UctDatWriteOnlyShadow . Read-only, Volatile. Used to pass the upper 16-bits for streaming messages. Not used in passing major messages.

UMC[00...11]CTLx00000804_address000D0040 (UMC::Phy::Dfi0DebugControl)

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0040; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address000D0041 (UMC::Phy::Dfi0DebugCapture0)

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0041; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:12	Reserved.
11:0	Dfi0DebugCapture0. Read-only, Volatile. Reserved for future use.

UMC[00...11]CTLx00000804_address000D0042 (UMC::Phy::Dfi0DebugCapture1)

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0042; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:10	Reserved.
9:0	Dfi0DebugCapture1. Read-only, Volatile. Reserved for future use.

UMC[00...11]CTLx00000804_address000D0043 (UMC::Phy::Dfi0DebugPerfCtrEn)

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0043; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	Dfi0DebugPerfCtrEn. Read-write. Reset: 0. Reserved for future use.

UMC[00...11]CTLx00000804_address000D0044 (UMC::Phy::Dfi0DebugPerfCtr)

Read-only, Volatile.

Reserved for future use.

_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0044; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	Dfi0DebugPerfCtr. Read-only, Volatile. Reserved for future use.

UMC[00...11]CTLx00000804_address000D0048 (UMC::Phy::Dfi1DebugControl)

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0048; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:0	RFU. Read-write. Reset: 0000h.

UMC[00...11]CTLx00000804_address000D0049 (UMC::Phy::Dfi1DebugCapture0)

Read-only, Volatile.

Reserved for future use.

_umc0_chiptype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0049; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:12 Reserved.

11:0 **Dfi1DebugCapture0**. Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000D004A (UMC::Phy::Dfi1DebugCapture1)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiptype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D004A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:10 Reserved.

9:0 **Dfi1DebugCapture1**. Read-only, Volatile. Reserved for future use.**UMC[00...11]CTLx00000804_address000D004B (UMC::Phy::Dfi1DebugPerfCtrEn)**

Read-write. Reset: 0000h.

Reserved for future use.

_umc0_chiptype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D004B; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**

15:1 Reserved.

0 **Dfi1DebugPerfCtrEn**. Read-write. Reset: 0. Reserved for future use.**UMC[00...11]CTLx00000804_address000D004C (UMC::Phy::Dfi1DebugPerfCtr)**

Read-only, Volatile.

Reserved for future use.

_umc0_chiptype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D004C; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]**Bits Description**15:0 **Dfi1DebugPerfCtr**. Read-only, Volatile. Reserved for future use.

UMC[00...11]CTLx00000804_address000D0099 [Controls reset and clock shutdown on the local microcontroller] (UMC::Phy::MicroReset)

Read-write. Reset: 0001h.

Controls reset and clock shutdown on the local microcontroller.

_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D0099; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:4	Reserved.
3	ResetToMicro. Read-write. Reset: 0. Description: - Set this bit to apply synchronous reset to the microcontroller. - Clear this bit to release reset to the microcontroller. This reset initializes the program counter to begin execution from the boot vector. This reset also clears the interrupt sticky bits.
2	RSVDMicro. Read-write. Reset: 0. RSVD
1	TestWakeup. Read-write. Reset: 0. Reserved. Must always be set to 0.
0	StallToMicro. Read-write. Reset: 1. Description: This field is programmed as follows: - Set this bit to stall the microcontroller by hardware. - Clear this bit to allow the microcontroller to continue executing from its current program counter location. Typically, this bit is used at power up to hold the program counter at the boot vector while BIOS loads the microcontroller program code. While stalled, the microcontroller clocks are gated off for power reduction.

UMC[00...11]CTLx00000804_address000D009A (UMC::Phy::MicroResetPIEEn)

Read-write. Reset: 0000h.

Gives control of the csrMicroReset function to the MicroResetPIE register

_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D009A; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	MicroResetPIEEn. Read-write. Reset: 0. When set, gives control of the csrMicroReset function to the MicroResetPIE register

UMC[00...11]CTLx00000804_address000D00E7 (UMC::Phy::SequencerOverride)

Read-write. Reset: 0080h.

Reserved for PHY training firmware use.

_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D00E7; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:11	Reserved.
10:0	RFU. Read-write. Reset: 080h.

UMC[00...11]CTLx00000804_address000D00FA (UMC::Phy::DfiInitCompleteShadow)

Read-only, Volatile.

dfi_init_complete - Controller Read-only Shadow

_umc0_chiplettype13_inst[UMCWPHY[11:0]]_aliasSMN; UMC[11:00]CTLx00000804_address000D00FA; UMC[11:00]CTL=00[B:0]5_1000h;
DataPortWrite=UMC::ApbCmd[Address]

Bits	Description
15:1	Reserved.
0	DfiInitCompleteShadow. Read-only, Volatile. Description: This csr presents a read-only view (a shadow) of the Register DfiInitComplete which is used by the sequencer to control the state of dfi_init_complete. The value in this Register is not affected by the BlockSeq0Back field of the SequencerOverride register.

